Informatica
from Salesforce

Informatica® Data Privacy Management 10.5.10

User Guide

Informatica Data Privacy Management User Guide
10.5.10
February 2026

© Copyright Informatica LLC 2015, 2026

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

Informatica, the Informatica logo, [and any other trademarks appearing in the document] are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html. Other company and product names may be trade names or trademarks of their respective owners.

Subject to your opt-out rights, the software will automatically transmit to Informatica in the USA information about the computing and network environment in which the Software is deployed and the data usage and system statistics of the deployment. This transmission is deemed part of the Services under the Informatica privacy policy and Informatica will use and otherwise process this information in accordance with the Informatica privacy policy available at https://www.informatica.com/in/privacy-policy.html. You may disable usage collection in Administrator tool.

Portions of this software and/or documentation are subject to copyright held by third parties, including without limitation: Copyright DataDirect Technologies. All rights reserved. Copyright © Sun Microsystems. All rights reserved. Copyright © RSA Security Inc. All Rights Reserved. Copyright © Ordinal Technology Corp. All rights reserved. Copyright © Aandacht c.v. All rights reserved. Copyright Genivia, Inc. All rights reserved. Copyright Isomorphic Software. All rights reserved. Copyright © Meta Integration Technology, Inc. All rights reserved. Copyright © Intalio. All rights reserved. Copyright © Oracle. All rights reserved. Copyright © Adobe Systems Incorporated. All rights reserved. Copyright © DataArt, Inc. All rights reserved. Copyright © ComponentSource. All rights reserved. Copyright © Microsoft Corporation. All rights reserved. Copyright © Rogue Wave Software, Inc. All rights reserved. Copyright © Teradata Corporation. All rights reserved. Copyright © Yahoo! Inc. All rights reserved. Copyright © Glyph & Cog, LLC. All rights reserved. Copyright © Thinkmap, Inc. All rights reserved. Copyright © Clearpace Software Limited. All rights reserved. Copyright © Information Builders, Inc. All rights reserved. Copyright © OSS Nokalva, Inc. All rights reserved. Copyright Edifecs, Inc. All rights reserved. Copyright Cleo Communications, Inc. All rights reserved. Copyright © International Organization for Standardization 1986. All rights reserved. Copyright © ej-technologies GmbH. All rights reserved. Copyright © Jaspersoft Corporation. All rights reserved. Copyright © International Business Machines Corporation. All rights reserved. Copyright © yWorks GmbH. All rights reserved. Copyright © Lucent Technologies. All rights reserved. Copyright © University of Toronto. All rights reserved. Copyright © Daniel Veillard. All rights reserved. Copyright © Unicode, Inc. Copyright IBM Corp. All rights reserved. Copyright © MicroQuill Software Publishing, Inc. All rights reserved. Copyright © PassMark Software Pty Ltd. All rights reserved. Copyright © LogiXML, Inc. All rights reserved. Copyright © 2003-2010 Lorenzi Davide, All rights reserved. Copyright © Red Hat, Inc. All rights reserved. Copyright © The Board of Trustees of the Leland Stanford Junior University. All rights reserved. Copyright © EMC Corporation. All rights reserved. Copyright © Flexera Software. All rights reserved. Copyright © Jinfonet Software. All rights reserved. Copyright © Apple Inc. All rights reserved. Copyright © Telerik Inc. All rights reserved. Copyright © BEA Systems. All rights reserved. Copyright © PDFlib GmbH. All rights reserved. Copyright © Orientation in Objects GmbH. All rights reserved. Copyright © Tanuki Software, Ltd. All rights reserved. Copyright © Ricebridge. All rights reserved. Copyright © Sencha, Inc. All rights reserved. Copyright © Scalable Systems, Inc. All rights reserved. Copyright © jQWidgets. All rights reserved. Copyright © Tableau Software, Inc. All rights reserved. Copyright© MaxMind, Inc. All Rights Reserved. Copyright © TMate Software s.r.o. All rights reserved. Copyright © MapR Technologies Inc. All rights reserved. Copyright © Amazon Corporate LLC. All rights reserved. Copyright © Highsoft. All rights reserved. Copyright © Python Software Foundation. All rights reserved. Copyright © BeOpen.com. All rights reserved. Copyright © CNRI. All rights reserved.

This product includes software developed by the Apache Software Foundation (http://www.apache.org/), and/or other software which is licensed under various versions of the Apache License (the "License"). You may obtain a copy of these Licenses at http://www.apache.org/licenses/. Unless required by applicable law or agreed to in writing, software distributed under these Licenses is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the Licenses for the specific language governing permissions and limitations under the Licenses.

This product includes software which was developed by Mozilla (http://www.mozilla.org/), software copyright The JBoss Group, LLC, all rights reserved; software copyright © 1999-2006 by Bruno Lowagie and Paulo Soares and other software which is licensed under various versions of the GNU Lesser General Public License Agreement, which may be found at http:// www.gnu.org/licenses/lgpl.html. The materials are provided free of charge by Informatica, "as-is", without warranty of any kind, either express or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose.

The product includes ACE(TM) and TAO(TM) software copyrighted by Douglas C. Schmidt and his research group at Washington University, University of California, Irvine, and Vanderbilt University, Copyright (©) 1993-2006, all rights reserved.

This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (copyright The OpenSSL Project. All Rights Reserved) and redistribution of this software is subject to terms available at http://www.openssl.org and http://www.openssl.org/source/license.html.

This product includes Curl software which is Copyright 1996-2013, Daniel Stenberg, <daniel@haxx.se>. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://curl.haxx.se/docs/copyright.html. Permission to use, copy, modify, and distribute this software for any purpose with or without fee is hereby granted, provided that the above copyright notice and this permission notice appear in all copies.

The product includes software copyright 2001-2005 (©) MetaStuff, Ltd. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://www.dom4j.org/ license.html.

The product includes software copyright © 2004-2007, The Dojo Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://dojotoolkit.org/license.

This product includes ICU software which is copyright International Business Machines Corporation and others. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://source.icu-project.org/repos/icu/icu/trunk/license.html.

This product includes software copyright © 1996-2006 Per Bothner. All rights reserved. Your right to use such materials is set forth in the license which may be found at http:// www.gnu.org/software/ kawa/Software-License.html.

This product includes OSSP UUID software which is Copyright © 2002 Ralf S. Engelschall, Copyright © 2002 The OSSP Project Copyright © 2002 Cable & Wireless Deutschland. Permissions and limitations regarding this software are subject to terms available at http://www.opensource.org/licenses/mit-license.php.

This product includes software developed by Boost (http://www.boost.org/) or under the Boost software license. Permissions and limitations regarding this software are subject to terms available at http:/ /www.boost.org/LICENSE_1_0.txt.

This product includes software copyright © 1997-2007 University of Cambridge. Permissions and limitations regarding this software are subject to terms available at http:// www.pcre.org/license.txt.

This product includes software copyright © 2007 The Eclipse Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http:// www.eclipse.org/org/documents/epl-v10.php and at http://www.eclipse.org/org/documents/edl-v10.php.

This product includes software licensed under the terms at http://www.tcl.tk/software/tcltk/license.html, http://www.bosrup.com/web/overlib/?License, http://www.stlport.org/doc/ license.html, http://asm.ow2.org/license.html, http://www.cryptix.org/LICENSE.TXT, http://hsqldb.org/web/hsqlLicense.html, http://httpunit.sourceforge.net/doc/ license.html, http://jung.sourceforge.net/license.txt , http://www.gzip.org/zlib/zlib_license.html, http://www.openldap.org/software/release/license.html, http://www.libssh2.org, http://slf4j.org/license.html, http://www.sente.ch/software/OpenSourceLicense.html, http://fusesource.com/downloads/license-agreements/fuse-message-broker-v-5-3- license-agreement; http://antlr.org/license.html; http://aopalliance.sourceforge.net/; http://www.bouncycastle.org/licence.html; http://www.jgraph.com/jgraphdownload.html; http://www.jcraft.com/jsch/LICENSE.txt; http://jotm.objectweb.org/bsd_license.html; . http://www.w3.org/

Consortium/Legal/2002/copyright-software-20021231; http://www.slf4j.org/license.html; http://nanoxml.sourceforge.net/orig/copyright.html; http://www.json.org/license.html; http://forge.ow2.org/projects/javaservice/, http://www.postgresql.org/about/licence.html, http://www.sqlite.org/copyright.html, http://www.tcl.tk/software/tcltk/license.html, http://www.jaxen.org/faq.html, http://www.jdom.org/docs/faq.html, http://www.slf4j.org/license.html; http://www.iodbc.org/dataspace/iodbc/wiki/iODBC/License; http://www.keplerproject.org/md5/license.html; http://www.toedter.com/en/jcalendar/license.html; http://www.edankert.com/bounce/index.html; http://www.net-snmp.org/about/license.html; http://www.openmdx.org/#FAQ; http://www.php.net/license/3_01.txt; http://srp.stanford.edu/license.txt; http://www.schneier.com/blowfish.html; http://www.jmock.org/license.html; http://xsom.java.net; http://benalman.com/about/license/; https://github.com/CreateJS/EaselJS/blob/master/src/easeljs/display/Bitmap.js; http://www.h2database.com/html/license.html#summary; http://jsoncpp.sourceforge.net/LICENSE; http://jdbc.postgresql.org/license.html; http://protobuf.googlecode.com/svn/trunk/src/google/protobuf/descriptor.proto; https://github.com/rantav/hector/blob/master/LICENSE; http://web.mit.edu/Kerberos/krb5-current/doc/mitK5license.html; http://jibx.sourceforge.net/jibx-license.html; https://github.com/lyokato/libgeohash/blob/master/LICENSE; https://github.com/hjiang/jsonxx/blob/master/LICENSE; https://code.google.com/p/lz4/; https://github.com/jedisct1/libsodium/blob/master/LICENSE; http://one-jar.sourceforge.net/index.php?page=documents&file=license; https://github.com/EsotericSoftware/kryo/blob/master/license.txt; http://www.scala-lang.org/license.html; https://github.com/tinkerpop/blueprints/blob/master/LICENSE.txt; http://gee.cs.oswego.edu/dl/classes/EDU/oswego/cs/dl/util/concurrent/intro.html; https://aws.amazon.com/asl/; https://github.com/twbs/bootstrap/blob/master/LICENSE; https://sourceforge.net/p/xmlunit/code/HEAD/tree/trunk/LICENSE.txt; https://github.com/documentcloud/underscore-contrib/blob/master/LICENSE, and https://github.com/apache/hbase/blob/master/LICENSE.txt.

This product includes software licensed under the Academic Free License (http://www.opensource.org/licenses/afl-3.0.php), the Common Development and Distribution License (http://www.opensource.org/licenses/cddl1.php) the Common Public License (http://www.opensource.org/licenses/cpl1.0.php), the Sun Binary Code License Agreement Supplemental License Terms, the BSD License (http:// www.opensource.org/licenses/bsd-license.php), the new BSD License (http://opensource.org/licenses/BSD-3-Clause), the MIT License (http://www.opensource.org/licenses/mit-license.php), the Artistic License (http://www.opensource.org/licenses/artistic-license-1.0) and the Initial Developer's Public License Version 1.0 (http://www.firebirdsql.org/en/initial-developer-s-public-license-version-1-0/).

This product includes software copyright © 2003-2006 Joe Walnes, 2006-2007 XStream Committers. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://xstream.codehaus.org/license.html. This product includes software developed by the Indiana University Extreme! Lab. For further information please visit http://www.extreme.indiana.edu/.

This product includes software Copyright (c) 2013 Frank Balluffi and Markus Moeller. All rights reserved. Permissions and limitations regarding this software are subject to terms of the MIT license.

See patents at https://www.informatica.com/legal/patents.html.

DISCLAIMER: Informatica LLC provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica LLC does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

See patents at https://www.informatica.com/legal/patents.html.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-05-08

Table of Contents

Preface 21

Informatica Resources 21

Informatica Network 21

Informatica Knowledge Base 21

Informatica Documentation 21

Informatica Product Availability Matrices 22

Informatica Velocity 22

Informatica Marketplace 22

Informatica Global Customer Support 22

Chapter 1: Introduction to Data Privacy Management 23

Data Privacy Management Overview 23

Data Privacy Management Use Cases 24

Data Privacy Process 26

Step 1. Define Locations 26

Step 2. Define Data Store Groups 26

Step 3. Define Data Stores 27

Step 4. Define Data Domains 27

Step 5. Define Classification Policies 27

Step 6. Define and Schedule Scans 27

Step 7. Monitor Scan Jobs 28

Step 8. View Scan Results 28

Step 9. Import User and User Access Metadata 28

Step 10. Define Security Policies 28

User Interface 29

Header 29

Workspaces 30

Logging in to Data Privacy Management 33

Part I: Configuration 35

Chapter 2: System Settings 36

Settings Workspace 36

Settings Workspace Properties 39

Changing the Default Background Color of the User Interface 40

Changing the Default Dashboard 41

Configuring Agent Scan Settings 41

Configuring Data Privacy Settings 41

Specifying Data Store Sensitivity Levels 42

Changing Risk Score Factor Weights 42

Adding the Maximum Risk Score to Data Store Details. . . . 43
Adding Custom Risk Score Factors. . . . 43
Changing Anomaly Factor Weights. . . . 44
Changing the Default Risk Cost. . . . 45
Changing the Default Conformance Ranges. . . . 45
Changing the Default Anomaly Severity Ranges. . . . 46
Configuring Axon Data Governance Integration. . . . 47
Configuring a Brand Logo. . . . 47
Adding a Custom Logo. . . . 48
Image File Specifications. . . . 48

Chapter 3: Locations. . . . 51

Locations Overview. . . . 51
Locations Workspace. . . . 51
Location Properties. . . . 52
Location Management. . . . 54
Creating a Location. . . . 54
Exporting Locations. . . . 54
Importing Locations. . . . 55
Editing a Location. . . . 58
Copying a Location. . . . 58
Changing Data Store Location Assignments. . . . 59
Deleting a Location. . . . 59

Chapter 4: Data Store Groups. . . . 60

Data Store Groups Overview. . . . 60
Data Store Groups Workspace. . . . 60
System-Defined Data Store Groups. . . . 61
Data Store Group Management. . . . 62
Creating Data Store Groups. . . . 62
Managing Data Store Groups. . . . 63
Exporting Data Store Groups. . . . 64

Chapter 5: Extensions. . . . 65

Extensions Overview. . . . 65
Extension Types. . . . 66
Supported Data Store Types for Protection Extension Plugins. . . . 66
Extensions Workspace. . . . 67
Extensions List Page. . . . 68
Extension Properties. . . . 68
Custom Extension Properties. . . . 69
Email Extension Properties. . . . 69
Encryption Protection Extension Properties. . . . 70

Persistent Data Masking - Big Data Protection Extension Properties. . . . 70
Persistent Data Masking - Remote Domain Protection Extension Properties. . . . 71
Service Management Extension Properties. . . . 72
System Log Extension Properties. . . . 73
Extension Management. . . . 73
Adding or Updating a Protection Extension on the Data Domain Details Page. . . . 74
Creating an Extension. . . . 74
Editing an Extension. . . . 75
Deleting an Extension. . . . 75

Chapter 6: Remote Agents. . . . 76

Remote Agents Overview. . . . 76
Guidelines for Remote Agents. . . . 77
Remote Agents Workspace. . . . 77
Remote Agents List Page. . . . 78
Remote Agent Filter Conditions. . . . 78
Protection Remote Agent Details Page. . . . 78
Subject Registry Remote Agent Details Page. . . . 79
Remote Agent Properties. . . . 80
Remote Agent Management. . . . 81
Creating Remote Agents. . . . 81
Editing Remote Agents. . . . 82
Copying Remote Agents. . . . 82
Exporting Remote Agents. . . . 82
Importing Remote Agents. . . . 84
Publishing Protection Remote Agents. . . . 84

Part II: Discovery. . . . 85

Chapter 7: Data Stores. . . . 86

Data Stores Overview. . . . 86
Data Stores Workspace 87
Data Stores List Page. . . . 87
Data Store Details Page. . . . 88
Data Store Status Filters. . . . 89
Data Store Summary. . . . 89
Data Store Management. . . . 90
Creating a Data Store. . . . 91
Copying a Data Store. . . . 93
Editing a Data Store. . . . 94
Fixing a Data Store. . . . 94
Resetting the Classification Results of a Data Store. . . . 95
Updating User Names and Passwords in Bulk. . . . 96

Synchronizing Users. 97
Importing Data Stores. 97
Importing Data Store Aliases. 106
Importing Data Owners. 108
Importing Protection Statuses. 109
Importing Custom Lineage. 112
Importing Connection Assignments. 114
Importing Catalog Metadata. 115
Importing Catalog Resources. 116
Exporting Data Breach Reports. 117
Exporting Data Store Aliases. 119
Exporting Data Store Configurations. 119
Merging Data Stores. 122
Deleting a Data Store. 124

Chapter 8: Data Store Properties. 126

Data Store Properties Overview. 127
Additional Properties. 128
Active Directory Connection Properties. 129
Amazon Redshift Connection Properties. 131
Amazon S3 Connection Properties. 133
Apache Cassandra Connection Properties. 136
Cloudera Navigator Connection Properties. 138
Database Management Connection Properties. 139
Office 365 Outlook. 144
File System Connection Properties. 145
Google BigQuery Connection Properties. 148
Google Drive Connection Properties. 149
Hadoop Distributed File System Connection Properties. 150
Hive Connection Properties. 155
Informatica Cloud Connection Properties. 158
Informatica Data Engineering Integration Properties. 159
Informatica PowerCenter on IBM Db2 Connection Properties. 162
Informatica PowerCenter on Microsoft SQL Server Connection Properties. 166
Informatica PowerCenter on Oracle Connection Properties. 169
Microsoft Azure Blob Storage Connection Properties. 173
Microsoft Azure Data Lake Connection Properties. 175
Microsoft Azure SQL Database Connection Properties. 178
Microsoft Azure SQL Data Warehouse Connection Properties. 181
Microsoft OneDrive Connection Properties. 183
Microsoft SharePoint Connection Properties. 186
Salesforce Connection Properties. 190
SAP Connection Properties. 191

SAP S/4HANA Data Store Connection Properties. . . . 194
Snowflake Connection Properties. . . . 196
Snowflake Advanced Scanner data store properties. . . . 197
SQL Server Integration Services Connection Properties. . . . 199

Chapter 9: Data Domains. . . . 202

Data Domains Overview. . . . 202
Data Domains Workspace. . . . 203
Data Domains List Page. . . . 204
Data Domain Details Page. . . . 204
Data Domain Properties. . . . 205
Metadata Match Condition. . . . 207
Pattern. . . . 207
Reference Table. . . . 208
Rule. . . . 208
Data Match Condition. . . . 209
Pattern. . . . 209
Reference Table. . . . 210
Rule. . . . 211
Conformance Score. . . . 212
Sensitivity Status for Fields in the Validation Range. . . . 213
Data Domains Management. . . . 213
Creating a Data Domain. . . . 214
Exporting Data Domains. . . . 215
Importing Data Domains. . . . 215
Editing a Data Domain. . . . 219
Copying Data Domains. . . . 219
Delinking Associations Between Data Domains and Data Stores. . . . 219
Synchronizing Data Domains. . . . 220
Deleting Data Domains. . . . 220

Chapter 10: Classification Policies. . . . 221

Classification Policies Overview. . . . 221
Classification Policies Workspace. . . . 221
Classification Policies List Page. . . . 222
Classification Policy Details Page. . . . 222
Classification Policy Properties. . . . 223
Data Domains in a Classification Policy. . . . 224
Data Domain Match Condition. . . . 225
Match Any. . . . 225
Match All. . . . 226
Custom. . . . 226
Classification Policy Usage in a Scan Job. . . . 227

Classification Policy Management. . . . 228
Creating a Classification Policy. . . . 229
Exporting Classification Policies. . . . 229
Importing Classification Policies. . . . 230
Editing a Classification Policy. . . . 233
Copying a Classification Policy. . . . 234
Deleting a Classification Policy. . . . 234

Chapter 11: Column Sensitivity 235

Column Sensitivity Overview. . . . 235
The Always Sensitive/Never Sensitive Workspace. . . . 236
Always Sensitive/Never Sensitive List Page. . . . 237
Always Sensitive/Never Sensitive List Management. . . . 237
Specifying a Column as Always Sensitive. . . . 238
Specifying a Column as Never Sensitive. . . . 238
Super Rules and Subset Rules. . . . 238
Guidelines for Always Sensitive/Never Sensitive Rules. . . . 239
Importing Global Sensitivity Status. . . . 240
Marking a Rule as Active. . . . 244
Removing a Rule from the Active Tab. . . . 244
Editing an Always Sensitive/Never Sensitive Rule. . . . 244
Exporting Global Sensitivity Status. . . . 245
Deleting Always Sensitive/Never Sensitive Rules. . . . 245

Chapter 12: Scans. . . . 246

Scans Overview. . . . 246
Compressed Files in Scans. . . . 247
Optical Character Recognition Files in Unstructured Scans. . . . 248
File Types and Specifications. . . . 248
Rules and Guidelines for Optical Character Recognition File Types. . . . 248
Scans Workspace. . . . 249
Scans List Page. . . . 249
Scan Details Page. . . . 250
Scan Properties. . . . 251
How Data Privacy Management Determines Sensitivity Status. . . . 252
Profiling or AgentProfiling Job Step. . . . 252
Global Sensitivity Status. . . . 253
User Updates. . . . 254
Scan Management. . . . 254
Creating a Scan. . . . 254
Running a Scan. . . . 255
Editing a Scan. . . . 258
Copying a Scan. . . . 258

Deleting a Scan. . . . 258
Exporting Scans. . . . 258

Chapter 13: Scan Options. . . . 260

Application Scan Options. . . . 260
Metadata Profiling Scan Option. . . . 261
Data Profiling Scan Option. . . . 261
Big Data Scan Options. . . . 263
Metadata Profiling Scan Option. . . . 264
Data Profiling Scan Option. . . . 264
Process for Cloudera Navigator. . . . 266
Cloud Scan Options. . . . 267
Metadata Profiling Scan Option. . . . 267
Data Profiling Scan Option. . . . 268
Data Integration Scan Options. . . . 270
Process for Informatica Cloud. . . . 271
Process for Informatica Data Engineering Integration. . . . 271
Process for Informatica PowerCenter. . . . 272
Process for SQL Server Integration Services. . . . 272
Database Management Scan Options. . . . 273
Metadata Profiling Scan Option. . . . 274
Data Profiling Scan Option. . . . 275
File Management Scan Options. . . . 276
Metadata Profiling Scan Option. . . . 277
Data Profiling Scan Option. . . . 277
NoSQL Scan Options. . . . 279
Metadata Profiling Scan Option. . . . 280
Data Profiling Scan Option (Cassandra). . . . 280

Chapter 14: Jobs. . . . 283

Jobs Overview. . . . 284
Jobs Workspace. . . . 284
Job Filters. . . . 285
Jobs List Page. . . . 286
Job Details Page. . . . 286
Job Types. . . . 287
Big Data Job. . . . 291
Rules and Guidelines for Big Data Scan Jobs. . . . 292
Cloud Job. . . . 293
Rules and Guidelines for Cloud Scan Jobs. . . . 293
Data Integration Job. . . . 294
Rules and Guidelines for Data Integration Scan Jobs. . . . 295
Database Management Job. . . . 296

Rules and Guidelines for Database Management Scan Jobs. . . . 297
Evaluate Classification Policies Job. . . . 297
Evaluate Risk Scores Job. . . . 298
Evaluate Security Policy Job. . . . 298
File Management Job. . . . 298
Rules and Guidelines for File Management Jobs. . . . 299
Import Job. . . . 299
Rules and Guidelines for Import Jobs. . . . 300
Import Catalog Results Job. . . . 300
Incremental Scan Job. . . . 301
Orchestration Job. . . . 301
Protection Job. . . . 302
Reset Classification Results Job. . . . 303
Salesforce Import job. . . . 303
Subject Data Report Purge Job. . . . 305
Subject Registry Scan Job. . . . 305
Privacy Dashboard Job. . . . 306
Sync Catalog Updates Job. . . . 307
Synchronize Users Job. . . . 307
Job Management. . . . 307
Pausing a Job. . . . 307
Resuming a Job. . . . 308
Stopping a Job. . . . 308
Terminating a Job. . . . 309
Exporting Jobs. . . . 309
Downloading Scan Job Reports. . . . 309
Downloading Rejected Records for an Import Job. . . . 310

Part III: Detection. . . . 312

Chapter 15: Users and User Access. . . . 313

Users and User Access Overview. . . . 313
Configure an Event Share Location. . . . 314
Users Workspace. . . . 315
Import Process for Users and User Access. . . . 315
Import Process from Supported LDAP Directory Services. . . . 316
Import Process from External Systems and Unsupported LDAP Directory Services. . . . 316
Import Process from Salesforce. . . . 317
Import and Synchronization from an LDAP Directory Service. . . . 317
LDAP Connection Properties. . . . 318
Creating an LDAP Connection. . . . 319
Running LDAP Synchronization. . . . 320
Editing an LDAP Connection. . . . 320

Copying an LDAP Connection. . . . 321
Deleting an LDAP Connection. . . . 322
Import and Synchronization from Salesforce. . . . 322
Creating a Salesforce Connection. . . . 323
Running Salesforce Synchronization. . . . 325
Editing a Salesforce Connection. . . . 325
Deleting a Salesforce Connection. . . . 325
User Import from a CSV File. . . . 326
CSV File Format to Import Users. . . . 326
Example CSV File to Import Users. . . . 327
Rules and Guidelines to Import Users. . . . 328
Steps to Import Users. . . . 328
User Group Import from a CSV File. . . . 329
CSV File Format to Import User Groups. . . . 330
Example of CSV File to Import User Groups. . . . 330
Rules and Guidelines to Import User Groups. . . . 331
Steps to Import Groups. . . . 331
User Group Membership Import from a CSV File. . . . 332
CSV File Format to Import Group Memberships. . . . 333
Example of CSV File to Import Group Memberships. . . . 334
Rules and Guidelines to Import Group Memberships. . . . 334
Steps to Import Group Memberships. . . . 335
User Access Import from a CSV File. . . . 335
CSV File Format to Import User Access. . . . 336
Example of CSV File to Import User Access. . . . 338
Rules and Guidelines to Import User Access. . . . 338
Steps to Import User Access. . . . 339
User Aliases Import from a CSV File. . . . 340
CSV File Format to Import User Aliases. . . . 340
Example of CSV File to Import User Aliases. . . . 341
Rules and Guidelines to Import User Aliases. . . . 341
Steps to Import User Aliases. . . . 342
Export of Users and User Access. . . . 342
Exporting Users. . . . 343
Exporting User Groups. . . . 343
Exporting User Group Memberships. . . . 344
Exporting User Access. . . . 344
Exporting User Aliases. . . . 345

Chapter 16: Anomaly Detection. . . . 346

Anomaly Detection Overview. . . . 346
Anomaly Detection Prerequisites. . . . 347
Anomaly Detection Workspace. . . . 348

Anomaly Summary. . . . 349
Top Anomalous Factors Indicator. . . . 350
Top Data Stores Indicator. . . . 351
Top Users Indicator. . . . 352
Anomalous Factors View. . . . 353
Data Stores View. . . . 354
Users View. . . . 355
Anomalies List. . . . 356
Anomaly Details. . . . 359
Anomalous Factors. . . . 361
Anomaly Management. . . . 362
Flagging or Unflagging An Anomaly 363
Marking an Anomaly as Read or Unread. . . . 363
Filtering Anomalies. . . . 365
Exporting Anomalies. . . . 366
Deleting an Anomaly. . . . 367
Suppressing an Anomaly. . . . 367

Chapter 17: Anomaly Suppression Rules. . . . 369

Suppression Rules Overview. . . . 369
Suppression Rules Workspace. . . . 369
Suppression Rules List. . . . 370
Suppression Rules Details. . . . 371
Suppression Rules Management. . . . 372
Filtering Suppression Rules. . . . 372
Editing Suppression Rules. . . . 373
Exporting Suppression Rules. . . . 373
Deleting a Suppression Rule. . . . 374

Chapter 18: Actions. . . . 375

Actions Overview. . . . 375
Actions Workspace. . . . 376
Actions List Page. . . . 376
Action Properties. . . . 377
Custom Action Properties. . . . 377
Email Action Properties. . . . 378
Service Management Action Properties. . . . 380
System Log Action Properties. . . . 382
Placeholders in Custom, Email, Service Management, and System Log Actions. . . . 383
Action Management. . . . 385
Creating an Action. . . . 385
Editing an Action. . . . 386
Copying an Action. . . . 386

Deleting an Action. 386

Chapter 19: Manual Actions. 388

Manual Actions Overview. 388
Creating a Service Management Ticket. 389
Creating a Service Management Ticket from the Anomaly Detection Workspace. 389
Creating a Service Management Ticket from the Proliferation Page. 391
Creating a Service Management Ticket from the Security Policy Violations Workspace. 393
Creating a Service Management Ticket from the Sensitive Fields Page. 395
Creating a Service Management Ticket from the Top Data Domains List Page. 398
Creating a Service Management Ticket from the Top Data Stores Grid Page. 400
Protecting Data. 403
Protecting Data from the Anomaly Detection Workspace. 403
Protecting Data from the Proliferation Page. 405
Protecting Data from the Security Policy Violations Workspace. 406
Protecting Data from the Sensitive Fields Page. 408
Protecting Data from the Top Data Domains List Page. 409
Protecting Data from the Top Data Stores Grid Page. 411
Running a Custom Script. 412
Running a Custom Script from the Anomaly Detection Workspace. 412
Running a Custom Script from the Proliferation Page. 414
Running a Custom Script from the Security Policy Violations Workspace. 416
Running a Custom Script from the Sensitive Fields Page. 418
Running a Custom Script from the Top Data Domains List Page. 420
Running a Custom Script from the Top Data Stores Grid Page. 422
Sending an Email. 423
Sending an Email from the Anomaly Detection Workspace. 424
Sending an Email from the Proliferation Page. 426
Sending an Email from the Security Policy Violations Workspace. 428
Sending an Email from the Sensitive Fields Page. 430
Sending an Email from the Top Data Domains List Page. 432
Sending an Email from the Top Data Stores Grid Page. 434
Writing a System Log Message. 436
Writing a System Log Message from the Anomaly Detection Workspace. 437
Writing a System Log Message from the Proliferation Page. 439
Writing a System Log Message from the Security Policy Violations Workspace. 441
Writing a System Log Message from the Sensitive Fields Page. 443
Writing a System Log Message from the Top Data Domains List Page. 445
Writing a System Log Message from the Top Data Stores Grid Page. 446

Chapter 20: Security Policies. 449

Security Policies Overview. 449
Security Policy Types. 450

Anomaly Policy Example. . . . 450
Data Store Policy Example. . . . 450
Decryption Policy Example. . . . 451
User Activity Policy Example. . . . 451
Security Policies Workspace. . . . 452
Security Policies List Page. . . . 452
Security Policy Details Page. . . . 452
Security Policy Properties. . . . 453
Security Policy Management. . . . 456
Creating a Security Policy. . . . 456
Editing a Security Policy. . . . 458
Copying a Security Policy. . . . 458
Deleting a Security Policy. . . . 458

Chapter 21: Security Policy Groups. . . . 459

Security Policy Groups Overview. . . . 459
Security Policy Groups Workspace. . . . 459
Security Policy Group List Panel. . . . 460
Security Policy Group Details Panel. . . . 460
Security Policy Groups Properties. . . . 461
Security Policy Groups Management. . . . 461
Creating a Security Policy Group. . . . 462
Viewing a Security Policy Group. . . . 462
Editing a Security Policy Group. . . . 462
Copying a Security Policy Group. . . . 463
Deleting a Security Policy Group. . . . 463
Exporting Security Policy Groups, Security Policies, and Actions. . . . 463
Importing Security Policy Groups, Security Policies, and Actions. . . . 464

Chapter 22: Security Policy Violations. . . . 465

Security Policy Violations Overview. . . . 465
Security Policy Violations Workspace. . . . 465
Security Policy Violations Summary. . . . 467
Security Policy Violation Indicators. . . . 468
Violations List. . . . 468
Security Policy Violation Details Page. . . . 471
Violation Details. . . . 472
Security Policy Details. . . . 477
Security Policy Violations Workspace Views. . . . 477
Security Policy Violation Workspace View Properties. . . . 477
Security Policy Violations Management. . . . 479
Flagging or Unflagging A Security Policy Violation. . . . 480
Marking a Violation as Read or Unread. . . . 480

Adding or Removing Security Policy Violation Tags. . . . 481
Exporting Violations. . . . 481
Deleting a Violation. . . . 482

Chapter 23: Tasks. . . . 483

Tasks Overview. . . . 483
Tasks Workspace. . . . 484
Tasks List Page. . . . 484
Task Filter Conditions. . . . 485
DSAR Task Properties. . . . 485
Protection, System, and Data Subject Task Properties. . . . 487
Custom Task Details Page. . . . 489
DSAR Task Details Page. . . . 489
Email Task Details Page. . . . 490
Protection Task Details Page. . . . 492
Service Management Task Details Page. . . . 493
System Log Task Details Page. . . . 494
Task Status. . . . 495
Task Management. . . . 497
Adding Attachments to a Task. . . . 498
Deleting a Task. . . . 499
Editing a Custom Task. . . . 499
Editing an Email Task. . . . 499
Editing a Service Management Task. . . . 500
Editing a System Log Task. . . . 500
Exporting a Task. . . . 500
Marking a Task as Closed. . . . 501
Reassigning a Task. . . . 501
Running a System or Data Subject Task. . . . 502
Protection Task Management. . . . 502
Configuring Protection Tasks. . . . 502
Decrypting a Closed Encryption Task. . . . 506
Editing a Protection Task. . . . 506
Marking a Protection Task as Completed. . . . 506
Marking a Protection Task as Configured. . . . 507
Reconfiguring a Protection Task. . . . 507
Rejecting a Protection Task. . . . 508
Scheduling a Protection Job for a Task. . . . 508
Viewing the Configuration for a Protection Task. . . . 511

Part IV: Analysis 512

Chapter 24: Security Dashboard 513

Security Dashboard Overview 513
Enterprise-Level Summary of Sensitive Data 514
Discovery Bar Indicator 515
Risk Score Indicator 515
Protection Status Indicator 517
Sensitivity Level Indicator 518
Residual Risk Indicator 518
Security Dashboard Indicators 519
Sensitive Data by Data Store Groups Indicator 520
Sensitive Data for Location Indicator 521
Sensitive Data Proliferation Indicator 522
Top Data Domains Indicator 523
Top Data Stores Indicator 525
Top Departments Indicator 527
Top Users Indicator 529
User Access Indicator 531
Pages Accessed from the Security Dashboard 532
Data Domain Details Page 532
Data Domains List Page 534
Data Stores Grid Page 536
Data Stores List Page 538
Departments Page 541
Proliferation Page 542
Sensitive Data by Data Store Group Page 545
Sensitive Data Proliferation by Classification Policy Page 546
Sensitive Data Proliferation by Data Store Group Page 547
Sensitive Data Locations Page 548
Sensitive Fields Page 549
Sensitive Files Page 558
User Access Page 564
User Activity Page 565
User Profile Page 567
Users Page 570
Security Dashboard Management 572
Customizing the Security Dashboard 572
Filtering Information on the Security Dashboard 573
Exporting Information on the Security Dashboard 579

Chapter 25: Summary Workspaces. 581

Summary Workspaces Overview. 581
Data Stores Summary Workspace. 581
Department Summary Workspace. 584
User Activity Summary Workspace. 587
User Activity Summary Indicators. 590
Summary Workspace Management. 590
Exporting Information from the Summary Workspaces. 590

Part V: Protection. 591

Chapter 26: Protection. 592

Protection Overview. 592
Protection Prerequisites. 593
Protection Process Flow. 594

Chapter 27: Encryption Rules. 597

Encryption Rules Overview. 597
Encryption Rules Workspace. 598
Encryption Rules List Page. 598
Encryption Rule Details Page. 598
Encryption Rule Properties. 599
Encryption Techniques. 601
Encryption Rule Management. 601
Creating an Encryption Rule. 601
Exporting Encryption Rules. 602
Importing Encryption Rules. 603
Copying an Encryption Rule. 603

Chapter 28: Risk Simulation Plans. 604

Risk Simulation Plans Overview. 604
Risk Simulation Plans Workspace. 604
Risk Simulation Plans List Page. 605
Risk Simulation Plan Details Page. 605
Risk Simulation Plan Properties. 606
Risk Simulation Plan Management. 607
Creating a Risk Simulation Plan. 607
Exporting a Risk Simulation Plan. 608
Copying a Risk Simulation Plan. 608
Editing a Risk Simulation Plan. 609

Part VI: Privacy . . . 610

Chapter 29: Subject Registry . . . 611

Subject Registry Overview . . . 611
Legal Hold . . . 612
Residency . . . 613
Category . . . 613
Purpose . . . 613
Subject Registry Workspace . . . 614
Subject Registry Search . . . 615
Subject Registry Summary . . . 616
Subject Registry Workspace Management . . . 616
Importing a Master List of Categories . . . 617
Importing a Master List of Purposes . . . 617
Subject Details Page . . . 618
Subject Registry Data Store Details Page . . . 619
DSAR Report Formats . . . 620
Customize the DSAR Report PDF Format . . . 622
Subject Details Page Management . . . 623
Applying and Removing Legal Holds . . . 624
Preparing a DSAR Report . . . 625
Downloading a DSAR Report . . . 625
Creating a Service Management Ticket . . . 626
Running a Custom Script . . . 627
Sending an Email . . . 628

Chapter 30: Privacy Dashboard . . . 629

Privacy Dashboard Overview . . . 629
Enterprise-Level Summary of Subject Data . . . 630
Subjects Indicator . . . 630
Subject Requests Indicator . . . 630
Subject Types Indicator . . . 631
Data Stores Indicator . . . 631
Privacy Dashboard Indicators . . . 631
Top Data Stores by Subjects . . . 631
Subject Requests by Type . . . 633
Subject Data by Location . . . 634
Subject Requests by Due Date . . . 636
Privacy Dashboard Management . . . 637
Filtering Information on the Privacy Dashboard . . . 638
Export Files From the Privacy Dashboard . . . 638

Appendix A: Troubleshooting 640

Data Stores 640

Jobs 641

Protection 645

Scans 646

Subject Registry 648

Appendix B: Updating Keystore and Truststore Certificates to Maintain Secure Communication 649

Overview 649

Updating the Keystore and Truststore with Informatica Certificates 649

Updating the Keystore and Truststore with OpenSSL-Generated Certificates 650

Index 652

Preface

Use the *Data Privacy Management User Guide* to learn how to use the Data Privacy Management application. You can discover, classify, and protect sensitive data. Detect user activity on sensitive data, view data subject information and generate data subject reports for compliance with data privacy regulations.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services and based on real-world experiences from hundreds of data management projects. Informatica Velocity represents the collective knowledge of Informatica consultants who work with organizations around the world to plan, develop, deploy, and maintain successful data management solutions.

You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

Introduction to Data Privacy Management

This chapter includes the following topics:

- Data Privacy Management Overview, 23
- Data Privacy Management Use Cases, 24
- Data Privacy Process, 26
- User Interface, 29
- Logging in to Data Privacy Management, 33

Data Privacy Management Overview

Data Privacy Management is a data security and privacy product that identifies sensitive and personal data across an organization. Use Data Privacy Management to comply with data security standards such as Payment Card Industry (PCI) and Personally Identifiable Information (PII), and data privacy regulations such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA).

You can use Data Privacy Management to monitor and track subject data that you store in data stores within an organization. Create a Subject Registry that consists of all data stores that contain subject data. To comply with governmental or organizational regulations, you might need to identify and provide a summary of all information that you store on a data subject. For example, GDPR, the right-to-be-forgotten, or other data privacy requests. To respond to and comply with such requests, you must be able to identify where you store information and how you share the information. You must also be able to track and delete all information on a subject if a request to do so is raised. Run scans to identify subject data and create a Subject Registry. You can raise subject requests and track the requests to completion. Monitor Subject Registry data and requests from the Data Privacy dashboard.

To identify sensitive data in your organization and sensitive information about a data subject, you scan data stores. A data store is a repository object that connects to the data source that you want to analyze. You can create data stores that connect to relational databases, Cloud applications, Cloudera Navigator, file systems, Hadoop, Informatica Data Engineering Integration, Informatica Intelligent Cloud Services (IICS), Informatica PowerCenter, and SQL Server Integration Services. You scan data stores based on data domain and classification policy configuration.

Data domains define the rules that identify specific columns in a data store. For example, you can create data domains to identify social security numbers or credit card numbers. You assign data domains to classification policies, which are collections of related data domains. You can define classification policies based on data security standards and data privacy regulation requirements. For example, you can create a

classification policy that identifies PII. You include data domains for social security number, first name, last name, and date of birth. You can also create a classification policy that identifies information about a data subject. For example, to comply with GDPR, you create a GDPR classification policy to automate the process of creating reports to satisfy data subject requests about the personal information that your enterprise holds for an individual.

You scan a data store to identify sensitive and private data. Data Privacy Management classifies the data stores based on the match to a classification policy in the scan. Data Privacy Management evaluates PowerCenter mappings to determine sensitive data proliferation. For example, you might have mappings that move sensitive data from one source to multiple targets. To identify if sensitive data is protected, Data Privacy Management verifies if a PowerCenter mapping contains a data masking transformation to mask data.

After you scan data stores, you view the scan results on the **Overview** workspace. In the Security Dashboard view, the **Overview** workspace displays a high-level summary of the total risk score and protection status of all data stores. The dashboard also includes predefined queries for a detailed analysis of the scan results. For example, you can view scan results by region, department, organizational hierarchy, and classification policy. In the Privacy Dashboard view, the **Overview** workspace displays a high-level summary of subject data and indicators that show information about data subjects, data subject access requests (DSARs), data subject types, and data stores associated with data subjects.

You can create security policies and alerts to automate the delegation of data subject reporting, identification of suspicious behavior such as unauthorized use, detection of policy violations, and sensitive data protection with Persistent Data Masking, Dynamic Data Masking, and encryption. You can also intervene manually to protect sensitive data or send notifications about potential data security risks or data privacy regulation compliance.

Example

You are a security officer for a financial company. The company has applications for investment banking, commercial lending, and retail. The data is stored in several relational databases. The CIO wants to conduct an internal audit to verify compliance for PCI data security standards across all applications. You need to identify the databases that contain PCI data such as account numbers, birth dates, and credit cards. You also need to verify if sensitive data is protected and which sensitive data is sent to other databases.

As a security officer, you also need to respond and take action on requests raised by data subjects. You identify what data and in which data stores the organization stores subject data. You provide a summary of that data, and can delete all data stored on a subject if a valid request is raised.

Data Privacy Management Use Cases

Identifying sensitive data across multiple applications and databases can be a time-consuming and inaccurate process. You might have hundreds of applications and databases due to different production, test, and development environments. You might have additional systems due to acquisitions and mergers. Use Data Privacy Management as the single point of access to identify, analyze, and rank sensitive data across an

organization. You can also use Data Privacy Management to identify who can access sensitive data and monitor their level of activity on sensitive data.

For example, you have the following data security goals:

- Simplify the compliance auditing process for sensitive data.

  To help determine the scope of the audit and evaluate compliance to regulatory standards, you need a tool to identify sensitive data, evaluate protection and proliferation status, and provide the credentials and activity of users who have access to the sensitive data.

- Plan and prioritize security and risk investment.

  To help devise a strategy and to prioritize and align resources, you need a tool that quantifies the risk of data stores and users.

- Monitor and identify high risk activity.

  To detect suspicious activity, you need a tool that ranks users by activity on sensitive data, a real-time log of user activity on sensitive data, and the details of the sensitive data accessed.

Use the following capabilities of Data Privacy Management to achieve the data security goals of your organization:

Identify sensitive data

Identify all sensitive data that exists in relational databases, cloud applications, and file systems. You might not know where sensitive data exists in all applications. You might have an internal catalog of applications that contain sensitive data. However, you are unsure of the accuracy and completeness of the catalog. Use Data Privacy Management to validate your internal catalog and to quickly identify where sensitive data exists.

Identify classifications of sensitive data as defined by data security standards

After you discover individual table columns that contain sensitive data, identify if you have groupings of sensitive data as defined by data security standards. If you have groups of data defined as sensitive, you must comply with industry requirements for security management. For example, if you are a merchant that accepts credit cards, you need to comply with PCI data security standards. Use Data Privacy Management to identify the classification of sensitive data.

Identify if sensitive data is protected

Data Privacy Management can identify sensitive data that is masked by a PowerCenter data masking transformation.

You can manually identify the sensitive data protection status for table columns that are not included in PowerCenter mappings.

Identify the proliferation of sensitive data

Sensitive data can originate in one system but can end up in multiple systems. For example, you use PowerCenter to extract, transform, and load data from a production system to various test and development systems. Use Data Privacy Management to identify the target systems to which sensitive data is extracted and loaded.

Identify how many and which users have access to sensitive data

After you discover sensitive data and import users, find out who can access sensitive data. Use Data Privacy Management to determine the number of users with access to sensitive data. You can view the data stores that a user can access, and the user groups that include a specific user.

Detect activity on sensitive data

Data Privacy Management displays details about each user and their activity, and ranks users by activity on sensitive data. You can determine the type of information the user accessed. You can also detect any unusual patterns of access such as increased activity on sensitive data during non-business hours.

Quantify cost and the risk of sensitive data

The dashboard displays the risk cost and risk score of sensitive data. The risk cost is the total monetary loss that the organization might have to bear if sensitive data is lost, exposed to an unauthorized user, or becomes unavailable for use in ongoing operations. The risk score is based on a variety of risk factors that together determine the potential of a security risk. Use the risk cost and risk score values to prioritize and remediate security risks.

Data Privacy Process

To discover and classify sensitive data, you run scans on data stores. A data store is a repository object that includes connection properties to a data source. View the scan results to analyze the identification and classification of sensitive data according to data security standards. View the scan results to analyze the proliferation and protection status of sensitive data.

Perform the following high-level steps to identify sensitive data:

1. Define locations to identify the geographic areas where your data sources exist.
2. Define data store groups to logically group similar data stores.
3. Define data stores to access data from data sources.
4. Define data domains to include the rules that Data Privacy Management uses to identify sensitive columns in a data store.
5. Define classification policies to classify sensitive data based on industry or organization data security standards.
6. Define scans to determine how and when Data Privacy Management discovers and classifies sensitive data for a data store.
7. Monitor the status of Scan jobs to verify that the jobs completed successfully.
8. View the scan results to analyze the discovery and classification of sensitive data.
9. Import user and user access metadata to display which users have access and activity on sensitive data.
10. Define security policies to detect unusual or high-risk situations.

Step 1. Define Locations

Define locations to identify the geographic areas of your data sources. For example, you can create locations for countries, states, or cities.

When you create a location, you specify a pattern of IP addresses or host names to identify the data sources that belong to the location. When you create a data store, Data Privacy Management tries to identify the data store location based on the data store IP address or host name. Alternatively, you can manually assign a location to the data store. After you scan data stores, you can view scan results for all data stores in the same location.

Step 2. Define Data Store Groups

Define data store groups to logically group related data stores and define the hierarchy of the groups. For example, you can group data stores by application type or line of business. When you create a data store, you assign a data store group to the data store. After you scan data stores, you can view scan results for all data stores in a data store group.

Step 3. Define Data Stores

Define data stores to access data sources such as relational databases, Cloud applications, Cloudera Navigator, file systems, Hadoop, Informatica Data Engineering Integration, Informatica Cloud, Informatica PowerCenter, and SQL Server Integration Services. A data store is a repository object that includes a connection to the data source that contains the data that you want to discover and classify. A data store also includes metadata, such as the data owner, department, location, data store group, and security group.

You can create or import data stores. You can import data stores from a parent repository, such as Informatica PowerCenter, or from a file. When you create a scan, you assign data stores to the scan. After you run the scan, you can view the scan results for the data stores. For example, you can scan results for data stores by department, location, or data store group.

Step 4. Define Data Domains

Define data domains to identify the semantics of sensitive columns based on the column name or column data. For example, define data domains to identify columns that contain Social Security numbers or credit card numbers.

Data Privacy Management includes predefined data domains. You can also create data domains or import data domains from a file.

You include data domains in classification policies. When you scan a data store, you select classification policies. Data Privacy Management uses the data domains in the classification policy to perform data domain discovery. The data domain discovery identifies the sensitive columns in the data store based on the metadata or data domain conditions.

Step 5. Define Classification Policies

Define classification policies to classify sensitive data according to industry or organizational data security standards. For example, you can create a classification policy that identifies personally identifiable information (PII). The classification policy includes data domains for social security number, first name, last name, and date of birth.

Data Privacy Management includes predefined classification policies. You can also create or import classification policies. When you create a classification policy, you assign data domains and define data domain match conditions. When you create a scan, you select the classification policies and data stores that you want to evaluate. Data Privacy Management uses the data domain match conditions to evaluate if the data stores include all of the sensitive data domains as defined in the classification policy.

After you scan data stores, you can view scan results for data stores that match a classification policy.

Step 6. Define and Schedule Scans

Define scans to configure how and when Data Privacy Management discovers and classifies sensitive data. For example, you want to scan a database to identify if the database contains sensitive data as defined in the PII classification policy.

When you create a scan, you specify the type of scan to run and the data stores to scan. You assign classification policies to the scan and schedule when the scan runs.

Data Privacy Management runs data domain discovery to identify the sensitive columns in the data store. Data Privacy Management compares the data domains in the classification policy against the metadata and data from the data store. Then, Data Privacy Management evaluates if the data store contains the sensitive data specified in the classification policy. Data Privacy Management uses the data domain match conditions in the classification policy to evaluate if the data store contains the sensitive data specified in the classification policy.

Step 7. Monitor Scan Jobs

Monitor the status of the scan job to verify that the job completed successfully. If the job did not complete successfully, view the job steps to see which step encountered an error. View the job log to get more information about any errors that the job encountered.

Step 8. View Scan Results

View the scan results in the dashboard. The dashboard contains a summary of all data stores that you scanned. For example, you can view the overall risk score and percentage of unprotected data for all data stores. The dashboard also includes predefined queries to analyze the data store results. For example, you can view scan results by locations, data store groups, classification policies, risk levels, or protection status.

Step 9. Import User and User Access Metadata

Import metadata about users and user access to display which users have access and activity on sensitive data.

You can import metadata about users, groups, and group memberships from LDAP directory services and from CSV files. You can import user access and user alias metadata from CSV files. You can import users, user access, and user activity from Salesforce.

When you import user metadata, you schedule an Import job. The Import job adds the user metadata to the Data Privacy Management repository. The Data Privacy Management Service augments scan results and user activity data with the imported user metadata. On the **Overview** workspace, you can view users that have permission to access sensitive data in data stores. When you view user activity on data stores, you can also view information about the user account that performed an event on the sensitive data.

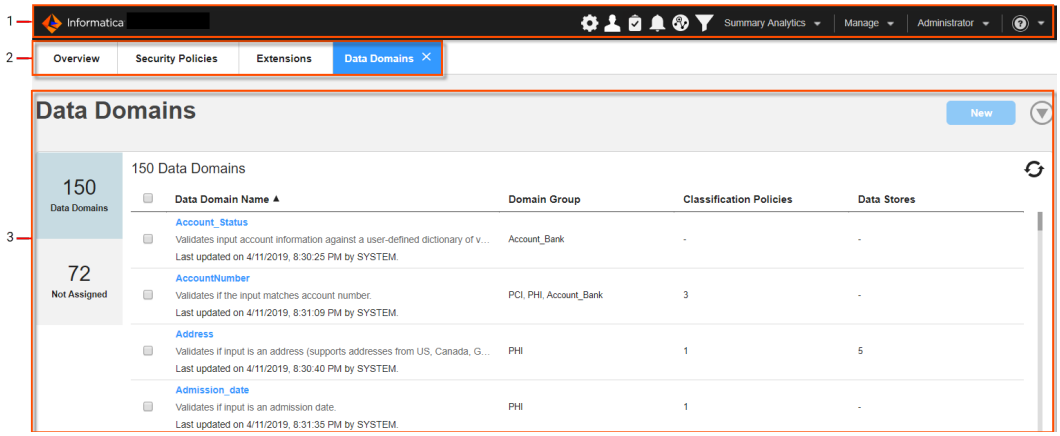
Step 10. Define Security Policies

Set up rules and notifications to identify or monitor specific situations. For example, you want to know when sensitive data moves across locations, when a table in a high risk data store is accessed, or when a user accesses a number of sensitive data records that far exceeds their normal patterns for the same period of time.

User Interface

The user interface includes a header, workspace tabs, and workspaces.

The following image shows the user interface:

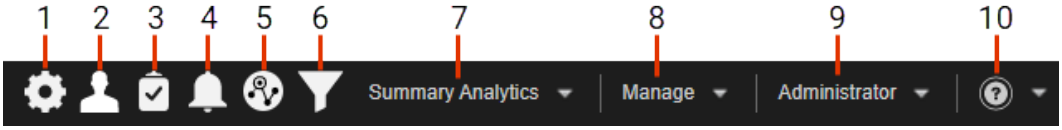


1. Header
2. Workspace tabs
3. Workspace

Header

The header appears at the top of the user interface. Use the header icons to define system settings, view subject registry, view tasks, view security policy violations, view anomalies, filter a workspace list page, and view online help. Use the header menus to view summaries of scan results, open workspaces, change passwords, and log out of Data Privacy Management.

The following image shows the header:



The following table describes the actions you can perform on the header:

Number	Name	Description
1	Settings icon	Open the **Settings** workspace and configure the background color for the user interface, sensitivity levels, risk score factor weights, anomaly factor weights, risk cost and conformance score criteria, anomaly severity levels, and an Axon resource name.
2	Subject Registry icon	Open the **Subject Registry** workspace and view subject details and history. Create an automated DSAR report for a data subject.
3	Tasks icon	Open the **Tasks** workspace and configure, manage, and run tasks.
4	Security Policy Violations icon	Open the **Security Policy Violations** workspace and view a list of incidents that met the conditions specified in security policies.

Number	Name	Description
5	Anomaly Detection icon	Open the **Anomaly Detection** workspace and view incidents of unusual activity.
6	Filter icon	Sort a workspace list page by specific properties.
7	Summary Analytics menu	Open a workspace that shows scan result summaries for data stores, departments, or user activity.
8	Manage menu	Open a workspace for actions, always sensitive/never sensitive fields, classification policies, data domains, data store groups, data stores, encryption rules. extensions, jobs, locations, remote agents, risk simulation plans, scans, security policies, security policy groups, suppression rules, and users.
9	<User name> menu	Change your password and log out of Data Privacy Management.
10	Help icon	Access online help for the current workspace and access the **Start Up** window.

Workspaces

A workspace is a page that contains all data and functions related to an aspect of sensitive data in your organization.

You can access workspaces from the main menu or from icons in the header. You can also access workspaces from tasks that you perform in other workspaces, such as in the following examples:

- When you select a data store on the **Data Store Groups** workspace, the **Data Store** workspace opens in a new tab.
- When you click a blue object in a column, such as a number in the **Sensitive Fields** column, a user's name, or a number of classification policies, the associated workspace and specific page or list opens in a new tab.

To view any workspace, you must have access privileges. To perform tasks in a workspace, you must have privileges for specific types of tasks. Multiple open workspaces display as tabs. Use the tabs to navigate between workspaces.

The following table describes the workspaces included in Data Privacy Management:

Workspace Name	Description
Actions	Create reusable custom, email, service management, and system log actions. After you create a reusable action, you can include it in security policies that allow the action type. Reusable actions standardize the task that Data Privacy Management must perform when security policy violations that include the actions occur or when the conditions of the policy are met. Reusable actions also decrease the amount of time it takes to create a security policy.
Always Sensitive/Never Sensitive	View and manage Always Sensitive and Never Sensitive rules that specify column sensitivity for a specific data domain, or for any data domain. Import or export the global sensitivity status of columns to the Data Privacy Management repository.
Anomaly Detection	View a generated list of unusual incidents that Data Privacy Management detects. For example, view a list of users who accessed a large volume of sensitive data during non-work hours.

Workspace Name	Description
Classification Policies	Create, import, view, and manage classification policies to classify groupings of sensitive data according to industry or organizational data security standards.
Data Domains	Create, import, view, and manage data domains to identify the semantics of sensitive columns based on the column name or column data. Manage data domains that exist in the Model repository.
Data Store Groups	Create, view, and manage data store groups. Logically group data stores into hierarchical groupings with parent and child levels, and then drag data stores into the groups.
Data Stores	Create data stores or import data stores from a file to access data from all supported data sources. View a list and a summary of all data stores, view or edit data store properties, test data store connections, merge duplicate data stores, synchronize users, update the username and password associated with a data store, upload configuration files to fix data stores that have incomplete or missing connection properties, import or export data store information, export data breach reports, import protection status details for sensitive fields in a data store, and import catalog metadata and resources.
Data Stores Summary	View data store analytics. For example, view the number of data stores by classification policy, department, sensitivity level, region, data store group, proliferation status, and location.
Department Summary	View data store information based on departments. For example, view the number of departments by risk score, protection status, sensitivity level, proliferation, classification policy, data value, and sensitive data volume.
Encryption Rules	Create and manage encryption rules and associated encryption keys to protect sensitive fields in Cloudera Hive data stores. Define encryption rules with the following encryption methods: - Change metadata - Preserve format and metadata - Preserve metadata Specify encryption rule definitions and keys in data domains and encryption protection task properties.
Extensions	View, configure, and manage extension connection properties to simplify the use of these properties in actions and tasks. Specify extensions in actions you attach to security policies that run tasks when the policies are violated or when the policy conditions are met. You must specify an extension when you manually configure tasks. You can only specify extensions if the extensions have an Active status and your role includes privileges to manage extensions. Data Privacy Management includes the following extensions: - Custom - Email - Protection: Encryption, Persistent Data Masking - Big Data, and Persistent Data Masking - Remote Domain - Service Management: ServiceNow - System Log
Jobs	View and manage past, present, and future jobs. For each job, view the job type, job properties, the status of jobs and job steps, and the user who scheduled the job. View or export job logs that contain informational, warning, and error messages that the job encountered. Pause, resume, and terminate jobs and job steps.

Workspace Name	Description
Locations	Create, view, and manage locations to identify the geographic areas for each data source. View a list and a summary of all locations. For example, view how many data stores are assigned to a location.
Overview	In the Privacy Dashboard, which is the default view when you log in to Data Privacy Management, view and access enterprise-level information about subject data in the Subject Registry such as the number of subjects, subject requests, subject types, and data stores. In the Security Dashboard, view enterprise-level metrics such as the percentage of scanned data stores, and the risk score, protection status, sensitivity level, number of policy impressions, and risk cost of scanned data stores. View summary results based on elements such as geography, sensitive data proliferation, departments, data domains, data stores, data store groups, user activity, and risk score. Drill down to view details that help identify and remediate sensitive and privacy data risks. To customize the indicators that display on the **Overview** workspace, select **Actions** > **Customize Dashboard**. To change the dashboard view from the Privacy Dashboard to the Security Dashboard, select the option on the **Settings** workspace.
Remote Agents	Create and manage remote connections to data sources with sensitive data you encrypt or with subject registry information. Associate a subject registry remote agent with unstructured data stores to perform scans on unstructured data sources. Configure the host and administrator port, associate the remote agent with data stores, test the connection, export connection details to a CSV file, and publish the data store connection details to the remote agent.
Risk Simulation Plans	Model the projected effects of data protection, including the estimated risk score, protection status, and estimated residual risk cost. For example, model the outcomes of supporting a budget request for a masking solution, create a risk simulation plan to model the projected ROI of prioritizing a data protection initiative, or estimate the financial risk of unprotected sensitive data in the event of a data breach.
Scans	Create, view, and manage scans to configure how and when Data Privacy Management discovers and classifies sensitive data. View a list and a summary of all scans, including scans that are in progress or scans that failed. View the scan properties, scan status, and a summary of the scan results.
Security Policies	Create, view, and manage security policy rules to identify specific situations. Data Privacy Management includes the following security policy types: - Anomaly - Data Store - Decryption - User Activity For example, create a security policy to determine when sensitive data moves across locations or when a user accessed a table in a high-risk data store.
Security Policy Groups	Create security policy groups to organize related security policies. For example, create a security policy group for all PCI related security policies. Then, use security policy groups as a filter option to quickly locate a violation on the **Security Policy Violations** workspace.
Security Policy Violations	View a list of violations that Data Privacy Management generates when the conditions in a security policy are met.

Workspace Name	Description
Settings	Specify the Security or Privacy dashboard view, the background color for the user interface, an Axon resource name, and the default settings for remote agent scans, data privacy, sensitivity levels, risk score factor weights, anomaly factor weights, risk cost and conformance score criteria, and anomaly severity levels.
Subject Registry	Locate and manage sensitive information related to a data subject. From the **Subject Details** page, prepare and download an automated DSAR report. View details about sensitive data held for a data subject, including whether the information is shared with third parties. Apply and clear legal holds for data subjects. Create custom, email, and service management tasks to fulfill data subject requests.
Suppression Rules	View and manage the rules that prevent specific anomalous factors from triggering an anomaly.
Tasks	View and manage tasks that are associated with data stores to which you have access. Custom, DSAR, email, encryption, protection, service management, and system log tasks run when you create the tasks, when you schedule the tasks to run, or when the conditions of a security policy that contains an action to perform the tasks are met. Configure and schedule protection tasks on the **Tasks** workspace, and optionally rollback/decrypt a closed encryption task that ran successfully. Run a DSAR task when you create a DSAR report for a data subject on the **Subject Details** page.
User Activity Summary	View scan results based on user activity. For example, view the number of sensitive data impressions that users accessed by data domain, sensitivity level, and day of the week.
Users	Import metadata about users and user access to display user activity on sensitive data. Import metadata about users, user groups, and group memberships from LDAP directory services and from CSV files. Import user access and user alias metadata from CSV files. Import users, user access, and user activity from Salesforce.

Logging in to Data Privacy Management

Use the Data Privacy Management URL to log in to the application. Contact the administrator to get the URL. The administrator configures the URL when the administrator creates the Data Privacy Management Service in Informatica Administrator.

The administrator can log in to Data Privacy Management directly from the service URL in Informatica Administrator. When you log in to Data Privacy Management, you enter a user name and password.

1. Start a Microsoft Internet Explorer or Google Chrome browser.
2. In the **Address** field, enter the following URL:

   `http[s]://<host name or host IP address>:<port number>`

   For example, `https://abc62dept:6243`.
3. Press **Enter**.

   The login page appears.
4. Enter your user name and password.

5. Click **Log In**.

   Data Privacy Management displays the **Overview** workspace.

   Note: The first time you log in to Data Privacy Management, the **Start Up** window appears.

To log out of Data Privacy Management, click **<User name>** menu **> Log Out**.

Part I: Configuration

This part contains the following chapters:

- System Settings, 36
- Locations, 51
- Data Store Groups, 60
- Extensions, 65
- Remote Agents, 76

CHAPTER 2

System Settings

This chapter includes the following topics:

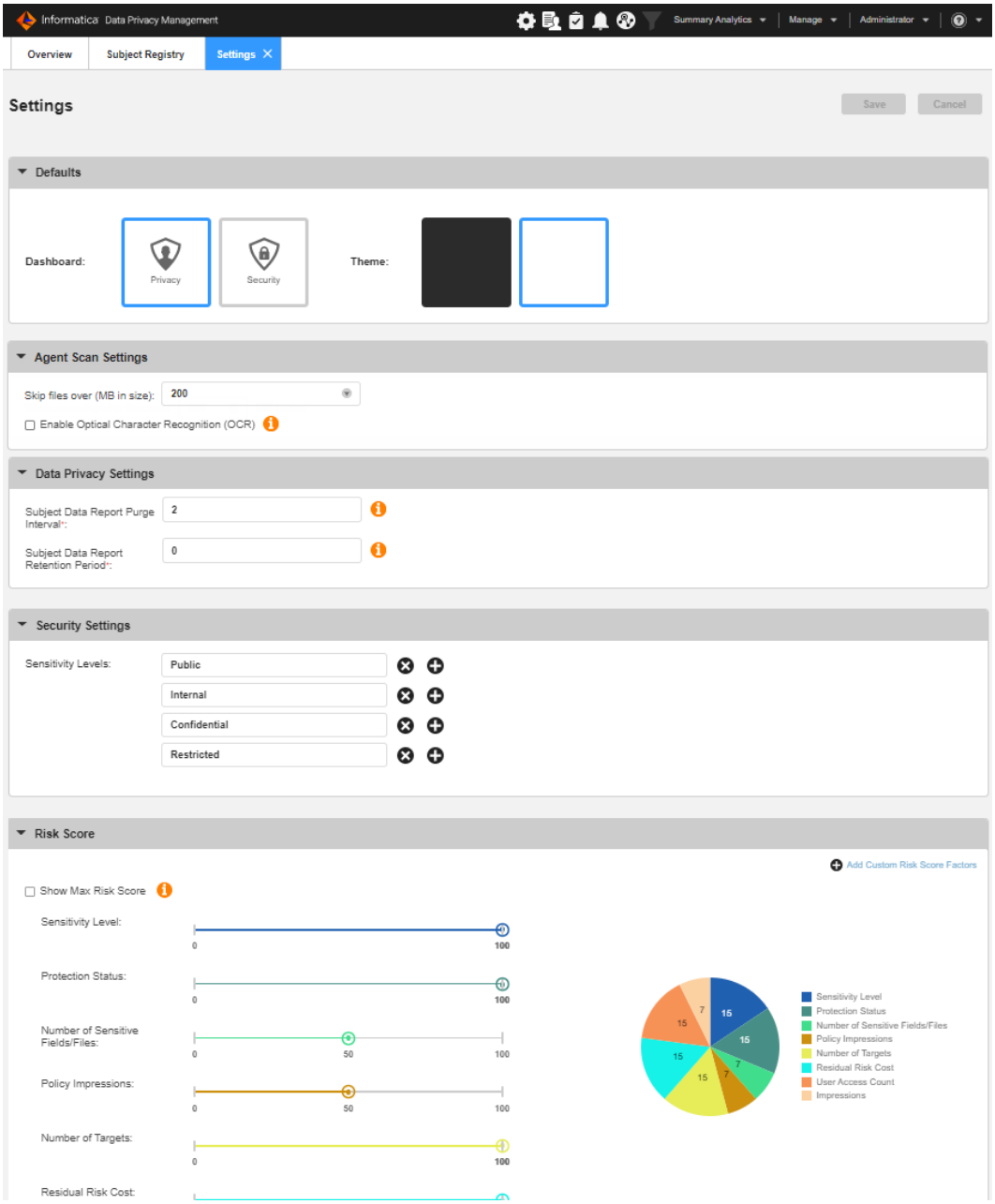
- Settings Workspace, 36
- Settings Workspace Properties, 39
- Changing the Default Background Color of the User Interface, 40
- Changing the Default Dashboard, 41
- Configuring Agent Scan Settings, 41
- Configuring Data Privacy Settings, 41
- Specifying Data Store Sensitivity Levels, 42
- Changing Risk Score Factor Weights, 42
- Adding the Maximum Risk Score to Data Store Details, 43
- Adding Custom Risk Score Factors, 43
- Changing Anomaly Factor Weights, 44
- Changing the Default Risk Cost, 45
- Changing the Default Conformance Ranges, 45
- Changing the Default Anomaly Severity Ranges, 46
- Configuring Axon Data Governance Integration, 47
- Configuring a Brand Logo, 47

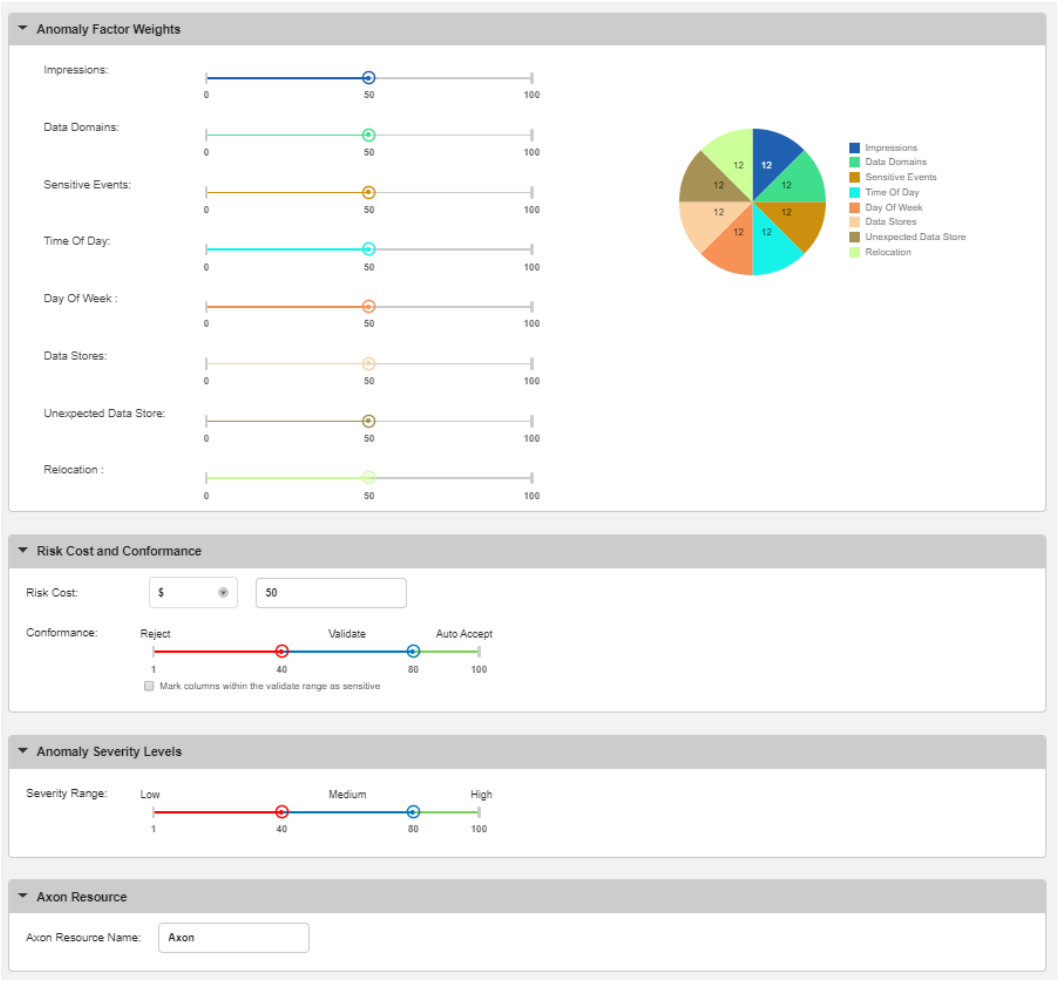
Settings Workspace

On the **Settings** workspace, you can customize system settings that determine the color theme, default dashboard, remote agent scan settings, data privacy settings, data store sensitivity levels, risk score and anomaly factor weighting criteria, risk cost and conformance ranges, and severity ranges for anomalous user behavior. You can also specify an Axon resource name.

To access the **Settings** workspace, click the **Settings** icon on the header.

The following image shows the default values on the **Settings** workspace:




Anomaly Factor Weights
Impressions:
Data Domains:
Sensitive Events:
Time Of Day:
Day Of Week :
Data Stores:
Unexpected Data Store:
Relocation :
0
50
100
12
Impressions
Data Domains
Sensitive Events
Time Of Day
Day Of Week
Data Stores
Unexpected Data Store
Relocation
Risk Cost and Conformance
Risk Cost:
$
50
Conformance:
Reject
Validate
Auto Accept
1
40
80
100
Mark columns within the validate range as sensitive
Anomaly Severity Levels
Severity Range:
Low
Medium
High
Axon Resource
Axon Resource Name:
Axon

Settings Workspace Properties

Customize the **Settings** workspace properties to specify global settings in Data Privacy Management that best meet the needs of your organization or role.

The following table describes the properties on the **Settings** workspace:

Category	Description
Defaults	**Dashboard**: The dashboard that opens when you login to Data Privacy Management. Default is Privacy Dashboard. **Theme**: The background color for the user interface. Options are black and white. Default is a black background.
Agent Scan Settings	The maximum file size that Data Privacy Management scans for unstructured data stores associated with a remote agent. When you scan an unstructured data store, Data Privacy Management skips files that exceed the specified size, in MB. Default is 200 MB. You can enable Optical Character Recognition in unstructured scans. Default is disabled. You cannot disable OCR after you enable and save the change.
Data Privacy Settings	In the **Subject Data Report Purge Interval** property, specify the number of days between each Subject Data Report Purge job. Default is 2. In the **Subject Data Report Retention Period** property, specify the number of days to retain DSAR reports. To always retain the subject data, enter `0`. Default is 0.
Security Settings	**Sensitivity Levels**: The degrees of data sensitivity. You can define a maximum of five sensitivity levels that you select when you define classification policies. If a data store matches a classification policy, Data Privacy Management assigns the sensitivity level of the classification policy to the data store. The default sensitivity levels from lowest to highest sensitivity are: Public, Internal, Confidential, and Restricted. You can rename the defaults, add one sensitivity level, or delete a sensitivity level if it is not associated with a classification policy.
Risk Score	When you scan data stores, Data Privacy Management calculates a risk score for each data store based on the weight of risk score factors. You can customize the risk score factor weights. At least one factor must have a weight that is greater than zero. A pie chart dynamically adjusts the weight of each factor when you move a slider. The **Settings** workspace includes the following default risk score factors and weights: - Sensitivity Level. Default weight is 15 percent. - Protection Status. Default weight is 15 percent. - Number of Sensitive Fields/Files. Default weight is 7 percent. - Policy Impressions. Default weight is 7 percent. - Number of Targets. Default weight is 15 percent. - Residual Risk Cost. Default weight is 15 percent. - User Access Count. Default weight is 15 percent. - Impressions. Default weight is 7 percent. To add a maximum of five new factors, click **Add Custom Risk Score Factor** and define settings for each new factor. Choose **View Max Risk Score** to view the maximum risk score value on data store details pages in addition to the average risk score. The average risk score appears by default.

Category	Description
Anomaly Factor Weights	Anomaly detection identifies irregular or unusual patterns of user activity on sensitive data. Data Privacy Management detects anomalies based on anomalous factors and assigns a score to each anomaly. The **Settings** workspace includes the following anomalous factors - Impressions - Data Domains - Sensitive Events - Time Of Day - Day Of Week - Data Stores. - Unexpected Data Store - Relocation The default weight of each factor is 12 percent. You can customize the anomaly score factor weights. At least one factor must have a weight that is greater than zero. A pie chart dynamically adjusts the weight of each factor when you move a slider.
Risk Cost and Conformance	**Risk Cost**: An estimate of the default monetary cost to an organization if sensitive data is lost, exposed to an unauthorized user, or becomes unavailable for use in ongoing operations. The default cost is $50 for each occurrence of sensitive data. You can change the default currency unit and amount. **Conformance**: When a Scan job runs data profiling, Data Privacy Management calculates a conformance score based on the percentage of column values that must match a data domain for fields to be identified as sensitive, not sensitive, or requiring validation. The default conformance matching ranges are: - Reject: 0-39% - Validate: 40-79% - Accept: 80-100% You can move the sliders to change the conformance limits. To automatically identify fields in the Validate range as sensitive when you scan a data store, select **Mark columns within the validate range as sensitive**. You can also change the default conformance limits in data domain properties.
Anomaly Severity Levels	**Severity Range**: Based on anomaly scores, Data Privacy Management identifies anomalies as high, medium, or low severity. Default anomaly severity ranges are: - Low. Anomaly scores of 1-39. - Medium. Anomaly scores of 40-79. - High. Anomaly scores of 80-100.
Axon Resource	**Axon Resource Name**: Integrates Data Privacy Management with Axon Data Governance. Enter the name of the Axon resource created in Enterprise Data Catalog.

Changing the Default Background Color of the User Interface

You can change the default background theme color of the user interface from black to white.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. In the **Defaults** pane, click the white square.
3. Click **Save**.

Changing the Default Dashboard

You can change the default dashboard that opens when you log in to Data Privacy Management.

To change the default dashboard, your user account must fulfil any of the following requirements:

- You belong to the Administrator group.
- You are assigned the domain Administrator role.
- You belong to a group that is assigned the domain Administrator role.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. In the **Defaults** pane, click the required dashboard.
3. Click **Save**.

Configuring Agent Scan Settings

You can configure agent scan settings such as the default maximum file size for unstructured data stores and inclusion of optical character recognition (OCR) files in scans.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. In the **Agent Scan** pane, select a file size from the list.
3. Click **Save**.

   When you scan unstructured data stores associated with a remote agent, Data Privacy Management skips files that exceed the specified file size.
4. Optional. To include OCR files in remote agent scans, select the Enable Optical Character Recognition (OCR) option and click **Save**.

   If you enable the option, you can choose to include image file types in scans. Scans process all PDF files as optical character files.

   Note: You cannot disable OCR after you enable and save the change. If you need to disable the option, contact Informatica Global Customer Support.

   You must have installed all prerequisite files on the remote agent machine. For information about prerequisite files for the remote machine, see the Informatica Data Privacy Management Installation and Configuration Guide.

Configuring Data Privacy Settings

You can change the default interval between Subject Data Report Purge jobs and the number of days to retain subject data.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.

2. Scroll to the **Data Privacy Settings** pane.
3. In the **Subject Data Report Purge Interval** property, enter the interval, in days, between Subject Data Report Purge jobs.

   Default is 2.
4. In the **Subject Data Report Retention Period** property, enter the number of days to retain DSAR reports.

   Default is 0, which indicates that Data Privacy Management always retains DSAR reports.
5. Click **Save**.

Specifying Data Store Sensitivity Levels

You can change the default sensitivity level names, add another sensitivity level, and delete sensitivity levels.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears. The **Security Settings** pane lists the default sensitivity levels from least to most sensitive.
2. In the **Security Settings** pane, enter the name of the least sensitive level in the first text box. Continue entering sensitivity level names, up to a maximum of five levels.

   Use the **Add** and **Delete** icons to manage sensitivity levels.
3. Click **Save**.

When you create a classification policy, you select one of the sensitivity levels defined on the **Settings** workspace. When a data store matches a classification policy, Data Privacy Management assigns the sensitivity level of the classification policy to the data store.

Related Topics:

- "Sensitivity Level Indicator" on page 518
- "Departments Page" on page 541
- "Creating a Classification Policy" on page 229

Changing Risk Score Factor Weights

You can customize the risk score calculation based on the importance of the default risk score factors to your organization.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. In the **Risk Score** pane, move the sliders to increase or decrease the default weight of each factor that Data Privacy Management uses to calculate the risk score.

   Note: At least one factor must have a weight above zero.
3. Click **Save**.

4. If you ran data store Scan jobs in the past, a confirmation prompt asks if you want to save the new risk score factor weights for previous and future risk score calculations. To confirm the new settings, click **Yes**.

   Data Privacy Management uses the new settings to calculate risk scores for data stores in future scans. The Recalculate Risk Scores job runs to recalculate and update previous risk scores.

After the job finishes, the new risk scores appear on the **Security Dashboard**, the **Scan Details** page, and the **Data Stores** list page.

Related Topics:

- "Risk Score Indicator" on page 515

Adding the Maximum Risk Score to Data Store Details

The **Risk Score** field that appears on data store widgets displays the average risk score. The average risk score might not provide a realistic picture if the scan includes multiple classification policies. You can update the configuration details to view the maximum risk score in addition to the average risk score.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. In the **Risk Score** pane, select **Show Max Risk Score**.
3. Click Save.

   Refresh the browser page to view the **Max Risk Score** column in the data store details.

Adding Custom Risk Score Factors

You can customize the default risk score calculation by defining new evaluation criteria.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. In the **Risk Score** pane, click **Add Custom Risk Score Factor**.

   The **Custom Factor 1** panel appears.



3. Move the slider to select a weight for the custom risk score factor.

4. Enter a unique name for the custom risk score factor. The name cannot match the other risk score factors or data store attributes such as data store name, user name, category, and data store type.

   Each custom risk score factor appears as a property on the **Data Store Details** page.

5. Select one of the following options:
   - **List of Values**. You can enter a minimum of two values and a maximum of five values for the custom risk factor. Each value must be unique within the factor.
   - **Range**. You can specify a minimum of two ranges and a maximum of five ranges for the custom risk factor. Each range must not overlap with any other range within the factor.
6. Enter a value or range. Use the **Add** or **Delete** icons as needed.
7. Move the slider to specify a weight for each value or range between 0 and 100.
8. Optionally, select a default value.

   For existing data stores, a default value does not display in the **Custom Risk Score Factor** property. When you create new data stores, the default value appears in the **Custom Risk Score Factor** property.

9. To create another custom risk factor, repeat steps 1-8.

   You can create a maximum of five custom risk score factors.

10. Click **Save**.

    If you ran Scan jobs in the past, a confirmation prompt asks if you want to save the new risk score factor setting for previous and future risk score calculations.

11. Click **Yes**.

Data Privacy Management performs the following actions:

- Calculates the risk scores of data stores in future Scan jobs using the new settings.
- Starts the Recalculate Risk Scores job to recalculate and update the risk scores of previously scanned data stores.
- Updates the new risk scores on the **Security Dashboard**, the **Scan Details** page, and the **Data Stores** list page.
- Displays the custom risk score factor on the **Data Store Details** page for existing and future data stores.
- Displays the default value of the custom risk score factor on the **Data Store Details** page for future data stores.

Related Topics:

- "Risk Score Indicator" on page 515

Changing Anomaly Factor Weights

You can customize the way that Data Privacy Management calculates anomaly scores based on the importance of the anomaly factors to your organization. You can change the default weight for each anomaly factor.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.

2. To change the default weight of an anomaly factor, move the corresponding slider in the **Anomaly Factor Weights** pane.

You must set at least one factor weight to a value greater than zero.

The pie chart automatically adjusts the percentage weight for all factors when you move a slider.

3. Click **Save**.

   Data Privacy Management calculates future anomaly scores based on the new settings.

Related Topics:

- "Anomalous Factors" on page 361

Changing the Default Risk Cost

You can change the default currency unit and amount for the estimated cost each time that sensitive data is at risk of exposure. The default risk cost is $50.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. Go to the **Risk Cost and Conformance** pane.
3. To change the default **Risk Cost** currency, select one of the following options:
   - €
   - ¥
   - £
   - ₹
4. To change the default **Risk Cost** amount, enter the value in the field.
5. Click **Save**.

When you create a new classification policy, the **Cost for Each Impression** property displays the default risk cost amount defined on the **Settings** workspace. You can change the risk cost value, but you cannot change the currency unit for the **Cost for Each Impression** property.

Related Topics:

- "Residual Risk Indicator" on page 518
- "Classification Policy Properties" on page 223

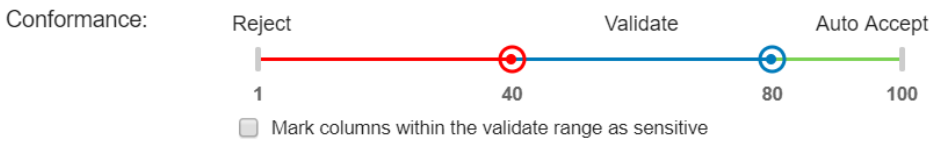
Changing the Default Conformance Ranges

When a Scan job runs data profiling, Data Privacy Management calculates a conformance score based on the percentage of column values that must match a data domain for fields to be identified as sensitive, not sensitive, or requiring validation. You can change the default conformance ranges that determine the sensitivity status of a field.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.

2. In the **Risk Cost and Conformance** pane, move the sliders on the **Conformance** band to change the Reject, Validate, or Auto Accept ranges.



3. To automatically identify fields in the Validate range as sensitive when you scan a data store, select **Mark columns within the validate range as sensitive**.
4. Click **Save**.

When you create a data domain, the conformance properties on the **Settings** workspace are the default settings on the **Data Match** tab. You can change the default settings for each data domain.

Related Topics:

- "Downloading Scan Job Reports" on page 309
- "Conformance Score" on page 212

Changing the Default Anomaly Severity Ranges

Based on anomaly scores, Data Privacy Management identifies anomalies as high, medium, or low severity. You can change the default anomaly score ranges for each severity level.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. In the **Anomaly Severity Levels** pane, move the sliders on the **Severity Range** band to change severity limits.
3. Click **Save**.

   A confirmation prompt asks you to confirm the changes.
4. Click **Yes**.

   The new anomaly score ranges will determine the severity level of future anomalies.

When Data Privacy Management detects anomalies, the **Anomaly Detection** workspace displays a list of the anomalies and details about each anomaly occurrence. The summary section of the workspace shows the total number of anomalies with high, medium, and low severity levels.

Related Topics:

- "Anomaly Summary" on page 349

Configuring Axon Data Governance Integration

Configure the Axon resource name to integrate Data Privacy Management with Axon Data Governance.

Before you enter the Axon resource name, log in to Enterprise Data Catalog and create an Axon resource.

1. Click the **Settings** icon in the header.

   The **Settings** workspace appears.
2. In the **Axon Resource** pane, enter the name of the Axon resource in Enterprise Data Catalog.

   Data Privacy Management integrates with the Axon Data Governance instance that you enter.

Log in to Axon Data Governance. From the Admin Panel, click **Customize and Configure** > **System Settings** > **Group**. Configure Enterprise Data Catalog and Data Privacy Management settings.

Configuring a Brand Logo

You can customize the Data Privacy Management logo. Customize the logo if you want your brand logo to appear in place of the Informatica logo in Data Privacy Management.

You can customize the logo that appears in the following locations:

- Favicon. The icon that appears with the Data Privacy Management web page.
- Login page
- Splash screen
- Startup screen
- Main page. The icon that appears on the main header on all pages.

You can choose to use the same image in all locations or use different images in specific locations. For example, if the background color differs on the splash screen or startup screen, an image created for a different background color might not appear properly on these screens. You can create different image files with appropriate colors to appear on these screens

Read the reference information in the Image File Specifications section to understand recommended file sizes for each location.

Related Topics:

- "Image File Specifications" on page 48

Adding a Custom Logo

Create the logo files in the required location to customize the logo.

All folder and file names are case sensitive.

1. In the `<Data Privacy Management installation directory>\tools` folder, create a `productbranding` folder.
2. In the `productbranding` folder, create a folder with the name of the Data Privacy Management Service that the logo refers to.

   For example, if the Data Privacy Management Service name is DPM_Service, then create the following folder path: `<Data Privacy Management installation directory>\tools\productbranding\DPM_Service`
3. Copy a `favicon` image file that contains the image for the favicon to the service folder created.

   The file can be an `.svg`, `.png`, or `.ico` file.
4. Copy a `logo` image file that contains the required image to the service folder created.

   The image is used on the main page, header, startup screen, and splash screen. The file can be either an `.svg` file or a `.png` file.
5. Optional. To use different images in specific locations, create the required files in the folder:
   - `loginPagelogo`. The image to use on the login page.
   - `mainHeaderLogo`. The image to use on the main header page.
   - `startupScreenLogo`. The image to use on the startup screen.
   - `splashScreenLogo`. The image to use on the splash screen.

   The specific file overrides the default `logo` file and appears in the specified location.

Image File Specifications

Identify supported file types and read the specifications and recommended file settings when you create image files to customize the logo.

Note: All file names are case sensitive.

Create files with the following file names and supported types in the corresponding location:

File Name	Location	File Type
favicon	Appears with the name of the web page.	You can use one of the following file types: - .ico - .png - .svg `.ico` is the recommended file type.
logo	Appears on the following pages unless an override image file is created for the page: - Login page - Main header on all pages - Startup screen - Splash screen	You can use one of the following file types: - .png - .svg `.svg` is the recommended file type.
loginPageLogo	Login page	You can use one of the following file types: - .png - .svg `.svg` is the recommended file type.
mainHeaderLogo	Main header on all pages	You can use one of the following file types: - .png - .svg `.svg` is the recommended file type.
startupScreenLogo	Startup screen	You can use one of the following file types: - .png - .svg `.svg` is the recommended file type.
splashScreenLogo	Splash screen	You can use one of the following file types: - .png - .svg `.svg` is the recommended file type.

The following specifications apply to image files in each location:

Location	Specification
Login page	The login page image has the following specifications: - Container Height: 30 px - Background: #000000(Black)
Splash screen	The splash screen image has the following specifications: - Container Height: 40 px - Background: #F2F2F2(Light)
Startup screen	The startup screen image has the following specifications: - Container Height: 33 px - Background: #F2F2F2(Light)
Main header	The main header image has the following specifications: - Container Height: 26 px - Background: #1d1d1d(Dark)

Read the following recommendations for file types:

File Type	Recommendations
.svg	For the Favicon image, follow the browser recommendations. For all other images, do not assign a width and height to the image. The image takes the container height and the width is calculated based on the aspect ratio.
.png	For the Favicon image, follow the browser recommendations. For all other images, allow at least 40 pixel height to ensure that you can use the logo in all locations.
.ico	Follow the browser recommendations.

CHAPTER 3

Locations

This chapter includes the following topics:

- Locations Overview, 51
- Locations Workspace, 51
- Location Properties, 52
- Location Management, 54

Locations Overview

A location is a geographic area of a data center, a geographic area from which user activity originates, or both. For example, you can create a location to indicate the country, state, or city of the data center or user activity.

Create a location for each data center that hosts the data sources that you want to scan. When you create a location, you specify a pattern of host names or IP addresses to identify the data sources that belong to the location. When you create or import a data store, Data Privacy Management tries to identify the data store location based on the location configuration. If there is a match, Data Privacy Management assigns the location to the data store. You can also manually assign a location to the data store when you update data store properties.

Create a location for each geographic area from which user activity originates. When you create a location, you specify an IP address subnet to identify a geographic origin of user activity. When evaluating user activity events for anomalous behavior, Data Privacy Management tries to identify the event location by matching the IP address in the event to the IP address range in the location configuration.

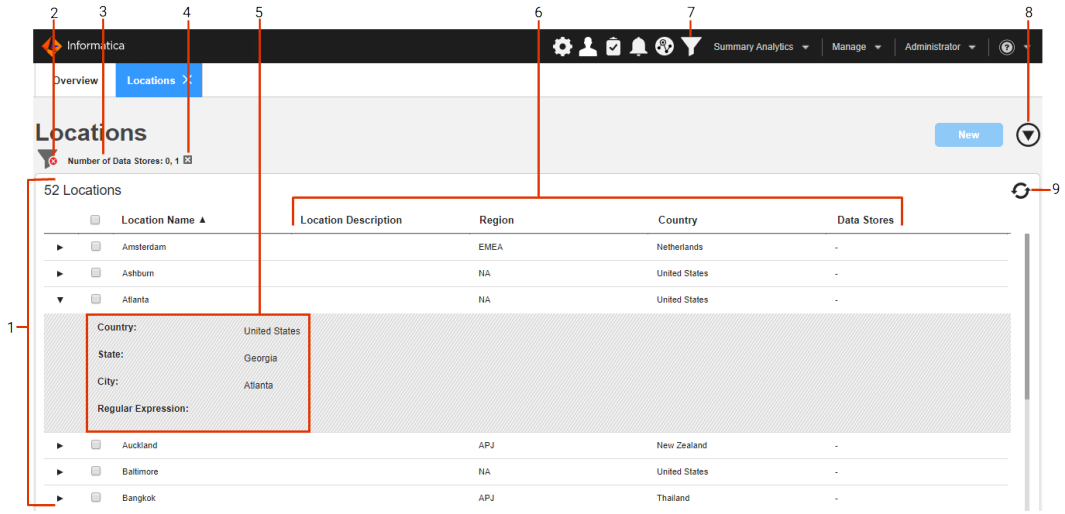
After you scan data stores, you can access the dashboard or the data store summary to view scan results for all data stores assigned to a location. For example, you can view the scan results for all data stores in the same country or state. You can view user activity anomalies based on the location or relocation anomalous factors on the **Anomaly Detection** workspace.

Locations Workspace

You can manage locations on the **Locations** workspace. The **Locations** workspace displays a list of all locations. You can expand a location to view the location details.

To access the **Locations** workspace, click **Manage** > **Locations**.

The following image shows an example of the **Locations** workspace:



1. Locations
2. Clear Filter icon
3. Filter conditions
4. Clear Filter Condition icon
5. Location details
6. Location summary
7. Filter
8. Actions menu
9. Refresh icon

Location Properties

Configure the location properties to identify the geographic areas of data sources and user activity.

A location includes the following properties:

Property	Description
Name	Required. The name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 255 characters, contain spaces, or include the following special characters: ~ ! $ % ^ & * ( ) +
Description	Optional. Enter a long description of the location that is a maximum of 255 characters.

Property	Description
Regular Expression	Optional. Enter a Java-based regular expression that describes a pattern of values for the host name or IP address of the data stores that belong to the location. Data Privacy Management validates the syntax of the expression when you save a location, and evaluates the expression to assign a default value for the data store location when you create, edit, or import a data store. If a data store matches more than one location, Data Privacy Management assigns the location of the first matched regular expression. If a data store does not match a location, the location is shown as Unknown. You can change the location if you do not want to use the matched location. For example, enter `10\\.65\\.123.*` to match data stores with an IP address that starts with `10.65.123`. Enter `injrlx.*` to match data stores with a host name that starts with `injrlx`. The host name match is not case sensitive. The regular expression only identifies the location of data stores. To specify the location for user activity, configure the **IP Address / Range** property. For information about creating regular expressions, see tutorials and documentation for regular expressions on the internet such as http://www.regular-expressions.info/tutorial.html.
Region	Required. Select one of the following values: - APJ. Asia Pacific. - EMEA. Europe, Middle East, and Africa. - LATAM. Latin America. - NA. North America.
Country	Optional. Select a country from the list of countries for the selected region.
State	Optional. For Canada and the United States, select a state from the list.
City	Optional. For Canada and the United States, select a city from the list.
IP Address / Range	Optional. Enter an IP address range or subnet that identifies a geographic location from which a user performs an event on a data store. Enter IP addresses in the CIDR format. For example, `10.10.10.0/6`. If you enter a format that is not CIDR, you receive an error when you save the location. To add multiple entries, click the **+** icon. Data Privacy Management evaluates the IP address range to assign a location to user activity events and detect anomalous user behavior. Data Privacy Management evaluates user activity events for anomalous behavior and attempts to identify the event location by matching the IP address in the event to the IP address range in locations. If there is a match, Data Privacy Management assigns a location to the event. If there are no matches, the event location is shown as Unknown. Locations help you detect anomalous behavior based on the origin of user activity. For example, a user performs an event from one location. An hour later, the user performs an event from a location that is too far to travel within an hour. Data Privacy Management detects an anomaly to indicate potential shared or compromised credentials. To specify the location of data stores, configure the **Regular Expression** property.
VPN	Optional. Specifies whether the **IP Address / Range** value is from a VPN provider. If enabled, Data Privacy Management does not detect anomalies based on distance for the Relocation anomaly factor. If disabled, Data Privacy Management detects anomalies based on distance for the Relocation anomaly factor.

Location Management

You can create and manage locations on the **Locations** workspace. Location information is stored in the Data Privacy Management repository.

You can create, view, edit, copy, or delete locations. You can import or export information about locations.

Creating a Location

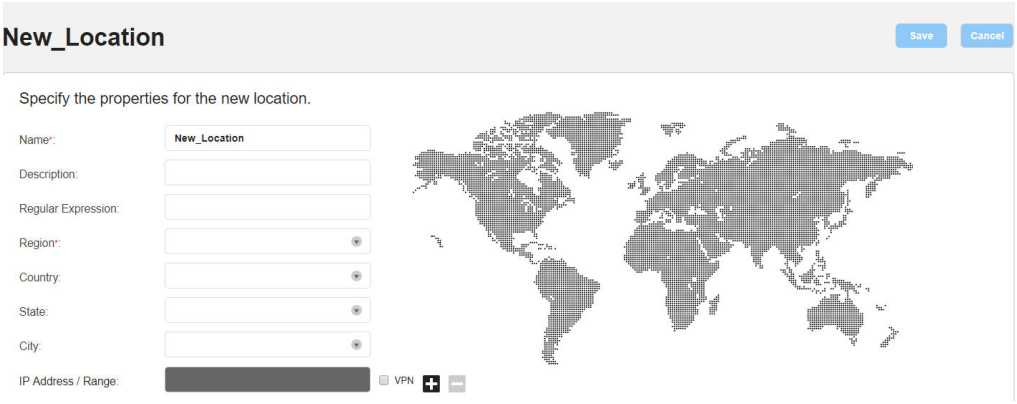
Create a location for each geographic area of a data center or geographic area from which user activity originates. For example, you can create a location to indicate the country, state, or city of the data center or user activity.

1. Click **Manage** > **Locations**.

   The **Locations** workspace appears.

2. Click **New**.

   The **New_Location** page appears.



3. Enter the location properties.
4. Click **Save**.

Exporting Locations

You can export a list of locations to a CSV file. The export file includes a list of all locations and the corresponding location properties. If you filtered the location list, the export file contains the locations that match the active filter conditions.

You can use the exported CSV file to create, update, or delete locations in the Data Privacy Management repository. The export file contains the same format that is required to import locations. You can edit and then import the file to update location information.

1. Click **Manage** > **Locations**.

   The **Locations** workspace appears.

2. From the **Actions** menu, select **Export**.

   A dialog box appears with an export file named `Locations.csv`.
   Save the file to a directory on your machine, and optionally edit the file name. If you export the file again, the default file name includes an incremental number such as `Locations (1).csv`.

3. Open the file to view or edit the location information.

For more information about importing locations from CSV files, see "Importing Locations" on page 55.

Importing Locations

When you import location information from a CSV file, the Import job runs immediately, using the details in the file to create, update, or delete a location in the Data Privacy Management repository.

1. Click **Manage** > **Locations**.

   The **Locations** workspace appears.

2. From the **Actions** menu, select **Import**.

   The **Import Locations** page appears.

3. Enable or disable the **Replace Duplicates with Items Imported** option.

   By default, the option is enabled. The Import job updates the location properties in the repository with the values in the CSV file.

   Warning: If a column in the CSV file contains a null value, the Import job replaces the value in the repository with a null value.

   If you disable the option, the Import job skips duplicate rows.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the Import job finishes, you can view a list of the imported locations on the **Locations** workspace.

CSV File Format to Import Locations

Create a CSV file to create, update, or delete information about locations in the Data Privacy Management repository. The content of the CSV file must be in the order and format that the Import job requires. The CSV file must contain all of the column headings. However, not all columns require values.

For more detailed descriptions of the column values, see "Location Properties" on page 52. The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Column Value Description
1	Name	Required. The name is not case sensitive and must be unique within the repository. The name cannot exceed 255 characters, contain spaces, or include the following special characters: ~ ! $ % ^ & * ( ) + If blank, the import job rejects the row.
2	Description	Optional. Enter a long description of the location that is a maximum of 255 characters.

Column Order	Column Heading	Column Value Description
3	Regular Expression	Optional. Enter a Java-based regular expression that describes a pattern of values for the host name or IP address of the data stores that belong to the location. Data Privacy Management validates the syntax of the expression when you save a location, and evaluates the expression to assign a default value for the data store location when you create, edit, or import a data store. If the syntax is not valid, the Import job rejects the row.
4	Region	Required. Region for the location. Enter one of the following values: - APJ - EMEA. - LATAM - NA If blank, the Import job rejects the row.
5	Country	Optional. Enter a valid country value. To view a list of valid values, create a new location in the user interface. If the entry is not valid, the Import job rejects the row. If blank, the Import job inserts an empty string.
6	State	Optional. For Canada and the United States, enter a valid state value. To view a list of valid values, create a new location in the user interface. If the entry is not valid, the Import job rejects the row. If blank, the Import job inserts an empty string.
7	City	Optional. For Canada and the United States, enter a valid city value. To view a list of valid values, create a new location in the user interface. If the entry is not valid, the Import job rejects the row. If blank, the Import job inserts an empty string.

Column Order	Column Heading	Column Value Description
8	IP Address Range/VPN	Optional. Enter an IP address range or subnet that identifies a geographic location from which a user performs an event on a data store. Data Privacy Management evaluates the IP address range to assign a location to user activity events and detect anomalous user behavior. Use the following syntax to enter values: *<IP address in CIDR format>*,*<VPN>* Where - *<IP address CIDR format>* includes the range of IP addresses. For example, `10.10.10.0/6`. - *<VPN>* specifies if the IP address or range is from a VPN provider. Enter `Y` to indicate the IP address is from a VPN provider. Enter `N` to indicate the IP address is not from a VPN provider. If the entry is a VPN address, anomalies are not detected based on distance for the Relocation anomaly factor. Use a semicolon (;) to separate multiple entries. For example: `100.100.0.1/32,Y;10.10.10.3/9,N;1.1.1.1/2,N` If the entry is not in CIDR format, the Import job rejects the row.
9	Action	Optional. Instructs the import job to create, update, or delete a location in the repository. Enter one of the following values: - U. Creates or updates a location in the repository. If the location does not exist, the Import job creates the location. If the location exists and the **Replace Duplicates with Items Imported** option is enabled, the Import job updates the location. If the location exists and the **Replace Duplicates with Items Imported** option is not enabled, the Import job skips the row. **Warning:** If a column in the CSV file contains a null value, the Import job replaces the value in the repository with a null value. - D. Deletes a location from the repository. If the location does not exist, the Import job rejects the row. If blank, default is U. Creates or updates a location in the repository.

Rules and Guidelines to Import Locations

Review the rules and guidelines to understand how to manage the CSV file and how the Import job processes data from the CSV file.

Consider the following rules and guidelines when you import locations from a CSV file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines a location. The Name column defines a unique location. If the Name column in one row is the same as the Name column in a second row, the Import job treats the second row as a duplicate of the first row.
- If the CSV file contains duplicate entries, the Import job processes the entries in the order in which the rows appear in the file.

- The column values cannot contain the following keywords or terms:

```
SELECT FROM
SELECT UNION
DELETE FROM
UPDATE SET
<SCRIPT
</SCRIPT>
ALERT(
"/>"
```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Editing a Location

When you edit a location, you can change all of the location properties.

Changes to the location properties do not retroactively change objects that are associated with the location. For example, if you change the regular expression, data stores that are assigned to the location do not change. If you change the IP address range, user activity events that matched the IP address do not change. Changes apply the next time you create or import a data store, or the next time Data Privacy Management evaluates anomalous behavior.

1. Click **Manage** > **Locations**.

   The **Locations** workspace appears.
2. Select the check box next to the location that you want to edit.
3. From the **Actions** menu, select **Edit**.

   The **Edit <location name>** page appears.
4. Edit the location details.
5. Click **Save**.

Copying a Location

To quickly create a location that has similar properties to an existing location, you can copy a location and then edit the location properties.

1. Click **Manage** > **Locations**.

   The **Locations** workspace appears.
2. Select the check box next to the location that you want to copy.
3. From the **Actions** menu, select **Duplicate**.

   The**Copy_of_<location name>** page appears.
4. Edit the location properties.
5. Click **Save**.

Changing Data Store Location Assignments

When you create a data store, you assign a location to the data store. When you import data stores from a parent data store, such as Informatica PowerCenter, or from a file, Data Privacy Management assigns a location to the data stores.

You can change location assignments for data stores manually or in bulk.

To update data store locations manually, go to the **Data Stores** workspace. Select the data store with a location you want to change. From the **Actions** menu, select **Edit** and select a different value in the **Location** property.

To update data store locations in bulk, go to the **Data Stores** workspace. From the **Actions** menu, select **Export** > **Data Stores**. The export CSV file includes a list of all data stores and the corresponding data store properties. Update the location assignment in the file. Then, select **Import** from the **Actions** menu to import the file.

Related Topics:

- "Editing a Data Store" on page 94
- "Importing Data Stores" on page 97

Deleting a Location

You can delete a location if the location is not assigned to any data stores. To delete a location that is assigned to data stores, first change the location assignment for the data stores associated with the location.

1. Click **Manage** > **Locations**.

   The **Locations** workspace appears.
2. Select the check box next to the location that you want to delete.
3. From the **Actions** menu, select **Delete**.

   A message asks you to confirm if the deletion.
4. Click **Yes**.

CHAPTER 4

Data Store Groups

This chapter includes the following topics:

- Data Store Groups Overview, 60
- Data Store Groups Workspace, 60
- System-Defined Data Store Groups, 61
- Data Store Group Management, 62

Data Store Groups Overview

A data store group is a collection of related data stores. You might create data store groups for departments, such as Finance, Sales, and Marketing. You might also create data store groups for lines of business, such as investment banking, wealth management, and commercial finance.

Data Privacy Management includes system-defined data store groups to help you manage data stores and data store group assignments.

You can also create data store groups and define the hierarchy of the groups. For example, you can group data stores by application type or line of business. When you create a data store, you can assign a data store group to the data store.

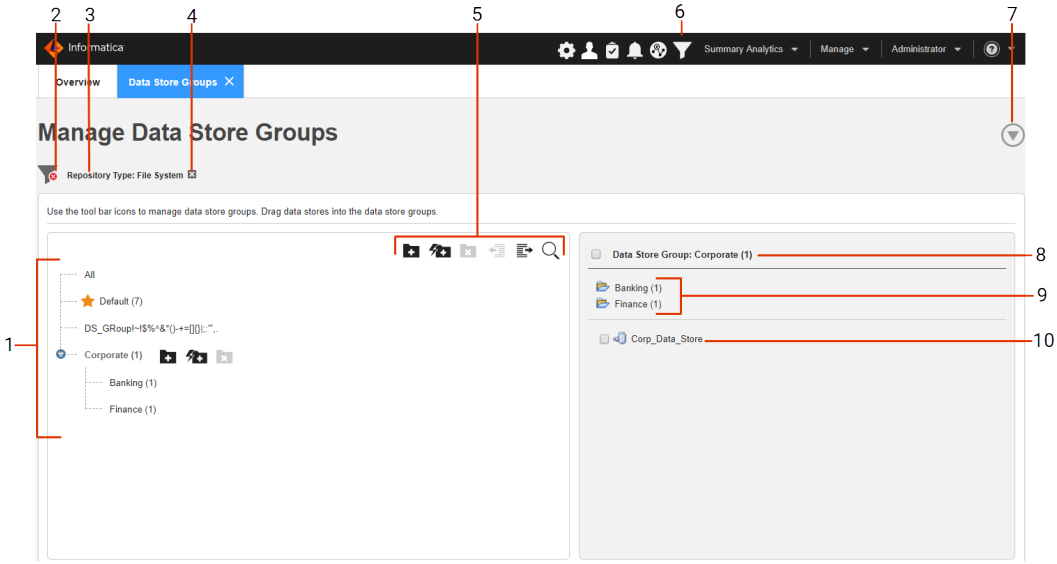
After you scan data stores, you can view the scan results for all data stores assigned to a data store group on the **Overview** workspace or the **Data Stores Summary** workspace. For example, you can view the scan results for all data stores in a Finance data store group.

Data Store Groups Workspace

You can manage data store groups and data store group assignments on the **Data Store Groups** workspace.

To access the **Data Store Groups** workspace, click **Manage** > **Data Store Groups**.

The following image shows an example of the **Data Store Groups** workspace:



1. Data store groups
2. Clear Filter icon
3. Filter conditions
4. Clear Filter Condition icon
5. Data store group toolbar icons
6. Filter
7. Actions menu
8. Selected data store group
9. Nested data store groups for the selected data store group
10Data stores assigned to the selected data store group

System-Defined Data Store Groups

System-defined data store groups help you manage data stores and data store group assignments. You can temporarily assign data stores to the Default data store group until you set up data store groups. You can use

the All data store group to view a list of data stores across all data store groups and to change data store group assignments.

The following table describes the system-defined data store groups:

System-Defined Data Store Group	Description
Default	When you create a data store, you must assign a data store group. You might want to temporarily assign a data store to the Default data store group until you set up data store groups. When you import data stores from a parent repository, such as Informatica PowerCenter, the data stores are assigned to the Default data store group. You can add or remove data stores from the Default data store group, but you cannot edit or delete the Default data store group. Also, you cannot nest another data store group within the Default data store group.
All	The All data store group displays a list of data stores across all data store groups. You can use the All data store group to view data stores or to change the data store group assignment for multiple data stores. When you click the All data store group, a list of all data stores appears. To change the data store group assignment, select multiple data stores. Then, drag the data stores to another data store group. The All data store group only appears in the **Data Store Groups** workspace. You cannot edit or delete the All data store group, assign data stores to the All data store group, or nest another data store group within the All data store group.

Data Store Group Management

On the **Data Store Groups** workspace, you can create and manage data store groups and data store assignments. You can also export a list of data store groups to a CSV file.

Creating Data Store Groups

On the **Data Store Groups** workspace, you can create a data store group for each logical grouping of data stores. You can nest a data store group within another data store group to create hierarchical groupings.

1. Click **Manage** > **Data Store Groups**.

   The **Data Store Groups** workspace appears.
2. Click the **Add Group** icon.

   A new data store group appears in the hierarchy as an empty text box.
3. Enter a data store group name in the text box and press **Enter**.

   The data store group name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot include a forward slash (/) or exceed 100 characters.

   The new data store group appears in the hierarchy in alphabetical order.

Managing Data Store Groups

On the **Data Store Groups** workspace, you can manage hierarchical groupings of data store groups. You can edit or delete data store groups except for the All and Default data store groups. You cannot delete a data store group if the group has a nested group or if the group has data store assignments.

You can use the data store group toolbar icons to perform the following tasks:

- Create a data store group at the same level as a selected group.
- Create a data store group nested within a selected group.
- Delete a data store group.
- Move a data store group one level up.
- Move a data store group one level down.
- Search for data store groups.

Managing Data Store Group Assignments

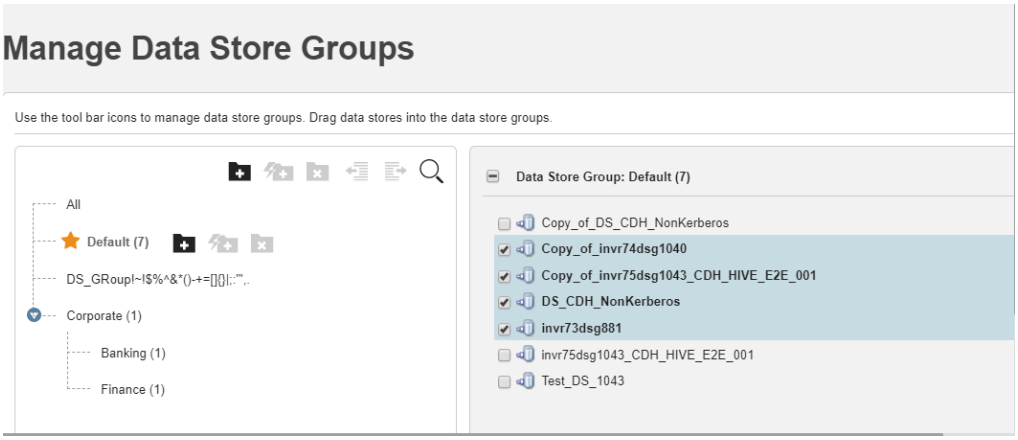
When you create a data store, you assign it to a data store group to the. When you import data stores from a parent data store, such as from Informatica PowerCenter or from a file, Data Privacy Management assigns the Default data store group to the data stores.

You can change data store group assignments manually on the **Data Store Groups** workspace. You can change data store group assignments manually or in bulk on the **Data Stores** workspace.

To change data store group assignments manually on the **Data Store Groups** workspace, perform the following steps:

1. Click **Manage** > **Data Store Groups**.

   The **Data Store Groups** workspace appears.



2. Select a data store group.

   When you select a data store group, a list of data stores that are assigned to the data store group appears. Select the All data store group to view all data stores across all data store groups.

3. Select one or more data stores.

4. Drag the data stores to the data store group to which you want to assign the data store.

To change data store group assignments manually on the **Data Stores** workspace, select a data store. From the **Actions** menu, select **Edit** and select a different value in the **Data Store Group** property.

To change data store group assignments in bulk on the **Data Stores** workspace, select **Export** > **Data Stores** from the **Actions** menu. The export CSV file includes a list of all data stores and the corresponding data store

properties. Update the data store group assignment in the file. Then, select **Import** from the **Actions** menu to import the file.

Exporting Data Store Groups

You can export a list of data store groups to a CSV file. The export file includes a list of all data store groups and the data store group properties. If you filtered the data store groups, the export file contains the data store groups that match the active filter conditions.

1. Click **Manage** > **Data Store Groups**.

   The **Data Store Groups** workspace appears.

2. From the **Actions** menu, select **Export**.

   A dialog box appears with an export file named `DataStoreGroups.csv`.

   Save the file to a directory on your machine, and optionally edit the file name. If you export the file again, the default file name includes an incremental number such as `DataStoreGroups (1).csv`.

3. Open the file.

CHAPTER 5

Extensions

This chapter includes the following topics:

- Extensions Overview, 65
- Extension Types, 66
- Supported Data Store Types for Protection Extension Plugins, 66
- Extensions Workspace, 67
- Extension Properties, 68
- Extension Management, 73

Extensions Overview

Data Privacy Management stores extension details so you can configure connection properties one time, and then re-use the connection properties when you specify the extension name in actions, security policies, or tasks.

When you add an action to a security policy or create an action to run a task, you specify an extension associated with the type of action. For example, if you add an action to send an email in a user activity security policy, you select a configured email extension.

Note: To specify an extension in actions and tasks, the extension must have an Active status.

You can create the following types of extensions:

- Custom
- Email
- Protection
- Service management
- System log

Data Privacy Management provides plugins that correspond to the extension types. You select the plugin when you create an extension.

Extension Types

Data Privacy Management includes extensions that you can use in custom, email, protection, service management, and system log actions and tasks.

You can create or edit the following types of extensions if your role includes privileges to manage extensions:

Extension Type	Plugin	Description
Custom	- Custom Plugin V1 - DSR Custom Plugin	Specifies the file path to a directory that contains an executable file, such as a script file, that performs a custom task when the task runs. The executable file must be located on the same server on which Data Privacy Management is installed.
Email	- DSR Email Plugin - Email plugin V1	Configures the server connection and security settings for actions that send an email to specified recipients when the email task runs.
Protection	- Encryption - Persistent Data Masking - Big Data - Persistent Data Masking - Remote Domain	Determines the way that tasks will protect sensitive data. Protection tasks are associated with one data store. You can add protection extensions to data domains to define the default protection rules for sensitive fields. For example, if you create a protection task for sensitive fields in a data domain and the sensitive fields exist in six data stores, six protection tasks appear on the **Tasks** workspace.
Service Management	- DSR ServiceNow Plugin - ServiceNow Plugin V1	Creates third-party service management tickets for issues that users or security policy violations identify or to fulfill data subject requests. Data Privacy Management supports service management extensions for the ServiceNow application.
System Log	System Log plugin V1	Creates a log message that Data Privacy Management writes to a remote system log server when system log tasks run.

Supported Data Store Types for Protection Extension Plugins

When you configure a protection task on the **Tasks** workspace or add a protection action in a data store, decryption, or user activity security policy, you must select an active protection extension plugin that Data Privacy Management supports for the data store specified in the task or security policy.

The following table lists the supported data store types for each protection extension plugin:

Protection Extension Plugin	Supported Data Store Types
Data store types not listed in this table are not supported for protection extension plugins.	
Encryption	Cloudera Hive

Protection Extension Plugin	Supported Data Store Types
Persistent Data Masking - Big Data	Hive
Persistent Data Masking - Remote Domain	- IBM Db2 - IBM Db2 for i5/OS - IBM Db2 for z/OS - JDBC - Microsoft Azure SqlDB - Microsoft Azure SqlDW - Microsoft SQL Server - MySQL - Netezza - Oracle - Sybase - Teradata

Extensions Workspace

On the **Extensions** workspace, you can view and manage custom, email, protection, service management, and system log extensions. The workspace includes a page that displays a list of extensions and a detail page for each extension.

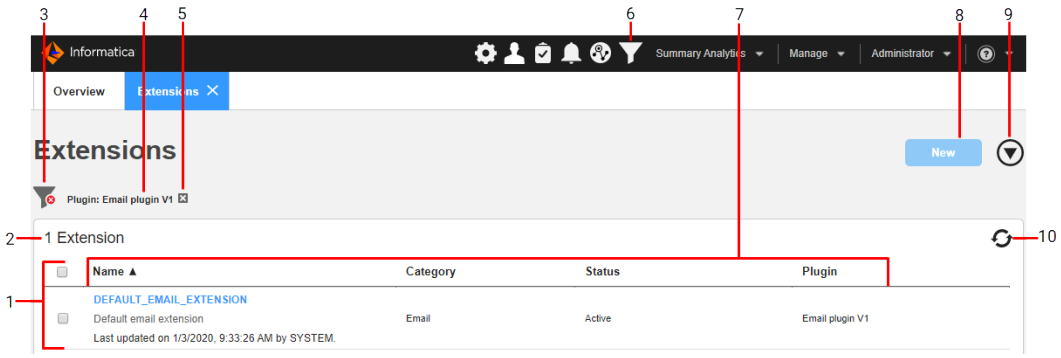
The workspace lists extensions by name, category, status, and plugin. You can view extensions that are associated with at least one data store that you can access. You can also create new extensions and edit or delete existing extensions if your role includes privileges to manage extensions.

To access the **Extensions** workspace, click **Manage** > **Extensions**.

Extensions List Page

The **Extensions** list page displays the extensions that you have access to view or manage. By default, you view the **Extensions** list page when you access the **Extensions** workspace.

The following image shows an example of the **Extensions** list page:



1. Extensions
2. Extension count
3. Clear Filter icon
4. Filter conditions
5. Clear Filter Condition icon
6. Filter icon
7. Extension properties
8. New button
9. Actions menu
10Refresh icon

You can access extension properties from this page by clicking an extension name.

Extension Properties

Extension properties configure details such as the extension category, plugin, status, connection information, and user credentials. Data Privacy Management re-uses the connection properties when you select an extension name in actions, security policies, and tasks.

The following table describes the common extension properties:

Property	Description
Name	Required. Unique name for the extension. The name cannot exceed 255 characters.
Description	Optional. Long description of the extension that cannot exceed 255 characters.
Category	Required. Select one of the following categories: - Custom - Email - Protection - Service Management - System Log

Property	Description
Plugin	Required. The name that specifies the sub-category for each extension. When you select an extension category, the plugin or list of plugins configured for the category appears. You cannot edit plugin names.
Active	Optional. Select to indicate that the extension is currently in use. To specify an extension in actions, security policies, and tasks, the extension must have an Active status. **Note:** You cannot delete extensions that are associated with Active tasks.

Custom Extension Properties

Configure the custom extension property to specify the path for an executable script file that runs when custom tasks run.

Important: Before you create a custom extension, you must first create a script file that you can specify in the extension properties.

You can specify custom extensions in actions, security policies, and custom tasks.

The following table describes the custom extension property:

Property	Description
Script File Path	Required. Enter the file path for the executable script file. For example: `tmp/updateLdapUserAtrribute.sh` The script file must be located on the same server where Data Privacy Management is installed. When you add a custom extension to a custom action in a security policy, Data Privacy Management will run the file to perform customized tasks in the event of a security policy violation. You can also add a custom extension to a new or existing custom action to run the file immediately, or you can save the action as a task to run manually later. For example, you create a script that adds users who trigger security policy violations to a monitored LDAP user group named HighRiskUsers. You provide the script file path in the custom extension properties. When a violation of a security policy that contains the custom action occurs, the file adds the user to the HighRiskUsers LDAP user group.

Email Extension Properties

Configure email extension properties for emails that Data Privacy Management sends when email tasks run.

You can specify email extensions in actions, security policies, and email tasks.

The following table describes the email extension properties:

Property	Description
Server Host Name	Required. The name of the host server that sends emails.
Server Port	Required. The port number for the server.
User Name	Required. The user name with credentials for the specified host server.
Password	Required. The password for the user at the specified host server. To change the password, select **Modify** and enter the new password.

Property	Description
Authentication Enabled	Optional. Select to indicate that secure authentication is required to connect to the specified server.
Use Security	Optional. Select to enable encryption for the connection to the specified server.
Security Protocol	Required if you select **Use Security**. Select SSL (Secure Sockets Layer) or TLS (Transport Layer Security).
Sender Email Address	Required. The email address that displays in the **From** field by default in email actions that use this extension.

Encryption Protection Extension Properties

Configure encryption protection extension properties for the Persistent Data Masking Service that uses encryption to mask sensitive data in Cloudera Hive data sources.

You can specify encryption protection extensions in data domains, decryption security policies, and protection tasks.

The following table describes the encryption protection extension properties:

Property	Description
Mark column as protected after execution	Optional. Select to mark sensitive fields in data domains as protected after a task configured with the extension runs.
Host	Required. The Persistent Data Masking Service host address.
Port	Required. The port number of the Persistent Data Masking Service.
Username	Required. The user name with credentials for the specified host server.
Password	Required. The password for the user specified for the Persistent Data Masking Service.
Is SSL Enabled	Optional. Select to enable Secure Sockets Layer (SSL) encryption for the connection to the specified Persistent Data Masking Service. Default is selected.
Max Parallel Sessions	Optional. The maximum number of parallel sessions of the Persistent Data Masking Service that can run concurrently. Default is 5.
Locale Code	Optional. Select a language and associated country. Default is English (United States).

Persistent Data Masking - Big Data Protection Extension Properties

Configure Persistent Data Masking - Big Data protection extension properties for the Persistent Data Masking Service that protects sensitive data in Hive data sources.

You can specify Persistent Data Masking - Big Data protection extensions in data domains, security policies, and protection tasks.

The following table describes the Persistent Data Masking - Big Data protection extension properties:

Property	Description
Mark column as protected after execution	Optional. Select to designate columns in data domains that use the extension as protected after an associated protection task runs.
Host	Required. The Persistent Data Masking Service host address.
Port	Required. The port number of the Persistent Data Masking Service.
Username	Required. The user name with credentials for the specified Persistent Data Masking Service.
Password	Required. The password for the user specified for the Persistent Data Masking Service.
Is SSL Enabled	Optional. Select to enable Secure Sockets Layer (SSL) encryption for the connection to the specified Persistent Data Masking Service.
Max Parallel Sessions	Optional. The maximum number of parallel sessions of the masking service that can run concurrently. Default is 5.
Locale Code	Optional. Select a language and associated country. Default is English (United States).

Persistent Data Masking - Remote Domain Protection Extension Properties

Configure Persistent Data Masking - Remote Domain protection extension properties for the Persistent Data Masking Service that protects sensitive data in relational database sources.

You can specify Persistent Data Masking - Remote Domain protection extensions in data domains, security policies, and protection tasks.

The following table describes the Persistent Data Masking - Remote Domain protection extension properties:

Property	Description
Mark column as protected after execution	Optional. Select to designate columns in data domains that use the extension as protected after an associated protection task runs.
Host	Required. The Persistent Data Masking service host address.
Port	Required. The port number of the Persistent Data Masking Service.
Username	Required. The username with credentials for the specified Persistent Data Masking Service.
Password	Required. The password for the user specified for the Persistent Data Masking Service.
Is SSL Enabled	Optional. Select to enable Secure Sockets Layer (SSL) encryption for the connection to the specified Persistent Data Masking service.
Stop on Error	Optional. Specifies whether a protection task based on this protection extension will stop running if it encounters the specified number of errors. Default is 1.

Property	Description
Suspend on Error	Optional. Select to indicate that a protection task based on this protection extension will suspend running if it encounters an error.
Rollback Transactions on Error	Optional. Select to indicate that a data source that uses this protection extension will rollback transactions if it encounters an error. Default is selected.
Enable High Availability on Recovery	Optional. Select to enable high availability on recovery of data sources associated with the protection extension.
Commit Interval	Optional. Specify the interval, in milliseconds, between commits on the data domains associated with the protection extension. Default is 10000.
Error Log Type	Optional. Select the error log file format. Default is None.
Error Log File Directory	Required if you select an error log type. The Data Privacy Management directory that contains the error log file for the data domains that use the protection extension.
Max Parallel Sessions	Optional. The maximum number of parallel sessions of the masking service that can run concurrently. Default is 5.
DTM Buffer Size	Optional. Specifies the amount of buffer memory that the Integration Service uses when the Data Transformation Manager (DTM) processes a session. Default is Auto.
Default Buffer Block Size	Optional. Specifies the amount of buffer memory used to move a block of data from the source to the target. Default is Auto.
Locale Code	Optional. Select a language and associated country. Default is English (United States).
Target Load Type	Optional. The method that the masking service uses to load data to targets. Options are: - Bulk. Loads data to targets without creating a database log. This option improves session performance but prevents the target database from recovering from an incomplete session. Target tables cannot contain key constraints in bulk loading. - Normal. Creates database logs to enable the target database to recover from an incomplete session, but slows performance. Default is Normal.
Batch Update	Optional. Indicates whether to update protected records in batches. Default is No.
Batch Size	Optional if you select to perform batch updates. The number of records to update in each batch update operation.

Service Management Extension Properties

Configure service management extension properties for ServiceNow tickets that Data Privacy Management creates when service management tasks run.

You can specify service management extensions in actions, security policies, and service management tasks.

The following table describes the service management extension properties:

Property	Description
Host	Required. The ServiceNow host address.
Username	Required. The user name with credentials for the specified host server.
Password	Required. The password for the user at the specified host server. To change the password, select **Modify** and enter the new password.
Table Name	Required. The name of the ServiceNow table in which Data Privacy Management creates a record of the service management ticket when tasks configured with the extension run. Enter one of the following table names in the format shown, based on the ticket type: - incident - problem - change_request Default is incident.

System Log Extension Properties

Configure system log extension properties for messages that Data Privacy Management writes when system log tasks run.

You can specify system log extensions in actions, security policies, and system log tasks.

The following table describes the system log extension properties:

Property	Description
Host	Required. The host address of the system log server.
Port	Required. The port number of the system log server.
Protocol	Required. The protocol that Data Privacy Management will use to connect to the system log server. Options are: **TCP**, **TCP/SSL**, and **UDP**. To use TCP/SSL protocol, first import the SSL certificate for the system log server to the Informatica domain truststore, and then restart the Informatica domain.

Extension Management

You can create and manage extensions on the **Extensions** workspace.

When you update an extension on the **Extensions** workspace, actions and tasks that include the extension also update with the new connection details.

Adding or Updating a Protection Extension on the Data Domain Details Page

You can add a new protection extension to a data domain or update the existing protection rules that are associated with a data domain.

1. Click **Manage** > **Data Domains**.

   The **Data Domains** workspace displays a list of data domains. You can use the navigation filters to narrow the list.
2. Click the name of the data domain for which you want to add or update a protection extension.

   The **Data Domain Details** page displays a read-only view of the data domain properties.
3. Click **Edit**.

   The **Update Properties** pane of the **Edit < DataDomainName >** page appears.
4. Click **2** to navigate to the **Update Protection Extension** pane of the **Edit < DataDomainName >** page.
5. To update protection rules for an existing protection extension, go to step 6. To add a protection extension to the data domain, click **+ Add Protection Extensions**.

   Select a protection extension.

   Note: You can add more than one protection extension to a data domain.
6. Add or update the protection rules for the data domain. The selected rules protect source columns in the data domain when protection tasks run.
   - For an Encryption protection extension, select encryption rules and associated encryption keys.
   - For a Persistent Data Masking protection extension, select protection rules and indicate the default rule.
7. Optional. To remove a protection extension from a data domain, click **Remove.**
8. Click **Save**.

Related Topics:

- "Data Domain Properties" on page 205

Creating an Extension

Create an extension to save the connection details for easy re-use in actions, security policies, and tasks.

1. Click **Manage** > **Extensions**.

   The **Extensions** workspace appears.
2. Click **New**.

   The **New_Extension** page appears.
3. Enter a name for the extension.
4. Optionally, enter a description.
5. In the **Category** field, select **Custom**, **Email**, **Protection**, **Service Management**, or **System Log**.
6. In the **Plugin** field, select a plugin from the list.

   The **Configure Settings** pane appears.
7. Select the **Active** check box if you want the extension to remain in use.

   Note: You cannot delete active extensions.

8. Configure the properties for the extension.
9. Click **Save**.

Editing an Extension

When you edit an extension, the tasks associated with the extension update with the new properties.

1. Click **Manage** > **Extensions**.

   The **Extensions** workspace displays a list of extensions.
2. Select the check box next to the extension that you want to edit.
3. From the **Actions** menu, click **Edit**.
4. Edit the extension properties that you want to change.
5. Click **Save**.

Deleting an Extension

You can delete an inactive extension if there are no tasks or jobs in progress that are associated with the extension. If jobs are running, go to the **Jobs** workspace. Terminate the jobs, delete the extension, and then configure any associated tasks with a different extension.

You cannot delete an active extension.

1. Click **Manage** > **Extensions**.

   The **Extensions** workspace displays a list of extensions.
2. Select the check box next to the extension that you want to delete.
3. From the **Actions** menu, select **Delete**.

   You receive a message to confirm if you want to delete the extension.
4. Click **Yes**.

CHAPTER 6

Remote Agents

This chapter includes the following topics:

- Remote Agents Overview, 76
- Remote Agents Workspace, 77
- Remote Agent Properties, 80
- Remote Agent Management, 81

Remote Agents Overview

Data Privacy Management stores connection details for remote agent servers, and protects the remote agent connections with two-way SSL authentication. You can associate remote agents with one or more data stores.

Data Privacy Management includes remote agents for protection and subject registry.

You can create, edit, delete, copy, export, import, test connections, and view remote agent details on the **Remote Agents** workspace.

Protection Remote Agents

Protection remote agents dynamically decrypt sensitive columns in Hive data stores that meet decryption security policy conditions. You can export protection remote agent connection details to a CSV file, save and edit the file, and import the updated information.

You can also publish associated data store details and port connection information to the protection remote agent.

Subject Registry Remote Agents

Subject registry remote agents provide remote connections to unstructured data stores that contain subject data. You can export subject registry remote agent connection details to a CSV file, save and edit the file, and import the updated information.

After you create a subject registry remote agent, create and run a scan on the **Scans** workspace. You can run a Subject Registry scan to build the index and identity map. You can run a Domain Discovery scan to identify sensitive fields and files that match data domain conditions and classification policies.

Guidelines for Remote Agents

Consider the following guidelines when you configure remote agents:

- Configure the machines that host remote agents to use 1 GB per thread memory share for each thread. Edit the following JVM parameter in the siagent.sh file: `-Xmx` to specify the correct amount of memory to use for each thread. You can view the result in the following file in the Tomcat Logs folder on the remote agent machine: `catalina.out`

  For example, if the remote agent runs on an 8-core machine, the recommended memory heap size setting is 10 GB.
- For Hive sources that you plan to associate with protection remote agents to encrypt and decrypt sensitive data, set the `Set hive.fetch.task.conversion` Hive parameter to `more` after you configure connectivity to the Hive source.
- For Subject Registry remote agents that connect to Microsoft SharePoint or OneDrive data stores, Scan jobs finish successfully but the jobs report an error scanning folders with more than 5,000 files. The Scan job error occurs because the file limit is a default SharePoint and OneDrive setting.

  To prevent the error during a scan, you can split the data store folder into multiple folders or site locations, move files into the new folders until each folder includes fewer than 5,000 files, and scan the data store again.
- Configure properties in the siagent.properties file to manage the agent job. You can enable the following properties:
  - si.fileProcessingTimeout. A time limit for file processing. If the processing does not complete, the specific file processing is canceled and reported in the Exception report. Enter the time in seconds. For example, si.fileProcessingTimeout=1800 indicates a timeout value of 30 minutes.
  - si.sentencePartitionLength. Block size of a sentence for file processing. Binary or unknown file types might take a long time to process because of missing delimiters in the content. Enable this property to ensure that sentences of a specific file type are divided into chunks based on the value specified in bytes. For example: si.sentencePartitionLength=1000

    Note: The property does not apply to file types that the agent supports. To include a file type, delete the extension from the si.sentencePartitionExcludeList property.
  - si.sentencePartitionExcludeList. A list of file types that do not apply to the si.sentencePartitionLength property. Enter the file extensions separated by commas. For example: si.sentencePartitionExcludeList=NO_EXTN,txt,zip,json,xml.

Remote Agents Workspace

On the **Remote Agents** workspace, you can view and manage protection and subject registry remote agents. The workspace includes a page that displays a list of remote agents and a detail page for each remote agent.

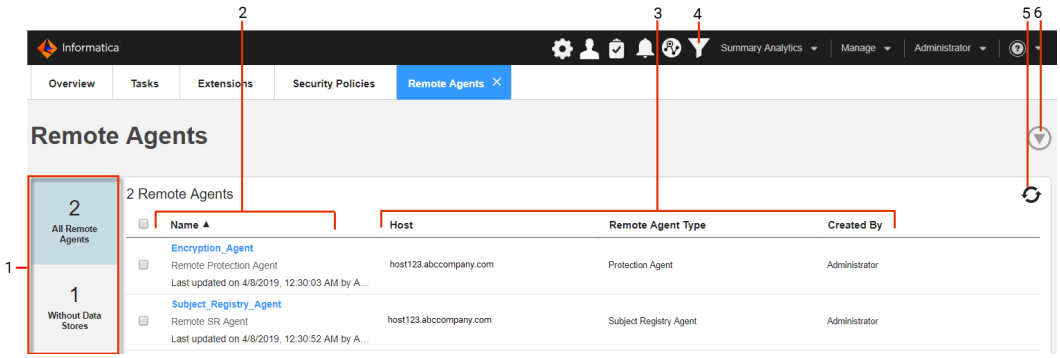
The workspace displays remote agents by name, host, remote agent type, and the user name of the user who created the remote agent. You can create new remote agents and view, edit, copy, or delete existing remote agents. You can also export and import remote agent details, test remote agent connections, and publish associated data store details and port connection information to protection remote agents.

To access the **Remote Agents** workspace, click **Manage** > **Remote Agents**.

Remote Agents List Page

The **Remote Agents** list page displays the remote agents that you have access to view or manage. By default, you view the **Remote Agents** list page when you access the **Remote Agents** workspace.

The following image shows the **Remote Agents** list page:



1. Remote agent status filters
2. Remote agent list
3. Remote agent properties
4. Filter icon
5. Refresh icon
6. Actions menu

Remote Agent Filter Conditions

To filter the list of remote agents on the **Remote Agents** workspace, click the **Filter** icon on the header and select filter conditions from the **Filter** pane.

The following table describes the filter conditions:

Filter Condition	Description
Name	Enter a full or partial remote agent name. The workspace displays a list of remote agents that match the name or name fragment.
Host	Enter a full or partial remote agent host address. The workspace displays a list of remote agents that match the host or host fragment.
Remote Agent Type	Select **Protection Agent** or **Subject Registry Agent**.
Created By	Select one or more users who created remote agents.
Data Store	Select one or more data stores to filter the list by remote agents associated with the selected data stores.

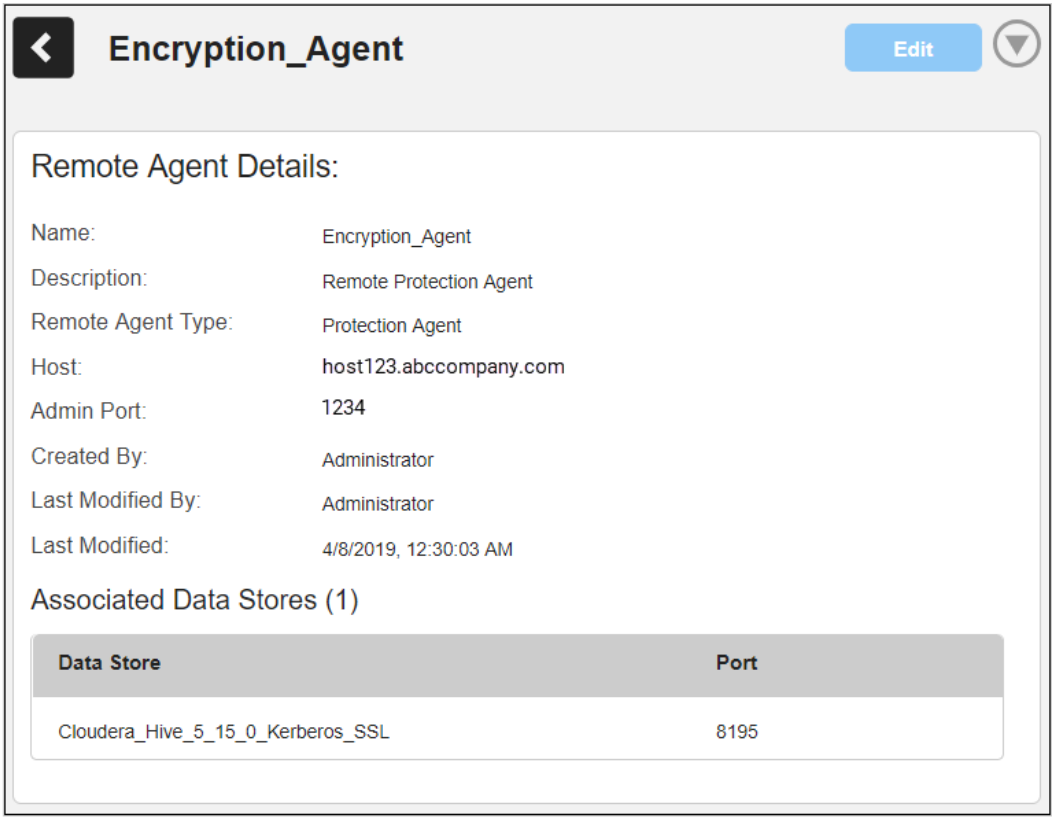
Protection Remote Agent Details Page

To view or edit the properties for a protection remote agent, access the **Remote Agent Details** page from the **Remote Agents** workspace.

- Click the name of a protection remote agent. To edit the properties, click **Edit** on the **Remote Agent Details** page.

- Select the check box next to a protection remote agent and then select **Open** or **Edit** from the **Actions** menu.

The following image shows a protection **Remote Agent Details** page:

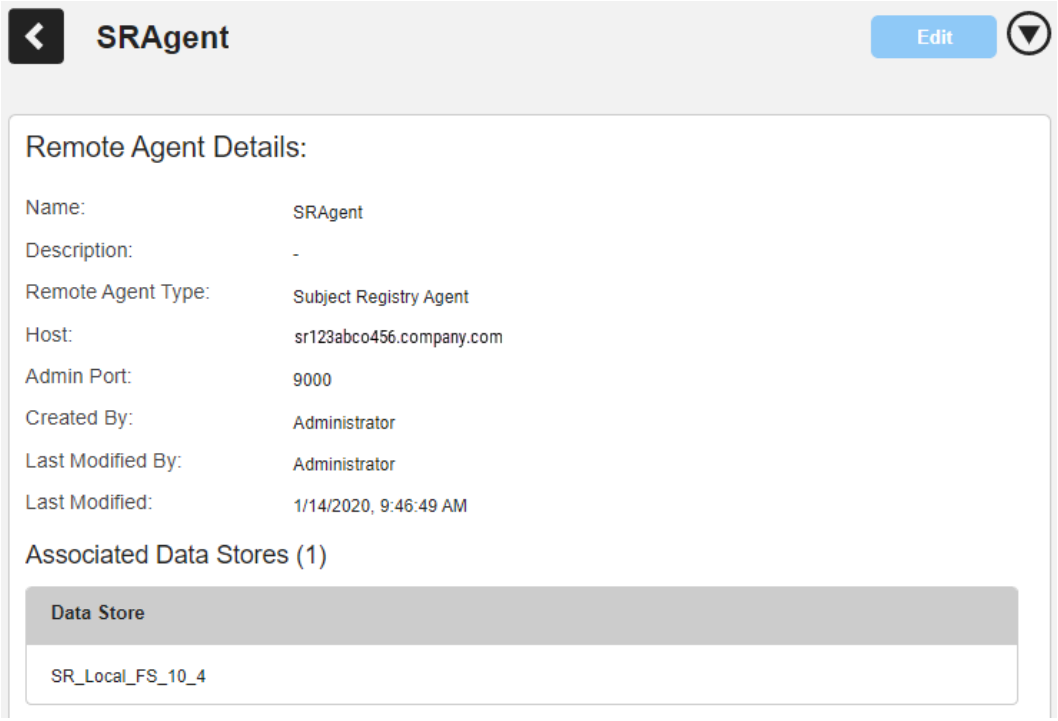


Subject Registry Remote Agent Details Page

To view or edit the properties for a subject registry remote agent, access the **Remote Agent Details** page from the **Remote Agents** workspace.

- Click the name of a subject registry remote agent. To edit the properties, click **Edit** on the **Remote Agent Details** page.
- Select the check box next to a subject registry remote agent and then select **Open** or **Edit** from the **Actions** menu.

The following image shows a subject registry **Remote Agent Details** page:



Remote Agent Properties

You can configure connection details for remote servers to create protection and subject registry remote agents.

Protection and subject registry remote agents include the following properties:

Property	Description
Name	Required. Unique name for the remote agent. The name cannot exceed 255 characters,
Description	Optional. Long description of the remote agent that contains no more than 255 characters.
Host	Required. The host address of the remote agent server.
Admin Port	Required. The admin port number of the remote agent at the specified host address.
Test Remote Agent	Optional. Verifies that the connection to the remote agent server is available and that the specified host and admin port properties successfully connect to the remote agent server. When you create or edit a remote agent, Data Privacy Management does not automatically validate the connection details.
Data Store	Optional. To associate a data store with the remote agent, select a data store. To associate additional data stores, click the plus sign (**+**).
Port	Required if you specify a data store. Enter the port number for the data store.

Remote Agent Management

On the **Remote Agents** workspace, you can create and manage remote agent connections for protection and subject registry.

You must publish a protection remote agent in the following situations:

- After you create a new protection remote agent.
- After you edit a protection remote agent.
- When you add a system log action to a decryption security policy.

Creating Remote Agents

You can create protection remote agents for Hive data stores and then protect the sensitive data with encryption extensions, rules, and keys. You can create subject registry remote agents to scan unstructured data stores or to build a Subject Registry index and identity map.

1. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace appears.
2. To create a protection remote agent, select **New** > **Protection Agent** from the **Actions** menu. To create a subject registry remote agent, select **New** > **Subject Registry Agent** from the **Actions** menu.

   For protection remote agents, the **New_Protection_Agent** page appears. For subject registry remote agents, the **New_Subject_Registry_Agent** page appears.
3. On the **Specify Properties** pane, enter a name for the remote agent.

   The name cannot exceed 255 characters.
4. Optionally, enter a description.

   The description cannot exceed 255 characters.
5. Enter the host address of the remote agent server.
6. Enter the admin port number of the remote agent at the specified host address.
7. Optionally, test the connection to the remote agent server.
8. Click the **Next** icon.

   The **Associate Data Stores** pane appears.
9. To associate a data store with the remote agent, select a data store from the list.
10. To associate another data stores with the remote agent, click the plus sign (**+**) and select a data store.

    Repeat this step to associate more data stores with the remote agent.
11. When finished, click **Save**.

After you create a protection remote agent, you must publish the agent.

After you create a subject registry remote agent, run a Subject Registry scan to build the index and identity map. You can also run a Domain Discovery scan to discover and classify sensitive data in unstructured data stores with the remote agent scanner.

Editing Remote Agents

You can edit a remote agent to change remote agent properties.

1. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace displays a list of remote agents.
2. Select the checkbox next to the remote agent that you want to edit.
3. From the **Actions** menu, select **Edit**.

   The **Edit_<Remote Agent Name>** page appears.
4. On the **Specify Properties** pane, you can edit the remote agent name. description, host, and admin port details. Optionally, test the remote agent connection if you edit the host or admin port.
5. Click the **2** icon.

   The **Associate Data Stores** pane appears.
6. You can associate more data stores with the remote agent or delete associated data stores.
7. When finished, click **Save**.

After you edit a protection remote agent, you must publish the agent.

Copying Remote Agents

You can copy a remote agent to create a new remote agent with similar properties.

1. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace displays a list of remote agents.
2. Select the checkbox next to the remote agent that you want to copy.
3. From the **Actions** menu, select **Duplicate**.

   The **Copy_of_<remote agent name>** page appears.
4. Edit the remote agent properties.

   The remote agent name and description cannot exceed 255 characters.
5. Click **Save**.

Exporting Remote Agents

To add new remote agents or to update remote agent properties in bulk, you can export remote agent details to a CSV file and edit the properties. After you save changes to the file, you can then import the CSV file on the **Remote Agents** workspace.

The export file includes a list of remote agents and the corresponding properties. You can export all remote agents, a filtered list of remote agents, or individually selected remote agents.

1. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace displays a list of remote agents.
2. From the **Actions** menu, select **Export**.

   A dialog box displays the exported CSV file with the following default file name: `ProxyDetails.csv`
3. Save the CSV file to a directory on your machine.
4. Open the CSV file.

The following table describes the columns in a remote agent export file and the corresponding property names on the **Remote Agent Details** page:

Export File Column	Export File Column Name	Remote Agent Property Name	Description
A	ProxyName	Name	Required. A unique name for the remote agent that cannot exceed 255 characters.
B	ProxyDescription	Description	Optional. Long description of the remote agent that cannot exceed 255 characters.
C	ProxyHostName	Host	Required. The host address of the remote agent server.
D	ProxyPort	Admin Port	Required. The admin port number of the remote agent at the specified host address.
E	ProxyType	Remote Agent Type	Required. Specify whether the remote agent is for protection or subject registry in the following format: - Protection remote agent: `DPNPROXY` - Subject registry remote agent: `SRAGENT`
F	AssociatedDataStores	Associated Data Stores. Includes data store name and port.	Optional. Specify one or more data stores configured on the **Data Stores** workspace in the following format: `[{repoName=`*`data_store_name`*`, servicePort=`*`data_store_port_number`*`}]` For protection remote agents, you can specify only Cloudera Hive data stores.
G	Action	-	Optional.

5. Update the remote agent properties and save the file.

Importing Remote Agents

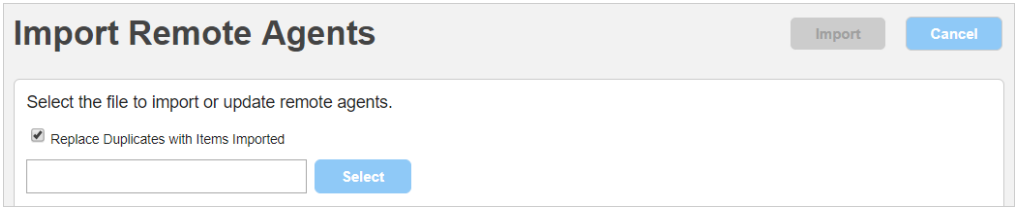
You can import remote agent details from a CSV file to add or update remote agent details in bulk.

1. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace displays a list of remote agents.

2. From the **Actions** menu, select **Import**.

   The **Import Remote Agents** page appears.



   By default, **Replace Duplicates with Items Imported** is enabled. This option prevents creating duplicate remote agents by updating existing agents with new details from the imported file.

3. Click in the empty field.

   The **Open** dialog box appears.

4. Browse to the directory that contains the CSV file with the remote agent details you want to import, select the file, and then click **Open**.

   The file name appears in the field on the **Import Remote Agents** page. A message indicates the number of remote agents that Data Privacy Management found in the CSV file.

5. Click **Select**.

6. Click **Import**.

Publishing Protection Remote Agents

You must publish a protection remote agent each time you create a new protection remote agent, edit a protection remote agent, and after you add a system log action to a decryption security policy that includes the data stores associated with the remote agent.

Before you can publish a protection remote agent, you must copy the keystore file from the Data Privacy Management instance to the remote agent machine, and then import the keystore file to the truststore that the remote agent uses.

When you publish a protection remote agent, Data Privacy Management pushes the connection details of data stores associated with the remote agent to the remote agent machine. You can publish one or more protection remote agents at a time.

1. Click **Manage** > **Remote Agents**.

   The **Remote Agents** workspace displays a list of remote agents.

2. Select the check box next to the protection remote agents you want to publish.

3. From the **Actions** menu, select **Publish**.

Related Topics:

- "Security Policy Properties" on page 453
- "Decryption Policy Example" on page 451

Part II: Discovery

This part contains the following chapters:

- Data Stores, 86
- Data Store Properties, 126
- Data Domains, 202
- Classification Policies, 221
- Column Sensitivity , 235
- Scans, 246
- Scan Options, 260
- Jobs, 283

CHAPTER 7

Data Stores

This chapter includes the following topics:

- Data Stores Overview, 86
- Data Stores Workspace , 87
- Data Store Management, 90

Data Stores Overview

A data store defines a connection to a data source. Create a data store for each data source that contains sensitive data that you want to discover, classify, analyze, protect, and for which you want to monitor user behavior and proliferation and ensure privacy regulation compliance.

You can create a data store that connects to one of the following data source types:

- SAP applications
- Big Data applications
- Cloud applications
- Data Integration services
- Databases
- File Management systems
- NoSQL
- Email Server

You can create data stores manually or you can import data stores from a parent repository, such as Informatica PowerCenter, and from CSV files.

When you create a data store, define or import data store metadata. For example, assign a data store group and security group to the data store, or import Catalog metadata or protection status from a CSV file for the data store from the **Actions** menu.

For most data stores, you perform a scan to identify sensitive data. When you scan a data store that connects to a parent repository, such as Informatica PowerCenter, a Scan job imports data stores.

If you scan a data store using Enterprise Data Catalog and then edit the data store to enable the **Scan with Remote Agent** property, the scan history and sensitive data information is deleted from Data Privacy Management but remains in Enterprise Data Catalog.

You can manage the data stores that the Scan job imported. For example, you might need to merge data stores that appear to connect to the same data source. Also, you might need to update data stores that have null or parameterized connection properties.

For Cloudera Hive data stores that you want to protect with encryption extensions, rules, and keys, you import Catalog metadata and protection status for the data store on the **Data Stores** workspace.

After you scan a data store, you can view the data store to see a summary of the scan results.

Data Stores Workspace

On the **Data Stores** workspace, you can view and manage data stores. The workspace includes a page that displays a list of data stores and a detail page for each data store.

The workspace displays the list of data stores by data store name, user name, data store type, risk score, the number of scans, and the percentage of protected sensitive fields. You can filter the list by all data stores, potential duplicates, incomplete data stores, data stores that are not valid, or data stores that are not scanned or partially scanned.

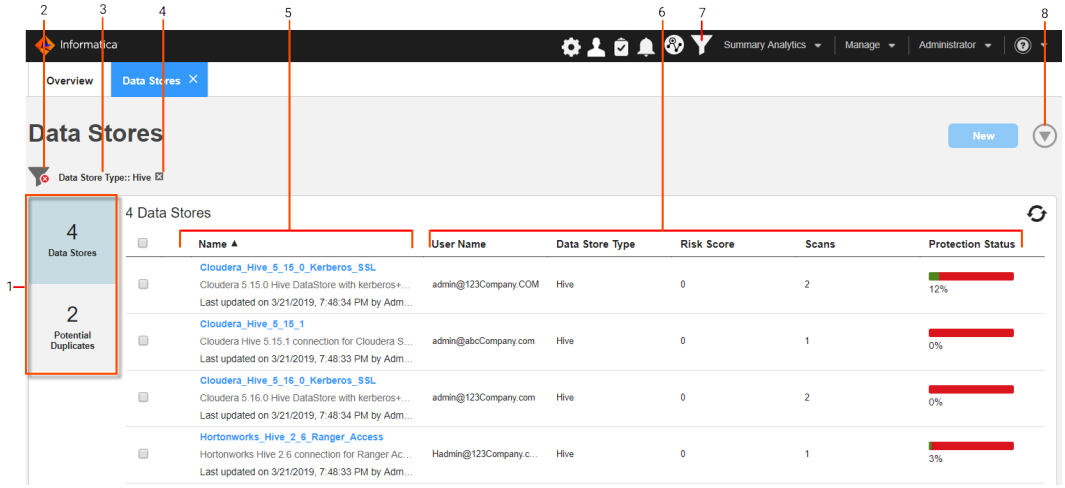
You can configure Risk Score settings on the dashboard to display the maximum risk score in addition to the average risk score that appears by default.

To access the **Data Stores** workspace, click **Manage** > **Data Stores**.

Data Stores List Page

The **Data Stores** list page displays the data stores that you have access to view or manage. By default, you view the **Data Stores** list page when you access the **Data Stores** workspace.

The following image shows the **Data Stores** list page:



1. Data store status filters
2. Clear Filter icon
3. Filter conditions
4. Clear Filter Condition icon
5. Data store list
6. Data store summary
7. Filter icon
8. Actions menu

Related Topics:

- "Discovery Bar Indicator" on page 515
- "Changing Risk Score Factor Weights" on page 42
- "Adding Custom Risk Score Factors" on page 43

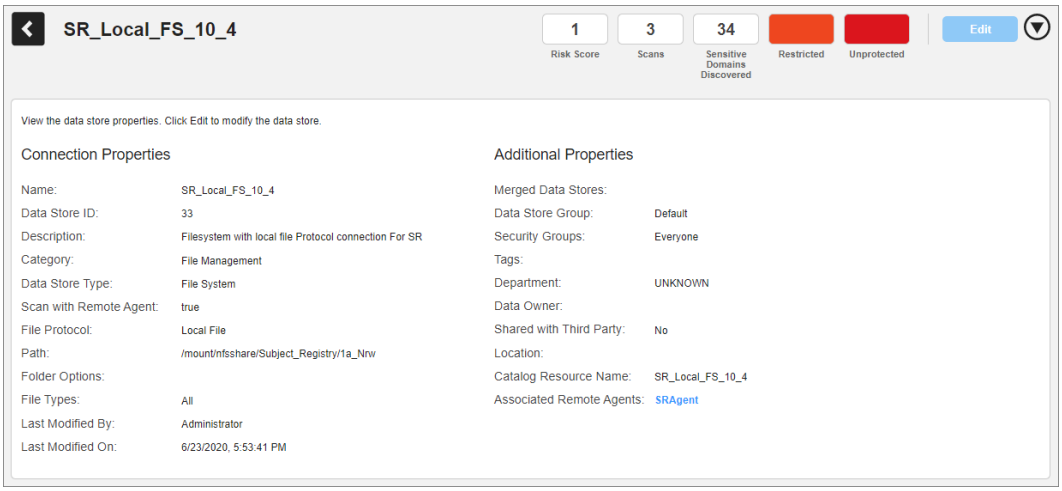
Data Store Details Page

To view or edit the properties for a data store, access the **Data Store Details** page from the **Data Stores** workspace.

Access the **Data Store Details** page in one of the following ways:

- Click the name of a data store. To edit the properties, click **Edit** on the **Data Store Details** page.
- Select the check box next to a data store and then select **Open** or **Edit** from the **Actions** menu.

The following image shows an example **Data Store Details** page for a file system data store:



If the **Associated Remote Agents** property lists a remote agent, you can click the remote agent name to view the details if you have the View Proxies privilege.

Related Topics:

- "Discovery Bar Indicator" on page 515
- "Proliferation Page" on page 542
- "Adding Custom Risk Score Factors" on page 43

Data Store Status Filters

The data store status filters indicate what action you might need to perform for the data stores that match the filter criteria.

The **Data Stores** workspace includes the following status filters:

Status Filter	Description
Data Stores	Default filter. Displays all data stores in the Data Privacy Management repository that you can access.
Potential Duplicates	Displays groups of data stores that might connect to the same data source because more than one data store contains the same connection details. Review the list of data stores and verify the connection details. Edit incorrect connection details, and merge data stores that are duplicates. If you do not merge or correct duplicate data stores, the **Data Stores Summary** page will contain an incorrect data store count.
Not Complete	Displays data stores with required connection properties that are not configured. You cannot scan data stores with incomplete connection properties. To fix an incomplete data store, edit the connection properties. For data stores that were imported from the PowerCenter repository, you can also select **Fix Data Stores** from the **Actions** menu on the **Data Stores** workspace to upload configuration files.
Invalid	Displays data stores to which Data Privacy Management cannot connect for one of the following reasons: - The data source machine or server is down for maintenance. - The user or password for the data source has changed. - Data Privacy Management cannot determine the data store type, and the type value is Any. You cannot scan data stores that are not valid. Determine the reason for the Invalid status, and edit the data store user, password, or type if required.
Not Scanned and Partially Scanned	Displays data stores that are not scanned and partially scanned. A data store is not scanned if Data Privacy Management has not completed at least one successful Scan job. A data store is partially scanned if an Informatica PowerCenter Scan job imported the data store.

Related Topics:

- "Data Stores Summary Workspace" on page 581

Data Store Summary

To quickly view the sensitivity and risk status for a data store, go to the **Data Store Details** page and view the data store summary metrics at the top of the page.

The Data Privacy Management Service updates the data store summary metrics when a Scan job finishes or when you import Catalog metadata or protection status information for a data store.

The following image shows an example of a data store summary:



The data store summary includes the following information:

Data Store Summary Metric	Description
Risk Score	Summarizes the risk level of the data store based on the **Risk Score** and **Anomaly Factor Weights** configurations on the **Settings** workspace. The Scan job determines the risk score based on the risk scoring configuration.
Scans	List the number of times the data store has been scanned. The number includes completed scans, scans that are in progress, scans that are scheduled to run in the future, and failed scans.
Sensitive Domains Discovered	Lists the number of sensitive data domains in the data store identified in a Scan job or from a manual import of Catalog metadata and protection status information on the **Data Stores** workspace. One data domain can include multiple columns in a data store. For this reason, the number of sensitive data domains can be different from the number of sensitive fields, or columns, in a data store.
Sensitivity Level	Indicates the sensitivity level of the classification policy that matches the sensitive fields in data domains that are included in the data store. The sensitivity levels are defined in the **Security Settings** pane on the **Settings** workspace. If a data store matches more than one classification policy, Data Privacy Management uses the most sensitive level.
Protection Status	Indicates whether the sensitive data domains in the data store are Unprotected, Partially Protected, or Protected.

Data Store Management

You can create and manage data stores on the **Data Stores** workspace.

After you create data stores, the next step is to discover sensitive columns using one of the following methods:

- On the **Scans** workspace, run a Scan job for the data store.
- On the **Data Stores** workspace, import Catalog metadata and then import protection status.

After you identify the sensitive fields in data stores and determine if the fields are protected or unprotected, you can perform the following tasks:

- Merge data stores that appear to connect to the same data source.
- Upload configuration files to fix data stores that have incomplete or missing connection properties.
- Update the user name and password associated with a data store.
- Synchronize user details.
- Import connection details, data store aliases, data owners, protection status, data lineage information, connection assignments, Catalog metadata, and Catalog resources.
- Export connection details, data store aliases, and breach reports as CSV files. Export data store configuration details as a text file to configure data store connectivity.
- View, copy, edit, or delete data stores.

- Reset classification results to delete column-to-domain match results but retain the connection properties.

Creating a Data Store

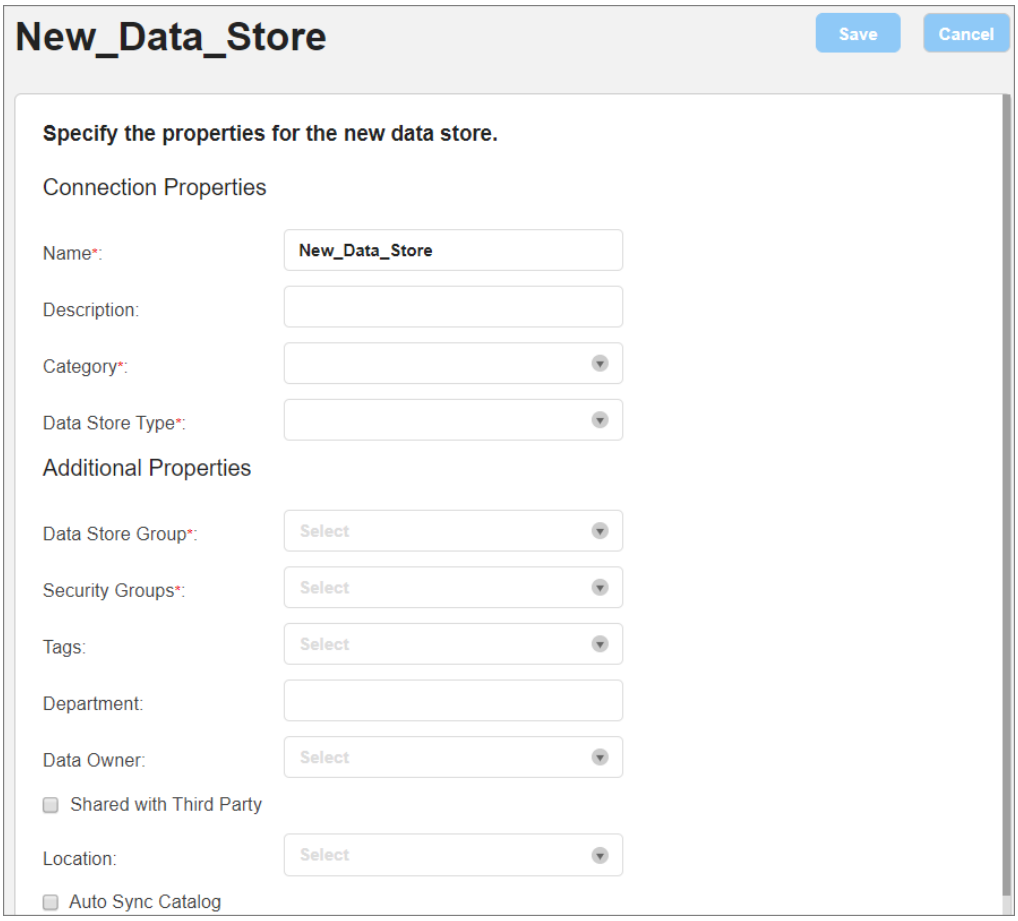
Configure the general properties and the connection properties for the data source. Also configure additional properties for reporting, grouping, and security.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. Click **New**.

   The **New_Data_Store** page appears.



3. Enter the connection properties for the data store.

   After you select the category and the data store type, additional connection properties appear.

4. Enter the connection properties.
5. Click **Test Connection** to verify that Data Privacy Management can connect to the data store with the connection details and log in.

   The connection test validates that the connection properties are configured, the connection is available, and the login credentials are valid.

If you create an unstructured data store and enable the **Scan with Remote Agent** option, you can test the connectivity to the data store after you associate the data store with a remote agent on the **Remote Agents** workspace.

Note: Data Privacy Management does not validate the connection details. You can save a data store with connection properties that are incomplete or not valid. For example, you might not have all of the connection details or login credentials when you create the data store. Or, the connection details and login credentials are correct, but the system is down for maintenance.

6. Enter the additional properties.
7. Click **Save**.

   After you save a data store, you cannot edit the category or the data store type. Also, if you enabled the **Scan with Remote Agent** option, you cannot disable the option when you copy or edit the data store.

Related Topics:

- "Data Store Properties Overview" on page 127

Before You Create a Data Store

Before you create a data store, complete the following tasks.

Define locations on the **Locations** workspace.

Define data store groups on the **Data Store Groups** workspace.

Verify that you configured connectivity to the data store. For example, if you create an Oracle data store, configure the `tnsnames.ora` file on the machine that hosts Enterprise Data Catalog. If you do not configure the connectivity, the connection test and the scan job will fail.

Testing a Remote Agent Connection to Unstructured Data Stores

When you create an unstructured data store to run domain discovery scans with a remote agent, you can click **Test Connection** to test the connection details.

If you create an unstructured data store and enable the **Scan with Remote Agent** option, you can test the connectivity to the data store after you associate the data store with a remote agent on the **Remote Agents** workspace.

If you configure an Amazon S3, File System, HDFS, Microsoft Azure Blob Storage, Microsoft OneDrive, or Microsoft Sharepoint data store to run scans with a remote agent, and you have not associated the data store with a remote agent, Data Privacy Management tests the connection with Enterprise Data Catalog.

If the connection test is successful, a success message indicates that the connection test was successful but that the data store must be associated with a remote agent before you can run a scan job.

When an unstructured data store is associated with multiple remote agents, Data Privacy Management tests the connection through all associated remote agents. If any of the associated agents can connect to the source, the connection test is successful.

If you configure a Google Drive or Office 365 Outlook data store to run scans with a remote agent, and you have not associated the data store with a remote agent, a message informs you that the connection test will not work until you associate the data store with a remote agent.

Unstructured File Extensions for Scans with Remote Agents

When you configure an unstructured data store to scan with a remote agent, you can include some or all supported file types in the scan.

The following table lists the file extensions that Data Privacy Management can scan with a remote agent for each file type:

File Type	File Extensions
Compressed Files	gz, tgz, xz, 7z, Z, gtar, zip, bz2, tbz2, tar, tar.bz2, tar.gz, tar.xz, tgz2
Delimited and Text	txt, csv
JSON and XML	json, xml
Microsoft Excel	xls, xlsx, xlsm, xlam, xltm, xltx
Microsoft PPT	pptm, potx, ppt, pot, pptx, ppsm, ppsx
Microsoft Word	doc, docm, docx, dotm, dotx, dot
Other	rtf
PDF	pdf
Webpage Files	chm, oth, xhtml, xht, mhtml, html, htm, ihtml
Image **Note:** Appears if you enable OCR for unstructured scans in the **System Settings**.	tiff, jpg. jpeg, png, bmp

Copying a Data Store

Copy a data store to create a new data store with similar properties. When you copy a data store, you can edit the data store properties except for the category and data store type.

Verify that the data store that you want to copy has the same category and data store type as the data store that you want to create.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. Select the checkbox next to the data store that you want to copy.
3. From the **Actions** menu, select **Duplicate**.

   The **Copy_of_<data_store_name>** page appears.
4. Edit the data store properties.
5. Click **Test Connection** to verify that Data Privacy Management can connect to the data store with the connection details and login.

   Data Privacy Management validates that the connection properties are configured, the connection is available, and that the login credentials are valid.

   Note: You can save a data store if the connection test fails. For example, you can configure a connection to a database if the database is down for maintenance. Data Privacy Management does not test the connection details when you save the data store.

6. Click **Save**.

   If there are no errors, Data Privacy Management saves the data store in the Data Privacy Management repository. An error can occur for the following reasons:

   - The connection properties are incomplete.
   - Another data store has an identical name.
   - Another data store has identical connection details.

If you selected **Scan with Remote Agent**, associate the data store with a remote agent on the **Remote Agents** workspace.

Editing a Data Store

Edit data store general properties, connection properties, or metadata for reporting, grouping, and security. Optionally, test the connection details.

You can use the **Not Valid** filter to find data stores that do not have valid connection properties, and you can use the **Not Complete** filter to find data stores with incomplete connection properties.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears. You can use the navigation filters to narrow the list of data stores.
2. Select the data store that you want to edit.
3. Click **Edit**.

   Edit the data store properties that you want to change.
4. Optionally, test the connection.
5. Click **Save**.

Fixing a Data Store

You can import database configuration files or PowerCenter parameter files to update data store connection properties.

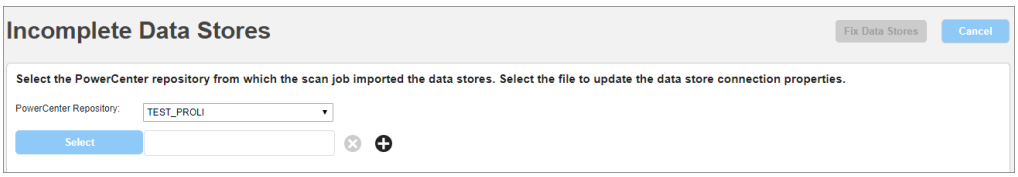
When you scan an Informatica PowerCenter data store, the Scan job imports data stores and associated connection strings from the PowerCenter repository. The Scan job uses values from the database configuration files or PowerCenter parameter files to update the remaining data store connection properties, such as the host name and port.

The Scan job reads the files from the machine on which the PowerCenter Repository Service runs. If the Scan job cannot access or read the files, the data stores can contain null or parameterized values for the connection properties. You cannot scan data stores that do not have complete connection properties.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. To check for data stores with incomplete connection properties, click **Not Complete** .
3. From the **Actions** menu, select **Fix Data Stores**.

   The **Incomplete Data Stores** page appears.



4. Select the Informatica PowerCenter repository from which the Scan job imported the data stores.
5. Click **Select** to navigate to the database configuration file or PowerCenter parameter file that contains the connection properties.
6. Click the **+** icon to add another file.

   Repeat this step for each database configuration file or PowerCenter parameter file that you want to import.
7. Click **Fix Data Stores**.

   The Data Privacy Management Service creates one Import job for each file. You can monitor the Import job on the **Jobs** workspace. After the job finishes, the data store properties are updated and the data stores no longer appear in the **Not Complete** list.

Rules and Guidelines to Fix a Data Store

When you fix a data store, you import database configuration files or PowerCenter parameter files. The Import job updates the data store connection properties in the Data Privacy Management repository.

Consider the following rules and guidelines when you fix a data store:

- You can fix data stores that a Scan job imported from a PowerCenter repository. You cannot fix data stores that you manually created or imported from a CSV file.
- When you fix data stores, you specify a PowerCenter repository from which a Scan job imported the data stores. You can specify one PowerCenter repository. Run the procedure again to fix data stores that a Scan job imported from another PowerCenter repository.
- When you fix data stores, you can specify multiple files. For example, you can select a `tnsname.ora` file and an `ODBC.ini` file. The Data Privacy Management Service creates one Import job for each file.
- To fix data stores that connect to IBM Db2, Oracle, and Sybase databases, import the corresponding database configuration files. To fix data stores that connect to all other supported relational databases, import the ODBC configuration files.
- When you fix a data store that connects to an IBM Db2 database, you must import the `db2cli.ini`, `db2.out`, and `node_db2.out` database configuration files in sequential order. Run the procedure once for each file. If you try to import the files at the same time, the Import job does not update the data store connection properties.
- If you specify multiple files that contain the same connection string, the Data Privacy Management Service updates the data store with the connection properties in the last file that the service imports.
- The Import job updates connection properties for data stores that exist in the repository and replaces existing connection properties with the values from the files.
- The Import job does not create data stores. If the files contain information for a data store that does not exist in the repository, the Import job skips the row.

Resetting the Classification Results of a Data Store

You can reset the classification results of a data store after you run a Scan job. The Scan job discovers and classifies sensitive data in the data store. The results match columns that the Scan job identifies as sensitive with the data domain to which the sensitive data belongs.

You must have the Delete Data Stores privilege to reset the classification results of a data store.

If you made changes to the data after you scanned the data store, you might want to run another scan to view the sensitive fields. A reset purges all sensitivity results and retains the data store information, including

connection properties, user access, activities, violations, and anomalies. The reset deletes the column-to data domain match results and sets the protection status of the data store to Not Classified.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears. You can use the navigation filters to narrow the list of data stores.
2. Select the check box next to the data store that you want to reset.
3. From the **Actions** menu, select **Reset Classification Results**.
4. Click **Yes, I'm Sure** to confirm the reset.

Rules and Guidelines to Reset the Classification Results of a Data Store

Consider the following rules and guidelines when you reset the classification results of a data store:

- You cannot reset a data store if it has a task in progress, failed, or completed state. The data store must be in closed state.
- You cannot reset an encrypted data store.
- You cannot stop, terminate, or pause a reset job.
- You cannot reset multiple data stores. Run a reset on individual data stores.
- You must have the Delete Data Stores privilege to reset a data store.

Updating User Names and Passwords in Bulk

You can update the user name and password for multiple data stores at the same time. You select one or more data stores to update. Then, enter a new user name and password. The changes apply to the selected data stores.

If a selected data store is included in an active Scan job, the update fails for all data stores. An active Scan job is a job that does not have a completed or terminated status.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace displays the list of data stores.
2. Select the data stores for which you want to change the user name and password.

   Tip: Click the **User Name** column to sort the list of data stores by user name, or click the **Filter** icon to filter the list of data stores by user name.
3. From the **Actions** menu, select **Update Username/Password**.

   The **Update User Name/Password** page appears.
4. Enter the new user name and password for the selected data stores.

   You must enter values for both properties. The properties cannot contain null values.
5. Click **Save**.

Synchronizing Users

To view and manage new users in Data Privacy Management, you can synchronize user details between the Data Privacy Management repository and the Enterprise Unified Metadata platform.

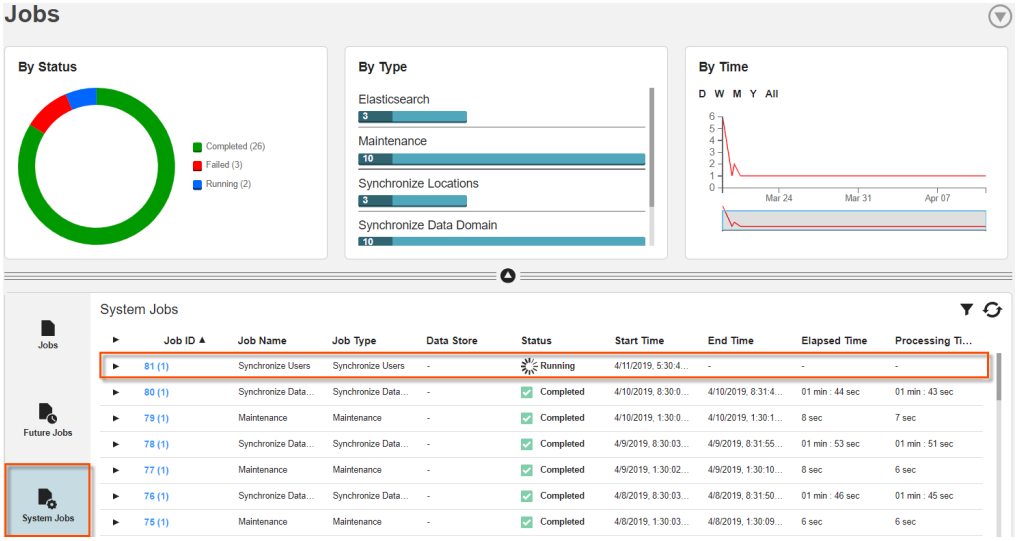
Add new users and configure roles, privileges, and assign the users to user groups in the Administrator tool before synchronizing users. For more information, see the *Informatica Data Privacy Management Administrator Guide*.

1. Click **Manage** > **Data Stores**.

   A list of data stores appears on the **Data Stores** workspace.

2. From the **Actions** menu, select **Synchronize** > **Users**.

3. To check the status of the Synchronize Users system job, go to the **Jobs** workspace and click the **System Jobs** filter.



After the Synchronize Users job completes successfully, you must log out of Data Privacy Management and log in again for the new user details to appear.

Importing Data Stores

To create or update a data store, you can import data store information from a CSV file. The file must be in the format that the import job requires. When you import the file, the Import job runs immediately and updates the Data Privacy Management repository with information from the file.

You can import information for Application, Big Data, Cloud, Database Management, and File Management data stores. You cannot import Cloudera Navigator or Data Integration data stores.

You might want to import data stores from a CSV file to analyze data sources that are not included in a parent repository. For example, you have an asset management system that contains connection details for all data sources in an organization. You can export the list of data sources in which you want to identify sensitive data. Data Privacy Management creates one data store for each entry in the CSV file.

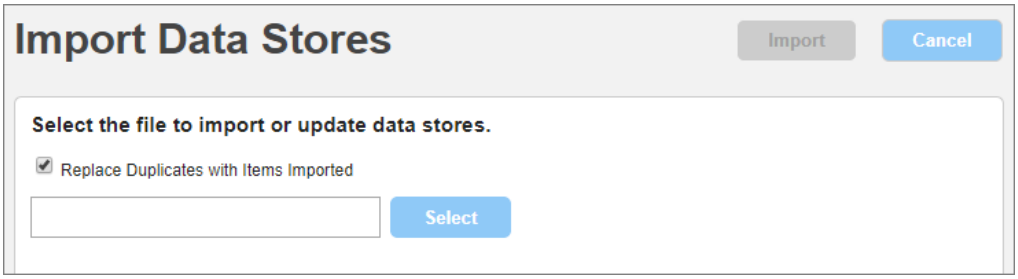
You can also create a CSV file to update an existing data store. Data Privacy Management replaces the details of the existing data store with the details in the CSV file.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. From the **Actions** menu, select **Import** > **Data Stores**.

   The **Import Data Stores** page appears.



3. Enable or disable the **Replace Duplicates with Items Imported** option.

   If you enable the option, the Import job updates the data store properties in the Data Privacy Management repository with the values in the CSV file.

   Warning: If a column in the file contains a null value, the Import job replaces the value in the repository with a null value.

   If you disable the option, the Import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

6. After the import job finishes, you can go to the **Data Stores** workspace to view the data store information that you imported.

   For security reasons, the CSV import file does not include columns for the database user and password. After you import new data stores, edit the data store to add the database user and password. If you don't add the database user and password, you cannot include the data store in a scan.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

Related Topics:

- "Import Job" on page 299

CSV File Format to Import Data Stores

Create a CSV file to create, update, or delete information about data stores in the Data Privacy Management repository. The content of the CSV file must be in the order and format that the Import job requires. The CSV file must contain all of the column headings. However, not all columns require values.

Each column is equivalent to a data store property. The description for the data store property depends on the category. For more information, see the *Data Store Properties* section to view the properties for the corresponding category and data store type.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Column Value Description
1	RepoName	Required. Short description of the data store. Name is case-sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 100 characters, contain spaces, or contain the following special characters: \ ~ ! $ % ^ & * ( ) +
2	RepoType	Required. Category to which you create a connection. Enter one of the following values: - Application. SAP - Cloud - DB. Database Management - FS. File Server - Hadoop. Big Data - NoSQL. Apache Cassandra You cannot import data stores for Cloudera Navigator or data store types included in the Data Integration category.
3	DBType	Required. Data store type for the data store category. The required entry depends on the value you specified for the RepoType property. - **Application** . Enter SAP. - **Cloud**. Enter one of the following values: Amazon Redshift, Azure Data Lake, Microsoft Azure SQL Data Warehouse, Microsoft Azure SQL Database, Salesforce, or Snowflake. - **Database Management**. Enter one of the following values: DB2, DB2i5OS, DB2zOS, JDBC, Microsoft SQL Server, Netezza, Oracle, SAP HANA, Sybase, or Teradata. - **File Server**. Enter one of the following values: Amazon S3, AzureBlob, Google Drive, HDFS, Microsoft OneDrive, Office365Outlook, or Microsoft SharePoint. - **Hadoop**. Enter Hive. - **NoSQL**. Enter Apache Cassandra.

Column Order	Column Heading	Column Value Description
4	Host/URL/Account	Optional. Description depends on the value you specified for the RepoType property. **Application**. The SAP host. **Cloud**: - Amazon Redshift. The host name or IP address for the Redhsift server. - Azure Data Lake. The OAuth 2.0 token endpoint URL in the Azure portal. - Microsoft Azure SQL Data Warehouse. The host name or IP address for the Azure server. - Microsoft Azure SQL Database. The host name or IP address for the Azure server. - Salesforce. The Salesforce Service URL. - Snowflake. The name of the Snowflake account. In the Snowflake URL, the account name is the first segment in the domain. **Database Management**. Host name or IP address of the database server for the following data store types: DB2, DB2i5OS, DB2zOS, JDBC, Microsoft SQL Server, Netezza, Oracle, SAP HANA, Sybase, and Teradata **File Management**: - Amazon S3. The Amazon Web Services bucket URL. - Azure Blob. The Microsoft Azure Blob Storage URL to access a container. - File Server. The primary name node URI. - Google Drive. The Client ID. - HDFS. The URI to the active HDFS NameNode. - Microsoft OneDrive. The URL to access OneDrive. - Microsoft SharePoint. The SharePoint URL. **Hadoop**. JDBC connection URL that accesses the Hadoop server. Use one of the following formats: - Non-Kerberos source: `jdbc:hive2://<host>:<port>/<DB>` - Kerberos source: `jdbc:hive2://<host>:<port>/<DB>;principal=<serviceName>/<host>@realm` **NoSQL**. The name of the Apache Cassandra host.
5	SchemaOption/ KeyspaceOption	Required for File Management data stores and Cassandra data stores. Specifies how you add schemas to the data store. Enter `SCHEMA` to enter a schema name. Leave the column blank to enter a schema path.
6	Schema/Keyspace/Path	Required for File Management data stores and Cassandra data stores. If you entered `SCHEMA` in the **SchemaOption** property, enter the schema name. If you left the property blank, enter the file path to the schema. For example: `/mount_cifs/Root_Folder_Unstructured/FileTypes`

Column Order	Column Heading	Column Value Description
7	Port	Optional. Port number for the data store server. Valid for the following data store types: - Amazon Redshift - Db2 - Db2 for z/OS - File Server - JDBC - Microsoft Azure SQL Database - Microsoft Azure SQL Data Warehouse - Microsoft SQL Server - Netezza - Oracle - SAP HANA - Sybase - Apache Cassandra
8	ConnectString	Optional. Native connection string used to access data from the data store. Valid for the following data store types: - Db2 - Db2 for i5/OS - JDBC - Microsoft Azure SQL Database - Microsoft SQL Server - Netezza - Oracle - SAP HANA - Sybase - Teradata
9	Role	Required for Snowflake data stores. Enter the name of the Snowflake role assigned to the user.
10	Warehouse	Required for Snowflake data stores. Enter the name of the Snowflake warehouse.
11	AdditionalParam	Required for Snowflake data stores. Specify one or more JDBC parameters in the following format: <param1>=<value>&<param2>=<value>&<param3>=<value>....
12	DatabaseName	Optional. Database name. Valid for the following data store types: - Amazon Redshift - Db2 - Db2 for z/OS - Microsoft Azure SQL Database - Microsoft Azure SQL Data Warehouse - Microsoft SQL Server - Netezza - SAP HANA - Sybase - Teradata
13	ServiceName	Required. Service name for Oracle databases.
14	CyberArkSafe	Optional. Name of the CyberArk safe that contains the database password.

Column Order	Column Heading	Column Value Description
15	DataOwnerSecurityDomain	Valid for Database Management data stores. The value of the **Profile Execution Engine** property on the **Data Store Details** page. Enter `Native` or `Hadoop`.
16	DataOwnerUserName	Optional. Individual, group, or organization name that is responsible for the accuracy and integrity of the source data on which you run a classification profiling scan. Name is case-sensitive and cannot exceed 100 characters.
17	ApplicationGroup	Optional. Name of an existing data store group. The data store group name cannot include a forward slash (/) or exceed 100 characters.
18	LocationName	Optional. Geographical mapping of the data store. You can assign one location to a data store.
19	UserName	Optional. User name that the Scan job uses to connect to the data source.
20	SecurityGroups	Optional. Grants users access to data stores. For example, the security group determines which data stores users see when users manage data stores and view scan results.
21	Tags	Optional. Keywords that add to the data store description. For example, to indicate the data store environment, you can assign tags such as production, development, or test to a data store.
22	Department	Optional. Department responsible for the maintenance or ownership of the data store. Name is case-sensitive and cannot exceed 100 characters.
23	StorageType	Type of data source in the selected category. Applicable for HDFS connections.
24	SourceDirectory	Directory path that contains the files to scan.
25	RegionName	Applicable for Amazon S3 connections. The name of the geographical region for the Amazon S3 bucket.
26	FileTypes	File types that the Scan job reads from the location configured in the **Path** property. Enter `ALL` or `SELECT`.
27	SelectedFileTypes	Applicable if you enter `Select` in the **File Types** property. Enter file types separated by commas. For example: `text,xml,json,pdf`
28	AWSBucketName	Applicable for Amazon S3 connections. Name of the Amazon S3 bucket that contains the semi-structured data to scan. Use the bucket name configured in the **AWS Bucket URL** property.
29	S3BucketName	Name of the Amazon S3 bucket that contains the semi-structured data to scan. If blank, the service sets the data store to Not Complete. You cannot scan incomplete data stores.
30	SourceConnectionName	Connection name in the Informatica domain. Do not enter a value for File Management data stores.

Column Order	Column Heading	Column Value Description
31	Memory	Memory that the Scan job uses when you scan the data store. Enter High, Medium, or Low.
32	Action	Optional. Instructs the Import job to create, update, or delete a data store in the Data Privacy Management repository. Enter one of the following values: - `U`. Creates or updates a data store in the repository. The action that the job performs depends on whether the data store exists in the repository and whether you select the **Replace Duplicates with Items Imported** option when you import the CSV file. If the data store does not exist, the Import job creates the data store. If the data store exists and the **Replace Duplicates with Items Imported** option is enabled, the Import job updates the data store. **Warning:** If a column in the CSV file contains a null value, the Import job replaces the value in the repository with a null value. If the data store exists and the **Replace Duplicates with Items Imported** option is not enabled, the Import job skips the row. - `D`. Deletes a data store from the repository. If the data store does not exist, the Import job rejects the row. If blank, default is `U`. Creates or updates a data store in the repository.
33	ThirdPartyShare	Optional. Specifies that the data in the data store is shared with a third party. Enter `TRUE` or `FALSE`.
34	ThirdPartyName	Optional. Applicable if the data store is shared with a third party. Enter the name of the third party.
35	RemoteAgentScanning	Required. Indicates if the data store is associated with a remote agent. Enter `TRUE` or `FALSE`.
36	AssociatedRemoteAgents	Do not enter a value for this property. The Import job does not import remote agent associations. After you import the data store and save the configuration details, associate the data store with a remote agent on the **Remote Agents** workspace.
37	FolderOption	Required for unstructured data stores that are associated with a remote agent. Enter `All`, `FolderSpecific`, or `Regex`.
38	FolderInclude	If you enter `FolderSpecific` in the **FolderOption** property, enter one or more folder names to include in the data store scan.
39	FolderExclude	If you enter `FolderSpecific` in the **FolderOption** property, optionally enter one or more folder names to exclude from the data store scan.
40	FolderRegex	If you enter `Regex` in the **FolderOption** property, enter a regular expression for the folders to include in the data store scan.
41	TrustorePath	Required for Cassandra data stores. Enter the location of the database Truststore.
42	LocalDatacenter	Required for Cassandra data stores. Enter the name of the datacenter that contains the required node.

Column Order	Column Heading	Column Value Description
43	KeystorePath	Required for Cassandra data stores. Enter the location of the database Keystore.
44	SSLEnabled	Required for Cassandra data stores. Option to enable SSL. Enter `TRUE` or `FALSE`.
45	ClientId	Applicable for Office 365 Outlook data stores. Client ID of the Azure Active Directory application.
46	TenantId	The unique ID of the tenant or domain.
47	IncludeDepartments	Applicable for Office 365 Outlook data stores. Enter the name of Office 365 departments that you want to include in scans. Use commas to separate multiple values.
48	IncludeUsers	Applicable for Office 365 Outlook data stores. Enter the email IDs of users or accounts that you want to scan. Use commas to separate multiple values.
49	ExcludeUsers	Applicable for Office 365 Outlook data stores. Enter the email IDs of users or accounts to exclude from the scan. Use commas to separate multiple values. The Exclude Users values takes precedence over Include Users.
50	EmailGroups	Applicable for Office 365 Outlook data stores. Enter the email groups to include. Use commas to separate multiple values.
51	EmailsWithAttachment	Applicable for Office 365 Outlook data stores. Enter TRUE or FALSE. Enter TRUE to include only emails with attachments. Scans ignore emails without attachments. If you enter TRUE, scans include the content body of all emails that have attachments and the attachments. If you enter TRUE and specify attachment types, the scan includes the content body of all emails that have attachments, and the attachment types that you specify. If you enter FALSE, scans include the content body of all emails with and without attachments and the attachments. If you enter FALSE and you specify attachment types, the scan includes the content body of all emails with and without attachments, and the attachment types that you specify.

Column Order	Column Heading	Column Value Description
52	AttachmentFileType	Applicable for Office 365 Outlook data stores. Specify attachment types to include. Use commas to separate multiple values. Choose from the following attachment types: - Avro - Compressed Files - Delimited and Text - Email - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, PDF content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.
53	IncludeMailFolders	Applicable for Office 365 Outlook data stores. Folders to include in the scan. You can enter top-level folder names. Use commas to separate multiple values.
54	ExcludeMailFolders	Applicable for Office 365 Outlook data stores. Folders to exclude in the scan. You can enter top-level folder names. Use commas to separate multiple values.

Rules and Guidelines to Import Data Stores

Review the rules and guidelines to understand how to manage the CSV file and how the Import job processes data from the file.

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines a data store.
- If the CSV file contains duplicate entries, the Import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

```
SELECT FROM
SELECT UNION
DELETE FROM
UPDATE SET
<SCRIPT
</SCRIPT>
ALERT(
"/>"
```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.

- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Importing Data Store Aliases

You can import data store alias information from a CSV file. The file must be in the format that the import job requires. When you import the file, the Import job runs immediately and updates the Data Privacy Management repository with information from the file.

User activity information is gathered from user actions taken on the source database or a parent repository such as Informatica PowerCenter. If a data store name differs from the original source name, you might want to give the data store an alias name. When you provide a data store alias name that matches the original source name, the user activity information for that source is linked to the corresponding data store. You can create an alias for any data store, whether you created the data store in Data Privacy Management or imported the data store from a parent repository.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. From the **Actions** menu, select **Import** > **Data Store Aliases**.

   The **Import Data Store Aliases** page appears.

3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the Import job handles updates for data store aliases that exist in the repository.
   If you enable the option, the Import job updates the data store alias properties in the repository with the values in the CSV file.

   Warning: If a column in the CSV file contains a null value, the Import job replaces the value in the repository with a null value.

   If you disable the option, the Import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

Related Topics:

- "Import Job" on page 299

CSV File Format to Import Data Store Aliases

Create a CSV file to create or delete information about data store aliases in the Data Privacy Management repository. The content of the file must be in the order and format that the Import job requires. The CSV file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Description
1	DataStoreName	Required. Name of the data store as it exists in the Data Privacy Management repository. The combination of the **DataStoreName** and **AliasName** columns identifies a unique data store for which to create or delete an alias. If blank, the Import job rejects the row.
2	AliasName	Required. Alternative name or alias for a data store. The combination of the **DataStoreName** and **AliasName** columns identifies a unique data store for which to create or delete an alias. You can specify one alias in a row. To specify multiple aliases for a data store, create one row for each alias. If blank, the Import job rejects the row.
3	Action	Optional. Instructs the Import job to create or delete an alias in the repository. Enter one of the following values: - U. Creates an alias in the repository. - D. Deletes an alias from the repository. If blank, default is U. Creates an alias in the repository.

Rules and Guidelines to Import Data Store Aliases

Review the rules and guidelines to understand how to manage the CSV file and how the Import job processes data from the file.

Consider the following rules and guidelines when you import data store aliases from a CSV file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines one alias for one data store.
  The combination of the **DataStoreName** and **AliasName** columns defines a unique data store. If two rows contain the same values in the **DataStoreName** and **AliasName** columns, the Import job creates two aliases for the data store.
- The column values cannot contain the following keywords or terms:

```
SELECT FROM
SELECT UNION
DELETE FROM
UPDATE SET
<SCRIPT
```

```
</SCRIPT>
ALERT(
"/>"
```

If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.

- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Importing Data Owners

To set or update the data owner for a data store, import data owner information from a CSV file.

Data Privacy Management sets or updates the data owner for each row in the CSV file that contains data owner details. If the file contains details for a data store that already exists, Data Privacy Management replaces the details of the existing data owner with the details in the file. To update the data owner, you must select the **Replace Duplicates with Items Imported** option when you upload the CSV file.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. From the **Actions** menu, select **Import** > **Data Owners**.

   The **Import Data Owners** page appears.
3. To update the data owner for an existing data store, select the **Replace Duplicates with Items Imported** option.
4. Select the file that you want to import.
5. Click **Import**.

   The Import job begins.
6. To check the status of the Import job, go to the **Jobs** workspace.
7. After the Import job finishes, you can return to the **Data Stores** workspace and view the data owner information that you imported.

Related Topics:

- "Import Job" on page 299

CSV File Format to Import Data Owners

The CSV file must contain data owner details in the order and format that the Data Privacy Management Service requires.

Create the case-sensitive column headings in the first row of the CSV file. Each subsequent row that contains values defines the data owner of one data store.

The following table describes the columns in the CSV file to import data owners:

Column Order	Column Heading	Description
1	RepoName	Short description of the data store. Required. The name is case-sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 100 characters, contain spaces, or include the following special characters: `\ ~ ! $ % ^ & * ( ) +` If the RepoName value contains a comma (,) character, you must enter the value surrounded by double quotation marks. For example, enter `"Repo1,Name"` for a data store named Repo1,Name. If the RepoName value contains a quotation mark ("), you must enter the value surrounded by double quotation marks with an additional double quotation mark placed after the quotation mark that exists in the value. For example, enter `"Repo1""Name"` for a data store named Repo"1Name.
2	DataOwnerSecurityDomain	Optional. Name of the collection of user accounts and groups in the Informatica domain that includes the specified data owner. Valid values are `Native` and `LDAP`. The Native security domain contains the users and groups created and managed in the Administrator tool. An LDAP security domain contains users and groups imported from an LDAP directory service.
3	DataOwnerUserName	Optional. Individual, group, or organization that is responsible for the accuracy and integrity of the source data on which you run a classification profiling scan. The name is case-sensitive and cannot exceed 100 characters. If you do not enter a value, Data Privacy Management deletes the current value. If the DataOwnerUserName value contains a comma (,) character, you must enter the value surrounded by double quotation marks. For example, enter `"Owner1,Name"` for a data store named Owner1,Name. If the DataOwnerUserName value contains a quotation mark ("), you must enter the value surrounded by double quotation marks, with an additional double quotation mark placed after the quotation mark that exists in the value. For example, enter `"Owner1""Name"` for a data store named Owner"1Name.

Note: To update an existing data owner, verify that the name of the data store in the **RepoName** column matches the name of the data store in the Data Privacy Management repository.

Importing Protection Statuses

You can import the protection status of columns in a data store from a CSV file that contains the format that the Import job requires. When you import the file, the Import job runs immediately.

Important: Before you import protection status for new sensitive fields in a data store, you must first import Catalog metadata for the sensitive fields.

If you know that a field in a data store contains sensitive information, you can import a CSV file that identifies the field as sensitive. For example, you scan a data store and the Scan job cannot automatically accept or reject the sensitive field as a match to a data domain. In this case, you must upload a CSV file to manually identify the field as sensitive.

You can also import information about whether or not the field is protected, and specify the protection method. For example, an Informatica PowerCenter Scan job can only verify that a sensitive field is masked if the field is included in a PowerCenter mapping with a data masking transformation. If you use a different protection product to mask sensitive data, you must upload a CSV file to manually identify the sensitive field as protected and specify the protection method.

To import protection status of fields in a data store, perform the following steps:

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. From the **Actions** menu, select **Import** > **Protection Status**.

   The **Import Protection Status** page appears.

3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the Import job handles updates for sensitivity and protection statuses that exist in the Data Privacy Management repository.
   If you enable the option, the Import job updates the sensitivity and protection status in the repository with the values in the CSV file.

   Warning: If a column in the CSV file contains a null value, the Import job replaces the value in the repository with a null value.

   If you disable the option, the Import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the Import job is finished, you can view the protection status information that you imported on the **Data Stores** workspace.

Related Topics:

- "Import Job" on page 299
- "Protection Status Indicator" on page 517

CSV File Format Importing Protection Status

Create a CSV file to create, update, or delete information about the protection status of sensitive fields. The content of the CSV file must be in the order and format that the Import job requires.

You can create the CSV file or update one of the following files that have the required format:

- The Scan report includes fields that the Scan job could not automatically reject or identify as sensitive. To download the report, go to the **Jobs** workspace and click the job ID of the Scan job. Then, click the icon in the **Download Report** column of the Profiling job step.
- The DataStoreDetails.csv file includes columns that the Scan job identified as sensitive.
  To download the file, navigate to the data store **Sensitive Fields** page, select **Actions** > **Export Data Store Details**, and download the .zip file. Then extract the DataStoreDetails.csv file.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file. The CSV file must contain all of the column headings. However, not all columns require values.

Column Order	Column Heading	Column Value Description
1	RepoName	Required. Short description of the data store. The name is case-sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 100 characters, contain spaces, or contain the following special characters: `\ ~ ! $ % ^ & * ( ) +`
2	SchemaName/ FolderName	Required. Name of the schema or folder that contains the field with protection status information.
3	Object	Required. Name of the table or object that contains the field with protection status information.
4	FieldName/FileType	Required. Name of the field that has protection status information.
5	IsSensitive	Optional. Indicates whether the field contains sensitive data. Enter one of the following values: - `Y` - `N`
6	DomainName	Optional. Name of the data domain that contains the sensitive field.
7	ConformanceMatch	Optional. Specifies the minimum percent of sensitive records for Data Privacy Management to automatically accept a field as sensitive.
8	ProtectionStatus	Optional. Indicates whether or not the field is protected. Enter one of the following values: - `Protected` - `Unprotected`
9	ProtectionTechnique	Optional. The method to protect the field.
10	Verified	Creates or updates protection status in Data Privacy Management. Indicates whether the field contains verified data. Enter one of the following values: - `Y` - `N` If the field in the CSV file is empty or you enter N, the Import job skips the row and does not import the protection status.

Guidelines for Importing Protection Statuses

Review the guidelines to understand how to manage the CSV file and how the Import job processes file data.

Consider the following guidelines when you import protection statuses:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines the sensitivity and protection status of one field in a data store.

- The column values cannot contain the following keywords or terms:

```
SELECT FROM
SELECT UNION
DELETE FROM
UPDATE SET
<SCRIPT
</SCRIPT>
ALERT(
"/>"
```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.
- For Snowflake databases, you must enter the values in the **SchemaName** column in the following format: `<database name>.<schema name>`
  This is required to differentiate between schemas with the same name across different databases. The rows are rejected if you do not enter the values in the correct format.

Importing Custom Lineage

You can import data store lineage information from a CSV file. The file must be in the format that the Import job requires. When you import the file, the Import job runs immediately and updates the Data Privacy Management repository with information from the file.

Data lineage describes the flow of data between data stores. Data Privacy Management augments the data lineage information you import with scan results for the data stores. To import custom data store lineage, perform the following steps:

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. From the **Actions** menu, select **Import** > **Custom Lineage**.

   The **Import Custom Lineage** page appears.
3. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.
4. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the Import job finishes, you can access the **Proliferation** page from the **Overview** workspace to view the lineage. The **Overview** workspace shows lineage for scanned data stores. If you imported lineage for an unscanned data store, scan the data store to view the lineage.

Related Topics:

- "Import Job" on page 299

CSV File Format to Import Lineage

Create a CSV file to import information about data lineage in the Data Privacy Management repository. The content of the CSV file must be in the order and format that the Import job requires. The file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Description
1	EtlTool	Required. Name of the tool from which you extracted the lineage. If blank, the Import job rejects the row.
2	FromDataStore	Required. Name of the source data store from which the data moves. Enter the exact name of the data store in the Data Privacy Management repository. The job rejects the row if the value is blank or if the data store name does not exist in the repository.
3	ToDataStore	Required. Name of the target data store to which data moves. Enter the exact name of the data store in the repository. The job rejects the row if the value is blank or if the data store name does not exist in the repository.
4	FromSchema	Optional. Name of the schema in the source data store from which data moves.
5	ToSchema	Optional. Name of the schema in the target data store to which data moves.
6	FromTable	Required. Name of the table in the source data store from which data moves. If blank, the Import job rejects the row.
7	ToTable	Required. Name of the table in the target data store to which data moves. If blank, the Import job rejects the row.
8	FromColumn	Required. Name of the column in the source data store from which data moves. If blank, the Import job rejects the row.
9	ToColumn	Required. Name of the column in the target data store to which data moves. If blank, the Import job rejects the row.
10	ProtectionStatus	Optional. Indicates if the column data is protected. Enter one of the following values: - `Yes`. Data is protected. - `No`. Data is not protected. If blank, default is `No`.

Rules and Guidelines to Import Lineage

Review the rules and guidelines to understand how to manage the CSV file and how the Import job processes custom lineage data from the file.

Consider the following rules and guidelines when you import lineage from a CSV file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines one level of lineage for a column. For example, the social security number column flows from data store A to data store B.
  The combination of the EtlTool, FromDataStore, ToDataStore, FromSchema, ToSchema, FromTable, ToTable, FromColumn, and ToColumn columns defines a unique row of lineage. If the columns two rows are identical, the Import job treats the rows as duplicates.
- If the CSV file contains duplicate entries, the Import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

  ```
  SELECT FROM
  SELECT UNION
  DELETE FROM
  UPDATE SET
  <SCRIPT
  </SCRIPT>
  ALERT(
  "/>"
  ```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.
- The data stores in the CSV file must exist in the Data Privacy Management repository. If you import lineage for a data store that does not exist, the Import job rejects the row.
- The data stores in the CSV file do not require scan results to import the lineage. However, scan results are required to view the data store proliferation on the **Overview** workspace.
- When you import the file, the Import job rejects the row if a data store is included in a running scan. Wait to import the file until the Scan job completes or terminates.
- To add new or changed lineage, append changes to the original file. When the Import job runs, the job deletes all previously imported lineage. Then, the job imports the information from the current file.

Importing Connection Assignments

To add or update connection assignments for a data store, you can import a CSV file on the **Data Stores** workspace.

Data Privacy Management sets or updates the connection assignments for each row in the CSV file that contains connection assignment details. If the CSV file contains details for a data store that already exists,

Data Privacy Management replaces the details of the existing connection assignment with the details in the CSV file.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. From the **Actions** menu, select **Import** > **Connection Assignment**.

   The **Import Connection Assignment** page appears.
3. Click in the **Select** field.

   The **Open** window appears.
4. Browse to the directory that contains the connection details file in CSV format, select the file, and then click **Open**.
5. Click **Select**.
6. To check the status of the Import job, go to the **Jobs** workspace.
7. To view the imported connection assignments after the Import job finishes, return to the **Data Stores** workspace and view the **Data Store Details** page.

Related Topics:

- "Import Job" on page 299

Importing Catalog Metadata

To add Catalog metadata to a data store, you can import a CSV file on the **Data Stores** workspace.

For Snowflake databases, you must enter the values in the **SchemaName** column in the following format:
`<database name>.<schema name>`
This is required to differentiate between schemas with the same name across different databases. The rows are rejected if you do not enter the values in the correct format.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. Select the data store for which you want to import Catalog metadata.
3. From the **Actions** menu, select **Import** > **Catalog Metadata**.

   The **Import Catalog Metadata** page appears.
4. Click in the **Select** field.

   The **Open** window appears.
5. Browse to the directory that contains the metadata file in CSV format, select the file, and then click **Open**.
6. Click **Select**.

After you import Catalog metadata for a Cloudera Hive data store, import the protection status for the sensitive fields in the data store. The sensitive fields listed in the protection status CSV file must contain Catalog metadata.

Related Topics:

- "Import Job" on page 299

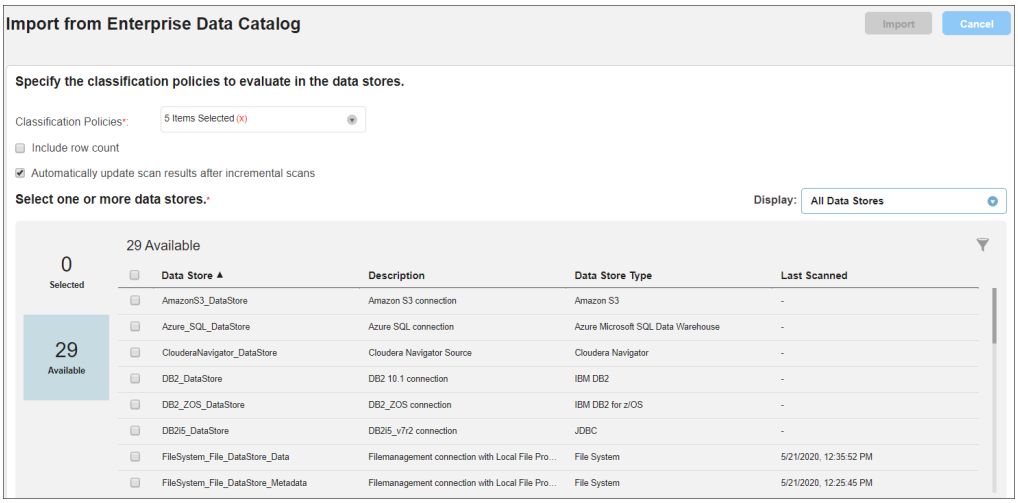
Importing Catalog Resources

You can add resources from Informatica Enterprise Data Catalog to a data store on the **Data Stores** workspace.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. Select the data store for which you want to import Enterprise Data Catalog resources.
3. From the **Actions** menu, select **Import** > **Catalog Resources**.

   The **Import from Enterprise Data Catalog** page appears.



4. In the **Display** menu, select **Missing Data Stores** or **All Data Stores**.
5. In the **Classification Policies** field, select the classification policies to evaluate in the data stores.
6. Optionally, select the **Include row count** check box.

   Note: Data Privacy Management includes row counts when you import Catalog resources only if you clear the **Automatically update scan results after incremental scans** check box.
7. Optionally, clear the **Automatically update scan results after incremental scans** check box. Default is selected.
   - When enabled, the Sync Catalog Updates job runs when you import the resources. The Sync Catalog Updates job imports only the updated columns that are included in the data domain resources that match the selected classification policies and in the data store scan options. The Sync Catalog Updates job does not include row counts.
   - When disabled, the Import Catalog Resources job runs when you import the resources. The Import Catalog Resources job imports the columns that are included in the data domain resources that match the selected classification policies and in the data store scan options.
8. Select one or more data stores to import Enterprise Data Catalog resources.

9. Click **Import**.

   Data Privacy Management imports the data domain resources that match the selected classification policies for the selected data stores.

   Note: If the data store resources contain lineage data, an Import Catalog Lineage job is automatically triggered to import the lineage data into Data Privacy Management. Import Catalog Lineage job incorporates checkpoints. You can resume a failed job from the point of failure.

After you import resources from Enterprise Data Catalog, update the connection string details for the selected data stores, if applicable, in the data store properties before you scan the data store.

Related Topics:

- "Import Job" on page 299

Exporting Data Breach Reports

If a security event occurs that compromises subject data in sensitive fields or files, you can export a data breach report as a CSV file from the **Data Stores** list page or the **Data Store Details** page. You can export a data breach report for one data store at a time. If a breach impacted multiple data stores, download a data breach report for each data store.

To export a breach report, you must have permissions to manage data stores. Also, a Subject scan and data profiling must complete successfully before breach reports are available to export.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. On the **Data Stores** list page, select the check box next to a data store that was included in the security event. Or, click the data store name to access the **Data Store Details** page.

3. From the **Actions** menu, select **Export** > **Data Breach Report**.

   The **Save As** dialog box appears with the CSV file. The breach report has the following default file name: `BreachReport_Summary.csv`

4. Optionally, edit the file name. For example, include the data store name or the date you export the file.

5. Save the CSV file to a directory on your machine.

6. To review the data breach details, open the CSV file.

CSV File Format for Data Breach Reports

When you export a data breach report, you export a CSV file that contains details about the sensitive information in the data store that the breach impacted.

If a data store does not contain information for a column, the column value is empty in the data breach report.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Description
1	Name	Name of the data store as it appears in the Name column on the **Data Stores** workspace.
2	Data Store Type	Type of data store as it appears in the Data Store Type column on the **Data Stores** workspace.
3	Data Owner	Name of the individual, group, or organization that is responsible for the accuracy and integrity of the source data.
4	Location	If the data store is associated with a location defined on the **Locations** workspace, the location name.
5	User	Name of the user account to log in to the data store.
6	Time	Date and time that you export the data breach report, in the India Standard Time (IST) zone. The column value uses the following format: *mm/dd/yyyy hh:mm:ss <AM_PM>* For example: `2/22/2020 12:10:16 AM`
7	Subjects Impacted	The number of data subjects impacted by the breach.
8	Fields/Files	The number of sensitive fields or files in the data store that were impacted by the breach.
9	Sensitive Fields protected	The percentage of sensitive fields in the data store that are protected. For data store types that protection extensions do not support, the value is Not Applicable.
10	Protection Status	Indicates whether the sensitive fields in the data store are Protected or Unprotected. For data store types that protection extensions do not support, the value is Unprotected.
11	Protection Extensions	If sensitive fields are protected, lists the names of the protection extensions configured on the **Extensions** workspace. Multiple values are separated by a comma (,). For data store types that protection extensions do not support, the value is Not Applicable.
12	Categories	Names of the categories associated with data domains in the data stores. Multiple values are separated by a comma (,). For unstructured data stores configured for Subject Registry, data domains are not associated with categories. The value for unstructured data stores is Not Applicable.
13	Purposes	The reason that you collected the sensitive data about the individual. Multiple values are separated by a comma (,). Unstructured data stores configured for Subject Registry cannot specify a purpose. The value for unstructured data stores is Not Applicable.

Column Order	Column Heading	Description
14	Shared with Third Party	Indicates that the data is shared with a third party such as a third-party organization or vendor. Values are Yes or No.
15	Third-Party Description	If the data is shared with a third party, lists the names of the third parties. Multiple values are separated by a comma (,).

Exporting Data Store Aliases

To review a list of data stores and the corresponding aliases, export the data store aliases to a CSV file.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. From the **Actions** menu, select **Export** > **Data Store Aliases**.

   A dialog box appears with the exported file.
   The export file is named `dataStoreAliases.csv`. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, another file downloads. The file name includes an incremental number such as dataStoreAliases (1).csv.
3. Open the file.
4. Update the data store aliases in the file.
5. Save the file.

After you export the file, you can update and then import the file to add, delete, or update data store alias details in the Data Privacy Management repository. The export file contains the same format that is required for the import.

Exporting Data Store Configurations

Export data store connection information to update the configuration files and run commands on the machine that hosts the data store. Update the machine for the services that are on the same domain as the Data Privacy Management Service.

Before you export data store connection information, scan the data store.

To configure the connectivity for all data stores except for IBM Db2, add an entry for the data stores in the corresponding configuration files. To configure the connectivity for IBM Db2 data stores, run a set of commands from a command prompt. The configuration export file includes the configuration entries to copy and the IBM Db2 commands to run.

For example, a Scan job imported data stores that connect to Oracle databases. Update the `tnsnames.ora` configuration file to include an entry for each data store that the Scan job imported.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. From the **Actions** menu, select **Export** > **Data Store Configurations**.

   The **Save As** dialog box prompts you to download the .txt configuration file. The default file name is `DbConfigFile.txt`. Save the file to a directory.
3. Open the file.

The following text shows an example of the configuration file content:

```
################################################################################
#                                  tnsnames.ora                                #
################################################################################

orasas =
  (DESCRIPTION =
   (ADDRESS = (PROTOCOL = TCP)(HOST = IRL64ILM04)(PORT = 1521))
   (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = sas.informatica.com)
   )
  )

INVR268 =
  (DESCRIPTION =
   (ADDRESS = (PROTOCOL = TCP)(HOST = INVR28ILM268)(PORT = 1521))
   (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = orcl.informatica.com)
   )
  )

################################################################################
#                                   odbc.ini                                   #
################################################################################

[ODBC Data Sources]
SQL Server Wire Protocol 1=DataDirect 7.1 SQL Server Wire Protocol

[SQL Server Wire Protocol 1]
Driver=$ODBC_HOME/lib/DWsqls27.so
Description=DataDirect 7.1 SQL Server Wire Protocol
AlternateServers=
AlwaysReportTriggerResults=0
AnsiNPW=1
ApplicationName=
ApplicationUsingThreads=1
AuthenticationMethod=1
BulkBinaryThreshold=32
BulkCharacterThreshold=-1
BulkLoadBatchSize=1024
BulkLoadFieldDelimiter=
BulkLoadOptions=2
BulkLoadRecordDelimiter=
ConnectionReset=0
ConnectionRetryCount=0
ConnectionRetryDelay=3
Database=source1
EnableBulkLoad=0
EnableQuotedIdentifiers=0
EncryptionMethod=0
FailoverGranularity=0
FailoverMode=0
FailoverPreconnect=0
FetchTSWTZasTimestamp=0
FetchTWFSasTime=1
GSSClient=native
HostName=IRW28ILM01
HostNameInCertificate=
InitializationString=
Language=
LoadBalanceTimeout=0
LoadBalancing=0
LoginTimeout=15
LogonID=
MaxPoolSize=100
MinPoolSize=0
PacketSize=-1
```

```
Password=
Pooling=0
PortNumber=1433
QueryTimeout=0
ReportCodePageConversionErrors=0
SnapshotSerializable=0
TrustStore=
TrustStorePassword=
ValidateServerCertificate=1
WorkStationID=
XMLDescribeType=-10
IANAAppCodePage=UTF-8

################################################################################
#                                      db2                                     #
################################################################################

db2 catalog tcpip <node_name> remote invrlx61ilm174 server 60003 port
remote_instance DB10X174
db2 catalog database DB10X174 as DB10X174 at node <node_name>

################################################################################
```

Note: The tnsnames.ora section in the file includes a set of configurations for each connection string. For example, if multiple data stores have the same connection string, but the data stores have different host and ports, the export file includes only one instance of the connection string.

4. Configure the following property for ODBC data sources that have dashes or hyphens in the table names or columns:

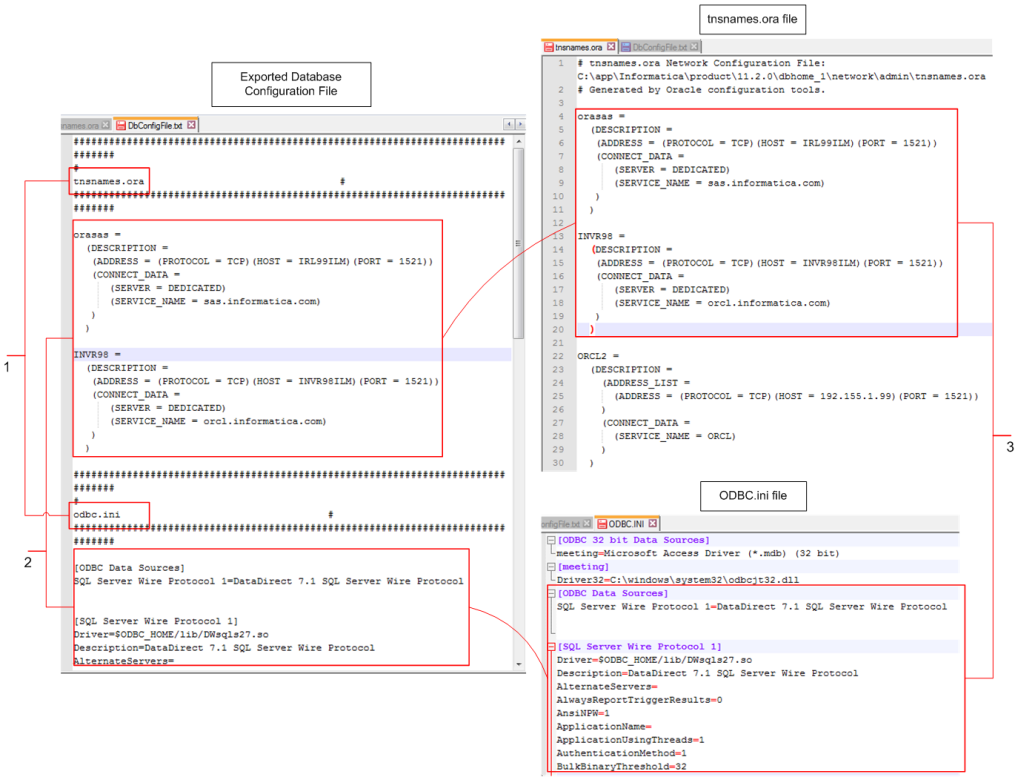
   ```
   EnableQuotedIdentifiers=1
   ```

   If you do not configure the property, the Scan job might fail.

5. Access the configuration files on the machine that hosts the Data Integration Service. If the services are hosted on different machines, update the configuration files on each machine.

6. Copy the entries from the export .txt file to the corresponding configuration file. Save the configuration file.

The following image shows an example of the entries that you add to the configuration files:



1. Names of the database configuration files on the machines that host the services.
2. Entries to add to the database configuration files on the machines that host the services.
3. Database configuration file entries that you copied from the export file.

In this example, you copy the `orasas` and `INVR98` entries from the export file to the `tnsnames.ora` file. You copy the `SQL Server Wire Protocol 1` entry from the export file to the `odbc.ini` file.

7. For IBM Db2 data stores, access the command line on the machine that hosts the Data Integration Service.
8. Copy the IBM Db2 commands from the export file and run the commands. If the services are hosted on different machines, run the commands on each machine.

   In this example, copy and run the following entries:

```
db2 catalog tcpip <node_name> remote invrlx61ilm174 server 60003 port
remote_instance DB10X174
db2 catalog database DB10X174 as DB10X174 at node <node_name>
```

Merging Data Stores

Use the data store filters to determine which data stores you need to merge. Merge data stores that connect to the same data source. Merge duplicate data stores to have an accurate data store count in the dashboard and to avoid scanning the same data source multiple times.

Duplicate data stores can have different metadata such as location, owner, and data store group. Data Privacy Management only uses the connection properties to determine duplicate data stores.

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.

2. In the navigation filters, click **Potential Duplicates**.

   The list of data stores shows groupings of data stores that have the same connection properties.

3. Select the data stores that you want to merge.

   You can merge data stores from multiple groups at the same time. You might want to verify that the data stores have the same security group. If you merge data stores that have different security groups, you receive an error.

4. Select the primary icon for the data store that you want use as the primary data store grouping.

5. Click the **Merge** icon.

   The remainder of the data stores that you merge become aliases of the primary data store. The **Overview** workspace only displays and counts the primary data stores. You can scan primary data stores, but you cannot scan data store aliases.

Example of Duplicate Data Stores

Data stores that connect to the same data source are duplicate data stores.

The following table lists sample data store connection properties for an Oracle database:

Data Store Name	Data Store Type	Host	Port	Service Name
Data Store 1	Oracle 11g	ilm600	1521	(null)
Data Store 2	Oracle 11g	ilm600	1521	(null)
Data Store 3	Oracle 11g	ilm600	1521	orcl
Data Store 4	Oracle 11g	ilm600	1521	orcl
Data Store 5	Oracle 11g	ilm600	1521	informatica

Data Privacy Management groups the following duplicate data stores:

- Two Oracle 11g Data Stores [ilm600:1521]. This group includes Data Store 1 and Data Store 2.
- Two Oracle 11g Data Stores [ilm600:1521:orcl]. This group includes Data Store 3 and Data Store 4.

Data Store 5 has the same data store type, host, and port as the other data stores. However, the service name is unique, so Data Store 5 is not a duplicate data store.

Use Cases for Merging Data Stores

After you create or import data stores, you might need to merge data stores.

You might merge data stores for one or more of the following purposes:

Consolidate duplicate data stores that you imported from a parent repository.

When a Scan job imports data stores from a parent repository, such as Informatica PowerCenter, the Scan job can create duplicate data stores. For example, when you scan an Informatica PowerCenter data store, the Scan job creates one data store for each source and target connection in the PowerCenter repository mappings. You can have duplicate data stores if a mapping includes the same connection as a source and a target.

Consolidate duplicate data stores that you imported from CSV files.

When you import data stores from CSV files, the Import job creates one data store for each connection in the file. You can have duplicate data stores if the CSV file includes multiple data stores with the same connection properties.

Consolidate duplicate data stores from multiple creation or import methods.

You might have duplicate data stores if you use a combination of import and creation methods to create data stores. For example, you manually create data stores, import data stores from a CSV file, and import data stores from Informatica PowerCenter. The CSV file might contain one or more data stores that you imported from Informatica PowerCenter or that you manually created.

Deleting a Data Store

You can delete data stores from the Data Privacy Management repository. You cannot delete data stores that are included in a completed scan, a scan that is currently running, or a scan that is scheduled to run in the future.

You might want to delete data stores that connect to the same database. When you create or import data stores, Data Privacy Management does not validate if a data store with the same connection properties exists. Data Privacy Management identifies data stores that contain the same connection properties as duplicate data stores.

Review rules and guidelines to learn when you can delete stores and what other options you might need to consider. You might want to merge data stores instead of deleting data stores.

To delete a data store, perform the following steps:

1. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
2. Select the check box next to the data store that you want to delete.
3. From the **Actions** menu, select **Delete**.

   A message appears to confirm that you want to delete the data store.
4. Click **Yes**.

Rules and Guidelines to Delete a Data Store

Consider the following rules and guidelines when you delete a data store:

- You cannot delete data stores that are included in a completed scan, a scan that is currently running, or a scan that is scheduled to run in the future.
- When you delete data stores, you delete the data store from the Data Privacy Management repository. You also delete the corresponding audit trail for the data store. If the data store that you want to delete is a duplicate data store, consider merging the data store instead. When you merge data stores, you keep the audit trail of all the data stores.
- Consider merging data stores that you imported from Informatica PowerCenter instead of deleting the data stores. If you delete data stores that an Informatica PowerCenter Scan job imported, the Scan job imports the data stores again the next time you scan the Informatica PowerCenter data store.

  When you delete data stores, you remove the data store from the repository. When you scan an Informatica PowerCenter data store, the Scan job imports data stores from Informatica PowerCenter. If the data store exists in the repository, the Scan job does not import the data store. If you merge data stores, all data stores remain in the repository.

- When you delete data stores, you remove the rules associated with the data store from the **Active** and **Suggestions** tabs on the **Always Sensitive /Never Sensitive** workspace.
- You can delete merged data stores. When you delete an alias data store, Data Privacy Management removes the alias reference from the primary data store. When you delete a primary data store, Data Privacy Management removes references from the alias data stores. If the primary data store included more than one alias data store, the alias data stores become duplicate data stores. Merge the duplicate data stores to select a new primary data store.

CHAPTER 8

Data Store Properties

This chapter includes the following topics:

- Data Store Properties Overview, 127
- Additional Properties, 128
- Active Directory Connection Properties, 129
- Amazon Redshift Connection Properties, 131
- Amazon S3 Connection Properties, 133
- Apache Cassandra Connection Properties, 136
- Cloudera Navigator Connection Properties, 138
- Database Management Connection Properties, 139
- Office 365 Outlook, 144
- File System Connection Properties, 145
- Google BigQuery Connection Properties, 148
- Google Drive Connection Properties, 149
- Hadoop Distributed File System Connection Properties, 150
- Hive Connection Properties, 155
- Informatica Cloud Connection Properties, 158
- Informatica Data Engineering Integration Properties, 159
- Informatica PowerCenter on IBM Db2 Connection Properties, 162
- Informatica PowerCenter on Microsoft SQL Server Connection Properties, 166
- Informatica PowerCenter on Oracle Connection Properties, 169
- Microsoft Azure Blob Storage Connection Properties, 173
- Microsoft Azure Data Lake Connection Properties, 175
- Microsoft Azure SQL Database Connection Properties, 178
- Microsoft Azure SQL Data Warehouse Connection Properties, 181
- Microsoft OneDrive Connection Properties, 183
- Microsoft SharePoint Connection Properties, 186
- Salesforce Connection Properties, 190
- SAP Connection Properties, 191
- SAP S/4HANA Data Store Connection Properties, 194
- Snowflake Connection Properties, 196

- Snowflake Advanced Scanner data store properties, 197
- SQL Server Integration Services Connection Properties, 199

Data Store Properties Overview

When you create a data store or import data stores from Informatica PowerCenter or Informatica Data Engineering Integration, you configure connection properties to the data source and additional properties that contain metadata about the data store.

When you view a **Data Store Details** page, the page contains sections for the following property types:

Property Type	Description
Connection properties	Contain the following information for all data store types: data store name, description, category, and data store type.
Additional properties	Contain the following information: metadata such as location and tags, third party share status, the remote agent associated with the data store, and whether the Scan job will automatically synchronize with Enterprise Data Catalog. The additional properties you can configure depend on the data store type.

The following table describes the common connection properties:

Property	Description
Name	Required. Short description of the data store. The name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 255 characters or contain spaces or the following special characters: `\~!$%^&*()+`
Description	Optional. Long description of the data store that does not exceed 255 characters.

Property	Description
Category	Required. Select one of the following options: - Application - Big Data - Cloud - Data Integration - Database Management - File Management - NoSQL - Email Server After you save the data store, you cannot change the category.
Data Store Type	Required. Select one of the following options, depending on the category: - Application: SAP, and Active Directory. - Big Data: Cloudera Navigator, Hadoop Distributed File System, and Hive. - Cloud: Amazon Redshift, Amazon S3, Azure Data Lake, Microsoft Azure Blob Storage, Microsoft Azure SQL Data Warehouse, Microsoft Azure SQL Database, Google BigQuery, Salesforce, and Snowflake. - Data Integration: Informatica Cloud, Informatica Data Engineering Integration, Informatica PowerCenter on IBM Db2, Informatica PowerCenter on Microsoft SQL Server, Informatica PowerCenter on Oracle, and SQL Server Integration Services. - Database Management: IBM Db2, IBM Db2 for i5/OS, IBM Db2 for z/OS, JDBC, Microsoft SQL Server, Netezza, Oracle, SAP HANA, Sybase, and Teradata. - File Management: File System, Google Drive, Microsoft OneDrive, and Microsoft SharePoint. - NoSQL: Apache Cassandra - Email Server: Office 365 Outlook After you save the data store, you cannot change the data store type.

Related Topics:

- "Creating a Data Store" on page 91

Additional Properties

Configure additional properties related to the data store metadata. Configure the data owner and department for the data store. Assign data stores to locations, data store groups, and security groups. You can use the additional properties for drill-down analysis of the scan results.

The following table describes the additional properties you can configure for all data store types:

Connection Property	Description
Data Store Group	Required. Select a logical grouping of related data stores from the list of groups defined on the **Data Store Groups** workspace. You can assign a data store to one data store group.
Security Groups	Required. Select a security group from the list of all security groups in the Informatica domain, including native and LDAP groups. Native groups display the group name. For example, Administrator. LDAP groups include the security domain and the group name. For example, *<security domain>/<LDAP group name>*.

Connection Property	Description
Tags	Optional. Keywords that add to the description. Create a new tag or select an existing tag from the list. You can assign multiple tags to a data store. After you assign a tag to a data store, you can use tags in alert rules to improve the specificity of the alert. For example, you want to be alerted when the risk score of any production data store increases by 20%. However, you do not want to be alerted for non-production data stores. You can assign the tag PROD to the production data stores. Then, configure the alert conditions to alert you when the risk score increases by 20% for data stores that have the PROD tag. When you delete a data store, if a tag was only assigned to the deleted data store, the Data Privacy Management Service deletes the tag from the repository.
Department	Optional. Enter the department that is responsible for the maintenance or ownership of the data store. The department can be a line of business, such as investment banking or wealth management, or a corporate department, such as Sales or Marketing. The name is case-sensitive and cannot exceed 100 characters. After you scan data stores, you can view a summary of scan results for all data stores by department. You can filter the scan results by location on the **Overview** workspace.
Data Owner	Optional. Select the individual, group, or organization name that is responsible for the accuracy and integrity of the source data on which you run a classification profiling scan.
Shared with Third Party	Optional. Indicates that the data in the data store is shared with a third party such as a third-party organization or vendor.
Third-Party Description	Optional. Appears if you select the **Shared with Third Party** check box. Enter a description of the third party. You can enter up to 2000 characters and use the following special characters: , _ @
Location	Optional. Select a location that is defined on the **Locations** workspace. If you do not select a location, Data Privacy Management assigns the Unknown location to the data store.
Auto Sync Catalog	Optional. Valid only for data stores that are not scanned with remote agents. Determines if the Scan job automatically synchronizes data source information in Enterprise Data Catalog with the corresponding data store information in Data Privacy Management. Default is disabled.
Custom Risk Score Factor	Optional. If you created a custom risk score factor on the **Settings** workspace, the factor appears as a property. For existing data stores, you can select a value or range. For new data stores, the property shows the default value or range that you specified on the **Settings** workspace. You can select another value or range from the list.

Active Directory Connection Properties

Configure connection properties to connect to an Active Directory data store. Enterprise Data Catalog does not support Active Directory resources. You cannot configure the data store to auto sync with Enterprise Data Catalog or import an Active Directory resource from Enterprise Data Catalog.

You cannot determine lineage of sensitive data in an Active Directory scan.

The following table describes the Active Directory connection properties:

Property	Description
Driver Class	Name of the JDBC driver class.
URL	Enter the JDBC connection URL used to access the server.
User	Name of the user account to log in to the database.
Password	Optional. Password for the database user name. When you save the data store, Data Privacy Management encrypts the password.
Agent URL	Optional. Enter the URL of the Enterprise Data Catalog agent in the following syntax: `http://<host name or IP address>:<port>/<directory>`
Connection String	Optional. Enter the value that you enter for the Agent URL.
Secure JDBC Parameters	Optional. Database parameters to access databases that are secured with the SSL protocol or Azure Key Vault. The parameter string cannot exceed 500 characters. When you save the data store, Data Privacy Management encrypts the parameter string. Enter the parameters as `name=value` pairs separated by semicolon characters (;). Data Privacy Management appends the secure JDBC parameters to the connection string.
Schema	Enter one or more schemas that Data Privacy Management reads when you scan the data store. The schema is case-sensitive. Use a comma to separate multiple entries. If empty, Data Privacy Management reads all schemas from the database.
Source Metadata Filter	Optional. Default is All. A filter to specify tables that you want to include or exclude in the scan. Enter a regular expression to include multiple filter parameters. For example: `Table1, NOT Table2, SUB%, %EMP, %DATA%`. The filter includes Table1, tables with names that start with SUB, names that end with EMP, and tables that include DATA anywhere in the name. The filter excludes Table2.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select High, Low, or Medium. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
Custom JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Agent Options	Optional. Enter the Agent configuration options to include in the scan.
CyberArk Safe	Optional. Name of the CyberArk safe that contains the database password. When you test the data store connection or you scan the data store, Data Privacy Management retrieves the password from CyberArk.

Property	Description
Include Views in the Scan	Optional. Determines whether a Scan job evaluates data in views. If enabled, a Scan job identifies sensitive data in views during metadata and data profiling. The Scan job evaluates data in all views for the schemas specified in the **Schema** property. If the property is blank, the Scan job evaluates data in all views of all schemas. If disabled, the Scan job does not evaluate data in views. If you disable the property after a Scan job completes, any scans that you run in the future will not evaluate data in views. However, the completed Scan job results remain.
Profile Execution Engine	Optional. Select one of the following profiling engine modes: - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. Default is Native.
Source Connection Name	Optional. Connection name in the Informatica domain that the scan job uses to connect to the source database. If the Scan job cannot connect to the source, the job fails.

Amazon Redshift Connection Properties

Configure properties to connect to Amazon Redshift. You can include an Amazon Redshift data store in a Cloud scan.

When you test a connection to Amazon Redshift, the Data Privacy Management Service tests the combination of the **User**, **Host**, **Port**, and **Database** properties. If any of the property values are not valid, the test connection fails.

The following table describes the Amazon Redshift connection properties:

Property	Description
User	Required. Name of the user account to log in to Amazon Redshift. The user must have read permission to the source files and have access to the IP address.
Password	Optional. Password for the user account to log in to Amazon Redshift. When you save the data store, Data Privacy Management encrypts the password.
Host	Required. Host name or IP address for the database server.
Port	Required. Port number for the database server.
Database	Required. The name of the database instance.
AWS Access Key ID	Required. Access key ID to access the Amazon account resources. For example, `AKIAIOSFODNN7EXAMPLE`. When you save the data store, the Data Privacy Management Service encrypts the value in the repository. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.

Property	Description
AWS Secret Access Key	Required. Secret access key to access the Amazon account resources. This value is associated with the access key ID and uniquely identifies the account. For example, `wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY`. When you save the data store, the Data Privacy Management Service encrypts the value in the repository. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
AWS Master Symmetric Key	Optional. Provide a 256-bit AES encryption key in the Base64 format. You can generate a key using a third-party tool.
Schema Option	Optional. Select one of the following options to determine how you add schemas to the data store: - **All**. Includes all schemas that you can access. - **Select From List**. Displays a list of schemas in the **Schema** property. - **Specify Regex**. Enables the **Regular Expression for Schema** property. Select this option if you want to enter schemas as a regular expression.
Regular Expression for Schema	Optional. Available if you select the **Specify Regex** option in the **Schema Option** property. Enter a regular expression to specify schemas that you want to include or exclude from the scan. Schema names are case-sensitive.
AWS Cluster Node Type	Optional. Node type of the Amazon Redshift cluster. Select one of the following options: - dc1.8xlarge - dc1.large - ds1.8xlarge - ds1.xlarge - ds2.8xlarge - ds2.xlarge For more information about nodes in the cluster, see the Amazon Redshift documentation.
AWS Total Nodes in Cluster	Required. Number of nodes in the Amazon Redshift cluster. For more information about nodes in the cluster, see the Amazon Redshift documentation.
Schema	Optional. Displays a list of available schemas to include in the scan if you chose **Select From List** in the **Schema Option** property.
S3 Bucket Name	Required. Name of the Amazon S3 bucket that contains the semi-structured data to scan. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
Case Sensitive	Optional. Select to indicate that the data store is configured for case sensitivity. Default is cleared.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.

Property	Description
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Source Connection Name	Optional. Connection name in the Informatica domain that the Scan job uses to connect to the Amazon Redshift source.

Amazon S3 Connection Properties

To connect to an Amazon Simple Storage Service (Amazon S3) data store, configure the connection properties on the **Data Stores** workspace.

Create a data store for each Amazon Web Services (AWS) bucket. You can create multiple data stores that connect to the same AWS bucket, but the **Source Directory** property must be unique for each data store.

When you test a connection to Amazon S3, Data Privacy Management tests the combination of the **AWS Bucket URL**, **AWS Access Key ID**, **AWS Secret Access Key**, **AWS Bucket Name**, and **Source Directory** properties. If any of the property values are not valid, the test connection fails.

The following table describes the Amazon S3 connection properties:

Property	Description
Scan with Remote Agent	Select to associate the data store with a remote agent when you run scans to discover personal and sensitive data or build a Subject Registry index. **Important:** If you select this option, you must associate the data store with a remote agent before running a scan. After you save the data store, you cannot change this property.
S3 Supported Type	Specify if the data store connects to a storage type that is compatible with Amazon S3 and can be accessed by Amazon Web Services REST APIs.
AWS Bucket URL	Required if you select No in the S3 Supported Type field. Virtual-hosted-style URL for the AWS bucket that contains the semi-structured data to scan. Use the following syntax for the URL: `<Bucket_Name>.s3.amazonaws.com` For example, if the bucket name is `DPM_QA`, enter `DPM_QA.s3.amazonaws.com`. Do not include the HTTP protocol in the value. If another data store has the same value for the AWS bucket URL, Data Privacy Management marks the data store as a potential duplicate.
REST EndPoint URL	Required if you select Yes in the S3 Supported Type field. Enter the REST endpoint URL of the Amazon S3 compatible source. For example, the Scality Ring REST endpoint URL is: `http://s3.isv.scality.com`

Property	Description
Are Credentials Temporary?	Specify whether the credentials are temporary or permanent.
AWS Session Token	Required if the credentials are temporary. Enter the AWS token for the session.
AWS Access Key ID	Optional. Access key ID to access the Amazon account resources. For example: `AKIAIOSFODNN7EXAMPLE`. When you save the data store, the Data Privacy Management Service encrypts the value in the repository. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
AWS Secret Access Key	Optional. Secret access key to access the Amazon account resources. The secret access key is associated with the access key ID and uniquely identifies the account. For example, `wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY`. When you save the data store, the Data Privacy Management Service encrypts the value in the repository. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
AWS Bucket Name	Optional. Name of the Amazon S3 bucket that contains the semi-structured data to scan. Use the bucket name configured in the **AWS Bucket URL** property. For example, if the AWS Bucket URL value is `DPM_QA.s3.amazonaws.com`, enter `DPM_QA`. If the bucket name does not match the bucket name in the **AWS Bucket URL** property, the Scan job fails. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
Source Directory	Optional. Directory path that contains the files to scan. To scan all root directories, enter a forward slash `/`. To scan a specific directory, use the following syntax: `/<Directory_Name>` For example, if the directory name is `Folder1`, enter `/Folder1`. Do not include a forward slash `/` at the end of the directory. The Scan job reads files from the last directory specified. For example, if you enter `/HumanResource/Employees/US`, the Scan job only reads files from the `US` directory. The scan reads files from the specified directory only. To enable a scan to read from subdirectories, enable the **Include Sub Directory** property. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
Region Name	Optional. The name of the geographical region for the Amazon S3 bucket. The scan uses the region name to create a connection in the Administrator tool. Select the same region that is associated with the Amazon S3 bucket in Amazon S3. If you select a different region, the Cloud job fails during the Profiling step. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
Folder Options	Required if you enable the **Scan with Remote Agent** property. Folders that the scan reads from the location configured in the **Source Directory** property. Select one of the following options: - All - Select Specific Folders - Use Regular Expression Default is All.
Include Folders	Appears when you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to include in the scan. You can also skip this property and list folders in the **Exclude Folders** property. In this case, Data Privacy Management scans the folders in the data store except the excluded folders.

Property	Description
Exclude Folders	Optional if you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to exclude from the scan.
Folder Regular Expression	Required if you enable the **Scan with Remote Agent** property and you choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder to include in the scan.
File Types	Required if you enable the **Scan with Remote Agent** property. File types that the scan reads from the location configured in the **Source Directory** property. Choose **All** to scan all file types. Choose **Select** to select specific file types to scan. Default is All.
Selected File Types	Required if you choose **Select** in the **File Types** property. Click **Select All** to choose all file types, or select one or more of the following options: - Avro - Compressed Files - Delimited and Text - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, the PDF file content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.
Enter File Delimiter	Optional. Available if you do not enable the **Scan with Remote Agent** property. Character that separates entries in the file. Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`. Default is a comma (,).
First Level Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Name of the folder that you want to scan. For example: `subfolderA`. The folder must be an immediate subfolder of the directory that you specified in the **Source Directory** property. If you do not provide a folder name, Data Privacy Management will scan the files in the source directory. To specify multiple folders, separate the folder names with a comma (,). For example: `subfolderA, subfolderB`
Include Sub Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. If enabled, the scan reads data from the directory and all subdirectories of the location configured in the **Source Directory** property or the **First Level Directory** property, if specified. Default is disabled.

Property	Description
Memory	Optional. Memory that the scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Source Connection Name	Optional. Connection name in the Informatica domain that the scan uses to connect to the Amazon Redshift source.

Apache Cassandra Connection Properties

Configure properties to connect to Apache Cassandra. You can include an Apache Cassandra data store in a NoSQL scan.

When you test a connection to Apache Cassandra, the Data Privacy Management Service tests the combination of the **User**, **password**, **Host**, **Port**, and **Data Center** properties. If any of the property values are not valid, the test connection fails.

The following table describes the Apache Cassandra connection properties:

Property	Description
Host	Host name or IP address for the Apache Cassandra server.
Port	Optional. Port number of the Apache Cassandra server. Default port number is 9042.
Local Data Center	Name of the datacenter that contains the required node.
User Name	Optional. Name of the user account to log in to the Apache Cassandra server. The user must have read permission to the source files and have access to the Cassandra server.
Password	Optional. Password for the user account to log in to the server. When you save the data store, Data Privacy Management encrypts the password.
SSL Enabled	Option to enable SSL.
Keystore Path	Location of the Apache Cassandra Keystore.

Property	Description
Keystore Password	Password to access the Apache Cassandra Keystore.
Truststore Path	Location of the Apache Cassandra Truststore.
Truststore Password	Password to access the Apache Cassandra Truststore.
Keyspace Option	Imports a particular database schema. Select one of the following options to determine how you add keyspaces to the data store: - **All**. Includes all keyspaces that you can access. - **Select From List**. Displays a list of keyspaces in the **Keyspace** property. - **Specify Regex**. Enables the **Regular Expression for Keyspace** property. Select this option if you want to enter schemas as a regular expression.
Regular Expression for Keyspace	Optional. Available if you select the **Specify Regex** option in the **Keyspace Option** property. Enter a regular expression to specify teh keyspace that you want to include or exclude from the scan.
Keyspace	Optional. Displays a list of available keyspaces to include in the scan if you chose **Select From List** in the **Keyspace Option** property.
Case Sensitive	Optional. Indicates that the data store is configured for case sensitivity. Default is cleared. **Note:** You cannot include keyspaces, tables, or columns that include uppercase characters in a domain discovery scan on an Apache Cassandra data store.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
Custom JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.

Cloudera Navigator Connection Properties

Configure the connection details to a Cloudera Navigator Metadata Server. When you scan a Cloudera Navigator data store, the Cloudera Navigator Scan job identifies sensitive data proliferation between scanned Hive data stores that are on the same host as Cloudera Navigator.

When you test a connection to Cloudera Navigator, the Data Privacy Management Service tests the combination of the **Navigator URL**, **User**, and **Password** properties. The Data Privacy Management Service uses the host name specified in the **Navigator URL** property to identify duplicate data stores.

The following table describes the Cloudera Navigator connection properties:

Property	Description
Navigator URL	Required. Enter the URL of the Cloudera Navigator Metadata Server. The Cloudera Navigator Scan job identifies proliferation between scanned Hive data stores that are on the same host as Cloudera Navigator.
User	Required. Name of the user account to log in to Cloudera Navigator. The Cloudera Navigator Scan job requires a Cloudera Manager user account that includes the Full Administrator or Navigator Administrator role. The roles include authorization to connect to and retrieve metadata from Cloudera Navigator. If the user account does not have one of these roles, the Scan job fails.
Password	Required. Password for the user account to log in to Cloudera Navigator. When you save the data store, Data Privacy Management encrypts the password.
Disable SSL Validation	Optional. Determines whether Data Privacy Management validates the SSL certificate, truststore, keytab file, and other SSL details for a Cloudera Navigator data store encrypted with the SSL protocol. When the check box is selected, Data Privacy Management does not validate the SSL details. When the check box is cleared, Data Privacy Management validates the SSL details. Default is cleared.
Hive Database	Optional. Hive databases for which the Cloudera Navigator Scan job identifies lineage. To determine lineage for all databases in scanned Hive data stores that are on the same host as Cloudera Navigator, do not enter a value in this property. The Hive database name is case-sensitive. Use a comma to separate multiple entries.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.

Property	Description
Enable Reference Resources	Optional. Select this option to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is disabled.
Retain Unresolved Reference Assets	Optional. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is enabled.

Database Management Connection Properties

Configure connection properties to connect to a relational database. You can connect to IBM Db2, JDBC, Microsoft SQL Server, Netezza, Oracle, SAP HANA, Sybase, and Teradata databases.

You can use the JDBC type with the corresponding CDATA connectors to create data stores for MongoDB, Workday applications, and REST Services. You can perform domain discovery and Subject Registry scans on MongoDB, Workday applications, and REST services.

The connection properties depend on the database. The following table describes the database management connection properties:

Property	Description
Authentication Mode	Required for Microsoft SQL Server data stores only. Select SQL Server Authentication or Windows Authentication. Default is SQL Server Authentication.
Driver Class	Required for JDBC data stores only. Name of the JDBC driver class.
URL	Required for JDBC data stores only. Enter the JDBC connection URL used to access the Hadoop server in one of the following formats: - Non-Kerberos source: `jdbc:hive2://<host>:<port>/<DB>` - Kerberos source: `jdbc:hive2://<host>:<port>/<DB>;principal=<serviceName>/<host>@realm` For Workday applications, the URL must contain the following: - User name and password - Tenant - Host - Service Optionally, you can add views in the URL.
User	Optional for SAP HANA data stores only. Required for other database management data store types. Name of the user account to log in to the database.
Password	Optional. Password for the database user name. When you save the data store, Data Privacy Management encrypts the password.
Sub System ID	Required for IBM Db2 for z/OS data stores only. Enter the subsystem ID (SSID) for the Db2 system.

Property	Description
Agent URL	Optional for IBM Db2 for i5/OS and JDBC data stores only. Enter the URL of the Enterprise Data Catalog agent in the following syntax: `http://<host name or IP address>:<port>/<directory>`
Host	Required for IBM Db2, IBM Db2 for i5/OS, Microsoft SQL Server, Netezza, Oracle, SAP HANA, Sybase, and Teradata data stores. Optional for IBM Db2 for z/OS data stores. Host name or IP address for the database server.
Port	Required for IBM Db2, IBM Db2 for i5/OS, Microsoft SQL Server, Netezza, Oracle, SAP HANA, and Sybase data stores. Optional for IBM Db2 for z/OS data stores. Port number for the database server.
DDF Location	Appears for IBM Db2 for z/OS. Optional. Location of the DB2 subsystem on IBM Db2 for z/OS. Use the DB2 'DISPLAY DDF' command to identify the value.
Database Option	Appears for IBM Db2 for z/OS. Optional. Select one of the following options to determine how you add IBM DB2 for z/OS databases to the data store: - **All**. Includes all databases that you can access. - **Select From List**. Displays a list of databases in the **Database** property. - **Specify Regex**. Enables the **Regular Expression for Database** property. Select this option if you want to enter databases as a regular expression.
Database	Required for IBM Db2, IBM Db2 for i5/OS, Microsoft SQL Server, Netezza, Oracle, SAP HANA, and Sybase data stores. Optional for IBM Db2 for z/OS and Teradata data stores. The name of the database instance.
Instance	Optional for Microsoft SQL Server data stores only.
Service	Oracle data stores only. Required. The name of the database service.
Jdbc Username	Optional for IBM Db2 for z/OS data stores only. The user name for the PowerExchange user account.
Jdbc Password	Optional for IBM Db2 for z/OS data stores only. The password for the PowerExchange user account.
Pwx Username	Optional for IBM Db2 for i5/OS data stores only. The user name for the PowerExchange user account.
Pwx Password	Optional for IBM Db2 for i5/OS data stores only. The password for the PowerExchange user account.
Connection String	Required for IBM Db2 for z/OS data stores. Optional for other database management data store types. The connection string cannot exceed 32 characters. Enter the connection string using the following syntax: - IBM: `<database_name>` - JDBC: `<JDBC_data_source_name>` **Note:** For MongoDB, Workday applications, and REST Services, do not enter a connection string. - Microsoft SQL Server: `<host>,<port>@<database_name>` or for named instances: `<host_name\instance_name>@<database_name>` - Oracle: The connection string is the TNSNAMES entry: `<database_name>` - Netezza: `<ODBC_data_source_name>` - Sybase: `<server_name>@<database_name>` - Teradata: `<ODBC_data_source_name>`

Property	Description
Secure JDBC Parameters	Optional. Database parameters to access databases that are secured with the SSL protocol or Azure Key Vault. The parameter string cannot exceed 500 characters. When you save the data store, Data Privacy Management encrypts the parameter string. Enter the parameters as `name=value` pairs separated by semicolon characters (;). You can enter the following secure database parameters: SSL Protocol: - EncryptionMethod. Required. Indicates whether data is encrypted when transmitted over the network. This parameter must be set to `SSL`. - HostNameInCertificate. Optional. Host name of the machine that hosts the secure database. If you specify a host name, Data Privacy Management validates the host name included in the connection string against the host name in the SSL certificate. - TrustStore. Required. Path and file name of the truststore file that contains the SSL certificate for the database. - TrustStorePassword. Required. Password for the truststore file. - KeyStore. Directory of the keystore file. - KeyStorePassword. Password used to access the keystore file. - KeyPassword. Password used to access individual keys in the keystore file. - ValidateServerCertificate. Optional. Set to True or False. If True, Data Privacy Management validates the certificate that the database server sends. If you specify the HostNameInCertificate parameter, Data Privacy Management also validates the host name in the certificate. If False, Data Privacy Management does not validate the certificate that the database server sends. Data Privacy Management ignores any truststore information that you specify. Data Privacy Management appends the secure JDBC parameters to the connection string. Use the following JDBC parameters if the database uses Azure Key Vault to encrypt the data: - *columnEncryption*. Indicates whether the driver is enabled for **Always Encrypted** functionality when accessing data from encrypted columns. - *AEKEYSTOREPRINCIPALID*. The principal ID used to authenticate against the Azure Key Vault. Required when you enable **Always Encrypted**. - *AEKEYSTORECLIENTSECRET*. The Client Secret used to authenticate against the Azure Key Vault. Required when you enable **Always Encrypted**. Data Privacy Management appends the secure JDBC parameters to the connection string.

Use DSN	Optional for Microsoft SQL Server data stores only. Determines whether Data Privacy Management uses the Data Source Name for the connection. If enabled, sets the **Database** and **Instance** properties to the database name and server name from the DSN. If you do not select this option, you must manually enter the database name and server name.
	Required if the database is encrypted using Azure Key Vault.
	Before you scan the data store encrypted using Azure Key Vault, add the following properties to the database entry in the `$ODBC_HOME/odbc.ini` file:
	`ColumnEncryption=<Enabled>`
	`AEKeystorePrincipalId=<principal ID>`
	`AEKeystoreClientSecret=<keystore client secret>`
	`AEKeyCacheTTL=<length of time in seconds>`
	The *AEKeyCacheTTL* property determines how long a column encryption key stays in the cache before the driver deletes it.

Property	Description
Schema Option	Optional for IBM Db2, IBM Db2 for z/OS, Microsoft SQL Server, Oracle, SAP HANA, Sybase, and Teradata data stores only. Select one of the following options to determine how you add schemas to the data store: - **All**. Includes all schemas that you can access. - **Select From List**. Displays a list of schemas in the **Schema** property. - **Specify Regex**. Enables the **Regular Expression for Schema** property. Select this option if you want to enter schemas as a regular expression.
Schema	Optional for IBM Db2 for i5/OS, JDBC, and Netezza data stores. Enter one or more schemas that Data Privacy Management reads when you scan the data store. The schema is case-sensitive. Use a comma to separate multiple entries. If empty, Data Privacy Management reads all schemas from the database. Required for IBM Db2, IBM Db2 for z/OS, Microsoft SQL Server, Oracle, SAP HANA, Sybase, and Teradata data stores if you select **Select From List** in the **Schema Option** property.
Regular Expression for Schema	Required for IBM Db2, IBM Db2 for z/OS, Microsoft SQL Server, Oracle, SAP HANA, Sybase, and Teradata data stores if you select **Specify Regex** in the **Schema Option** property. Enter a regular expression to specify schemas that you want to include or exclude from the scan. Schema names are case-sensitive.
Fetch Views Data Types	Optional for Teradata data stores only. Select this option to enable the Scan job to retrieve the data types in source columns from Teradata views.
Source Metadata Filter	Optional. Default is All. A filter to specify tables that you want to include or exclude in the scan. Enter a regular expression to include multiple filter parameters. For example: `Table1, NOT Table2, SUB%, %EMP, %DATA%`. The filter includes Table1, tables with names that start with SUB, names that end with EMP, and tables that include DATA anywhere in the name. The filter excludes Table2.
Stored Procedures, Functions	Optional for Microsoft SQL Server data stores only. The names of stored procedures or functions to include in the scan.
Case Sensitive	Optional for IBM Db2, IBM Db2 for z/OS, Microsoft SQL Server, Netezza, Oracle, SAP HANA, Sybase, and Teradata data stores. Select to indicate that the data store is configured for case sensitivity. Default is cleared.
Case Sensitivity	Required for IBM Db2 for i5/OS data stores only. Select one of the following options: Auto, Case Insensitive, or Case Sensitive. Default is Auto.
Import Private and Public Synonyms	Optional for Oracle data stores only. Select this option to import Oracle private and public synonyms for the schemas included in the data store scan.
Import Synonyms	Optional for Microsoft SQL Server data stores only. Select this option to import table synonyms from the source data store.
Import Database Names	Optional for Oracle data stores only. Select to import database names from Enterprise Data Catalog when the scan job runs.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select High, Low, or Medium. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.

Property	Description
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Auto Assign Connections	Optional for Microsoft SQL Server data stores only. Select to automatically assign the schemas in Enterprise Data Catalog to the data store. Default is cleared.
Enable Reference Resources	Optional for Microsoft SQL Server data stores only. Select to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is cleared.
Retain Unresolved Reference Assets	Optional for Microsoft SQL Server data stores only. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is enabled.
Agent Options	Optional for IBM Db2 for i5/OS, JDBC, Microsoft SQL Server, Sybase, and Teradata data stores. Optional. Enter the Agent configuration options to include in the scan.
CyberArk Safe	Optional. Name of the CyberArk safe that contains the database password. When you test the data store connection or you scan the data store, Data Privacy Management retrieves the password from CyberArk.
Include Views in the Scan	Optional. Determines whether a Scan job evaluates data in views. If enabled, a Scan job identifies sensitive data in views during metadata and data profiling. The Scan job evaluates data in all views for the schemas specified in the **Schema** property. If the property is blank, the Scan job evaluates data in all views of all schemas. If the data store was imported from a parent data store, such as Informatica PowerCenter, the Scan job for the parent data store can identify the lineage of sensitive data in views. For example, you ran an Informatica PowerCenter Scan job that imported a data store that included views. You ran a Database Scan job on the data store that identified sensitive data in the views. When you run an Informatica PowerCenter Scan job to identify data proliferation, the Scan job identifies lineage for the sensitive data in the views. If disabled, the Scan job does not evaluate data in views. If you disable the property after a Scan job completes, any scans that you run in the future will not evaluate data in views. However, the completed Scan job results remain.
Profile Execution Engine	Optional. Select one of the following profiling engine modes: - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. Default is Native.

Property	Description
Hadoop Connection Name	If you select **Hadoop** in the **Profile Execution Engine** property, enter the name of the Hadoop connection listed in the Administrator tool, **Connections** tab.
Source Connection Name	Optional. Connection name in the Informatica domain that the scan job uses to connect to the source database. If the Scan job cannot connect to the source, the job fails.

Office 365 Outlook

You can connect to and scan an Office 365 Outlook server. Configure the properties and then associate the data store with a remote agent to perform a scan. Office 365 Outlook data stores do not connect to Enterprise Data Catalog. When you scan the data store, the remote agent scanner maps sensitive files to data domains.

The following table describes the Office 365 Outlook connection properties:

Property	Description
Client ID	Client ID of the Azure Active Directory application.
Client Secret	Client secret of the Azure Active Directory application.
Tenant ID	The unique ID of the tenant or domain.
Include Departments	A filter to include departments. Enter the name of Office 365 departments that you want to include in scans. Use commas or press **Enter** to separate multiple values.
Include Users	A filter to include users. Enter the email IDs of users or accounts that you want to scan. Use commas or press **Enter** to separate multiple values.
Exclude Users	A filter to exclude users. Enter the email IDs of users or accounts to exclude from the scan. Use commas or press **Enter** to separate multiple values. **Note:** The Exclude Users values takes precedence over Include Users.
Include Email Groups	A filter to include users based on group membership. Enter the group names. Use commas or press **Enter** to separate multiple values.
Include Mail Folders	A filter to include specific mail folders during scan. You can enter top-level folder names.
Exclude Mail Folders	A filter to exclude specific mail folders during scan. You can enter top-level folder names.

Property	Description
Include Only Emails with Attachments	Choose this option to include only emails with attachments. Scans ignore emails without attachments. If you select the option, scans include the content body of all emails that have attachments and the attachments. If you select the option and specify attachment types, the scan includes the content body of all emails that have attachments, and the attachment types that you specify. If you do not select the option, scans include the content body of all emails with and without attachments and the attachments. If you do not select the option and you specify attachment types, the scan includes the content body of all emails with and without attachments, and the attachment types that you specify.
Attachment Types	You can choose to include specific types of attachments in a scan. Choose from the following attachment types: - Avro - Compressed Files - Delimited and Text - Email - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, PDF content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.

File System Connection Properties

Configure connection properties to connect to a file system data source. You can scan data from structured file types such as JSON, XML, text, and delimited files. You can scan data from unstructured files such as Microsoft Excel, Microsoft PowerPoint, Microsoft Word, and PDF.

When you test a connection to a file system data source, the Data Privacy Management Service verifies if the path you enter in the **Path** property exists on the Domain node.

The following table describes the file system connection properties:

Property	Description
Scan with Remote Agent	Select to associate the data store with a remote agent when you run Scan jobs to discover personal and sensitive data or build a Subject Registry index. **Important:** If you select this option, you must associate the data store with a remote agent before running a scan. After you save the data store, you cannot change this property.
File Protocol	Required. Select one of the following file transfer protocols: **Local File**, **SFTP**, and **SMB/CIFS**. Default is Local File.
User	Required if you select **SFTP** or **SMB/CIFS** in the **File Protocol** property. Name of the user account to access the data store.
Password	Required if you select **SFTP** or **SMB/CIFS** in the **File Protocol** property. Password for the user account that connects to the data store.
Host	Required if you select **SFTP** or **SMB/CIFS** in the **File Protocol** property. Host name for the data store server.
Port	Required if you select **SFTP** in the **File Protocol** property. Port number for the data store server.
Physical Location	Optional if you select **SMB/CIFS** in the **File Protocol** property.
Path	Required. The CIFS or NFS mount directory you want to scan. For example: `/home/infa/Folder1` If you selected **Scan with Remote Agent**, make sure that the agent machine and Data Privacy Management can access the directory. **Note:** You must mount the directory on a path present on the Data Privacy Management domain and on the cluster nodes. Use the CIFS mount option to mount a Windows directory. Use the NFS mount option to mount a Linux or network directory.
Folder Options	Required if you enable the **Scan with Remote Agent** property. Folders that the Scan job reads from the location configured in the **Path** property. Select one of the following options: - All - Select Specific Folders - Use Regular Expression Default is All.
Include Folders	Appears if you if you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to include in the scan. You can also skip this property and list folders in the **Exclude Folders** property. In this case, Data Privacy Management scans the folders in the data store except the excluded folders.
Exclude Folders	Optional if you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to exclude from the scan.
Folder Regular Expression	Required if you enable the **Scan with Remote Agent** property and choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder that you want to include in the scan.
File Types	Required. File types that the scan reads from the location configured in the **Path** property. Select **All** to scan all file types. Choose **Select** to select specific file types to scan. Default is All.

Property	Description
Selected File Types	Required if you choose **Select** in the **File Types** property. Click **Select All** to choose all file types, or select one or more of the following options: - Avro - Compressed Files - Delimited and Text - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, the PDF file content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.
Enter File Delimiter	Optional. Available if you do not enable the **Scan with Remote Agent** property. Character that separates entries in the file. Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`. Default is a comma (,).
First Level Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Name of the folder that you want to scan. For example: `subfolderA`. To specify multiple folders, separate the folder names with a comma (,). For example: `subfolderA, subfolderB`
Include Sub Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. If enabled, the scan reads data from the directory and all subdirectories of the location configured in the **Path** property or the **First Level Directory** property, if specified. Default is disabled.
Memory	Optional. Memory to use when you scan the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.

Property	Description
Profile Execution Engine	Optional. Select one of the following profiling engine modes: - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. Default is Native.
Hadoop Connection Name	If you select **Hadoop** in the **Profile Execution Engine** property, enter the name of the Hadoop connection listed in the Administrator tool, **Connections** tab.
Source Connection Name	The connection name in the Informatica domain. Do not enter a value for File Management scans.

Google BigQuery Connection Properties

Configure connection properties to connect to a Google BigQuery data source.

The following table describes the Google BigQuery connection properties:

Property	Description
Project ID	Name of the Google Cloud Platform project that you want to access.
Private Key	The private key associated with the service account.
Client Email	The client email address associated with the service account.
Dataset Option	Datasets that the scan job reads from the location configured in the Project ID property. Select one of the following options: - **All** - **Select from list** - **Specify Regex** Default is All.
Dataset	Available if you select the **List** option in the **Dataset Option** property. Select the datasets that you want to use to include in the scan from the list.
Dataset Regular Expression	Available if you select the **Specify Regex** option in the **Dataset Option** property. Enter a regular expression to specify datasets that you want to include or exclude from the scan. Dataset names are case-sensitive.
Scan Hidden Datasets	Select the option to include hidden datasets in scans.
Source Metadata Filter	You can include or exclude tables and views from the resource run. Enter the filter query as a Java regular expression. For example, to include all tables that have CONF or DEMO in the name, enter the following expression: `.*CONF.*|.*DEMO.*`

Property	Description
Memory	Specifies the memory required to run the Enterprise Data Catalog scanner job. Select one of the following values based on the data set size that you plan to create: - Low - Medium - High
Custom JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Source Connection Name	The connection name in the Informatica domain.

Google Drive Connection Properties

Configure connection properties to connect to a Google Drive data source.

To scan a Google Drive data store, you must associate the data store with a remote agent on the **Remote Agents** workspace after you save the data store. Google Drive data stores do not connect to Enterprise Data Catalog. When you scan the data store, the remote agent scanner maps sensitive files to data domains and adds subject data in the subject registry.

The following table describes the Google Drive connection properties:

Property	Description
Scan with Remote Agent	Read-only. Selected. Indicates that you will associate the data store with a remote agent to run the Scan job .
Client ID	Required. The Client ID from Google Developer Console.
Folder Options	Required. Folders that the File Management job reads from the location configured in the **Path** property. Select one of the following options: - All - Select Specific Folders - Use Regular Expression Default is All.
Include Folders	Required if you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to include in the scan.
Exclude Folders	Optional if you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to exclude from the scan.

Property	Description
Folder Regular Expression	Required if you choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder that you want to include in the scan.
File Types	Required. File types that the File Management job reads from the location configured in the **Path** property. Select **All** to scan all file types. Choose **Select** to select specific file types to scan. Default is All.
Selected File Types	Required if you choose **Select** in the **File Types** property. Click **Select All** to choose all file types, or select one or more of the following options: - Avro - Compressed Files - Delimited and Text - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, the PDF file content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.
Client Secret	Required. The Client Secret from Google Developer Console.
Access Token	Required. The Access Token received after you exchange the authorization code from the Google Authorization Server.
Expires In	Required. The remaining lifetime of the access token in seconds.
Refresh Token	Required. The Refresh Token received after you exchange the authorization code from the Google Authorization Server.
Path	Required. The path of the directory you want to scan.

Hadoop Distributed File System Connection Properties

Configure properties to connect to a Hadoop Distributed File System (HDFS) on a cluster.

You can create multiple data stores that connect to HDFS on the same cluster, but the **Source Directory** property must be unique for each data store.

When you test the connection to HDFS, Data Privacy Management tests the combination of the **Name Node URI 1**, **User Name**, and **Source Directory** properties. If any property values are not valid, the connection test fails.

The following table describes the Hadoop Distributed File System connection properties:

Property	Description
Scan with Remote Agent	Select to associate the data store with a remote agent when you run scan jobs to discover personal and sensitive data or build a Subject Registry index. Default is selected. **Important:** If you select this option, you must associate the data store with a remote agent before running a scan. After you save the data store, you cannot change this property.
Cluster Configuration Details	Required. Select one of the following options to provide configuration details about the cluster: - **Load From Configuration Archive File**. You must select this option if you want to specify the Microsoft Azure Data Lake Store storage type. You can select this option for other storage types. - **Provide Configuration Details**. Select this option if you want to specify a storage type of Distributed File System or Windows Azure Blob Storage. Default is Provide Configuration Details.
Storage Type	Required if you select **Provide Configuration Details** in the **Cluster Configuration Details** property. Select one of the following options: - **ABFS**. Azure Blob File System. - **DFS**. Distributed File System. - **WASB**. Windows Azure Blob Storage. Default is DFS.
Distribution Type	Required if you select **Load From Configuration Archive File** in the **Cluster Configuration Details** property. Select one of the following options: - Amazon Emr - Azure HDInsight - Cloudera - Hortonworks - IBM BigInsights - MapR FS Default is Azure HDInsight.
Azure Storage Account URI	Required if you select **ABFS** or **WASB** in the **Storage Type** property. The fully qualified URI to access data stored in ABFS or WASB. You can get the URI from the from the `fs.defaultFS` property in the `core-site.xml` file. Example: `wasb://secureatsource-2017-08-08t06-13-20-522z@secureatsource.blob.core.windows.net`
Azure Storage Account Name	Required if you select **ABFS** or **WASB** in the **Storage Type** property. Name of the Microsoft Azure Storage account to stage the files. You can get the name from the FS URI. The name is the first term after the forward slash and before the time stamp. For example, in the following URI, the name of the storage account is `dataprivacy`: `wasb://dataprivacy-2019-08-08t06-13-20-522z@dataprivacy.blob.core.windows.net`

Property	Description
Azure Storage Account Key	Required if you select **ABFS** or **WASB** in the **Storage Type** property. Key to access the Microsoft Azure Storage account. Use the decrypted version of the key. To get the decrypted version of the key, perform the following steps: 1. Get the key from the `fs.azure.account.key.secureatsource.blob.core.windows.net Key` property in the `core-site.xml` file. The value is the encrypted version of the key. 2. Get the file path of the decrypt script from the `fs.azure.shellkeyprovider.script` property in the `core-site.xml` file. Example of the decrypt script file path: `/usr/lib/hdinsight-common/scripts/decrypt.sh` 3. Navigate to the decrypt script file path. 4. Run the `decrypt.sh` file with the encrypted key as an argument. The value returned is the decrypted version of the key. 5. Enter the decrypted version of the key.
Name Node URI 1	Required if you select **Provide Configuration Details** in the **Cluster Configuration Details** property. URI to the active HDFS NameNode. The active HDFS NameNode manages all the client operations in the cluster. Use the following format to specify the NameNode URI: `hdfs://<NameNode>:<port>` - *<NameNode>* is the host name or IP address of the NameNode. - *<port>* is the port on which the NameNode listens for remote procedure calls (RPC). Use the same value that is configured for the property `fs.defaultFS` in the following file on the Hadoop cluster: `core-site.xml` If another data store has the same value, Data Privacy Management identifies the data store as a potential duplicate. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
HA Cluster	Optional. Appears if you select **Provide Configuration Details** in the **Cluster Configuration Details** property. Indicates whether the HDFS cluster is highly available (HA). Select **Yes** or **No**. Default is Yes.
Name Node URI 2	Required if you select **Yes** in the **HA Cluster** property. URI to access the secondary NameNode for the HDFS. Use the following syntax: `hdfs://<SecondaryNameNode>:<port>` - <SecondaryNameNode> is the host name or IP address of the Secondary NameNode. - <port> is the port on which the NameNode listens for remote procedure calls (RPC). You can find the host name, IP address, and port in property `dfs.namenode.rpc-address.<ClusterName>.<Name_NodeID>` in file `hdfs-site.xml` on the Hadoop cluster. Property `dfs.ha.namenodes.<ClusterName>` includes a list of the NameNode IDs. In Ambari, you can get the host name or IP address from the Standby NameNode property for HDFS.
HDFS Service Name	Required if you select **Yes** in the **HA Cluster** property. Logical name of the NameNode cluster. Use the same value that is configured for property `dfs.nameservices` in file `hdfs-site.xml` on the Hadoop cluster.
User Name/ User Principal	Required. User that connects to the Hadoop cluster. The user must have read permission to the directory configured in the Source Directory property and to all files in the directory. For highly available clusters, specify the User Name. For Kerberos-enabled clusters, specify the User Principal.
Source Directory	Required. Directory path that contains the files to scan. To scan all root directories, enter a forward slash `/`. To scan a specific directory, use the following syntax: `/<Directory_Name>` For example, if the directory name is `Folder1`, enter `/Folder1`. Do not include a forward slash `/` at the end of the directory. The scan job reads files from the last directory specified. For example, if you enter `/HumanResource/Employees/US`, the scan job only reads files from the `US` directory. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.

Property	Description
HDFS Transparent Encryption	Required. Select **Yes** to indicate that transparent encryption is enabled for HDFS. Select **No** to indicate that transparent encryption is not enabled for HDFS.
Key Management Server Provider URI	Required if you select **Yes** in the **HDFS Transparent Encryption** property. The fully qualified URI to the Key Management Server key provider. Used to interact with encryption keys while reading and writing to an encryption zone. You can get the URI from the `dfs.encryption.key.provider.uri` property in the `hdfs-site.xml` file.
Kerberos Cluster	Optional. Select **Yes** to indicate that the cluster uses Kerberos authentication. Select **No** to indicate that the cluster does not use Kerberos authentication. Default is No.
Hadoop RPC Protection	Appears if you select **Yes** in the **Kerberos Cluster** property. Security infrastructure for Hadoop Remote Procedure Call (RPC). Select one of the following options: - **Authentication**. Authentication only. The client and server mutually authenticate during connection setup. - **Integrity**. Authentication and integrity. Guarantees the integrity of data exchanged between the client and server and the integrity of authentication. - **Privacy**. Authentication, integrity, and confidentiality. Guarantees that data exchanged between the client and server is encrypted. Default is Authentication.
HDFS Service Principal	Required if you select **Yes** in the **Kerberos Cluster** property. Service Principal Name (SPN) of the HDFS service. If blank, the Scan job fails.
KeyTab File	Required if you select **Yes** in the **Kerberos Cluster** property. Path and file name of the Service Principal Name (SPN) keytab file for the user account to impersonate when connecting to the Kerberos-enabled cluster. The keytab file must be generated from the Kerberos-enabled cluster and copied to a directory on the machine where the Data Privacy Management Service runs. For example: */<Data Privacy Management installation directory>*/hdfskeytab/admin.keytab When you save the data store, the Data Privacy Management Service encrypts the value. If blank, the Scan job fails.
Cluster Configuration ID	Optional. Identifier used to identify the nodes in the cluster. To get the cluster configuration ID, see the value for the `ID` property of the cluster configured in the Administrator tool. Enter the ID in lower case.
Folder Options	Required if you enable the **Scan with Remote Agent** property. Folders that the scan job reads from the location configured in the **Source Directory** property. Select one of the following options: - **All** - **Select Specific Folders** - **Use Regular Expression** Default is All.
Include Folders	Required if you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to include in the scan.
Exclude Folders	Optional if you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to exclude from the scan.

Property	Description
Folder Regular Expression	Required if you enable the **Scan with Remote Agent** property and choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder that you want to include in the scan.
File Types	Required. File types that the scan job reads from the location configured in the **Source Directory** property. Select **All** to scan all file types. Choose **Select** to select specific file types to scan. Default is All.
Selected File Types	Required if you choose **Select** in the **File Types** property. Click **Select All** to choose all file types, or select one or more of the following options: - Avro - Compressed Files - Delimited and Text - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, the PDF file content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.
Enter File Delimiter	Optional. Available if you do not enable the **Scan with Remote Agent** property. Character that separates entries in the file. Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`. Default is a comma (,).
First Level Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Name of the folder that you want to scan. For example: `subfolderA`. To specify multiple folders, separate the folder names with a comma (,). For example: `subfolderA, subfolderB`
Include Sub Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. If enabled, the scan reads data from the directory and all subdirectories of the location configured in the **Source Directory** property or the **First Level Directory** property, if specified. Default is disabled.
Memory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.

Property	Description
JVM Options	Optional. Available if you do not enable the **Scan with Remote Agent** property. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Source Connection Name	Optional. Available if you do not enable the **Scan with Remote Agent** property. The connection name in the Informatica domain that the Scan job uses to connect to HDFS. To determine the name, go to the Administrator tool and click the cluster configuration object that created the HDFS connections automatically. The name of the HDFS connection is the source connection name. For example: `HDFS_cco` If you create multiple data stores that connect to the same cluster, you can use the same connection name for all data stores. You must enter a value for **Source Connection Name** if you selected the **Load from Configuration Archive File** option in the **Cluster Configuration Details** property.

Hive Connection Properties

After you create a Hive data store, configure the connection details to access Hive data on a Cloudera or Hortonworks Hadoop distribution.

When you test a data store connection, the Data Privacy Management Service tests the combination of the **Hadoop Distribution** and **JDBC URL** properties. When Kerberos is used for Hive authentication, the Data Privacy Management Service also tests the **Keytab File**, **Kerberos Configuration File Path**, and **User Proxy** properties.

When you save a data store, the Data Privacy Management Service uses the host and port specified in the **URL** and **User** properties to identify duplicate data stores.

For Cloudera Hive data stores you want to protect with encryption extensions, rules, and keys, Data Privacy Management does not perform Scan jobs. After you configure the connection to a Cloudera Hive data store for encryption, instead of running a scan, you import Catalog metadata and protection status for the data store on the **Data Stores** workspace.

The following table describes the Hive connection properties:

Property	Description
Hadoop Distribution	Required. Select one of the following options: - Amazon EMR - Azure HDInsight - Cloudera - Hortonworks - MapR Default is Cloudera. If blank, the data store status is Not Complete. You cannot scan incomplete data stores.
MapR Home	Required if you select **MapR** in the **Hadoop Distribution** property. The file path for the MapR home directory for the specified user.
URL	Required. JDBC connection URL used to access the Hadoop server. Use one of the following formats: - Non-Kerberos source: `jdbc:hive2://<host>:<port>/<DB>` - Kerberos source: `jdbc:hive2://<host>:<port>/<DB>;principal=<serviceName>/<host>@realm` If blank, the data store status is Not Complete. You cannot scan incomplete data stores.
User	Optional. The Hive user name for the data store connection. The Big Data Scan job requires a user account that has read, write, and execute privileges. If the user does not have the required privileges, the Scan job fails.
Password	Optional. Password for the Hive user account.
Kerberos Configuration File	Required for Hive sources that use Kerberos authentication. Common path of the krb5.conf file located in both the domain and the Hadoop cluster.
Keytab File	Required for Hive sources that use Kerberos authentication. Common path of the keytab file located in both the domain and the Hadoop cluster.
User Proxy	Required for Hive sources that use Kerberos authentication. Specifies a user proxy setting.
Include Views in the Scan	Optional. Determines whether a Scan job evaluates data in views. If enabled, a Scan job identifies sensitive data in views during metadata and data profiling. The Scan job evaluates data in all views for the schemas specified in the **Schema** property. If the property is blank, the Scan job evaluates data in all views of all schemas. The Cloudera Navigator Scan job can identify the lineage of sensitive data in views. For example, you ran a Hadoop Scan job on the data store that identified sensitive data in views. When you run a Cloudera Navigator Scan job to identify data proliferation, the Scan job identifies the lineage for the sensitive data in views. If disabled, the Scan job does not evaluate data in views. If you disable the property after a Scan job completes, any scans that you run in the future will not evaluate data in views. However, the completed Scan job results remain.
Server Name for Sentry Authorization	Optional. The symbolic name for the service that enables Sentry authorization for the Hive service. This property appears if you select **Cloudera** in the **Hadoop Distribution** property. To get the server name for Sentry authorization, perform the following steps: 1. In Cloudera Manager, go to the Hive cluster. 2. Click **Configuration** in the header menu. 3. In the **Filters** pane, expand **Scope** and select **Hive (Service-Wide)**. 4. Expand **Category** and select **Advanced**. The name appears in the **Server Name for Sentry Authorization** field.

Property	Description
Ranger Service Name	Name of the service configured in Ranger Service Manager for the Hive service properties. This property appears if you select **Hortonworks** in the **Hadoop Distribution** property.
Cluster Configuration ID	Optional. Identifier used to identify the nodes in the cluster. To get the cluster configuration ID, see the value for the `ID` property of the cluster configured in the Administrator tool. Enter the ID in lower case.
Schema Option	Optional. Select one of the following options to determine how you add schemas to the data store: - **All**. Includes all schemas that you can access. - **Select From List**. Displays a list of schemas in the **Schema** property. - **Specify Regex**. Enables the **Regular Expression for Schema** property. Select this option to enter schemas as a regular expression.
Schema	Displays a list of available schemas to include in the scan if you select **Select From List** in the **Schema Option** property. A Hadoop Scan job can run metadata or data profiling on multiple schemas in the same data store. However, a Cloudera Navigator Scan job determines lineage of sensitive data between Hive data stores. If one Hive data store contains multiple schemas and sensitive data moves between the schemas, the Cloudera Navigator Scan job cannot show sensitive data lineage between schemas in the same data store. To show complete lineage between each schema, create one Hive data store for each schema.
Regular Expression for Schema	Displays if you select **Specify Regex** in the **Schema Option** property. Enter a regular expression to specify schemas that you want to include or exclude from the scan. Schema names are case-sensitive.
Table	Specifies a Hive table to scan. Enter one table or leave blank to scan all tables. If you enter more than one table, the Scan job fails.
SerDe JARs List	Specifies a list of fully-qualified path names, separated by semicolons, to the SerDe JAR files that the bridge uses to run remotely on the Hive system. If the Hive database contains customized tables, you must provide a custom SerDe JAR file list that will be used to serialize and deserialize the customized tables.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Auto Assign Connections	Optional. Select to automatically assign the schemas in Enterprise Data Catalog to the data store. Default is cleared.

Property	Description
Enable Reference Resources	Optional. Select to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is cleared.
Retain Unresolved Reference Assets	Optional. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is enabled.
Profile Execution Engine	Optional. Select one of the following profiling engine modes: - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. - **Hive (Deprecated)**. Data Privacy Management uses the Hive engine to run profiles. Informatica does not recommend this option because it slows performance. - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. Default is Native.
Hadoop Connection Name	If you select **Hadoop** in the **Profile Execution Engine** property, enter the name of the Hadoop connection listed in the Administrator tool, **Connections** tab. The Hadoop connection is created in the Administrator tool when you create a cluster configuration. Enter the name of the Hadoop connection that corresponds to the cluster configuration name. You must enter the Hadoop connection name for Cloudera Hive data stores that you want to protect with encryption extensions and rules.
Hive Connection Name	If you select **Hive** in the **Profile Execution Engine** property, enter the name of the Hive connection. You can find the Hive connection name in the Administrator tool, **Connections** tab.
Source Connection Name	The Hive connection name in the Informatica domain that Data Privacy Management uses to connect to Hive. Run the Hadoop Configuration Manager to create the Hive connection in the Informatica domain. Specify the connection name that was configured in the utility. If you create multiple data stores that connect to the same cluster, you can use the same connection for all data stores. If blank, Data Privacy Management creates a Hive connection in the Informatica domain and uses the user name specified in the data store connection properties.

Informatica Cloud Connection Properties

Configure the connection details to the Informatica Cloud instance that you want to scan as a data source.

When you test a connection to Informatica Cloud, Data Privacy Management tests the combination of the **URL**, **User Name**, and **Password** properties.

The following table describes the Informatica Cloud connection properties:

Property	Description
Cloud URL	Required. Base URL of the Informatica Cloud server. For example, `https://sts1.company.com/`
User Name	Required. User name for the Informatica Cloud server.

Property	Description
Password	Required. Password for the Informatica Cloud server.
Detailed Lineage	Optional. Select this option to enable scanning the complete data store lineage in Enterprise Data Catalog. Default is disabled.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Enable Reference Resources	Optional. Select to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is cleared.
Retain Unresolved Reference Assets	Optional. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is enabled.

Informatica Data Engineering Integration Properties

Configure connection details to the Data Engineering Integration instance that you want to scan as a data source. Configure the connection details to the Informatica domain and node on which the Model Repository

Service for Data Engineering Integration runs. Do not confuse with the Model Repository Service that exists on the same domain as the Data Privacy Management Service.

The following table describes the Informatica Data Engineering Integration connection properties:

Property	Description
Execute On	Required. Select a server where the Data Engineering Integration data store runs. Default is Default Catalog Server.
Target Version	Required. Select one of the following Data Engineering Integration product versions: - 10.0 - 10.1 - 10.1.1 - 10.1.1.HF1 - 10.2 - 10.2.1 - 10.2.2 - 10.2.2.HF1 - 10.2.HF1 - 10.2 HF2 - 10.4.0 - 10.4.1 - 10.5.0 - 9.6.1.HF3 - 9.6.1.HF4 Default is 9.6.1 HF3. **Note:** For Snowflake databases, select 10.4.0 or later. On versions earlier than 10.4.0, lineage information is not extracted.
MRS Domain Name	Required. Name of the Informatica domain on which the Model Repository Service runs.
Data Integration Service Name	Required. Name of the Data Integration Service.
MRS User Name	Required. User name to access the Model Repository Service database.
MRS Password	Required. Password to access the Model Repository Service database.
MRS Security Domain	Optional. Name of the security domain to which the Model Repository Service user belongs. Default is Native.
MRS Domain Host	Required. Host name or IP address of the machine that hosts the Model Repository Service database.
MRS Domain Port	Required. Port number of the machine that hosts the Model Repository Service database.
Os Profile Name	Optional. Name of the operating system profile if the Data Integration Service is enabled to use operating system profiles.
Application Name	Optional. Name of the application that contains parameter sets.
Param Set For Mappings in Application	Optional. Name of the parameter set in the Model Repository that is mapped to the application specified in the property.

Property	Description
Select Parameter Set or Parameter File	Required. Select **None**, **Parameter Set**, or **Parameter File**. Default is None.
Parameter Sets	Optional. Name of parameter sets that run mappings and workflows when the associated application is deployed.
Parameter Sets File	Optional. Name of the file that lists parameter sets.
Parameter File	Optional. Name of the XML file that lists user-defined parameters and their assigned values.
Node Name	Optional. Name of the node on which the Model Repository Service runs.
MRS Node Host Name	Optional. Host name or IP address of the node on which the Model Repository Service runs.
MRS Node Port	Optional. Port number of the node on which the Model Repository Service runs.
MRS Service Name	Optional. Name of the Model Repository Service.
MRS is SSL Enabled	Optional. If selected, indicates that the Model Repository Service uses the Secure Sockets Layer security protocol.
Detailed Lineage (Applicable for Target version 10.2.1 onwards)	Optional. If you selected 10.2.1 or a later version in the **Target Version** property, you can select this option to scan the complete data store lineage in Enterprise Data Catalog. Default is disabled.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Medium. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Enable Reference Resources	Optional. For Data Engineering Integration version 10.4.0 or later, select this option to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is cleared.

Property	Description
Retain Unresolved Reference Assets	Optional. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is selected.
Preferred DIS	Optional. Select a preferred Data Integration Service from the list.
Source Connection Name	Optional. The connection name in the Informatica domain that the Scan job uses to connect to HDFS. Use the same name that was configured as the connection name in the Administrator tool. If you create multiple data stores that connect to the same cluster, you can use the same connection name for all data stores. If blank, the Scan job creates a connection to HDFS on a cluster that is not highly available or Kerberos-enabled. The connection might not work for a highly available or Kerberos-enabled cluster. If the Scan job cannot connect to the source, the job fails.

Informatica PowerCenter on IBM Db2 Connection Properties

Configure connection details to the PowerCenter repository that you want to scan as a data source. You can connect to a PowerCenter repository that is hosted on an IBM Db2 database. Configure the connection details to the Informatica domain and node on which the PowerCenter Repository Service runs.

Configure connection details for the PowerCenter repository that includes workflows to extract, load, and transform data between databases. When you run an Informatica PowerCenter scan, the Scan job identifies sensitive data, sensitive data protection, and sensitive data proliferation for columns in PowerCenter workflows. Do not confuse with the PowerCenter repository that exists on the same domain as the Data Privacy Management Service.

The following table describes the Informatica PowerCenter on IBM Db2 connection properties:

Property	Description
Gateway Host Name or Address	Required. Host name or IP address of the machine that hosts the PowerCenter repository database.
Gateway Port Number	Required. Port number of the machine that hosts the PowerCenter repository database.
Informatica Security Domain	Optional. A collection of user accounts and groups in the Informatica domain. Enter the LDAP security domain name, if one exists. Otherwise, enter: `Native`. Default is Native.
Repository Name	Required. PowerCenter repository service name.
Repository User Name	Required. User that accesses the PowerCenter repository. The user must have PowerCenter Repository Service privileges for the folders, runtime objects, sources and targets, and design objects privilege groups.

Property	Description
Repository User Password	Required. Password for the PowerCenter repository user. When you save the data store, Data Privacy Management encrypts the password.
PowerCenter Version	Required. Select one of the following PowerCenter product versions: - 10.0.0 - 10.1.0 - 10.1.1 - 10.1.1.HF1 - 10.1.1.HF2 - 10.2.0 - 10.2.0HF1 - 10.2.0HF2 - 10.4.0 - 10.4.1 - 10.5.0 - 9.1.0 - 9.1.0.HF1 - 9.1.0.HF2 - 9.1.0.HF3 - 9.1.0.HF4 - 9.1.0.HF5 - 9.1.0.HF6 - 9.5.0 - 9.5.1 - 9.5.1.HF1 - 9.5.1.HF2 - 9.5.1.HF3 - 9.5.1.HF4 - 9.6.0 - 9.6.1 - 9.6.1.HF1 - 9.6.1.HF2 - 9.6.1.HF3 - 9.6.1.HF4 **Note:** If you choose version 10.2.0HF2, ensure that you have applied Informatica EBF-20976. The host, port, and service name values do not appear if you have not applied the EBF.
Connection String	Optional. Native connection string the PowerCenter Repository Service uses to access the PowerCenter repository database. The connection string cannot exceed 32 characters. Enter the connection string using the following syntax: `<database_name>`

Property	Description
Secure JDBC Parameters	Required database parameters to access databases that are secured with the SSL protocol. The parameter string cannot exceed 500 characters. When you save the data store, Data Privacy Management encrypts the parameter string. Enter the parameters as `name=value` pairs separated by semicolon characters (;). You can enter the following secure database parameters: - EncryptionMethod. Required. Indicates whether data is encrypted when transmitted over the network. This parameter must be set to `SSL`. - HostNameInCertificate. Optional. Host name of the machine that hosts the secure database. If you specify a host name, Data Privacy Management validates the host name included in the connection string against the host name in the SSL certificate. - TrustStore. Required. Path and file name of the truststore file that contains the SSL certificate for the database. - TrustStorePassword. Required. Password for the truststore file. - KeyStore. Directory of the keystore file. - KeyStorePassword. Password used to access the keystore file. - KeyPassword. Password used to access individual keys in the keystore file. - ValidateServerCertificate. Optional. Set to True or False. If True, Data Privacy Management validates the certificate that the database server sends. If you specify the HostNameInCertificate parameter, Data Privacy Management also validates the host name in the certificate. If False, Data Privacy Management does not validate the certificate that the database server sends. Data Privacy Management ignores any truststore information that you specify. Data Privacy Management appends the secure JDBC parameters to the connection string.
Database Host	Optional. Database host name for the PowerCenter repository service.
Database Port	Optional. Database port number for the PowerCenter repository service.
Database User	Optional. User name for the PowerCenter repository database. The user account must have the CREATE_VIEW privilege. If you use a read-only user, the user account must have the CREATE_SYNONYM privilege.
Database Password	Optional. Password for the PowerCenter repository database user. When you save the data store, Data Privacy Management encrypts the password.
Informatica Domain	Optional. Name of the Informatica domain on which the PowerCenter Repository Service runs.
PCIS Name	Optional. Name of the PowerCenter Integration Service (PCIS) that is assigned to the PowerCenter repository.
PowerCenter Folder Name	Optional. Name of the folder that Data Privacy Management creates in the PowerCenter repository. The folder includes a workflow that reads connection properties from database configuration files and PowerCenter parameter files. Data Privacy Management requires the workflow to import the properties for all connections in the PowerCenter repository.

Property	Description
Site Key Location	Full directory path of the encryption key for the Informatica domain on which the PowerCenter Repository Service runs. The encryption key is stored in a file named *siteKey*. When you scan a PowerCenter repository, the Scan job imports connections and connection properties. The Scan job creates a data store in the Data Privacy Management repository for each connection in the PowerCenter repository. To read encrypted passwords in the PowerCenter repository, Data Privacy Management uses the encryption key for the Informatica domain on which the PowerCenter Repository Service runs. To encrypt the passwords in the Data Privacy Management repository, Data Privacy Management uses the encryption key for the Information domain on which the Data Privacy Management Service runs. Required for PowerCenter repositories on Informatica version 9.6x or later. If the encryption key is not on the machine that hosts the Data Privacy Management installation, create a mount point. If you do not configure the encryption key location, Data Privacy Management cannot read the passwords from the PowerCenter repository. After you scan the PowerCenter repository, you must manually add the passwords for all created data stores. Not required for PowerCenter repositories on Informatica version 9.5x or earlier. By default, Data Privacy Management uses a static key to read the passwords in the PowerCenter repository.
PCIS OS Profile	Optional. Name of the operating system (OS) profile for the PowerCenter Integration Service (PCIS). An operating system profile is a type of security that the PowerCenter Integration Service uses to isolate the runtime user environment.
Database	Optional. Name of the database that hosts the PowerCenter repository.
Parameter File	Optional. Select a database configuration file that contains parameters to use when the data store scan runs.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Enable Reference Resources	Optional. Select this option to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is cleared.
Retain Unresolved Reference Assets	Optional. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is selected.

Informatica PowerCenter on Microsoft SQL Server Connection Properties

Configure connection details to the PowerCenter repository that you want to scan as a data source. You can connect to a PowerCenter repository that is hosted on a Microsoft SQL Server database. Configure the connection details to the Informatica domain and node on which the PowerCenter Repository Service runs.

Configure connection details for the PowerCenter repository that includes workflows to extract, load, and transform data between databases. When you run an Informatica PowerCenter scan, the Scan job identifies sensitive data, sensitive data protection, and sensitive data proliferation for columns in PowerCenter workflows. Do not confuse with the PowerCenter repository that exists on the same domain as the Data Privacy Management Service.

The following table describes the Informatica PowerCenter on Microsoft SQL Server connection properties:

Property	Description
Gateway Host Name or Address	Required. Host name or IP address of the machine that hosts the PowerCenter repository database.
Gateway Port Number	Required. Port number of the machine that hosts the PowerCenter repository database.
Informatica Security Domain	Optional. A collection of user accounts and groups in the Informatica domain. Enter the LDAP security domain name, if one exists. Otherwise, enter: `Native`. Default is Native.
Repository Name	Required. PowerCenter repository service name.
Repository User Name	Required. User that accesses the PowerCenter repository. The user must have PowerCenter Repository Service privileges for the folders, runtime objects, sources and targets, and design objects privilege groups.
Repository User Password	Required. Password for the PowerCenter repository user. When you save the data store, Data Privacy Management encrypts the password.

Property	Description
PowerCenter Version	Required. Select one of the following PowerCenter product versions: - 10.0.0 - 10.1.0 - 10.1.1 - 10.1.1.HF1 - 10.1.1.HF2 - 10.2.0 - 10.2.0HF1 - 10.2.0HF2 - 10.4.0 - 10.4.1 - 10.5.0 - 9.1.0 - 9.1.0.HF1 - 9.1.0.HF2 - 9.1.0.HF3 - 9.1.0.HF4 - 9.1.0.HF5 - 9.1.0.HF6 - 9.5.0 - 9.5.1 - 9.5.1.HF1 - 9.5.1.HF2 - 9.5.1.HF3 - 9.5.1.HF4 - 9.6.0 - 9.6.1 - 9.6.1.HF1 - 9.6.1.HF2 - 9.6.1.HF3 - 9.6.1.HF4
Connection String	Optional. Native connection string the PowerCenter Repository Service uses to access the PowerCenter repository database. The connection string cannot exceed 32 characters. Enter the connection string in the following syntax: `<server_name>@<database_name>`
Secure JDBC Parameters	Required database parameters to access databases that are secured with the SSL protocol. The parameter string cannot exceed 500 characters. When you save the data store, Data Privacy Management encrypts the parameter string. Enter the parameters as `name=value` pairs separated by semicolon characters (;). You can enter the following secure database parameters: - EncryptionMethod. Required. Indicates whether data is encrypted when transmitted over the network. This parameter must be set to `SSL`. - HostNameInCertificate. Optional. Host name of the machine that hosts the secure database. If you specify a host name, Data Privacy Management validates the host name included in the connection string against the host name in the SSL certificate. - TrustStore. Required. Path and file name of the truststore file that contains the SSL certificate for the database. - TrustStorePassword. Required. Password for the truststore file. - KeyStore. Directory of the keystore file. - KeyStorePassword. Password used to access the keystore file. - KeyPassword. Password used to access individual keys in the keystore file. - ValidateServerCertificate. Optional. Set to True or False. If True, Data Privacy Management validates the certificate that the database server sends. If you specify the HostNameInCertificate parameter, Data Privacy Management also validates the host name in the certificate. If False, Data Privacy Management does not validate the certificate that the database server sends. Data Privacy Management ignores any truststore information that you specify. Data Privacy Management appends the secure JDBC parameters to the connection string.

Property	Description
Database Host	Optional. Database host name for the PowerCenter repository service.
Database Port	Optional. Database port number for the PowerCenter repository service.
Database User	Optional. User name for the PowerCenter repository database. The user account must have the CREATE_VIEW privilege. If you use a read-only user, the user account must have the CREATE_SYNONYM privilege.
Database Password	Optional. Password for the PowerCenter repository database user. When you save the data store, Data Privacy Management encrypts the password.
Informatica Domain	Optional. Name of the Informatica domain on which the PowerCenter Repository Service runs.
PCIS Name	Optional. Name of the PowerCenter Integration Service (PCIS) that is assigned to the PowerCenter repository.
PowerCenter Folder Name	Optional. Name of the folder that Data Privacy Management creates in the PowerCenter repository. The folder includes a workflow that reads connection properties from database configuration files and PowerCenter parameter files. Data Privacy Management requires the workflow to import the properties for all connections in the PowerCenter repository.
Site Key Location	Full directory path of the encryption key for the Informatica domain on which the PowerCenter Repository Service runs. The encryption key is stored in a file named *siteKey*. When you scan a PowerCenter repository, the Scan job imports connections and connection properties. The Scan job creates a data store in the Data Privacy Management repository for each connection in the PowerCenter repository. To read encrypted passwords in the PowerCenter repository, Data Privacy Management uses the encryption key for the Informatica domain on which the PowerCenter Repository Service runs. To encrypt the passwords in the Data Privacy Management repository, Data Privacy Management uses the encryption key for the Information domain on which the Data Privacy Management Service runs. Required for PowerCenter repositories on Informatica version 9.6x or later. If the encryption key is not on the machine that hosts the Data Privacy Management installation, create a mount point. If you do not configure the encryption key location, Data Privacy Management cannot read the passwords from the PowerCenter repository. If you select PowerCenter version 10.5 and do not enter a site key location, the scan fails.
PCIS OS Profile	Optional. Name of the operating system (OS) profile for the PowerCenter Integration Service (PCIS). An operating system profile is a type of security that the PowerCenter Integration Service uses to isolate the runtime user environment.
Database	Optional. Name of the database that hosts the PowerCenter repository.
Parameter File	Optional. Select a database configuration file that contains parameters to use when the data store scan runs.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.

Property	Description
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Enable Reference Resources	Optional. Select this option to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is cleared.
Retain Unresolved Reference Assets	Optional. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is selected.

Informatica PowerCenter on Oracle Connection Properties

Configure connection details to the PowerCenter repository that you want to scan as a data source. You can connect to a PowerCenter repository that is hosted on an Oracle database. Configure the connection details to the Informatica domain and node on which the PowerCenter Repository Service runs.

Configure connection details for the PowerCenter repository that includes workflows to extract, load, and transform data between databases. When you run an Informatica PowerCenter scan, the Scan job identifies sensitive data, sensitive data protection, and sensitive data proliferation for columns in PowerCenter workflows. Do not confuse with the PowerCenter repository that exists on the same domain as the Data Privacy Management Service.

The following table describes the Informatica PowerCenter on Oracle connection properties:

Property	Description
Gateway Host Name or Address	Required. Host name or IP address of the machine that hosts the PowerCenter repository database.
Gateway Port Number	Required. Port number of the machine that hosts the PowerCenter repository database.
Informatica Security Domain	Optional. A collection of user accounts and groups in the Informatica domain. Enter the LDAP security domain name, if one exists. Otherwise, enter: `Native`. Default is Native.
Repository Name	Required. PowerCenter repository service name.

Property	Description
Repository User Name	Required. User that accesses the PowerCenter repository. The user must have PowerCenter Repository Service privileges for the folders, runtime objects, sources and targets, and design objects privilege groups.
Repository User Password	Required. Password for the PowerCenter repository user. When you save the data store, Data Privacy Management encrypts the password.
PowerCenter Version	Required. Select one of the following PowerCenter product versions: - 10.0.0 - 10.1.0 - 10.1.1 - 10.1.1.HF1 - 10.1.1.HF2 - 10.2.0 - 10.2.0HF1 - 10.2.0HF2 - 10.4.0 - 10.4.1 - 10.5.0 - 9.1.0 - 9.1.0.HF1 - 9.1.0.HF2 - 9.1.0.HF3 - 9.1.0.HF4 - 9.1.0.HF5 - 9.1.0.HF6 - 9.5.0 - 9.5.1 - 9.5.1.HF1 - 9.5.1.HF2 - 9.5.1.HF3 - 9.5.1.HF4 - 9.6.0 - 9.6.1 - 9.6.1.HF1 - 9.6.1.HF2 - 9.6.1.HF3 - 9.6.1.HF4 **Note:** If you choose version 10.2.0HF2, ensure that you have applied Informatica EBF-20976. The host, port, and service name values do not appear if you have not applied the EBF.
Connection String	Optional. Native connection string the PowerCenter Repository Service uses to access the PowerCenter repository database. The connection string cannot exceed 32 characters. Enter the connection string using the following syntax: `<database_name>`

Property	Description
Secure JDBC Parameters	Required database parameters to access databases that are secured with the SSL protocol. The parameter string cannot exceed 500 characters. When you save the data store, Data Privacy Management encrypts the parameter string. Enter the parameters as `name=value` pairs separated by semicolon characters (;). You can enter the following secure database parameters: - EncryptionMethod. Required. Indicates whether data is encrypted when transmitted over the network. This parameter must be set to `SSL`. - HostNameInCertificate. Optional. Host name of the machine that hosts the secure database. If you specify a host name, Data Privacy Management validates the host name included in the connection string against the host name in the SSL certificate. - TrustStore. Required. Path and file name of the truststore file that contains the SSL certificate for the database. - TrustStorePassword. Required. Password for the truststore file. - KeyStore. Directory of the keystore file. - KeyStorePassword. Password used to access the keystore file. - KeyPassword. Password used to access individual keys in the keystore file. - ValidateServerCertificate. Optional. Set to True or False. If True, Data Privacy Management validates the certificate that the database server sends. If you specify the HostNameInCertificate parameter, Data Privacy Management also validates the host name in the certificate. If False, Data Privacy Management does not validate the certificate that the database server sends. Data Privacy Management ignores any truststore information that you specify. Data Privacy Management appends the secure JDBC parameters to the connection string.
Database Host	Optional. Database host name for the PowerCenter repository service.
Database Port	Optional. Database port number for the PowerCenter repository service.
Database Service	Optional. TNS service name for the Oracle database in the PowerCenter repository.
Database User	Optional. User name for the PowerCenter repository database. The user account must have the CREATE_VIEW privilege. If you use a read-only user, the user account must have the CREATE_SYNONYM privilege.
Database Password	Optional. Password for the PowerCenter repository database user. When you save the data store, Data Privacy Management encrypts the password.
Informatica Domain	Optional. Name of the Informatica domain on which the PowerCenter Repository Service runs.
PCIS Name	Optional. Name of the PowerCenter Integration Service (PCIS) that is assigned to the PowerCenter repository.
PowerCenter Folder Name	Optional. Name of the folder that Data Privacy Management creates in the PowerCenter repository. The folder includes a workflow that reads connection properties from database configuration files and PowerCenter parameter files. Data Privacy Management requires the workflow to import the properties for all connections in the PowerCenter repository.

Property	Description
Site Key Location	Full directory path of the encryption key for the Informatica domain on which the PowerCenter Repository Service runs. The encryption key is stored in a file named *siteKey*. When you scan a PowerCenter repository, the Scan job imports connections and connection properties. The Scan job creates a data store in the Data Privacy Management repository for each connection in the PowerCenter repository. To read encrypted passwords in the PowerCenter repository, Data Privacy Management uses the encryption key for the Informatica domain on which the PowerCenter Repository Service runs. To encrypt the passwords in the Data Privacy Management repository, Data Privacy Management uses the encryption key for the Information domain on which the Data Privacy Management Service runs. Required for PowerCenter repositories on Informatica version 9.6x or later. If the encryption key is not on the machine that hosts the Data Privacy Management installation, create a mount point. If you do not configure the encryption key location, Data Privacy Management cannot read the passwords from the PowerCenter repository. After you scan the PowerCenter repository, you must manually add the passwords for all created data stores. Not required for PowerCenter repositories on Informatica version 9.5x or earlier. By default, Data Privacy Management uses a static key to read the passwords in the PowerCenter repository.
PCIS OS Profile	Optional. Name of the operating system (OS) profile for the PowerCenter Integration Service (PCIS). An operating system profile is a type of security that the PowerCenter Integration Service uses to isolate the runtime user environment.
Parameter File	Optional. Select a database configuration file that contains parameters to use when the data store scan runs.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Enable Reference Resources	Optional. Select this option to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is cleared.
Retain Unresolved Reference Assets	Optional. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is selected.

Microsoft Azure Blob Storage Connection Properties

Configure properties to connect to Microsoft Azure Blob Storage data stores.

When you test the data store connection, the Data Privacy Management Service tests the combination of the **Blob Endpoint URL** and **Container Name** properties. If any of the property values are not valid, the test connection fails.

The following table describes the Microsoft Azure Blob Storage connection properties:

Property	Description
Scan with Remote Agent	Select to associate the data store with a remote agent when you run Scan jobs to discover personal and sensitive data or build a Subject Registry index. **Important:** If you select this option, you must associate the data store with a remote agent before running a scan. After you save the data store, you cannot change this property.
Blob Endpoint URL	Required. Microsoft Azure Blob Storage URL to access a container. Use the value listed for the Primary BLOB Service endpoint URL in the Azure portal.
Account Name	Optional. Name of the Microsoft Azure Blob Storage account.
Container Name	Required. Name of the container that contains the blobs.
Source Directory	Optional. Directory path that contains the files to scan. To scan all root directories, enter a forward slash `/`. To scan a specific directory, use the following syntax: `/<Directory_Name>` For example, if the directory name is `Folder1`, enter `/Folder1`. Do not include a forward slash `/` at the end of the directory. The Scan job reads files from the last directory specified. For example, if you enter `/HumanResource/Employees/US`, the Scan job only reads files from the `US` directory. The Scan job reads files from the specified directory only. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
Shared Access Signature Token	Required. A token that provides access to the Microsoft Azure Blob Storage resource. Use the value for the Shared Access Signature in the Azure portal.
Folder Options	Required if you enable the **Scan with Remote Agent** property. Folders that the Scan job reads from the location configured in the **Source Directory** property. Select one of the following options: - All - Select Specific Folders - Use Regular Expression Default is All.
Include Folders	Required if you enable the **Scan with Remote Agent** property and you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the directory for at least one folder that you want to include in the scan. The directory must begin with a forward slash (/).
Exclude Folders	Optional if you enable the **Scan with Remote Agent** property and you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the directory for at least one folder that you want to exclude from the scan. The directory must begin with a forward slash (/).

Property	Description
Folder Regular Expression	Required if you enable the **Scan with Remote Agent** property and you choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder that you want to include in the scan.
Blob Prefix	Optional. To sort the blobs based on the prefix of the Microsoft Azure Blob name, enter the prefix. The prefix is case sensitive.
File Types	Required. File types that the Scan job reads from the location configured in the **Source Directory** property. Select **All** to scan all file types. Choose **Select** to select specific file types to scan. Default is All.
Selected File Types	Required if you choose **Select** in the **File Types** property. Click **Select All** to choose all file types, or select one or more of the following options: - Avro - Compressed Files - Delimited and Text - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, the PDF file content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.
Enter File Delimiter	Optional. Available if you do not enable the **Scan with Remote Agent** property. Character that separates entries in the file. Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`. Default is a comma (,).
Extract File Metadata from other File Types	Optional. Select to scan metadata from the files specified in the **File Types** property.
First Level Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Name of the folder that you want to scan. For example: `subfolderA`. To specify multiple folders, separate the folder names with a comma (,). For example: `subfolderA, subfolderB`
Include Sub Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. If enabled, the scan reads data from the directory and all subdirectories of the location configured in the **Source Directory** property or the **First Level Directory** property, if specified. Default is disabled.

Property	Description
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Azure Blob Account Key	Microsoft Azure Blob Storage access key. If you do not enter a key, the data store is incomplete. You cannot scan incomplete data stores.
Source Connection Name	Optional. Connection name in the Informatica domain that the Scan job uses to connect to the Microsoft Azure Blob Storage source. If you do not enter a name, Data Privacy Management creates the source connection name.

Microsoft Azure Data Lake Connection Properties

Configure properties to connect to Microsoft Azure Data Lake.

When you test the data store connection, Data Privacy Management tests the combination of the **Client Id**, **Client Key**, **Directory Name**, and **Auth Endpoint URL** properties. If any of the property values are not valid, the test connection fails.

Prerequisite: Before you create the data store, you must merge the certificates contained in the `<INFA_HOME>/java/jre/lib/security/cacerts` directory in the `infa_truststore.jks` file, which is located in the following directory: `<INFA_HOME>/services/shared/security/`.

The following table describes the Microsoft Azure Data Lake connection properties:

Property	Description
Scan with Remote Agent	Select to associate the data store with a remote agent when you run Scan jobs to discover personal and sensitive data or build a Subject Registry index. **Important:** If you select this option, you must associate the data store with a remote agent before running a scan. After you save the data store, you cannot change this property.
Account Name	Optional. Name of the Microsoft Azure Data Lake account that you created on the Azure portal.
ADLS Source Type	Required. Azure Data Lake Storage (ADLS) source type for big data analytic workloads. Select **Data Lake Store Gen 1** or **Data Lake Store Gen 2**. Default is Data Lake Store Gen 1.

Property	Description
Client Id	Required if the ADLS Source Type is Data Lake Store Gen 1. Client ID to connect to the Microsoft Azure Data Lake Store. Use the value listed for the application ID on the Azure portal.
Client Key	Required if the ADLS Source Type is Data Lake Store Gen 1. Client key to connect to the Microsoft Azure Data Lake Store. Use the Azure Active Directory application key value in the Azure portal as the client key.
ADLS Gen2 Authentication Method	Required if the ADLS Source Type is Data Lake Store Gen 2. The authentication method used to grant access to to files and directories in the data store. Select **Shared Key** or **Azure Active Directory**. Default is Shared Key.
Authentication via Proxy	Required if the ADLS Source Type is Data Lake Store Gen 2. Indicates whether users can access the data store remotely. Select **Disabled** or **Enabled**. Default is Disabled. If you selected the **Scan with Remote Agent** check box, do not change the default value.
Directory Name	Required. Directory of the Azure Data Lake Store. To scan all root directories, enter a forward slash `/`. To scan a specific directory, use the following syntax: `/<Directory_Name>` For example, if the directory name is `Folder1`, enter `/Folder1`. Do not include a forward slash `/` at the end of the directory. The Scan job reads files from the last directory specified. For example, if you enter `/HumanResource/Employees/US`, the Scan job only reads files from the `US` directory. If blank, the data store has a status of Not Complete. You cannot scan incomplete data stores.
Tenant ID	Required if the ADLS Source Type is Data Lake Store Gen 2 and the ADLS Gen2 Authentication Method is Azure Active Directory. The ID of the Azure Active Directory.
Auth Endpoint URL	Required if the ADLS Source Type is Data Lake Store Gen 1. The OAuth 2.0 token endpoint URL in the Azure portal.
Storage Account Key	Required if the ADLS Source Type is Data Lake Store Gen 2 and the ADLS Gen2 Authentication Method is Shared Key. The 512-bit primary storage account access key.
Folder Options	Required if you enable the **Scan with Remote Agent** property. Folders that the Scan job reads from the location configured in the **Source Directory** property. Select one of the following options: - All - Select Specific Folders - Use Regular Expression Default is All.
Include Folders	Required if you enable the **Scan with Remote Agent** property and you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the directory for at least one folder that you want to include in the scan. If the ADLS Source Type is Data Lake Store Gen 2, the directory must begin with a forward slash (`/`).
Exclude Folders	Optional if you enable the **Scan with Remote Agent** property and you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the directory for at least one folder that you want to exclude from the scan. If the ADLS Source Type is Data Lake Store Gen 2, the directory must begin with a forward slash (`/`).
Folder Regular Expression	Required if you enable the **Scan with Remote Agent** property and you choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder that you want to include in the scan.

Property	Description
File Types	Required. File types that the Scan job reads from the location configured in the **Directory Name** property. Select **All** to scan all file types. Choose **Select** to select specific file types to scan. Default is All.
Selected File Types	Required if you choose **Select** in the **File Types** property. Click **Select All** to choose all file types, or select one or more of the following options: - Avro - Compressed Files - Delimited and Text - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, the PDF file content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.
Enter File Delimiter	Optional. Available if you do not enable the **Scan with Remote Agent** property. Character that separates entries in the file. Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`. Default is a comma (,).
First Level Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Name of the folder that you want to scan. For example: `subfolderA`. To specify multiple folders, separate the folder names with a comma (,). For example: `subfolderA, subfolderB`
Include Sub Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. If enabled, the scan reads data from the directory and all subdirectories of the location configured in the **Directory Name** property or the **First Level Directory** property, if specified. Default is disabled.
Memory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.

Property	Description
JVM Options	Optional. Available if you do not enable the **Scan with Remote Agent** property. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Source Connection Name	Optional. Available if you do not enable the **Scan with Remote Agent** property. Connection name in the Informatica domain that the scan job uses to connect to the Microsoft Azure Data Lake source. If you do not enter a name, Data Privacy Management creates the source connection name.

Microsoft Azure SQL Database Connection Properties

Configure properties to connect to a Microsoft Azure SQL Database data source.

When you test the data store connection, the Data Privacy Management Service tests the combination of the **User**, **Host**, **Port**, and **Database** properties. If any of the property values are not valid, the test connection fails.

The following table describes the Microsoft Azure SQL Database connection properties:

Property	Description
User	Required. User name to connect to the Microsoft Azure SQL Database account. The user must have the Administrator role or read permission to the table data and views. The user must also have access to the IP address.
Password	Optional. Password to connect to the Microsoft Azure SQL Database account.
Host	Required. Host name or IP address for the database server.
Port	Required. Port number for the database server.
Database	Required. Name of the Microsoft Azure SQL database.
Instance	Optional. The instance name of the Microsoft Azure SQL Database. You can alternatively specify the port number of the instance. Informatica recommends that you specify the port number of the instance.

Property	Description

Connection String	Required. Native connection string to access data from the database.
	Use the following syntax: `<server_name>@<database_name>`
	Note:
	- The server name and database entry must exist in the `odbc.ini` file.
	- You must set the property EncryptionMethod to 1 in the `odbc.ini` file. For example:
	`[azuresqldbqe]`
	`Driver=/home/opt/infa/10.2.2/ODBC7.1/lib/DWsqls28.so`
	`Description=SQL Server Connection with encryption`
	`HostName=infaserver.database.windows.net`
	`PortNumber=1433`
	`Database=azuresqldbqe`
	`EncryptionMethod=1`
	`ValidateServerCertificate=0`

Secure JDBC Parameters	Required database parameters to access databases that are secured with the SSL protocol. The parameter string cannot exceed 500 characters. When you save the data store, Data Privacy Management encrypts the parameter string. Enter the parameters as `name=value` pairs separated by semicolon characters (;). You can enter the following secure database parameters: - EncryptionMethod. Required. Indicates whether data is encrypted when transmitted over the network. This parameter must be set to `SSL`. - HostNameInCertificate. Optional. Host name of the machine that hosts the secure database. If you specify a host name, Data Privacy Management validates the host name included in the connection string against the host name in the SSL certificate. - TrustStore. Required. Path and file name of the truststore file that contains the SSL certificate for the database. - TrustStorePassword. Required. Password for the truststore file. - KeyStore. Directory of the keystore file. - KeyStorePassword. Password used to access the keystore file. - KeyPassword. Password used to access individual keys in the keystore file. - ValidateServerCertificate. Optional. Set to True or False. If True, Data Privacy Management validates the certificate that the database server sends. If you specify the HostNameInCertificate parameter, Data Privacy Management also validates the host name in the certificate. If False, Data Privacy Management does not validate the certificate that the database server sends. Data Privacy Management ignores any truststore information that you specify. Data Privacy Management appends the secure JDBC parameters to the connection string.
Schema Option	Optional. Select one of the following options to determine how you add schemas to the data store: - **All**. Includes all schemas that you can access. - **Select From List**. Displays a list of schemas in the **Schema** property. - **Specify Regex**. Enables the **Regular Expression for Schema** property. Select this option if you want to enter schemas as a regular expression.
Schema	Displays a list of available schemas to include in the scan if you select **Select From List** in the **Schema Option** property.
Regular Expression for Schema	Displays if you select **Specify Regex** in the **Schema Option** property. Enter a regular expression to specify schemas that you want to include or exclude from the scan. Schema names are case-sensitive.

Property	Description
Source Metadata Filter	Optional. A filter to specify tables that you want to include or exclude in the scan. Enter a regular expression to include multiple filter parameters. Default is All. For example: `Table1, NOT Table2, SUB%, %EMP, %DATA%`. The filter includes Table1, tables with names that start with SUB, names that end with EMP, and tables that include DATA anywhere in the name. The filter excludes Table2.
Case Sensitive	Optional. Select to indicate that the data store is configured for case sensitivity. Default is cleared.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
CyberArk Safe	Optional. Name of the CyberArk safe that contains the database password. When you test the data store connection or you scan the data store, Data Privacy Management retrieves the password from CyberArk.
Include Views in the Scan	Optional. Determines whether a Scan job evaluates data in views. If enabled, a Scan job identifies sensitive data in views during metadata and data profiling. The Scan job evaluates data in all views for the schemas specified in the **Schema** property. If the property is blank, the Scan job evaluates data in all views of all schemas. If the data store was imported from a parent data store, such as Informatica PowerCenter, the Scan job for the parent data store can identify the lineage of sensitive data in views. For example, you ran an Informatica PowerCenter Scan job that imported a data store that included views. You ran a Database Scan job on the data store that identified sensitive data in the views. When you run an Informatica PowerCenter Scan job to identify data proliferation, the Scan job identifies lineage for the sensitive data in the views. If disabled, the Scan job does not evaluate data in views. If you disable the property after a Scan job completes, any scans that you run in the future will not evaluate data in views. However, the completed Scan job results remain.
Source Connection Name	Optional. Connection name in the Informatica domain that the Scan job uses to connect to the Microsoft Azure SQL Database source.

Microsoft Azure SQL Data Warehouse Connection Properties

Configure properties to connect to a Microsoft Azure SQL Data Warehouse data source.

When you test the data store connection, the Data Privacy Management Service tests the combination of the **User**, **Host**, **Port**, and **Database** properties. If any of the property values are not valid, the test connection fails.

The following table describes the Microsoft Azure SQL Data Warehouse connection properties:

Property	Description
User	Required. User name to connect to the Microsoft Azure SQL Data Warehouse account. The user must have read permission to the source files. The user must also have access to the IP address.
Password	Optional. Password to connect to the Microsoft Azure SQL Data Warehouse account.
Host	Required. Host name or IP address for the database server.
Port	Required. Port number for the database server.
Database	Required. Name of the Microsoft Azure SQL Data Warehouse database.
Instance	Optional. The instance name of the Microsoft Azure SQL Data Warehouse database. You can alternatively specify the port number of the instance. Informatica recommends that you specify the port number of the instance.
Secure JDBC Parameters	Required database parameters to access databases that are secured with the SSL protocol. The parameter string cannot exceed 500 characters. When you save the data store, Data Privacy Management encrypts the parameter string. Enter the parameters as `name=value` pairs separated by semicolon characters (;). You can enter the following secure database parameters: - EncryptionMethod. Required. Indicates whether data is encrypted when transmitted over the network. This parameter must be set to `SSL`. - HostNameInCertificate. Optional. Host name of the machine that hosts the secure database. If you specify a host name, Data Privacy Management validates the host name included in the connection string against the host name in the SSL certificate. - TrustStore. Required. Path and file name of the truststore file that contains the SSL certificate for the database. - TrustStorePassword. Required. Password for the truststore file. - KeyStore. Directory of the keystore file. - KeyStorePassword. Password used to access the keystore file. - KeyPassword. Password used to access individual keys in the keystore file. - ValidateServerCertificate. Optional. Set to True or False. If True, Data Privacy Management validates the certificate that the database server sends. If you specify the HostNameInCertificate parameter, Data Privacy Management also validates the host name in the certificate. If False, Data Privacy Management does not validate the certificate that the database server sends. Data Privacy Management ignores any truststore information that you specify. Data Privacy Management appends the secure JDBC parameters to the connection string.
Azure Blob Account Key	Optional. Microsoft Azure Storage access key to stage the files.

Property	Description
Schema	Schemas from which Data Privacy Management reads when you scan the data store. The schema is case-sensitive. Use a comma to separate multiple entries. If empty, Data Privacy Management reads all schemas from the database. When you test the connection, the Data Privacy Management Service does not test if the schema exists. If the schema does not exist, the Scan job fails.
Source Metadata Filter	Optional. A filter to specify tables that you want to include or exclude in the scan. Enter a regular expression to include multiple filter parameters. Default is All. For example: `Table1, NOT Table2, SUB%, %EMP, %DATA%`. The filter includes Table1, tables with names that start with SUB, names that end with EMP, and tables that include DATA anywhere in the name. The filter excludes Table2.
Azure Blob Container Name	Required. Name of the container in Microsoft Azure Storage to use for staging before extracting data from Microsoft Azure SQL Data Warehouse.
Case Sensitive	Optional. Select to indicate that the data store is configured for case sensitivity. Default is cleared.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
CyberArk Safe	Optional. Name of the CyberArk safe that contains the database password. When you test the data store connection or you scan the data store, Data Privacy Management retrieves the password from CyberArk.

Property	Description
Include Views in the Scan	Optional. Determines whether a Scan job evaluates data in views. If enabled, a Scan job identifies sensitive data in views during metadata and data profiling. The Scan job evaluates data in all views for the schemas specified in the **Schema** property. If the property is blank, the Scan job evaluates data in all views of all schemas. If the data store was imported from a parent data store, such as Informatica PowerCenter, the Scan job for the parent data store can identify the lineage of sensitive data in views. For example, you ran an Informatica PowerCenter Scan job that imported a data store that included views. You ran a Database Scan job on the data store that identified sensitive data in the views. When you run an Informatica PowerCenter Scan job to identify data proliferation, the Scan job identifies lineage for the sensitive data in the views. If disabled, the Scan job does not evaluate data in views. If you disable the property after a Scan job completes, any scans that you run in the future will not evaluate data in views. However, the completed Scan job results remain.
Source Connection Name	Optional. Connection name in the Informatica domain that the Scan job uses to connect to the Microsoft Azure SQL Data Warehouse source.

Microsoft OneDrive Connection Properties

Configure connection properties to a Microsoft OneDrive data source. You can configure the data store to include either multiple user accounts in a tenant or a single user account. When you test the connection to the data source, Data Privacy Management verifies the connection properties.

The following table describes the common Microsoft OneDrive connection properties:

Property	Description
Scan with Remote Agent	Select to associate the data store with a remote agent when you run scan jobs to discover personal and sensitive data or build a Subject Registry index. Default is selected. Required if you configure the data store with tenant credentials. **Important:** If you select this option, you must associate the data store with a remote agent before running a scan. After you save the data store, you cannot change this property.
Use Tenant Credentials	Select to configure the data store to include multiple user accounts in a scan. Available if you choose to scan with remote agent. After you save the data store, you cannot change this property.
Folder Options	Required if you enable the **Scan with Remote Agent** property. Folders that the Scan job reads from the location configured in the **Source Directory** property. Select one of the following options: - All - Select Specific Folders - Use Regular Expression Default is All.
Include Folders	Required if you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to include in the scan.

Property	Description
Exclude Folders	Optional if you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to exclude from the scan.
Folder Regular Expression	Required if you enable the **Scan with Remote Agent** property and choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder that you want to include in the scan.
File Types	Required. File types that the scan job reads from the location configured in the **Source Directory** property. Select **All** to scan all file types. Choose **Select** to select specific file types to scan. Default is All.
Selected File Types	Required if you choose **Select** in the **File Types** property. Click **Select All** to choose all file types, or select one or more of the following options: - Avro - Compressed Files - Delimited and Text - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, the PDF file content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.

The following table describes the properties that appear if you configure the data store with tenant credentials:

Property	Description
Client ID	Client ID of the Azure Active Directory application. Required if you configure the data store with tenant credentials.
Client Secret	Client secret of the Azure Active Directory application. Required if you configure the data store with tenant credentials.
Tenant ID	The unique ID of the Azure tenant. Required if you configure the data store with tenant credentials.
Department	A filter to include users based on the user departments they belong to. Enter the name of the departments that you want to include in scans. Use commas or press **Enter** to separate multiple values.
Include Users	A filter to include users. Enter the email IDs of users or accounts that you want to scan. Use commas or press **Enter** to separate multiple values.

Property	Description
Exclude Users	A filter to exclude users. Enter the email IDs of users or accounts to exclude from the scan. Use commas or press **Enter** to separate multiple values. **Note:** The Exclude Users values takes precedence over Include Users.
Email groups	A filter to include users based on group membership. Enter the group names. Use commas or press **Enter** to separate multiple values.

The following table describes the properties that appear if you configure the data store with user credentials:

Property	Description
OneDrive URL	Required for single user account configuration. URL to access OneDrive.
Client ID	Client ID of the Azure Active Directory application. Required to authenticate the Microsoft OneDrive data source with user credentials grant type.
Tenant ID	Unique ID of the tenant or domain. Required to authenticate Microsoft OneDrive data source with user credentials grant type.
User Name	Required if you do not use tenant credentials. The user name for the OneDrive account.
Password	Required if you do not use tenant credentials. The password for the OneDrive account.
Source Directory	Required if you do not use tenant credentials. The source directory that contains the files you want to scan. You must enter `/Documents` before the file path. Example: `/Documents/<userFolder_for_Scan>`
Enter File Delimiter	Optional. Available if you do not enable the **Scan with Remote Agent** property. Character that separates entries in the file. Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`. Default is a comma (,).
Extract File Metadata from other File Types	Available if you do not enable the **Scan with Remote Agent** property. Select to scan metadata from the file types not specified in the **Selected File Types** property. Data Privacy Management extracts the metadata for the file types that the Profiling job step does not support.
First Level Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Name of the folder that you want to scan. For example: `subfolderA`. To specify multiple folders, separate the folder names with a comma (,). For example: `subfolderA, subfolderB`
Include Sub Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. If enabled, the scan reads data from the directory and all subdirectories of the location configured in the **Source Directory** property or the **First Level Directory** property, if specified. Default is disabled.

Property	Description
Memory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Memory to use when you scan the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. Available if you do not enable the **Scan with Remote Agent** property. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Profile Execution Engine	Optional. Available if you do not enable the **Scan with Remote Agent** property. Select one of the following profiling engine modes: - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. Default is Native.
Hadoop Connection Name	Available if you do not enable the **Scan with Remote Agent** property. If you select **Hadoop** in the **Profile Execution Engine** property, enter the name of the Hadoop connection listed in the Administrator tool, **Connections** tab.
Source Connection Name	Available if you do not enable the **Scan with Remote Agent** property. The connection name in the Informatica domain. Do not enter a value for File Management scans.

Microsoft SharePoint Connection Properties

Configure connection properties to a Microsoft SharePoint data source. You can configure the data store to include a tenant or a specific user account. When you test the connection to the data source, the Data

Privacy Management Service verifies the combination of the **SharePoint URL**, **User Name**, and **Password** properties.

The following table describes the common Microsoft SharePoint connection properties:

Property	Description
Scan with Remote Agent	Select to associate the data store with a remote agent when you run Scan jobs to discover personal and sensitive data or build a Subject Registry index. Default is selected. Required if you configure the data store with tenant credentials. **Important:** If you select this option, you must associate the data store with a remote agent before running a scan. After you save the data store, you cannot change this property.
Use Tenant Credentials	Select to configure the data store to include multiple user accounts in a scan. Available if you choose to scan with remote agent.
Enable Subsite Scan	Select **Yes** to scan subsites from the SharePoint site. Default is No.
Include Nested Subsites	If you select **Yes** in the **Enable Subsite Scan** property, this option appears. Select the option to include nested subsites when scanning subsites from the SharePoint site.
Folder Options	Required if you enable the **Scan with Remote Agent** property. Folders that the Scan job reads from the location configured in the **SharePoint URL** property. Select one of the following options: - All - Select Specific Folders - Use Regular Expression Default is All. **Note:** The names that appear in the list of folders might differ from the internal names. For tenant-level data stores, scan results include the internal names and not the display names. If the scan results display a different name, check the SharePoint web URL of the folder to identify the folder it refers to.
Include Folders	Required if you enable the **Scan with Remote Agent** property and you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to include in the scan. If you enabled subsite scans, enter a space as a prefix to the path. Use the following syntax: `/ <space><subsite name>/<folder name>` For example: `/ finance/year2020`
Exclude Folders	Optional if you enable the **Scan with Remote Agent** property and you choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to exclude from the scan. If you enabled subsite scans, enter a space as a prefix to the path. Use the following syntax: `/ <space><subsite name>/<folder name>` For example: `/ finance/year2020`
Folder Regular Expression	Required if you enable the **Scan with Remote Agent** property and you choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder that you want to include in the scan. If you enabled subsite scans, enter a space as a prefix to the path. Use the following syntax: `/ <space><subsite name>/<RegEx>` For example: `/ finance/year*`

Property	Description
File Types	Required. File types that the Scan job reads from the location configured in the **SharePoint URL** property. Select **All** to scan all file types. Choose **Select** to select specific file types to scan. Default is All.
Enter File Delimiter	Optional. Available if you do not enable the **Scan with Remote Agent** property. Character that separates entries in the file. Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma (,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Verify that you enclose the delimiter in single quotes. For example, `'$'`. Use a comma to separate multiple delimiters. For example, `'$','%','&'`. Default is a comma (,).
Selected File Types	Required if you choose **Select** in the **File Types** property. Click **Select All** to choose all file types, or select one or more of the following options: - Avro - Compressed Files - Delimited and Text - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - PDF - Parquet - Webpage Files - XML **Note:** If you enable Optical Character Recognition (OCR) and you select the PDF file type, the PDF file content is processed as optical characters. To choose the Image file type, OCR must be enabled for unstructured scans in the **System Settings**.

The following table describes the properties that appear if you configure the data store with tenant credentials:

Property	Description
SharePoint URL	Required. The root URI for the data source exposed through the Open Data Protocol layer. All requests are extensions of this URI. For example: `https://infasharepoint.abcd.com/Site/_vti_bin/Data.svc`
Client ID	Client ID of the Azure Active Directory application. Required if you configure the data store with tenant credentials.
Client Secret	Client secret of the Azure Active Directory application. Required if you configure the data store with tenant credentials.
Tenant ID	The unique ID of the tenant or domain. Required if you configure the data store with tenant credentials.

Property	Description
Enable Subsite Scan	Select the option to include subsites from the SharePoint site in the scan. By default, the option is selected.
Include Sites	Enter the sites to include in the scan. By default, the scan includes all sites. Enter sites to scan specific sites and exclude all other sites.

The following table describes the properties that appear if you configure the data store with user credentials:

Property	Description
SharePoint URL	Required. The SharePoint URl that you extract. For example: `https://infasharepoint.abcd.com/Site/_vti_bin/Data.svc`
Client ID	Client ID of the Azure Active Directory application. Required to authenticate the Microsoft Sharepoint data source with user credentials grant type.
Tenant ID	Unique ID of the tenant or domain. Required to authenticate the Microsoft Sharepoint data source with user credentials grant type.
User Name	Required. The user name for the SharePoint account.
Password	Required. The password for the SharePoint account.
SharePoint Content Type	Required. Specifies the type of SharePoint content. Select one of the following options: - **All**. Scans contents from SharePoint lists and libraries. - **SharePoint List**. Scans content from attached files in the SharePoint list, such as calender events, Wikipedia pages, announcements, and links. - **SharePoint Library**. Scans content from the SharePoint library. Default is All.
Extract File Metadata from other File Types	Select to scan metadata from the file types not specified in the **Selected File Types** property. Data Privacy Management extracts the metadata for the file types that the Profiling job step does not support.
First Level Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. Name of the folder that you want to scan. For example: `subfolderA`. To specify multiple folders, separate the folder names with a comma (,). For example: `subfolderA, subfolderB`
Include Sub Directory	Optional. Available if you do not enable the **Scan with Remote Agent** property. If enabled, the scan reads data from the directory and all subdirectories of the location configured in the **SharePoint URL** property or the **First Level Directory** property, if specified. Default is disabled.
Memory	Optional. Memory used when you scan the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.

Property	Description
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Profile Execution Engine	Optional. Select one of the following profiling engine modes: - **Hadoop**. Data Privacy Management uses the Blaze engine to run profiles. - **Native**. Data Privacy Management uses the Informatica Data Integration Service engine to run profiles. Default is Native.
Hadoop Connection Name	If you select **Hadoop** in the **Profile Execution Engine** property, enter the name of the Hadoop connection listed in the Administrator tool, **Connections** tab.
Source Connection Name	The connection name in the Informatica domain. Do not enter a value for File Management scans.

Salesforce Connection Properties

Configure properties to connect to a Salesforce application.

The following table describes the Salesforce connection properties:

Property	Description
User	Required. Name of the user account to log into Salesforce. To import user metadata from Salesforce, the user account must have the View All Data and View Event Log Files privileges. The Salesforce org must have the Bulk API feature enabled.
Password	Required. Password combined with the security token for the Salesforce user name. Salesforce issues tokens through email during user registration. Enter the Salesforce password followed by the token, without a space in between. For example, if the Salesforce password is `Infa2020` and the security token is `vK2VVTty2bw1V2K9zUj0KEzr`, enter the password as `Infa2020vK2VVTty2bw1V2K9zUj0KEzr`.
Service URL	Required. By default, this property displays the following URL: `https://$YourInstance$.salesforce.com/services/Soap/u/34.0` Replace `$YourInstance$` with the instance name of the Salesforce account that you want to access. For example, if your account is served by "ap2.salesforce.com," enter the following URL: `http://ap2.salesforce.com/services/Soap/u/34.0`

Property	Description
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Source Connection Name	Optional. Connection name in the Informatica domain that the scan job uses to connect to the Salesforce application.

SAP Connection Properties

Configure the connection details to an SAP application. When you scan an SAP data store, the scan job connects to the data store and scans it for sensitive data. To perform Subject Registry scans, use CDATA connectors to configure the data store.

When you test a connection to SAP, Data Privacy Management tests the combination of the user name and password, and the system and client numbers. Data Privacy Management uses the application server host name to determine duplicate data stores.

Note: You can connect to the SAP application server through FTP or non-FTP mode. If you do not use the FTP mode, you must create an NFS mount between the Staging Directory in the SAP server and the Source Directory in the Data Privacy Management server.

The following table describes the SAP connection properties:

Property	Description
User Name	Required. Name of the user account to connect to the SAP application server.
Password	Required. Password for the user account to connect to the SAP application server. When you save the data store, Data Privacy Management encrypts the password.
Application Server Host	Required. Host name or IP address of the SAP application server. Example: `sapappserver.domain.com`
System Number	Required. SAP system number. Example: `00`

Property	Description
Client	Required. SAP client number. Example: `800`
Language	Optional. Language code that corresponds to the SAP language. Example: `EN`
Enable Streaming for Data Access	Select to access the source SAP data through streaming rather than through FTP.
Encoding	Required. Code page compatible with the SAP server. Must also correspond to the language code. Select **UTF-8 encoding of Unicode**.
Connection Type	SAP connection type. Select **Application**.
Staging Directory	Required for data profiling. Path in the SAP application server where the SAP user can read, write, and delete the temporary files. Example: `C:\usr\sap\sapsid\default` You do not need to set the property for metadata profiling.
Source Directory	For data profiling, path to the directory on the Data Privacy Management server where the Data Integration Service can read, write, and delete files. Example: `/INFA/shared` You do not need to set the property for metadata profiling.
Use FTP	Select to enable FTP access to SAP. You do not need to enable the property for metadata profiling.
FTP Host	If you select **Use FTP**, enter the host name or IP address of the FTP server. Example: `ftp.domain.com`
FTP User	If you select **Use FTP**, enter the FTP user name to connect to the FTP server. Example: `ftpuser`
FTP Password	If you select **Use FTP**, enter the password for the FTP user.
Retry Period	If you select **Use FTP**, enter the number of seconds the Data Privacy Management Service attempts to reconnect to the FTP host if the connection fails. To indicate an infinite retry period, enter `0`.
Use SFTP	Appears if you select **Use FTP**. Enables secure FTP access to SAP.
Public Key File Name	If you select **Use SFTP**, enter the public key file path and filename. Required if the SFTP server uses public key authentication.
Private Key File Name	If you select **Use SFTP**, enter the private key file path and filename. Required if the SFTP server uses private key authentication.
Private Key File Password	If you select **Use SFTP**, enter the private key file password used to decrypt the private key file. Required if the SFTP server uses private key authentication and the private key is encrypted.
Scan with Remote Agent	Select to associate the data store with a remote agent when you run scan jobs to discover personal and sensitive data or build a Subject Registry index. **Important:** If you select this option, you must associate the data store with a remote agent before running a scan. After you save the data store, you cannot change this property.

Property	Description
Folder Options	Required if you enable the **Scan with Remote Agent** property. Folders that the scan job reads from the location configured in the **Staging Directory** or **Source Directory** property. Select one of the following options: - All - Select Specific Folders - Use Regular Expression
Include Folders	Required if you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to include in the scan.
Exclude Folders	Optional if you enable the **Scan with Remote Agent** property and choose the **Select Specific Folders** option in the **Folder Options** property. Enter the file paths for at least one folder that you want to exclude from the scan.
Folder Regular Expression	Required if you enable the **Scan with Remote Agent** property and choose the **Use Regular Expression** option in the **Folder Options** property. Enter a regular expression for at least one folder that you want to include in the scan.
Repository Objects	Optional. Select one or more SAP repository objects, such as ABAP programs, classes, or CDS objects from the list. Each repository object is assigned to a package that contains related tables. Data Privacy Management will scan the tables in the selected repository objects. If you plan to scan the entire SAP system, do not select repository objects.
Memory	Optional. Available if you enable the **Scan with Remote Agent** property. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Options are: **High**, **Low**, and **Medium**. Select High. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. Available if you enable the **Scan with Remote Agent** property. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Source Connection Name	Optional. Available if you enable the **Scan with Remote Agent** property. Connection name in the Informatica domain that the Scan job uses to connect to the SAP source.

SAP S/4HANA Data Store Connection Properties

Configure connection details for an SAP S/4HANA application. During a scan, the scan job connects to the data store to find sensitive data. For Subject Registry scans, use CDATA connectors to configure the data store.

When you test the SAP S/4HANA connection, Data Privacy Management verifies the user name and password, along with the system and client numbers. Data Privacy Management uses the application server host name to determine duplicate data stores.

Note: You can connect to the SAP S/4HANA application server through FTP or non-FTP mode. If you do not use the FTP mode, create an NFS mount between the Staging Directory on the SAP S/4HANA server and the Source Directory on the Data Privacy Management server.

The following table describes the SAP S/4HANA connection properties:

Property	Description
SAP Server Type	Required. Select **SAP S/4HANA** server type. Default is SAP Central Instance.
Server Host	Required. Host name or IP address of the SAP S/4HANA application server. Example: `saphanaappserver.domain.com`
System Number	Required. SAP S/4HANA system number. Example: `00`
SAP SID	Required. SAP S/4HANA system ID. Example: `S41`
Client	Required. SAP S/4HANA client number. Example: `800`
Language	Optional. Language code that corresponds to the SAP S/4HANA language. Example: `EN`
Connection Type	Required. SAP S/4HANA source connection type. Default is basic.
User Name	Required. User account name to connect to SAP S/4HANA.
Password	Required. Password for the user account to connect to the SAP S/4HANA application server. When you save the data store, Data Privacy Management encrypts the password.
Enable Streaming for Data Access ENG	Required. Select to access the source SAP S/4HANA data through streaming instead of FTP.
Connection Type	SAP S/4HANA connection type. Select **Application**.
Staging Directory	Required for data profiling. Path on the SAP S/4HANA application server where the SAP S/4HANA user can manage the temporary files. Example: `C:\usr\sap\sapsid\default` You do not need to set the property for metadata profiling scans.
Source Directory	For data profiling, path to the directory on the Data Privacy Management server where the Data Integration Service manages files. Example: `/INFA/shared` You do not need to set the property for metadata profiling.

Property	Description
Use FTP	Select to enable FTP access to SAP S/4HANA. Not required for metadata profiling.
FTP Host	Required if you selected **Use FTP**. Enter the host name or IP address of the FTP server. Example: `ftp.domain.com`
FTP User	Required if you selected **Use FTP**. Enter the FTP user name to connect to the FTP server. Example: `ftpuser`
FTP Password	Required if you selected **Use FTP**. Enter the password for the FTP user.
Retry Period	Required if you selected **Use FTP**. Enter the number of seconds the Data Privacy Management Service attempts to reconnect to the FTP host if the connection fails. Enter `0` for infinite retries.
Use SFTP	Appears if you select **Use FTP**. Enables secure FTP access to SAP S/4HANA.
Public Key File Name	Required if you selected **Use SFTP**. Enter the public key file path and filename for SFTP authentication.
Private Key File Name	Required if you selected **Use SFTP**. Enter the private key file path and filename for SFTP authentication.
Private Key File Password	Required if you selected **Use SFTP**. Enter the private key file password used to decrypt the private key file for SFTP authentication.
Repository Objects	Optional. Select one or more SAP S/4HANA repository objects, such as ABAP programs, classes, or CDS objects to scan. Each repository object is assigned to a package that contains related tables. Data Privacy Management will scan the tables in the selected repository objects. Omit to scan the entire system.
Enable Reference Resource	Select to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is No.
Memory	Optional. Scanner memory for scan jobs. Options are: **High**, **Low**, and **Medium**. Select High. Increase scanner memory for large scans or multiple parallel scans. Additionally, update the memory settings in the Yarn Resource Manager to match any scanner memory changes.
Custom JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. When `pmem` is enabled, the configuration increases the scanner container memory. Default is 1. - `DDeliveryClassFilter=<A,C,L,G,E,S,W>`. Filters tables based on delivery class. Use a comma-separated delimiter for the multiple delivery classes.
Source Connection Name	Optional. Connection name in the Informatica domain that the scan job uses to connect to the SAP S/4HANA source.

Snowflake Connection Properties

Configure properties to connect to a Snowflake database.

The following table describes the Snowflake connection properties:

Property	Description
Account	Name of the Snowflake account. In the Snowflake URL, your account name is the first segment in the domain. For example, `123abc` is your account name in the URL `https://123abc.snowflakecomputing.com`.
User	The user name to connect to the Snowflake account.
Password	Password associated with the Snowflake user account.
Role	The Snowflake role assigned to the user.
Warehouse	The Snowflake warehouse name.

Additional Parameters	Specify one or more JDBC parameters in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example:
	`user=jon&warehouse=mywh&db=mydb&schema=public`

Case Sensitive	Specifies that the data store is configured for case sensitivity. Select the check box to specify that the data store is case sensitive. Clear the check box to specify that the data store is case insensitive. Default is selected.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Filter Depth	Specifies the number of child levels to return.
Schema	Name of the Snowflake schema.

Property	Description
Include Views in the Scan	Optional. Use to evaluate data included in views If enabled, a scan job identifies sensitive data in views during metadata and data profiling. The scan job evaluates data in all views for the schemas specified in the **Schema** property. If the property is blank, the scan job evaluates data in all views of all schemas. If disabled, the scan job does not evaluate data in views. If you disable the property after a scan job completes, any scans that you run in the future do not evaluate data in views. However, the completed scan job results remain. **Note:** You must apply the Informatica 10.5.2.1 Service Pack to include views in a scan.
Source Connection Name	Optional. Connection name in the Informatica domain that the Cloud job uses to connect to the Snowflake database.

Snowflake Advanced Scanner data store properties

You can't create a Snowflake Advanced Scanner data store in Data Privacy Management, but you can view and modify the data store properties after you import the resource from the catalog. Properties that you modify in Data Privacy Management are updated in the Catalog Administrator tool. If you configure the data store to sync with the catalog, updates made to the data store in the catalog are visible in Data Privacy Management after the Sync Catalog job runs.

To work with a Snowflake Advanced Scanner data store, you must apply the Informatica 10.5.2.1 service pack. For information about how to view domain discovery scan results and run subject registry scans with the Snowflake advanced scanner in Data Privacy Management, see the How-To article: *Using the Snowflake Advanced Scanner in Data Privacy Management*.

The following table describes the Snowflake Advanced Scanner connection properties:

Property	Description
Name	Required. Short description of the data store. The name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 255 characters or contain spaces or the following special characters: `\~!$%^&*()+`
Description	Optional. Long description of the data store that does not exceed 255 characters.
Category	Read only. Indicates the data store type as Cloud.
Data Store Type	Read only. Snowflake Advanced Scanner.
Account	Name of the Snowflake account. In the Snowflake URL, your account name is the first segment in the domain. For example, `123abc` is your account name in the URL `https://123abc.snowflakecomputing.com`.
User	The user name to connect to the Snowflake account.

Property	Description
Password	Password associated with the Snowflake user account.
Source Connection Name	Connection name in the Informatica domain that the Cloud job uses to connect to the Snowflake database.

You can configure the following additional properties:

Property	Description
Data Store Group	Required. Select a logical grouping of related data stores from the list of groups defined on the **Data Store Groups** workspace. You can assign a data store to one data store group.
Security Groups	Required. Select a security group from the list of all security groups in the Informatica domain, including native and LDAP groups. Native groups display the group name. For example, Administrator. LDAP groups include the security domain and the group name. For example, <*security domain*>/<*LDAP group name*>.
Tags	Optional. Keywords that add to the description. Create a new tag or select an existing tag from the list. You can assign multiple tags to a data store. After you assign a tag to a data store, you can use tags in alert rules to improve the specificity of the alert. For example, you want to be alerted when the risk score of any production data store increases by 20%. However, you do not want to be alerted for non-production data stores. You can assign the tag PROD to the production data stores. Then, configure the alert conditions to alert you when the risk score increases by 20% for data stores that have the PROD tag. When you delete a data store, if a tag was only assigned to the deleted data store, the Data Privacy Management Service deletes the tag from the repository.
Department	Optional. Enter the department that is responsible for the maintenance or ownership of the data store. The department can be a line of business, such as investment banking or wealth management, or a corporate department, such as Sales or Marketing. The name is case-sensitive and cannot exceed 100 characters. After you scan data stores, you can view a summary of scan results for all data stores by department. You can filter the scan results by location on the **Overview** workspace.
Data Owner	Optional. Select the individual, group, or organization name that is responsible for the accuracy and integrity of the source data on which you run a classification profiling scan.
Shared with Third Party	Optional. Indicates that the data in the data store is shared with a third party such as a third-party organization or vendor.
Third-Party Description	Optional. Appears if you select the **Shared with Third Party** check box. Enter a description of the third party. You can enter up to 2000 characters and use the following special characters: , _ @
Location	Optional. Select a location that is defined on the **Locations** workspace. If you do not select a location, Data Privacy Management assigns the Unknown location to the data store.
Auto Sync Catalog	Optional. Determines if the Scan job automatically synchronizes data source information in the catalog with the corresponding data store information in Data Privacy Management. Default is disabled.

The following read-only properties appear on the data store details page. The values do not impact the data store:

Property	Description
Auto Assign Connections	Select to automatically assign the schemas in Enterprise Data Catalog to the data store. Default is No.
Case Sensitive	Specifies that the data store is configured for case sensitivity. Select the check box to specify that the data store is case sensitive. Clear the check box to specify that the data store is case insensitive. Default is selected.
Memory	Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
Custom JVM Options	JVM parameters that configure the scanner container. Configure the following parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key value>`. Key pair value that you need to set in the YARN environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Enable Reference Resources	Select to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is No.
Retain Unresolved reference Assets	Appears when you select the Enable Reference Resources property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is Yes.

SQL Server Integration Services Connection Properties

Create a data store and configure properties to scan data in Microsoft SQL Server Integration Services packages. The type of data store that you create depends on the server that hosts the packages.

When you scan an SQL Server Integration Services data store, Data Privacy Management creates a Data Integration job. Each data store should connect to a different package. The connections in the mappings should also be unique. Data Privacy Management does not validate if the data store connects to different packages.

For file-based data stores, when you test the connection, Data Privacy Management tests the **Agent URL** property. If the property value is not valid, the test connection fails.

For repository-based data stores, when you test the connection, Data Privacy Management tests the combination of the **Agent URL**, **SQL Server Version**, **Package/Repository Name**, and **Password** properties. If any of the property values are not valid, the test connection fails.

The following table describes the SQL Server Integration Services connection properties:

Property	Description
SSIS Scanner Type	Required. Select one of the following options: - **FILE**. A file-based data store. Select this option to scan packages that you created in Microsoft Visual Studio, but do not exist in Microsoft SQL Server Management Studio. The package files must exist on the machine that hosts the Data Privacy Management Service. - **Repository Server**. A repository-based data store. Select this option to scan packages that exist in the Microsoft SQL Server Management Studio repository.
Agent URL	Required. URL of the Enterprise Data Catalog agent. Use the following syntax to enter the URL: `http://<host name or IP address>:<port>/<directory>` For file-based data stores, if the package file is not password protected, you can provide a URL to an Enterprise Data Catalog agent that is installed on any machine. If the package file is password protected, provide a URL to an Enterprise Data Catalog agent that exists on the same machine that hosts Microsoft Visual Studio or the Microsoft SQL Server Management Studio. The Microsoft Visual Studio version must support the development of Integration Services packages. For repository-based data stores, provide a URL to an Enterprise Data Catalog agent that is on the same machine that hosts the Microsoft SQL Server Management Studio.
File	Required if you select **FILE** in the **SSIS Scanner Type** property. Full file path and file name for the package .dtsx or .zip file. To scan multiple package files, add the files to one .zip file. The file must exist on the same machine that hosts the Data Privacy Management Service. For example, enter `/test/Package.dtsx` if the package .dtsx file exists in the test folder. If the file path is not valid or does not exist, the Scan job fails.
Encoding	Required if you select **FILE** in the **SSIS Scanner Type** property. Code page for the extracted metadata. Default is Western European (iso-8859-1).
SQL Server Version	Required if you select **Repository Server** in the **SSIS Scanner Type** property. Select one of the following options: - SQL Server 2012 - SQL Server 2014 - SQL Server 2016 If blank, the data store status is Not Complete. You cannot scan incomplete data stores.
Host Name	Required if you select **Repository Server** in the **SSIS Scanner Type** property. Host name or IP address for the server. If blank, the data store status is Not Complete. You cannot scan incomplete data stores.
Password	Password for the package. Required if the package is encrypted with a password. When you save the data store, Data Privacy Management encrypts the value in the repository. If the package is encrypted and the password is not valid or blank, the Scan job fails.

Property	Description
Package/ Repository Name	If you select **FILE** in the **SSIS Scanner Type** property, enter the Microsoft Visual Studio package name. If you select **Repository Server** in the **SSIS Scanner Type** property, enter the directory path and package name in the Microsoft SQL Server Management Studio from which the Data Integration job reads. Specify the directory structure starting after the Stored Packages MSDB directory. The Data Integration job connects only to stored packages in MSDB subdirectories. - For example, you want to scan a package named TestPackage. The full path for the package in the Microsoft SQL Server Management Studio is `<Server>/Stored Packages/MSDB/TestFolder/TestPackage` - Enter `/TestFolder/TestPackage` - Use a comma to separate multiple values. If you include multiple packages in a data store, the packages must use the same password. You can also use a Java-based regular expression. If blank, the Data Integration job reads from all packages in the MSDB subdirectories.
Variable Values File	Optional. Full directory path and file name of the text file that defines the values for the user-defined variables in the package. The variables value file must be on the same machine where the Data Privacy Management Service runs. For example, enter `/home/test/variables.txt` if the variables text file exists in the `/home/test` directory. If the data flow includes variables and you do not provide a variable values file, the Data Integration job cannot identify the sensitive data proliferation between the connections in the mapping. If the file path does not exist or is not valid, the Data Integration job fails. The Data Integration job does not validate the content of the text file.
Working Directory Path	Optional. File path to the working directory of the SQL Server Integration Services database.
Memory	Optional. Memory that the Enterprise Data Catalog scanner uses when you scan the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
JVM Options	Optional. JVM parameters that configure the scanner container. Use the following arguments to configure the parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key=value>`. Key pair value that you need to set in the Yarn environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Enable Reference Resources	Optional. Select this option to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is cleared.
Retain Unresolved Reference Assets	Optional. Appears when you select the **Enable Reference Resources** property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is disabled.
Agent Options	Optional. Enter the SQL Server Agent configuration options to include in the scan.

CHAPTER 9

Data Domains

This chapter includes the following topics:

- Data Domains Overview, 202
- Data Domains Workspace, 203
- Data Domain Properties, 205
- Metadata Match Condition, 207
- Data Match Condition, 209
- Conformance Score, 212
- Data Domains Management, 213

Data Domains Overview

A data domain defines the rules that identify and protect sensitive data in a data store.

The Data Privacy Management installation includes a set of commonly used data domains that you can use to identify and protect sensitive data in a data store. You can also create data domains. Data Privacy Management stores data domains in the Model repository.

When you create a data domain, you specify a rule to identify data by metadata or data values and a protection extension to protect sensitive data.

For example, you want to find columns that contain U.S. Social Security numbers. A data store might have Social Security numbers in a column called `SSN` in one table and a column called `Social_Security` in a different table. The data store might also contain Social Security numbers in a column called `Comments`. To identify all columns with Social Security numbers, you can create a metadata match condition and a data match condition. The metadata match condition describes possible column names for Social Security numbers such as `SSN` and `Social`. The data match condition can specify a reference table, rule, or pattern that the data values must match. In this example, you can use a data match condition that specifies the pattern used for U.S. Social Security numbers: `999-99-9999`

When you specify a data match condition, you also specify a conformance range. During a data scan, Data Privacy Management calculates the conformance score of a column. The conformance score is the percentage of values in a column that match the data domain. Based on the conformance range you specified, Data Privacy Management uses the conformance score to reject or identify a column as sensitive and to determine when to report a column for validation.

When you add protection extensions to a data domain, you also specify the default protection rules that Data Privacy Management applies to the sensitive fields in each data store that is included in the data domain.

When you configure a protection task, you can apply the default protection rules from the data domains to the sensitive fields in the data store for which the protection task is created.

You assign data domains to classification policies. When you define a scan, you assign a classification policy to the scan. Data Privacy Management uses the information in the data domains to discover sensitive columns.

The discovery of sensitive columns in a data store forms the basis of determining critical security information such as policy classification, data risk cost, sensitive data proliferation, and protection status.

Related Topics:

- "Extensions Overview" on page 65

Data Domains Workspace

On the **Data Domains** workspace, you can view, create, and manage data domains.

The **Data Domains** workspace provides the following information:

List of data domains

When you first access the **Data Domains** workspace, you see a list of the predefined data domains available in the Model repository. From the **Actions** menu, you can import, create, view, edit, copy, and delete data domains.

Data Domain details

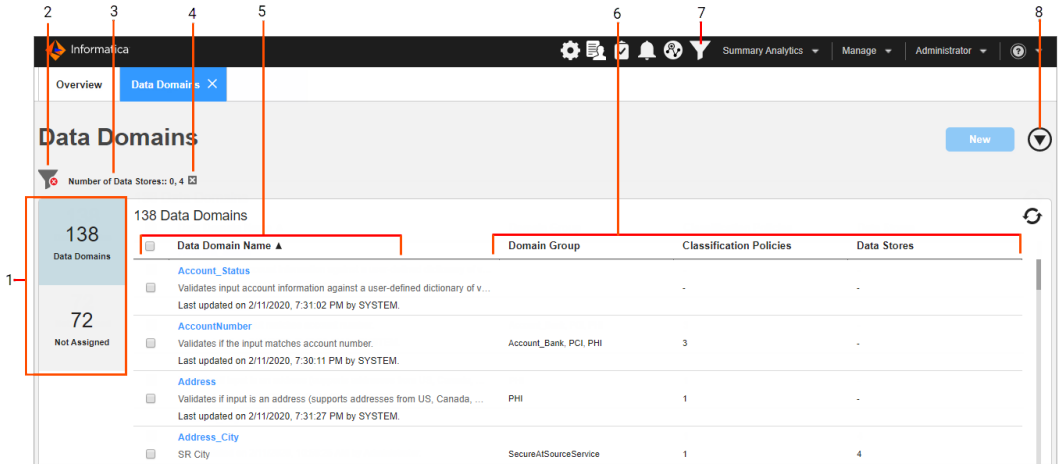
When you select a data domain from the list of data domains, the **Data Domain Details** page appears. The **Data Domain Details** page displays the name, description, match condition, and protection extensions for a data domain. The match condition specifies the rules that determine whether a column in the data store matches the data domain. Protection extensions specify the protection rules that Data Privacy Management applies to the sensitive fields identified in the data store.

To access the **Data Domains** workspace, click **Manage** > **Data Domains**.

Data Domains List Page

The **Data Domains List** page lists the data domains in the Model repository. By default, the list of data domains appears when you access the **Data Domains** workspace.

The following image shows an example of the **Data Domains** workspace with a list of data domains:



1. Data domain status filters
2. Clear Filter icon
3. Filter conditions
4. Clear Filter Condition icon
5. Data domain list
6. Data domain summary
7. Filter
8. Actions menu

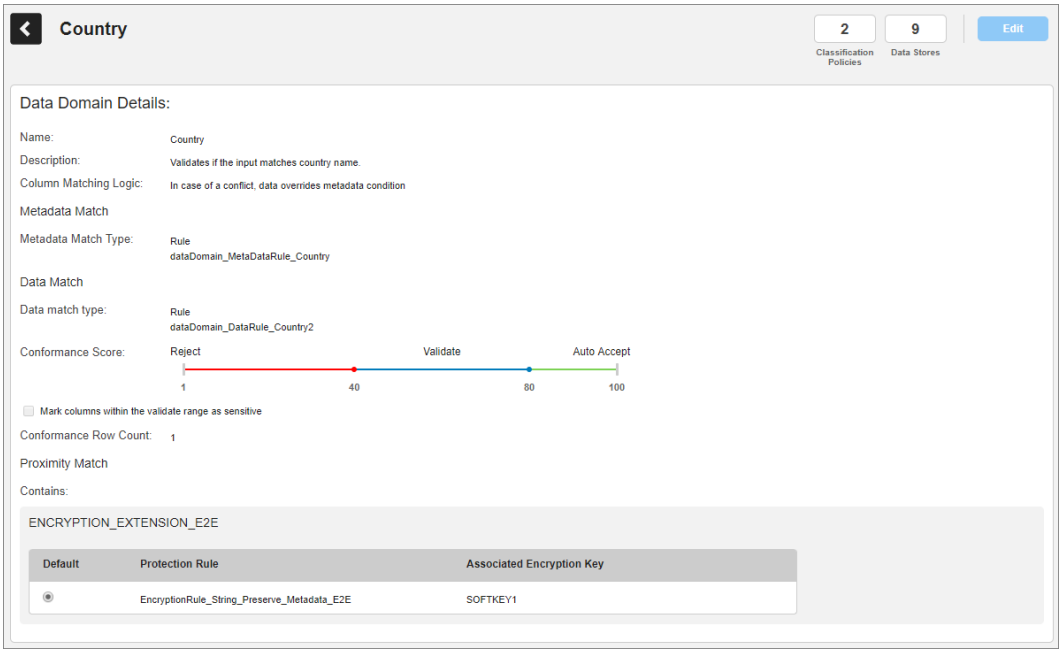
Data Domain Details Page

The **Data Domain Details** page shows the properties and summary information for a data domain.

To view or edit the properties for a data domain, access the **Data Domain Details** page from the **Data Domains** list page in one of the following ways:

- Click a data domain name. To edit the properties, click **Edit** on the **Data Domain Details** page.
- Select the check box next to a data domain and then select **Open** or **Edit** from the **Actions** menu.

The following image shows the **Data Domain Details** page for a Country data domain:



Data Domain Properties

Data domain properties identify the data domain, specify criteria for a column to match the data domain, and optionally specify how Data Privacy Management protects the sensitive fields that the data domain identifies.

The following table describes the data domain properties:

Property	Description
Name	Required. Unique name for the data domain. The name is not case sensitive and cannot exceed 255 characters. The name cannot contain spaces or the following special characters: ~ ! $ % ^ & * ( ) - + = [ ] { } | ; : ' " , . / < > ?
Description	Optional. Long description of the data domain that cannot exceed 255 characters.
Column Matching Logic	Required. Determines how Data Privacy Management handles conflicts that might occur between data match conditions and metadata match conditions defined for the data domain. One of the conditions can override the other, or you can specify that both conditions must match to include a data store column in the data domain.
Metadata Match Condition	Identifies columns to include in the data domain. Matches column name metadata to a regular pattern expression, a reference table in the Model repository, or a metadata rule in the Model repository.
Data Match Condition	Identifies columns to include in the data domain. Matches column values to a regular pattern expression, a reference table in the Model repository, or a metadata rule in the Model repository.

Property	Description
Conformance Score	For data match conditions, specifies the minimum percentage of values in a column that must match the data domain for Data Privacy Management to identify the column as sensitive. Also specifies the lower and upper limit of percentages to reject, validate, or automatically accept a column as sensitive. The Scan job calculates conformance scores when the Scan job runs data profiling. Define the default conformance score ranges on the **Settings** workspace.
Mark columns within the validate range as sensitive	For data match conditions, Scan jobs identify fields as sensitive if they meet the validation criteria in the conformance score setting. If you change the default validation range in a data domain, the Scan job identifies sensitive fields based on the data domain setting.
Conformance Row Count	For data match conditions, specifies the minimum number of values in a column that must match the data domain to identify the column as sensitive. **Important:** The sampling technique for a data store scan must include at least the same number of rows as the conformance row count. If the sampling technique contains fewer rows than the conformance row count, Data Privacy Management will not identify the column as sensitive, and the sensitive field might remain unprotected.
Proximity Match Condition	Identifies columns to include in the data domain. Matches column values to other selected data domains that contain nearly identical columns or fields.
Add Protection Extension	Optional. Applies one or multiple protection extensions to sensitive fields in the data domain for data stores that the protection extension supports. Protection extensions are defined on the **Extensions** workspace and must have an Active status. Options are: - Encryption - Persistent Data Masking - Big Data - Persistent Data Masking - Remote Domain
Protection Rules	Required if you add a protection extension. Defines the default protection rules for sensitive fields in the data domain for data stores that the protection extension supports. Protection rule options are: - Encryption. Applies one or multiple encryption rules that are defined on the **Encryption Rules** workspace. - Persistent Data Masking - Big Data. Specifies the default masking rule to protect the sensitive fields. - Persistent Data Masking - Remote Domain. Specifies the default masking rule to protect the sensitive fields.
Associated Encryption Key	Required if you add an encryption protection extension. Associates a selected encryption key to each encryption rule.

Related Topics:

- "Supported Data Store Types for Protection Extension Plugins" on page 66
- "Encryption Rules Overview" on page 597
- "Settings Workspace Properties" on page 39

Metadata Match Condition

The metadata match condition defines the rule to identify a column by column name. Data Privacy Management uses this rule to discover columns in the Profiling job step of a scan.

To specify a metadata match condition, use one of the following methods to describe a column name:

- Pattern
- Reference table
- Rule

Pattern

To use the pattern method for a metadata match condition, enter a regular expression that describes the pattern of characters in a column name.

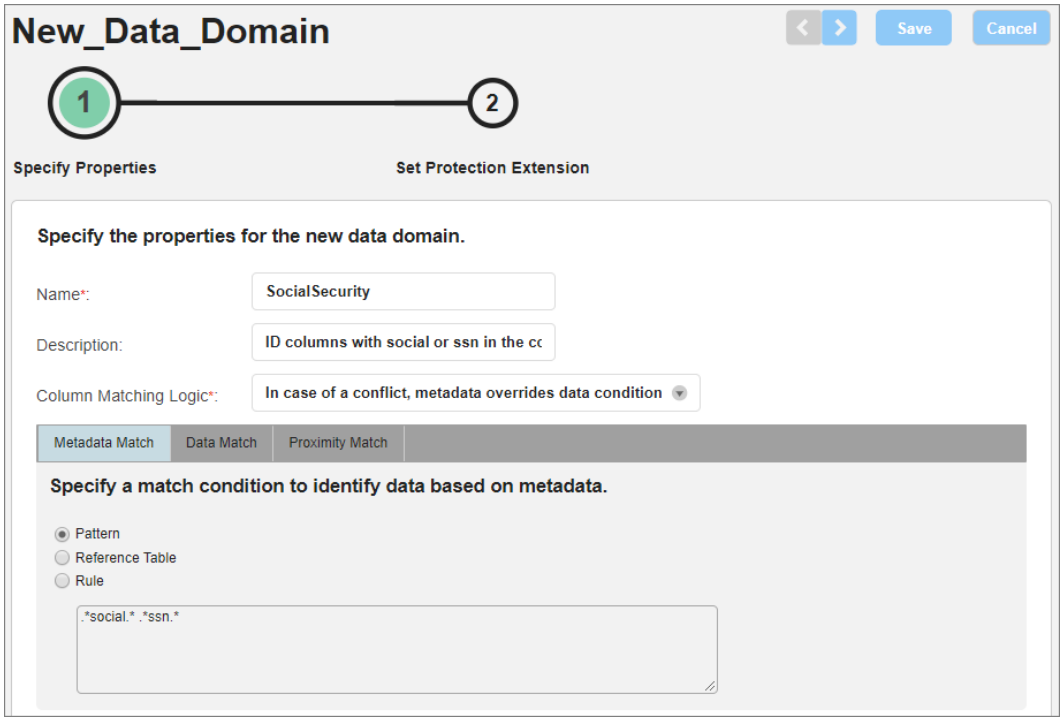
For example, a name for a column with Social Security numbers might include the term `SSN` or `Social`. You can use the following expressions to find columns that have either `SSN` or `Social` in the column name:

```
.*social.*
.*ssn.*
```

A data domain can contain multiple pattern expressions. Data Privacy Management identifies a column as sensitive if the column name matches any one expression in the pattern.

For information about creating regular expressions, see tutorials and documentation for regular expressions on the internet such as http://www.regular-expressions.info/tutorial.html.

The following image shows a metadata match condition that is defined by two regular expressions:



Reference Table

A metadata reference table contains a collection of standard and alternative versions of column names for a specific type of column.

For example, a reference table for credit card column names might contain the following values:

- ccn
- credit_card
- credit_card_number
- credit_card_no.

Data Privacy Management discovers columns with a column name that matches any one of the strings in the reference table.

To use a reference table, select one from the Model repository. You can use Informatica Developer to view the contents of the reference table. For a list and descriptions of the metadata reference tables that are available with the Data Privacy Management installation, see the *System-Defined Data Domains* appendix in the *Informatica Data Privacy Management Administrator Guide*.

To create a reference table, use Informatica Analyst or Informatica Developer. For information about creating reference tables, see the *Informatica Data Quality Getting Started Guide*.

Rule

A metadata rule defines the logic to discover a column by column name.

For example, to discover columns with birth dates, the metadata rule might specify the following expressions that describe the column names `DOB`, `Date of Bir*`, and `Birth Da*`:

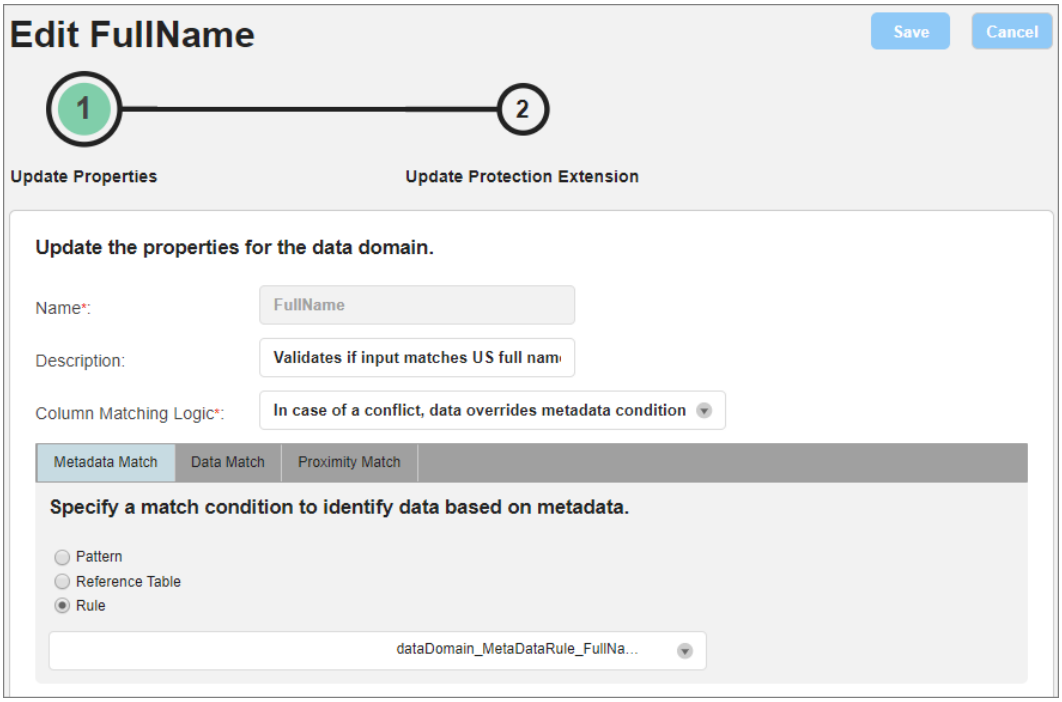
- `^[dD][oO][bB]$`
- `^[dD][aA][tT][eE].*[oO][fF].*[bB][iI][rR].*$`
- `^[bB][iI][rT][tT][hH].*[dD][aA].*`

Data Privacy Management discovers columns with a column name that matches any one of the expressions in the rule.

To use a rule, select one from the Model repository. You can use Informatica Developer to view the expressions in a rule. For a list and description of the metadata rules that are available with the Data Privacy Management installation, see the *System-Defined Data Domains* appendix in the *Informatica Data Privacy Management Administrator Guide*.

To create a rule use Informatica Analyst or Informatica Developer. For information about creating rules, see the *Informatica Data Quality Getting Started Guide*.

The following image shows a data domain with a metadata match condition that is defined by a rule:



Data Match Condition

The data match condition defines the rule to identify a column by column data. Data Privacy Management uses this rule to discover columns in the Profiling job step of a scan.

To specify a data match condition, use one of the following methods to describe column values:

- Pattern
- Reference Table
- Rule

Pattern

To use the pattern method for a data match condition, enter a regular expression that describes the values in a column.

For example, you want to discover columns that contain Social Security numbers. Social Security numbers have a specific pattern such as the `999-99-9999` pattern used for U.S. Social Security numbers.

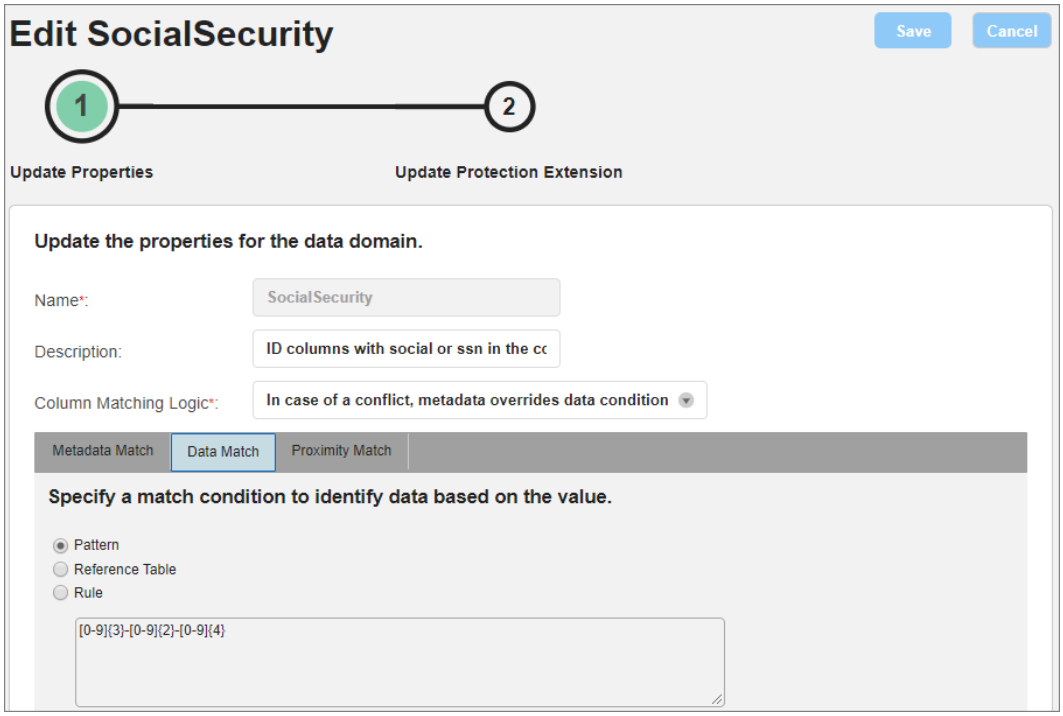
You can create the following expression to describe the data pattern for U.S. Social Security numbers:

```
[0-9]{3}-[0-9]{2}-[0-9]{4}
```

A data domain can contain multiple data pattern expressions. Data Privacy Management discovers columns that match any expression in the pattern.

For information about creating regular expressions, see tutorials and documentation for regular expressions on the internet such as http://www.regular-expressions.info/tutorial.html.

The following image shows a data match condition that is defined by a data pattern:

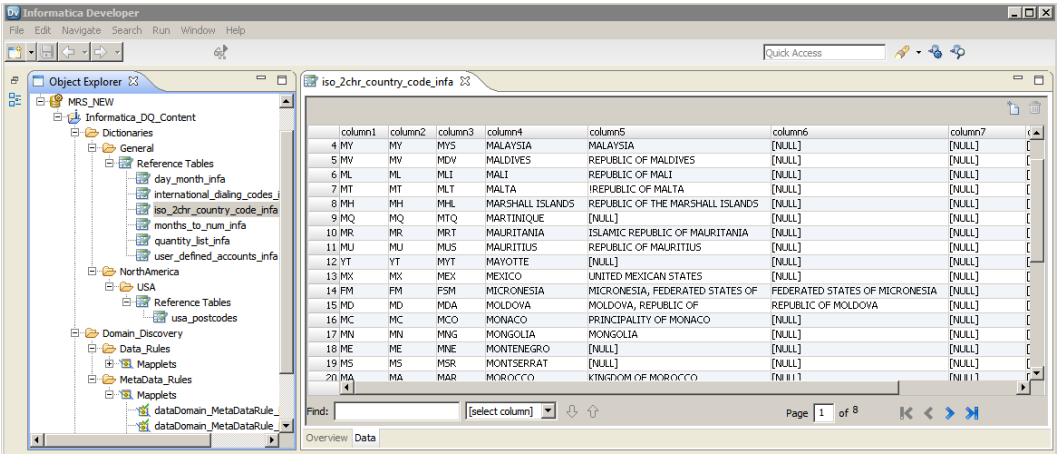


Reference Table

A data reference table contains a collection of the standard and alternative versions of column data for a specific type of column.

For example, you want to discover columns that contain country names in your data store. You can use a reference table that lists the ISO country code and the different versions of names for each country.

The following image shows a data reference table in the Model repository for country names:



Data Privacy Management discovers columns with values that match at least one of the values in the reference table.

To use a reference table, select one from the Model repository. You can use Informatica Developer to view the contents of the reference table. For a list and descriptions of the data reference tables that are available

with the Data Privacy Management installation, see the *System-Defined Data Domains* appendix in the *Informatica Data Privacy Management Administrator Guide*.

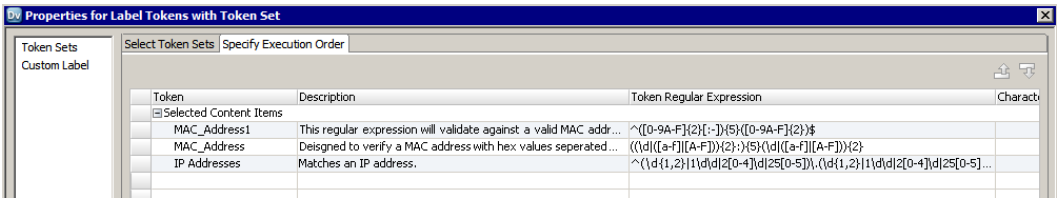
To create a reference table, use Informatica Analyst or Informatica Developer. For information about creating reference tables, see the *Informatica Data Quality Getting Started Guide*.

Rule

A data rule defines the logic to discover a column by column values.

For example, you want to discover columns with IP addresses. You can use a rule from the Model repository that defines the patterns for IP addresses.

The following image shows the expressions in a data rule for IP addresses:



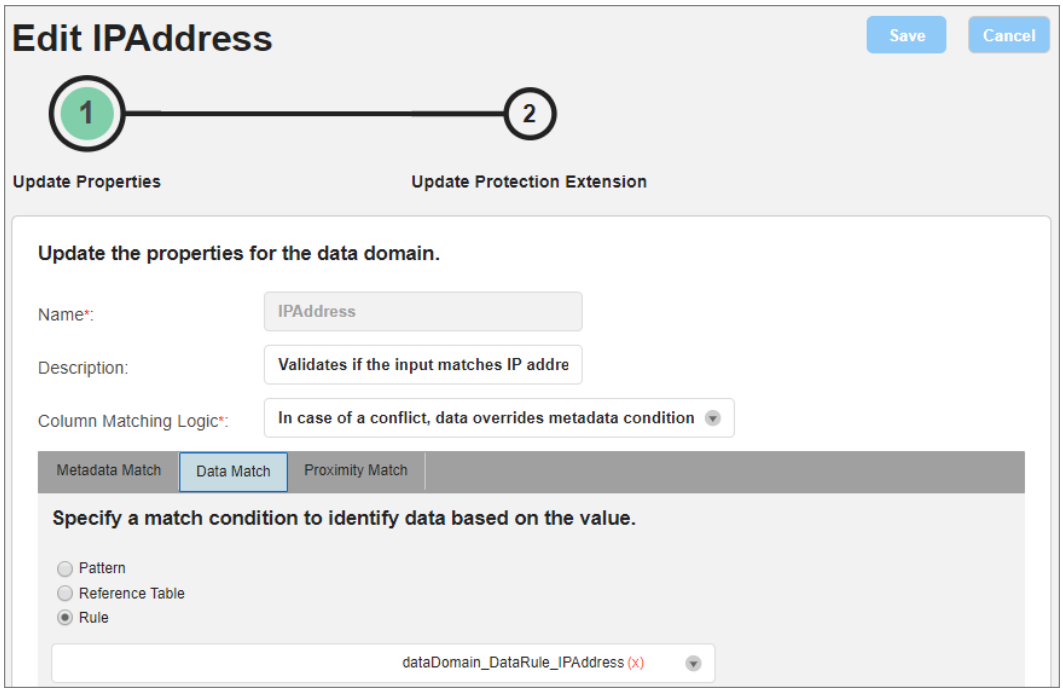
Data Privacy Management discovers columns with values that match at least one of the expressions in the data rule.

To use a rule, select one from the Model repository. You can use Informatica Developer to view the expressions in a rule. For a list and description of the data rules that are available with the Data Privacy Management installation, see the *System-Defined Data Domains* appendix in the *Informatica 2*

Administrator Guide.

You can also create a rule using Informatica Analyst or Informatica Developer. For information about creating rules, see the *Informatica Data Quality Getting Started Guide*.

The following image shows a data domain with a data match condition that is defined by a rule:



Conformance Score

The conformance score specifies the criteria that identifies a field as sensitive in Application, Big Data, Cloud, Data Integration, and Database Management data store types.

During the Profiling job step of a Scan job, Data Privacy Management calculates the percentage of field values that match each data domain. You can exclude null values from the calculation when you create the scan.

Note: Data Privacy Management does not calculate conformance scores for File Management data store types.

Depending on the percentage of field values that match the data domain, Data Privacy Management performs one of the following actions for each field in the data store:

- Rejects a field as sensitive.
- Reports a field in the validation range. The data domain configuration determines if the field is identified as sensitive or reported in a scan report.
- Identifies a field as sensitive.

To set the default conformance ranges, specify the following settings on the **Settings** workspace. You can customize the settings when you create or edit a data domain.

Setting	Description
Maximum Reject Score	The maximum data domain match percentage at or below which Data Privacy Management rejects a field as sensitive. For example, to reject fields as sensitive when 40% or fewer of the values match the data domain, set the value to 40%.
Minimum Accept Score	The minimum data domain match percentage at or above which Data Privacy Management accepts a field as sensitive. For example, to accept fields as sensitive when 80% or more of the values match the data domain, set the value to 80%.
Mark columns within the validate range as sensitive	If selected, Data Privacy Management identifies the fields in the validation range as sensitive. If cleared, Data Privacy Management identifies the fields in the validation range as not sensitive and includes the fields in a scan report. The data store owner reviews the scan report and determines if the fields are sensitive. Default is cleared.

If a field returns a data match percentage between the Maximum Reject Score and the Minimum Accept Score, Data Privacy Management flags the field for validation.

Example

You want to determine the sensitivity status of a field based on the following criteria:

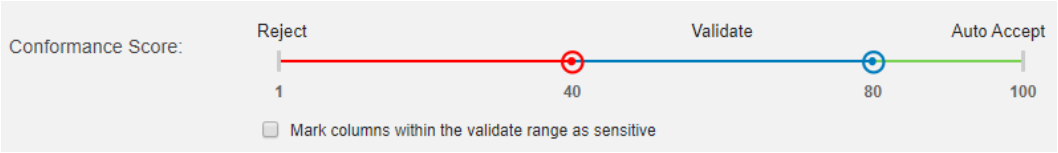
- If at least 80% of the field values match the data domain, identify the field as sensitive.
- If 41-79% of the field values match the data domain, flag the field for the data owner to validate the data and determine the sensitivity status of the field.
- If 0-40% of the field values match the data domain, reject the field as sensitive.

On the **Data Match** tab of the **New_Data_Domain** page, use the slider buttons to set the following values:

- Maximum Reject Score = 40%

- Minimum Accept Score = 80%

The following image shows these settings:



Related Topics:

- "Changing the Default Conformance Ranges" on page 45
- "Downloading Scan Job Reports" on page 309

Sensitivity Status for Fields in the Validation Range

You can configure how Data Privacy Management identifies the sensitivity status for fields in the validation range.

When you create a data domain and specify a data match condition, the **Mark columns within the validation range as sensitive** check box is cleared by default. If you keep the default setting, the data store scan results include only the fields in the auto-accept range in the sensitive field count. If you select the check box, the data store scan results include fields in the auto-accept and validation ranges.

To validate the sensitivity status of the fields in the scan results, view the list of fields and corresponding conformance scores on the **Sensitive Fields** page accessed from the **Security Dashboard**. You can update the sensitivity status in one of the following ways:

- Manually. On the **Sensitive Fields** page, select a row to view the **Properties** pane for the sensitive field. Edit the **Sensitive** property.
- In bulk. Export the list of sensitive fields as a CSV file and change the sensitivity status for the fields.

After you update the sensitivity status, import the file on the **Data Stores** workspace.

Data Domains Management

Create, view, and manage data domains to identify the semantics of sensitive columns based on the column name or column data. A data domain is a repository object that specifies the columns that you want to find and secure in a data store. When you create or import a data domain, Data Privacy Management creates a repository object in Enterprise Data Catalog (EDC). When you edit a data domain, Data Privacy Management updates the data domain in EDC.

To find and secure sensitive data, you scan a data store with a classification policy that contains data domains. Create a data domain for each type of sensitive data that you want to discover and, optionally, include protection extensions to protect the sensitive data. Create the data domain before you create the classification policy.

You can create data domains in one of the following ways:

- Create data domains manually.
- Import data domains.
- Copy an existing data domain.

After you create data domains, you can view, edit, export, or delete data domains. You can also remove the link between a data domain and the data stores associated with it.

Creating a Data Domain

When you create a data domain, you enter a name and define the rule to match a field or file in the data store to the data domain. Optionally, include one or more protection extensions to protect sensitive data in data stores that match the data domain.

1. Click **Manage** > **Data Domains**.

   The **Data Domains** workspace appears.
2. Click **New**.

   The **New_Data_Domain** page appears.
3. Enter the name for the data domain.

   The name cannot contain spaces or the following special characters: `~!$%^&*()-+=[]{}\|;:'",./<>?`
4. Optionally, enter a description.
5. In the **Column Matching Logic** field, select one of the following options:
   - **In case of a conflict, metadata overrides data condition**.
   - **In case of a conflict, data overrides metadata condition**.
   - **Both metadata and data conditions match**.
   - **Either the metadata or data condition matches**.
6. On the **Metadata Match** tab, select a metadata match condition.
7. On the **Data Match** tab, perform the following steps:
   a. Select a data match condition.
   b. Optionally, specify the conformance score ranges that specify how Data Privacy Management identifies a column as sensitive, not sensitive, or requiring validation. Move the sliders along the conformance score bar to set the ranges.

      Default conformance score ranges are specified on the **Settings** workspace.
   c. If you define a data match condition, select or clear the**Mark columns within the validate range as sensitive** check box to specify whether Data Privacy Management identifies fields in the validation range as sensitive:
      - If you select the check box, Data Privacy Management identifies fields in the validation range as sensitive. The scan results include the fields that are in the auto-accept and validation ranges when calculating the sensitive field count.
      - If you clear the check box, Data Privacy Management identifies fields in the validation range as not sensitive and includes the fields in a scan report. The scan results include only the fields that are in the auto-accept range when calculating the sensitive field count.

      The default setting is configured on the **Settings** workspace. If you change the default setting, the Scan job uses the data domain configuration to determine if the fields are sensitive.
8. On the **Proximity Match** tab, select data domains to include in a proximity match condition.
9. Click the **Next** icon.

   The **Set Protection Extension** pane appears.
10. Optionally, click **Add Protection Extensions** and select a protection extension name.

    The protection rule options for the selected extension appear.

11. Select the protection rules for the protection extension.
12. To add additional protection extensions to the data domain, repeat steps 10 and 11.

    Data Privacy Management applies protection extensions to the sensitive fields in each data store, based on the protection task that is configured for the data store.
13. Click **Save**.

Exporting Data Domains

You can export a list of data domains to a CSV file. The export file includes a list of all data domains and the corresponding data domain properties. If you filtered the data domain list, the export file contains the data domains that match the active filter conditions.

The export file contains the same format as the file required to import data domains. You can use the export file to add data domains or update data domain properties in bulk. After you save the updates to the export file, import the file. When you import the file, the Import job adds and updates the data domains in the Model repository.

1. Click **Manage** > **Data Domains**.

   The **Data Domains** workspace appears.
2. From the **Actions** menu, select **Export**.

   A dialog box appears with the exported file.
   The export file is named `DataDomains.csv`. Data Privacy Management saves the file to the default download directory on your machine. If you export the file again, a new file downloads. The file name includes an incremental number such as `DataDomains (1).csv`.
3. Open the file.
4. Optionally, add data domains or update existing data domain properties in the file.
5. Save the file.

After you export a list of the data domains and update the file, import the file in to the Model repository.

Importing Data Domains

To create or update a data domain, you can import data domain information from a CSV file. The file must be in the format that the Import job requires. When you import the file, the import job runs immediately.

1. Click **Manage** > **Data Domains**.

   The **Data Domains** workspace appears.
2. From the **Actions** menu, select **Import**.

   The **Import Data Domains** page appears.
3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the Import job handles updates for existing data domains. If you enable the option, the Import job updates the data domain properties with the values in the CSV file. If you disable the option, the Import job skips the row.
4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the Import job finishes, you can go to the **Data Domains** workspace to see the data domains that you imported. Data Privacy Management saves the data domain in the Model repository.

Related Topics:

- "Import Job" on page 299

CSV File Format to Import Data Domains

Create a CSV file to create, update, or delete information about data domains in the Data Privacy Management repository. The content of the file must be in the order and format that the Import job requires. The CSV file must contain all of the column headings. However, not all columns require values.

The following table describes the columns, column headings, and the values in the CSV file:

Column Order	Column Heading	Description
1	DomainName	Required. Enter the name of the data domain. You can use up to 100 characters. Do not use backslashes. To update a data domain, ensure that the name of the data domain in the DomainName column matches the name of the data domain in the Model repository.
2	Description	Optional. Enter unique details of the data domain. The description cannot exceed 254 characters.
3	DataMatchType	Required. Specify one of the following values: - To indicate the metadata match condition, enter: `METADATA_MATCH` - To indicate the data match condition, enter: `DATA_MATCH` - To indicate both the metadata and data match conditions, enter: `BOTH`
4	MetaDataMatchOption	Required if you entered `METADATA_MATCH` or `BOTH` for the `DataMatchType` property. Enter one of the following metadata match conditions: - `PATTERN` - `REFERENCE_TABLE` - `RULE`
5	MetaDataRule	Required if you entered `RULE` for the `MetaDataMatchOption` property. Enter the name of a metadata rule that is located in the Model repository.
6	MetaDataPattern	Required if you entered `PATTERN` for the `MetaDataMatchOption` property. Enter an expression to describe the column name. For example, to discover a column with account numbers, you can enter `.*acc.*`. Data Privacy Management discovers columns with `acc` in the column name.

Column Order	Column Heading	Description
7	MetaDataReferenceTable	Required if you entered `REFERENCE_TABLE` for the `MetaDataMatchOption` property. Enter the name of a reference table that is located in the Model repository.
8	DataMatchOption	Required if you entered `DATA_MATCH` or `BOTH` for the `DataMatchType` property. Enter one of the following metadata match conditions: - `PATTERN` - `REFERENCE_TABLE` - `RULE`
9	DataRule	Required if you entered `RULE` for the `DataMatchOption` property. Enter the name of a data rule that is located in the Model repository.
10	DataPattern	Required if you entered `PATTERN` for the `DataMatchOption` property. Enter an expression to describe the column data. For example, to discover a column with U.S. Social Security numbers, enter `999-99-9999`. Data Privacy Management discovers columns with values that match this pattern.
11	DataReferenceTable	Required if you entered `REFERENCE_TABLE` for the `DataMatchOption` property. Enter the name of a reference table that is located in the Model repository.
12	MaxRejectScore	Optional. Specifies the percent of matched column values at or below which Data Privacy Management must reject a column as sensitive. Enter a number from 0 to 100. Default is 40.
13	MinAcceptanceScore	Optional. Specifies the percent of matched column values at or above which Data Privacy Management must reject a column as sensitive. Enter a number between the value you set for the `MaxRejectScore` to 100. Default is 80.
14	MinConformanceRowcount	Specifies the minimum number of values in a column that must match the data domain to identify the column as sensitive. Default is 1.
15	DomainTrumpRule	Specifies how Data Privacy Management handles conflicts that might occur between data match conditions and metadata match conditions defined for the data domain. Enter one of the following rules: - `DATA_OVERRIDES_METADATA` - `METADATA_OVERRIDES_DATA` - `METADATA_AND_DATA` - `METADATA_OR_DATA`

Column Order	Column Heading	Description
16	Tool Association	Optional.
17	Action	Optional. Instructs the import job to create, update, or delete a data domain in the model repository. Enter one of the following values: - U. Creates or updates a data domain in the Data Privacy Management repository. The action that the job performs depends on if the data domain exists in the repository and on the **Replace Duplicates with Items Imported** option. You configure the **Replace Duplicates with Items Imported** option when you import the CSV file. If the data domain does not exist, the Import job creates the data domain. If the data domain exists and the **Replace Duplicates with Items Imported** option is enabled, the Import job updates the data domain. **Warning:** If a column in the CSV file contains a null value, the Import job replaces the value in the repository with a null value. If the data domain exists and the **Replace Duplicates with Items Imported** option is not enabled, the Import job skips the row. - D. Deletes a data domain from the repository. If the data domain does not exist, the Import job rejects the row. If blank, default is U. Creates or updates a data domain in the Data Privacy Management repository.

Rules and Guidelines to Import Data Domains

Review the rules and guidelines to understand how to manage the CSV file and how the Import job processes data from the file.

Consider the following rules and guidelines when you import data domains from a CSV file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines a data domain.
- If the CSV file contains duplicate entries, the Import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

```
SELECT FROM
SELECT UNION
DELETE FROM
UPDATE SET
<SCRIPT
</SCRIPT>
ALERT(
"/>"
```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.

- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Editing a Data Domain

You can edit the properties of a data domain.

1. Click **Manage** > **Data Domains**.

   The **Data Domains** workspace appears. You can use the navigation filters to narrow the list of data domains.
2. Click the name of the data domain that you want to edit.
3. Click **Edit**.
4. Edit the data domain properties that you want to change.

   The data domain name cannot contain spaces or the following special characters: `~!$%^&*()-+=[]{}\|;:'",./<>?`
5. Click **Save**.

Copying Data Domains

You can copy a data domain to create a data domain with similar properties. After you copy the data domain, edit the data domain properties.

1. Click **Manage** > **Data Domains**.
2. Select the checkbox next to the data domain that you want to copy.
3. From the **Actions** menu, select **Duplicate**.

   The **Data Domain Details** page appears.
4. Edit the data domain properties.

   The data domain name cannot contain spaces or the following special characters: `~!$%^&*()-+=[]{}\|;:'",./<>?`
5. Click **Save**.

Data Privacy Management creates a data domain with the following naming convention:

```
Copy of <data domain name>
```

Delinking Associations Between Data Domains and Data Stores

You can remove the links between a data domain and any data store associated with the data domain.

You must have the Add or Modify Data Domain privilege to disassociate data domains from data stores.

Verify that the data domains are not associated with a scan that is in progress, a protection task, or user access data.

1. Click **Manage** > **Data Domains**.

   The **Data Domains** workspace appears.
2. Select one or more data domains that you want to delink from all associated data stores.

3. From the **Actions** menu, select **Delink Associations**.

   A confirmation message appears asking you to confirm the action.
4. Confirm the action.

   A success message appears.

Synchronizing Data Domains

You can synchronize data domain details between the Data Privacy Management repository and the Enterprise Unified Metadata platform.

1. Click **Manage** > **Data Domains**.

   The **Data Domains** workspace appears.
2. Select one, more, or all data domains that you want to synchronize.
3. From the **Actions** menu, select **Synchronize Data Domains**.

   A message informs you that Data Privacy Management has started the Synchronize Data Domains system job.
4. To check the status of the system job, go to the **Jobs** workspace and click the **System Jobs** filter.

Deleting Data Domains

You can delete data domains from the Model repository. You cannot delete a data domain if a past, present, or future scan includes the data domain.

Note: When you delete data domains, you remove the rules associated with the data domain from the **Active** and **Suggestions** tabs on the **Always Sensitive /Never Sensitive** workspace.

1. Click **Manage** > **Data Domains**.

   The **Data Domains** workspace appears.
2. Select the check box next to the data domain that you want to delete.
3. From the **Actions** menu, select **Delete**.

   You receive a message to confirm if you want to delete the data domain.
4. Click **Yes**.

CHAPTER 10

Classification Policies

This chapter includes the following topics:

- Classification Policies Overview, 221
- Classification Policies Workspace, 221
- Classification Policy Properties, 223
- Data Domains in a Classification Policy, 224
- Data Domain Match Condition, 225
- Classification Policy Usage in a Scan Job, 227
- Classification Policy Management, 228

Classification Policies Overview

A classification policy defines the rules that identify sensitive data and classify a data store to meet data security standards and comply with data privacy regulations.

You can use a system-defined classification policy or create, copy, or import a classification policy. When you create a classification policy, you assign data domains to the classification policy and specify a data domain match condition, the cost for each record with sensitive data, and a data sensitivity level. You assign one or more classification policies and data stores to include in a scan.

A data domain match condition describes the criteria that classifies the data store as a match for the classification policy. If a data store matches a classification policy, Data Privacy Management assigns the sensitivity level to the data store.

Scan jobs analyze data stores to discover sensitive and personal data based on the classification policies included in the scan.

Classification Policies Workspace

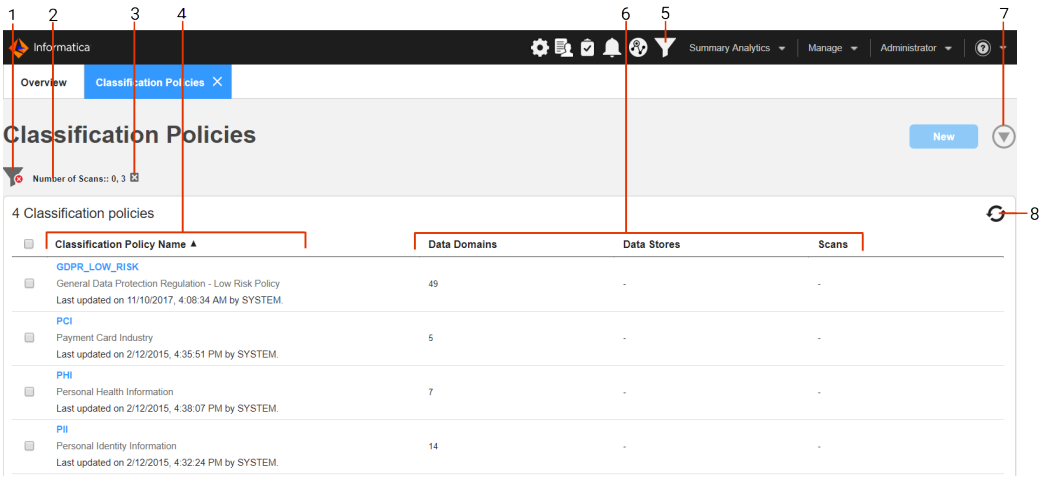
Use the **Classification Policies** workspace to create, view, and manage classification policies.

To access the **Classification Policies** workspace, click **Manage** > **Classification Policies**.

Classification Policies List Page

The **Classification Policies** list page lists the classification policies in the Data Privacy Management repository. By default, the list of classification policies appears when you access the **Classification Policies** workspace.

The following image shows an example of the **Classification Policies** workspace with the classification policies list page:



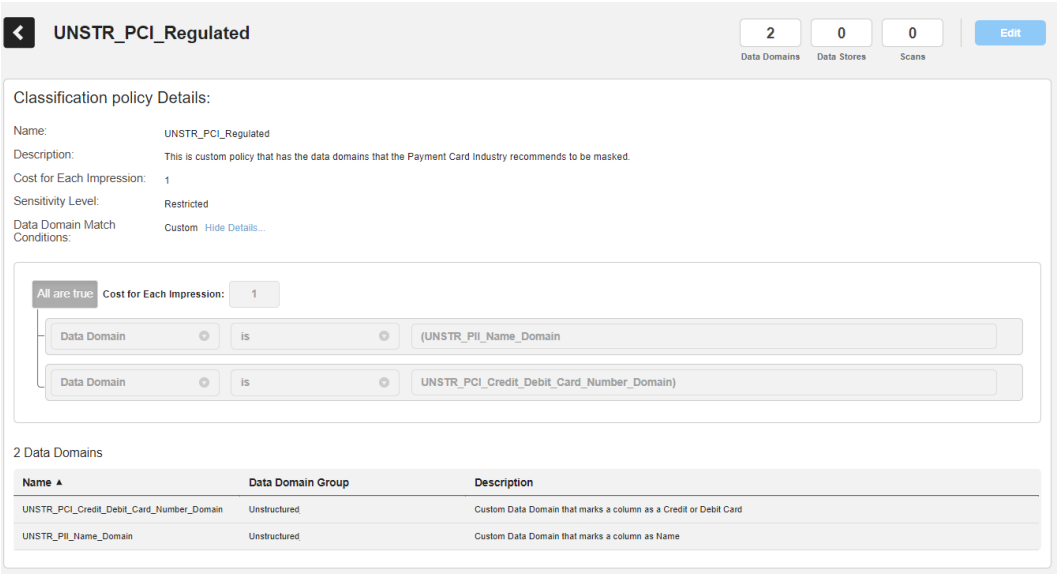
1. Clear Filter icon
2. Filter condition
3. Clear Filter Condition icon
4. Classification policies list
5. Filter
6. Classification policies summary
7. Actions menu
8. Refresh icon

Classification Policy Details Page

The **Classification Policy Details** page shows the classification policy properties, the data domains that the classification policy includes, and the data domain match conditions.

Access the classification policy details page to view or edit a classification policy. You access the classification policy details page when you click a classification policy name from the classification policies list page or when you view a classification policy from the **Actions** menu.

The following image shows a **Classification Policy Details** page:



Classification Policy Properties

Classification policy properties identify the classification policy, specify the cost of sensitive data, and specify the data sensitivity level.

The following table describes the classification policy properties:

Property	Description
Name	Name of the classification policy. The name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 255 characters, contain a space, or contain the following special characters: ~ ! $ % ^ & * ( ) +
Description	Optional. The description cannot exceed 255 characters. Use this property to specify unique details to differentiate between similarly named classification policies.

Property	Description
Cost for Each Impression	The monetary cost that your organization would incur for each occurrence of sensitive data when exposed to an unauthorized user. Data Privacy Management provides a default value. You can set the cost to a maximum value of 999. To estimate the cost for each occurrence of sensitive data, consider the direct and indirect costs that your organization might incur if sensitive data is exposed, lost, or unavailable. For example, direct costs might include the following expenses: - Hiring experts and purchasing applications to help resolve a data breach. - Providing customers with hotline support. - Providing customers with free credit monitoring subscriptions and discounts for future products and services. Indirect costs might include the following expenses: - Investigation and communication. - Loss of critical data for use in primary applications that serve employees, customers, or partners. - Potential loss of customers. The scan results show the total cost of sensitive data in the data stores evaluated in the scan.
Sensitivity Level	Degrees of data sensitivity based on industry or organization data security standards. You can select from up to five data sensitivity levels. For example, you can specify the following data sensitivity levels: - **Public.** Indicates that data can be shared with anyone. For instance, data that you share on the company website. - **Internal.** Indicates that data can be shared with employees only. For instance, a marketing and sales strategy. - **Confidential.** Indicates that data can be shared with a select group of users. For instance, details for a patent. - **Restricted.** Indicates that data can be shared with key personnel. For instance, personal information.

Related Topics:

- "Residual Risk Indicator" on page 518
- "Changing the Default Risk Cost" on page 45

Data Domains in a Classification Policy

Select the data domains that you want to include in the classification policy.

A data domain identifies the semantics of sensitive data and specifies the rules that Data Privacy Management must use to match a field or file name or value to the data store.

You can include predefined data domains in classification policies, or you can create and import data domains. For example, to scan a data store for PII data, you can include data domains for Social Security number, first name, last name, and birth date in the classification policy.

When you scan a data store, Data Privacy Management discovers the field or file names or values that you specified in the data domains included in the policy.

Data Domain Match Condition

The data domain match condition specifies the rule that Data Privacy Management uses to classify sensitive data during the Evaluate Classification Policies job step of the scan. If the sensitive fields or files in the data store meet the data domain match condition, Data Privacy Management designates the data store as a match for the classification policy.

Specify one of the following data domain match conditions:

- **Match Any**. The data store must have one or more fields or files that match at least one data domain in the classification policy.
- **Match All**. The data store must have one or more fields or files that match all the data domains in the classification policy.
- **Custom**. The data store must have fields or files that match a data domain or file tag match condition that you create.

Match Any

When you select the **Match Any** data domain match condition, the data store must contain a field or file that matches at least one data domain in the classification policy for Data Privacy Management to designate the data store as a match for the classification policy.

When to Use the Match Any Condition

Consider using the Match Any condition when you have multiple data domains in the classification policy. If you have more than one data domain in the classification policy, only one data domain match is required to classify the data store.

For example, you want to evaluate a data store for PCI data. You create a classification policy with the credit card number and magnetic stripe data domains. You specify the Match Any condition and scan the data store with the classification policy. Data Privacy Management classifies the data store as a match for PCI if the data store contains a field for either credit card or magnetic stripe data.

When you create a classification policy with multiple data domains, Match Any is the default data domain match condition.

When Not to Use the Match Any Condition

When a classification policy contains a large number of data domains, the Match Any condition might cause a false positive result.

For example, you want to evaluate a data store for PII data. You create a classification policy that includes many data domains such as age, birth date, birth place, drivers license number, first name, last name, gender, IP address, and Social Security number. You scan the data store with the classification policy. Data Privacy Management classifies the data store as PII if the data store matches one of the data domains.

For instance, Data Privacy Management would classify the data store as PII even if the data store matched only the first name data domain. The scan results would not be meaningful or accurate because a first name by itself is not a unique identifier.

Match All

When you select the **Match All** data domain match condition, a data store must contain fields or files that match all data domains in the classification policy for Data Privacy Management to designate the data store as a match.

When to Use the Match All Condition

Consider using the Match All condition when you have a single data domain or multiple related data domains in a classification policy and you want the fields to match all data domains.

For example, you want to evaluate a data store for PHI data. You create a classification policy with the account number, first name, and last name data domains. You specify the Match All condition and scan the data store with the classification policy. Data Privacy Management classifies the data store as a match for PHI if the data store contains fields for account number, first name, and last name.

When you create a classification policy with a single data domain, Match All is the default data domain match condition.

When Not to Use the Match All Condition

When a classification policy contains a large number of data domains, the Match All condition might be too stringent.

For example, you want to evaluate a data store for PII data. You create a classification policy with seven data domains that include first name, last name, and Social Security number. You scan the data store with the classification policy. The scan classifies the data store as PII only if the data store contains fields that match all seven data domains. However, if a data store matches only the first name, last name, and Social Security number data domains, Data Privacy Management does not classify the data store as PII.

Custom

Create Custom match conditions if the classification policy contains many data domains or you want to include Google Drive or Microsoft SharePoint file tags.

The condition pallette enables you to create a detailed condition set. You can create a suite of conditions, group and nest the conditions, and use the **All are true** and **Any are true** operators to relate the conditions and groups of conditions.

For every condition group, you can use the default **Cost for Each Impression** value or you can specify a different cost.

Data Domain Match

You can create a simple condition that specifies data domains that a data store must match to be identified as sensitive for the data domains. Or, you can specify the number of values that must match one or more data domains for a data store to be identified as sensitive for that data domain.

File Tag Match

You can also create custom match conditions that specify file tags that must match for Google Drive or Microsoft SharePoint data stores. If you use a file tag in a condition group, the domain match condition must also exist in the condition group. Create a data domain match condition with the **Is** or **IsNumberOfValues** operator and select a data domain from the list. Then, add file tag conditions for the data domain within the condition group.

You can enter one file tag for each condition that uses the following operators:

- **Is**
- **Is not**

- **Contains**: single value
- **Does not contain**: single value
- **Any of**: multiple values
- **None of**: multiple values

If you specify multiple file tags with the **Any of** or **None of** operators, separate each value with a comma (,).

If you include more than one condition group with file tag match conditions, and the groups have different costs for each impression, the scan job uses the highest cost per impression value to calculate the risk cost for the data stores that match the classification policy.

If a file matches a custom match condition that specifies file tags, Data Privacy Management increases the policy impression value for the file by 1.

Example: Custom Data Domain Match Condition

For example, you want to evaluate a data store for PCI data. You create a classification policy that includes the `LastName`, `CreditCardNumber`, and `SSN` data domains. According to your company's standard, a data store is PCI if the data store contains a minimum of 30 values that match the `CreditCardNumber` data domain. If the data store matches all three data domains, the cost of each occurrence of sensitive data triples.

When you scan the data store, Data Privacy Management classifies the data store as PCI if the sensitive fields or files match either condition group.

Rules and Guidelines for Custom Match Conditions

Consider the following rules and guidelines for custom match conditions:

- You can export and import classification policies that use custom file tag match conditions.
- If you create a scan that includes a classification policy with custom file tag match conditions, and the policy matches data stores other than Google Drive and Microsoft SharePoint, the scan job returns unexpected values for the **Policy Impressions** property on the **Sensitive Fields** page.
- Data Privacy Management calculates the **Confidence** level and displays the value on the **Sensitive Fields** page for files that match the classification policy condition.
  - For unstructured data stores that do not scan with a remote agent, the **Sensitive Fields** page does not list these columns.

Classification Policy Usage in a Scan Job

Data Privacy Management uses the information in a classification policy to discover and classify sensitive data in a data store.

Sensitive Data Discovery

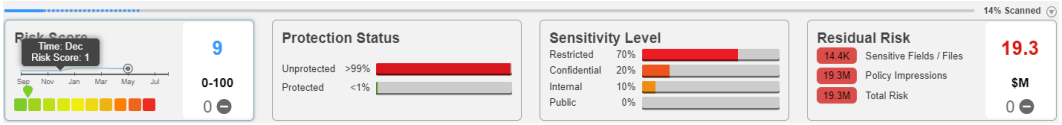
During the sensitive data discovery process, Data Privacy Management determines if information in a data store matches a data domain match condition in the classification policy. A data domain defines the metadata, data, or proximity match condition for a specific field or file. If a field or file in the data store matches the match condition, Data Privacy Management identifies the field or file as sensitive.

If the data store contains sensitive information, Data Privacy Management performs the following tasks:

- Calculates the number of fields or files and impressions with sensitive data. Updates the results on the **Security Dashboard** and on the **Data Store Details** page.

- Calculates the risk score for the data store. Updates the results on the **Security Dashboard** and on the **Data Store Details** page.
- Calculates the residual risk cost of unprotected sensitive data. Updates the results on the **Security Dashboard** and on the **Data Store Details** page.

The following image shows some of the dashboard indicators that Data Privacy Management updates after the sensitive data discovery process:

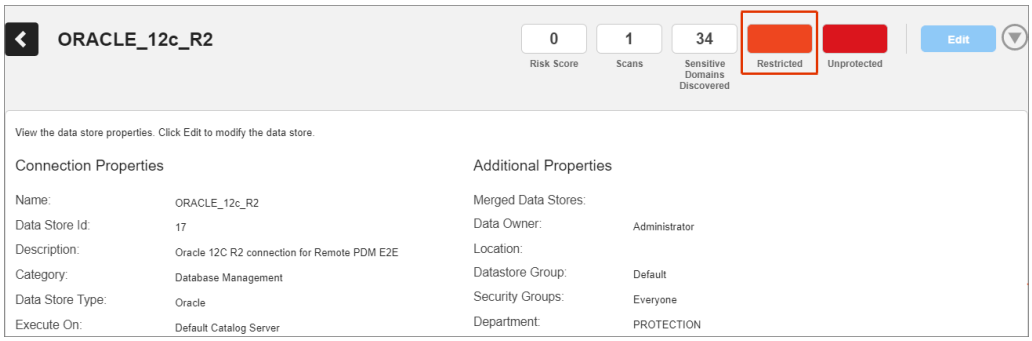


Sensitive Data Classification

During the sensitive data classification process, Data Privacy Management evaluates the data store to determine if it meets the data domain match condition defined in the classification policy.

If there is a match, Data Privacy Management performs the following tasks:

- Classifies the data store as a match for the classification policy. If the data store matches more than one classification policy, Data Privacy Management assigns the most severe data-sensitivity level from the matched classification policies to the data store. Data Privacy Management displays the sensitivity level on the **Data Store Details** page.



- Calculates the percentage of scanned data stores for each data-sensitivity level. Data Privacy Management updates the results on the **Security Dashboard**.

Classification Policy Management

You can create, import, export, view, edit, copy, and delete classification policies on the **Classification Policies** workspace. The Data Privacy Management repository stores classification policies. When you create, manage, or use classification policies in a scan, the Data Privacy Management Service accesses and updates the information in the repository.

To identify and classify sensitive data, you scan a data store based on the classification policies included in the scan. Create, copy, or import a classification policy for each category of sensitive data that you want to identify in the data store.

Creating a Classification Policy

When you create a classification policy, you configure the policy properties, include data domains, and specify the data domain match condition.

1. Click **Manage** > **Classification Policies**.

   The **Classification Policies** workspace appears.
2. Click **New**.

   The **New_Classification_Policy** page appears.
3. On the **Specify Properties and Assign Data Domains** panel, enter values for the properties.
4. Select the data domains that you want to include in the classification policy.
5. Click the **Next** icon.

   The **Data Domain Match Conditions** panel appears.
6. Specify a data domain match condition.
7. Click **Save**.

Related Topics:

- "Specifying Data Store Sensitivity Levels" on page 42
- "Sensitivity Level Indicator" on page 518

Exporting Classification Policies

You can export a list of classification policies to a CSV file. The export file includes a list of all classification policies and the corresponding classification policy properties. If you filtered the classification policy list, the export file contains the classification policies that match the active filter conditions.

The export file contains the same format as the file required to import classification policies. You can use the export file to add classification policies or to update policy properties in bulk. After you save the export file, import the file. When you import the file, the Import job adds and updates the classification policies in the Data Privacy Management repository.

1. Click **Manage** > **Classification Policies**.

   The **Classification Policies** workspace appears.
2. From the **Actions** menu, click **Export**.

   A dialog box appears with the exported file.
   The export file is named `ClassificationPolicies.csv`. Save the file to a directory on your machine. If you export the file again, a new file downloads. The file name includes an incremental number such as `ClassificationPolicies (1).csv`.
3. Open the file.
4. Optionally, add classification policies or update existing policy properties in the file.
5. Save the file.

Importing Classification Policies

To create or update a classification policy, you can import policy information from a CSV file. The CSV file must be in the format that the Import job requires. When you import the file, the Import job runs immediately. The import job updates the Data Privacy Management repository with information from the CSV file.

1. Click **Manage** > **Classification Policies**.

   The **Classification Policies** workspace appears.

2. From the Actions menu, select **Import**.

   The **Import Classification Policies** page appears.

3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the Import job handles classification policy updates in the repository.
   If you enable the option, the Import job updates the classification policy properties in the repository with the values in the CSV file.

   Warning: If a column in the CSV file contains a null value, the Import job replaces the repository value with a null value.

   If you disable the option, the Import job skips the row. By default, the option is enabled.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the Import job finishes, you can return to the **Classification Policies** workspace to see the classification policies that you imported.

Related Topics:

- "Import Job" on page 299

CSV File Format to Import Classification Policies

Create a CSV file to create, update, or delete information about classification policies in the Data Privacy Management repository. The content of the CSV file must be in the order and format that the Import job requires. The CSV file must contain all of the column headings. However, not all columns require values.

The following table describes the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Description
1	ClassificationPolicyName	Required. Enter a classification policy name using up to 100 characters. Do not use a space or special characters.
2	Description	Optional.
3	DomainNames	Enter the names of the data domains that you want to include in the classification policy. The names of the data domains must match the names of the data domains in the Data Privacy Management repository. This field is case sensitive. Use the following format to enter a list of domains: `<data domain1>,<data domain2>,...`
4	DomainMatchExpressionJson	Enter a data domain match condition. Enter the condition as a JSON expression.
5	Severity	Enter a data sensitivity level. The name of the data sensitivity level must match the name of one of the defined data sensitivity levels. For example, if you defined Public, Internal, Confidential, or Restricted as data sensitivity levels, you can only enter one of these levels. You must use uppercase letters.

Column Order	Column Heading	Description
6	CostPerRow	Optional. Enter a positive value up to 999. If you do not enter a value, the Data Privacy Management Service assigns the default cost for each row.
7	Action	Optional. Instructs the Import job to create, update, or delete a classification policy in the repository. Enter one of the following values: - U. Creates or updates a classification policy in the repository. The action that the job performs depends on whether the classification policy exists in the repository and on the **Replace Duplicates with Items Imported** option. If the classification policy does not exist, the Import job creates the classification policy. If the classification policy exists and the **Replace Duplicates with Items Imported** option is enabled, the Import job updates the classification policy. **Warning:** If a column in the CSV file contains a null value, the Import job replaces the value in the repository with a null value. If the classification policy exists and the **Replace Duplicates with Items Imported** option is not enabled, the Import job skips the row. - D. Deletes a classification policy from the repository. If the classification policy does not exist, the Import job rejects the row. If blank, default is U.

Note: To update an existing classification policy, ensure that the name of the policy in the **ClassificationPolicyName** column matches the name of the classification policy in the Data Privacy Management repository.

Rules and Guidelines to Import Classification Policy Information

Review the rules and guidelines to understand how to manage the CSV file and how the Import job processes data from the file.

Consider the following rules and guidelines when you import classification policies from a CSV file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines a classification policy.
- If the CSV file contains duplicate entries, the Import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

```
SELECT FROM
SELECT UNION
DELETE FROM
UPDATE SET
<SCRIPT
</SCRIPT>
ALERT(
"/>"
```

If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.

- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Editing a Classification Policy

After you create or import a classification policy, you might need to update the policy properties, the list of data domains, or the data domain match condition.

Review the rules and guidelines before you edit a classification policy.

Rules and Guidelines for Editing a Classification Policy

Consider the following rules and guidelines when you edit a classification policy:

- Before you edit a classification policy, verify if the policy is included in a scan.
- If a classification policy is associated with a completed scan, the Evaluate Classification Policies job step starts when you save the policy after one or more of the following edits:
  - Change the cost for each row
  - Change the severity level
  - Add a data domain
  - Remove a data domain
  - Change the data domain match condition
- If a classification policy is associated with a completed scan and you add or remove data domains, you must repeat the scan to identify sensitive columns and classify the sensitive data.
- If a classification policy is associated with a completed scan and your edits do not include changes to the list of data domains, no further action is necessary. The Evaluate Classification Policies job step runs and updates the results of the previous scan.
- You can change the name of a classification policy. Data Privacy Management identifies classification policies by the policy ID instead of the policy name.
- When a Scan job starts, the job uses the classification policy properties, list of domains, and the data domain match condition that exist at the time the job starts. If you edit a classification policy that is associated with a Scan job that is currently running, the job does not include the classification policy changes.

Steps to Edit a Classification Policy

Edit classification policy properties, add or delete data domains, or change the data domain match condition.

1. Click **Manage** > **Classification Policies**.

   A list of classification policies appears. You can use the filter and sort options to narrow the list of classification policies.
2. Select the check box next to the classification policy that you want to edit.

3. From the **Actions** menu, select **Edit**

   The **Edit <classification policy name>** page appears.
4. Edit the classification policy details.
5. Click **Save**.

Copying a Classification Policy

You can copy a classification policy to create a policy with similar policy information. After you copy the classification policy, you can edit the classification policy properties, add or delete data domains, and change the data domain match condition.

1. Click **Manage** > **Classification Policies**.

   A list of classification policies appears. You can use the filter and sort options to narrow the list of classification policies.
2. Select the check box next to the classification policy that you want to copy.
3. From the **Actions** menu, select **Duplicate**.

   The **Classification Policy Details** page appears.
4. Edit the classification policy details.
5. Click **Save**.

Data Privacy Management creates a classification policy with the following naming convention: `Copy of <classification policy name>`

Deleting a Classification Policy

When you delete a classification policy, you delete the policy from the Data Privacy Management repository. You cannot delete a classification policy that is included in a completed scan or a scan that is currently running. If a scan that is scheduled to run in the future includes the classification policy, remove the classification policy from the scan before you delete the classification policy.

1. Click **Manage** > **Classification Policies**.

   A list of classification policies appears.
2. Select the check box next to the classification policy that you want to delete.
3. From the **Actions** menu, select **Delete**.

   You receive a message to confirm if you want to delete the classification policy.
4. Click **Yes**.

CHAPTER 11

Column Sensitivity

This chapter includes the following topics:

- Column Sensitivity Overview, 235
- The Always Sensitive/Never Sensitive Workspace, 236
- Always Sensitive/Never Sensitive List Management, 237

Column Sensitivity Overview

An Always Sensitive list identifies columns that are always sensitive. A Never Sensitive list identifies columns that are never sensitive. You can specify whether a column is always sensitive or never sensitive by placing it on an Always Sensitive list or a Never Sensitive list. If a column is always sensitive for a particular domain, place it on the Always Sensitive list. If a column is never sensitive, place it on the Never Sensitive list. You can add a column to an Always Sensitive or Never Sensitive list for any data domain.

From the **Always Sensitive/Never Sensitive** workspace, you can view and manage Always Sensitive and Never Sensitive rules. Each rule specifies column sensitivity for a specific data domain, or for any data domain.

Based on your knowledge of column names and column data, you can import the global sensitivity status of columns to the Data Privacy Management repository. Global sensitivity has the following statuses:

- A column is always sensitive for a data domain.
- A column is never sensitive for a data domain.
- A column is never sensitive for any data domain.

Global sensitivity is applicable for all data stores, schemas, and objects.

After you import the global sensitivity status, you can edit the scope of the Always Sensitive or Never Sensitive rules, mark rules as active or inactive, or delete rules.

You can also export the global sensitivity status of columns to a CSV file. After you export the file, you can update and then import the file to add, delete, or update the global sensitivity status to the Data Privacy Management repository.

The export file includes a list of column names, the corresponding data domain name, and whether the column is an Always Sensitive column or a Never Sensitive column. An Always Sensitive column indicates that the column is always sensitive for a specific data domain. A Never Sensitive column indicates that the column is never sensitive for a specific data domain or for any data domain.

For more information about sensitivity status, see "How Data Privacy Management Determines Sensitivity Status" on page 252.

Note: You can only add structured data sources to an Always Sensitive or a Never Sensitive list.

The Always Sensitive/Never Sensitive Workspace

To view and manage Always Sensitive and Never Sensitive rules, use the **Always Sensitive/Never Sensitive** workspace.

The **Always Sensitive/Never Sensitive** workspace displays a list of active and suggested rules by data store, schema, object, field, domain, type, and last updated. You can click a column header to sort the list by the selected column.

The workspace contains the following tabs:

Active

The **Active** tab lists active rules. From the **Actions** menu, you can import and export rules. When you select a rule, you can edit or delete the rule, or move the rule to the **Suggestions** tab to inactivate the rule. Default is the **Active** tab.

Suggestions

The **Suggestions** tab list inactive rules that you might want to make active. Data Privacy Management gets the rules from inactive Always Sensitive and Never Sensitive rules and data that Data Privacy Management generates when you perform one of the following actions:

- Change the sensitive status of fields on the **Sensitive Fields** page.
- Change the sensitive status of exported fields by exporting to a CSV file, modifying the file, and then importing the modified CSV file.
- Add sensitive fields and import them using **Datastore** > **Protection Status**.

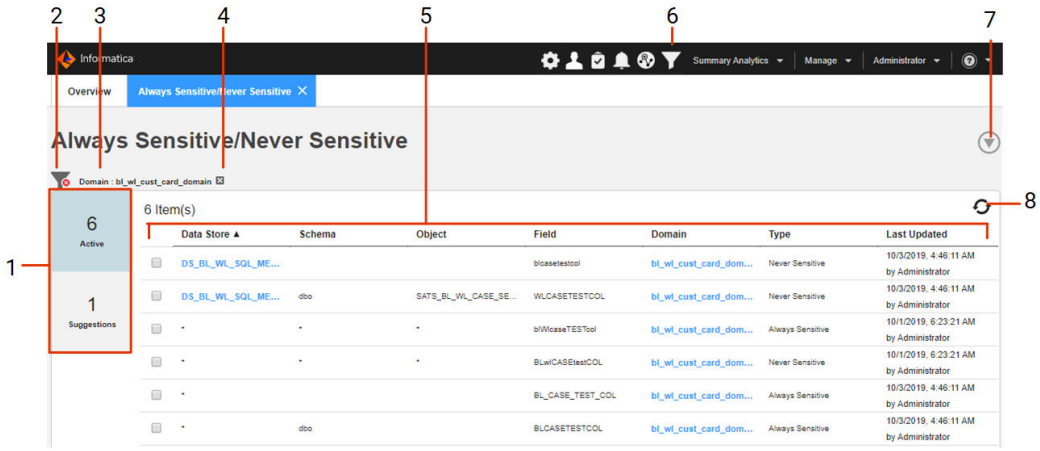
From the Actions menu, you can export suggested Always Sensitive and Never Sensitive rules. When you select a rule, you can edit or delete the rule, or mark the rule as active.

To access the **Always Sensitive/Never Sensitive** workspace, click **Manage** > **Always Sensitive/Never Sensitive**.

Always Sensitive/Never Sensitive List Page

The **Always Sensitive/Never Sensitive** list page displays active and suggested rules. By default, the active Always Sensitive/Never Sensitive list appears when you access the **Always Sensitive/Never Sensitive** workspace.

The following image shows an example of the **Always Sensitive/Never Sensitive** workspace with a list of active rules:



1. Active and Suggested rules tabs
2. Clear Filter icon
3. Filter condition
4. Clear Filter Condition icon
5. Rule list
6. Filter
7. Actions menu
8. Refresh icon

To display the **Data Domain Details** page for a data domain, click the data domain name. To display the **Sensitive Fields** page for a data store, click the data store name.

Always Sensitive/Never Sensitive List Management

Use the **Always Sensitive/Never Sensitive** workspace to create and manage Always Sensitive and Never Sensitive lists.

After you identify the sensitive columns, you can perform the following tasks:

- View, edit, or delete Always Sensitive and Never Sensitive rules.
- Mark an Always Sensitive or Never Sensitive rule as active.
- Remove a rule from the Active tab to make the rule inactive and place it on the **Suggestions** tab.
- Import global Always Sensitive and Never Sensitive lists.
- Export global Always Sensitive and Never Sensitive lists.

Specifying a Column as Always Sensitive

You can specify whether a column is Always Sensitive for a data domain in several ways.

The following table shows how you can specify a column as Always Sensitive for a data domain:

You can specify:	Is Always Sensitive for a data domain and is applicable for:
A column.	All tables, schemas, and data stores or, a specified table, all schemas, and all data stores.
A column of one or more tables of a specified schema.	Specified tables and schemas and all data stores.
A column of one or more tables of a specified schema of a data store.	Specified tables, schemas, and data stores.

Specifying a Column as Never Sensitive

You can specify a column is Never Sensitive for a data domain in several ways.

The following table shows how you can specify a column as Never Sensitive for a data domain:

You can specify:	Is never sensitive for a data domain and is applicable for:
A column.	All tables, schemas, and data stores, or a specified table, all schemas, and all data stores.
A column of one or more tables of a specified schema.	Specified tables and schemas and all data stores.
A column of one or more tables of a specified data store schema.	Specified tables, schemas, and data stores.

Super Rules and Subset Rules

A super rule is a generic rule. A subset rule is a more specific rule that shares a common data domain and field.

The following image shows an example of a super rule and eight subset rules. The rule in row 1 is the most generic rule and it applies to all data stores, schemas, and objects for the **DOB** field for the data domain BirthDay. The rules in rows 2 through 9 are subset rules of the rule in row 1.

Row	DataStoreName	SchemaName	ObjectName	FieldName	DataDomainName	WhiteList_or_BlackList
1				DOB	BirthDay	W
2	DS1	SCH1	OBJ1	DOB	BirthDay	W
3	DS2		OBJ2	DOB	BirthDay	W
4	DS3			DOB	BirthDay	W
5	DS4		OBJ3	DOB	BirthDay	W
6	DS5	SCH2		DOB	BirthDay	W
7		SCH3	OBJ4	DOB	BirthDay	W
8			OBJ5	DOB	BirthDay	W
9		SCH4		DOB	BirthDay	W

Super rules and subset rules cannot coexist in the **Active** tab. Super rules and subset rules can coexist in the **Suggestions** tab.

Guidelines for Always Sensitive/Never Sensitive Rules

Review the following guidelines to understand how to manage Always Sensitive and Never Sensitive rules.

As you create and edit Always Sensitive and Never Sensitive rules, keep the following guidelines in mind:

- You cannot have duplicate rules on the same tab.
- Super rules and their subset rules cannot exist on the same tab unless they are on the **Suggestions** tab.

Consider two rules: Rule 1 and Rule 2. The following table shows the rule combinations Data Privacy Management permits:

Rule Properties	Rule 1 and Rule 2 are listed as Active	Rule 1 and Rule 2 are listed as Suggestions	Rule 1 listed as Active and Rule 2 listed as a Suggestion	Description
Same rules; both rules are Always Sensitive.	N	N	Y	Data Privacy Management does not allow duplicate rules in the same tab.
Same rules; both rules are Never Sensitive.	N	N	Y	Data Privacy Management does not allow duplicate rules in the same tab.
Same rules; one rule is Always Sensitive and one rule is Never Sensitive.	N	N	Y	The rules are the same. One rule is Always Sensitive and the other rule is Never Sensitive. Data Privacy Management does not allow this combination in the same tab.

Rule Properties	Rule 1 and Rule 2 are listed as Active	Rule 1 and Rule 2 are listed as Suggestions	Rule 1 listed as Active and Rule 2 listed as a Suggestion	Description
Rule 1 is a subset of Rule 2 and both rules are Always Sensitive.	N	Y	Y	Data Privacy Management does not allow subset rules on the **Active** tab when a super rule exists. Data Privacy Management allows subset rules on the **Suggestions** tab when a super rule exists.
Rule 1 is a subset of Rule 2 and both rules are Never Sensitive.	N	Y	Y	Data Privacy Management does not allow subset rules on the **Active** tab when a super rule exists. Data Privacy Management allows subset rules on the **Suggestions** tab when a super rule exists.

Importing Global Sensitivity Status

You can import the global sensitivity status of columns to the Data Privacy Management repository. In the import file, you specify a column as W (whitelist) if the column is Always Sensitive for a data domain. You specify a column as B (blacklist) if the column is Never Sensitive for a data domain or for any data domain.

Example of a globally sensitive column

The columns in your data stores that contain Social Security numbers are named SSA. You want the SSA columns to always be considered sensitive for the SSN data domain. To specify that the SSA column is Always Sensitive for SSN, import a CSV file with the column name SSA, the data domain SSN, and specify that it is whitelist. In all future scans, if the data stores contain the SSA column and the classification policy contains the SSN data domain, the scan job identifies the SSA column as sensitive for SSN.

Example of a column that is globally not sensitive

Your data stores have columns named DOB. The DOB columns do not contain sensitive information. To specify that the DOB column is Never Sensitive, import a CSV file with the column name DOB and specify that it is blacklist. In all future scans, if the data stores contain the DOB column, regardless of the data domains, the scan job identifies the DOB column as not sensitive.

CSV File Format to Import Global Sensitivity Status

To import the global sensitivity status of fields to the Data Privacy Management repository, create a CSV file that contains the order and format that the Import job requires. The CSV file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Column Value Description
1	DataStoreName	Name of the data store. Enter the exact name of the data store in the Model repository. Values are not case sensitive.
2	SchemaName	Name of the schema. Enter the exact name of the data schema in the Model repository. Values are not case sensitive.
3	ObjectName	Name of the object. Enter the exact name of the object in the Model repository. Values are not case sensitive.
4	FieldName	Name of the field. Enter the exact name of the field in the Model repository. Values are not case sensitive.
5	DataDomainName	Required for Always Sensitive columns. Name of the data domain that contains the Always Sensitive or Never Sensitive column. Enter the exact name of the data domain in the Model repository. Values are not case sensitive. For updates to Never Sensitive columns, use the following guidelines: - If you previously imported a Never Sensitive column without a data domain, you must continue to leave the data domain column blank in subsequent imports. - If you previously imported a Never Sensitive column with a data domain, you must continue to add a data domain name in the data domain column in subsequent imports.

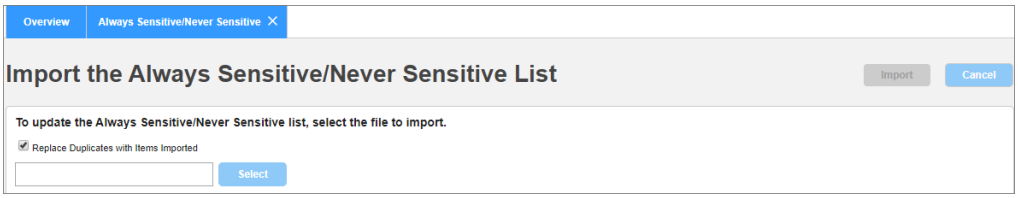
Column Order	Column Heading	Column Value Description
6	WhiteList_or_BlackList	Required. If the column is Always Sensitive for a data domain, enter `W` or `Whitelist`. Values are not case sensitive. If the column is Never Sensitive, enter `B` or `Blacklist`. Values are not case sensitive. The Import job rejects the row if the value is blank.
7	Action	Optional. Instructs the Import job to create, update, or delete the global sensitivity status of a column in the Data Privacy Management repository. Enter one of the following values: - `U`. Adds or updates the global sensitivity status of a column to the Data Privacy Management repository. The action that the job performs depends on whether the global sensitivity for a column is already specified in the repository and whether you select the **Replace Duplicates with Items Imported** option when you import the CSV file. If the global sensitivity of the column is not already specified in the repository, the Import job adds the column to the list. If the global sensitivity of the column is already specified in the repository and you do not enable the **Replace Duplicates with Items Imported** option, the Import job skips the row. If the global sensitivity of the column is already specified in the repository and you enable the **Replace Duplicates with Items Imported** option, the Import job updates the information in the repository. - `D`. Deletes the global sensitivity status of a column from the repository. If blank, default is `U`. The Import job adds or updates the global sensitivity status of a column in the Data Privacy Management repository.

Steps to Import Global Sensitivity Status

You import global sensitivity status for columns from a CSV file in the format that the Import job requires. When you import the file, the Import job runs immediately. The Import job updates the Data Privacy Management repository with information from the CSV file.

1. From the **Actions** menu on the **Active** tab, click **Import**.

   The **Import the Always Sensitive/Never Sensitive List** page appears.



2. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the Import job handles updates for users that exist in the Data Privacy Management repository.
   If you enable the option, the Import job updates the global sensitivity status in the repository with the values in the CSV file.

   If you disable the option, the Import job skips the row.

3. Select the file from which you want to import data.

The Data Privacy Management Service validates the import file, such as the file extension and column headers.
If the file is valid, the user interface displays a count of how many records are in the file.

If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

4. Click **Import**.

   The Import job begins.

The import job processes the entries in the order in which the rows appear in the file. If the columns in one row are the same as the columns in a second row, the Import job treats the second row as a duplicate of the first row.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the Import job finishes, scan the data stores with the Always Sensitive and Never Sensitive columns to update the dashboard results.

Rules and Guidelines to Import Global Sensitivity Status

Review the rules and guidelines to understand how to manage the CSV file and how the Import job processes data from the file.

Consider the following rules and guidelines when you import global sensitivity status of columns from a CSV file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines the global sensitivity status for a column. For example, the column with name `ssa` is always sensitive for the `SSN` data domain.
- The column values cannot contain the following keywords or combination of characters:

  ```
  SELECT FROM
  SELECT UNION
  DELETE FROM
  UPDATE SET
  <SCRIPT
  </SCRIPT>
  ALERT(
  "/>"
  ```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.
- The data domains in the CSV file must exist in the Model repository.
- If you previously imported a Never Sensitive column without a data domain, you must continue to leave the data domain column blank in subsequent imports.

- If you previously imported a Never Sensitive column with a data domain, you must continue to add a data domain name in the data domain column in subsequent imports.
- After you import the CSV file, the results of previous scans do not change automatically. You must scan the data store again to apply the updates.

Marking a Rule as Active

To implement a rule so it is included in subsequent scans, mark a rule as active.

1. Click **Manage** > **Always Sensitive/Never Sensitive**.
2. Click **Suggestions**.

   The list of suggested rules appears.
3. Select the check box next to the rule you want to make active.
4. From the **Actions** menu, select **Mark as Active**.

   The selected rule moves to the **Active** tab.

Removing a Rule from the Active Tab

To inactivate a rule and place it on the **Suggestions** tab, remove the rule from the **Active** tab.

1. Click **Manage** > **Always Sensitive/Never Sensitive**.
2. Click **Active**.

   The list of active rules appears.
3. Select the check box next to the rule you want to remove.
4. From the **Actions** menu, select **Remove from Active**.

   The selected rule moves to the **Suggestions** tab.

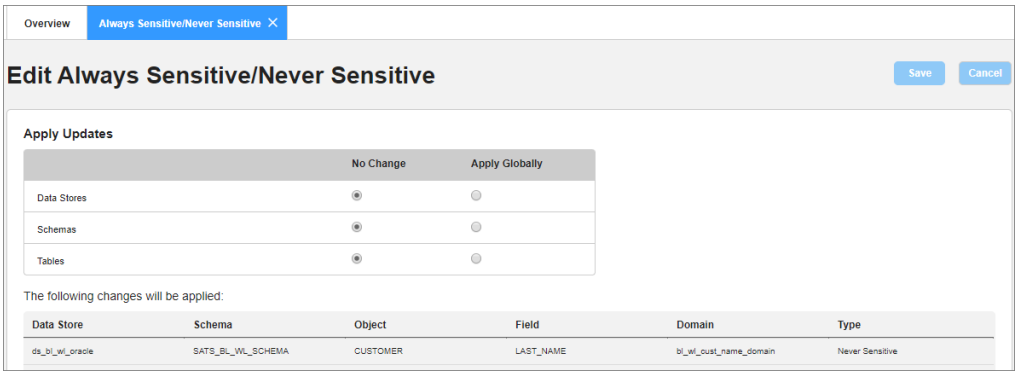
Editing an Always Sensitive/Never Sensitive Rule

You can edit an Always Sensitive/Never Sensitive rule from the **Always Sensitive/Never Sensitive** workspace. You can edit the scope of an Always Sensitive or Never Sensitive rule to be specific to a data store, schema, and table or apply it globally at each of these levels.

1. Click **Manage** > **Always Sensitive/Never Sensitive**.

   The **Always Sensitive/Never Sensitive** workspace appears.
2. Click the tab that contains the rule you want to edit. You can use the filters to narrow the list of rules.
3. Select the check box next to the rule you want to edit.
4. From the **Actions** menu, select **Edit**.

The **Edit Always Sensitive/Never Sensitive** page appears:



5. Specify **No Change** or **Apply Globally** for each item, such as Data Stores, Schemas, and Tables. Default is **No Change**.
6. Click **Save**.

Exporting Global Sensitivity Status

The Data Privacy Management repository maintains a list of columns that you previously specified as always sensitive for a data domain or never sensitive. To review whether a column is Always Sensitive for a data domain or Never Sensitive, export the global sensitivity status to a CSV file.

1. Click **Manage** > **Always Sensitive/Never Sensitive**.

   The **Always Sensitive/Never Sensitive** workspace appears.
2. Click the tab that contains the rules you want to export. You can select one or more rules to export. If you do not select any rules, Data Privacy Management exports all of the rules.
3. From the **Actions** menu, click **Export**.

   If you are exporting rules from the **Active** tab, the file name is `GlobalList_Active.csv`. If you are exporting rules from the **Suggestions** tab, the file name is `GlobalList_Recommendations.csv`.

   The file downloads to the default download directory on your machine. If you export the file again, the file name includes an incremental number such as `GlobalList_Active(1).csv`.
4. Open the file.
5. You can update and then import the file to add, delete, or update the global sensitivity status to the repository.

Deleting Always Sensitive/Never Sensitive Rules

Perform the following steps to delete rules from the Always Sensitive or Never Sensitive lists:

1. Click **Manage** > **Always Sensitive/Never Sensitive**.
2. Click the tab that contains the rules that you want to delete.
3. Select the check box next to the rules you want to delete.
4. From the **Actions** menu, select **Delete**.

   A confirmation message appears.
5. Confirm that you want to delete the selected rules.

CHAPTER 12

Scans

This chapter includes the following topics:

- Scans Overview, 246
- Compressed Files in Scans, 247
- Optical Character Recognition Files in Unstructured Scans, 248
- Scans Workspace, 249
- Scan Properties, 251
- How Data Privacy Management Determines Sensitivity Status, 252
- Scan Management, 254

Scans Overview

A scan defines how to discover and classify sensitive and subject registry information in data stores. When you run a scan, the Data Privacy Management Service creates one Scan job for each data store in the scan.

You can create, run, and manage scans on the **Scans** workspace. You create and run scans after you define data stores, data domains, and classification policies. When you create a scan, you configure properties and options for the Scan job. You specify the data store category and the data stores that you want to scan. You create a scan for one category. You add one or more data stores in the same category for which you want to identify sensitive or subject data. You assign one or more classification policies to the scan.

Run scans to create and build your subject registry. A Subject Registry scan can include Domain Discovery scans to map fields to data domains. You also run scans on golden data stores to identify subjects, and on transaction data stores to link subjects in the data stores with subjects identified in golden data stores. This ensures that the subject registry includes all subject data.

You schedule when to run the scan. You can run the scan immediately or in the future. You can run the scan once or on a recurring basis. When you save the scan, the Data Privacy Management Service creates a Scan job for each data store in the scan.

You can monitor the Scan job status on the **Jobs** workspace. After the Scan job finishes, you can view Domain Discovery scan results on the **Security Dashboard**. You can view Subject Registry scan results on the **Privacy Dashboard**.

Compressed Files in Scans

Data Privacy Management can scan compressed files with specific extensions. You can include compressed files in scans on unstructured sources.

A compressed file contains individual files. An individual file inside a compressed file with a supported extension might in turn be a compressed file with a supported or unsupported extension.

A Data Privacy Management scan handles compressed files in the following ways based on the extension:

As a single file

Considers the compressed file to be a single file and identifies and lists sensitive data as one file.

As a folder with individual files

Considers a compressed file as a folder and lists each file in it as a separate file. Each compressed file nested in the main file is also listed as a folder and the files in it as separate files. If the file contains unstructured file types that Data Privacy Management does not support, the scan skips the file and adds the file name to the Exception report.

The process can drill down multiple levels until it encounters a compressed file with an extension that is not supported for drill down. This file is considered a single file with no further drill down.

You can configure the number of levels that a scan drills down to in the Data Privacy Management Service properties.

The following supported compressed file extensions are handled as single files or folders with individual files:

Single File	Folder with Individual Files
xz	zip
bz2	tar
tbz2	tar.bz2
gz	tar.gz
-	tar.xz
-	tgz2
-	tgz
-	gtar
-	7z

The agent adds entries to the Exception report in the following cases:

- For compressed files with bz2, gz2, or xz extensions without tar archive and with a compression ratio greater than 100, the agent generates zip bomb exceptions during a domain discovery scan. The agent writes the exceptions to the Exception report and continues the scan.
- For compressed files with 7z extension that are very large or highly compressed, the agent generates a heap space exception. The exception also occurs if a very large or highly compressed 7z file is present in another compressed file. The agent writes the exceptions to the Exception report and continues the scan.

To include the files in the scan, you can extract the files and run the scan.

Optical Character Recognition Files in Unstructured Scans

You can include scanned PDFs and image files in a scan of unstructured sources. Include the file types when you create the data store to include them in a scan.

You can include specific image file types and files with content in English. Files can include specific font types and must be within specific dimensions and specifications.

Read the rules and guidelines and verify the prerequisites before you include image files or scanned files in a scan.

For information about the prerequisites, see the Install the Informatica Discovery Agent chapter in the *Informatica Data Privacy Management Installation and Configuration Guide*.

File Types and Specifications

You can use the following file types with the corresponding specifications:

File Type	Resolution	Image Dimension
PNG	>= 300 dpi	600x400
JPG	>= 150 dpi	600x400
JPEG	>= 150 dpi	600x400
TIFF	>= 150 dpi	600x400
BMP	>= 150 dpi	600x400
PDF	>= 120 dpi	NA

The following specifications apply to all file types:

- Font styles can include Times New Roman, Verdana, Arial, and Calibri.
- The font size must be >=10 pt.
- The background color must be white.
- If a border is present, it must be a Solid border.
- Images must be clear and without noise.

Rules and Guidelines for Optical Character Recognition File Types

Read the following rules and guidelines before you include the files in a scan:

- Scans can detect content with font sizes of 8pt and 9pt, but there might be inaccuracies of up to 3% to 8% deviation in extracted text based on the data.

- Dark font colors have a higher accuracy against a white background. Accuracy reduces as the contrast reduces.
- You cannot scan blurred images and images with colored backgrounds.
- You can scan images that are skewed by 90, 180, or 270 degrees, but a scan does not capture table content that is skewed by 180 degrees.
- You can scan images skewed by +/- 5 degrees from 0 degrees.
- There might be some data loss in a scan of rotated content.
- There might be differences in the extracted text for the same content and resolution in different file types.

Scans Workspace

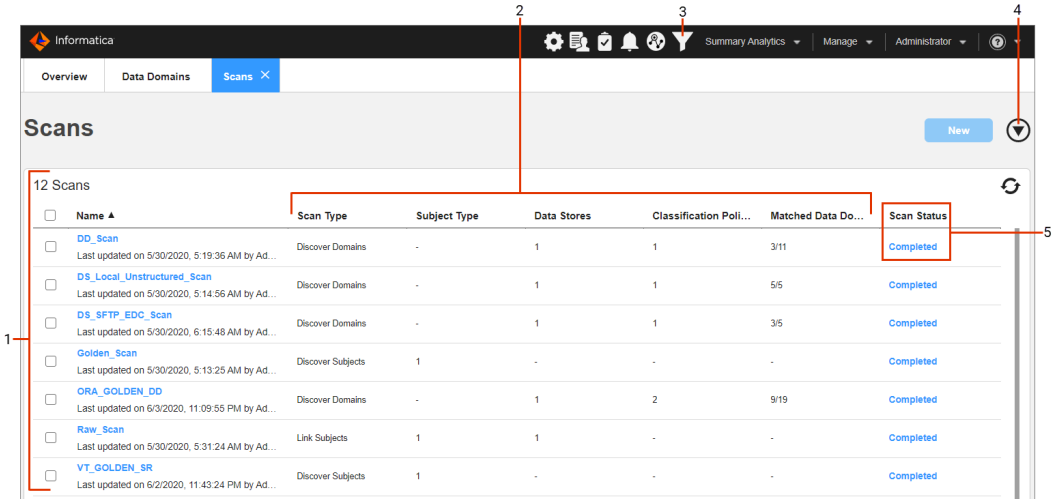
Use the Scans workspace to create, run, and manage scans. The Scans workspace includes a page that displays a list of all scans and a page that displays the properties for a scan. You can use the scan filters to navigate to the scans that you want to manage. You can view a summary of the scan results.

To access the Scans workspace, click **Manage** > **Scans**.

Scans List Page

The **Scans** list page shows all of the scans in the Data Privacy Management repository. The scans in the list depend on the active filter. By default, the list of scans appears when you access the **Scans** workspace.

The following image shows an example of the **Scans** list page:

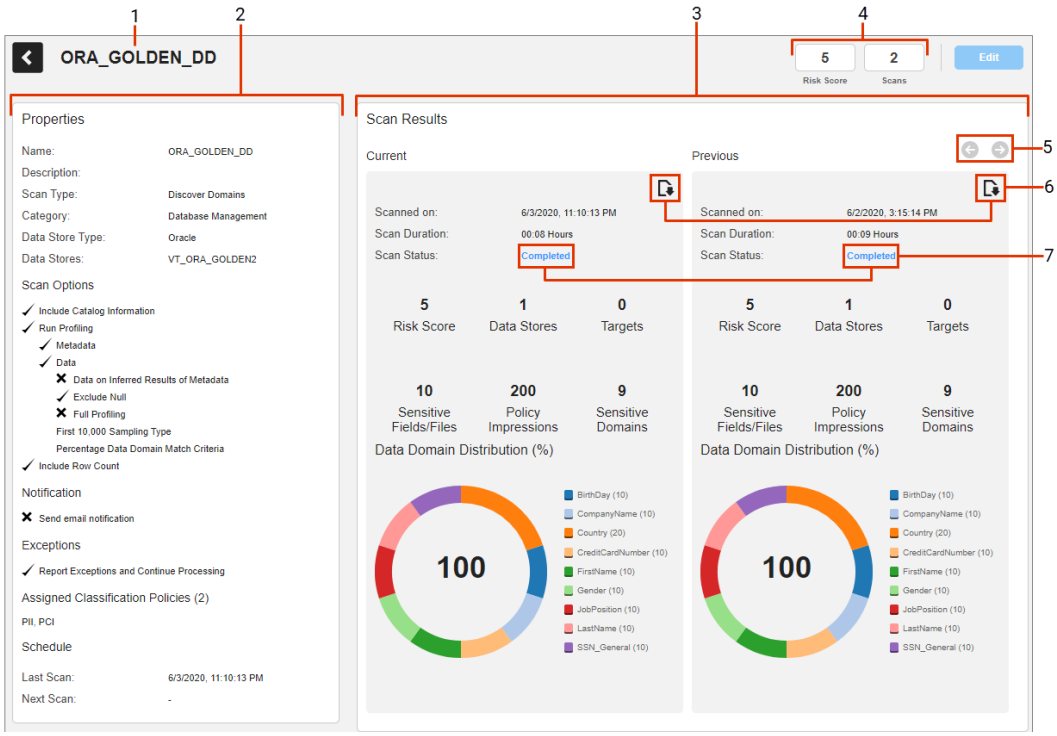


1. Scan list. Click a name to view the scan details.
2. Scan summary
3. Filter
4. Actions menu
5. Scan status. Click to view the jobs for the scan.

Scan Details Page

The scan details page shows the scan properties and scan summary. Access the scan details page to view or edit the scan properties. You access the scan details page when you click a scan name from the scan list page, or when you view a scan from the Actions menu.

The following image shows an example of a scan details page:



1. Scan name
2. Scan properties
3. Scan results
4. Scan summary
5. Scan status. Click to view the jobs for the scan.

The scan summary shows the combined risk score of all the data stores in the scan. The Secure@Source Service updates the combined risk score when the scan job completes. The scan summary also shows the number of times the scan job ran.

The scan results include key statistics for all data stores in the scan. The Secure@Source Service updates the scan results when the scan job completes. Each time you run a scan, the Secure@Source Service generates a new set of scan results. If you run the scan multiple times, you can view and compare how the scan results change over time. Use the arrow icons to scroll though all of the results for the scan. For more detailed scan results, view the dashboard or the data store.

The status is displayed as a link that opens the jobs workspace. A list of jobs appears. The list shows jobs that are associated with one set of scan results. For example, if you click the scan status for the current scan results, the list shows the jobs for the current scan only. The list does not show the jobs for previous runs of the same scan.

Related Topics:

- "Changing Risk Score Factor Weights" on page 42
- "Adding Custom Risk Score Factors" on page 43

Scan Properties

Scan properties determine the type of Scan job that the Data Privacy Management Service creates and the data stores to scan. The scan options depend on the data store type that you select for the scan and the **Scan with Remote Agent** setting for the data stores included in the scan.

The following table describes the general properties for all scans:

Property	Description
Name	Short description of the scan. The name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 255 characters and cannot contain spaces or the following special characters: `~ ! $ % ^ & * ( ) +` When you run a scan, Data Privacy Management creates a Scan job. The Scan job name is the same as the scan name.
Description	Long description of the scan. The description cannot exceed 255 characters.
Scan Type	The type of scan that you want to run. Select one or more of the following options: - **Domain Discovery**: Maps columns in data stores to data domains. - **Discover Subjects**: Scans golden data stores to identify data subjects in Subject Registry scans. **Note:** If you configure a scan with the Discover Subjects option only, you do not choose a category, data store type, or data store in the scan configuration. - **Link Subjects**: Scans transaction data stores to link subjects in the data stores with subjects identified in golden data stores.
Subject Type	For Discover Subjects or Link Subjects scans, the entity files that you want to include in a subject registry scan. An entity file contains properties that determine how subject details appear in the subject registry. Each entity file represents a subject type and can include one or more golden data stores.
Category	A logical group of data store types. Select one of the following options: - Application - Big Data - Cloud - Data Engineering - Database Management - File Management - NoSQL
Data Store Type	Select one of the following options, depending on the value of the **Category** property: - **Application** scans include SAP data stores. - **Big Data** scans include Cloudera Navigator, Hadoop Distributed File System, and Hive data stores. - **Cloud** scans include Amazon Redshift, Amazon S3, Azure Data Lake, Microsoft Azure Blob Storage, Microsoft Azure SQL Data Warehouse, Microsoft Azure SQL Database, Salesforce, and Snowflake data stores. - **Data Integration** scans include Informatica Cloud, Informatica Data Engineering Integration, Informatica PowerCenter on IBM Db2, Informatica PowerCenter on Microsoft SQL Server, Informatica PowerCenter on Oracle, and SQL Server Integration Services data stores. - **Database Management** scans include IBM DB2, IBM DB2 for i5/OS, IBM Db2 for z/OS, JDBC, Microsoft SQL Server, Netezza, Oracle, SAP HANA, Sybase, and Teradata data stores. - **File Management** scans include File System, Google Drive, Microsoft OneDrive, and Microsoft SharePoint data stores. - **NoSQL** scans include Apache Cassandra.
Data Stores	Data stores to include in the scan. The list shows data stores for the selected data store type and that include complete connection properties. The list does not show incomplete data stores.

Property	Description
Scan Options	Determines the job steps that the Scan job includes. The scan options that you configure depend on the data store type. If the scan includes only data stores that scan with remote agent connections, the scan job performs data profiling, but the options do not appear in the user interface.
Subject Scan Options	Determines subject registry scan job options. Includes the following option: - **Full Subject Scan.** Enables complete scans on repeat scan runs. Default is selected. Uncheck the option to enable incremental scans on repeat scan runs. **Note:** Applicable for structured sources. You cannot configure incremental scans on unstructured sources.
Notification	Enable the property to send an email notification when the Scan job status changes. You can enter multiple recipient email addresses, including distribution lists. Use a comma to separate multiple entries. The Data Privacy Management Service validates the format of the email addresses when you navigate to the next panel. The Data Privacy Management Service uses the email server configuration defined in the Data Privacy Management Service properties.
Exceptions	Determines how a Scan job behaves if it encounters errors. If you want the scan to ignore profiling-related errors, enable the **Report Exceptions and Continue Processing** exception option. Default is selected. You can download a report from the scan **Jobs** page. The report contains the warning messages. **Note:** The **Report Exceptions and Continue Processing** option applies only to scans that include the **Run Profiling** or **Include Row Count** scan option, or that contain data stores that run scans with remote agents. You cannot resume the job to scan the tables that fail with warnings. To scan the failed tables, run the scan again or create and run a scan for the failed tables.

How Data Privacy Management Determines Sensitivity Status

Data Privacy Management evaluates the information from the Scan job and any user-specified information to determine whether a field or file is sensitive for a specific data domain.

Data Privacy Management processes the information in the following order of precedence to determine column sensitivity for a data domain:

1. Profiling or AgentProfiling job step in the data store Scan job
2. Global sensitivity status set on the **Always Sensitive/Never Sensitive** workspace
3. User updates

Profiling or AgentProfiling Job Step

During a data store scan with Enterprise Data Catalog, the Profiling job step determines if a field or file is sensitive for a data domain in the classification policy that you included in the scan. During an unstructured data store scan with a remote agent, the AgentProfiling job step determines the file sensitivity.

The Profiling job step determines sensitivity based on the following factors:

- The data domain match conditions for metadata, data, or both.

- The override rule in case of a conflict between the metadata profile results and the data profile results. You specify the override rule when you configure the data domain.
- For data stores scanned with Enterprise Data Catalog, the conformance score range that specifies whether data is sensitive, potentially sensitive, or not sensitive based on the percentage of fields that meet the data domain match conditions. The conformance score calculation is affected by the sampling technique and whether you want to include or exclude null values in the sample.
- The confidence level that specifies the likelihood that an unstructured file scanned with a remote agent contains data that matches sensitive data domain definitions.
- The data profiling method: metadata, data, or metadata with data.

In the absence of user-specified information, Data Privacy Management identifies fields in structured data stores as sensitive, potentially sensitive, or rejects the fields as sensitive. Sensitive fields in structured and semi-structured data stores appear on the **Sensitive Fields** page with a value of Auto in the **Classification Mechanism** property.

Sensitive files in unstructured data stores appear on the **Sensitive Fields** page with a value of Yes in the **Sensitive** column. The **Properties** pane lists the sensitive data domains in the file.

Global Sensitivity Status

For structured and semi-structured data stores, the global sensitivity status of a field takes precedence over the results from the Profiling job step, if the field is included in a list on the **Always Sensitive/Never Sensitive** workspace.

The following table describes how Data Privacy Management assigns sensitivity according to the global sensitivity status:

Global Sensitivity Status	Description
Column is always sensitive for a data domain	Data Privacy Management identifies the field as sensitive for the data domain regardless of the results of the Profiling job step. The field appears on the **Sensitive Fields** page with the **Auto (Always Sensitive)** classification mechanism.
Column is never sensitive for a data domain	Data Privacy Management identifies the field as not sensitive for the data domain. If the Profiling job step found that the field is sensitive for the data domain, Data Privacy Management lists the field on the **Sensitive Fields** page. The sensitivity status is set to Not Sensitive. The classification mechanism displays **Auto (Never Sensitive)**. If the Profiling job step did not find the field sensitive for the data domain, Data Privacy Management does not list the field on the **Sensitive Fields** page.
Column is never sensitive for any data domain	Data Privacy Management identifies the field as not sensitive regardless of the data domain in the scan classification policy. If the Profiling job step found that the field is sensitive for a data domain, Data Privacy Management lists the field on the **Sensitive Fields** page. The sensitivity status is set to Not Sensitive. The classification mechanism displays **Auto (Never Sensitive)**. If the Profiling job step did not find the field sensitive for any data domain, Data Privacy Management does not list the field on the **Sensitive Fields** page.

User Updates

Data Privacy Management assigns the highest level of precedence to any changes you make to the sensitivity status of a field or file. The sensitivity status that you specify applies only to the field or file in a specific data store.

Data Privacy Management saves the updates you make. If a future scan job includes the same data store and a classification policy with the same data domain, Data Privacy Management assigns the sensitivity status you specified.

To specify sensitivity status, you can import a CSV file to perform the following tasks:

- Change the existing sensitivity status of one or more fields or files.
- For structured data stores, add one or more fields or files to the list of sensitive data on the **Sensitive Fields** page.

When you import the sensitivity status, the value of the **Classification Mechanism** property for structured and semi-structured fields changes to Import on the **Sensitive Fields** page.

You can also change the existing sensitivity status of a field or data domain directly on the **Sensitive Fields** page. In the **Properties** pane for a field or file, click **Edit** to change the sensitivity selection for a field or data domain.

Scan Management

You can create, run, and manage scans on the **Scans** workspace.

A scan is a template for a scan job definition. You can create a scan once. Then, you can run the scan multiple times. When you run a scan, the Data Privacy Management Service creates a scan job.

You can view, copy, edit, or delete scans. You can view and export a list of all scans, a scan summary, and the scan status. For example, you can see how many data stores and classification policies a scan includes.

You can edit a scan to change the scan options. For example, if you configure all the scan options for a data store scan, the scan job performance might be impacted. Instead, run the scan multiple times with a different scan option selected.

Creating a Scan

Create a scan to configure how and when the scan job discovers and classifies sensitive data. When you create a scan, you specify the type of scan, configure the options, and schedule when the scan runs.

1. Click **Manage** > **Scans**.

   The **Scans** workspace appears.
2. Click **New**.

   The **Define & Connect** panel appears.
3. Enter a name and optionally, enter a description of the scan.
4. Choose the scan type. Select one or more of the following options:
   - **Discover Domains**. Maps fields in data stores to data domains that match the classification policies you include in the scan.
   - **Discover Subjects**. Scans golden data stores to identify subjects. If you select

- **Link Subjects**. Links subjects in the data stores with subjects identified in golden data stores to ensure that the subject registry includes all subject data

5. The options that appear depend on the scan type:
   - If you selected **Discover Domains** or **Link Subjects**, select one of the following categories: Application, Big Data, Cloud, Data Integration, Database Management, or File Management. Then select the data store type and the data stores to include in the scan.
   - If you selected **Discover Subjects** or **Link Subjects**, select one or more subject types that appear in the list.
6. If you selected **Discover Subjects** or **Link Subjects**, you can choose to configure repeat scans as incremental or full. Select the **Full Subject Scan** option to configure repeat subject registry scans as full scans. Clear the option to configure repeat scans as incremental. Default is selected.
7. If you selected **Discover Domains** or **Link Subjects**, select the scan options.

   Note: If you selected **Discover Domains** and the scan includes only data stores that scan with a remote agent, the scan options do not appear in the user interface.
8. Click **Next**.

   The **Assign Classification Policy** panel appears.
9. If you selected **Discover Subjects** only, you cannot assign classification policies. Go to step 9. For all other scans, select the classification policies to assign to the scan. You can use the following filters:
   - **Classification Policies Selected** . Displays a list of the classification policies that are assigned to the scan. Use the filter to view or remove classification policies.
   - **Assign Classification Policies**. Displays a list of all classification policies in the Data Privacy Management repository. Use the filter to add or remove classification policies.

   Note: If a classification policy includes custom match conditions based on file tags, include the policy only if the scan includes Microsoft SharePoint data stores.
10. Click **Next**.

    The **Schedule** panel appears.
11. Schedule when the scan runs.

    You can schedule the job to run immediately, in the future, or on a recurring schedule.
12. Click **Save**.

Related Topics:

- "Application Scan Options" on page 260
- "Big Data Scan Options" on page 263
- "Cloud Scan Options" on page 267
- "Data Integration Scan Options" on page 270
- "Database Management Scan Options" on page 273
- "File Management Scan Options" on page 276

Running a Scan

When you run a scan, Data Privacy Management creates a scan job and runs the job according to the scan schedule.

1. Click **Manage** > **Scans**.

2. Select the check box next to the scan that you want to run.
3. From the **Actions** menu, select **Execute**.

   You can monitor the status of the scan job on the **Jobs** workspace.

Rules and Guidelines for Running Scans

Consider the following rules and guidelines when you run a scan:

- You can create a scan once and run the scan multiple times. For example, you create a scan and schedule the scan to run every month. One week later, you want to run the scan. You do not have to create another scan.
- You can run scans that have a completed or terminated status. To run scans that have a failed, paused, scheduled, or stopped status, terminate the job. Then, run the scan.
- You can include a data store in multiple scans. However, you cannot run multiple scans on the same data store simultaneously.

  The scan fails if another job that includes the same data store is in progress. For example, a scan job, Import job, or Evaluate Classification Policies job might include the same data store.
- You can run a scan multiple times. However, before you run a scan again, ensure that all jobs associated with the scan are completed or terminated.

  For example, you included 10 data stores in a scan and ran the scan. Data Privacy Management created 10 scan jobs. Before you can run the scan again, all 10 jobs must be completed or terminated.
- When you schedule a recurring scan, Data Privacy Management performs the same validation as when you schedule a scan to run immediately. If the service cannot run a scan, the service writes an error to the system log. The service tries to run the scan at the next scheduled time.
- In a subject registry scan, you cannot use the same data domain for more than one column in an SQL query. Map each column in an SQL file to a different data domain.
- Domain discovery scans for subject registry must include schemas and objects that are marked as sensitive in an SQL file. If the SQL file does not include the schemas and objects that are marked as sensitive, information in DSAR reports for data subjects might be incorrect.

Rules and Guidelines for Remote Agent Scans

Consider the following rules and guidelines for remote agent scans:

- When you run a scan on a data store configured to scan with a remote agent, the remote agent associated with the data store must be up and running.
- If the remote agent associated with the data store is processing another job when you run the scan, the scan job will run after the other job finishes.
- Data Privacy Management performs domain discovery through the remote agent. You can also create and use a custom data domain with a pattern data match condition that includes a regular expression. For more information about data domains in remote agent scans, see the How-To Library Article Configuring Data Domains for Unstructured Sources in Data Privacy Management, on the Informatica documentation portal.
- Remote agents do not process data domains configured with a metadata match condition only.

Rules and Guidelines for Delta Scans

Consider the following rules and guidelines for delta scans:

- To run a domain discovery scan that analyzes only the files that are new or changed since the last time the scan job ran, the data store must meet the following criteria:
  - The scan definition and associated data stores, classification policies, and data domains configured in the initial scan are unchanged since the last scan finished successfully or completed with a warning.
  - You can perform a delta scan for files that you create or edit after a previous scan finishes successfully or the Browse job step ends with a warning in the current scan.
- The first data store scan is a Full scan. A delta scan will become a full scan if any of the following conditions apply:
  - The data store, classification policies, data domains, or scan definitions changed at any time.
  - When you run a Reset Classification Results job on a data store, the next data store scan will be a Full scan.
  - If you change the **Scan with Remote Agent** data store property from disabled to enabled, the next scan you run on the data store will perform a full scan.
- If files added to a data store have a timestamp that is before the timestamp for the previous successful scan or a previous scan that ended with a warning, the Delta Scan job skips the files.

Rules and Guidelines for Incremental Subject Registry Scans

Consider the following rules and guidelines for incremental scans:

- To enable incremental scans on structured sources, you must include a field with the `IncrementalScanField` property set to true in the entity file. Incremental scans of unstructured sources do not require the field.
- The field must include a timestamp value in the following format: `yyyy-MM-dd HH24:mm:ss`
- A scan fails with errors if you configure an incremental scan on the scan UI but do not configure the property in the entity file.
- The field must contain a timestamp value. The scan uses the value to verify if the data was added or updated after the last scan.
- The scan uses the values in this column to determine records updated or added after the last scan.
- For unstructured sources, if you add files after a domain discovery scan and then run an incremental Subject Registry scan, the scan skips files added after the last domain discovery scan. To avoid loss of Subject Registry information, run a domain discovery scan before you run an incremental Subject Registry scan on unstructured sources. Alternatively, you can include domain discovery in the Subject Registry scan configuration.
- If you set the *IncrementalScanField* value to true for more than one field, only the first field is considered.
- The scan considers the last updated time and adds records that have a last updated time greater than the time of previous scan.
- The scan does not consider the last updated time on records that you delete after the last scan. So an incremental scan does not remove records that you delete after the last scan from the golden record table. Run a full scan to remove deleted records.
- Search results on the UI and DSAR reports that you generate after an incremental scan might therefore include incorrect data for deleted records or fail.
- If you update a record such that it does not match with a subject, an incremental scan does not delete the link to the subject. This can result in incorrect data included in search results and in DSAR reports.

Editing a Scan

You can edit the scan name, data stores, scan options, notification settings, classification policies, or schedule. You cannot edit a scan if the scan is running.

1. Click **Manage** > **Scans**.

   A list of all scans appears. You can use the navigation filters to narrow the list of scans.
2. Select the check box next to the scan that you want to edit.
3. From the **Actions** menu, select **Edit**.

   The **Edit <scan name>** page appears.
4. Navigate to the scan properties that you want to edit.
5. Click **Save**.

Copying a Scan

You can copy a scan to create a scan with similar properties. When you copy a scan, you can edit the scan properties.

1. Click **Manage** > **Scans**.

   A list of scans appears.
2. Select the check box next to the scan that you want to edit.
3. From the **Actions** menu, select **Duplicate**.

   The **Duplicate <scan name>** page appears.
4. Use the panel tabs to navigate to the properties that you want to edit.
5. Click **Save**.

Deleting a Scan

You can delete a scan if the scan has a completed, scheduled, or terminated status. You cannot delete a scan that has a failed, paused, or stopped status.

1. Click **Manage** > **Scans**.

   A list of scans appears.
2. Select the check box next to the scan that you want to delete.
3. From the **Actions** menu, select **Delete**.

   A message appears to confirm that you want to delete the scan.
4. Click **Yes**.

Exporting Scans

You can export a list of scans to a CSV file. The export file includes a list of all scans and the corresponding scan summaries. If you filtered the scan list, the export file contains the scans that match the active filter conditions.

1. Click **Manage** > **Scans**.

   The **Scans** workspace appears.

2. From the **Actions** menu, click **Export**.

   A dialog box appears with the exported file.

   The export file is named `Scans.csv`. Save the file to a directory on your machine. If you export the file again, a new file downloads. The file name includes an incremental number such as `Scans (1).csv`.

3. Open the file.

CHAPTER 13

Scan Options

This chapter includes the following topics:

- Application Scan Options, 260
- Big Data Scan Options, 263
- Cloud Scan Options, 267
- Data Integration Scan Options, 270
- Database Management Scan Options, 273
- File Management Scan Options, 276
- NoSQL Scan Options, 279

Application Scan Options

Create an Application scan to scan an SAP data source. When you define an Application scan, you configure scan options to extract metadata from the database, perform data domain discovery, and calculate the total row count of sensitive columns. The scan options determine the job steps that the Application Scan job includes.

You can configure the following scan options:

Option	Description
Include Catalog Information	Extracts a metadata catalog from the data store. The Data Privacy Management repository must have at least one scan result that includes the catalog information before the Scan job can run data domain discovery or determine the row count.
Run Profiling	Determines the type of profiling that the Scan job runs to perform data domain discovery: - **Metadata**. Matches data domains with the source metadata. - **Data**. Matches data domains with the source data. You can also specify options for running a data profile on the data store.
Include Row Count	Calculates the total number of rows for each sensitive field that the Scan job identifies. This option might affect the Scan job performance. To optimize performance, run a scan with this option only. If you do not include the row count, the **Security Dashboard** does not display the risk cost for the data stores in the scan.

You can enable one or more of the options for a scan. However, some of the options depend on scan results from other options. For example, if you enable the Run Profiling option and the Data Privacy Management repository does not include scan results for the Include Catalog Information option, the Scan job fails.

You can run a scan once with all options enabled, or you can run a scan multiple times with different options enabled. For example, you can run a scan with the Include Catalog Information option enabled. Then, run a scan with the Run Profiling option enabled. Then, run a scan with the Include Row Count option enabled.

Related Topics:

- "Creating a Scan" on page 254

Metadata Profiling Scan Option

The metadata profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source metadata.

The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Metadata profiling is typically the fastest method of profiling. However, metadata profiling can generate more false positives than data profiling or metadata profiling with data profiling. A false positive can occur if a data domain uses a broad regular expression for the data domain metadata match condition.

For example, you configure a data domain metadata match condition to use a general expression that searches for "name" in column metadata. The Scan job identifies the FirstName, LastName, ProductName, and DepartmentName columns as a match for the data domain. However, the ProductName and DepartmentName columns are not sensitive. To reduce the chance of false positives, use the data profiling scan option or metadata profiling with data profiling scan option.

Data Profiling Scan Option

The **Data** profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source data.

The Scan job uses the data match conditions in a data domain to determine if a column matches the data domain based on the column data. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Data profiling is typically the most accurate and slowest method of profiling. You can choose the following options:

Data on Inferred Results of Metadata

The option enables a Scan job to run data domain discovery profiles. The profiles match data domains with the source metadata and data. The Scan job evaluates the data domains in the classification policies that you include in the scan.

The Scan job first performs metadata profiling, and then runs a data domain discovery profile that matches data domains with the source metadata. The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is no metadata match, the Scan job does not identify the column as sensitive. If there is a metadata match, the Scan job performs data profiling on the column.

The Scan job performs data profiling for the columns that the scan job identified as a metadata match. The Scan job runs a data domain discovery profile that matches data domains with the source data. The Scan job uses the data match conditions in a data domain to determine if a column matches a data

domain based on the column data. If there is a data match, the Scan job identifies the column as sensitive. If there is no data match, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling is typically faster than data profiling and slower than metadata profiling. Compared to metadata profiling, metadata profiling with data profiling reduces the amount of false positives in the scan results. A false positive occurs when a column is not sensitive, but the Scan job identifies the column as sensitive.

For example, a DepartmentName column might match a metadata match condition for a name string in a data domain. However, the data within the DepartmentName column does not match the data match conditions in the data domain. If you run a scan with the Metadata profiling option, the Scan job identifies the column as sensitive. If you run a scan with the Data profiling option or the Metadata profiling with Data profiling scan option, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling might increase the amount of false negatives in the scan results. A false negative occurs when a column is sensitive, but the Scan job does not identify the column as sensitive. A false negative can occur if the metadata contains poorly labeled columns.

Exclude Null

The Scan job ignores rows with null values within the sample set of rows in a column to calculate the conformance score. When you enable this option, the sensitivity status of a column might be more accurate.

For example, you enable the Exclude Null property and specify that the Scan job runs a profile on the first 10,000 rows of each column. However, a column has 1,000 null values in the first 10,000 rows. The Profiling job step bases the conformance score calculation on the 9,000 rows that have values. The Profiling job step does not increase the size of the sample.

Full Profiling

The Scan job performs data profiling for all data domain matches in the data store, including matches discovered in previous data store scans.

Sampling Technique

The following table describes the sampling options that determine the number of rows on which the Scan job runs a profile.

Sampling Option	Description
Auto Random	Runs the profile on a random sample size. The random sample size is based on the number of rows in the data store.
First <number>	Runs the profile on the number of rows that you select in sequential order. The scan starts with the first row of the table and reads the rows in sequential order up to the number of rows you selected. You can select 10,000, 100,000, or 500,000 rows.
No Sampling	Runs the profile on all rows of the data store. If the data store contains a large number of rows, the Scan job performance might be impacted.
Random <number>	Runs the profile on the number of rows that you select in random order. You can select 10,000, 100,000, or 500,000 rows.

Default is First 10,000.

Data Domain Match Criteria

The unit to measure the quantity of rows that must match the data domain for Data Privacy Management to identify the column as sensitive. Select one of the following options:

- **Percentage**. Evaluates the percentage of matched rows in the column with the minimum percentage for sensitivity specified in the **Conformance Score** property of the data domain. If the percentage of matched rows is equal to or exceeds the minimum percentage for sensitivity in the conformance score, Data Privacy Management identifies the column as sensitive.
- **Rows**. Evaluates the number of matched rows in the column with the number specified in the **Conformance Row Count** property of the data domain. If the number of matched rows is equal to or exceeds the conformance row count, Data Privacy Management identifies the column as sensitive.

Default is Percentage.

Big Data Scan Options

Create a Big Data scan to scan Cloudera Navigator, Hadoop Distributed File System (HDFS), or Hive data sources. The scan options determine the job steps that the Big Data Scan job includes.

You can configure the following scan options:

Option	Description
Identify Proliferation and Data Protection	For Cloudera Navigator data stores, identifies the lineage of sensitive data between scanned Hive data stores and identifies the protection status of sensitive data based on transformation details. For example, if a sensitive column is included in a Data Masking transformation, the Scan job determines that the column is protected.
Include Catalog Information	For HDFS and Hive data stores, extracts a metadata catalog from the data store. The Data Privacy Management repository must have at least one scan result that includes the catalog information before the Scan job can run data domain discovery or determine the row count.
Run Profiling	For HDFS and Hive data stores, determines the type of profiling that the Scan job runs to perform data domain discovery: - **Metadata**. Matches data domains with the source metadata. - **Data**. Matches data domains with the source data. You can also specify options for running a data profile on the data store.
Include Row Count	For Hive data stores, calculates the total number of rows for each sensitive field that the Scan job identifies. This option might affect the Scan job performance. To optimize performance, run a scan with this option only. If you do not include the row count, the **Security Dashboard** does not display the risk cost for the data stores in the scan.

You can enable one or more of the options for a scan. However, some of the options depend on scan results from other options. For example, if you enable the Run Profiling option and the Data Privacy Management repository does not include scan results for the Include Catalog Information option, the Scan job fails.

You can run a scan once with all options enabled, or you can run a scan multiple times with different options enabled. For example, you can run a scan with the Include Catalog Information option enabled. Then, run a scan with the Run Profiling option enabled. Then, run a scan with the Include Row Count option enabled.

Related Topics:

- "Creating a Scan" on page 254

Metadata Profiling Scan Option

The metadata profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source metadata.

The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Metadata profiling is typically the fastest method of profiling. However, metadata profiling can generate more false positives than data profiling or metadata profiling with data profiling. A false positive can occur if a data domain uses a broad regular expression for the data domain metadata match condition.

For example, you configure a data domain metadata match condition to use a general expression that searches for "name" in column metadata. The Scan job identifies the FirstName, LastName, ProductName, and DepartmentName columns as a match for the data domain. However, the ProductName and DepartmentName columns are not sensitive. To reduce the chance of false positives, use the data profiling scan option or metadata profiling with data profiling scan option.

Data Profiling Scan Option

The **Data** profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source data.

The Scan job uses the data match conditions in a data domain to determine if a column matches the data domain based on the column data. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Data profiling is typically the most accurate and slowest method of profiling. You can choose the following options:

Data on Inferred Results of Metadata

The option enables a Scan job to run data domain discovery profiles. The profiles match data domains with the source metadata and data. The Scan job evaluates the data domains in the classification policies that you include in the scan.

The Scan job first performs metadata profiling, and then runs a data domain discovery profile that matches data domains with the source metadata. The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is no metadata match, the Scan job does not identify the column as sensitive. If there is a metadata match, the Scan job performs data profiling on the column.

The Scan job performs data profiling for the columns that the scan job identified as a metadata match. The Scan job runs a data domain discovery profile that matches data domains with the source data. The Scan job uses the data match conditions in a data domain to determine if a column matches a data domain based on the column data. If there is a data match, the Scan job identifies the column as sensitive. If there is no data match, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling is typically faster than data profiling and slower than metadata profiling. Compared to metadata profiling, metadata profiling with data profiling reduces the amount of false positives in the scan results. A false positive occurs when a column is not sensitive, but the Scan job identifies the column as sensitive.

For example, a DepartmentName column might match a metadata match condition for a name string in a data domain. However, the data within the DepartmentName column does not match the data match conditions in the data domain. If you run a scan with the Metadata profiling option, the Scan job identifies the column as sensitive. If you run a scan with the Data profiling option or the Metadata profiling with Data profiling scan option, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling might increase the amount of false negatives in the scan results. A false negative occurs when a column is sensitive, but the Scan job does not identify the column as sensitive. A false negative can occur if the metadata contains poorly labeled columns.

Exclude Null

The Scan job ignores rows with null values within the sample set of rows in a column to calculate the conformance score. When you enable this option, the sensitivity status of a column might be more accurate.

For example, you enable the Exclude Null property and specify that the Scan job runs a profile on the first 10,000 rows of each column. However, a column has 1,000 null values in the first 10,000 rows. The Profiling job step bases the conformance score calculation on the 9,000 rows that have values. The Profiling job step does not increase the size of the sample.

Full Profiling

The Scan job performs data profiling for all data domain matches in the data store, including matches discovered in previous data store scans.

Sampling Technique

The following table describes the sampling options that determine the number of rows on which the Scan job runs a profile.

Sampling Option	Description
Auto Random	Runs the profile on a random sample size. The random sample size is based on the number of rows in the data store.
First <number>	Runs the profile on the number of rows that you select in sequential order. The scan starts with the first row of the table and reads the rows in sequential order up to the number of rows you selected. You can select 10,000, 100,000, or 500,000 rows.
No Sampling	Runs the profile on all rows of the data store. If the data store contains a large number of rows, the Scan job performance might be impacted.
Random <number>	Runs the profile on the number of rows that you select in random order. You can select 10,000, 100,000, or 500,000 rows.

Default is First 10,000.

Data Domain Match Criteria

The unit to measure the quantity of rows that must match the data domain for Data Privacy Management to identify the column as sensitive. Select one of the following options:

- **Percentage**. Evaluates the percentage of matched rows in the column with the minimum percentage for sensitivity specified in the **Conformance Score** property of the data domain. If the percentage of matched rows is equal to or exceeds the minimum percentage for sensitivity in the conformance score, Data Privacy Management identifies the column as sensitive.

- **Rows**. Evaluates the number of matched rows in the column with the number specified in the **Conformance Row Count** property of the data domain. If the number of matched rows is equal to or exceeds the conformance row count, Data Privacy Management identifies the column as sensitive.

Default is Percentage.

Process for Cloudera Navigator

The sequence in which you create and run scans for Cloudera Navigator data stores determines the accuracy and completeness of the scan results.

Perform the following steps to create and run scans:

Step	Description	Details
1	Create Hive data stores	To show complete lineage between each schema, create one Hive data store for each schema and configure connection details. A Big Data Scan job determines lineage of sensitive data between scanned Hive data stores. If one Hive data store contains multiple schemas and sensitive data moves between the schemas, the Big Data Scan job cannot show sensitive data lineage between schemas in the same data store.
2	Scan the Hive data stores to identify sensitive data	Run a Domain Discovery scan for the Hive data stores.
3	Create a Cloudera Navigator data store	Configure the connection details to the Cloudera Navigator Metadata Server that is on the same host as the Hive data stores.
4	Scan the Cloudera Navigator data store to identify the proliferation of sensitive data	The scan job identifies sensitive data proliferation between Hive data stores that are on the same host as the Cloudera Navigator data store.

Cloud Scan Options

Create a Cloud scan to scan Amazon Redshift, Amazon S3, Azure Data Lake, Microsoft Azure Blob Storage, Microsoft Azure SQL Data Warehouse, Microsoft Azure SQL Database, Salesforce, or Snowflake data stores.

You can configure the following scan options:

Option	Description
Include Catalog Information	Extracts a metadata catalog from the data store. The Data Privacy Management repository must have at least one scan result that includes the catalog information before the scan job can run data domain discovery or determine the row count.
Run Profiling	Determines the type of profiling that the scan job runs to perform data domain discovery: - **Metadata**. Matches data domains with the source metadata. - **Data**. Matches data domains with the source data. You can also specify options for running a data profile on the data store. If a Cloud scan includes Amazon S3 data stores that scan with a remote agent, the scan job performs data profiling, but you do not select options when you create the scan. If a Cloud scan includes data stores that scan with a remote agent and data stores that do not scan with a remote agent, you select profiling options. The scan job applies the data profiling options to the data stores that do not scan with a remote agent when you run the scan.
Include Row Count	For Amazon Redshift, Microsoft Azure SQL Data Warehouse, Microsoft Azure SQL Database, Salesforce, and Snowflake data stores, calculates the total number of rows for each sensitive field that the Scan job identifies. This option might affect the Scan job performance. To optimize performance, run a scan with this option only. If you do not include the row count, the **Security Dashboard** does not display the risk cost for the data stores in the scan.

You can enable one or more of the options for a scan. However, some of the options depend on scan results from other options. For example, if you enable the Run Profiling option and the Data Privacy Management repository does not include scan results for the Include Catalog Information option, the Scan job fails.

You can run a scan once with all options enabled, or you can run a scan multiple times with different options enabled. For example, you can run a scan with the Include Catalog Information option enabled. Then, run a scan with the Run Profiling option enabled. Then, run a scan with the Include Row Count option enabled.

Related Topics:

- "Creating a Scan" on page 254
- "Cloud Job" on page 293

Metadata Profiling Scan Option

The metadata profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source metadata.

The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Metadata profiling is typically the fastest method of profiling. However, metadata profiling can generate more false positives than data profiling or metadata profiling with data profiling. A false positive can occur if a data domain uses a broad regular expression for the data domain metadata match condition.

For example, you configure a data domain metadata match condition to use a general expression that searches for "name" in column metadata. The Scan job identifies the FirstName, LastName, ProductName, and DepartmentName columns as a match for the data domain. However, the ProductName and DepartmentName columns are not sensitive. To reduce the chance of false positives, use the data profiling scan option or metadata profiling with data profiling scan option.

Data Profiling Scan Option

The **Data** profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source data.

The scan job uses the data match conditions in a data domain to determine if a column matches the data domain based on the column data. If there is a match, the scan job identifies the column as sensitive. The scan job evaluates the data domains in the classification policies that you include in the scan.

Data profiling is typically the most accurate and slowest method of profiling. You can choose the following options:

Data on Inferred Results of Metadata

The option enables a Scan job to run data domain discovery profiles. The profiles match data domains with the source metadata and data. The Scan job evaluates the data domains in the classification policies that you include in the scan.

The Scan job first performs metadata profiling, and then runs a data domain discovery profile that matches data domains with the source metadata. The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is no metadata match, the Scan job does not identify the column as sensitive. If there is a metadata match, the Scan job performs data profiling on the column.

The Scan job performs data profiling for the columns that the scan job identified as a metadata match. The Scan job runs a data domain discovery profile that matches data domains with the source data. The Scan job uses the data match conditions in a data domain to determine if a column matches a data domain based on the column data. If there is a data match, the Scan job identifies the column as sensitive. If there is no data match, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling is typically faster than data profiling and slower than metadata profiling. Compared to metadata profiling, metadata profiling with data profiling reduces the amount of false positives in the scan results. A false positive occurs when a column is not sensitive, but the Scan job identifies the column as sensitive.

For example, a DepartmentName column might match a metadata match condition for a name string in a data domain. However, the data within the DepartmentName column does not match the data match conditions in the data domain. If you run a scan with the Metadata profiling option, the Scan job identifies the column as sensitive. If you run a scan with the Data profiling option or the Metadata profiling with Data profiling scan option, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling might increase the amount of false negatives in the scan results. A false negative occurs when a column is sensitive, but the Scan job does not identify the column as sensitive. A false negative can occur if the metadata contains poorly labeled columns.

Exclude Null

The Scan job ignores rows with null values within the sample set of rows in a column to calculate the conformance score. When you enable this option, the sensitivity status of a column might be more accurate.

For example, you enable the Exclude Null property and specify that the Scan job runs a profile on the first 10,000 rows of each column. However, a column has 1,000 null values in the first 10,000 rows. The

Profiling job step bases the conformance score calculation on the 9,000 rows that have values. The Profiling job step does not increase the size of the sample.

Full Profiling

The Scan job performs data profiling for all data domain matches in the data store, including matches discovered in previous data store scans.

Sampling Technique

The following table describes the sampling options that determine the number of rows on which the scan job runs a profile.

Sampling Option	Description
Auto Random	Runs the profile on a random sample size. The random sample size is based on the number of rows in the data store. This option is not available for Amazon S3 data stores.
First <number>	Runs the profile on the number of rows that you select in sequential order. The scan starts with the first row of the table and reads the rows in sequential order up to the number of rows you selected. You can select 10,000, 100,000, or 500,000 rows.
No Sampling	Runs the profile on all rows of the data store. If the data store contains a large number of rows, the Scan job performance might be impacted.
Random <number>	Runs the profile on the number of rows that you select in random order. You can select 10,000, 100,000, or 500,000 rows. This option is not available for Cloud scans for Amazon S3 data stores.

Default is First 10,000.

Data Domain Match Criteria

The unit to measure the quantity of rows that must match the data domain for Data Privacy Management to identify the column as sensitive. Select one of the following options:

- **Percentage**. Evaluates the percentage of matched rows in the column with the minimum percentage for sensitivity specified in the **Conformance Score** property of the data domain. If the percentage of matched rows is equal to or exceeds the minimum percentage for sensitivity in the conformance score, Data Privacy Management identifies the column as sensitive.
- **Rows**. Evaluates the number of matched rows in the column with the number specified in the **Conformance Row Count** property of the data domain. If the number of matched rows is equal to or exceeds the conformance row count, Data Privacy Management identifies the column as sensitive.

Default is Percentage.

Data Integration Scan Options

Create a Data Integration scan to scan Informatica Cloud, Informatica Data Engineering Integration, Informatica PowerCenter, or SQL Server Integration Services sources. The scan options determine the job steps that the Data Integration Scan job includes.

You can configure the following scan options:

Option	Description
Collect Data Store Information	For Informatica Cloud and Informatica Data Engineering Integration data stores, imports connections from the data source and creates data stores in the Data Privacy Management repository. The Scan job creates one data store for each connection in Informatica Cloud. Select one of the following options: - **Retrieve Security Groups**. The imported data stores will have the same security group as the parent data store - **Select Security Groups**. Select one or more security groups for the imported data stores.
Include Data Stores and Catalog Information	For Informatica PowerCenter data stores, imports connections from the PowerCenter repository and creates data stores in the Data Privacy Management repository. The Scan job extracts a metadata catalog and identifies sensitive data in the PowerCenter repository for each data store. The Scan job identifies connection-level data lineage to show data flow between the data stores. For example, the Scan job identifies that data moves from data store A to data store B. The Scan job evaluates the combination of sensitive data in each data store to determine if a data store matches a classification policy. Select one of the following options: - **Retrieve Security Groups**. The imported data stores will have the same security group as the parent data store - **Select Security Groups**. Select one or more security groups for the imported data stores.
Identify Proliferation and Data Protection	Extracts a metadata catalog from the source database, identifies the lineage of sensitive data, and identifies the protection status of sensitive data based on transformation details. For example, if a sensitive column is included in a Data Masking transformation, the Scan job determines that the column is protected. If you enable the option and the Data Privacy Management repository does not have scan results for the Collect Data Store Information option, the Data Integration Scan job fails.

You can enable one or more of the options for a scan. However, some of the options depend on scan results from other options. For example, if you enable the Run Profiling option and the Data Privacy Management repository does not include scan results for the Include Catalog Information option, the Scan job fails.

You can run a scan once with all options enabled, or you can run a scan multiple times with different options enabled. For example, you can run a scan with the Include Catalog Information option enabled. Then, run a scan with the Run Profiling option enabled. Then, run a scan with the Include Row Count option enabled.

Related Topics:

- "Creating a Scan" on page 254
- "Data Integration Job" on page 294

Process for Informatica Cloud

The sequence in which you create and run scans determines the accuracy and completeness of the scan results. Before you create scans, consider the data stores that you want to scan and the sequence in which to run scans.

Complete the following steps to create and run scans:

Step 1. Create an Informatica Cloud data store.

Create a data store and configure connection details to Informatica Cloud.

Step 2. Create and run a Cloud scan to import data stores.

Create and run a Cloud scan to import data stores from Informatica Cloud.

After you run the scan, monitor the scan job. Edit the imported data stores to update the properties that the scan job did not complete. For example, the scan job does not import passwords or connections strings. The scan job does not import any connection properties for Amazon S3 data stores. Verify if there are duplicate data stores that you need to merge. Then, configure connectivity to the imported data stores.

Step 3. Create and run scans to identify sensitive data for the imported data stores.

Scan the data stores that the Cloud Scan job imported. The type of scan you run depends on the repository type of the imported data store. For example, if the scan job imported an Oracle data store, run a Database Management scan. The scan job identifies and classifies sensitive data in each data store.

After you run the scan, monitor the scan job and curate the profile results.

Step 4. Run a Cloud scan to identify the proliferation of sensitive data in the imported data stores.

Run a Cloud scan to identify proliferation of sensitive data in mappings. The Scan job identifies column-level lineage of sensitive data between the imported data stores. The Scan job identifies if Data Masking transformations mask the sensitive data.

After you run the scan, monitor the scan job.

Process for Informatica Data Engineering Integration

The sequence in which you create and run scans determines the accuracy and completeness of the scan results. Before you create scans, consider the data stores that you want to scan and the sequence in which to run scans.

Complete the following steps to create and run scans:

Step 1. Run a Data Integration scan to import data stores.

Run a Data Integration scan to import data stores from Data Engineering Integration. The Scan job creates one data store in the Data Privacy Management repository for each connection in Informatica Data Engineering Integration.

After you run the scan, monitor the Scan job and verify the imported data stores. For example, verify if there are duplicate or incomplete data stores that you need to manage. Configure connectivity to the imported data stores.

Step 2. Scan the imported data stores to identify sensitive data.

Scan the data stores that the Data Integration Scan job imported. The type of scan you run depends on the repository type of the imported data store. For example, if the Data Integration Scan job imported a Hive data store, run a Big Data scan. The Scan job identifies and classifies sensitive data in each data store.

After you run the scan, monitor the Scan job and curate the profile results.

Step 3. Run a Data Integration scan to identify the proliferation and protection status of sensitive data.

Run a DataIntegration scan to identify proliferation of sensitive data in mappings. The Scan job identifies column-level lineage of sensitive data between sources and targets. The Scan job identifies if Data Masking transformations mask the sensitive data.

After you run the scan, monitor the Scan job.

Process for Informatica PowerCenter

The sequence in which you create and run scans determines the accuracy and completeness of the scan results. Before you create scans, consider the data stores that you want to scan and the sequence in which to run scans.

Complete the following steps to create and run scans:

Step 1. Run a Data Integration Scan to import data stores.

Run a Data Integration scan to import data stores from a PowerCenter repository. The Scan job identifies and classifies sensitive data in PowerCenter mappings. The Scan job identifies an approximate connection-level lineage of sensitive data between sources and targets.

After you run the scan, monitor the Scan job and verify the imported data stores. For example, verify if there are duplicate or incomplete data stores that you need to manage. Configure connectivity to the imported data stores.

Step 2. Run a Database Management scan to identify sensitive data for the imported data stores.

Run a Database Management scan to scan the data stores that the Data Integration Scan job imported. The Database Management Scan job identifies and classifies sensitive data in each database. The Scan job evaluates data that is not included in PowerCenter mappings.

After you run the scan, monitor the Scan job and curate the profile results.

Step 3. Run a Data Integration scan to identify the proliferation of sensitive data.

Run a Data Integration scan to identify proliferation of sensitive data in PowerCenter mappings. The Scan job identifies column-level lineage of sensitive data between sources and targets. The Scan job identifies if Data Masking transformations mask the sensitive data.

After you run the scan, monitor the Scan job.

Process for SQL Server Integration Services

The sequence in which you create and run scans determines the accuracy and completeness of the scan results. Before you create scans, consider the data stores that you want to scan and the sequence in which to run scans.

Perform the following steps to create and run scans:

Step 1. Create a data store for each connection in a SQL Server Integration Services mapping.

Create a data store and configure connection details to each supported connection in a SQL Server Integration Services mapping. Create a data store for each OLE DB Source and OLE DB Destination that connects to SQL Server Native Client. When you create the data store, use the category **Database Management** and the data store type **Microsoft SQL Server**.

The data stores will be the child data stores of the SQL Server Integration Services data store that you create in step 3.

Step 2. Create and run a scan to identify sensitive data in the child data stores.

Create and run a scan on the data stores that you created in the previous step. The Scan job identifies and classifies sensitive data in the data stores.

After you run the scan, monitor the Scan job and curate the profile results.

Step 3. Create a data store that connects to the SQL Server Integration Services package.

Create a data store and configure connection details to the SQL Server Integration Services package that includes the mapping from step 1. The mapping in the package should include the connections for which you created data stores.

The type of data store that you create depends on the server that hosts the packages. You can create a file-based data store to scan packages that you created in Microsoft Visual Studio, but do not exist in Microsoft SQL Server Management Studio. You can create a repository-based data store to scan packages that exist in the Microsoft SQL Server Management Studio repository.

Step 4. Create and run a scan on the SQL Server Integration Services data store to identify the proliferation of sensitive data in the child data stores.

Create and run a scan on the SQL Server Integration Services data store. The Data Integration Scan job identifies the proliferation of sensitive data between the connections in the mapping.

After you run the scan, monitor the Scan job.

Important: After you run the first SQL Server Integration Services scan, you must run the scan again after you scan a child data store again. For example, you scan a child data store again because you added more columns in the mapping. Another example, you run a scan on a child data store every month to identify sensitive data. After the Scan job identifies sensitive data in the child data store, you must run the SQL Server Integration Services scan again to update the proliferation of the sensitive data.

Database Management Scan Options

Create a Database Management scan to scan IBM Db2, JDBC, Microsoft SQL Server, Netezza, Oracle, SAP HANA, Sybase, and Teradata data stores. When you define a Database Management scan, you configure

options to extract metadata from the database, perform data domain discovery, and calculate the total row count of sensitive columns.

You can configure the following scan options:

Option	Description
Include Catalog Information	Extracts a metadata catalog from the data store. The Data Privacy Management repository must have at least one scan result that includes the catalog information before the Scan job can run data domain discovery or determine the row count.
Run Profiling	Determines the type of profiling that the Scan job runs to perform data domain discovery: - **Metadata**. Matches data domains with the source metadata. - **Data**. Matches data domains with the source data. You can also specify options for running a data profile on the data store.
Include Row Count	Calculates the total number of rows for each sensitive field that the Scan job identifies. This option might affect the Scan job performance. To optimize performance, run a scan with this option only. If you do not include the row count, the **Security Dashboard** does not display the risk cost for the data stores in the scan.

You can enable one or more of the options for a scan. However, some of the options depend on scan results from other options. For example, if you enable the Run Profiling option and the Data Privacy Management repository does not include scan results for the Include Catalog Information option, the Scan job fails.

You can run a scan once with all options enabled, or you can run a scan multiple times with different options enabled. For example, you can run a scan with the Include Catalog Information option enabled. Then, run a scan with the Run Profiling option enabled. Then, run a scan with the Include Row Count option enabled.

Related Topics:

- "Creating a Scan" on page 254
- "Database Management Job" on page 296

Metadata Profiling Scan Option

The metadata profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source metadata.

The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Metadata profiling is typically the fastest method of profiling. However, metadata profiling can generate more false positives than data profiling or metadata profiling with data profiling. A false positive can occur if a data domain uses a broad regular expression for the data domain metadata match condition.

For example, you configure a data domain metadata match condition to use a general expression that searches for "name" in column metadata. The Scan job identifies the FirstName, LastName, ProductName, and DepartmentName columns as a match for the data domain. However, the ProductName and DepartmentName columns are not sensitive. To reduce the chance of false positives, use the data profiling scan option or metadata profiling with data profiling scan option.

Data Profiling Scan Option

The **Data** profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source data.

The Scan job uses the data match conditions in a data domain to determine if a column matches the data domain based on the column data. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Data profiling is typically the most accurate and slowest method of profiling. You can choose the following options:

Data on Inferred Results of Metadata

The option enables a Scan job to run data domain discovery profiles. The profiles match data domains with the source metadata and data. The Scan job evaluates the data domains in the classification policies that you include in the scan.

The Scan job first performs metadata profiling, and then runs a data domain discovery profile that matches data domains with the source metadata. The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is no metadata match, the Scan job does not identify the column as sensitive. If there is a metadata match, the Scan job performs data profiling on the column.

The Scan job performs data profiling for the columns that the scan job identified as a metadata match. The Scan job runs a data domain discovery profile that matches data domains with the source data. The Scan job uses the data match conditions in a data domain to determine if a column matches a data domain based on the column data. If there is a data match, the Scan job identifies the column as sensitive. If there is no data match, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling is typically faster than data profiling and slower than metadata profiling. Compared to metadata profiling, metadata profiling with data profiling reduces the amount of false positives in the scan results. A false positive occurs when a column is not sensitive, but the Scan job identifies the column as sensitive.

For example, a DepartmentName column might match a metadata match condition for a name string in a data domain. However, the data within the DepartmentName column does not match the data match conditions in the data domain. If you run a scan with the Metadata profiling option, the Scan job identifies the column as sensitive. If you run a scan with the Data profiling option or the Metadata profiling with Data profiling scan option, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling might increase the amount of false negatives in the scan results. A false negative occurs when a column is sensitive, but the Scan job does not identify the column as sensitive. A false negative can occur if the metadata contains poorly labeled columns.

Exclude Null

The Scan job ignores rows with null values within the sample set of rows in a column to calculate the conformance score. When you enable this option, the sensitivity status of a column might be more accurate.

For example, you enable the Exclude Null property and specify that the Scan job runs a profile on the first 10,000 rows of each column. However, a column has 1,000 null values in the first 10,000 rows. The Profiling job step bases the conformance score calculation on the 9,000 rows that have values. The Profiling job step does not increase the size of the sample.

Full Profiling

The Scan job performs data profiling for all data domain matches in the data store, including matches discovered in previous data store scans.

Sampling Technique

The following table describes the sampling options that determine the number of rows on which the Scan job runs a profile.

Sampling Option	Description
Auto Random	Runs the profile on a random sample size. The random sample size is based on the number of rows in the data store.
First <number>	Runs the profile on the number of rows that you select in sequential order. The scan starts with the first row of the table and reads the rows in sequential order up to the number of rows you selected. You can select 10,000, 100,000, or 500,000 rows.
No Sampling	Runs the profile on all rows of the data store. If the data store contains a large number of rows, the Scan job performance might be impacted.
Random <number>	Runs the profile on the number of rows that you select in random order. You can select 10,000, 100,000, or 500,000 rows.

Default is First 10,000.

Data Domain Match Criteria

The unit to measure the quantity of rows that must match the data domain for Data Privacy Management to identify the column as sensitive. Select one of the following options:

- **Percentage**. Evaluates the percentage of matched rows in the column with the minimum percentage for sensitivity specified in the **Conformance Score** property of the data domain. If the percentage of matched rows is equal to or exceeds the minimum percentage for sensitivity in the conformance score, Data Privacy Management identifies the column as sensitive.
- **Rows**. Evaluates the number of matched rows in the column with the number specified in the **Conformance Row Count** property of the data domain. If the number of matched rows is equal to or exceeds the conformance row count, Data Privacy Management identifies the column as sensitive.

Default is Percentage.

File Management Scan Options

Create a File Management scan to scan File System, Google Drive, Microsoft OneDrive, or Microsoft SharePoint data stores.

For data stores that enable the **Scan with Remote Agent** option, the scan job performs data profiling but you do not select profiling options when you create the scan. If you create a scan that includes data stores that scan with a remote agent and data stores that do not scan with a remote agent, you select profiling options. The scan job applies the data profiling options to the data stores that do not scan with a remote agent when you run the scan.

For File System, Microsoft OneDrive, and Microsoft SharePoint data stores that do not scan with a remote agent, you can configure the following scan options:

Option	Description
Include Catalog Information	Extracts a metadata catalog from the data store. The Data Privacy Management repository must have at least one scan result that includes the catalog information before the Scan job can run data domain discovery or determine the row count.
Run Profiling	Determines the type of profiling that the Scan job runs to perform data domain discovery: - **Metadata**. Matches data domains with the source metadata. - **Data**. Matches data domains with the source data. You can also specify options for running a data profile on the data store.

You can enable one or more of the options for a scan. However, some of the options depend on scan results from other options. For example, if you enable the Run Profiling option and the Data Privacy Management repository does not include scan results for the Include Catalog Information option, the Scan job fails.

You can run a scan once with all options enabled, or you can run a scan multiple times with different options enabled. For example, you can run a scan with the Include Catalog Information option enabled. Then, run a scan with the Run Profiling option enabled.

Related Topics:

- "Creating a Scan" on page 254
- "File Management Job" on page 298

Metadata Profiling Scan Option

The metadata profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source metadata.

The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Metadata profiling is typically the fastest method of profiling. However, metadata profiling can generate more false positives than data profiling or metadata profiling with data profiling. A false positive can occur if a data domain uses a broad regular expression for the data domain metadata match condition.

For example, you configure a data domain metadata match condition to use a general expression that searches for "name" in column metadata. The Scan job identifies the FirstName, LastName, ProductName, and DepartmentName columns as a match for the data domain. However, the ProductName and DepartmentName columns are not sensitive. To reduce the chance of false positives, use the data profiling scan option or metadata profiling with data profiling scan option.

Data Profiling Scan Option

The **Data** profiling scan option enables the scan job to run a data domain discovery profile that matches data domains with the source data.

The scan job uses the data match conditions in a data domain to determine if a field matches the data domain based. If there is a match, the scan job identifies the field as sensitive. The scan job evaluates the data domains in the classification policies that you include in the scan.

Data profiling is typically the most accurate and slowest method of profiling. You can choose the following options:

Data on Inferred Results of Metadata

The option enables a Scan job to run data domain discovery profiles. The profiles match data domains with the source metadata and data. The Scan job evaluates the data domains in the classification policies that you include in the scan.

The Scan job first performs metadata profiling, and then runs a data domain discovery profile that matches data domains with the source metadata. The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is no metadata match, the Scan job does not identify the column as sensitive. If there is a metadata match, the Scan job performs data profiling on the column.

The Scan job performs data profiling for the columns that the scan job identified as a metadata match. The Scan job runs a data domain discovery profile that matches data domains with the source data. The Scan job uses the data match conditions in a data domain to determine if a column matches a data domain based on the column data. If there is a data match, the Scan job identifies the column as sensitive. If there is no data match, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling is typically faster than data profiling and slower than metadata profiling. Compared to metadata profiling, metadata profiling with data profiling reduces the amount of false positives in the scan results. A false positive occurs when a column is not sensitive, but the Scan job identifies the column as sensitive.

For example, a DepartmentName column might match a metadata match condition for a name string in a data domain. However, the data within the DepartmentName column does not match the data match conditions in the data domain. If you run a scan with the Metadata profiling option, the Scan job identifies the column as sensitive. If you run a scan with the Data profiling option or the Metadata profiling with Data profiling scan option, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling might increase the amount of false negatives in the scan results. A false negative occurs when a column is sensitive, but the Scan job does not identify the column as sensitive. A false negative can occur if the metadata contains poorly labeled columns.

Exclude Null

The Scan job ignores rows with null values within the sample set of rows in a column to calculate the conformance score. When you enable this option, the sensitivity status of a column might be more accurate.

For example, you enable the Exclude Null property and specify that the Scan job runs a profile on the first 10,000 rows of each column. However, a column has 1,000 null values in the first 10,000 rows. The Profiling job step bases the conformance score calculation on the 9,000 rows that have values. The Profiling job step does not increase the size of the sample.

Sampling Technique

You can configure sampling options for faster profiling on the source data. The following table describes the sampling options that determine the number of rows on which the Scan job runs a profile.

Sampling Option	Description
First <number>	Runs the profile on the number of rows that you select in sequential order. The scan starts with the first row of the table and reads the rows in sequential order up to the number of rows you selected. You can select 10,000, 100,000, or 500,000 rows. Not valid if the file system data store is enabled for JSON and XML file types.
No Sampling	Runs the profile on all rows of the data store. If the data store contains a large number of rows, the Scan job performance might be impacted. **Note:** If the scan includes unstructured data stores, you must select this option for the Scan job to finish successfully.

Data Domain Match Criteria

The unit to measure the quantity of rows that must match the data domain for Data Privacy Management to identify the column as sensitive.

You must select **Rows**. Data Privacy Management evaluates the number of matched rows in the column with the number specified in the Conformance Row Count property of the data domain. If the number of matched rows is equal to or exceeds the conformance row count, Data Privacy Management identifies the column as sensitive.

NoSQL Scan Options

Create a NoSQL scan to scan Apache Cassandra data stores. When you define a NoSQL scan, you configure options to extract metadata from the database, perform data domain discovery, and calculate the total row count of sensitive columns.

You can configure the following scan options:

Option	Description
Include Catalog Information	Extracts a metadata catalog from the data store. The Data Privacy Management repository must have at least one scan result that includes the catalog information before the Scan job can run data domain discovery or determine the row count.
Run Profiling	Determines the type of profiling that the Scan job runs to perform data domain discovery: - **Metadata**. Matches data domains with the source metadata. - **Data**. Matches data domains with the source data. You can also specify options for running a data profile on the data store.
Include Row Count	Calculates the total number of rows for each sensitive field that the Scan job identifies. This option might affect the Scan job performance. To optimize performance, run a scan with this option only. If you do not include the row count, the **Security Dashboard** does not display the risk cost for the data stores in the scan.

You can enable one or more of the options for a scan. However, some of the options depend on scan results from other options. For example, if you enable the Run Profiling option and the Data Privacy Management repository does not include scan results for the Include Catalog Information option, the Scan job fails.

You can run a scan once with all options enabled, or you can run a scan multiple times with different options enabled. For example, you can run a scan with the Include Catalog Information option enabled. Then, run a scan with the Run Profiling option enabled. Then, run a scan with the Include Row Count option enabled.

Note: You cannot include keyspaces, tables, or columns that include uppercase characters in a domain discovery scan on an Apache Cassandra data store.

Metadata Profiling Scan Option

The metadata profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source metadata.

The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Metadata profiling is typically the fastest method of profiling. However, metadata profiling can generate more false positives than data profiling or metadata profiling with data profiling. A false positive can occur if a data domain uses a broad regular expression for the data domain metadata match condition.

For example, you configure a data domain metadata match condition to use a general expression that searches for "name" in column metadata. The Scan job identifies the FirstName, LastName, ProductName, and DepartmentName columns as a match for the data domain. However, the ProductName and DepartmentName columns are not sensitive. To reduce the chance of false positives, use the data profiling scan option or metadata profiling with data profiling scan option.

Data Profiling Scan Option (Cassandra)

The **Data** profiling scan option enables the Scan job to run a data domain discovery profile that matches data domains with the source data.

The Scan job uses the data match conditions in a data domain to determine if a column matches the data domain based on the column data. If there is a match, the Scan job identifies the column as sensitive. The Scan job evaluates the data domains in the classification policies that you include in the scan.

Data profiling is typically the most accurate and slowest method of profiling. You can choose the following options:

Data on Inferred Results of Metadata

The option enables a Scan job to run data domain discovery profiles. The profiles match data domains with the source metadata and data. The Scan job evaluates the data domains in the classification policies that you include in the scan.

The Scan job first performs metadata profiling, and then runs a data domain discovery profile that matches data domains with the source metadata. The Scan job uses the metadata match conditions in a data domain to determine if a column matches a data domain based on the column name. If there is no metadata match, the Scan job does not identify the column as sensitive. If there is a metadata match, the Scan job performs data profiling on the column.

The Scan job performs data profiling for the columns that the scan job identified as a metadata match. The Scan job runs a data domain discovery profile that matches data domains with the source data. The Scan job uses the data match conditions in a data domain to determine if a column matches a data

domain based on the column data. If there is a data match, the Scan job identifies the column as sensitive. If there is no data match, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling is typically faster than data profiling and slower than metadata profiling. Compared to metadata profiling, metadata profiling with data profiling reduces the amount of false positives in the scan results. A false positive occurs when a column is not sensitive, but the Scan job identifies the column as sensitive.

For example, a DepartmentName column might match a metadata match condition for a name string in a data domain. However, the data within the DepartmentName column does not match the data match conditions in the data domain. If you run a scan with the Metadata profiling option, the Scan job identifies the column as sensitive. If you run a scan with the Data profiling option or the Metadata profiling with Data profiling scan option, the Scan job does not identify the column as sensitive.

Metadata profiling with data profiling might increase the amount of false negatives in the scan results. A false negative occurs when a column is sensitive, but the Scan job does not identify the column as sensitive. A false negative can occur if the metadata contains poorly labeled columns.

Exclude Null

The Scan job ignores rows with null values within the sample set of rows in a column to calculate the conformance score. When you enable this option, the sensitivity status of a column might be more accurate.

For example, you enable the Exclude Null property and specify that the Scan job runs a profile on the first 10,000 rows of each column. However, a column has 1,000 null values in the first 10,000 rows. The Profiling job step bases the conformance score calculation on the 9,000 rows that have values. The Profiling job step does not increase the size of the sample.

Full Profiling

The Scan job performs data profiling for all data domain matches in the data store, including matches discovered in previous data store scans.

Sampling Technique

The following table describes the sampling options that determine the number of rows on which the Scan job runs a profile.

Note: You cannot use random sampling options.

Sampling Option	Description
First <number>	Runs the profile on the number of rows that you select in sequential order. The scan starts with the first row of the table and reads the rows in sequential order up to the number of rows you selected. You can select 10,000, 100,000, or 500,000 rows.
No Sampling	Runs the profile on all rows of the data store. If the data store contains a large number of rows, the Scan job performance might be impacted.

Default is First 10,000.

Data Domain Match Criteria

The unit to measure the quantity of rows that must match the data domain for Data Privacy Management to identify the column as sensitive. Select one of the following options:

- **Percentage**. Evaluates the percentage of matched rows in the column with the minimum percentage for sensitivity specified in the **Conformance Score** property of the data domain. If the percentage of matched rows is equal to or exceeds the minimum percentage for sensitivity in the conformance score, Data Privacy Management identifies the column as sensitive.
- **Rows**. Evaluates the number of matched rows in the column with the number specified in the **Conformance Row Count** property of the data domain. If the number of matched rows is equal to or exceeds the conformance row count, Data Privacy Management identifies the column as sensitive.

Default is Percentage.

CHAPTER 14

Jobs

This chapter includes the following topics:

- Jobs Overview, 284
- Jobs Workspace, 284
- Job Types, 287
- Big Data Job, 291
- Cloud Job, 293
- Data Integration Job, 294
- Database Management Job, 296
- Evaluate Classification Policies Job, 297
- Evaluate Risk Scores Job, 298
- Evaluate Security Policy Job, 298
- File Management Job, 298
- Import Job, 299
- Import Catalog Results Job, 300
- Incremental Scan Job, 301
- Orchestration Job, 301
- Protection Job, 302
- Reset Classification Results Job, 303
- Salesforce Import job, 303
- Subject Data Report Purge Job, 305
- Subject Registry Scan Job, 305
- Privacy Dashboard Job, 306
- Sync Catalog Updates Job, 307
- Synchronize Users Job, 307
- Job Management, 307

Jobs Overview

Data Privacy Management includes a job framework to manage and automate tasks.

A job is a collection of steps that the Data Privacy Management Service performs to complete a task. For example, the Data Privacy Management Service runs a job to discover and classify sensitive data in data stores.

Data Privacy Management includes job types to manage alerts, import data, evaluate classification policies, run tasks, generate subject data reports, synchronize information, and scan data stores to build the subject registry and discover and classify sensitive data.

Data import and classification policy evaluation jobs do not include scheduling options and run immediately. For example, if you edit a classification policy, the evaluate classification policies job runs when you save the classification policy.

Alert, Protection, Scan, and Subject Data Report Purge jobs run according to scheduling options that you define. For example, when you create an alert rule, you define the scheduling options for the Alert job. You can schedule a job to run immediately, in the future, or on a recurring schedule.

The Data Privacy Management Service can run multiple jobs in parallel. You define the maximum number of parallel jobs when you configure the Data Privacy Management Service.

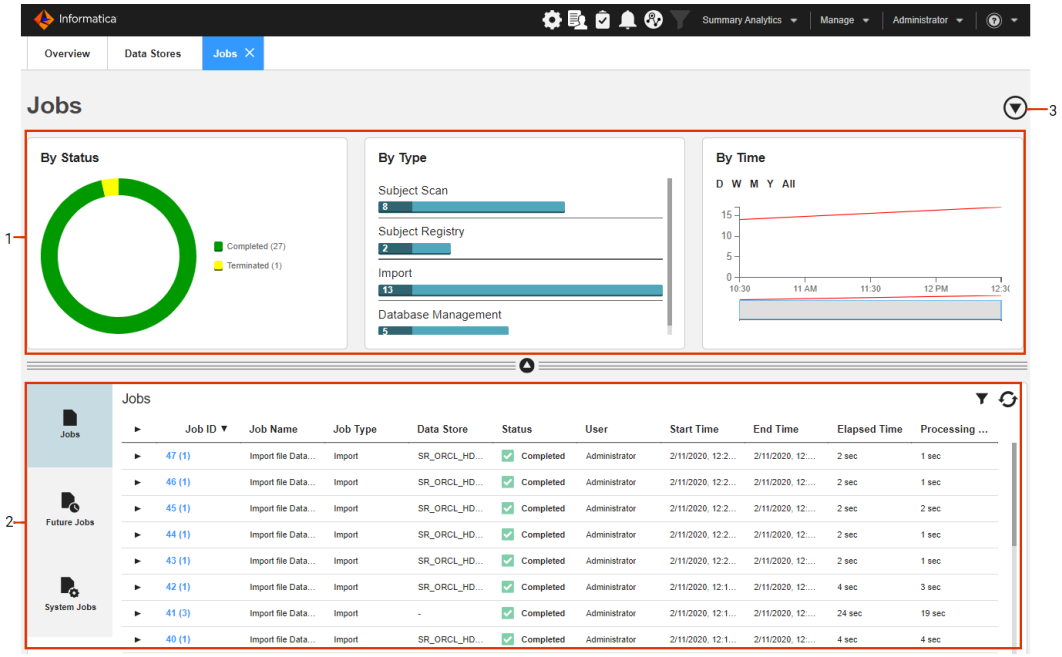
After you run or schedule a job, you can monitor the job status to verify if the job completed. You can view the job details and the status of the job steps. You can view and export the job logs to find out more information about the tasks that the job performs or the errors that the job encountered. You can pause, resume, stop, or terminate a job.

Jobs Workspace

Use the **Jobs** workspace to monitor and manage jobs. The **Jobs** workspace includes a list of jobs and graphical filters to filter the job list. You can view completed, failed, paused, stopped, or terminated jobs. You can view jobs that are in progress or jobs that are scheduled to run in the future.

To access the **Jobs** workspace, click **Manage** > **Jobs**.

The following image shows an example of the **Jobs** workspace:



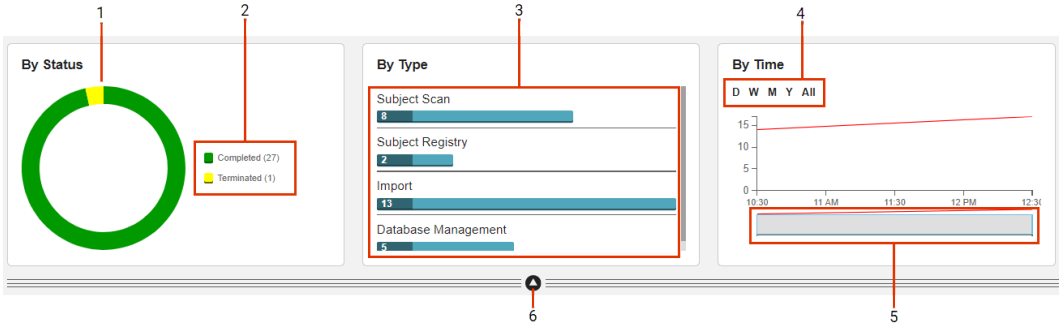
1. Job filters
2. Jobs list
3. Actions menu

Job Filters

Use the job filters to quickly navigate to jobs. You can filter jobs based on the job status, the job type, and the job start time.

When you click a filter, the **Jobs** list page shows the jobs that match the filter condition. To clear the filter conditions, click the filter again. You can hide and unhide the job filters.

The following image shows an example of the job filters:

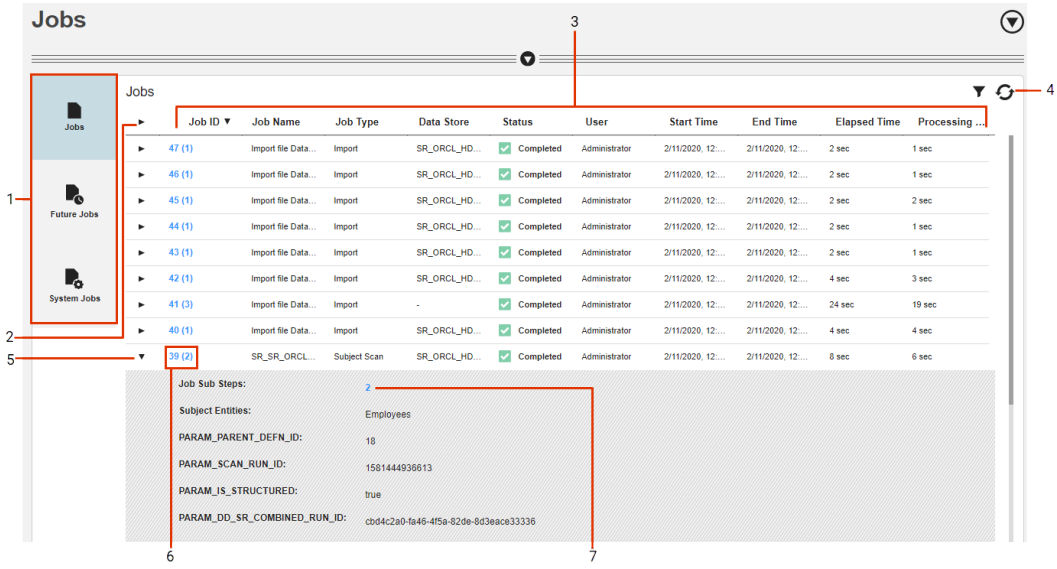


1. Click a color in the graph to filter based on job status.
2. Click the text to filter based on job status.
3. Click a job type. Use the arrows to scroll to more job types.
4. Click the text to filter based on the job start time.
5. Resize the rectangle to filter within a time range.
6. Click the arrow to hide or unhide the job filters.

Jobs List Page

The **Jobs** workspace displays a list of jobs in the Data Privacy Management repository. You can view the job properties and the job parameters. You can navigate to the job steps and job logs.

The following image shows an example of the **Jobs** list page that shows the job parameters for a Subject Scan job:



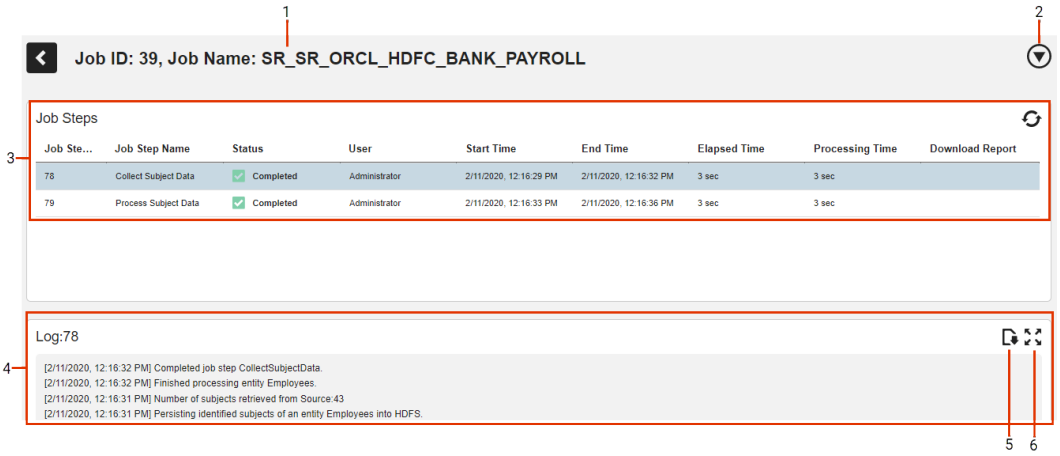
1. Job filters
2. Job list
3. Job properties
4. Refresh icon
5. Click the arrow next to a job to display the job parameters.
6. Click the Job ID to view the job steps and job logs.
7. Click the Job Sub Steps to view the job steps and job logs.

Job Details Page

The Job details page shows the job steps that the job includes, the status of the job steps, and the job logs for each job step.

Access the Job details page to view the job steps or job logs. You access the Job details page when you click a **Job ID** from the **Jobs** list page or when you view a job from the **Actions** menu.

The following image shows an example of the Job details page for the Job ID 39 with Job Name SR_SR_ORCL_HDFC_BANK_PAYROLL :



1. Job ID and job name
2. Actions menu
3. Job steps
4. Job log
5. Download job log icon
6. Expand job log icon

Job Types

A job type is a template that includes a predefined workflow of one or more job steps. Data Privacy Management includes job types for tasks related to alerts, data evaluation, data import, risk score calculations, scans, subject data reports, tasks, and users.

Data Privacy Management creates a job when you perform a task that is associated with a job type, or when system jobs are configured to run. The job type determines when the job runs. Jobs run according to scheduling options that you define or immediately after you perform an action. System jobs run at preconfigured intervals.

For scan jobs that you configure with both the **Include Catalog Information** and **Run Profiling** options, the job steps are interdependent. If the Profiling job step fails, the Catalog information is not available and the Load Catalog job step fails.

Data Privacy Management includes the following job types that run when you perform a task or schedule the job to run:

Job Name	Description
Big Data	Analyzes metadata in Cloudera Navigator and identifies the proliferation of sensitive data between scanned Hive data stores. Identifies sensitive data proliferation between Hive data stores that are on the same host as the Cloudera Navigator data store. A Big Data Scan job runs when you scan a data store that connects to a big data application and schedule the scan.
Cloud	Analyzes metadata, data, or both in an Amazon Redshift, Amazon S3, Azure Data Lake, Microsoft Azure Blob Storage, Microsoft Azure SQL Data Warehouse, Microsoft Azure SQL Database, Salesforce, or Snowflake data store. Identifies sensitive data and classifies the sensitive data according to the classification policies in the scan. A Cloud Scan job runs when you scan a data store that connects to a cloud application and schedule the scan.
Data Integration	Analyzes metadata and imports data stores from an Informatica Cloud, Informatica Data Engineering Integration, Informatica PowerCenter on IBM Db2, Informatica PowerCenter on Microsoft SQL Server, Informatica PowerCenter on Oracle, or SQL Server Integration Services repository. The Scan job identifies the proliferation of sensitive data and verifies if the sensitive data is protected. A Data Integration Scan job runs when you scan a data store that connects to a data integration repository and schedule the scan.
Database Management	Analyzes metadata, data, or both in an IBM Db2, JDBC, Microsoft SQL Server, Oracle, Netezza, SAP HANA, Sybase, or Teradata data store. Identifies sensitive data according to the classification policies in the scan. A Database Management Scan job runs when you scan a data store that connects to a database and schedule the scan.
Evaluate Classification Policies	Evaluates whether a scanned data store matches a classification policy. If the data store is a match, the job calculates and updates the risk score, residual risk cost, sensitivity level, and the number of sensitive fields for the data store. You create an Evaluate Classification Policies job when you edit a classification policy that is associated with a completed scan. The job runs immediately after you save the classification policy or when you select **Actions** > **Evaluate Classification Policies** from the **Classification Policies** workspace.

Job Name	Description
Evaluate Security Policy	Evaluates data store properties and the data store scan results. If the data store properties and scan results meet the conditions in the data store security policy, the job creates a security policy violation. An Evaluate Security Policy job runs when you create or edit a data store security policy and configure the schedule.
File Management	Analyzes metadata, data, or both in a file system, Microsoft OneDrive, or Microsoft SharePoint data store. The Scan job identifies sensitive data and classifies the sensitive data according to the classification policies in the scan. A File Management Scan job runs when you scan a data store that connects to a file management data store type and schedule the scan.
Import	Uses data from CSV files to create, update, or delete information about classification policies, data stores, data store aliases, data store Catalog metadata and resources, data store custom lineage, data store protection status, data store owners, data domains, encryption rules, locations, remote agents, subject registry categories, and subject registry purposes. You can also import information about users, user groups, user group memberships, user access, and user aliases. The Data Privacy Management Service creates an Import job that runs immediately when you import data from CSV files.
Import Catalog Results	Imports information about data stores from Enterprise Data Catalog to Data Privacy Management. An Import Catalog Results job runs when you import Catalog resources on the **Data Stores** workspace.
Incremental Scan	Uses the scan options from an Application, Big Data, Cloud, Data Integration, Database Management, or File Management scan that completed to run the scan again on the data stores, but only performs profiling on the changed fields or files included in the scan job. An Incremental Scan job runs when you select **Scan for Missing Data Domains** from the **Actions** menu of a **Data Store Details** page.
LDAP Import	Imports users, groups, and group memberships from an LDAP directory service in to the Data Privacy Management repository. An LDAP Import job runs when you save an LDAP connection and configure the schedule.

Job Name	Description
Orchestration	Creates service management tickets, runs custom script files, sends emails, and writes system log messages to a remote system log server. Orchestration jobs run when tasks that are configured with a custom, email, service management, or system log extension run. If a custom, email, service management, or system log action is associated with a security policy, the job runs automatically when Data Privacy Management detects a violation of the security policy.
Protection	Applies the protection rules that are configured in a protection task to protect sensitive fields in a data store. You schedule when the job runs after you configure a protection task on the **Tasks** workspace.
Recalculate Risk Scores	Uses the risk score setting on the **Settings** workspace to calculate and update the risk score of scanned data stores. Data Privacy Management creates a Recalculate Risk Scores job when you change risk factor weights. The job runs immediately after you confirm that you want to recalculate the existing risk scores.
Reset Classification Status	Deletes the column-to-data domain match results of a scan job for a data store. Data Privacy Management creates a Reset Classification Status job when you select the option from the **Actions** menu on the **Data Store Details** page. The job runs immediately after you confirm that you want to reset the results.
Subject Registry Scan	When you run a subject registry scan, the service creates one master Subject Registry Scan job and individual Subject Scan jobs for every golden and transaction data store included in the scan. The master Subject Registry Scan job includes child jobs that track all scans that you include in the subject registry scan. Based on how you configure the scan, the job can contain the following child jobs: - Poll Domain Discovery. Tracks the status of all domain discovery scans. Applicable if you include the Domain Discovery scan type. - Scan Golden Data Stores. Tracks the status of all subject scans on golden data stores. Applicable if you include the Discover Subjects scan type. - Scan Transaction Data Stores. Tracks the status of all subject scans on transaction data stores. Applicable if you include the Link Subjects scan type. Data Privacy Management creates an individual Subject Scan job for each golden and transaction data store scan that you include in a subject registry scan.

Job Name	Description
Privacy Dashboard	When a master subject registry scan job completes, it triggers a job that updates the data of the Privacy dashboard.
Sync Catalog Updates	Synchronizes the information for a data store that the Import Catalog Results job found. A Sync Catalog Updates job runs when you select **Import Results from Enterprise Data Catalog** from the **Actions** menu of a **Data Store Details** page.

Data Privacy Management includes the following system job types that you can view on the **System Jobs** tab on the **Jobs** workspace:

Job Name	Descriptions
Evaluate Risk Scores	Calculates and updates the risk score of scanned data stores. The Data Privacy Management Service runs the Evaluate Risk Scores job once a day at the scheduled time. The default scheduled time is 1:30 am UTC.
Subject Data Report Purge	Automatically removes subject data reports according to the interval that you specify in the **Data Privacy Settings** pane on the **Settings** workspace.
Synchronize Users	Synchronizes user details between the Data Privacy Management repository and the Enterprise Unified Metadata platform. After you create a user in the Administrator tool, you run the Synchronize Users job. To run the job, select **Synchronize** > **Users** from the **Actions** menu on the **Data Stores** workspace. To view and manage new users, you must log out of Data Privacy Management and log in again after you add the users in the Administrator tool and run the Synchronize Users job.

Related Topics:

- "Changing Risk Score Factor Weights" on page 42
- "Adding Custom Risk Score Factors" on page 43

Big Data Job

The Big Data job runs when you scan a Cloudera Navigator, Hadoop Distributed File System (HDFS), or Hive data store.

The Big Data job includes the following job steps:

Job Step	Description
Load and Link	For Cloudera Navigator, extracts a metadata catalog from the source.
Load Catalog	For HDFS and Hive, extracts a metadata catalog from the source if the scan is configured to include catalog metadata.

Job Step	Description
Update Augmentation	For Cloudera Navigator, determines column-level data lineage for sensitive columns between scanned Hive data stores that are on the same host as the Cloudera Navigator data store.
Profiling	For HDFS and Hive, creates and runs a profile to perform data domain discovery on source metadata, data, or both if the scan is configured for data profiling. The job uses the profile results to determine sensitivity status.
Collect Row Count	For Hive data stores, calculates the row count for each table that contains sensitive data if the scan is configured to include a row count.
Copy Augmentation	Copies scan results from temporary tables to the Data Privacy Management repository. The job stores scan results in temporary tables while running. After the scan results are stored in the history tables, the job commits the scan results to the repository in one database commit.
Evaluate Policies	Evaluates if the data store contains the classification of sensitive data as defined in the policy. Calculates the risk score, residual risk cost, and sensitivity levels. Calculates the total number of sensitive columns that each data store contains.

Rules and Guidelines for Big Data Scan Jobs

Consider the following information when you run a Big Data Scan job on a Hive data store:

- The scan job can connect to Hadoop clusters that use Kerberos authentication.
- The auto random and random sampling techniques are not supported. You can use the First N sampling techniques for data profiling.
- The Scan job cannot run profiling for columns that include the following datatypes:

  ```
  ARRAYS
  BINARY
  MAPS
  STRUCT
  ```

  If the source contains the datatypes, the Scan job skips profiling for columns that contain the unsupported datatypes.
- The Big Data Scan job identifies and classifies sensitive data in Hive data stores. To identify the proliferation of sensitive data between Hive data stores and to identify the protection status of sensitive data, run a Cloudera Navigator scan.

Cloud Job

A Cloud job runs when you scan an Amazon Redshift, Amazon S3, Azure Data Lake, Microsoft Azure Blob Storage, Microsoft Azure SQL Data Warehouse, Microsoft Azure SQL Database, Salesfoece, or Snowflake data store.

The Cloud job includes the following job steps:

Job Step	Description
Load Catalog	Extracts a metadata catalog from the source if the scan is configured to include catalog metadata.
Profiling	Creates and runs a profile to perform data domain discovery on source metadata, data, or both if the scan is configured for data profiling. The job uses the profile results to determine sensitivity status.
Collect Row Count	Calculates the row count for each table that contains sensitive data if the scan is configured to include a row count.
Copy Augmentation	Copies scan results from temporary tables to the Data Privacy Management repository. The job stores scan results in temporary tables while running. After the scan results are stored in the history tables, the job commits the scan results to the repository in one database commit.
Evaluate Classification Policies	If the scan is configured to include catalog information, run profiling, or include a row count, the job evaluates if the data store contains the classification of sensitive data as defined in the classification policy. Calculates the risk score, residual risk cost, and sensitivity levels. Calculates the total number of sensitive fields that each data store contains.

Related Topics:

- "Cloud Scan Options" on page 267

Rules and Guidelines for Cloud Scan Jobs

Consider the following information when you run a Cloud scan job on a Salesforce data store:

- The scan job cannot run profiling or determine row count for the following Salesforce tables:

```
ActivityHistory
AggregateResult
AttachedContentDocument
CollaborationGroupRecord
CombinedAttachment
ContentDocumentLink
DatacloudCompany
DatacloudContact
DatacloudDandBCompany
DatacloudSocialHandle
EmailStatus
FeedLike
FeedTrackedChange
IdeaComment
Name
NoteAndAttachment
OpenActivity
OwnedContentDocument
ProcessInstanceHistory
UserRecordAccess
Vote
```

The scan job excludes the tables because Salesforce does not allow an application to run a select query on the tables.

- When you pause and resume the Load Catalog, Profiling, or Collect Row Count job steps, the scan job restarts the step. The scan job cannot resume the job step from the point in which the job step stopped.

Data Integration Job

The Data Integration job runs when you scan an Informatica Cloud, Informatica Data Engineering Integration, Informatica PowerCenter on Db2, Informatica PowerCenter on Microsoft SQL Server, Informatica PowerCenter on Oracle, or SQL Server Integration Services data store.

The Data Integration job includes the following job steps:

Job Step	Description
Collect Connections	For Informatica Data Engineering Integration data store scans configured to collect data store information, imports connections and connection properties to the Data Privacy Management repository. The Scan job creates one data store in the repository for each connection.
Collect Connections from Informatica Cloud	For Informatica Cloud data store scans configured to collect data store information, connects to Informatica Cloud using the connection properties in the data store. Imports connections and connection properties from Informatica Cloud to the Data Privacy Management repository for Amazon S3, Microsoft SQL Server, MySQL, Oracle, and Salesforce data store types. The Scan job creates one data store for each supported Informatica Cloud connection. For MySQL connections, the Scan job creates a data store with data store type JDBC. The Scan job uses the following naming convention to create a data store: *<Data Privacy Management Informatica Cloud Data Store Name>/<Informatica Cloud Org ID>/<Informatica Cloud Connection Name>* The Scan job does not import the password or connection string properties. The Scan job does not import connection properties for Amazon S3. You must update the password and connection string in the imported data stores after the Scan job finishes.
Collect Connections and Catalog	For Informatica PowerCenter data store scans configured to include data stores and catalog information, imports relational database and ODBC connections from the PowerCenter repository to the Data Privacy Management repository. Imports the metadata from the PowerCenter repository for each connection. Determines an approximate connection-level data lineage for the data stores.
Load and Link	For Informatica Cloud data store scans configured to identify proliferation and data protection, extracts a metadata catalog from the source database. For Informatica PowerCenter data store scans configured to identify proliferation and data protection, determines column-level data lineage for the sensitive columns and identifies the protection status of sensitive data. For SQL Server Integration Services data store scans configured to identify proliferation and data protection, extracts a metadata catalog from the source database. Identifies the number of scanned child data stores in the mapping between which sensitive data proliferates. For example, if a mapping has three connections and sensitive data moves between two connections, the job log shows the assignable connection size as two.

Job Step	Description
Profiling	For Informatica PowerCenter data store scans configured to include data stores and catalog information, runs a metadata profile job to identify sensitive data and runs data domain discovery to match metadata with the data domains in the scan. The Scan job uses the profile results to determine the metadata sensitivity status. The Scan job updates the Data Privacy Management repository with the results of the sensitive data identification.
Collect Database Configuration Files	For Informatica PowerCenter data stores scans configured to include data stores and catalog information, updates the data store connection properties in the Data Privacy Management repository. The Scan job runs a PowerCenter workflow that reads connection properties from database configuration files and from PowerCenter parameter files. The Scan job updates the repository with the connection properties for the data stores that the Scan job imported from the PowerCenter repository.
Generate Connection Lineage	For Informatica PowerCenter data stores scans configured to include data stores and catalog information, determines connection-level data lineage for sensitive columns. The Scan job analyzes the PowerCenter repository to determine data lineage for the data stores that the Scan job imported. The Scan job determines forward lineage from sources to targets only.
Update Augmentation	If the scan is configured to identify proliferation and data protection, determines column-level data lineage for sensitive columns and identifies the protection status of sensitive data.
Copy Augmentation	If the scan is configured to collect data store information and identify proliferation and data protection, copies scan results from temporary tables to the Data Privacy Management repository. The Scan job stores scan results in temporary tables while running. After the scan results are stored in the history tables, the Scan job commits the scan results to the Data Privacy Management repository in one database commit.
Evaluate Classification Policies	If the scan is configured to collect data store information and identify proliferation and data protection, evaluates if the data store contains the classification of sensitive data as defined in the scan policies. Calculates the risk score, residual risk cost, and sensitivity levels. Calculates the total number of sensitive columns that each data store contains.

Related Topics:

- "Data Integration Scan Options" on page 270

Rules and Guidelines for Data Integration Scan Jobs

Consider the following information when you run a Data Integration Scan job on Informatica Data Engineering Integration data stores:

- The Scan job uses the connections details in the associated Data Integration Service of the Data Privacy Management Service to connect to Informatica Data Engineering Integration. Because the Scan job does not connect to the database directly, the Scan job does not require user privileges.
- The Scan job cannot connect to Informatica Data Engineering Integration if the Model repository is on an Informatica domain that uses Kerberos authentication.

- The Scan job uses the following transformations to determine data lineage for sensitive columns:

```
Aggregator
Expression
Joiner
Rank
Router
Union
Merge
Decision
Classifier
TokenStandardiser
CaseConverter
Association
WeightBasedAnalyser
Group
Consolidation
Parser
Labeler
StringDistance
Matcher
Exception
Lookup
DataMasking
```

- When you pause and resume the Collect Connections or Load and Link job steps, the Scan job restarts the job step. The Scan job cannot resume the job step from the point at which the job step stopped.

Consider the following information when you run a Data Integration Scan job on SQL Server Integration Services data stores:

- The Scan job does not import connections from the SQL Server Integration Services package.
- The Scan job does not determine if sensitive data is protected. All sensitive columns show as unprotected unless you manually mark the columns as protected.

Database Management Job

The Database Management job runs when you scan an IBM Db2, JDBC, Microsoft SQL Server, Oracle, Netezza, SAP HANA, Sybase, or Teradata data store.

The Database Management job includes the following job steps:

Job Step	Description
Load Catalog	Extracts a metadata catalog from the source. The job includes the job step if the scan is configured to include catalog information.
Profiling	Creates and runs a profile to perform data domain discovery. The scan options determine if the data domain discovery runs on the source metadata, data, or both. The job uses the profile results to determine the sensitivity status. The job includes the job step if the scan is configured to run profiling.
Collect Row Count	Calculates the row count for each table that contains sensitive data. The job includes the job step if the scan is configured to include a row count.

Job Step	Description
Copy Augmentation	Copies scan results from temporary tables to the Data Privacy Management repository. The job stores scan results in temporary tables while running. After the scan results are stored in the history tables, the job commits the scan results to the repository. The job includes the job step if the scan is configured to include catalog information, to run profiling, or to include a row count.
Evaluate Classification Policies	Evaluates if the data stores contain the classification of sensitive data as defined in the classification policy. Calculates the risk score, residual risk cost, and sensitivity levels. Calculates the total number of sensitive columns that each data store contains. The job includes the job step if the scan is configured to include catalog information, to run profiling, or to include a row count.

Related Topics:

- "Database Management Scan Options" on page 273

Rules and Guidelines for Database Management Scan Jobs

Consider the following information when you run a Database Management Scan job:

- The Scan job runs most job steps in sequential order. A job step must complete before the next job step starts.
- If you configure a scan to include catalog information and to run data profiling, the Scan job runs the Load Catalog and Profiling job steps in parallel. If you configure a scan to include catalog information and to run metadata profiling or metadata with data profiling, the job runs the job steps sequentially.
- You can pause, stop, terminate, and resume a Scan job that runs the Load Catalog and Profiling job steps in parallel. The action affects the job steps that are running. The behavior is the same as when the job steps run in sequential order. For example, if you pause the job after the Load Catalog job step completes and while the Profiling job step is running, the job only pauses the Profiling job step. When you resume the job, the job resumes the Profiling job step.

  When a Scan job runs the Load Catalog and Profiling job steps in parallel and the Load Catalog job step fails, the Profiling job step also fails. When you resume the job, the job resumes both job steps.

Evaluate Classification Policies Job

The evaluate classification policies job calculates and updates the risk score, residual risk cost, sensitivity level, and sensitive field count for scanned data stores.

The Data Privacy Management Service creates the job when you edit a classification policy that is associated with a completed scan. The job runs immediately after you save the classification policy or trigger the Evaluate Classification Policies action from the **Classification Policies** workspace.

The evaluate classification policies job includes the following job step:

Evaluate classification policies

Evaluates if data stores contain the classification of sensitive data as defined in the classification policy. Calculates the risk score, residual risk cost, and sensitivity level. Calculates the total number of sensitive columns that each data store contains.

Evaluate Risk Scores Job

The evaluate risk scores job calculates and updates the risk score of scanned data stores.

The Data Privacy Management Service runs the evaluate risk scores job once a day at the scheduled time. By default, the scheduled time is 1:30 am UTC.

The evaluate risk scores job includes the following job step:

Evaluate risk scores

Calculates and updates the risk score of scanned data stores.

Evaluate Security Policy Job

The evaluate security policy job evaluates data stores that the user can access. The job determines if a data store meets the conditions in the data store security policy. If the data store meets the security policy conditions, the job creates a security policy violation.

The Data Privacy Management Service creates the job when you create or edit a data store security policy. You schedule when the job runs when you configure the data store security policy.

The evaluate security policy job includes the following job step:

Evaluate security policies

Evaluates the properties and scan results of data stores that the user can access. If the properties or scan results meet the conditions in the data store security policy, the job creates a security policy violation.

File Management Job

The File Management job runs when you scan a file system, Google Drive, Microsoft OneDrive, or Microsoft SharePoint data store..

The File Management job includes the following job steps:

Job Step	Description
Load Catalog	For data stores that scan with Enterprise Data Catalog, extracts a metadata catalog from the source. The job includes the job step if the scan is configured to include catalog information.
Browse	For data stores that scan with a remote agent, loads the metadata of files in the data store based on the value in the **Agent Scan Settings** pane on the **Settings** workspace and the folders and file types included in the data store configuration. If the Browse job step discovers zero files, the File Management job has a Warning status, and the remaining job steps have a Warning message and the job ends.
Profiling	For data stores that scan with Enterprise Data Catalog, creates and runs a profile to perform data domain discovery. The scan options determine if the data domain discovery runs on the source metadata, data, or both. The job uses the profile results to determine the sensitivity status. The job includes the job step if the scan is configured to run profiling.

Job Step	Description
AgentProfiling	For data stores that scan with a remote agent, creates and runs a profile to perform data domain discovery through a remote agent on the list of files that the Browse job step returns.
Evaluate Rules	For data stores that scan with Enterprise Data Catalog, evaluates the sensitivity status of fields. The job step is included if the scan is configured to run profiling.
Copy Augmentation	Copies scan results from temporary tables to the Data Privacy Management repository. The job stores scan results in temporary tables while running. After the scan results are stored in the history tables, the job commits the scan results to the repository. The job includes the job step if the scan is configured to include catalog information or run profiling.
Evaluate Classification Policies	Evaluates if the data stores contain the classification of sensitive data as defined in the classification policy. Calculates the risk score, residual risk cost, and sensitivity levels. Calculates the total number of sensitive fields or files that each data store contains. The job includes the job step if the scan is configured to include catalog information or run profiling.

Related Topics:

- "File Management Scan Options" on page 276

Rules and Guidelines for File Management Jobs

Consider the following information when you run a File Management job:

- The File Management job does not determine an actual or estimated row count. The job ignores the row count factor when calculating the risk score.
- The File Management jobs sets the residual risk cost to zero.
- For unstructured file management data stores that scan with a remote agent, if the Browse job step encounters files that exceed the size specified in the **Agent Scan Settings** pane on the **Settings** workspace, the job skips the files and continues processing.

  You can download a report as a CSV file from the **Download Report** column for the Browse job step on the **Job Details** page. The report contains information about the current maximum file size setting, the folders and file types that the data store includes and excludes, and a list of files that returned errors.

  You can download reports as CSV files from the **Download Report** column for the Profiling job step on the **Job Details** page. The reports contains information about the files that the Scan job profiled successfully and the files that returned errors in the Profiling job step.
- If you scan a file management data store with Enterprise Data Catalog and later edit the data store to enable the **Scan with Remote Agent** setting, you must scan the data store again after you associate the data store with a remote agent. The scan skips files larger than the setting specified on the **Settings** workspace.

Import Job

The Import job uses data from CSV files to create, update, or delete information about classification policies, data stores, data store aliases, data store protection status, data store owners, data domains, encryption

rules, locations, and remote agents. You can also import information about users, user groups, user group memberships, user access, and user aliases.

The Data Privacy Management Service creates an Import job that runs immediately when you import data from CSV files.

The Import job includes the following job steps:

Job Step	Description
Import File	Processes data from the CSV file and stores data in temporary tables in the Data Privacy Management repository.
Save to Repository	Moves data from the temporary tables to the main tables in the Data Privacy Management repository. Then, the job step deletes data in the temporary tables.

Rules and Guidelines for Import Jobs

Consider the following information when you run an Import job:

- If the job fails during the Import File job step, download the rejected records from the job log. Fix the issues and import the file again.
- If you terminate the job during the Import File job step, the job deletes all data from the temporary tables. The job does not save any data in the Data Privacy Management repository. To continue, import the CSV file again.
- If you pause or stop the job during the Import File job step, the job pauses or stops at the end of the job step. When you resume the job, the job starts at the Save to Repository job step.
- If the job fails or you terminate the job during the Save to Repository job step, the job might not save all of the data to the main tables in the Data Privacy Management repository. You can export data from the repository to view what information the job imported. Then, import a file that contains the data that the job did not process.
- If you pause or stop the job during the Save to Repository job step, the job pauses or stops at the end of the job step. The job finishes at the end of the Save to Repository job step.

Import Catalog Results Job

The Import Catalog Results job runs when you import Catalog resources on the **Data Stores** workspace.

The Import Catalog Results job includes the following job steps:

Job Step	Description
Import Catalog Resources	Imports classification policies from Enterprise Data Catalog that you select on the **Data Stores** workspace when configuring the resources to import.
Import Profiling Results	Imports the data store profile from Enterprise Data Catalog.

Job Step	Description
Copy Augmentation	Copies profiling results from temporary tables to the Data Privacy Management repository.
Evaluate Classification Policies	Evaluates if the data stores contain the classification of sensitive data as defined in the classification policies imported from Enterprise Data Catalog. Calculates the risk score, residual risk cost, and sensitivity levels. Calculates the total number of sensitive fields or files that each data store contains.

Incremental Scan Job

The Incremental Scan job runs when you select **Scan for Missing Data Domains** from the **Actions** menu of a **Data Store Details** page.

The Incremental Scan job includes the following job steps:

Job Step	Description
Profiling	For data stores that scan with Enterprise Data Catalog, creates and runs a profile to perform data domain discovery. The scan options determine if the data domain discovery runs on the source metadata, data, or both. The job uses the profile results to determine the sensitivity status. The job includes the job step if the scan is configured to run profiling.
Collect Row Count	For Application, Hive, Cloud, Data Integration, and Database Management data stores, calculates the row count for each table that contains sensitive data if the scan is configured to include a row count.
Copy Augmentation	Copies scan results from temporary tables to the Data Privacy Management repository. The job stores scan results in temporary tables while running. After the scan results are stored in the history tables, the job commits the scan results to the repository. The job includes the job step if the scan is configured to include catalog information or run profiling.
Evaluate Classification Policies	Evaluates if the data stores contain the classification of sensitive data as defined in the classification policy. Calculates the risk score, residual risk cost, and sensitivity levels. Calculates the total number of sensitive fields or files that each data store contains. The job includes the job step if the scan is configured to include catalog information or run profiling.

Orchestration Job

The Orchestration job performs the actions that are configured in custom, email, service management, and system log tasks.

You run the job when you create a manual action or on demand from the **Tasks** workspace. If a custom, email, service management, or system log action is associated with a security policy, the orchestration job runs automatically when Data Privacy Management detects a violation of the security policy. The

Orchestration job steps depend on the extension associated with the custom, email, service management, or system log task.

The Orchestration job includes the following job steps:

OrchestrationJobStep

Performs the actions that are specified in the custom, email, service management, or system log task and the extensions that are configured for each task.

Protection Job

The Protection job applies the rules to protect sensitive fields in a data store that are configured in a protection task. The Protection job steps depend on the protection extension that is specified in the task.

You schedule when the job runs after you configure a protection task on the **Tasks** workspace. If an action to protect data is associated with a data store or user activity security policy, Data Privacy Management creates the protection task when a violation of the security policy occurs. If an action to protect data is associated with a decryption security policy, Data Privacy Management creates the protection task when the policy conditions are met.

The Protection job includes the following job steps:

Job Step Name	Description
Export Connection XML	For tasks configured with an encryption or Persistent Data Masking protection extension, creates a connection from Data Privacy Management to the Test Data Management service. The Protection job builds a connection XML file, exports the file from Data Privacy Management, and then imports the file into Persistent Data Masking.
Export Encryption Rule	For tasks configured with an encryption protection extension, builds an XML file that contains the encryption rules and associated encryption keys for the data domains included in the task. The Protection job then runs the ExportProject XML job step.
Export Project XML	For tasks configured with an encryption or Persistent Data Masking protection extension, creates a project in the Test Data Management service. The Protection job builds a project XML file that includes metadata from the data sources and assignments in the task. The job then exports the project XML file from Data Privacy Management and imports the file into Persistent Data Masking.
Generate Workflow	For tasks configured with an encryption or Persistent Data Masking protection extension, creates the workflow that masks the sensitive columns according to the rules specified in the task.
Execute Workflow	For tasks configured with an encryption or Persistent Data Masking protection extension, runs the workflow that masks the sensitive columns according to the rules specified in the task.
Update Encryption Configuration	For tasks configured with an encryption protection extension, encrypts the sensitive columns with the associated encryption rules and keys specified for each data domain in the data store.
Update Protection Status	For tasks configured with a protection extension that enables the **Mark column as protected after execution** option, changes the status of sensitive columns from Unprotected to Protected after the job completes successfully.

Related Topics:

- "Scheduling a Protection Job for a Task" on page 508
- "Configuring Protection Tasks" on page 502

Reset Classification Results Job

The Reset Classification Results job resets the column-to-data domain match results of a data store.

The job runs immediately after you confirm that you want to reset the results.

The Reset Classification Results job includes the following job steps:

Job Step	Description
Reset Protection and Sensitivity Results	Resets the results of a protection job and resets the Protection status and Sensitivity results.
Reset Catalog Results	Catalog results reset to remove domain association information. For unstructured data stores associated with a subject registry remote agent, the job skips this step.
Update Classification Status	Updates the classification status of the data store to Not Analyzed.

Salesforce Import job

The Salesforce Import job imports users, user access, and user activity from Salesforce. The Data Privacy Management Service creates a Salesforce Import job when you save or run a Salesforce server connection in the Users workspace.

In the connection, you define the type of information the job imports and set up a schedule to periodically synchronize Data Privacy Management with Salesforce. You can schedule the job to run immediately, in the future, or set up a job that runs on a regular basis.

The Import job includes multiple job steps. The job steps depend on the options configured in the Salesforce server connection. The connection includes an option to import users, user access, and user activity.

The Import job includes the following job steps:

Import Salesforce Users

Connects to the Salesforce application using the connection properties in the Salesforce data store. The job sends a query through the Bulk API to collect users from Salesforce. The first time you run the job for the connection, the job collects all users from Salesforce. For subsequent runs, the Import job queries changes since the time the last Import job completed.

The Bulk API returns a .csv file that includes a list of users. The job processes the records in the CSV file and imports the data into a temporary table in the Data Privacy Management repository. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

If the job fails or if you terminate the job during the job step, the job deletes data from the temporary tables. The job does not save data to the repository.

When you run a synchronization, the Import job always includes the job step to import users.

Import Salesforce User Access

Connects to the Salesforce application using the connection properties in the Salesforce data store. The job sends a query through the Bulk API to collect user groups, user group memberships, and user access from Salesforce. The Import job retrieves user groups from Salesforce permission sets and retrieves user group memberships from the Salesforce permission set assignments. The Import job queries object permissions for sensitive tables and field permissions for sensitive columns. Every time the job runs, the job gets user access details that are valid at the time the job runs.

The Bulk API returns four CSV files that include a list of permission sets, permission set assignments, object permissions, and field permissions. The job processes the records in the CSV files and imports the data into a temporary table in the Data Privacy Management repository. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated for each CSV file. If the job rejected one or more rows in a CSV file, you can download a list of the rejected rows. If a table in the object permissions CSV file is not identified as sensitive in the repository, the Import job rejects the row.

If the job fails or if you terminate the job during the job step, the job deletes data from the temporary tables. The job does not save data to the repository.

The Import job includes the job step if the connection is configured to import user access.

Save to Repository

Copies user and user access data from the temporary tables to the main tables and deletes data from the temporary tables. If the job fails or if you terminate the job during the Import File job step, the job deletes data from the temporary tables. The job does not save data to the Data Privacy Management repository.

The Import job includes the job step if the connection is configured to import users and user access.

Evaluate Risk Scores

Evaluates the user access count in the risk score for the Salesforce data store.

The Import job includes the job step if the connection is configured to import user access.

Import Salesforce User Activity

Connects to the Salesforce application using the connection properties in the Salesforce data store. The job sends a query through the Bulk API to collect event logs from Salesforce. The Import job retrieves user activity from the EventLogFile object for the API, REST API, and Bulk API event types. The job uses the scheduling options in the connection to determine the time range in which to retrieve the user activity. The job saves user activity directly to the user activity store. The retention period configured in the Data Privacy Management Service determines how long the events remain in the user activity store.

If the Import job fails during the job step, any of the events that the job saves in the user activity store remain. You can terminate the job and run the job again. The job will not save duplicate events in the user activity store.

The Import job includes the job step if the connection is configured to import user activity.

Related Topics:

- "Creating a Salesforce Connection" on page 323

Subject Data Report Purge Job

The Subject Data Report Purge job removes subject data in DSAR reports according to the interval specified in the **Data Privacy Settings** pane on the **Settings** workspace.

The Data Privacy Management Service runs the job at the interval specified for purging DSAR reports in the **Subject Data Report Purge Interval** property, if the **Subject Data Report Retention Period** value is greater than 0. A 0 value indicates that you always retain DSAR reports.

The **Subject Details** page retains the history of reports created for each data subject on the **Requests** pane. The **Download Options** column displays one of the following messages for each DSAR report created for a data subject:

- Available for download. You can download the DSAR report as a CSV file from the associated DSAR **Task Properties** page, or by clicking the **Download DSAR Report** icon on the **Subject Details** page and selecting a CSV or PDF report format.
- Available for download until *mm/dd/yyyy*. You can download the DSAR report until the specified date. The Subject Data Report Purge job is scheduled to run on the date listed in the column.
- Not available for download. The DSAR report was purged when the Subject Data Report Purge Job ran, and you cannot download the DSAR report.

The Subject Data Report Purge job includes the following job step:

Job Step	Description
Purge DSAR	Automatically removes DSAR reports according to the **Data Privacy Settings** on the **Settings** workspace. For example, the **Subject Data Report Retention Period** value is 2 and the **Subject Data Report Purge Interval** value is 7. The Subject Data Report Purge job runs every seven days, and the job step purges DSAR reports from Data Privacy Management that users created at least two days before the job runs. If the job runs today, Data Privacy Management retains DSAR reports that users created yesterday and purges the reports when the job runs again in seven days.

Subject Registry Scan Job

You run a Subject Registry scan to configure and create the Subject Registry. A Subject Registry scan job runs scans on all data stores that you include in the scan.

When you run a Subject Registry scan, the service creates a master Subject Registry scan job. The job includes multiple child jobs based on how you configure the scan. For example, you can choose to run golden and transaction data store scans in the same scan or different scans. You can choose to include the required domain discovery scan in the subject registry scan or run the domain discovery scan as a separate scan.

A Subject Registry job can include the following child jobs:

Poll Domain Discovery

Tracks all the domain discovery jobs that run in the scan. Created if you include the Domain Discovery scan type in the scan. If you have not run a domain discovery scan on data stores in the subject registry, you can configure domain discovery properties in the subject registry scan. All domain discovery scans must finish before other jobs can run.

Scan Golden Data Stores

Tracks subject scans of all golden data stores that you include in the scan. Created if you include the Discover Subjects scan type in the scan.

Scan Transaction Data Stores

Tracks subject scans of all transaction data stores that you include in the scan. Created if you include the Link Subjects scan type in the scan.

Note: A child job is triggered after the previous child job finishes successfully.

When the service creates a master Subject Registry job, it also triggers individual Subject Scan jobs. Subject scans are scan jobs that run on each golden and transaction data store included in the scan. A separate scan job runs for each data store.

Subject scan jobs run in the following order:

1. Domain discovery scans
2. Subject scans on golden data stores
3. Subject scans on transaction data stores

Deleting a master job deletes all its child jobs and the subject scans that it creates. If you delete the master job, a message appears to confirm that you want to delete the job and all its child jobs and subject scans.

On the **Jobs** workspace, you can filter and view jobs by the **Subject Registry** and **Subject Scan** job types.

Privacy Dashboard Job

When a master subject registry scan job completes, it triggers a job that updates the data of the Privacy dashboard.

The Data Privacy dashboard job fetches the updated data from the Subject Registry tables and refreshes the data counts of the Privacy dashboard.

The job must complete successfully for updated data to reflect when you refresh the dashboard.

You cannot start, stop, or pause a Privacy Dashboard job. You can resume a job that pauses or fails.

The job is listed in the Jobs workspace as: `Privacy_Dashboard_<subject registry scan name>`.

Sync Catalog Updates Job

The Sync Catalog Updates job runs when you select **Import Results from Enterprise Data Catalog** from the **Actions** menu of a **Data Store Details** page.

The Sync Catalog Updates job includes the following job steps:

Job Step	Description
Sync Catalog Updates	Synchronizes the information from Enterprise Data Catalog that the Import Catalog Updates job imported with the information about the data store in Data Privacy Management.
Evaluate Classification Policies	Evaluates if the data stores contain the classification of sensitive data as defined in the classification policies imported from Enterprise Data Catalog. Calculates the risk score, residual risk cost, and sensitivity levels. Calculates the total number of sensitive fields or files that each data store contains.

Synchronize Users Job

The Synchronize Users job synchronizes user details between the Data Privacy Management repository and the Enterprise Unified Metadata platform. After you add a new user in the Administrator tool, you run the Synchronize Users job.

The job runs when you select **Synchronize** > **Users** from the **Actions** menu on the **Data Stores** workspace.

The Synchronize Users job includes the following job step:

Sync Users

The Sync Users job step queries the list of domain users in the Enterprise Unified Metadata platform and copies the user details to the repository.

To view and manage new users in Data Privacy Management, you must log out of Data Privacy Management and log in again after you add the users in the Administrator tool and run the Synchronize Users job.

Job Management

On the **Jobs** workspace, you can monitor and manage jobs. You can also download reports for scan and Import jobs.

Pausing a Job

You can pause a job that is in progress. When you pause a job, the job pauses at the next logical point within the current job step or at the end of the current job step. For example, if you pause an Informatica

PowerCenter scan job during the Load and Link job step, the job pauses after the Load and Link job step completes.

1. Click **Manage** > **Jobs**.

   The **Jobs** workspace appears.

2. Select a job.

   To select a job, click any job property except for the Job ID.

3. From the **Actions** menu, select **Pause**.

   When you select the **Actions** menu, the Data Privacy Management Service validates if the option is supported for the job. If the Pause option is not enabled, the Data Privacy Management Service cannot pause the job.

You can resume a paused job.

Resuming a Job

You can resume a job that has a failed, paused, or stopped status.

1. Click **Manage** > **Jobs**.

   The **Jobs** workspace appears.

2. Select a job.

   To select a job, click any job property except for the Job ID.

3. From the **Actions** menu, select **Resume**.

   When you select the **Actions** menu, the Data Privacy Management Service validates if the option is supported for the job. If the Resume option is not enabled, the Data Privacy Management Service cannot resume the job.

Stopping a Job

You can stop a job that is in progress. When you stop a job, the Data Privacy Management Service aborts the job immediately. If you stop an alert or import job, the job stops after the current job step completes. To stop the job at the next logical step, pause the job instead.

1. Click **Manage** > **Jobs**.

   The **Jobs** workspace appears.

2. Select a job.

   To select a job, click any job property except for the Job ID.

3. From the **Actions** menu, select **Stop**.

   When you select the **Actions** menu, the Data Privacy Management Service validates if the option is supported for the job. If the Stop option is not enabled, the Data Privacy Management Service cannot stop the job.

You can resume a stopped job.

Terminating a Job

You can terminate a job that you no longer want to run. You can terminate jobs that have a failed, paused, or stopped status. You cannot resume a terminated job.

1. Click **Manage** > **Jobs**.

   The **Jobs** workspace appears.

2. Select a job.

   To select a job, click any job property except for the Job ID.

3. From the **Actions** menu, select **Terminate**.

   When you select the **Actions** menu, the Data Privacy Management Service validates if the option is supported for the job. If the Terminate option is not enabled, the Data Privacy Management Service cannot terminate the job.

Exporting Jobs

You can export a list of jobs to a CSV file. The export file includes a list of all jobs and the corresponding job properties. If you filtered the jobs list, the export file contains the jobs that match the active filter conditions.

1. Click **Manage** > **Jobs**.

   The **Jobs** workspace appears.

2. From the **Actions** menu, click **Export**.

   A dialog box appears with the exported file.

   The export file is named `Jobs.csv`. Save the file to a directory on your machine. If you export the file again, a new file downloads. The file name includes an incremental number such as `Jobs (1).csv`.

3. Open the file.

Downloading Scan Job Reports

You can download reports from the AgentProfiling or Profiling job step of a scan job.

After you scan data stores, you can view the scan results. You download the reports in a .zip file from the **Job Details** page.

1. Click **Manage** > **Jobs**.

   The **Jobs** workspace appears.

2. Navigate to a data store scan job. Click the Job ID to view the job steps.

   The **Job Steps** panel appears.

3. Click the icon in the **Download Report** column of the AgentProfiling or Profiling job step.

   If the AgentProfiling or Profiling job step does not include a **Download Report** icon, there are no profile results to curate. You can skip this task.

   A dialog box appears with the downloaded file named `<job step ID>.zip`. Save the file to a directory on your machine and extract the CSV files.

4. Open the CSV files.

The report is a CSV file in the format required to import sensitivity and protection status. Verify the sensitivity and protection status of the fields in the report. You can change the sensitivity and protection status for fields in a data store in one of the following ways:

- Manually update individual fields/files on the **Sensitive Fields** or **Sensitive Files** page accessed from the **Security Dashboard**.
- Update fields and then import the CSV file on the **Data Stores** workspace. From the **Actions** menu, select **Import** > **Protection Status** and import a CSV file.

Related Topics:

- "Changing the Default Conformance Ranges" on page 45
- "Conformance Score" on page 212

Downloading Remote Agent Scan Reports

You can download a report from the Browse job step and the AgentProfiling job step for unstructured data store scans that use a remote agent.

After you scan unstructured data stores with a remote agent, you can view the scan results. You download the Browse job step report in a CSV file from the **Job Details** page. You download the AgentProfiling job step report in a ZIP file from the **Job Details** page.

1. Click **Manage** > **Jobs**.

   The **Jobs** workspace appears.

2. Navigate to a remote agent scan job. Click the Job ID to view the job steps.

   The **Job Steps** panel appears.

3. Click the icon in the **Download Report** column of the Browse job step or the AgentProfiling job step.

   From the Browse job step, a dialog box appears with the downloaded file named `<job step ID>.csv`. From the AgentProfiling job step, a dialog box appears with the downloaded file named `<job step ID>.zip`. Save the file to a directory on your machine.

4. Open the CSV file to view the following information about the scan job: the maximum file size setting configured on the **Settings** workspace, the folders included and excluded in the scan options, and a list of any files that the Browse job step skipped.

   Extract the ZIP file to view a scan exception report that lists the exceptions that the AgentProfiling job step discovered while performing domain discovery on a file.

Downloading Rejected Records for an Import Job

You can download a report of records that an Import job rejected. The report includes all records that the Import job could not process and the reason why the job rejected the records. You download the report from the Import File job step.

You can use the report to process the rejected records. The report is a CSV file that contains the same file format that the Import job requires. Correct the records in the file and remove the column that contains the rejection reason. Then, import the file on the workspace that corresponds to the type of information that the job imported.

1. Click **Manage** > **Jobs**.

   The **Jobs** workspace appears.

2. Navigate to the Import job and click the Job ID.

   The job steps appear.

3. Click the **Download Report** icon from the Import File job step.

   After you click the icon, a dialog box appears with the downloaded file.

   The download file is named `<job step ID>.csv`. Save the file to a directory on your machine. If you download the file again, the file name includes an incremental number such as 40(1).csv.
4. Open the file.
5. Review the reasons why the Import job rejected the records and correct the entries in the file.
6. Delete the `ReasonForRejection` column.
7. Import the file.

Part III: Detection

This part contains the following chapters:

- Users and User Access, 313
- Anomaly Detection, 346
- Anomaly Suppression Rules, 369
- Actions, 375
- Manual Actions, 388
- Security Policies, 449
- Security Policy Groups, 459
- Security Policy Violations, 465
- Tasks, 483

CHAPTER 15

Users and User Access

This chapter includes the following topics:

- Users and User Access Overview, 313
- Configure an Event Share Location, 314
- Users Workspace, 315
- Import Process for Users and User Access, 315
- Import and Synchronization from an LDAP Directory Service, 317
- Import and Synchronization from Salesforce, 322
- User Import from a CSV File, 326
- User Group Import from a CSV File, 329
- User Group Membership Import from a CSV File, 332
- User Access Import from a CSV File, 335
- User Aliases Import from a CSV File, 340
- Export of Users and User Access, 342

Users and User Access Overview

After you scan data stores, you can import metadata about users and user access. The Data Privacy Management Service augments scan results and user activity data with the imported user metadata. When you view the dashboard, you can view users that have permissions to access sensitive data in data stores. When you view user activity on data stores, you can view information about the user account that performed an event on the sensitive data.

You can import the following types of user metadata:

Users

A user is an account that accesses data in a data store. A user can be an application user, a database user, or an operating system user. You manage users in external systems or LDAP directory services. You import metadata related to the user account such as full name, department, and location. You can import users from an LDAP directory service, Salesforce, and from a CSV file.

Groups

A group is a set of related users or groups that have the same data store access levels. You can import groups from an LDAP directory service, Salesforce, and from a CSV file.

Group memberships

A group membership is an assignment of a user to a group or an assignment of a group to a group. You can import group memberships from an LDAP directory service, Salesforce, and from a CSV file.

User access

User access specifies users or groups that have access to data stores and the type of access. You can import user access from Salesforce and from a CSV file.

User activity

User activity is a set of actions that a user performs on sensitive data. You can import user activity from Salesforce.

User aliases

A user alias is an alternative account name for a user. If a user has multiple accounts, you can import a list of user aliases to resolve multiple accounts to one user. You import user aliases from a CSV file.

You can import data from a csv file. You can set up a recurring job that synchronizes Data Privacy Management with user metadata from an LDAP directory service or Salesforce. When you import user metadata, you schedule an import job. The import job saves users, groups, group memberships, user access, and user aliases metadata in the Data Privacy Management repository. The import job saves user activity metadata in the user activity store.

If user information changes in the source system, you must update Data Privacy Management to reflect the changes. Import changes from a .csv file or run a synchronization job. Use the same method to create and update user metadata. For example, if you imported users from an LDAP directory service, run an LDAP synchronization job. If you imported users from a .csv file, import changes in a .csv file.

You can export information about users, user groups, user group membership, user access, and user aliases to a CSV file. You can use the exported information for reporting or analysis. You can also use the exported CSV file to update user metadata in the Data Privacy Management repository.

Configure an Event Share Location

Before you can view and analyze user activity data in Data Privacy Management, you must provide a location to store event messages streamed from the database.

Create a folder on the domain machine. Enter the folder location when you configure the Data Privacy Management Service during install. You can also update the information in the Data Privacy Management Service service properties from the Administrator Tool after you install.

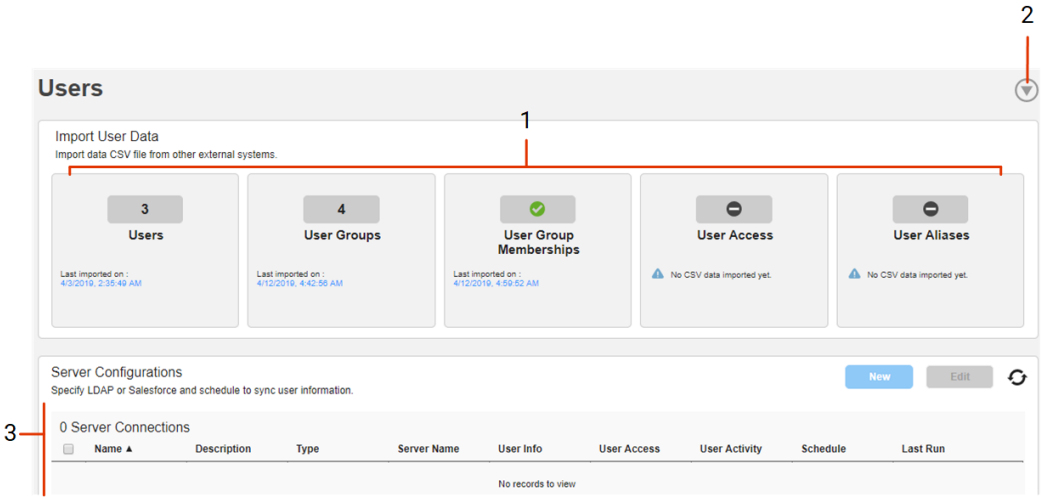
Mount the folder that you create on the domain machine to a location on the Spark node. In a multinode setup, mount the location to all Spark nodes. The path to the mounted location must be identical on the domain machine and all Spark nodes.

Create the folder on the domain machine and then create identical folder paths on all Spark nodes. Create the mount on each node after you create the folder paths.

Users Workspace

Use the Users workspace to import metadata about users and user access. The Users workspace includes a panel that displays a count of user objects in the Data Privacy Management repository and a panel to manage server connections. You can create connections to synchronize user information with LDAP directory services or Salesforce.

The following image shows an example of the Users workspace:



1. Imported user data summary
2. Actions menu
3. Server connections

To access the Users workspace, click **Manage** > **Users**.

Import Process for Users and User Access

You can import several types of user metadata from multiple sources. The process you follow depends on what you want to import and from which source. To import user metadata from multiple sources, follow the process for each source.

The following table lists the supported import types for each source:

Import Type	Supported LDAP Directory Services	Salesforce	CSV Files
Users	Yes	Yes	Yes
User groups	Yes	Yes	Yes
User group memberships	Yes	Yes	Yes
User access	No	Yes	Yes

Import Type	Supported LDAP Directory Services	Salesforce	CSV Files
User activity	No	Yes	No
User aliases	No	No	Yes

Related Topics:

- "Import and Synchronization from an LDAP Directory Service" on page 317
- "Import and Synchronization from Salesforce" on page 322
- "User Import from a CSV File" on page 326
- "User Group Import from a CSV File" on page 329
- "User Group Membership Import from a CSV File" on page 332
- "User Access Import from a CSV File" on page 335
- "User Aliases Import from a CSV File" on page 340

Import Process from Supported LDAP Directory Services

If you use manage users in Microsoft Active Directory or IBM Tivoli Directory Service, you set up a connection to an LDAP server to import and synchronize the users.

Use the following process to import metadata about users and user access:

1. Create a connection to an LDAP directory service and schedule when the synchronization occurs.
   The synchronization imports users, groups, and group memberships from the LDAP directory service based on the criteria that you specify in the connection.
2. Import user access metadata from a CSV file.
   Import the level of data store access for a user or group. When you import user access metadata for a group, all users that are members of the group and the nested groups inherit the same user access levels.
3. Optionally, import user aliases from a CSV file.
   If a user has multiple user accounts, import a list of user aliases to resolve multiple user accounts to one user.

Import Process from External Systems and Unsupported LDAP Directory Services

You can import metadata about users and user access from CSV files if you manage users in a user management system or in an LDAP directory service other than Microsoft Active Directory or IBM Tivoli Directory Service. Each import requires a separate CSV file.

Use the following process to import metadata about users and user access:

1. Import users from a CSV file.
2. Import user groups from a CSV file.
   Required to import user access for groups. Not required if you import user access only for users.
3. Import user group memberships from a CSV file.
   You can assign a user to a group, or assign a group to a group. Required to import user access for groups. Not required if you import user access only for users.

4. Import user access metadata from a CSV file.
   Import the level of data store access for a user or group. When you import user access metadata for a group, all users that are members of the group and the nested groups inherit the same user access levels.
5. Optionally, import user aliases from a CSV file.
   If a user has multiple user accounts, import a list of user aliases to resolve multiple user accounts to one user.

Import Process from Salesforce

You can set up a server definition to import and synchronize users, user groups, user group memberships, user access, and user activity from Salesforce.

Use the following process to import user metadata from Salesforce:

1. Create a Salesforce data store.
2. Create and run a Cloud scan on the Salesforce data store to identify sensitive data in Salesforce.
3. Create a Salesforce server definition. Specify the import type, select the Salesforce data store from the first step, and schedule when the synchronization occurs.
4. Optionally, import user aliases from a CSV file.
   If a user has multiple user accounts, import a list of user aliases to resolve multiple user accounts to one user.

Import and Synchronization from an LDAP Directory Service

You can import users, groups, and memberships from an LDAP directory service into the Data Privacy Management repository. You can set up a recurring job that synchronizes changes in the LDAP directory services with the Data Privacy Management repository.

You can import from Microsoft Active Directory and IBM Tivoli Directory Service. If you have users in other directory services, you must import users, groups, and group memberships from CSV files.

To import users, groups, and memberships from an LDAP directory service, you create a connection to an LDAP server. You configure the connection to the LDAP server that contains the directory service from which you want to import. You specify a base distinguished name and a search filter to define the set of users and groups that the import job imports into the Data Privacy Management repository. The base distinguished name is the point from which the LDAP server starts the search with the search filter.

You define when the import occurs and set up a schedule to periodically synchronize the Data Privacy Management repository with the LDAP directory service. When you save the connection, the Data Privacy Management Service creates an import job. You can schedule the job to run immediately, in the future, or set up a job that runs on a regular basis.

During the synchronization, the import job uses the base distinguished name and search filter to read users, groups, and group memberships from the LDAP directory service. The import job updates the Data Privacy Management repository with additions, changes, and deletions for objects that are included in the LDAP search results. The import job deletes an object from the Data Privacy Management repository if the object is no longer available in the search results for an import job with the same base distinguished name and search filter.

You can create connections to multiple LDAP servers. You can create multiple connections to the same LDAP server if each connection uses different search filters.

LDAP Connection Properties

Configure the connection to the server that contains the LDAP directory service from which you want to import users, groups, and group memberships. You can configure connection properties to connect to a Microsoft Active Directory or an IBM Tivoli Directory Service.

The connection properties cannot exceed 255 characters and are not case sensitive.

An LDAP connection includes the following properties:

LDAP Connection Name

Required. Name of the LDAP connection that is stored in the Data Privacy Management repository. The name must be unique.

Description

Optional. Description for the LDAP connection name.

Server Name

Required. Host name or IP address of the machine that hosts the LDAP directory service.

Port

Required. Listening port for the LDAP server. This is the port number to communicate with the LDAP directory service.

LDAP Directory Service

Required. Type of LDAP directory service. Select from the following directory services:

- Microsoft Active Directory
- IBM Tivoli Directory Service

User Name

Optional. LDAP user name. Leave blank for anonymous login.

Password

Required if you specify the user name. Password for the LDAP user name. Leave blank for anonymous login.

Base DN

Required. Distinguished name (DN) of the entry that serves as the starting point to search for users and groups in the LDAP directory service. The LDAP directory service finds an object in the directory according to the path in the distinguished name of the object. The import job uses the search base and filters to define the set of users and groups to import from the LDAP directory service.

For example, in Microsoft Active Directory, the distinguished name of a user object might be cn=UserName,ou=OrganizationalUnit,dc=DomainName, where the series of relative distinguished names denoted by dc=DomainName identifies the DNS domain of the object.

For more information about search filters, see the documentation for the LDAP directory service.

Search Filter

Optional. An LDAP query string that specifies the criteria to search for users and groups in the LDAP directory service. The filter can specify attribute types, assertion values, and matching criteria. The import job uses the combination of the base distinguished name and search filter to define the set of users and groups to import from the LDAP directory service.

For example, the filter (&(objectClass=user)(!(cn=susan))) searches all user objects except “susan.”

If blank, the import job uses the filter (objectclass=*) to import all user and groups in the base distinguished name.

For more information about search filters, see the documentation for the LDAP directory service.

Use SSL Certificate

Determines if the LDAP server uses the Secure Socket Layer (SSL) protocol. Required to connect to an SSL enabled directory service.

If enabled, the LDAP server uses the Secure Socket Layer (SSL) protocol.

If not enabled, the LDAP server does not use the Secure Socket Layer (SSL) protocol. Use anonymous login or simple authentication instead.

Default is not enabled.

Creating an LDAP Connection

Create an LDAP connection to configure how and when the import job synchronizes users, groups, and memberships from an LDAP directory service. When you create an LDAP connection, you specify the properties to connect to an LDAP server and schedule when the synchronization occurs. When you save the connection, the Data Privacy Management Service creates an import job.

Create an LDAP connection for each set of user and groups to import from an LDAP directory service. Set up search bases and filters to define the set of users and groups that the import job imports in to the Data Privacy Management repository. You can create multiple connections to the same directory service if each connection uses a different search filter.

1. Click **Manage** > **Users**.

   The Users page appears.

2. In the Server Configuration panel, click **New**.

   The **Define & Connect** page appears.

3. In the **Resource Type**, select **LDAP**.

   The connection properties appear.

4. Enter the LDAP connection properties.

5. Optionally, click **Test Connection** to test the connection to the LDAP server.

   The Data Privacy Management Service validates if all of the connection properties are configured, if the connection is available, and if the log in credentials are valid. A message indicates the success or failure of the validation.

   Note: The Data Privacy Management Service does not test the connection details when you save the connection. You can save an LDAP connection even if the test connection fails. For example, you can configure a connection to an LDAP server even if the server is down for maintenance.

6. Click **Next**.

   The **Schedule** page appears.

7. Schedule when the import job runs.

   You can schedule the job to run immediately, in the future, or on a recurring schedule.

8. Click **Save**.

   The Data Privacy Management Service validates that you specified all required properties and validates that the connection properties to an LDAP server are unique. The Data Privacy Management Service uses the combination of the server name, port, user, and search filter properties to determine a unique connection. If there are no errors, the Data Privacy Management Service saves the LDAP connection in the Data Privacy Management repository and creates an import job. The job runs according to the schedule that you configured. Use the Jobs workspace to monitor the status of the job.

Running LDAP Synchronization

You can trigger an import job that immediately synchronizes users, groups, and memberships from an LDAP directory service with the Data Privacy Management repository. When you define an LDAP connection, you specify when the synchronization occurs. You can synchronize once or schedule the synchronization to occur on a regular basis. To synchronize outside of the scheduled time frame, you can manually trigger the synchronization. The synchronization runs immediately.

For example, you defined a connection to an LDAP directory service and scheduled the synchronization to run every Monday. You had major changes to users on a Wednesday and do not want to wait for the next synchronization to occur the following Monday. You run the synchronization today. The next synchronization occurs as defined in the LDAP connection.

You can synchronize one LDAP connection at a time. To synchronize multiple LDAP connections, repeat the procedure for each LDAP connection.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.
2. Select the LDAP connection for which you want to synchronize with the Data Privacy Management repository.
3. From the Actions menu, click **Execute**.

   The Data Privacy Management Service creates an import job and runs the job immediately. Use the Jobs workspace to monitor the status of the job.

Editing an LDAP Connection

Edit an LDAP connection to change the connection properties or the import job scheduling options. Optionally, test the connection details. When you save the LDAP connection, the Data Privacy Management Service creates an import job.

When an import job starts, the job uses the LDAP connection properties that exist at the time the job starts. If you edit an LDAP connection that is included in a running import job, the job does not include the changes.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.
2. Select the LDAP connection to edit.
3. From the Actions menu, select **Edit**.

   The LDAP connection properties panel appears.
4. Edit the LDAP connection properties.
5. Click **Modify Password** to change the password.

   Data Privacy Management encrypts and saves the password in the Data Privacy Management repository when you save the connection.

6. Optionally, click **Test Connection** to test the connection to the LDAP server.

   The Data Privacy Management Service validates if the required connection properties are configured, if the connection is available, and if the log in credentials are valid. The Data Privacy Management Service uses the Server Name, Port, User Name, and Password properties to determine if the connection is available. The Data Privacy Management Service does not validate the BaseDN and Search Filter properties. The test connection fails if you try to connect to an SSL-enabled directory service and you did not enable the Use SSL Certificate property.

   Note: The Data Privacy Management Service does not test the connection details when you save the connection. You can save an LDAP connection even if the test connection fails. For example, you can configure a connection to an LDAP server even if the server is down for maintenance. The Import job tests the connection when the job runs. If the connection is not valid, the Import job fails.

7. Click **Save**.

   The Data Privacy Management Service validates that you specified all required properties and validates that the connection properties to an LDAP server are unique. The Data Privacy Management Service uses the combination of the server name, port, user, and search filter properties to determine a unique connection.

   If there are no errors, the Data Privacy Management Service saves the LDAP connection in theData Privacy Management repository and creates an Import job. The job runs according to the schedule that you configured. Monitor the status of the job on the **Jobs** workspace.

Copying an LDAP Connection

You can copy an LDAP connection to create an LDAP connection with similar properties. Specify the properties to connect to an LDAP server and schedule when the synchronization occurs. Optionally, test the connection details. When you save the connection, the Data Privacy Management Service creates an import job.

You can create multiple connections to the same LDAP directory service as long as the search base and filters are unique for each connection.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. Select the LDAP connection to copy.
3. From the Actions menu, select **Duplicate**.

   The LDAP connection properties panel appears.

4. Edit the LDAP connection properties.
5. Click **Modify Password** to change the password.

   Data Privacy Management encrypts and saves the password in the Data Privacy Management repository when you save the connection.

6. Optionally, click **Test Connection** to test the connection to the LDAP server.

   The Data Privacy Management Service validates if the required connection properties are configured, if the connection is available, and if the log in credentials are valid. The Data Privacy Management Service uses the Server Name, Port, User Name, and Password properties to determine if the connection is available. The Data Privacy Management Service does not validate the BaseDN and Search Filter properties. The test connection fails if you try to connect to an SSL-enabled directory service and you did not enable the Use SSL Certificate property.

   Note: The Data Privacy Management Service does not test the connection details when you save the connection. You can save an LDAP connection even if the test connection fails. For example, you can

configure a connection to an LDAP server even if the server is down for maintenance. The Import job tests the connection when the job runs. If the connection is not valid, the Import job fails.

7. Click **Save**.

   The Data Privacy Management Service validates that you specified all required properties and validates that the connection properties to an LDAP server are unique. The Data Privacy Management Service uses the combination of the server name, port, user, and search filter properties to determine a unique connection.

   If there are no errors, the Data Privacy Management Service saves the LDAP connection in theData Privacy Management repository and creates an Import job. The job runs according to the schedule that you configured. Monitor the status of the job on the **Jobs** workspace.

Deleting an LDAP Connection

You can delete an LDAP connection from the Data Privacy Management repository. When you delete an LDAP connection, the Data Privacy Management Service deletes all associated import jobs that are scheduled to run in the future. The users, groups, and group memberships that were imported from the LDAP directory service remain in the repository.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. Select the LDAP connection to delete.
3. From the Actions menu, select **Delete**.

   The Data Privacy Management Service deletes the LDAP connection from the repository.

Import and Synchronization from Salesforce

You can import users, user groups, user group memberships, user access, and user activity from Salesforce. You can set up a recurring job that synchronizes changes in Salesforce with Data Privacy Management. After the import completes, you can view results in the dashboard. You can import user metadata if your Salesforce org has the Bulk API feature enabled.

To import user metadata from Salesforce, you create a connection to a Salesforce instance. You add the Salesforce data store from which to import user metadata. You define when the import occurs and set up a schedule to periodically synchronize Data Privacy Management with Salesforce. When you save the connection, the Data Privacy Management Service creates an Import job. You can schedule the job to run immediately, in the future, or set up a job that runs on a regular basis.

During the synchronization, the Import job uses the Salesforce Bulk REST API to query the following information from Salesforce:

Users

The Import job retrieves active and inactive users. The job retrieves the user ID, department, email, first and last name, active status, manager ID, title, and user name.

User groups

The Import job retrieves user groups from Salesforce permission sets when you import user access.

User group memberships

The Import job retrieves user group memberships from the Salesforce permission set assignments when you import user access.

User access

The Import job queries the Salesforce ObjectPermission and FieldPermission objects to retrieve user access information for user groups. All users that are members of the group and the nested groups inherit the same user access. The job imports user access only for data that is identified as sensitive in the Data Privacy Management repository.

User activity

The Import job retrieves user activity from the Salesforce EventLogFile object for the API, REST API, and Bulk API event types. The job stores the events in the user activity store. The job imports user activity from a predefined time range. The time range that the job uses depends on the import synchronization schedule. The time frame does not include the current day when the job runs. The time frame starts the day before the job runs.

For example, if you run the job immediately or schedule the job to run monthly, the job imports user activity from the last 30 days. If you schedule the job to run daily, the job imports user activity from the last day. If you schedule the job to run weekly, the job imports user activity from the last week.

The Users workspace counts and the dashboard include all imported active and inactive users. The risk score only considers active users in the user access count factor. The risk score considers active and inactive users in the user activity count factor.

Creating a Salesforce Connection

Create a Salesforce connection to configure how and when the import job synchronizes user metadata from Salesforce. When you create a connection, you specify the metadata to import and the Salesforce data store from which to import the data. You schedule when the synchronization occurs. When you save the connection, the Data Privacy Management Service creates an import job.

You can create multiple connections that import metadata from the same Salesforce data store. For example, you create one connection to import users and user access on one synchronization schedule, such as on a daily basis. You create another connection to import users and user activity on a different synchronization schedule, such as on a weekly basis.

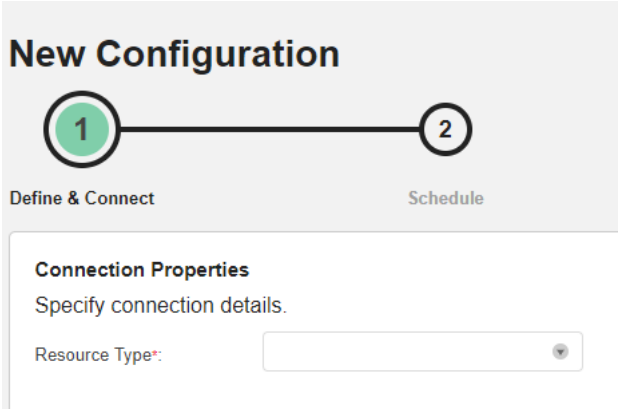
The import job only imports user access for data that is identified as sensitive in the Data Privacy Management repository. Before you create a Salesforce connection, scan the Salesforce data store to identify sensitive data.

1. Click **Manage** > **Users**.

   The **Users** page appears.

2. In the **Server Configurations** panel, click **New**.

The **Define & Connect** page appears.



3. In the **Resource Type**, select **Salesforce**.

   The connection properties appear.

4. Enter the following connection properties:

   Name

   Required. Name of the Salesforce synchronization connection to create in the Data Privacy Management repository. The name must be unique.

   Connection

   Required. Name of the Salesforce data store from which you want to import data. You can choose an existing data store or create a data store that connects to Salesforce. The list shows the Salesforce data stores that exist in the Data Privacy Management repository and that include complete connection properties. The list does not show data stores with incomplete connection properties as the job cannot connect to an incomplete data store.

   URL

   Read-only. When you select a connection, the Data Privacy Management Service populates the value with the Salesforce Service URL configured in the data store.

5. Specify the objects to synchronize.

   You can select one or more options in the same synchronization job. By default, the job always synchronizes users.

6. Click **Next**.

   The **Schedule** page appears.

7. Schedule when the import job runs.

   You can schedule the job to run immediately, in the future, or on a recurring schedule.
   The scheduling options determine from which time range the job imports user activity. The time range does not include the current day when the job runs. If you run the job immediately or schedule the job to run monthly, the job imports user activity information from the last 30 days. If you schedule the job to run daily, the job imports user activity from the last day. If you schedule the job to run weekly, the job imports user activity from the last week.

8. Click **Save**.

   The Data Privacy Management Service saves the connection in the Data Privacy Management repository and creates an import job. The job runs according to the schedule that you configured. Use the Jobs workspace to monitor the status of the job.

Related Topics:

- "Salesforce Import job" on page 303

Running Salesforce Synchronization

You can trigger an import job that immediately synchronizes user metadata from Salesforce with the Data Privacy Management repository. When you define a Salesforce connection, you specify when the synchronization occurs. You can synchronize once or schedule the synchronization to occur on a regular basis. To synchronize outside of the scheduled time frame, you can manually trigger the synchronization. The synchronization runs immediately.

For example, you defined a connection and scheduled the synchronization to run every Monday. You had major changes to users on a Wednesday and do not want to wait for the next synchronization to occur the following Monday. You run the synchronization today. The next synchronization occurs as defined in the connection.

You can synchronize one connection at a time. To synchronize multiple connections, repeat the procedure for each connection.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.
2. Select the connection to synchronize with the Data Privacy Management Service repository.
3. From the Actions menu, click **Execute**.

   The Data Privacy Management Service creates an import job and runs the job immediately. Use the Jobs workspace to monitor the status of the job.

Editing a Salesforce Connection

Edit a connection to change the connection properties, the import options, or the import job scheduling options. When you save the connection, the Data Privacy Management Service creates an import job.

When an import job starts, the job uses the connection properties that exist at the time the job starts. You cannot edit a connection that is included in a running import job.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.
2. Select the connection to edit.
3. From the Actions menu, click **Edit**.
4. Click **Save**.

   The Data Privacy Management Service saves the connection in the Data Privacy Management repository and creates an import job. The job runs according to the schedule that you configured. Use the Jobs workspace to monitor the status of the job.

Deleting a Salesforce Connection

You can delete a connection from the Data Privacy Management repository. When you delete a connection, the Data Privacy Management Service deletes all associated import jobs that are scheduled to run in the

future. The users, user access, and user activity that were imported from the connection remain in the repository.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.
2. Select the connection to delete.
3. From the Actions menu, select **Delete**.

   The Data Privacy Management Service deletes the connection from the repository.

User Import from a CSV File

To create, update, or delete a user, you can import information from a .csv file. The import job uses the details in the .csv file to create, update, or delete a user in the Data Privacy Management repository. The action that the import job performs depends on the action that you specify in the .csv file.

CSV File Format to Import Users

Create a CSV file to create, update, or delete information about users in the Data Privacy Management repository. The content of the CSV file must be in the order and format that the import job requires. The CSV file must contain all of the column headings. However, not all columns require values.

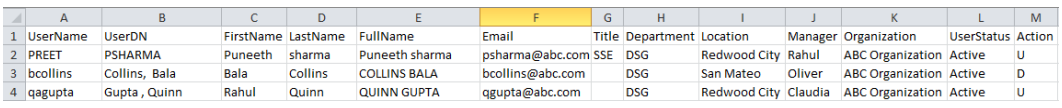
The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Column Value Description
1	UserName	Required. Name of a user account that accesses information in a data store. The combination of the UserName and UserDN columns identifies a unique user to create, update, or delete. If the UserDN is blank, the UserName must be unique. If blank, the import job rejects the row.
2	UserDN	Optional. Distinguished name for the principal user. The combination of the UserName and UserDN columns identifies a unique user to create, update, or delete. If blank, the UserName identifies a unique user.
3	FirstName	Optional. First name of the user.
4	LastName	Optional. Last name of the user.
5	FullName	Optional. Full name of the user. If blank, the import job populates the property with the values from the FirstName and LastName columns.
6	Email	Optional. Email address of the user.
7	Title	Optional. Job title of the user.

Column Order	Column Heading	Column Value Description
8	Department	Optional. Department of the user. This department is not related to the data store departments in the Data Privacy Management repository. If blank, default is UNKNOWN.
9	Location	Optional. Location of the user. This location is not related to the data store locations in the Data Privacy Management repository. If blank, default is UNKNOWN.
10	Manager	Optional. Manager of the user.
11	Organization	Optional. Organization of the user.
12	UserStatus	Optional. Status of the user. Enter one of the following values: - A. Active - I. Inactive If blank, default is A (active).
13	Action	Optional. Instructs the import job to create, update, or delete a user in the Data Privacy Management repository. Enter one of the following values: - U. Creates or updates a user in the Data Privacy Management repository. The action that the job performs depends on if the user exists in the Data Privacy Management repository and on the Replace Duplicates option. You configure the Replace Duplicates option when you import the .csv file. If the user does not exist, the import job creates the user. If the user exists and the Replace Duplicates option is enabled, the import job updates the user. **Warning:** If a column in the .csv file contains a null value, the import job replaces the value in the Data Privacy Management repository with a null value. If the user exists and the Replace Duplicates option is not enabled, the import job skips the row. - D. Deletes a user from the Data Privacy Management repository. The user activity data related to the user remains. If the user does not exist, the import job rejects the row. If blank, default is U. Creates or updates a user in the Data Privacy Management repository.

Example CSV File to Import Users

The following image shows the spreadsheet view of a sample .csv file where the first row contains the column headers and each subsequent row contains the details for a user:



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	UserName	UserDN	FirstName	LastName	FullName	Email	Title	Department	Location	Manager	Organization	UserStatus	Action
2	PREET	PSHARMA	Puneeth	sharma	Puneeth sharma	psharma@abc.com	SSE	DSG	Redwood City	Rahul	ABC Organization	Active	U
3	bcollins	Collins, Bala	Bala	Collins	COLLINS BALA	bcollins@abc.com		DSG	San Mateo	Oliver	ABC Organization	Active	D
4	qagupta	Gupta , Quinn	Rahul	Quinn	QUINN GUPTA	qgupta@abc.com		DSG	Redwood City	Claudia	ABC Organization	Active	U

The following image shows the text view of a sample .csv file where the first row contains the column headers and each subsequent row contains the details for a user:

```
1  UserName,UserDN,FirstName,LastName,FullName,Email,Title,Department,Location,Manager,Organization,UserStatus,Action
2  PREET,PSHARMA,Puneeth,sharma,Puneeth sharma,psharma@abc.com,SSE,DSG,Redwood City,Rahul,ABC Organization,Active,U
3  bcollins,"Collins,  Bala",Bala,Collins,COLLINS BALA,bcollins@abc.com,,DSG,San Mateo,Oliver,ABC Organization,Active,D
4  qagupta,"Gupta , Quinn",Rahul,Quinn,QUINN GUPTA,qgupta@abc.com,,DSG,Redwood City,Claudia,ABC Organization,Active,U
```

Rules and Guidelines to Import Users

Review the rules and guidelines to understand how to manage the .csv file and how the import job processes data from the .csv file.

Consider the following rules and guidelines when you import users from a .csv file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines a user.
  The combination of the UserName and UserDN columns defines a unique user. If the UserName and UserDN in one row is the same as the UserName and UserDN in a second row, the import job treats the second row as a duplicate of the first row.
- If the .csv file contains duplicate entries, the import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

  ```
  SELECT FROM
  SELECT UNION
  DELETE FROM
  UPDATE SET
  <SCRIPT
  </SCRIPT>
  ALERT(
  "/>"
  ```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

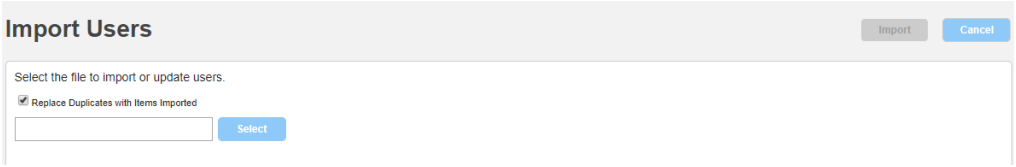
Steps to Import Users

You import user information from a CSV file. The file must be in the format that the import job requires. When you import the file, the import job runs immediately. The import job updates the Data Privacy Management repository with information from the CSV file.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.
2. From the Actions menu, click **Import** > **Users**.

The **Import Users** page appears.



3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the import job handles updates for users that exist in the Data Privacy Management repository.
   If you enable the option, the import job updates the user properties in the Data Privacy Management repository with the values in the .csv file.

   Warning: If a column in the .csv file contains a null value, the import job replaces the value in the Data Privacy Management repository with a null value.

   If you disable the option, the import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the import job completes, you can view a list of the imported users in the User Access page of the dashboard.

Related Topics:

- "Import Job" on page 299
- "Downloading Rejected Records for an Import Job" on page 310
- "Top Users Indicator" on page 529

User Group Import from a CSV File

To create, update, or delete a user group, you import information from a CSV file. The import job uses the details in the CSV file to create, update, or delete a group in the Data Privacy Management repository. The action that the import job performs depends on the action that you specify in the CSV file.

CSV File Format to Import User Groups

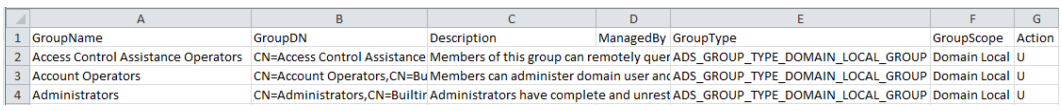
Create a CSV file to create, update, or delete information about user groups in the Data Privacy Management repository. The content of the CSV file must be in the order and format that the import job requires. The CSV file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Column Value Description
1	GroupName	Required. Name of a group. The combination of the GroupName and GroupDN columns identifies a unique group to create, update, or delete. If the GroupDN is blank, the GroupName must be unique. If blank, the import job rejects the row.
2	GroupDN	Optional. Distinguished name for the group. The combination of the GroupName and GroupDN columns identifies a unique group to create, update, or delete. If blank, the GroupName determines a unique group.
3	Description	Optional. Description of the group.
4	ManagedBy	Optional. Owner of the group.
5	GroupType	Optional. Type of group.
6	GroupScope	Optional. Scope of group.
7	Action	Optional. Instructs the import job to create, update, or delete a group in the Data Privacy Management repository. Enter one of the following values: - U. Creates or updates a group in the Data Privacy Management repository. The action that the job performs depends on if the group exists in the repository and on the Replace Duplicates option. You configure the Replace Duplicates option when you import the CSV file. If the group does not exist, the import job creates the group. If the group exists and the Replace Duplicates option is enabled, the import job updates the group. **Warning:** If a column in the CSV file contains a null value, the import job replaces the value in the Data Privacy Management repository with a null value. If the group exists and the Replace Duplicates option is not enabled, the import job skips the row. - D. Deletes a group and associated group memberships from the repository. If the group does not exist, the import job rejects the row. If blank, default is U. Creates or updates a group in the repository.

Example of CSV File to Import User Groups

The following image shows the spreadsheet view of a sample .csv file where the first row contains the column headers, and each subsequent row contains details for a group:



	A	B	C	D	E	F	G
1	GroupName	GroupDN	Description	ManagedBy	GroupType	GroupScope	Action
2	Access Control Assistance Operators	CN=Access Control Assistance	Members of this group can remotely quer		ADS_GROUP_TYPE_DOMAIN_LOCAL_GROUP	Domain Local	U
3	Account Operators	CN=Account Operators,CN=Bu	Members can administer domain user and		ADS_GROUP_TYPE_DOMAIN_LOCAL_GROUP	Domain Local	U
4	Administrators	CN=Administrators,CN=Builtir	Administrators have complete and unrest		ADS_GROUP_TYPE_DOMAIN_LOCAL_GROUP	Domain Local	U

The following image shows the text view of a sample .csv file where the first row contains the column headers, and each subsequent row contains details for a group:

```
1  GroupName,GroupDN,Description,ManagedBy,GroupType,GroupScope,Action
2  Access Control Assistance Operators,"CN=Access Control Assistance Operators,CN=Bui
   for resources on this computer.,,ADS_GROUP_TYPE_DOMAIN_LOCAL_GROUP,Domain Local,U
3  Account Operators,"CN=Account Operators,CN=Builtin,DC=secsrc,DC=com",Members can a
4  Administrators,"CN=Administrators,CN=Builtin,DC=secsrc,DC=com",Administrators have
```

Rules and Guidelines to Import User Groups

Review the rules and guidelines to understand how to manage the .csv file and how the import job processes data from the .csv file.

Consider the following rules and guidelines when you import user groups from a .csv file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines a group.
  The combination of the GroupName and GroupDN columns defines a unique group. If the GroupName and GroupDN in one row is the same as the GroupName and GroupDN in a second row, the import job treats the second row as a duplicate of the first row.
- If the CSV file contains duplicate entries, the Import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

  ```
  SELECT FROM
  SELECT UNION
  DELETE FROM
  UPDATE SET
  <SCRIPT
  </SCRIPT>
  ALERT(
  "/>"
  ```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

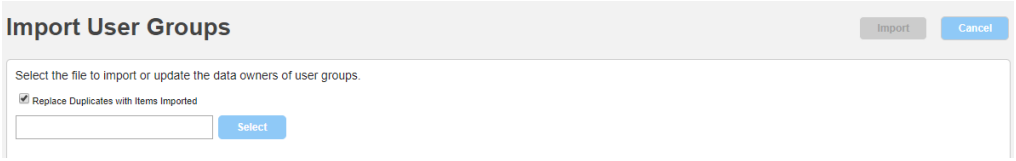
Steps to Import Groups

You import group information from a CSV file. The CSV file must be in the format that the import job requires. When you import the file, the import job runs immediately. The import job updates the Data Privacy Management repository with information from the CSV file.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.
2. From the Actions menu, click **Import** > **User Groups**.

The **Import User Groups** page appears.



3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the import job handles updates for groups that exist in the Data Privacy Management repository.
   If you enable the option, the import job updates the group properties in the repository with the values in the CSV file.

   Warning: If a column in the CSV file contains a null value, the import job replaces the value in the repository with a null value.

   If you disable the option, the import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the import job completes, you can view a list of the imported groups in the User Access page of the dashboard.

Related Topics:

- "Import Job" on page 299
- "Downloading Rejected Records for an Import Job" on page 310

User Group Membership Import from a CSV File

To create, update, or delete a user group membership, you import information from a CSV file. The import job uses the details in the CSV file to create, update, or delete a group membership in the Data Privacy Management repository. The action that the import job performs depends on the action that you specify in the CSV file.

When you import group membership, you specify the users and groups that are assigned to a parent group. The users and groups must exist in the repository before you begin the import.

CSV File Format to Import Group Memberships

Create a CSV file to create, update, or delete information about group membership in the Data Privacy Management repository. The content of the file must be in the order and format that the import job requires. The file must contain all of the column headings. However, not all columns require values.

Import group membership after you import users and groups.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Column Value Description
1	MemberName	Required. Name of the user or group that is a member of a group. The combination of the MemberName and MemberDN columns identifies a unique member for which to create, update, or delete a membership. If blank, the import job rejects the row.
2	MemberDN	Distinguished name of the member. The combination of the MemberName and MemberDN columns identifies a unique member for which to create, update, or delete a membership. Required if the member has a member distinguished name in the Data Privacy Management repository.
3	MemberType	Optional. Indicates the type of member to assign to a group. Enter one of the following values: - U. The member is a user. - G. The member is a group. If blank, default is U (User).
4	MemberOf	Required. Name of the group to which the member is assigned. If blank or the group does not exist in the repository, the import job skips the row.
5	MemberOfDN	Optional. Distinguished name of the parent group.
6	Action	Optional. Instructs the import job to create, update, or delete a group membership in the repository. Enter one of the following values: - U. Creates or updates a group membership in the repository. The action that the job performs depends on if the group membership exists in the repository and on the Replace Duplicates option. You configure the Replace Duplicates option when you import the .csv file. If the group membership does not exist, the import job creates the group membership. If the group membership exists and the Replace Duplicates option is enabled, the import job updates the group membership. If the group membership exists and the Replace Duplicates option is not enabled, the import job skips the row. - D. Deletes a group membership from the repository. If the group membership does not exist, the import job rejects the row. If blank, default is U. Creates or updates a group membership in the repository.

Example of CSV File to Import Group Memberships

The following image shows the spreadsheet view of a sample .csv file where the first row contains the column headers, and each subsequent row contains the details for a group membership:

	A	B	C	D	E	F
1	MemberName	MemberDN	MemberType	MemberOf	MemberOfDN	Action
2	PREET	PSHARMA	U	Access Control Assist	CN=Access Control A:	U
3	bcollins	Collins, Bala	U	Account Operators	CN=Account Operato	U
4	qagupta	Gupta , Quinn	U	Administrators	CN=Administrators,C	D

The following image shows the text view of a sample .csv file where the first row contains the column headers, and each subsequent row contains the details for a group membership:

```
1  MemberName,MemberDN,MemberType,MemberOf,MemberOfDN,Action
2  PREET,PSHARMA,U,Access Control Assistance Operators,"CN=Access Control Assistance Operators,CN=Builtin
3  bcollins,"Collins,Bala",U,Account Operators,"CN=Account Operators,CN=Builtin,DC=secsrc,DC=com",U
4  qagupta,"Gupta,Quinn",U,Administrators,"CN=Administrators,CN=Builtin,DC=secsrc,DC=com",D
```

Rules and Guidelines to Import Group Memberships

Review the rules and guidelines to understand how to manage the .csv file and how the import job processes data from the .csv file.

Consider the following rules and guidelines when you import group memberships from a .csv file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines a group membership.
  The combination of the MemberName and MemberDN defines a unique member for which to create or update memberships. A member can be a user or a group. If the MemberName and MemberDN in one row is the same as the MemberName and MemberDN in a second row, the import job treats the second row as a duplicate of the first row.
- If the CSV file contains duplicate entries, the Import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

  ```
  SELECT FROM
  SELECT UNION
  DELETE FROM
  UPDATE SET
  <SCRIPT
  </SCRIPT>
  ALERT(
  "/>"
  ```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Steps to Import Group Memberships

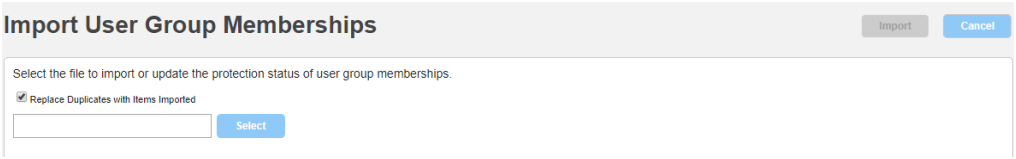
You import group membership information from a CSV file. The CSV file must be in the format that the import job requires. When you import the file, the import job runs immediately. The import job updates the Data Privacy Management repository with information from the file.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the Actions menu, click **Import** > **User Group Memberships**.

   The **Import User Group Memberships** page appears.



3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the import job handles updates for group memberships that exist in the repository.
   If you enable the option, the import job updates the group membership properties in the repository with the values in the CSV file.

   Warning: If a column in the CSV file contains a null value, the import job replaces the value in the repository with a null value.

   If you disable the option, the import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

Related Topics:

- "Import Job" on page 299
- "Downloading Rejected Records for an Import Job" on page 310

User Access Import from a CSV File

To create, update, or delete information about user access, you import information from a CSV file. The import job uses the details in the file to create, update, or delete user access information in the Data Privacy

Management repository. The action that the import job performs depends on the action that you specify in the file.

You import user access details into the Data Privacy Management repository from a CSV file if you manage user access in an access management system such as SailPoint. You import the level of access that a user or group has to a data store. You can specify access at the data store, schema, table, or column levels.

You also import the type of permissions the user or group has to the data store, such as read, write, or delete permissions. For example, user jdoe has read and write permissions on the employee table in the payroll schema. When you import user access information for a group, all users that are members of the group and the nested groups inherit the same user access levels.

When you import user access, you specify users or groups that have access to data stores. The users, groups, and data stores must exist in the repository before you begin the import. To import user access to a schema, table, or column in a data store, the repository must include scan results for the data store.

CSV File Format to Import User Access

Create a CSV file to create, update, or delete information about user access information in the Data Privacy Management repository. The content of the file must be in the order and format that the import job requires. The file must contain all of the column headings. However, not all columns require values.

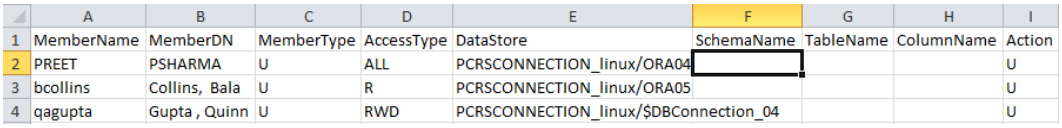
The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Column Value Description
1	MemberName	Required. User or group name for which you want to import access levels. The combination of the MemberName and MemberDN columns identifies a unique user to create, update, or delete. If blank, the import job rejects the row.
2	MemberDN	Distinguished name of the user or group. The combination of the MemberName and MemberDN columns identifies a unique user to create, update, or delete. Required if the member has a member distinguished name in the Data Privacy Management repository.
3	MemberType	Optional. Determines the type of member for the member name. Enter one of the following values: - U. The member name is a user. - G. The member name is a group. If blank, default is U (User).
4	AccessType	Optional. Determines the level of access that the user or group has to a data store. Enter one of the following values: - R. Read access. - W. Write access - D. Delete access. - RWD. Read, write, and delete access. - All. Includes all administrator privileges, such as modify schema and tables. If blank, default is All.

Column Order	Column Heading	Column Value Description
5	DataStore	Required. Name of the data store to which the user has access. You can specify one data store in a row. To specify access to multiple data stores for a member, create one row for each data store. If the data store name is blank or does not exist in the repository, the import job rejects the row. If you do not enter a SchemaName, TableName, or ColumnName, the import job sets access to all schemas and tables for the data store.
6	SchemaName	Optional. Name of the schema that identifies the table to which the user or group has access. The name is case sensitive. If you specify a SchemaName, you must specify a TableName. If blank, the import job sets access to all schemas that Data Privacy Management identified as sensitive for the data store. If the table is not identified as sensitive in the repository or if the case does not match, the import job rejects the row.
7	TableName	Required if you specified a SchemaName or a ColumnName. Name of the table in the data store to which the user or group has access. The name is case sensitive. If blank and you did not specify a SchemaName, the import job sets access to all tables that Data Privacy Management identified as sensitive for the data store. If blank and you did specify a SchemaName, the import job rejects the row. If the table is not identified as sensitive in the repository or if the case does not match, the import job rejects the row.
8	ColumnName	Optional. Name of the column to which the user or group has access. The name is case sensitive. If you specify a ColumnName, you must specify a TableName. If blank, the import job sets access to all columns that Data Privacy Management identified as sensitive for the table in the data store. If the column is not identified as sensitive in the repository or if the case does not match, the import job rejects the row.
9	Action	Optional. Instructs the import job to create, update, or delete user access in the repository. Enter one of the following values: - U. Creates or updates user access in the repository. The action that the job performs depends on if the user access exists in the repository and on the Replace Duplicates option. You configure the Replace Duplicates option when you import the CSV file. If the user access does not exist, the import job creates the user access. If the user access exists and the Replace Duplicates option is enabled, the import job updates the user access. **Warning:** If a column in the .csv file contains a null value, the import job replaces the value in the repository with a null value. If the user access exists and the Replace Duplicates option is not enabled, the import job skips the row. - D. Deletes user access from the repository. If the user access does not exist, the import job rejects the row. If blank, default is U. Creates or updates user access in the repository.

Example of CSV File to Import User Access

The following image shows the spreadsheet view of a sample .csv file where the first row contains the column headers, and each subsequent row contains the details for user access:



	A	B	C	D	E	F	G	H	I
1	MemberName	MemberDN	MemberType	AccessType	DataStore	SchemaName	TableName	ColumnName	Action
2	PREET	PSHARMA	U	ALL	PCRSCONNECTION_linux/ORA04				U
3	bcollins	Collins, Bala	U	R	PCRSCONNECTION_linux/ORA05				U
4	qagupta	Gupta , Quinn	U	RWD	PCRSCONNECTION_linux/$DBConnection_04				U

The following image shows the text view of a sample .csv file where the first row contains the column headers and each subsequent row contains the details for user access:

```
1  MemberName,MemberDN,MemberType,AccessType,DataStore,SchemaName,TableName,ColumnName,Action
2  PREET,PSHARMA,U,ALL,PCRSCONNECTION_linux/ORA04,,,,U
3  bcollins,"Collins,Bala",U,R,PCRSCONNECTION_linux/ORA05,,,,U
4  qagupta,"Gupta,Quinn",U,RWD,PCRSCONNECTION_linux/$DBConnection_04,,,,U
```

Rules and Guidelines to Import User Access

Review the rules and guidelines to understand how to manage the .csv file and how the import job processes data from the .csv file.

Consider the following rules and guidelines when you import user access from a .csv file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines user access.
  The combination of the MemberName and MemberDN columns defines one unique member for which to create or update access. A member can be a user or a group. If the MemberName and MemberDN in one row is the same as the MemberName and MemberDN in a second row, the import job treats the second row as a duplicate of the first row.
- If the .csv file contains duplicate entries, the import job processes the entries in the order in which the rows appear in the file.
- The column values cannot contain the following keywords or terms:

  ```
  SELECT FROM
  SELECT UNION
  DELETE FROM
  UPDATE SET
  <SCRIPT
  </SCRIPT>
  ALERT(
  "/>"
  ```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Steps to Import User Access

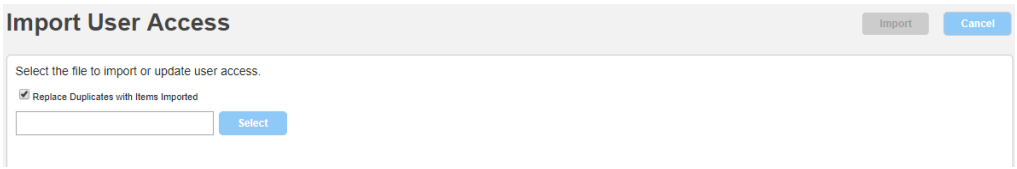
You import user access information from a CSV file. The file must be in the format that the import job requires. When you import the file, the import job runs immediately. The import job updates the Data Privacy Management repository with information from the file.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the Actions menu, click **Import** > **User Access**.

   The **Import User Access** page appears.



3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the import job handles updates for user access details that exist in the Data Privacy Management repository.
   If you enable the option, the import job updates the user access details in the repository with the values in the CSV file.

   Warning: If a column in the file contains a null value, the import job replaces the value in the repository with a null value.

   If you disable the option, the import job skips the row.

4. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

5. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

After the import job completes, you can view how many users have access to sensitive fields in the User Access indicator of the dashboard. You can also view the access levels for a user in the User Access page.

Related Topics:

- "Import Job" on page 299
- "Downloading Rejected Records for an Import Job" on page 310
- "User Access Indicator" on page 531

User Aliases Import from a CSV File

To create or delete information about user aliases, you import information from a CSV file. The import job uses the details in the file to create or delete user alias information in the Data Privacy Management repository. The action that the import job performs depends on the action that you specify in the file.

For example, John Doe has a user account `jdoe` in an LDAP directory service and a user account `johnd` in an external user management system. You can import johnd as a user alias for jdoe. When you view user access and user activity information in the dashboard, data for both user accounts is consolidated into one account.

When you import user aliases, you specify alias names for users. The users must exist in the Data Privacy Management repository before you begin the import.

CSV File Format to Import User Aliases

Create a CSV file to create or delete information about user aliases in the Data Privacy Management repository. The content of the file must be in the order and format that the import job requires. The file must contain all of the column headings. However, not all columns require values.

The following table lists the order of the columns, column headings, and the format for the values in the CSV file:

Column Order	Column Heading	Column Value Description
1	UserName	Required. Name of a user account that accesses information in a data store. The combination of the UserName and UserDN columns identifies a unique user for which to create or delete an alias. If blank, the import job rejects the row.
2	UserDN	Distinguished name for the principal user. The combination of the UserName and UserDN columns identifies a unique user for which to create or delete an alias. Required if the user has a user distinguished name in the repository.
3	AliasName	Required. Alternative name or alias for a user account. You can specify one alias in a row. To specify multiple aliases for a user, create one row for each alias. If blank, the import job rejects the row.
4	Action	Optional. Instructs the import job to create or delete an alias in the repository. Enter one of the following values: - U. Creates an alias in the repository. - D. Deletes an alias from the repository. If blank, default is U. Creates an alias in the repository.

Example of CSV File to Import User Aliases

The following image shows the spreadsheet view of a sample .csv file where the first row contains the column headers and each subsequent row contains the details for an alias:

	A	B	C	D
1	UserName	UserDN	AliasName	Action
2	PREET	PSHARMA	Sharma, Puneeth	U
3	bcollins	Collins@abc.com	Collins, Bala	U
4	qagupta	Gupta, Q	Gupta, Quinn	U

The following image shows the text view of a sample .csv file where the first row contains the column headers and each subsequent row contains the details for an alias:

```
1  UserName,UserDN,AliasName,Action
2  PREET,PSHARMA,"Sharma, Puneeth",U
3  bcollins,Collins@abc.com,"Collins, Bala",U
4  qagupta,"Gupta, Q","Gupta, Quinn",U
```

Rules and Guidelines to Import User Aliases

Review the rules and guidelines to understand how to manage the .csv file and how the import job processes data from the .csv file.

Consider the following rules and guidelines when you import user aliases from a .csv file:

- The first row of the CSV file must contain case-sensitive column headings in the required sequence. If the CSV file does not contain all of the column headings in the required sequence and case, the Import job fails.
- Each subsequent row after the column headings defines one alias for one user.
  The combination of the UserName and UserDN columns defines a unique user. If the UserName and UserDN in one row is the same as the UserName and UserDN in a second row, the import job creates two aliases for the user.
- The column values cannot contain the following keywords or terms:

  ```
  SELECT FROM
  SELECT UNION
  DELETE FROM
  UPDATE SET
  <SCRIPT
  </SCRIPT>
  ALERT(
  "/>"
  ```

  If a row contains keywords or characters that are not valid, the Import job rejects the row. The Import job logs an error in the job log and processes the remaining rows.
- The column values cannot exceed 255 characters. If a row contains more than 255 characters in any column, the Import job rejects the row.
- Enclose names that include a dot (.) or comma (,) within quotes for any column value that includes a name. If a value for a name column includes a dot or comma not enclosed within quotes, the Import job might not process the column values correctly. For example, the Import job processes a comma as a delimiter between column values.

Steps to Import User Aliases

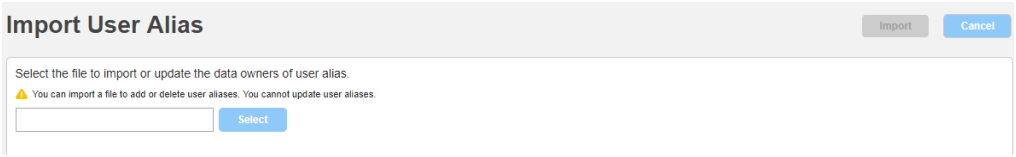
You import user alias information from a CSV file. The file must be in the format that the import job requires. When you import the file, the import job runs immediately. The import job updates the Data Privacy Management repository with information from the file.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the Actions menu, click **Import** > **User Aliases**.

   The **Import User Alias** page appears.



3. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

4. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

Related Topics:

- "Import Job" on page 299
- "Downloading Rejected Records for an Import Job" on page 310

Export of Users and User Access

You can export information about users, user groups, user group membership, user access, and user aliases to a CSV file. You can use the exported information for reporting or analysis. You can also use the exported CSV file to update user information in the Data Privacy Management repository.

You cannot create, edit, or delete user and user access information from a workspace. To add, edit, or delete user information, you must import details from a file. You can export the type of user information that you want to edit. The export file contains the same format that is required for the import. Add changes to the file. Then, import the file. When you import the file, the import job updates user information in the repository.

Exporting Users

You can export a list of users to a CSV file. The export file contains a list of all user names and properties. You can export a list of users to analyze the data offline. You can also export a list to update user data in the Data Privacy Management repository.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the Actions menu, click **Export** > **Users**.

   A dialog box appears with the exported file. The export file is named `Usermanagement_Users.csv`. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, the service downloads another file. The file name includes an incremental number such as `Usermanagement_Users (1).csv`.

3. Open the file. The following image shows the spreadsheet view of a sample export file:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	UserName	UserDN	FirstName	LastName	FullName	Email	Title	Department	Location	Manager	Organization	UserStatus	Action
2	PREET	PSHARMA	Puneeth	sharma	Puneeth sharma	psharma@abc.com	SSE	DSG	Redwood City	Rahul	ABC Organization	Active	U
3	bcollins	Collins, Bala	Bala	Collins	COLLINS BALA	bcollins@abc.com		DSG	San Mateo	Oliver	ABC Organization	Active	D
4	qagupta	Gupta , Quinn	Rahul	Quinn	QUINN GUPTA	qgupta@abc.com		DSG	Redwood City	Claudia	ABC Organization	Active	U

   The following image shows the text view of a sample export file:

```
1  UserName,UserDN,FirstName,LastName,FullName,Email,Title,Department,Location,Manager,Organization,UserStatus,Action
2  PREET,PSHARMA,Puneeth,sharma,Puneeth sharma,psharma@abc.com,SSE,DSG,Redwood City,Rahul,ABC Organization,Active,U
3  bcollins,"Collins,  Bala",Bala,Collins,COLLINS BALA,bcollins@abc.com,,DSG,San Mateo,Oliver,ABC Organization,Active,D
4  qagupta,"Gupta ,  Quinn",Rahul,Quinn,QUINN GUPTA,qgupta@abc.com,,DSG,Redwood City,Claudia,ABC Organization,Active,U
```

Exporting User Groups

You can export a list of user groups to a CSV file. The export file contains a list of all user groups and properties. You can export a list of user groups to analyze the data offline. You can also export a list to update user group data in the Data Privacy Management repository.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the Actions menu, click **Export** > **User Groups**.

   A dialog box appears with the exported file. The export file is named `Usermanagement_Groups.csv`. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, the service downloads a new file. The file name includes an incremental number such as `Usermanagement_Groups (1).csv`.

3. Open the file. The following image shows the spreadsheet view of a sample export file:

	A	B	C	D	E	F	G
1	GroupName	GroupDN	Description	ManagedBy	GroupType	GroupScope	Action
2	Access Control Assistance Operators	CN=Access Control Assistance	Members of this group can remotely quer		ADS_GROUP_TYPE_DOMAIN_LOCAL_GROUP	Domain Local	U
3	Account Operators	CN=Account Operators,CN=Bu	Members can administer domain user and		ADS_GROUP_TYPE_DOMAIN_LOCAL_GROUP	Domain Local	U
4	Administrators	CN=Administrators,CN=Builtir	Administrators have complete and unrest		ADS_GROUP_TYPE_DOMAIN_LOCAL_GROUP	Domain Local	U

   The following image shows the text view of a sample export file:

```
1  GroupName,GroupDN,Description,ManagedBy,GroupType,GroupScope,Action
2  Access Control Assistance Operators,"CN=Access Control Assistance Operators,CN=Bui
   for resources on this computer.,,ADS_GROUP_TYPE_DOMAIN_LOCAL_GROUP,Domain Local,U
3  Account Operators,"CN=Account Operators,CN=Builtin,DC=secsrc,DC=com",Members can a
4  Administrators,"CN=Administrators,CN=Builtin,DC=secsrc,DC=com",Administrators have
```

Exporting User Group Memberships

You can export a list of user group memberships to a CSV file. The export file contains a list of all user group memberships and properties. You can export a list of user group memberships to analyze the data offline. You can also export a list to update user group membership data in the Data Privacy Management repository.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the Actions menu, click **Export** > **User Group Memberships**.

   A dialog box appears with the exported file. The export file is named `Usermanagement_Groupmembership.csv`. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, the service downloads a new file. The file name includes an incremental number such as `Usermanagement_Groupmembership (1).csv`.

3. Open the file. The following image shows the spreadsheet view of a sample export file:

	A	B	C	D	E	F
1	MemberName	MemberDN	MemberType	MemberOf	MemberOfDN	Action
2	PREET	PSHARMA	U	Access Control Assist	CN=Access Control A:	U
3	bcollins	Collins, Bala	U	Account Operators	CN=Account Operato	U
4	qagupta	Gupta , Quinn	U	Administrators	CN=Administrators,C	D

The following image shows the text view of a sample export file:

```
1  MemberName,MemberDN,MemberType,MemberOf,MemberOfDN,Action
2  PREET,PSHARMA,U,Access Control Assistance Operators,"CN=Access Control Assistance Operators,CN=Builtin
3  bcollins,"Collins,Bala",U,Account Operators,"CN=Account Operators,CN=Builtin,DC=secsrc,DC=com",U
4  qagupta,"Gupta,Quinn",U,Administrators,"CN=Administrators,CN=Builtin,DC=secsrc,DC=com",D
```

Exporting User Access

You can export information about user access to a CSV file. The export file contains a list of all member names with the associated access type and other details. You can export a list of user access information to analyze the data offline. You can also export a list to update user access data in the Data Privacy Management repository.

1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the Actions menu, click **Export** > **User Access**.

   A dialog box appears with the exported file. The export file is named `Usermanagement_UserAccess.csv`. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, the service downloads a new file. The file name includes an incremental number such as `Usermanagement_UserAccess (1).csv`.

3. Open the file. The following image shows the spreadsheet view of a sample export file:

	A	B	C	D	E	F	G	H	I
1	MemberName	MemberDN	MemberType	AccessType	DataStore	SchemaName	TableName	ColumnName	Action
2	PREET	PSHARMA	U	ALL	PCRSCONNECTION_linux/ORA04				U
3	bcollins	Collins, Bala	U	R	PCRSCONNECTION_linux/ORA05				U
4	qagupta	Gupta , Quinn	U	RWD	PCRSCONNECTION_linux/$DBConnection_04				U

The following image shows the text view of a sample export file:

```
1  MemberName,MemberDN,MemberType,AccessType,DataStore,SchemaName,TableName,ColumnName,Action
2  PREET,PSHARMA,U,ALL,PCRSCONNECTION_linux/ORA04,,,,U
3  bcollins,"Collins,Bala",U,R,PCRSCONNECTION_linux/ORA05,,,,U
4  qagupta,"Gupta,Quinn",U,RWD,PCRSCONNECTION_linux/$DBConnection_04,,,,U
```

Exporting User Aliases

You can export a list of user aliases to a CSV file. The export file contains a list of all user aliases and properties. You can export a list of user alias information to analyze the data offline. You can also export a list to update user alias data in the Data Privacy Management repository.

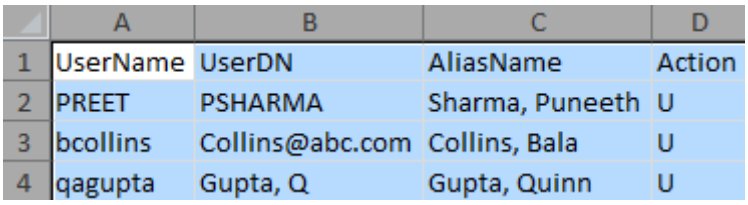
1. Click **Manage** > **Users**.

   The **Users** workspace appears.

2. From the Actions menu, click **Export** > **User Alias**.

   A dialog box appears with the exported file. The export file is named `UserAliases.csv`. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, the service downloads a new file. The file name includes an incremental number such as `UserAliases (1).csv`.

3. Open the file. The following image shows the spreadsheet view of a sample export file:



	A	B	C	D
1	UserName	UserDN	AliasName	Action
2	PREET	PSHARMA	Sharma, Puneeth	U
3	bcollins	Collins@abc.com	Collins, Bala	U
4	qagupta	Gupta, Q	Gupta, Quinn	U

The following image shows the text view of a sample export file:

```
1  UserName,UserDN,AliasName,Action
2  PREET,PSHARMA,"Sharma, Puneeth",U
3  bcollins,Collins@abc.com,"Collins, Bala",U
4  qagupta,"Gupta, Q","Gupta, Quinn",U
```

CHAPTER 16

Anomaly Detection

This chapter includes the following topics:

- Anomaly Detection Overview, 346
- Anomaly Detection Prerequisites, 347
- Anomaly Detection Workspace, 348
- Anomalous Factors, 361
- Anomaly Management, 362

Anomaly Detection Overview

Anomaly detection is a process that identifies irregular or unusual patterns of user activity on sensitive data. Irregular or unusual user activity can indicate malicious behavior, such as a data breach or stolen credentials.

Data Privacy Management analyzes user activity events to determine baseline behavior for a user and for the user's peer group. To determine a baseline, it tracks trends in behavior over evaluation intervals for a set of system-defined factors. When user activity behavior significantly deviates from the baseline, Data Privacy Management detects an anomaly.

An anomaly is a set of one or more user activity events that deviate from the baseline behavior. For example, Data Privacy Management detects an anomaly when a user downloads a high amount of sensitive data outside of the user's normal working day and time. The anomaly contains the observed and expected values for the anomalous factors used to detect the anomaly.

An anomalous factor is a parameter that Data Privacy Management uses to determine different types of irregular behavior. For example, the time of day anomalous factor identifies irregular behavior based on the time of the day that a user performs activity. A user normally performs activity from 8AM through 6PM. If the user performs activity at 3AM, Data Privacy Management detects an anomaly.

Use the anomaly detection workspace to view the anomalies detected. The workspace includes predefined views that display anomaly information in a variety of ways. For example, you can view a list of users that had the most anomalies. You can view the data stores and data domains that were involved with the most anomalies. You can mark an anomaly as read or flag an anomaly to review at a later time. You can also delete anomalies or suppress certain types of anomalies from appearing on the UI.

The anomaly detection workspace is a dashboard that displays all anomalies. To generate security policy violations, email notifications, or custom actions when Data Privacy Management detects an anomaly, you must configure a security policy.

Anomaly Detection Prerequisites

Data Privacy Management uses several components to detect anomalous behavior on sensitive data and to display information related to the anomaly.

Verify the following prerequisites for anomaly detection:

Configure Data Privacy Management to receive user activity events from external data sources.

Required to detect anomalous user behavior. Data Privacy Management analyzes user activity events to determine baseline behavior for a user and for the user's peer group. When user activity behavior deviates from the baseline, Data Privacy Management detects an anomaly. You can import user activity events from system logs in CEF, LEEF, or JSON formats. You can also import user activity events from Salesforce sources.

Set the default retention period for how long you want to store user activity events in the user activity store.

The anomaly detection workspace displays anomalies that exist in the user activity store. The Data Privacy Management Service purges events from the user activity store based on the retention period. Determine the period of time in which you want users to analyze anomalies. For example, if you want users to view anomalies detected up to one month ago, set the retention period to 31 days. To set the retention period, configure the Event Details Retention Period property in the Data Privacy Management Service.

Set up data store access for Data Privacy Management user accounts.

Required for users to view anomalies in the anomaly detection workspace. Users can see anomalies that are associated to the data stores to which they have access. To set up access to data stores:

- Create user accounts and Informatica domain groups in Informatica administrator.
- Assign Informatica domain groups to the user accounts in Informatica administrator.
- Assign security groups to data stores in Data Privacy Management. A security group is the same as an Informatica domain group.

Note: The user accounts are for users that log in to Data Privacy Management and not for users that perform activity on data stores.

Import metadata about users from external user management systems into the Data Privacy Management repository.

Required for the anomaly details to show information related to the user that performed the anomalous behavior. For example, the user full name, department, title, and office location. Required to determine the baseline behavior of the user peer group. Data Privacy Management uses the user department to determine user peer group behavior.

Note: A user is an account that accesses sensitive data in a data store, and not a user account that logs in to Data Privacy Management.

If you do not import user metadata, then the anomaly details only show the user name that performed the anomalous user activity event.

Create a location for each geographic area from which user activity originates.

Required for Data Privacy Management to detect anomalies and security policy violations based on the location of the user activity.

When you create a location, you specify an IP address subnet to identify a geographic origin of user activity. When Data Privacy Management evaluates user activity events for anomalous behavior, it tries to identify the event location. To identify the event location, it matches the IP address in the event to the IP address range in the location configuration. You can specify if the IP address range is from a VPN provider. If enabled, Data Privacy Management does not detect anomalies based on distance for the Relocation anomalous factor.

If you do not configure an IP address to identify the location of user activity, then Data Privacy Management cannot use the Location and Relocation anomalous factors to detect anomalies.

Run scans on data stores to identify sensitive data in data stores.

Required to detect anomalous behavior on sensitive data. Data Privacy Management identifies anomalous behavior for activity on data that is identified as sensitive in the repository.

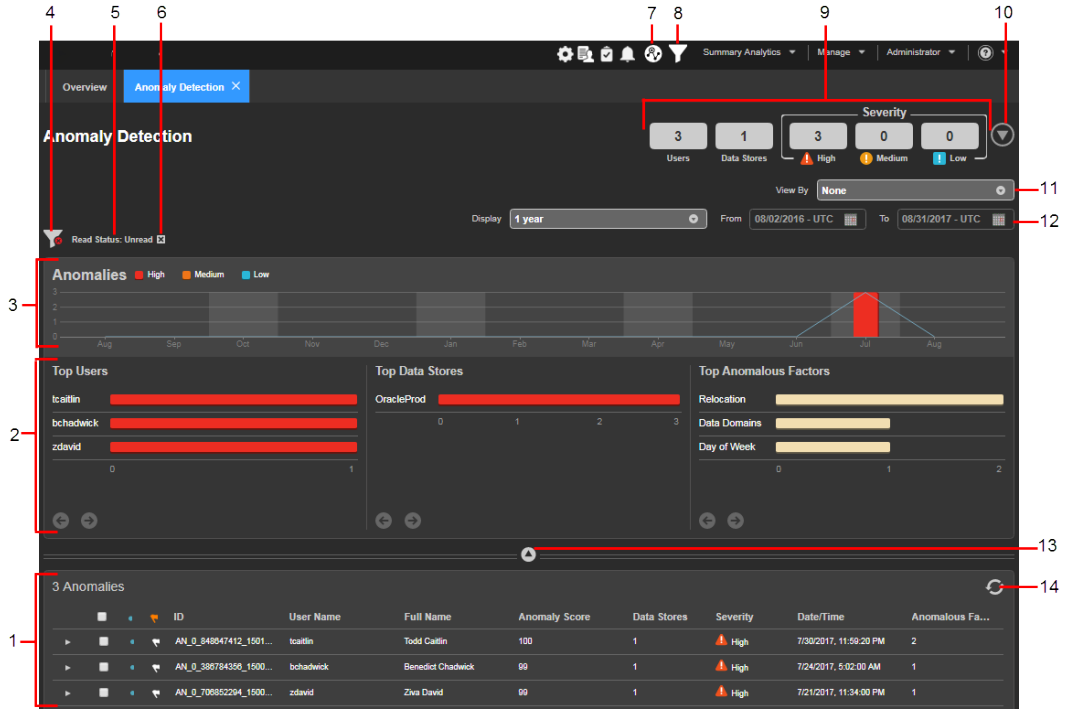
Anomaly Detection Workspace

Use the Anomaly Detection workspace to view the anomalies that Data Privacy Management detects. The workspace includes a summary of anomaly counts, an anomaly trend graph, anomaly indicators, and a list of anomalies. The information in the workspace is dynamic and changes based on the selected view, time period, and filter conditions.

The workspace displays anomalies based on the event retention period for the user activity store and based on data store access for the user. The event retention period that you configure in the Data Privacy Management Service properties determines how long anomalies remain in the user activity store. The data store security groups determine the users that have access to the anomalies. You can see anomalies that are associated with at least one data store to which you have access. You have access to a data store if your user account belongs to the same security group that is assigned to the data store.

To access the Anomaly Detection workspace, click the **Anomalies** icon in the header.

The following image shows a sample Anomaly Detection workspace:



The Anomaly Detection workspace includes the following sections:

1. Anomalies
2. Anomaly indicators

3. Anomalies graph
4. Clear filter icon
5. Filter conditions
6. Clear filter condition icon
7. Anomalies icon
8. Filter
9. Anomaly summary
10. Actions menu
11. Views
12. Time period
13. Expand/collapse
14. Refresh icon

Anomaly Summary

Use the anomaly summary to quickly assess the number and severity of anomalies across your enterprise. You can view the number of users and data stores associated to anomalies and the number of anomalies in each severity level. The anomaly summary shows information for all users based on the selected time period and filter conditions in the workspace.

The following image shows an example of the anomaly summary:



The anomaly summary includes the following information:

Users

Aggregated number of users associated to the anomalies based on the selected time period and filter conditions. The user count from the anomaly summary is not necessarily the same as the anomaly count from the anomalies list. For example, you can have 14 users and 100 anomalies. An anomaly can only include one user. However, one user can have multiple anomalies.

To view a list of all users that are associated to anomalies, click the Top Users indicator title, or select View By > User.

Data Stores

Aggregated number of data stores associated to the anomalies based on the selected time period and filter conditions. A data store is associated with an anomaly when at least one user activity event in the anomaly included the data store. The count only includes the data stores to which you have access. An anomaly can include multiple data stores. A data store can be included in multiple anomalies. The relocation speed anomalous factor is not associated to data stores.

Note: The data store filter does not affect the data store count. For example, if you filter the workspace by one data store, the data store count might show more than one data store. The anomaly that includes the filtered data store can also include other data stores. The count includes all data stores for the anomaly regardless of the data store filter.

To view a list of all data stores that are associated to anomalies, click the Top Data Stores indicator title, or select View By > Data Store.

Severity

Total number of anomalies in each severity level based on the selected time period and filter conditions. Data Privacy Management uses an internal algorithm to calculate the severity level for an anomaly. The severity level is mainly based on the number of anomalous factors used to detect the anomaly. An anomaly can only have one severity level.

To view a list of all anomalies for each severity level, view the anomalies list panel, or select View By > None. Then, filter the workspace by the severity level.

Related Topics:

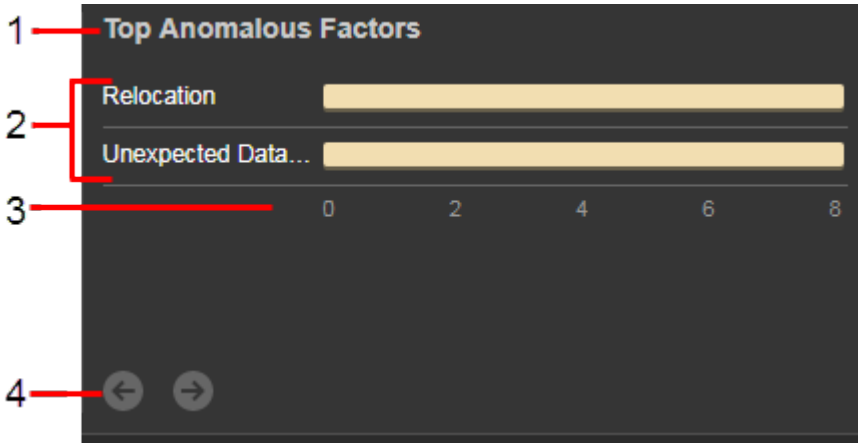
- "Changing the Default Anomaly Severity Ranges" on page 46

Top Anomalous Factors Indicator

The Top Anomalous Factors indicator displays a list of anomalous factors that have the highest number of anomalies based on the selected time period and filter conditions.

By default, the list is sorted by an internal anomaly score in descending order, then by anomalous factor name in ascending order.

The following figure shows an example of the Top Anomalous Factors indicator:



The Top Anomalous Factors indicator includes the following information:

1. Title

Click the title to open a list of all anomalous factors that are associated to at least one anomaly. You can then drill down on the users, data stores, or anomalies for each anomalous factor.

2. Anomalous factors

Anomalous factors that are associated with anomalies. Each row contains a bar that represents the number of anomalies that are associated to the anomalous factor.

Click an anomalous factor row to filter the workspace to show information related to the selected anomalous factor. A filter condition with the selected anomalous factor name appears at the top of the workspace. The anomaly summary, anomalies graph, anomaly indicators, and anomalies list display information only for the selected anomalous factor.

3. Anomaly count

Displays the number of anomalies that are associated to the anomalous factors.

4. Page icons

Click the back and forward arrows to scroll through the top 25 anomalous factors.

Top Data Stores Indicator

The Top Data Stores indicator displays a list of up to 25 data stores that have the highest number of anomalies based on the selected time period and filter conditions. The indicator only displays the data stores to which you have access. You have access to a data store if your user account belongs to the same security group that is assigned to the data store.

By default, the list is sorted by an internal anomaly score in descending order, then by data store name in ascending order.

Note: When you apply a filter to the workspace, the workspace includes information for all anomalies that meet the filter conditions. If you filter the workspace by specific data stores, the Top Data Stores indicator might show additional data stores than the ones you specified in the filter condition. The anomalies that include the filtered data stores can also include other data stores. For example, you filter the workspace by ORA1 data store. The anomalies that are associated with ORA1 are also associated with data stores PCRS_Linux and ORA2. The Top Data Stores indicator includes data stores ORA1, PCRS_Linux, and ORA2.

The following figure shows an example of the Top Data Stores indicator:



The Top Data Stores indicator includes the following information:

1. Title

Click the title to open a list of all data stores that are associated to at least one anomaly. You can then drill down on the users or anomalies for each data store.

2. Data stores

Data stores that are associated with anomalies. A data store is associated with an anomaly when at least one user activity event in the anomaly included the data store.

Each row contains a bar that represents the number of anomalies that are associated to the data store. The bar contains different colors based on the anomaly severity level.

Click a data store row to filter the workspace to show information related to the selected data store. A filter condition with the selected data store name appears at the top of the workspace. The anomaly

summary, anomalies graph, anomaly indicators, and anomalies list display information related to the selected data store.

3. Anomaly count

Displays the number of anomalies that are associated to the data stores.

4. Page icons

Click the back and forward arrows to scroll through the top 25 data stores.

Top Users Indicator

The Top Users indicator displays a list of up to 25 users that have the highest number of anomalies based on the selected time period and filter conditions.

By default, the list is sorted by an internal anomaly score in descending order, then by user name in ascending order.

The following figure shows an example of the Top Users indicator:



The Top Users indicator includes the following information:

1. Title

Click the title to open a list of all users that are associated to at least one anomaly. You can then drill down on the data stores or anomalies for each user.

2. Users

Name of the user that performed the user activity events that are associated with the anomaly. If the user account is associated with a user in the Data Privacy Management repository, the indicator displays the user's full name. If the user account is not associated with a user, the indicator displays the user account name from the source of the user activity information.

Each row contains a bar that represents the number of anomalies that are associated to the user. The bar contains different colors based on the anomaly severity level.

Click a user row to filter the workspace to show information related to the selected user. A filter condition with the selected user name appears at the top of the workspace. The anomaly summary, anomalies graph, anomaly indicators, and anomalies list display information related to the selected user.

3. Anomaly count

Displays the number of anomalies that are associated with the users.

4. Page icons

Click the back and forward arrows to scroll through the top 25 users.

Anomalous Factors View

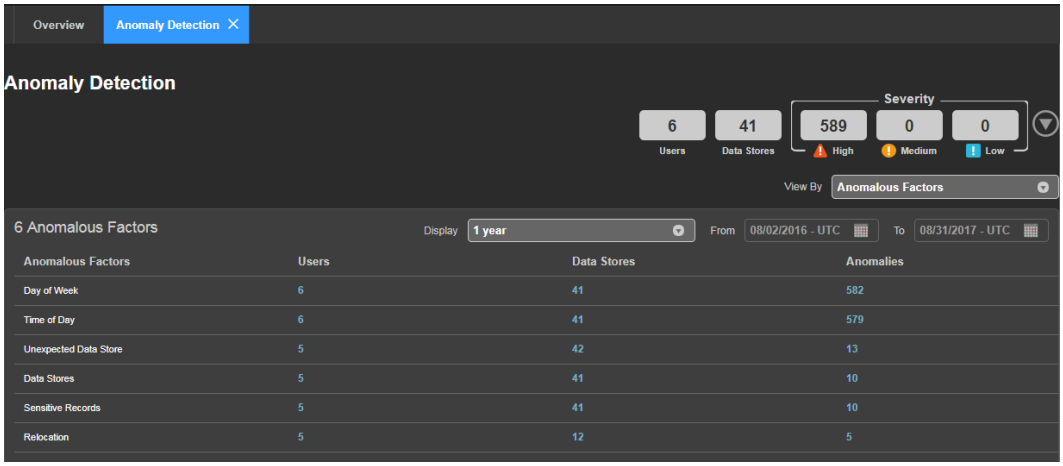
The anomalous factors view displays a list of all anomalous factors that have at least one or more anomalies within the selected time period and filter conditions.

To open the anomalous factors view from the Anomaly Detection workspace, select View By > Anomalous Factors, or click the Top Anomalous Factors indicator title.

You can drill down on the counts in the Users, Data Stores, or Anomalies columns for each anomalous factor. You can click a column header to sort the list by the selected column.

By default, the list is sorted by an internal anomaly score in descending order, then by anomalous factor name in ascending order.

The following figure shows an example of the anomalous factors view:



The anomalous factors view includes the following information:

Anomalous Factors

Anomalous factors that are associated to anomalies within the selected time period and filter conditions.

Users

Aggregated number of users that are associated to the anomalous factor. Click the user count to view a list of users for the selected anomalous factor. The users view opens with a filter condition for the selected anomalous factor. The view includes the same time period that you selected in the anomalous factor view.

You can click a user name to view more details about the user's activity on the User Profile page.

Data Stores

Aggregated number of data stores that are associated to the anomalous factor. Click the data store count to view a list of data stores for the selected anomalous factor. The data stores view opens with a filter condition for the selected anomalous factor. The view includes the same time period that you selected in the anomalous factor view.

Note: The count includes all of the data stores that are associated with the anomaly, regardless of the data stores to which you have access. When you click the count to open the data stores view, you only see the data stores to which you have access.

Anomalies

Total number of anomalies that Data Privacy Management detected for the anomalous factor. Click the anomaly count to view information related to the selected anomalous factor. The default workspace view opens with a filter condition for the selected anomalous factor. The view includes the same time period that you selected in the anomalous factor view.

To go back to the default workspace view, select View By > None.

Related Topics:

- "User Profile Page" on page 567

Data Stores View

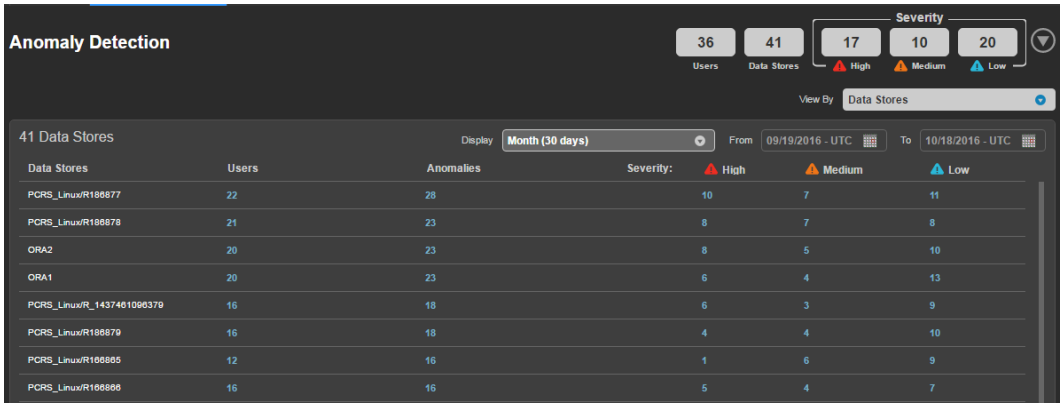
The data stores view displays a list of all data stores that have at least one or more anomalies within the selected time period and filter conditions. The view only includes the data stores to which you have access. You have access to a data store if your user account belongs to the same security group that is assigned to the data store.

To open the data stores view from the Anomaly Detection workspace, select View By > Data Store, or click the Top Data Stores indicator title.

You can drill down on the counts in the Users, Anomalies, or Severity columns. You can click a column header to sort the list by the selected column.

By default, the list is sorted by an internal anomaly score in descending order, then by data store name in ascending order.

The following figure shows an example of the data stores view:



The data stores view includes the following information:

Data Stores

List of data stores that are associated to anomalies within the selected time period and filter conditions. When you click a count in a data store row, a view appears with the selected data store as the active filter condition.

Users

Aggregated number of users that have anomalies for the data store. You can click the user count to view the users that had anomalies for the data store. When you click the user count, the users view opens. The workspace includes a filter condition for the selected data store and includes the same time period that you selected in the data stores view.

You can click a user name to view more details about the user's activity on the User Profile page.

Anomalies

Total number of anomalies in all severity levels detected for the data store. You can click the anomaly count to view anomaly information related to the selected data store. When you click the anomaly count, the default workspace view opens. The workspace includes a filter condition for the selected data store and includes the same time period that you selected in the data stores view.

Severity

Total number of anomalies in the selected severity level detected for the data store. Severity level is based on the anomaly score. From the Settings page, you can specify the range of anomaly scores for each severity level.

You can click the anomaly count to view anomaly information related to the selected data store and anomaly severity level. When you click the anomaly count, the default workspace view opens. The workspace includes filter conditions for the selected data store and anomaly severity level. The view includes the same time period that you selected in the data stores view.

To go back to the default workspace view, select View By > None.

Users View

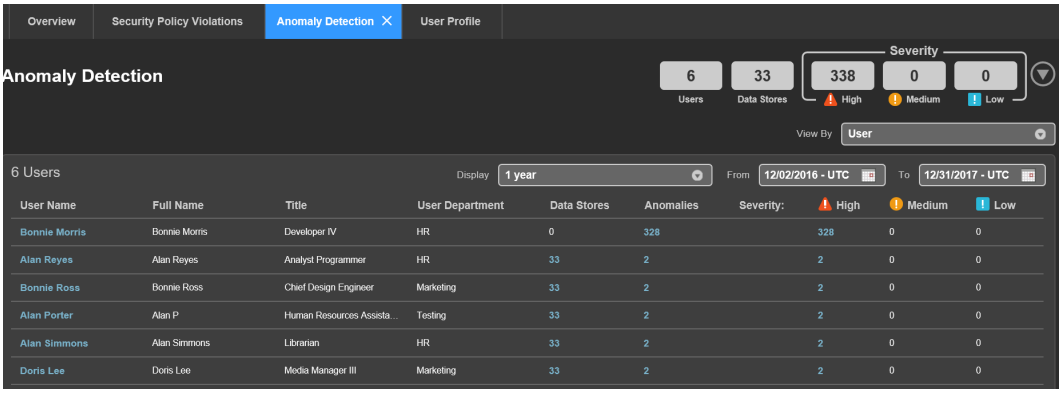
The users view displays a list of all users that have at least one or more anomalies within the selected time period and filter conditions.

To open the users view from the Anomaly Detection workspace, select View By > User, or click the Top Users indicator title.

You can drill down on each count in the Data Stores, Anomalies, or Severity columns. You can click a column header to sort the list by the selected column.

By default, the list is sorted by an internal anomaly score in descending order, then by user name in ascending order.

The following image shows an example of the users view:



The users view includes the following information:

User Name

Name of the user account that performed the user activity events that are associated with the anomaly. Data Privacy Management shows the user account name from the source of the user activity information, such as user activity logs.

To retrieve the rest of the user information in the anomaly details, Data Privacy Management matches the user account name to user names in the Data Privacy Management repository. If there is a match, the anomaly details include values for the full name, title, group, and user location.

You can click the user name to view more details about the user's activity on the User Profile page.

Full Name

Full name that is associated with the user account name in the Data Privacy Management repository. If the full name is null in the Data Privacy Management repository, the value is a concatenation of the first name and last name. If the first name and last name is null, the value is the user account name. If the user account is not in the repository, the value is the user account name.

Title

Title that is associated with the user account name in the repository. If the user account is not in the repository, the title is blank.

User Department

User department that is associated with the user account name in the repository. When Data Privacy Management analyzes user activity events for anomalous behavior, it compares behavior of the user and the user's peer group to detect an anomaly. Data Privacy Management uses the department for the user peer group. Any user that has the same department is included in the user peer group. If the user account is not in the repository, the user department is blank.

Data Stores

Total number of data stores that are associated to anomalies for the user. Click the data store count to view a list of data stores for the selected user. The data stores view opens with a filter condition for the selected user. The view includes the same time period that you selected in the users view.

Note: The count includes all of the data stores that are associated with the anomaly, regardless of the data stores to which you have access. When you click the count to open the data stores view, you only see the data stores to which you have access.

Anomalies

Total number of anomalies in all severity levels detected for the user. Click the anomaly count to view anomaly information related to the selected user. The default workspace view opens with a filter condition for the selected user. The view includes the same time period that you selected in the users view.

Severity

Total number of anomalies in the selected severity level detected for the user. Severity level is based on the anomaly score. From the Settings page, you can specify the range of anomaly scores for each severity level.

You can click the anomaly count to view anomaly information related to the selected user and anomaly severity level. When you click the anomaly count, the default workspace view opens. The workspace includes filter conditions for the selected user and anomaly severity level. The view includes the same time period that you selected in the data stores view.

To go back to the default workspace view, select View By > None.

Anomalies List

The anomalies list panel includes a list of all anomalies and the corresponding anomaly details. The list of anomalies is dynamic and can change based on the time period and filter conditions that you select in the

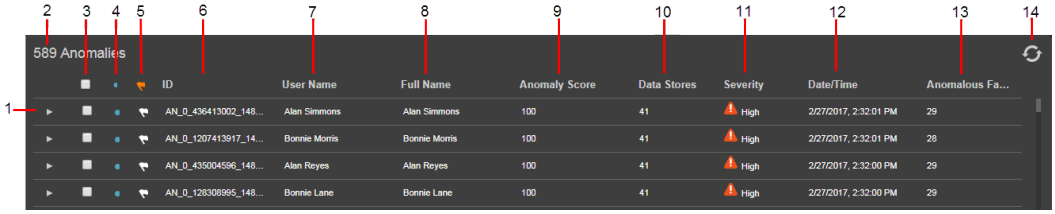
workspace. The list displays anomalies based on your data store access. You can see an anomaly if the anomaly is associated with at least one data store to which you have access.

To open the Anomaly Detection workspace, click the Anomalies icon in the header. When you select one of the views in the Anomaly Detection workspace, the anomalies list disappears. Select View By > None to display the anomalies list again.

By default, the list of anomalies is sorted by unread status, then by the highest severity level, then by the anomaly ID. You can click a column header to sort the list of anomalies by the column.

Tip: Click the filter icon to filter the list of anomalies. For example, you can filter the list to show only high severity anomalies.

The following figure shows an example of the anomalies list:



The anomalies list includes the following information:

1. Expand/collapse

Click to show or hide the anomaly details. For more information, see "Anomaly Details" on page 359.

2. Anomaly count

Total number of anomalies based on the selected time period and filter conditions.

3. Selection

You can perform actions on one or more anomalies. Select one, multiple, or all anomalies. Then, click the Actions menu and select the action. For example, you can mark all anomalies as read or unread.

4. Read/Unread

Read or unread status of an anomaly. You can mark an anomaly as read or unread. Changes to the status affects the sort order of the anomalies. The list of anomalies is first sorted by unread anomalies. You can click the column header to sort by the status. You can filter the workspace by the status to show only the unread or read anomalies.

By default, an anomaly is marked as unread until you mark the anomaly as read. For more information, see "Marking an Anomaly as Read or Unread" on page 363.

5. Flag/Unflag

Flagged or unflagged status of the anomaly. Click the flag to change the status. A clear flag indicates an anomaly that is not flagged. A filled flag indicates an anomaly that is flagged. You might want to flag the anomalies that you need to review further or to flag for follow-up. You can click the column header to sort by flag status. You can use the flag status as a filter condition for the workspace.

By default, an anomaly is not flagged until you mark the anomaly as flagged. The flag status does not affect the default anomaly sort order. For more information, see "Flagging or Unflagging An Anomaly " on page 363.

6. Anomaly ID

Anomaly identification. Data Privacy Management uses the following syntax to assign an ID to an anomaly:

```
AN_<Node ID>_<User hash code>_<Anomaly timestamp in Epoch format>
```

7. User Name

Name of the user account that performed the user activity events that are associated with the anomaly. Data Privacy Management shows the user account name from the source of the user activity information, such as user activity logs.

You can click the user name to view more details about the user's activity on the User Profile page.

8. Full Name

Full name that is associated with the user account name in the Data Privacy Management repository. If the full name is null in the Data Privacy Management repository, the value is a concatenation of the first name and last name. If the first name and last name is null, the value is the user account name. If the user account is not in the repository, the value is the user account name.

9. Anomaly Score

A calculated value based on the weight of each anomalous factor specified on the Settings page. You can use the score to sort and filter anomalies. You can also use the score to create a security policy. For example, you can create a security policy to send an email alert when an anomaly with a score greater than 75 is triggered.

10. Data Stores

Total number of data stores that the user accessed during the evaluation interval up until the anomalous behavior. A data store is included in the count when at least one user activity event accessed sensitive data in the data store during the evaluation interval. Expand the anomaly details to view a list of the data store names.

Note:

- The count includes all of the data stores that the user accessed regardless of the data stores to which you have access. When you view the anomaly details, you only see the data stores names to which you have access.
- The number of data stores might be different than the number of data stores in the observed value for the anomalous factor. The observed value shows the number of data stores that are associated with the anomalous behavior. The expected value shows the number of data stores that the user typically accesses.

11. Severity

Severity level of the anomaly. Data Privacy Management determines the severity level for each anomaly based on the anomaly score. An anomaly can have a high, medium, or low severity level. From the Settings page, you can specify the range of anomaly scores for each severity level.

12. Date/Time

Date and time of the anomalous behavior that is associated with the anomaly. This is not the date and time when Data Privacy Management detected the anomaly. All time periods are in the UTC time zone.

13. Anomalous Factors

Number of system-defined anomalous factors that are associated to the anomaly. Expand the anomaly details to view a list of the anomalous factors that are associated to the anomaly. For more information, see “Anomalous Factors” on page 361.

14. Refresh

Click to refresh the list of anomalies based on the selected time period and filter conditions.

Related Topics:

- "User Profile Page" on page 567

Anomaly Details

The anomaly details panel shows information related to the user that performed the anomalous activity, the data stores and data domains that are associated to the anomaly, and the anomalous factors that detected the anomaly. The user information, such as the full name, title, and group, originates from data that was imported into the Data Privacy Management repository.

You can expand each entry to show all details related to the anomaly.

The following image shows an example of anomaly details:



The anomaly details panel includes the following information:

User Name

Name of the user account that performed the user activity events that are associated with the anomaly. Data Privacy Management shows the user account name from the source of the user activity information, such as user activity logs.

To retrieve the rest of the user information in the anomaly details, Data Privacy Management matches the user account name to user names in the Data Privacy Management repository. If there is a match, the anomaly details include values for the full name, title, group, and user location.

You can click the user name to view more details about the user's activity on the User Profile page.

Full Name

Full name that is associated with the user account name in the Data Privacy Management repository. If the full name is null in the Data Privacy Management repository, the value is a concatenation of the first name and last name. If the first name and last name is null, the value is the user account name. If the user account is not in the repository, the value is the user account name.

Click the View Associated User Activity link to open a view that lists all user activity events that occurred within a close proximity to the anomalous behavior. You can increase or decrease the time range to further analyze the user activity events.

Title

Title that is associated with the user account name in the repository. If the user account is not in the repository, the title is blank.

Group

Peer group that Data Privacy Management uses to compare user behavior. When Data Privacy Management analyzes user activity events for anomalous behavior, it compares behavior of the user and the user's peer group to detect an anomaly. It uses the department that is associated with the user account name in the Data Privacy Management repository for the peer group. Any user that has the

same department is included in the peer group. If the user account is not in the repository, the group is blank.

User Office Location

Location that is associated with the user account in the repository. This is the location from which the user works. For example, the corporate office that the user is assigned to. Note that this is not necessarily the same location from which the user performs the user activity events.

Data Stores

List of data stores that the user accessed during the evaluation interval up until the anomalous behavior. A data store appears in the list when at least one user activity event accessed sensitive data in the data store during the evaluation interval. If you do not have authorization for a data store in the list, Data Privacy Management masks the data store name with a question mark. By default, the list is sorted by data store name in ascending order.

Note: The number of data stores in the list might be different than the number of data stores in the observed value for the anomalous factor. The observed value shows the number of data stores that are associated with the anomalous behavior. The expected value shows the number of data stores that the user typically accesses.

When Data Privacy Management detects an anomaly based on the relocation speed anomalous factor, it does not associate data stores to the anomalous factor. The data store list is blank if the anomaly only contains the relocation speed anomalous factor.

Data Domains

List of data domains that the user accessed during the evaluation interval up until the anomalous behavior. A data domain appears in the list when at least one user activity event accessed sensitive data that matches the data domain. By default, the list is sorted by data domain name in ascending order.

Note: The number of data domains in the list might be different than the number of data domains in the observed value for the anomalous factor. The observed value in the anomalous factor shows the number of data domains that are associated with the anomalous behavior. The expected value in the anomalous factor shows the number of data domains that the user typically accesses.

Anomalous Factors

List of system-defined anomalous factors used to detect the anomaly. Most anomalous factors include a degree, an observed value, and an expected value. To view the observed values for the relocation speed factor, click the information icon next to the factor. For more information about each anomalous factor, see "Anomalous Factors" on page 361.

Degree

System-defined degree that Data Privacy Management detects for the anomaly. Data Privacy Management evaluates the observed and expected values and assigns a degree based on the deviation from the expected value. An anomaly can either include an unusual degree or a highly unusual degree.

Suppressed

Indicates whether an anomalous factor is suppressed or not.

Observed

Count or value from the user activity events for the anomalous factor. This is the observed value during the anomalous behavior. The exact value depends on the anomalous factor. For example, the observed value for the sensitive fields anomalous factor is the number of sensitive fields that the user accessed in one or more events. Data Privacy Management uses the observed and expected values to determine if an anomaly is triggered. If Data Privacy Management triggers an anomaly, it compares the observed and expected values to determine the anomaly degree.

Expected

Count or value that Data Privacy Management determines is the normal baseline for the anomalous factor. The exact value depends on the anomalous factor. For example, the expected value for the sensitive fields anomalous factor is the number of sensitive fields that a user normally accesses. Data Privacy Management determines the expected number based on a pattern of behavior from the user over a period of time. It uses the observed and expected values to determine if an anomaly is triggered. If it triggers an anomaly, it compares the observed and expected values to determine the anomaly degree.

The relocation speed anomalous factor does not include an expected value. To view details related to the relocation speed anomalous factor, click the information icon next to the observed value.

Related Topics:

- "User Profile Page" on page 567

Anomalous Factors

An anomalous factor is a parameter that Data Privacy Management uses to identify unusual behavior and to detect an anomaly. Data Privacy Management can use one or more anomalous factor to detect an anomaly. An anomaly can include multiple factors.

You can view a list of anomalies that are associated with each anomalous factor in the Top Anomalous Factors indicator. When you view the details for an anomaly in the anomaly list panel, you can view a list of the anomalous factors that detected the anomaly. You can also view the observed and expected values for each anomalous factor.

Data Privacy Management detects anomalies based on the following system-defined anomalous factors:

Sensitive Records

Detects anomalies based on the number of sensitive records a user accessed that match a classification policy. For example, a user typically accesses 10,000 rows of sensitive data. If the user accesses 1,000,000 rows of sensitive data, then Data Privacy Management detects an anomaly.

Data Domains

Detects anomalies based on the number of data domains that a user accesses. For example, a user typically retrieves sensitive data that matches 10 data domains. If the user retrieves sensitive data that matches 1000 data domains, then Data Privacy Management detects an anomaly.

Note that the factor detects the number of matched data domains, regardless of how many columns matched the data domain. The Sensitive Field factor detects the actual number of columns that matched a data domain. Multiple sensitive columns can match the same data domain. For example, a user can access 10 sensitive columns that matched one credit card data domain.

Sensitive Events

Detects anomalies based on the number of user activity events that a user performs. For example, a user typically performs 5 events that access sensitive data a day. If the user performs 1000 events in one day, then Data Privacy Management detects an anomaly.

Time of Day

Detects anomalies based on the time of day that a user performs activity. For example, a user typically performs activity from 8 AM to 6 PM. If the user performs activity at 3 AM, then Data Privacy Management detects an anomaly.

When Data Privacy Management detects an anomaly based on the Time of Day anomalous factor, the anomaly can only be associated with one user activity event. The observed value for the anomalous factor shows the time of the day on which the anomalous user activity event occurred.

Day of Week

Detects anomalies based on the day of the week that a user performs activity. For example, a user typically performs activity from Monday through Friday. If the user performs activity on Saturday or Sunday, then Data Privacy Management detects an anomaly.

When it detects an anomaly based on the Day of Week anomalous factor, the anomaly can only be associated with one user activity event. The observed value for the anomalous factor shows the day of the week on which the anomalous user activity event occurred.

Data Stores

Detects anomalies based on the number of data stores that a user accesses. For example, a user typically retrieves sensitive data from 10 data stores. If the user retrieves sensitive data from 1000 data stores, then Data Privacy Management detects an anomaly.

Unexpected Data Store

Detects anomalies based on the data stores that a user typically accesses. For example, a user typically accesses the PC1, PC2, and PC3 data stores on a regular basis. If a user accesses an ORACLE1 data store, then Data Privacy Management detects an anomaly.

Relocation Speed

Detects anomalies based on the practicality of traveling between two locations according to geographic distance and land speed. For example, a user performs activity from a location in the United States. An hour later, the user performs activity from a location in India. The user cannot possibly travel the distance between the two locations within an hour. This factor helps to identify compromised or shared user credentials.

When Data Privacy Management detects an anomaly based on the relocation speed anomalous factor, it does not associate data stores to the anomalous factor. The anomaly details only include the observed value for the anomalous factor. Data Privacy Management does not provide an expected value.

Related Topics:

- "Changing Anomaly Factor Weights" on page 44

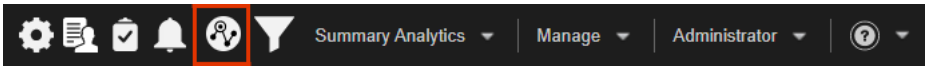
Anomaly Management

Use the Anomaly Detection workspace to view all of the anomalies that Data Privacy Management detected.

You can perform the following tasks in the Anomaly Detection workspace:

- Flag or unflag an anomaly for follow-up.
- Mark an anomaly as read or unread.
- Filter and view anomalies.
- Export information about anomalies to a CSV file.
- Delete an anomaly from the user activity store.
- Create a protection task to protect the data stores associated with an anomaly.

The following image shows the Anomalies icon in the header:



Flagging or Unflagging An Anomaly

By default, Data Privacy Management creates an anomaly with an unflagged status. You can flag or unflag anomalies. For example, you flag anomalies that you need to review further.

After you flag or unflag an anomaly, you can sort on the flag status. You can use the status to filter the workspace. For example, you can filter the workspace to only show information for flagged anomalies. Changes to the flag status do not affect the anomaly sort order in the anomalies list panel.

Note: When you flag or unflag an anomaly, Data Privacy Management updates the anomaly in the user activity store. The change impacts all users that can see the anomaly, not just the logged-in user.

1. Click the **Anomalies** icon in the header to open the **Anomaly Detection** workspace.
2. Navigate to the anomalies list panel and perform one of the following steps:
   - To flag or unflag one anomaly, click the icon in the flag status column for the associated anomaly.

     The following image shows an example of the flag status column:



   - To flag or unflag multiple anomalies, select the check box next to the anomalies. From the Actions menu, select **Flag** or **Unflag**.

     The following image shows an example of selected anomalies:



Marking an Anomaly as Read or Unread

By default, Data Privacy Management creates an anomaly with an unread status. You can mark an anomaly as read or unread. When you change the read/unread status of an anomaly, the sort order of the anomalies list panel changes. By default, the anomalies list is first sorted by anomalies that have an unread status.

After you mark an anomaly as read or unread, you can use the status to filter the workspace. For example, you can filter the workspace to only show information for unread anomalies.

Note: When you mark an anomaly as read or unread, Data Privacy Management updates the anomaly in the user activity store. The change impacts all users that can see the anomaly, not just the logged-in user.

1. Click the **Anomalies** icon in the header to open the **Anomaly Detection** workspace.
2. Navigate to the anomalies list panel and perform one of the following steps:
   - To mark one anomaly as read or unread, click the icon in the read/unread status column for the associated anomaly.

The following image shows an example of the read/unread status column:



- To mark multiple anomalies as read or unread, select the check box next to the anomalies. From the Actions menu, select **Mark as Read** or **Mark as Unread**.

  The following image shows an example of selected anomalies:



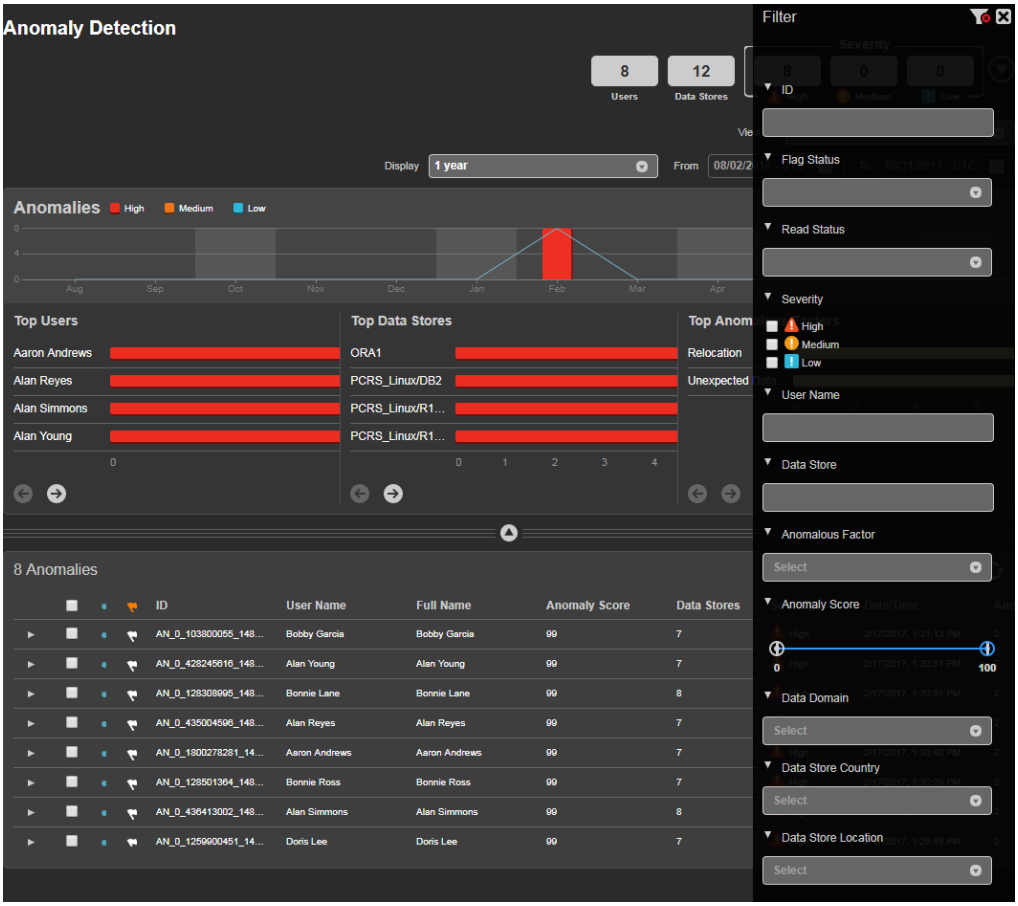
Data Privacy Management marks the anomaly as read or unread and refreshes the sort order based on the status.

Filtering Anomalies

You can filter information on the Anomaly Detection workspace and on the views that you access from the workspace. The filter conditions that you specify apply to the graph, summary, top indicators, and list of anomalies.

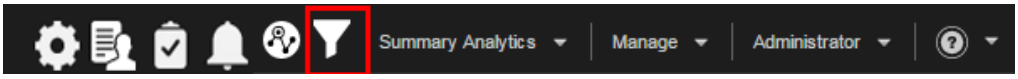
1. Click the **Anomalies** icon in the header to open the **Anomaly Detection** workspace and then click the Filter icon.

   The filter panel appears in the workspace. The following image shows the filter panel:



2. Click the filter icon in the header.

   The following image shows the filter icon:



3. Select one or more filters from the filter panel.

   The Data Privacy Management Service uses an AND operator if you specify more than one filter parameter. The service uses an OR operator if you specify multiple values within the same filter parameter.
   For example, you select LastName for the Data Domain parameter. You enter ORA1, ORA2 for the Data Stores parameter. The service filters the workspace to show anomalies that are associated with Data Domain LastName and Data Store ORA1 or Data Store ORA2.

   The service refreshes the information based on the selected filters. The filter conditions appear at the top of the workspace. For more information about the filter icons, see "Anomaly Detection Workspace" on page 348.

Exporting Anomalies

You can export information about anomalies to CSV files and use the exported files for reporting analysis. You can export information from the Anomaly Detection workspace and from each view in the workspace.

Anomaly Detection Export Files

When you export information from the Anomaly Detection workspace, the Data Privacy Management Service downloads a compressed file named Data.zip to the default download directory on your machine. The Data.zip file includes CSV files that contain information related to the page from which you exported. The information in the CSV files matches the selected filter conditions and time periods at the time of the export.

The following table lists the CSV files in the Data.zip file that you export from the Anomaly Detection workspace:

File Name	Description
UbaTopUsers.csv	Contains information from the Top Users indicator. The file includes a list of up to 25 users that have the highest number of anomalies.
UbaTopDataStores.csv	Contains information from the Top Data Stores indicator. The file includes a list of up to 25 data stores that have the highest number of anomalies.
UbaTopAnomalyFactors.csv	Contains information from the Top Anomalous Factors indicator. The file includes a list of up to 25 anomalous factors that have the highest number of anomalies.
UbaSummary.csv	Contains information from the anomaly summary. The file includes the total number of users and data stores associated with anomalies and the number of anomalies in each severity level.
UbaAnomalyRelocationDetails.csv	Contains information for the relocation anomalous factor.
UbaAnomalyGraph.csv	Contains information from the anomaly graph. The file includes a count of anomalies for each time period.
UbaAnomalyDetailsDataStores.csv	Contains information from the Data Stores view. The file contains a list of all data stores that have at least one or more anomalies.
UbaAnomalyDetailsDataDomains.csv	Contains a list of all data domains that are associated with at least one or more anomalies.
UbaAnomalyDetailsAnomalyFactors.csv	Contains information from the Anomalous Factors view. The file contains a list of all anomalous factors that have at least one or more anomalies and the corresponding observed and expected values.
UbaAnomaly.csv	Contains information from the anomaly list panel.

Steps to Export Anomalies

You can export information from the Anomaly Detection workspace and from each view in the workspace. The export file contains the anomalies that match the selected filter conditions and time periods at the time of the export.

1. Click the **Anomalies** icon in the header to open the **Anomaly Detection** workspace.
2. Optionally, set the filter conditions, time period, and view.

3. From the Actions menu, select **Export**.

   A dialog box appears with the exported file. The export file name is Data.zip. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, the service downloads a new file. The file name includes an incremental number such as `Data (1).zip`.

Deleting an Anomaly

You can delete anomalies in the Anomaly Detection workspace.

Warning: When you delete an anomaly, no other users can view the anomaly. You cannot restore deleted anomalies.

1. Click the **Anomalies** icon in the header to open the **Anomaly Detection** workspace.
2. Select the check box next to the anomaly that you want to delete.

   You can select one, multiple, or all anomalies.
3. From the Actions menu, select **Delete**.

   A message appears to confirm that you want to delete the anomaly.

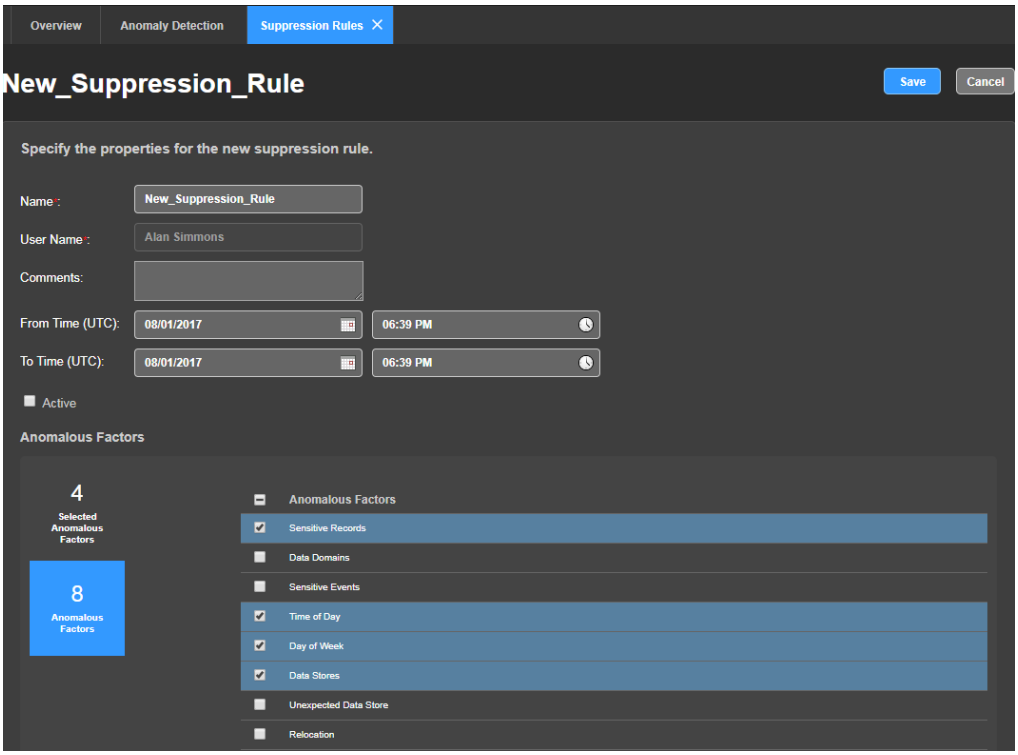
The Data Privacy Management Service deletes the anomaly from the user activity store.

Suppressing an Anomaly

To minimize the number of unnecessary anomalies created, create a suppression rule that Data Privacy Management can use to ignore scenarios that would normally trigger an anomaly. In the rule, you must specify the anomalous factors that you want Data Privacy Management to ignore. You can also specify the period of time that you want the suppression rule to be in effect.

1. Click the **Anomalies** icon in the header to open the **Anomaly Detection** workspace.
2. Select the check box next to an anomaly.
3. Click **Actions** > **Suppress**.

The **Create Suppression Rule** page in the **Suppression Rules** workspace appears.



4. Enter a name for the suppression rule.
5. Optionally, enter a comment and a time frame.
6. If you want the Data Privacy Management Service to start using the suppression rule, enable the **Active** check box.
7. Select one or more anomalous factors that you want the service to ignore during anomaly detection.
8. Click **Save**.

   The suppressed factor and associated details appear as a rule on the **Suppression Rules** workspace.

Data Privacy Management will not display anomalies that are associated solely with one or more of the suppressed anomalous factors. Data Privacy Management displays anomalies if an unsuppressed factor is also responsible for the anomaly. The anomaly details panel will show the suppressed and unsuppressed anomalous factors for the anomaly.

CHAPTER 17

Anomaly Suppression Rules

This chapter includes the following topics:

- Suppression Rules Overview, 369
- Suppression Rules Workspace, 369
- Suppression Rules Management, 372

Suppression Rules Overview

You can suppress certain types of anomalies from appearing on the UI.

To suppress anomalies, you must create a suppression rule. In the rule you specify the anomalous factors that you do not want to view anomalies for. After anomaly detection, Data Privacy Management does not display anomalies that are solely based on the suppressed factors.

For example, an employee from an office in the U.S. is visiting a branch office in Australia for a week. Typically, Data Privacy Management will create an anomaly when users access data outside of their normal business hours. To prevent such an anomaly, create a rule to suppress the Time of Day anomalous factor for the duration that the user is working in Australia.

You create suppression rules on the Anomaly Detection workspace. After you create the suppression rule, you can view, filter, sort, edit, export, and delete the suppression rule on the Suppression Rules workspace.

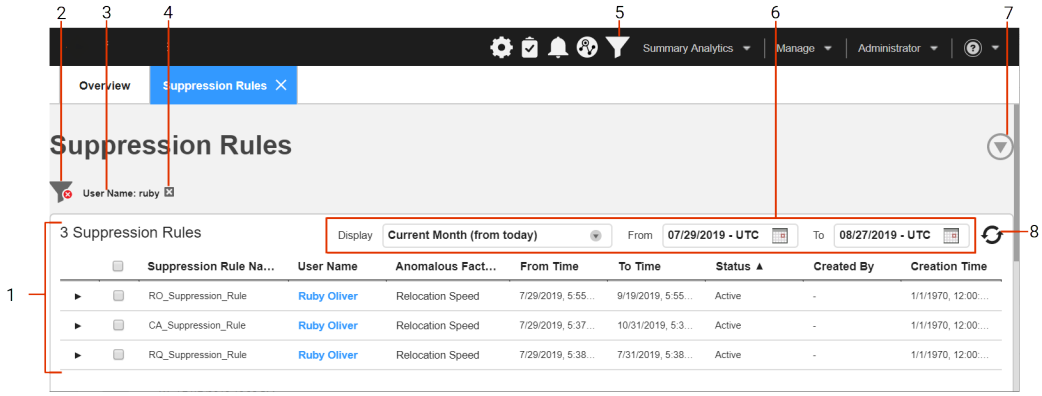
Suppression Rules Workspace

Use the Suppression Rules workspace to view the rules to suppress specific anomalous factors from triggering an anomaly. The information on the workspace is dynamic and changes based on the time period and filter conditions.

The workspace displays suppression rules for anomalies. When the rule is in effect, Data Privacy Management uses the rule to ignore unusual behavior associated with the anomalous factors specified in the rule.

To access the Suppression Rules workspace, click **Manage** > **Suppression Rules**.

The following image shows a sample Suppression Rules workspace:



1. Suppression rules
2. Clear filter icon
3. Filter conditions
4. Clear filter condition icon
5. Filter
6. Time period
7. Actions menu
8. Refresh icon

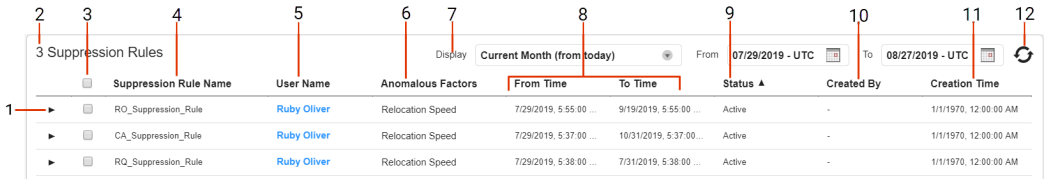
Suppression Rules List

The suppression rules list panel includes a list of all rules and corresponding rule details. The list of suppression rules is dynamic and can change based on the time period and filter conditions that you select in the workspace.

By default, the list of rules is sorted by status. You can click a column header to sort the list of rules by the column.

Tip: Click the filter icon to filter the list of suppression rules. For example, you can filter the list to show only suppression rules associated with the Unexpected Data Store factor.

The following figure shows an example of the suppression rules list:



The following table describes the suppression rules list:

Property	Description
1. Expand/collapse	Click to show or hide the rule details. For more information, see "Suppression Rules Details" on page 371.
2. Suppression rules count	Total number of rules based on the selected time period and filter conditions.

Property	Description
3. Selection	You can perform actions on one or more rules. To edit or delete a suppression rule, select the check box next to the rule. Click the **Actions** menu and select the action. You can select one, multiple, or all rules to delete.
4. Suppression Rule Name	Name of the suppression rule.
5. User Name	Name of the user account that created the suppression rule. Click the user name to view more details about the user's activity on the User Profile page.
6. Anomalous Factors	The anomalous factors associated with the suppression rule.
7. Display Time Period	Filters the list to show anomalies with either a From or To date that is within the time period. Time periods are in the UTC time zone. **Note:** For time periods in days, the workspace displays information for full days. A full day is from 0:00 through 23:59 UTC. The time period you select might not include information from the current day. For example, you select 30 days. The workspace shows suppression rules that will be active in the next 30 days. The count begins from the last completed day in the UTC time zone, which can be yesterday.
8. From and To Time Period	The time period that the rule is in effect. Time is in the UTC time zone.
9. Status	Indicates if the rule is active or inactive. Data Privacy Management only uses Active rules in anomaly detection.
10. Created By	User name of the user that created the suppression rule.
11. Creation Time	Date and time that the rule was created. Time is in UTC time zone.
12. Refresh	Click to refresh the list of rules based on the selected time period and filter conditions.

Related Topics:

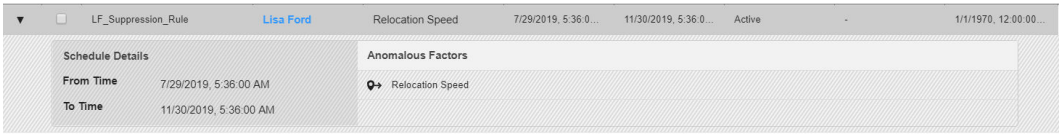
- "User Profile Page" on page 567

Suppression Rules Details

The suppression rule details panel shows the time period that the rule is in effect and the anomalous factors that are suppressed.

You can expand each entry to show the details related to the rule.

The following image shows an example of suppression rule details:



The following table describes the information on the suppression rule details panel:

Property	Description
From Time	The start date and time for the rule to be enforced if the status is also set to Active.
To Time	The end date and time for the rule to be enforced.
Anomalous Factors	List of system-defined anomalous factors that you do not want Data Privacy Management to use to detect anomalies.

Suppression Rules Management

Use the Suppression Rules workspace to view all of the anomaly-suppression rules that you created. You can filter to view rules, and edit or delete a suppression rule. You can export information about suppression rules to a .csv file.

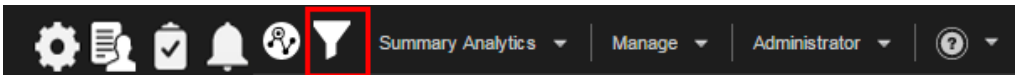
Filtering Suppression Rules

You can filter information on the Suppression Rules workspace to view rules based on criteria that matches your requirement.

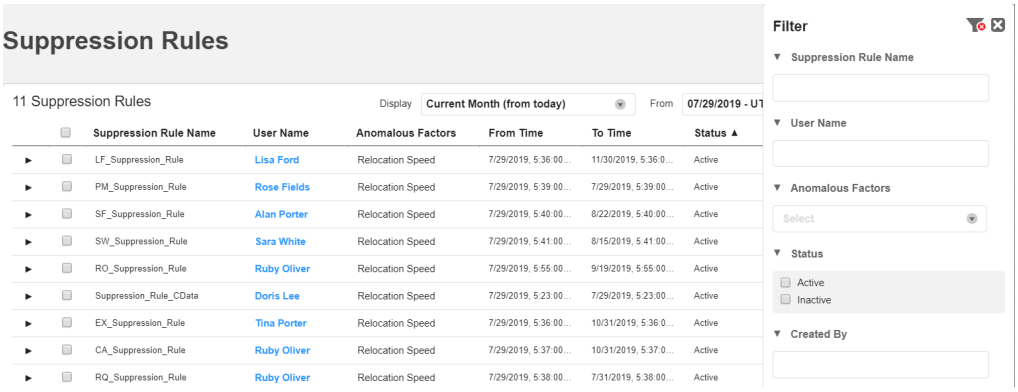
1. Click **Manage** > **Suppression Rules**.
2. Click the filter icon in the header.

   The filter pane appears with a list of properties.

   The following image shows the filter icon:



   The filter panel appears on the workspace. The following image shows the filter panel:



3. Select one or more filters from the filter panel.

   You can set the properties to filter information based on suppression rule name, user name, anomalous factors, rule status, and by the user who created the rule. When you apply a filter, the workspace displays the active filter conditions. The Data Privacy Management Service uses an AND operator if you specify more than one filter parameter. The service uses an OR operator if you specify multiple values within the same filter parameter.

Editing Suppression Rules

You can edit the properties of a suppression rule if you want to change the anomalous factors that you want to suppress. Edit the properties of a suppression rule on the Suppression Rules workspace.

1. Click that you do not want to view anomalies for.**Manage** > **Suppression Rules**.

   A list of all the suppression rules appears. You can use the filter options to narrow the list of rules.
2. Click the check box next to the suppression rule you want to edit.
3. From the **Actions** menu, select **Edit**.

   Edit the suppression rule properties that you want to change.
4. Click **Save**.

Exporting Suppression Rules

You can export information about suppression rules from the Suppression Rules workspace. Data Privacy Management exports the information to CSV files.

Suppression Rules Export Files

When you export information from the Suppression Rules workspace, the Data Privacy Management Service downloads a compressed file named Data.zip to the default download directory on your machine. The Data.zip file includes CSV files that contain information related to the page from which you exported. The information in the CSV files matches the selected filter conditions and time periods at the time of the export.

The following table lists the CSV files in the Data.zip file that you export from the Suppression Rules workspace:

File Name	Description
UbaAnomalySuppressionRule.csv	Contains details about the suppression rules.
UbaAnomalySuppressionRuleAnomalyFactor.csv	Contains information about the anomaly factors in the suppression rules.

Steps to Export Suppression Rules

You can export information from the Suppression Rules workspace. The export file contains the suppression rules that match the selected filter conditions and time periods at the time of the export.

1. Click **Manage** > **Suppression Rules**.

   The Suppression Rules workspace appears.
2. Optionally, set the filter conditions and time period.
3. From the Actions menu, select **Export**.

   A dialog box appears with the exported file. The export file name is Data.zip. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, the service downloads a new file. The file name includes an incremental number such as `Data (1).zip`.

Deleting a Suppression Rule

You can delete suppression rules on the Suppression Rules workspace.

You cannot delete a suppression rule that is in the Active state.

1. Click **Manage** > **Suppression Rules**.

   The Suppression Rules workspace appears.
2. Select the check box next to the suppression rule that you want to delete.

   You can select one, multiple, or all suppression rules.
3. From the Actions menu, select **Delete**.

   A message appears to confirm that you want to delete the suppression rule.

The SData Privacy Management Service deletes the suppression rule from the repository.

CHAPTER 18

Actions

This chapter includes the following topics:

- Actions Overview, 375
- Actions Workspace, 376
- Action Properties, 377
- Placeholders in Custom, Email, Service Management, and System Log Actions, 383
- Action Management, 385

Actions Overview

You can configure reusable actions and include them in security policies or in tasks that you run on demand.

You can create the following action types on the **Actions** workspace:

- **Custom**. Custom actions run a custom script when the conditions specified in the action are met.
- **Email**. Email actions send an email to designated recipients if a security policy violation occurs, to automate data subject requests, or when you discover information in Data Privacy Management and you want to notify other users.
- **Service Management**. Service management actions create a ServiceNow ticket in the event of a security policy violation, to satisfy a data subject request, or when you identify another issue in Data Privacy Management that requires creating a ticket.
- **System Log**. System log actions create a log message that Data Privacy Management can send to a remote system log server in the event of a security policy violation.

You can include multiple actions in a security policy, and you can include the same action in multiple security policies.

When you specify actions manually to run tasks, you can create an action or specify an existing action. You create manual actions from the **Actions** > **Take Action** menu option on the following views:

- **Anomaly Detection** workspace
- **Proliferation** page
- **Security Policy Violations** workspace
- **Sensitive Fields** page
- **Subject Details** page
- **Top Data Domains** list page

- **Top Data Stores** grid page

You can use placeholders in actions as a variable for a specific detail. When an action runs an associated task, Data Privacy Management substitutes each placeholder with the value for the variable.

Actions Workspace

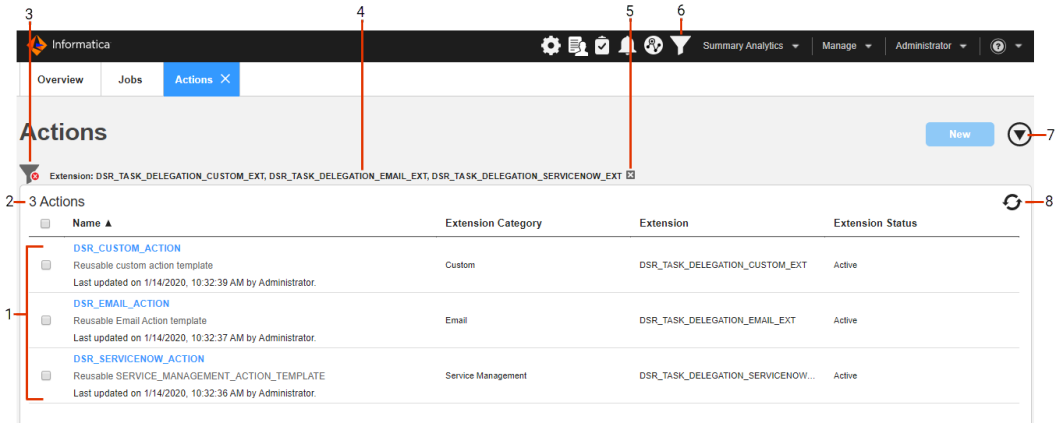
On the **Actions** workspace, you can view and manage custom, email, service management, and system log actions. The workspace includes a page that displays a list of actions and a detail page for each action.

The workspace lists actions by name, extension category, extension name, and extension status. To access the **Actions** workspace, click **Manage** > **Actions**.

Actions List Page

The **Actions** list page displays the actions that you have access to view or manage. By default, you view the **Actions** list page when you access the **Actions** workspace.

The following image shows an example of the **Actions** workspace:



1. Actions
2. Action count
3. Clear filter icon
4. Filter conditions
5. Clear filter condition icon
6. Filter icon
7. Actions menu
8. Refresh icon

You can access action properties from this page by clicking an action name.

Action Properties

Action properties define the details for a task that Data Privacy Management runs when an action is triggered by a security policy violation or manual intervention.

The following table describes the common action properties:

Property	Description
Name	Required. The name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 255 characters, contain spaces, or include the following special characters: `\~!$%^&*()+`
Description	Optional. Long description of the action that does not exceed 255 characters.
Extension Category	Required. Select one of the following options: - Custom - Email - Service Management - System Log
Extension	Required. Select an extension name from the list. For an extension to appear in the list, the extension must be configured on the **Extensions** workspace and have an Active status. After you select an extension, Data Privacy Management displays the following information: - The **Extension Status** field with a value of `Active`. - The **Plugin** field with the name of the plugin configured for the selected extension.

Custom Action Properties

Configure the custom action properties for the script that Data Privacy Management runs.

The following table describes the custom action properties:

Property	Description
Parameter	Provides a text field. To add a placeholder, enter a dollar sign (`$`) and select a placeholder from the list that appears. The script uses the values to produce the desired output.
Action Type	Optional. For actions configured with extensions that use the DSR Custom Plugin, select a subject request type from the list.
Report	Optional. For actions configured with extensions that use the DSR Custom Plugin, select one of the following options for the subject information the task provides: - Attributes. The task produces results that contain only the types of information that you hold about data subjects. - Attributes and Values. The task produces results that contain both the types of information that you hold about data subjects and the values of the personal data.
Manually close task after job completion	Optional. Select to indicate that the user assigned to the task marks the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management automatically updates the task status to Closed after the job completes successfully.
Context File Type	Required. Select CSV or JSON as the file type in the **Attachments** field on the **Task Properties** page after the associated task runs.

Property	Description
Execute separate action for each data store	For actions configured with extensions that use Custom Plugin V1, select to create a separate task for each data store that is included in the action when the custom script runs.
Add Context to Task Details as Attachment	Optional. Select to attach the details of the violation, data subject request, or incident that prompted the custom action. If selected, the context details appear in the **Attachments** field on the **Task Properties** page.
Notify assignee via email	Optional. Select to send an email to notify the assignee when tasks resulting from the action are assigned to them.
Tags	Optional. Select one or more tags that identify custom action characteristics.
Notes	Optional. A text field that contains descriptive notes about the custom action. Notes cannot exceed 255 characters.
Due in Days	Optional. The number of days from the date the custom action runs that the task must complete.

Email Action Properties

Configure email action properties for emails that you want Data Privacy Management to send.

The following table describes the email action properties:

Property	Description
From	The email address of the sender.
To	The email addresses of the recipients of the email. Use a comma to separate email addresses.
Cc	Optional. The email addresses of the recipients copied on the email. Use a comma to separate email addresses.
Bcc	Optional. The email addresses of the recipients blind copied on the email. Recipients in the **Bcc** field are not visible to the recipients in the **To** and **Cc** fields. Use a comma to separate email addresses.
Subject	Optional. A short description that highlights the message. To add a placeholder, enter a dollar sign ($) and select a placeholder from the list that appears. Before sending the email, Data Privacy Management replaces the placeholders with specific values.

Property	Description

Message	Optional. A message regarding the reason for sending the email. To add a placeholder, enter a dollar sign ($) and select a placeholder from the list that appears. Before sending the email, Data Privacy Management replaces the placeholders with specific values.
	For example, if you add the following text:
	`A violation that included $ViolationDataStoreCount data store(s) occurred.`
	`Violation detected: $ViolationDateTime`
	`User's login: $LoginUserName`
	`Data store names: $DataStoreName`
	Data Privacy Management replaces the placeholders with values and send the email with a message such as the following example:
	`A violation that included 1 data store occurred.`
	`Violation detected: 11/3/2019, 10:25:21 AM`
	`User's login: JaneJohn`
	`Data store names: DEI_US_DailyTrans`

Export selected items as email attachment	Select to attach a CSV file to the email that contains the details of the violation, subject request, or other incident that prompted the email action.
Attachments	Optional. Add files that you want to attach to emails that Data Privacy Management sends when the email action runs.
Action Type	Optional. For actions configured with extensions that use the DSR Email Plugin, select a subject request type from the list.
Report	Optional. For actions configured with extensions that use the DSR Email Plugin, select one of the following options for the subject information the task provides: - Attributes. The task produces results that contain only the types of information you hold about data subjects. - Attributes and Values. The task produces results that contain both the types of information you hold about data subjects and the values of the personal data.
Attach DSAR Report if Available	Optional. For actions configured with extensions that use the DSR Email Plugin, select to attach a DSAR report, if available to the email that Data Privacy Management sends when the task runs.
Manually close task after job completion	Optional. Select to indicate that the user assigned to the task marks the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management automatically updates the task status to Closed after the job completes successfully.
Context File Type	Required. Select CSV or JSON as the file type that appears in the **Attachments** field of the **Task Properties** page after the associated task runs.
Execute separate action for each data store	For actions configured with extensions that use Email plugin V1, select to indicate that Data Privacy Management creates separate tasks for each data store that is included in the action when the email task runs.
Add Context to Task Details as Attachment	Optional. Select to attach the details of the violation, data subject request, or incident that prompted the email action. If selected, the context details appear in the **Attachments** field of the **Task Properties** page.
Notify assignee via email	Optional. Select to send an email to notify the assignee when tasks resulting from the action are assigned to them.

Property	Description
Tags	Optional. Select one or more tags that identify email action characteristics.
Notes	Optional. A text field that contains descriptive notes about the email action. Notes cannot exceed 255 characters.
Due in Days	Optional. The number of days from the date the email action runs that the task must complete.

Service Management Action Properties

Configure service management action properties to specify details about how Data Privacy Management runs tasks that create ServiceNow tickets.

The following table describes the service management action properties:

Property	Description
Short Description	A short description of the action. You can add placeholders to include specific details in the description. To add a placeholder, enter a dollar sign (`$`) and select a placeholder from the list that appears.
Description	Optional. A more detailed description of the action. You can add placeholders to add more details in the description. To add a placeholder, enter `$` and select a placeholder from the list that appears.
Urgency	Optional. Select one of the following urgency levels for the ServiceNow ticket that tasks based on the action create: High, Low, or Medium.
Impact	Optional. Select one of the following options to indicate the level of impact that the ServiceNow ticket that tasks based on the action create: High, Low, or Medium.
Category	Optional. Select one of the following options to specify the product category with a reported issue or the nature of the subject request that prompted the creation of a ServiceNow ticket: - Database - Hardware - Inquiry - Network - Software
Attachments	Optional. Add files that you want to attach to ServiceNow tickets that Data Privacy Management creates when the service management action runs.
Wait for ServiceNow Ticket Status Change	Optional. Select to indicate that the ServiceNow ticket status must change to In Progress before the service management task can close.
Action Type	Optional. For actions configured with extensions that use the DSR ServiceNow Plugin, select a subject request type from the list.
Attach DSAR Report if Available	Optional. For actions configured with extensions that use the DSR ServiceNow Plugin, select to attach a DSAR report, if available, to the ServiceNow ticket.
Manually close task after job completion	Optional. Select to indicate that the user assigned to the task marks the task as closed on the **Tasks** workspace after the task job is successful. If unselected, Data Privacy Management automatically updates the task status to Closed after the job completes successfully.

Property	Description
Context File Type	Required. Select CSV or JSON as the file type in the **Attachments** field on the **Task Properties** page after the associated task runs.
Execute separate action for each data store	For actions configured with extensions that use ServiceNow Plugin V1, select to indicate that Data Privacy Management creates separate tasks for each data store included in the action when the service management task runs.
Add Context to Task Details as Attachment	Optional. Select to attach the details of the violation, data subject request, or incident that prompted the service management action. If selected, the context details appear in the **Attachments** field on the **Task Properties** page.
Notify assignee via email	Optional. Select to send an email to notify the assignee when tasks resulting from the action are assigned to them.
Tags	Optional. Select one or more tags that identify service management action characteristics.
Notes	Optional. A text field that contains descriptive notes about the service management action. Notes cannot exceed 255 characters.
Due in Days	Optional. The number of days from the date the service management action runs that the task must complete.

System Log Action Properties

Configure system log action properties for the system log message that you want Data Privacy Management to write to the specified server.

The following table describes the system log action properties:

Property	Description

Message	Required. The contents of the system log message that Data Privacy Management sends to the server when a security policy violation triggers the action. To add a placeholder, enter a dollar sign (`$`) and select a placeholder from the list that appears. In the event of a security policy violation, Data Privacy Management replaces the placeholders with specific values.
	Add the message contents in a text format, such as Common Event Format (CEF) or Log Event Extended Format (LEEF).
	For example, if you add the following CEF-formatted text:
	`CEF:0|security|threatmanager|1.5|100|Discovered high-risk data stores with CreditCard_Visa domain |10|CS1Label=ApplicationName CS1=S@S CS2Label=DataStoreNames CS2=$DataStoreName`
	Data Privacy Management replaces the placeholders with values and sends a system log message such as the following example:
	`CEF:0|security|threatmanager|1.5|100|Discovered high-risk data stores with CreditCard_Visa domain |10|CS1Label=ApplicationName CS1=S@S CS2Label=DataStoreNames CS2=PCRS_Linux/ORA05`

Log Level	Optional. Select one of the following log levels for the message: - Alert (1) - Critical (2) - Debug (7) - Emergency (0) - Error (3) - Informational (6) - Notice (5) - Warning (4)
Manually close task after job completion	Optional. Select to indicate that the user assigned to the task marks the task as closed on the **Tasks** workspace after the task job is successful. If unselected, Data Privacy Management automatically updates the task status to Closed after the job completes successfully.
Execute separate action for each data store	Optional. Select to indicate that Data Privacy Management creates separate tasks for each data store included in the action when the system log task runs.
Add Context to Task Details as Attachment	Optional. Select to attach the details of the violation or incident that prompted the system log action. If selected, the context details appear in the **Attachments** field on the **Task Properties** page.
Notify assignee via email	Optional. Select to send an email to notify the assignee when tasks resulting from the action are assigned to them.
Tags	Optional. Select one or more tags that identify system log action characteristics.
Notes	Optional. A text field that contains descriptive notes about the system log action. Notes cannot exceed 255 characters. .
Due in Days	Optional. The number of days from the date the system log action runs that the task must complete.

Placeholders in Custom, Email, Service Management, and System Log Actions

You can use placeholders in the custom, email, service management, and system log actions to specify a variable that populates with the correct value when a security policy violation occurs or a user takes manual action to run a task. To add a placeholder, enter a dollar sign ($) and select a placeholder.

A list of placeholders appears when you enter a dollar sign ($) in the **Parameter** text box of a custom action, in the **Subject** or **Message** field of an email action, in the **Short Description** or **Description** field of a service management action, or in the **Message** field of a system log action.

The following table lists the placeholders that you can use when you create or edit an action on the **Actions** workspace, create a new manual action, or when you add a custom, email, service management, or system log action to a security policy. The table also indicates the type of security policy for which the placeholder is valid, if applicable.

Placeholder	Description	Anomaly Security Policy	Data Store Security Policy	User Activity Security Policy
ActivityStoreCount	Includes the number of data stores that are associated to the user activity violation.	-	-	x
AnomalousFactorCount	Includes the number of system-defined anomalous factors that are associated to the anomaly.	x	-	-
AnomalyDateTime	Lists the date and time that the anomaly occurred.	x	-	-
DataStoreName	Lists the data stores involved in the event that caused the security policy violation.	x	x	x
DataStoreOwner	Lists the owners of each data store involved in the event that caused the security policy violation.	x	x	x
DataStoreProtectionStatus	Lists the protection status of data stores involved in the event that caused the security policy violation.	x	x	x
DataStoreRiskScore	Lists the risk score of data stores involved in the event that caused the security policy violation.	x	x	x
DataStoreSensitiveFieldNames	Lists the names of all sensitive fields in the data stores involved in the event that caused the security policy violation. .	x	x	x
DataStoreSensitiveFieldsCount	Lists the number of sensitive fields in the data stores involved in the event that caused the security policy violation.	x	x	x

Placeholder	Description	Anomaly Security Policy	Data Store Security Policy	User Activity Security Policy
DataStoreTargetNames	Lists the names of the target data stores involved in the event that caused the security policy violation.	x	x	x
DataStoreTargetsCount	Lists the number of target data stores involved in the event that caused the security policy violation.	x	x	x
DataStoreType	Lists the types of data stores involved in the event that caused the security policy violation.	x	x	x
DomainImpressions	Lists the data domain impressions involved in the event that caused the security policy violation.	x	x	x
FirstName	The first name of a data subject in the Subject Registry.	-	-	-
FullName	The full name of a data subject in the Subject Registry.	-	-	-
ImpressionsCount	Lists the number of data domain impressions involved in the event that caused the security policy violation.	x	-	x
LastName	The last name of a data subject in the Subject Registry.	-	-	-
LoginUserEmail	For email actions only, includes the email of the user who was logged in when the event that triggered the anomaly or security policy violation occurred.	x	x	x
LoginUserName	Includes the username of the user who was logged in when the event that triggered the anomaly or security policy violation occurred.	x	x	x
SecurityPolicyDescription	Includes the description provided in the security policy when the policy was created or last updated.	x	x	x
SecurityPolicyName	Includes the name of the security policy that Data Privacy Management used to detect the violation.	x	x	x
SecurityPolicyType	Includes the type of the security policy that Data Privacy Management used to detect the violation. A security policy type can be an anomaly, a data store, or user activity.	x	x	x

Placeholder	Description	Anomaly Security Policy	Data Store Security Policy	User Activity Security Policy
SensitiveFilesCount	Includes the number of sensitive files in the data stores that are involved in the secuirty policy violation.	x	x	x
SourceIP	Includes the IP address of the source data store that is involved in the security policy violation.	-	-	x
TargetIP	Includes the IP address of the target data store that is involved in the security policy violation.	-	-	x
TaskName	Includes the name of the task that Data Privacy Management generates when the action runs as a result of the security policy violation.	x	x	x
TaskURL	Includes the link to the Task Details and Properties page in Data Privacy Management.	x	x	-
ViolationDataStoreCount	Includes the number of data stores accessed in the event that triggered the security policy violation.	-	x	-
ViolationDateTime	Includes the time stamp of the data store event that triggered the security policy violation.	-	x	-
ViolationID	Includes the system issued ID of the security policy violation.	-	x	-

Action Management

You can create and manage actions on the **Actions** workspace.

When you edit an action on the **Actions** workspace, the changes apply immediately to security policies and active tasks that include the action.

Creating an Action

Create an action to configure details that Data Privacy Management requires to run associated tasks.

To create a system log action that uses TCP/SSL protocol, first import the SSL certificate for the system log server to the Informatica domain truststore, and then restart the Informatica domain. For more information, see the H2L article Updating Keystore and Truststore Certificates to Maintain Secure Communication.

1. Click **Manage** > **Actions**.

   The **Actions** workspace appears.

2. Click **New**.

   The **New_Action** page appears.
3. Enter the name for the action.
4. Optionally, enter a description.
5. In the **Extension Category** field, select **Custom**, **Email**, **Service Management**, or **System Log**.
6. In the **Extension** field, select an extension from the list.

   The **Extension Status** property displays with a value of Active. The **Plugin** property shows the name of the plugin configured for the extension. The **Configure Settings** pane appears.
7. Configure the properties for the action.
8. Click **Save**.

Editing an Action

When you edit an action, the changes apply immediately to security policies and active tasks that include the action.

1. Click **Manage** > **Actions**.

   The **Actions** workspace displays a list of actions.
2. Select the check box next to the action that you want to edit.
3. From the **Actions** menu, click **Edit**.
4. Edit the action properties that you want to change.
5. Click **Save**.

Copying an Action

You can copy an action to create a new action with similar properties. After you copy the action, edit the action properties.

1. Click **Manage** > **Actions**.

   The **Actions** workspace displays a list of actions.
2. Select the checkbox next to the action that you want to copy.
3. From the **Actions** menu, select **Duplicate**.

   The **Action Details** page appears with the following name: `Copy of <action name>`
4. Edit the action properties.
5. Click **Save**.

Deleting an Action

You can delete actions that are not included in a security policy.

1. Click **Manage** > **Actions**.

   The **Actions** workspace displays a list of actions.
2. Select the check box next to the action that you want to delete.
3. From the **Actions** menu, select **Delete**.

A message asks you to confirm if you want to delete the action.

4. Click **Yes**.

CHAPTER 19

Manual Actions

This chapter includes the following topics:

- Manual Actions Overview, 388
- Creating a Service Management Ticket, 389
- Protecting Data, 403
- Running a Custom Script, 412
- Sending an Email, 423
- Writing a System Log Message, 436

Manual Actions Overview

You can create tasks on demand from several views in Data Privacy Management to create a service management ticket, protect data, run a custom script, send an email, and write a system log message.

You create manual actions from the **Actions** > **Take Action** menu option on the following views in Data Privacy Management:

- **Anomaly Detection** workspace.
- **Data Domains** list page accessed from the **Overview** workspace
- **Data Stores** grid page accessed from the **Overview** workspace
- **Data Stores** list page accessed from the **Overview** workspace
- **Proliferation** page
- **Security Policy Violations** workspace.
- **Sensitive Fields** page accessed from the **Overview** workspace

The **Actions** > **Take Action** menu provides the following options:

- **Create a Service Management Ticket**
- **Protect Data**
- **Run a Custom Script**
- **Send an Email**
- **Write a System Log Message**

Manual actions to protect data create a protection task with a status of New and a prefix of `PRT-` on the **Tasks** workspace.

The remaining manual actions allow you to create a new action or use an existing action as a template. These manual actions create system tasks with a prefix of `SYS-`. You can run a system task immediately, or you can save the manual action as a New system task on the **Tasks** workspace and manually run the task later.

In Data Privacy Management, you can add system actions to anomaly, data store, decryption, and user activity security policy types. You can add Persistent Data Masking protection actions to data store and user activity security policies only. You can add encryption protection actions to decryption security policies only.

For more information, see "Security Policy Properties" on page 453.

Related Topics:

- "Data Domains List Page" on page 534
- "Data Stores Grid Page" on page 536
- "Data Stores List Page" on page 538
- "Proliferation Page" on page 542
- "Sensitive Fields Page" on page 549

Creating a Service Management Ticket

You can create a ServiceNow service management ticket on demand when you identify an issue in Data Privacy Management that requires a ticket to be created.

Create a new action or select an existing service management action to use as a template for creating the ticket. When you create a new action, specify the service management extension and all required properties. When you select an existing service management action, the extension and other properties such as placeholders appear automatically. Include a description of the action, which can contain placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.

Optionally, specify the urgency level, impact, and category of the issue requiring a service management ticket. You can also attach the full details of the violation to the ticket, add tags and other notes, and specify the number of days from now that the task is due.

Creating a Service Management Ticket from the Anomaly Detection Workspace

You can create a ServiceNow ticket for an anomaly from the **Anomaly Detection** workspace.

Important: Data Privacy Management can only perform actions or create tasks for anomalies that are associated with at least one data store. Verify that each anomaly lists at least one data store in the **Data Store** column before creating a ServiceNow ticket for the anomaly.

1. Click the **Anomalies** icon in the Data Privacy Management header.

   The **Anomaly Detection** workspace appears.
2. On the **Anomalies** list page, select the check box next to one or more anomalies for which you want to create a ServiceNow ticket.
3. From the **Actions** menu, select **Take Action** > **Create a Service Management Ticket**.

Note: The **Create a Service Management Ticket** option might be disabled if you do not have privileges to create a service management task or you do not have access to the data stores in the selected anomalies.

The **Create a Service Management Ticket** page displays on the **Tasks** workspace.

4. Select whether to create a new service management action or to use an existing service management action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the service management extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

5. Optionally, enter a unique description that will display under the task number on the **Tasks** workspace.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Enter or edit a short description of the action. You can add placeholders to include specific details in the description. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. In the **Description** field, optionally add or edit a more detailed description of the action. You can add placeholders to include specific details in the description.
9. Optionally, select or edit an urgency level, level of impact, and product categotry for the ServiceNow ticket.
10. To attach the anomaly details to the ServiceNow ticket, select **Add Context to Service Now Ticket**.
11. Optionally, add attachments.
12. Select **Manually close task after job completion** to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If unselected, Data Privacy Management automatically updates the task status to Closed after the job completes successfully.
13. Select **Execute separate action for each data store** to indicate that Data Privacy Management will create separate tasks for each data store included in the action when the ServiceNow ticket is created.
14. Optionally, select **Add Context to Task Details as Attachment** to attach the details of the anomaly to the tasks associated with the action.
15. To indicate that Data Privacy Management will notify the user assigned to the service management task, select **Notify assignee via email**.
16. Optionally, select tags for the service management task, add notes, and select a due date for the task.
17. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a ServiceNow task. The task appears on the **Tasks** workspace. Saved tasks have a status of New. Tasks you run immediately display the success or failure of the job in the **Status** column.

Related Topics:

- "Anomaly Detection Workspace" on page 348

Creating a Service Management Ticket from the Proliferation Page

You can create service management tickets for data stores on the **Proliferation** page.

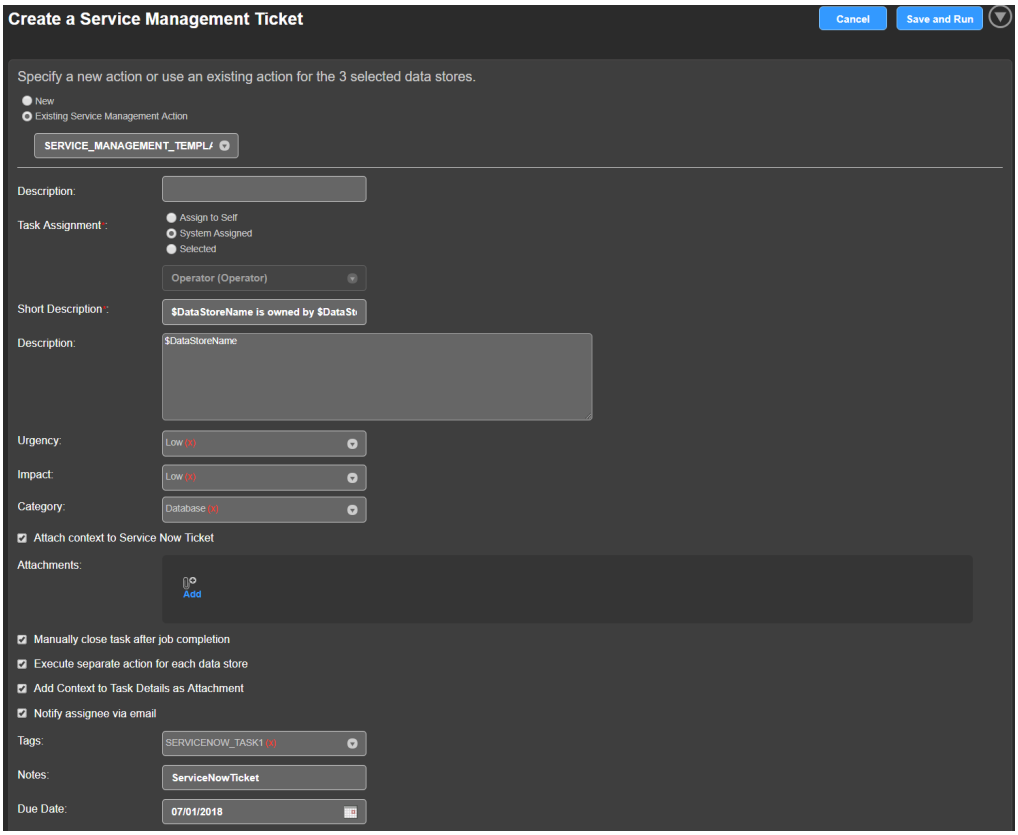
1. On the **Overview** workspace, click a data store name from the **Top Data Stores** indicator.

   The **Proliferation** page appears.
2. Select the check box next to one or more data stores on the **Proliferation** page.
3. From the **Actions** menu, select **Take Action** > **Create a Service Management Ticket**.

   Note: The **Create a Service Management Ticket** option might be disabled for one of the following reasons: you do not have privileges to create a service management task or a compatible service management extension does not exist for the selected data stores.

   The **Create a Service Management Ticket** page displays on the **Tasks** workspace.
4. Select whether to create a new service management action or to use an existing service management action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the service management extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Create a Service Management Ticket** page after selecting three data stores and an existing service management action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Enter or edit a short description of the action. You can add placeholders to include specific details in the description. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. In the **Description** field, optionally add or edit a more detailed description of the action. You can add placeholders to include specific details in the description.
9. Optionally, select or edit one of the following urgency levels for the ServiceNow ticket that tasks based on the action will create:
   - **High**
   - **Low**
   - **Medium**
10. Optionally, select or edit one of the following options to indicate the level of impact that the ServiceNow ticket that tasks based on the action will create:
    - **High**
    - **Low**
    - **Medium**
11. Optionally, select or edit one of the following options to specify the product category with a reported issue that prompted the creation of a ServiceNow ticket:
    - **Database**
    - **Hardware**
    - **Inquiry**
    - **Network**
    - **Software**
12. To attach the data store details to the ServiceNow ticket, select the **Add Context to Service Now Ticket** check box.
13. Optionally, add attachments.
14. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
15. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the ServiceNow ticket is created.
16. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
17. To indicate that Data Privacy Management will notify the user assigned to the service management task once you create it, select the **Notify assignee via email** check box.
18. Optionally, select tags for the service management task and add any notes.

19. Optionally, select a task due date.
20. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to create the ServiceNow ticket. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

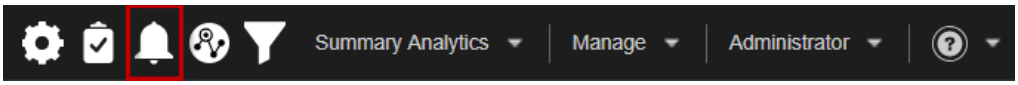
Related Topics:

- "Security Dashboard Overview" on page 513
- "Proliferation Page" on page 542
- "Top Data Stores Indicator" on page 525

Creating a Service Management Ticket from the Security Policy Violations Workspace

You can create service management tickets for security policy violations on the **Security Policy Violations** workspace.

1. Click the **Security Policy Violations** icon in the Data Privacy Management header. The following image shows the **Security Policy Violations** icon:



   The **Security Policy Violations** workspace appears.
2. In the **Violations** list, select the check box next to one or more security policy violations for which you want to create a service management ticket.

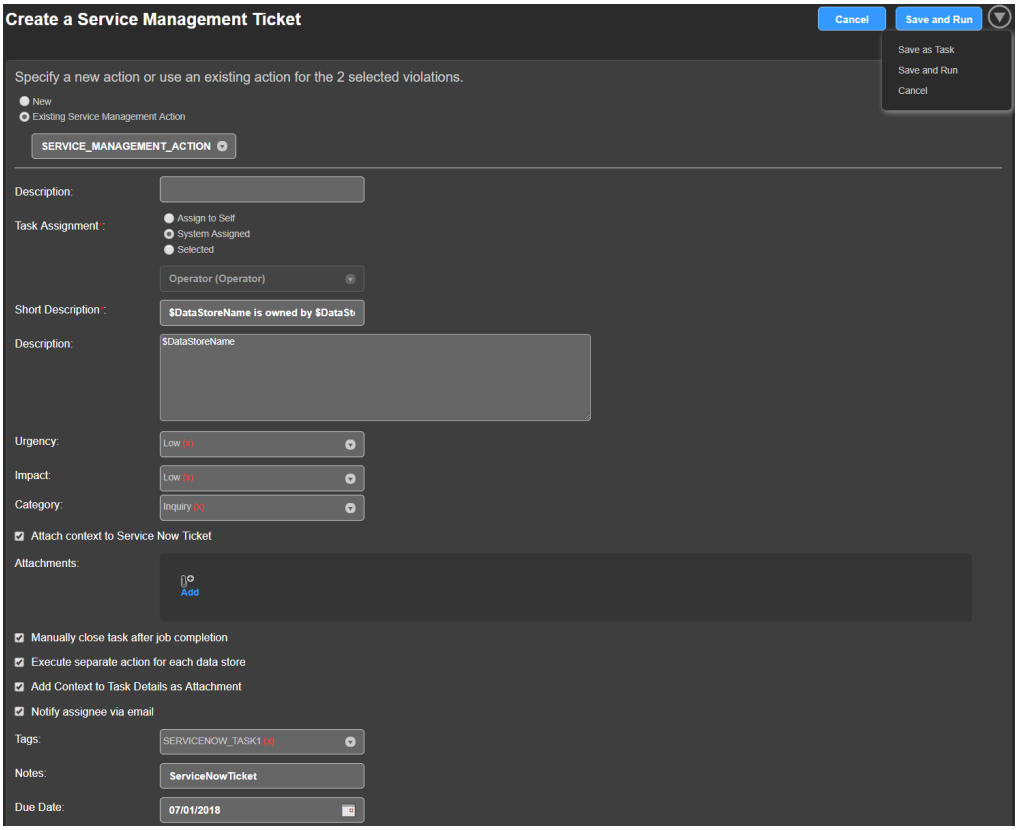
   Important: Data Privacy Management can only perform actions or create tasks for violations that are associated with at least one data store. If you select an anomaly violation, verify that the violation is associated with at least one data store before creating a service management ticket.
3. From the **Actions** menu, select **Take Action** > **Create a Service Management Ticket**.

   Note: The **Create a Service Management Ticket** option might be disabled if you do not have privileges to create a service management task or you do not have access to the data stores in the selected violations.

   The **Create a Service Management Ticket** page displays on the **Tasks** workspace.
4. Select whether to create a new service management action or to use an existing service management action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the service management extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Create a Service Management Ticket** page after selecting two low-severity anomaly violations and an existing service management action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Enter or edit a short description of the action. You can add placeholders to include specific details in the description. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. In the **Description** field, optionally add or edit a more detailed description of the action. You can add placeholders to include specific details in the description.
9. Optionally, select or edit one of the following urgency levels for the ServiceNow ticket that tasks based on the action will create:
   - **High**
   - **Low**
   - **Medium**
10. Optionally, select or edit one of the following options to indicate the level of impact that the ServiceNow ticket that tasks based on the action will create:

- **High**
- **Low**
- **Medium**

11. Optionally, select or edit one of the following options to specify the product category with a reported issue that prompted the creation of a ServiceNow ticket:
    - **Database**
    - **Hardware**
    - **Inquiry**
    - **Network**
    - **Software**
12. To attach the violation details to the ServiceNow ticket, select the **Add Context to Service Now Ticket** check box.
13. Optionally, add attachments.
14. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Managementwill automatically update the task status to Closed after the job completes successfully.
15. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the ServiceNow ticket is created.
16. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the violation details to the tasks associated with the action.
17. To indicate that Data Privacy Management will notify the user assigned to the service management task once you create it, select the **Notify assignee via email** check box.
18. Optionally, select tags for the service management task and add any notes.
19. Optionally, select a task due date.
20. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to create the ServiceNow ticket. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Policy Violations Workspace" on page 465

Creating a Service Management Ticket from the Sensitive Fields Page

You can create a service management ticket from the **Sensitive Fields** page.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.
2. Click **Grid** in the left column of the page.

The **Data Stores** page displays the list of data stores.

3. Click a value in the **Sensitive Fields** column.

   The **Sensitive Fields** page appears.

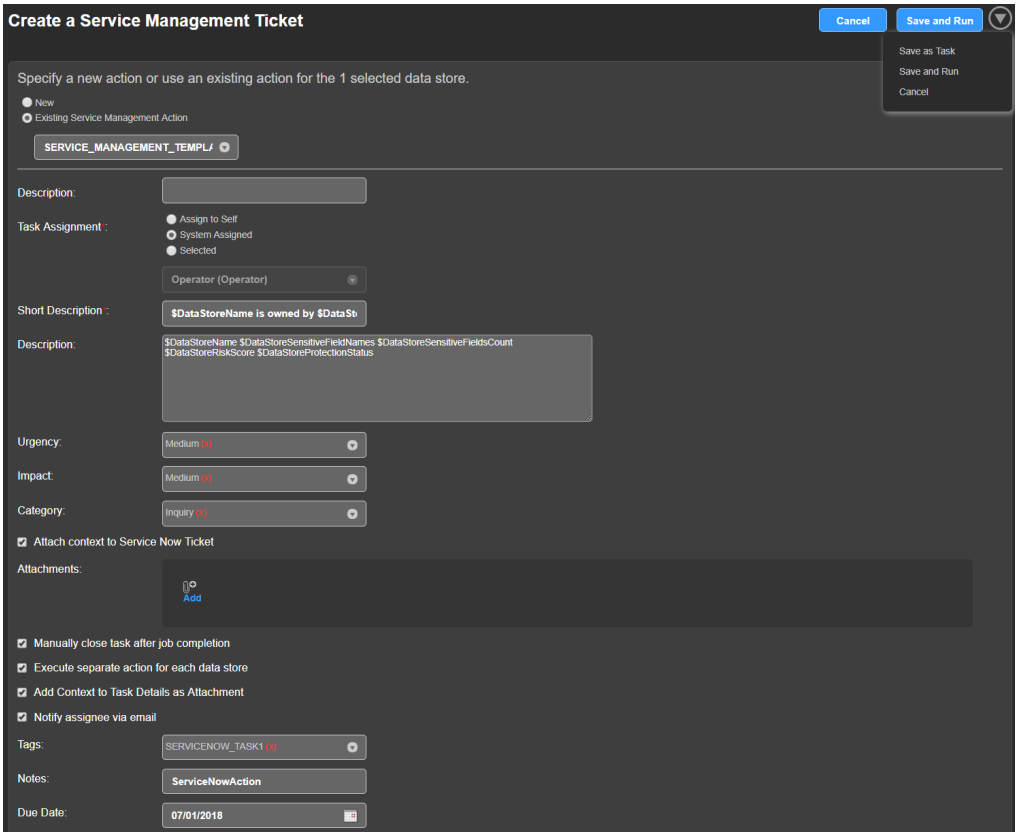
4. From the **Actions** menu, select **Take Action** > **Create a Service Management Ticket**.

   Note: The **Create a Service Management Ticket** option might be disabled if you do not have privileges to create a service management task or you do not have access to the data store that contains the sensitive fields.

   The **Create a Service Management Ticket** page displays on the **Tasks** workspace.

5. Select whether to create a new service management action or to use an existing service management action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the service management extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Create a Service Management Ticket** page after selecting a data store that has 119 sensitive fields and a risk score of 7:



6. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.

7. Select or change one of the following task assignments:

   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a Data Privacy Management user from the list.

8. Enter or edit a short description of the action. You can add placeholders to include specific details in the description. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
9. In the **Description** field, optionally add or edit a more detailed description of the action. You can add placeholders to include specific details in the description.
10. Optionally, select or edit one of the following urgency levels for the ServiceNow ticket that tasks based on the action will create:
    - **High**
    - **Low**
    - **Medium**
11. Optionally, select or edit one of the following options to indicate the level of impact that the ServiceNow ticket that tasks based on the action will create:
    - **High**
    - **Low**
    - **Medium**
12. Optionally, select or edit one of the following options to specify the product category with a reported issue that prompted the creation of a ServiceNow ticket:
    - **Database**
    - **Hardware**
    - **Inquiry**
    - **Network**
    - **Software**
13. To attach the sensitive field, data store, risk score, and protection status details to the ServiceNow ticket, select the **Add Context to Service Now Ticket** check box.
14. Optionally, add attachments.
15. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
16. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the ServiceNow ticket is created.
17. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the sensitive field details to the tasks associated with the action.
18. To indicate that Data Privacy Management will notify the user assigned to the service management task once you create it, select the **Notify assignee via email** check box.
19. Optionally, select tags for the service management task and add any notes.
20. Optionally, select a task due date.
21. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to create the ServiceNow ticket. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Sensitive Fields Page" on page 549
- "Top Data Stores Indicator" on page 525

Creating a Service Management Ticket from the Top Data Domains List Page

You can create a service management ticket from the **Top Data Domains** list page.

1. On the **Overview** workspace, click the **Top Data Domains** indicator label.

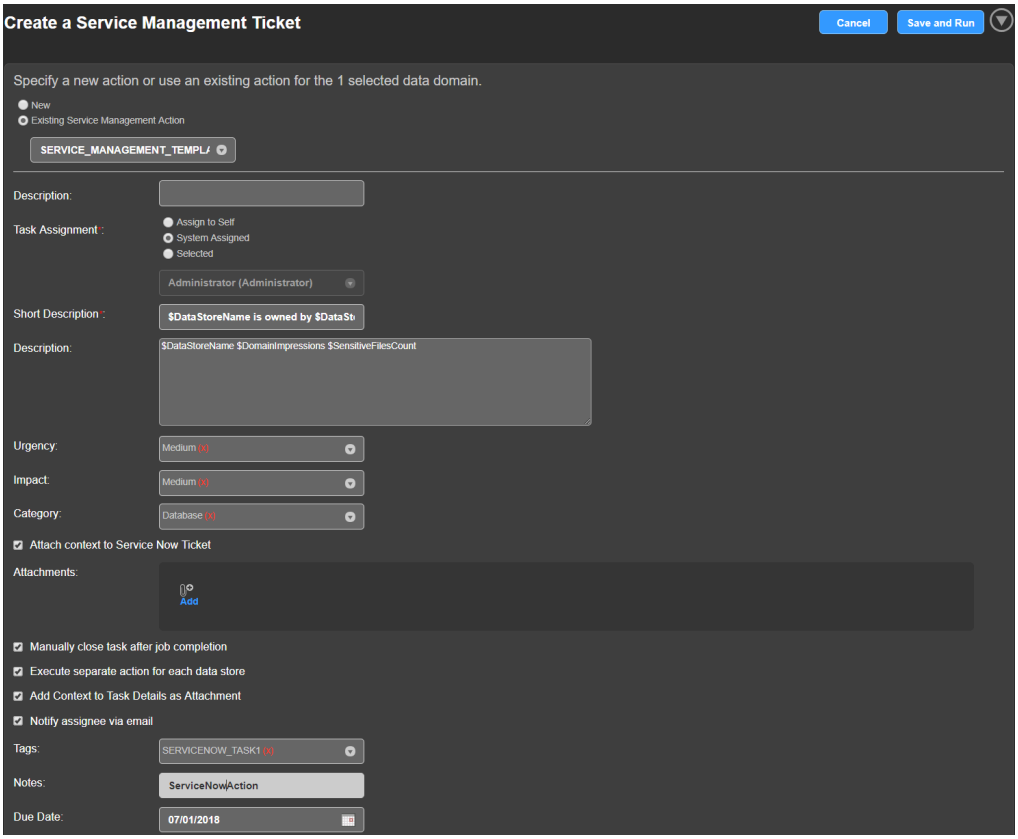
   The **Data Domains** list page appears.
2. Select the check box next to one or more data domains.
3. From the **Actions** menu, select **Take Action** > **Create a Service Management Ticket**.

   Note: The **Create a Service Management Ticket** option might be disabled if you do not have privileges to create a service management task or you do not have access to the data stores in the selected data domains.

   The **Create a Service Management Ticket** page displays on the **Tasks** workspace.
4. Select whether to create a new service management action or to use an existing service management action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the service management extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Create a Service Management Ticket** page after selecting a data domain that has two data stores:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Enter or edit a short description of the action. You can add placeholders to include specific details in the description. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. In the **Description** field, optionally add or edit a more detailed description of the action. You can add placeholders to include specific details in the description.
9. Optionally, select or edit one of the following urgency levels for the ServiceNow ticket that tasks based on the action will create:
   - **High**
   - **Low**
   - **Medium**
10. Optionally, select or edit one of the following options to indicate the level of impact that the ServiceNow ticket that tasks based on the action will create:

- **High**
- **Low**
- **Medium**

11. Optionally, select or edit one of the following options to specify the product category with a reported issue that prompted the creation of a ServiceNow ticket:
    - **Database**
    - **Hardware**
    - **Inquiry**
    - **Network**
    - **Software**
12. To attach the data domain details to the ServiceNow ticket, select the **Add Context to Service Now Ticket** check box.
13. Optionally, add attachments.
14. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
15. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the ServiceNow ticket is created.
16. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data domain details to the tasks associated with the action.
17. To indicate that Data Privacy Management will notify the user assigned to the service management task once you create it, select the **Notify assignee via email** check box.
18. Optionally, select tags for the service management task and add any notes.
19. Optionally, select a task due date.
20. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to create the ServiceNow ticket. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Creating a Service Management Ticket from the Top Data Stores Grid Page

You can create a service management ticket from the **Top Data Stores** grid page.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.
2. Click **Grid** in the left column of the page.

   The **Data Stores** page displays the list of data stores.
3. Select the check box next to one or more data stores.

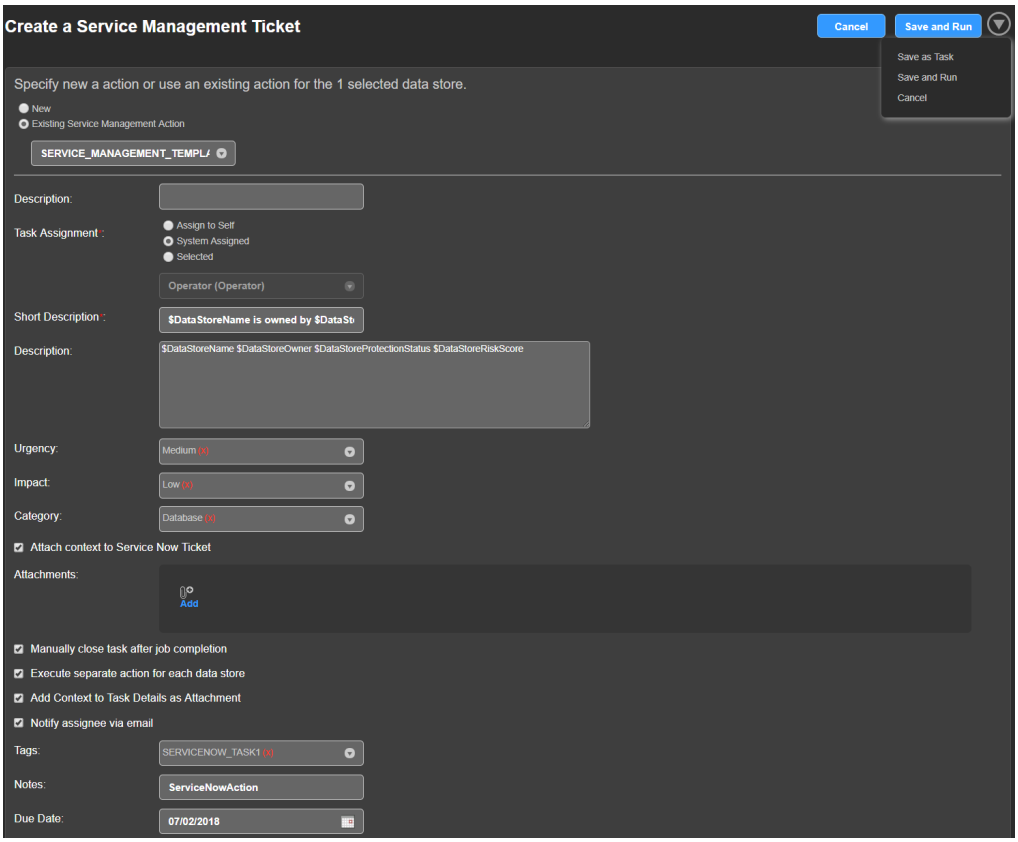
4. From the **Actions** menu, select **Take Action** > **Create a Service Management Ticket**.

   Note: The **Create a Service Management Ticket** option might be disabled if you do not have privileges to create a service management task or you do not have access to the selected data stores.

   The **Create a Service Management Ticket** page displays on the **Tasks** workspace.

5. Select whether to create a new service management action or to use an existing service management action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the service management extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Create a Service Management Ticket** page after selecting a data store with a risk score of 13:



6. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
7. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. Enter or edit a short description of the action. You can add placeholders to include specific details in the description. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.

9. In the **Description** field, optionally add or edit a more detailed description of the action. You can add placeholders to include specific details in the description.
10. Optionally, select or edit one of the following urgency levels for the ServiceNow ticket that tasks based on the action will create:
    - **High**
    - **Low**
    - **Medium**
11. Optionally, select or edit one of the following options to indicate the level of impact that the ServiceNow ticket that tasks based on the action will create:
    - **High**
    - **Low**
    - **Medium**
12. Optionally, select or edit one of the following options to specify the product category with a reported issue that prompted the creation of a ServiceNow ticket:
    - **Database**
    - **Hardware**
    - **Inquiry**
    - **Network**
    - **Software**
13. To attach the data store details to the ServiceNow ticket, select the **Add Context to Service Now Ticket** check box.
14. Optionally, add attachments.
15. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
16. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the ServiceNow ticket is created.
17. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
18. To indicate that Data Privacy Management will notify the user assigned to the service management task once you create it, select the **Notify assignee via email** check box.
19. Optionally, select tags for the service management task and add any notes.
20. Optionally, select a task due date.
21. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to create the ServiceNow ticket. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Top Data Stores Indicator" on page 525

Protecting Data

You can create a protection task on demand when you identify data stores or sensitive fields in Data Privacy Management that need to be protected.

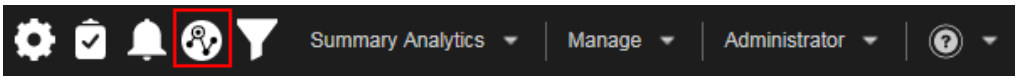
The new protection task appears on the **Tasks** workspace, where you configure the protection extension and protection rules for the task. For more information, see "Configuring Protection Tasks" on page 502.

Protecting Data from the Anomaly Detection Workspace

You can create a protection task for an anomaly from the **Anomaly Detection** workspace. Data Privacy Management creates a separate protection task for each data store.

Important: Data Privacy Management can only perform actions or create tasks for anomalies that are associated with at least one data store. Verify that each anomaly lists at least one data store in the **Data Store** column before creating a protection task for the anomaly.

1. Click the **Anomalies** icon in the Data Privacy Management header. The following image shows the **Anomalies** icon:

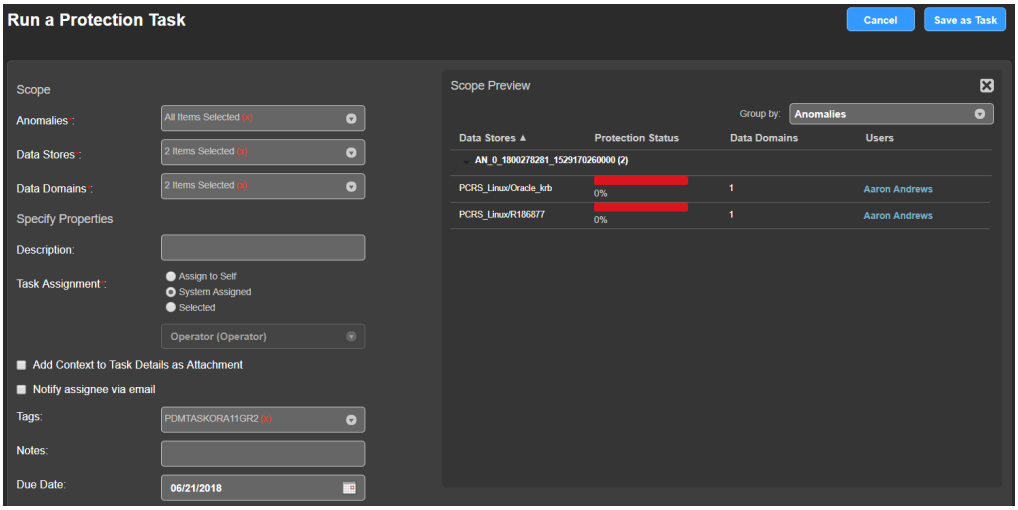
   

   The **Anomaly Detection** workspace appears.
2. In the **Anomalies** list page, select the check box next to one or more anomalies for which you want to create a protection task.
3. From the **Actions** menu, select **Take Action** > **Protect Data**.

   Note: The **Protect Data** option might be disabled for one of the following reasons: you do not have privileges to create a protection task, a compatible protection extension does not exist for the data store, or the data store does not contain any sensitive fields.

   The **Run a Protection Task** page displays on the **Tasks** workspace. The **Scope** pane lists the selected anomalies and associated data stores and data domains, which are all selected by default. The **Scope Preview** pane provides options for grouping the list of data stores and data domains so that you can quickly view the current protection status and scope of affected data stores that will be included in the protection task or tasks.

The following image shows this view:



4. In the **Data Stores** field, deselect the check box next to any associated data stores in the anomaly or anomalies for which you do not want to create a protection task.
5. In the **Data Domains** field, select the data domains in the data stores to protect.
6. Optionally, add a description for the task.
7. Select one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. To attach the anomaly details to the protection task, select the **Add Context to Task Details as Attachment** check box.
9. To indicate that Data Privacy Management will notify the user assigned to configure the protection task once you create it, select the **Notify assignee via email** check box.
10. Optionally, select tags for the protection task and add any notes.
11. Optionally, select a task due date.
12. Click **Save as Task**.

    Data Privacy Management creates a separate protection task for each selected data store. The tasks appear on the **Tasks** list page with a status of New and the extension is Unconfigured. A message indicates the number of tasks the Data Privacy Management Servicesaved successfully and the number of tasks that encountered an error. The service saves successful protection tasks in the Data Privacy Management repository.
    The user assigned to the tasks can now configure protection from the **Tasks** workspace.

Related Topics:

- "Anomaly Detection Workspace" on page 348

Protecting Data from the Proliferation Page

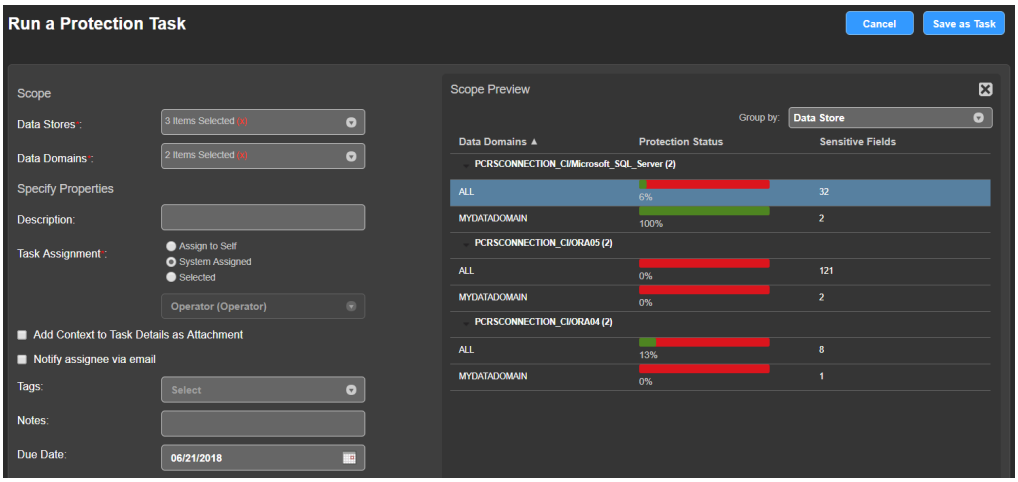
You can create new protection tasks for data stores from the **Proliferation** page. Data Privacy Management creates a separate protection task for each data store.

1. On the **Overview** workspace, click a data store name from the **Top Data Stores** indicator.

   The **Proliferation** page appears.
2. Select the check box next to one or more data stores on the **Proliferation** page.
3. From the **Actions** menu, select **Take Action** > **Protect Data**.

   Note: The **Protect Data** option might be disabled for one of the following reasons: you do not have privileges to create a protection task, a compatible protection extension does not exist for the selected data stores, or the data stores do not contain any sensitive fields.

   The **Run a Protection Task** page displays on the **Tasks** workspace. The **Scope** pane lists the selected data stores and associated data domains, which are all selected by default. The **Scope Preview** pane provides options for grouping the list of data stores and data domains so that you can quickly view the current protection status and scope of the data stores and data domains that will be included in the protection task.
   The following image shows this view after selecting three data stores on the **Proliferation** page:



4. Optionally, in the **Data Stores** field, deselect the check box next to any associated data stores for which you do not want to create a protection task.
5. Optionally, In the **Data Domains** field, deselect the check box next to any associated data domains you do not want to protect.
6. Optionally, add a description for the task.
7. Select one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.

8. To attach the data store proliferation details to the protection task, select the **Add Context to Task Details as Attachment** check box.
9. To indicate that Data Privacy Management will notify the user assigned to configure the protection task once you create it, select the **Notify assignee via email** check box.
10. Optionally, select tags for the protection task and add any notes.
11. Optionally, select a task due date.
12. Click **Save as Task**.

    Data Privacy Management creates a separate protection task for each selected data store. The tasks appear on the **Tasks** list page with a status of New and the extension is Unconfigured. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful protection tasks in the Data Privacy Management repository.
    The user assigned to the tasks can now configure protection from the **Tasks** workspace.

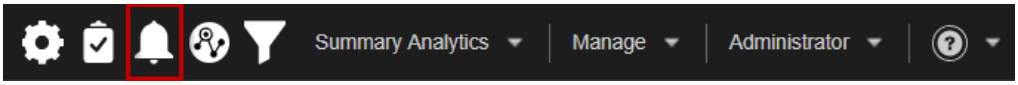
Related Topics:

- "Security Dashboard Overview" on page 513
- "Proliferation Page" on page 542
- "Top Data Stores Indicator" on page 525

Protecting Data from the Security Policy Violations Workspace

You can create new protection tasks for user activity and data store security policy violations on the **Security Policy Violations** workspace. Data Privacy Management creates a separate protection task for each data store.

1. Click the **Security Policy Violations** icon in the Data Privacy Management header. The following image shows the **Security Policy Violations** icon:

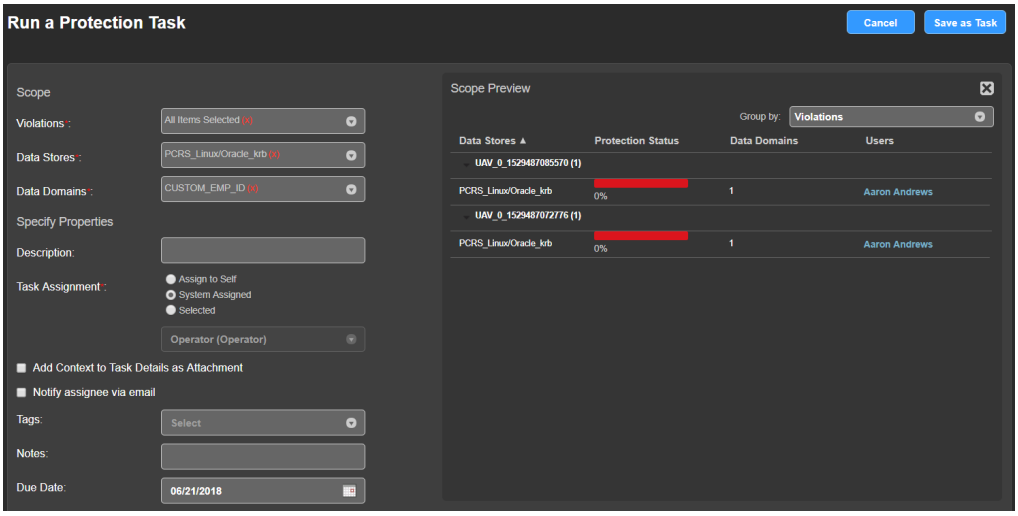


   The **Security Policy Violations** workspace appears.
2. In the **Violations** list panel, select the check box next to one or more user activity or data store security policy violations for which you want to create a protection task.
3. From the **Actions** menu, select **Take Action** > **Protect Data**.

   Note: The **Protect Data** option might be disabled for one of the following reasons: you do not have privileges to create a protection task, you selected an anomaly security policy violation, a compatible protection extension does not exist for the data store, or the data store does not contain any sensitive fields.

   The **Run a Protection Task** page displays on the **Tasks** workspace. The **Scope** pane lists the selected data stores and associated data domains, which are all selected by default. The **Scope Preview** pane provides options for grouping the list of violations so that you can quickly view the current protection status of the data stores and data domains that will be included in the protection task.

The following image shows this view after selecting two user activity security policy violations for the same data store on the **Security Policy Violations** workspace:



4. Optionally, in the **Data Stores** field, deselect the check box next to any associated data stores for which you do not want to create a protection task.
5. Optionally, In the **Data Domains** field, deselect the check box next to any associated data domains you do not want to protect.
6. Optionally, add a description for the task.
7. Select one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. To attach the security policy violation details to the protection task, select the **Add Context to Task Details as Attachment** check box.
9. To indicate that Data Privacy Management will notify the user assigned to configure the protection task once you create it, select the **Notify assignee via email** check box.
10. Optionally, select tags for the protection task and add any notes.
11. Optionally, select a task due date.
12. Click **Save as Task**.

    Data Privacy Management creates a separate protection task for each selected data store. The tasks appear on the **Tasks** list page with a status of New and the extension is Unconfigured. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful protection tasks in the Data Privacy Management repository.
    The user assigned to the tasks can now configure protection from the **Tasks** workspace.

Related Topics:

- "Security Policy Violations Workspace" on page 465

Protecting Data from the Sensitive Fields Page

You can create a protection task for a data store from the **Sensitive Fields** page. Data Privacy Management creates a separate protection task for each data store.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.

2. Click **Grid** in the left column of the page.

   The **Data Stores** page displays the list of data stores.

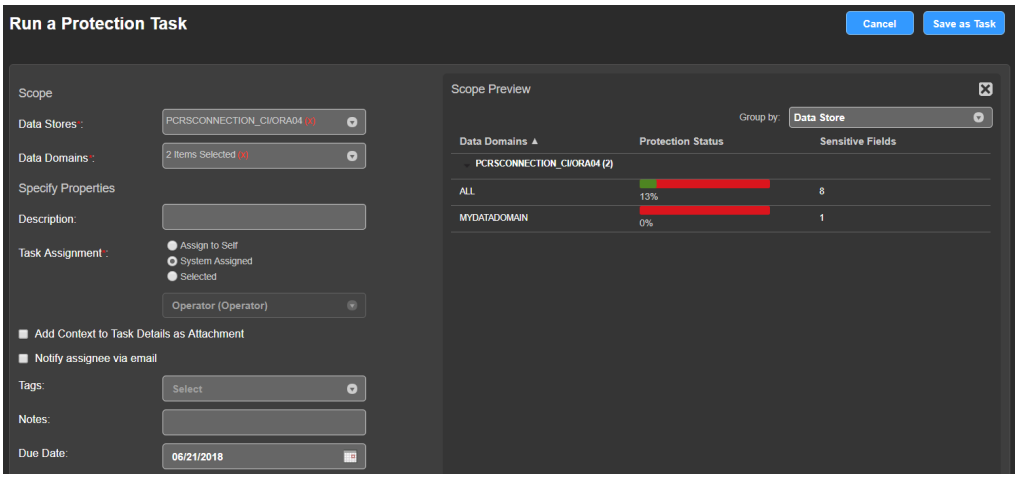
3. Click a value in the **Sensitive Fields** column.

   The **Sensitive Fields** page appears.

4. From the **Actions** menu, select **Take Action** > **Protect Data**.

   Note: The **Protect Data** option might be disabled for one of the following reasons: you do not have privileges to create a protection task, a compatible protection extension does not exist for the data stores that contain the sensitive fields, or the data store has not been scanned and has a value of Not Analyzed in the **Protection Status** column.

   The **Run a Protection Task** page displays on the **Tasks** workspace. The **Scope** pane lists the data stores and data domains associated with the sensitive fields, which are all selected by default. The **Scope Preview** pane provides options for grouping the list of sensitive fields so that you can quickly view the current protection status of the data stores and data domains that will be included in the protection task, as well as the number of sensitive fields in each data store and data domain.
   The following image shows this view after selecting a sensitive field number on the **Sensitive Fields** page:



5. Optionally, in the **Data Stores** field, deselect the check box next to any associated data stores for which you do not want to create a protection task.
6. Optionally, In the **Data Domains** field, deselect the check box next to any associated data domains you do not want to protect.
7. Optionally, add a description for the task.
8. Select one of the following task assignments:

- **Assign to Self**. Assign the task to yourself.
- **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
- **Selected**. Select a user from the list.

9. To attach the sensitive field details to the protection task, select the **Add Context to Task Details as Attachment** check box.
10. To indicate that Data Privacy Management will notify the user assigned to configure the protection task once you create it, select the **Notify assignee via email** check box.
11. Optionally, select tags for the protection task and add any notes.
12. Optionally, select a task due date.
13. Click **Save as Task**.

    Data Privacy Management creates a separate protection task for each data store. The tasks appear on the **Tasks** list page with a status of New and the extension is Unconfigured. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful protection tasks in the Data Privacy Management repository.
    The user assigned to the tasks can now configure protection from the **Tasks** workspace.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Sensitive Fields Page" on page 549
- "Top Data Stores Indicator" on page 525

Protecting Data from the Top Data Domains List Page

You can create a protection task for data domains from the **Top Data Domains** list page. Data Privacy Management creates a separate protection task for each data store.

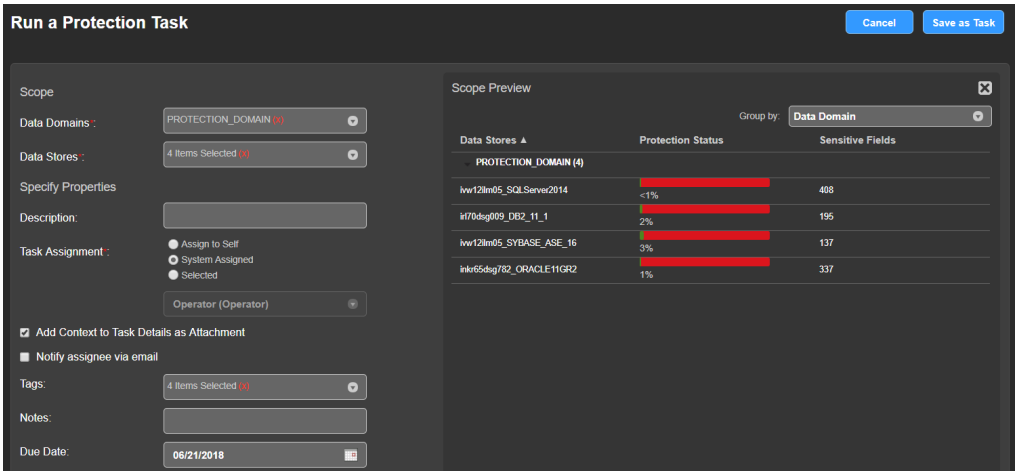
1. On the **Overview** workspace, click the **Top Data Domains** indicator label.

   The **Data Domains** list page appears.
2. Select the check box next to one or more data domains.
3. From the **Actions** menu, select**Take Action** > **Protect Data**.

   Note: The **Protect Data** option might be disabled for one of the following reasons: you do not have privileges to create a protection task, a compatible protection extension does not exist for the data stores in the selected data domains, or the data store has not been scanned and has a value of Not Analyzed in the **Protection Status** column.

   The **Run a Protection Task** page displays on the **Tasks** workspace. The **Scope** pane lists the data stores and data domains associated with the sensitive fields, which are all selected by default. The **Scope Preview** pane provides options for grouping the list of sensitive fields so that you can quickly view the current protection status of the data stores and data domains that will be included in the protection task, as well as the number of sensitive fields in each data store and data domain.

The following image shows this view after selecting a data domain that includes four data stores on the **Data Domains** page:



4. The **Data Domains** field lists the data domain or domains with sensitive fields that you selected to protect. Optionally, if you selected more than one data domain, deselect any data domains you do not want to include in the protection task.
5. Optionally, in the **Data Stores** field, deselect the check box next to any associated data stores for which you do not want to create a protection task.
6. Optionally, add a description for the task.
7. Select one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. To attach the data domain details to the protection task, select the **Add Context to Task Details as Attachment** check box.
9. To indicate that Data Privacy Management will notify the user assigned to configure the protection task once you create it, select the **Notify assignee via email** check box.
10. Optionally, select tags for the protection task and add any notes.
11. Optionally, select a task due date.
12. Click **Save as Task**.

    Data Privacy Management creates a separate protection task for each data store. The tasks appear on the **Tasks** list page with a status of New and the extension is Unconfigured. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful protection tasks in the Data Privacy Management repository.
    The user assigned to the tasks can now configure protection from the **Tasks** workspace.

Protecting Data from the Top Data Stores Grid Page

You can create protection tasks for data stores from the **Top Data Stores** grid page. Data Privacy Management creates a separate protection task for each data store.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.

2. Click **Grid** in the left column of the page.

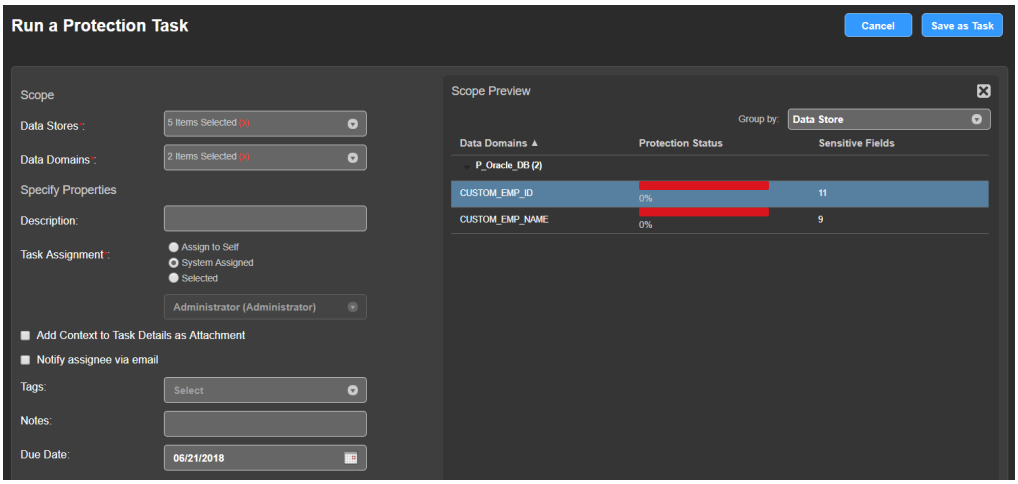
   The **Data Stores** page displays the list of data stores.

3. Select the check box next to one or more data stores.

4. From the **Actions** menu, select**Take Action** > **Protect Data**.

   Note: The **Protect Data** option might be disabled for one of the following reasons: you do not have privileges to create a protection task, a compatible protection extension does not exist or is not supported for the selected data stores, or the data store has not been scanned and has a value of Not Analyzed in the **Protection Status** column.

   The **Run a Protection Task** page displays on the **Tasks** workspace. The **Scope** pane lists the data stores and data domains associated with the sensitive fields, which are all selected by default. The **Scope Preview** pane provides options for grouping the list of sensitive fields so that you can quickly view the current protection status of the data stores and data domains that will be included in the protection task, as well as the number of sensitive fields in each data store and data domain.
   The following image shows this view after selecting five data stores that include two data domains on the **Top Data Stores** grid page:



5. Optionally, in the **Data Stores** field, deselect the check box next to any data stores for which you do not want to create a protection task.

6. Optionally, in the **Data Domains** field, deselect the check box next to any data stores for which you do not want to create a protection task.

7. Optionally, add a description for the task.

8. Select one of the following task assignments:

   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.

9. To attach the data store details to the protection task, select the **Add Context to Task Details as Attachment** check box.
10. To indicate that Data Privacy Management will notify the user assigned to configure the protection task once you create it, select the **Notify assignee via email** check box.
11. Optionally, select tags for the protection task and add any notes.
12. Optionally, select a task due date.
13. Click **Save as Task**.

    Data Privacy Management creates a separate protection task for each data store. The tasks appear on the **Tasks** list page with a status of New and the extension is Unconfigured. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful protection tasks in the Data Privacy Management repository.
    The user assigned to the tasks can now configure protection from the **Tasks** workspace.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Top Data Stores Indicator" on page 525

Running a Custom Script

You can run a custom script on demand when you identify an issue in Data Privacy Management.

Create a new action or select an existing custom action to use as a template for running the script. When you create a new action, specify the custom extension and all required properties. When you select an existing custom action, the extension and other properties such as placeholders appear automatically. Include a description of the action, which can contain placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.

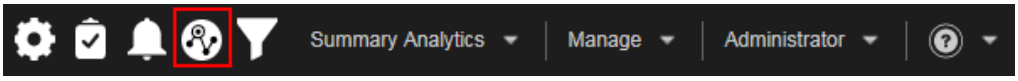
You can also edit the parameter and specify how Data Privacy Management will run and close the task. You can also add tags and other notes, and edit the due date.

Running a Custom Script from the Anomaly Detection Workspace

You can run a custom script from the **Anomaly Detection** workspace.

Important: Data Privacy Management can only perform actions or create tasks for anomalies that are associated with at least one data store. Verify that each anomaly lists at least one data store in the **Data Store** column before running a custom script for the anomaly.

1. Click the **Anomalies** icon in the Data Privacy Management header. The following image shows the **Anomalies** icon:



   The **Anomaly Detection** workspace appears.
2. In the **Anomalies** list page, select the check box next to one or more anomalies for which you want to run a custom script.

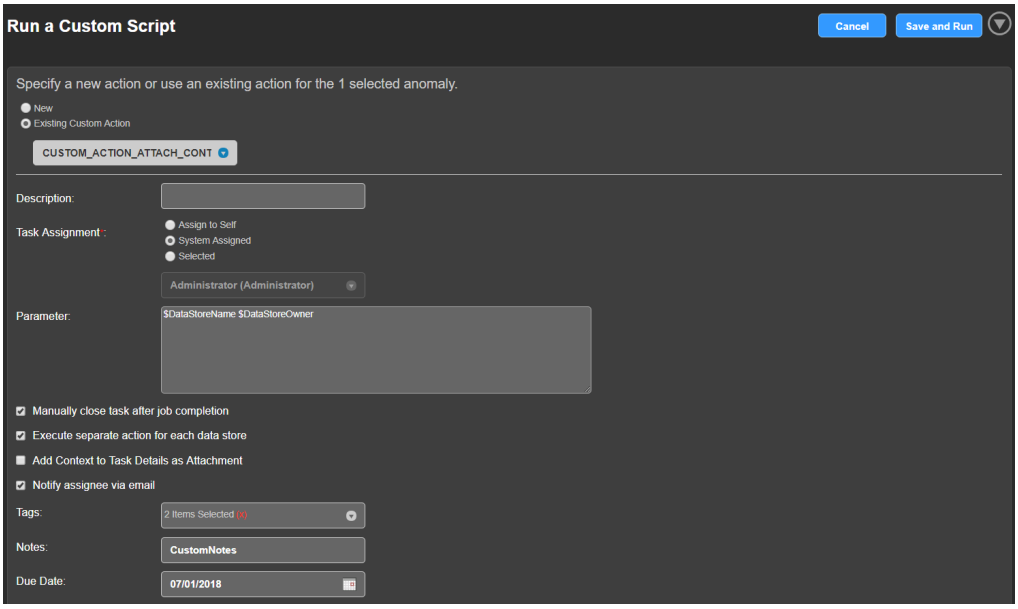
3. From the **Actions** menu, select **Take Action** > **Run a Custom Script**.

   Note: The **Run a Custom Script** option might be disabled if you do not have privileges to create a custom task or you do not have access to the data stores in the selected anomalies.

   The **Run a Custom Script** page displays on the **Tasks** workspace.

4. Select whether to create a new custom action or to use an existing custom action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the custom extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Run a Custom Script** page after selecting an existing custom action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Optionally, enter or edit a parameter for the action. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
9. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the script runs.
10. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the details of the anomaly to the tasks associated with the action.

11. To indicate that Data Privacy Management will notify the user assigned to the custom task once you create it, select the **Notify assignee via email** check box.
12. Optionally, select tags for the custom task and add any notes.
13. Optionally, select a task due date.
14. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to run the custom script. A message indicates the number of tasks the service saved successfully and the number of tasks that encountered an error. The Data Privacy Management Service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Anomaly Detection Workspace" on page 348

Running a Custom Script from the Proliferation Page

You can run a custom script from the **Proliferation** page.

1. On the **Overview** workspace, click a data store name from the **Top Data Stores** indicator.

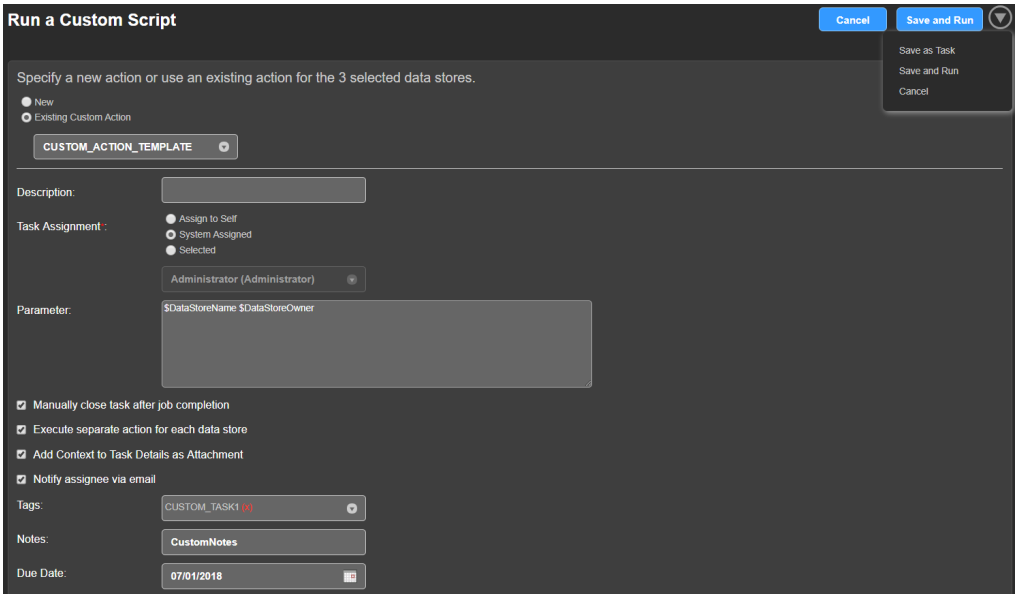
   The **Proliferation** page appears.
2. Select the check box next to one or more data stores on the **Proliferation** page.
3. From the **Actions** menu, select **Take Action** > **Run a Custom Script**.

   Note: The **Run a Custom Script** option might be disabled if you do not have privileges to create a custom task or you do not have access to the selected data stores.

   The **Run a Custom Script** page displays on the **Tasks** workspace.
4. Select whether to create a new custom action or to use an existing custom action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the custom extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Run a Custom Script** page after selecting three data stores and an existing custom action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Optionally, enter or edit a parameter for the action. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
9. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the script runs.
10. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
11. To indicate that Data Privacy Management will notify the user assigned to the custom task once you create it, select the **Notify assignee via email** check box.
12. Optionally, select tags for the custom task and add any notes.
13. Optionally, select a task due date.

14. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to run the custom script. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

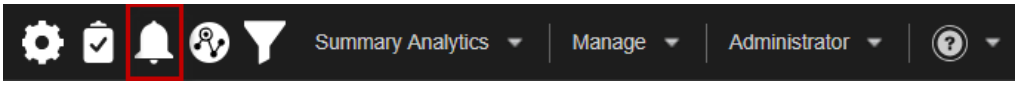
Related Topics:

- "Security Dashboard Overview" on page 513
- "Proliferation Page" on page 542
- "Top Data Stores Indicator" on page 525

Running a Custom Script from the Security Policy Violations Workspace

You can run a custom script for security policy violations on the **Security Policy Violations** workspace.

1. Click the **Security Policy Violations** icon in the Data Privacy Management header. The following image shows the **Security Policy Violations** icon:

    The **Security Policy Violations** workspace appears.

2. In the **Violations** list panel, select the check box next to one or more security policy violations for which you want to run a custom script.

    Important: Data Privacy Management can only perform actions or create tasks for violations that are associated with at least one data store. If you select an anomaly violation, verify that the violation is associated with at least one data store before running a custom script.

3. From the **Actions** menu, select **Take Action** > **Run a Custom Script**.

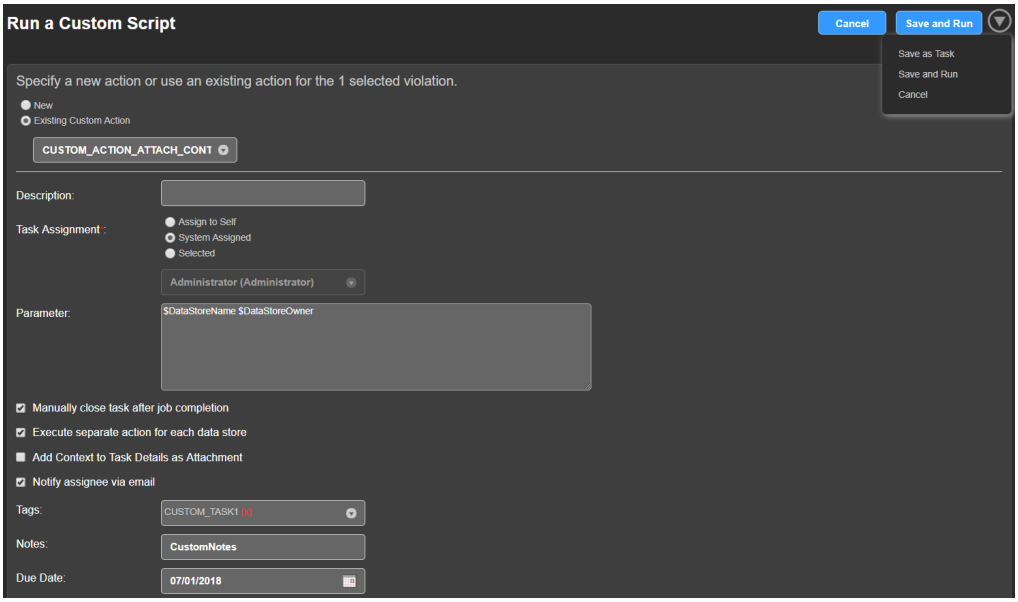
    Note: The **Run a Custom Script** option might be disabled if you do not have privileges to create a custom task or you do not have access to the data stores in the selected violations.

    The **Run a Custom Script** page displays on the **Tasks** workspace.

4. Select whether to create a new custom action or to use an existing custom action that is listed on the **Actions** workspace.

    If you select a new action, you first choose the custom extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Run a Custom Script** page after selecting a data store violation and an existing custom action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Optionally, enter or edit a parameter for the action. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
9. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the script runs.
10. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the violation details to the tasks associated with the action.
11. To indicate that Data Privacy Management will notify the user assigned to the custom task once you create it, select the **Notify assignee via email** check box.
12. Optionally, select tags for the custom task and add any notes.
13. Optionally, select a task due date.

14. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to run the custom script. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Policy Violations Workspace" on page 465

Running a Custom Script from the Sensitive Fields Page

You can run a custom script from the **Sensitive Fields** page.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.
2. Click **Grid** in the left column of the page.

   The **Data Stores** page displays the list of data stores.
3. Click a value in the **Sensitive Fields** column.

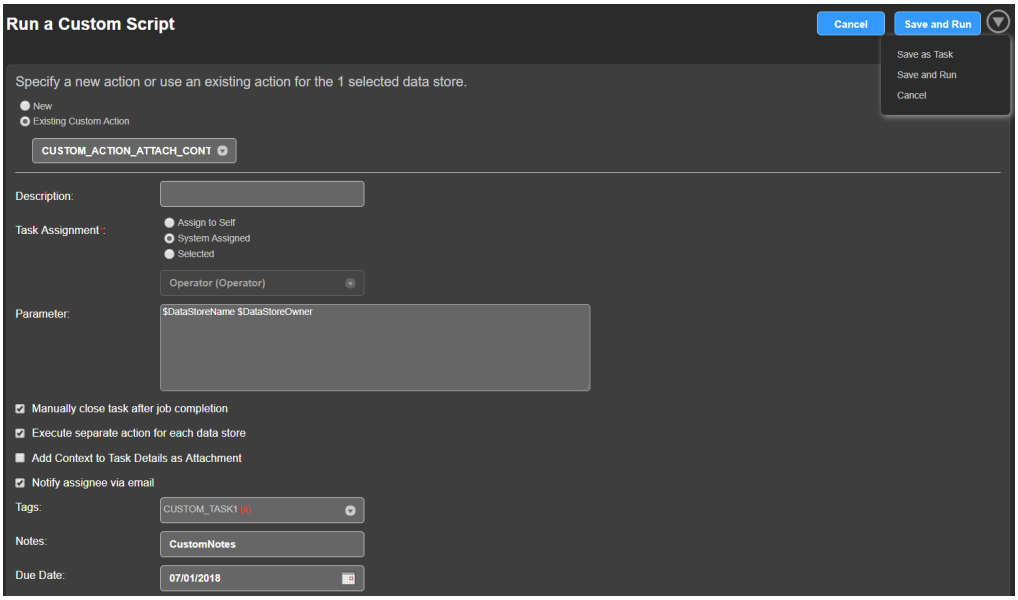
   The **Sensitive Fields** page appears.
4. From the **Actions** menu, select **Take Action** > **Run a Custom Script**.

   Note: The **Run a Custom Script** option might be disabled if you do not have privileges to create a custom task or you do not have access to the data store that contains the sensitive fields.

   The **Run a Custom Script** page displays on the **Tasks** workspace.
5. Select whether to create a new custom action or to use an existing custom action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the custom extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Run a Custom Script** page after selecting a data store that has 35 sensitive fields and a risk score of 1:



6. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
7. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. Optionally, enter or edit a parameter for the action. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
9. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
10. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the script runs.
11. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the sensitive field details to the tasks associated with the action.
12. To indicate that Data Privacy Management will notify the user assigned to the custom task once you create it, select the **Notify assignee via email** check box.
13. Optionally, select tags for the custom task and add any notes.
14. Optionally, select a task due date.

15. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to run the custom script. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Sensitive Fields Page" on page 549
- "Top Data Stores Indicator" on page 525

Running a Custom Script from the Top Data Domains List Page

You can run a custom script from the **Top Data Domains** list page.

1. On the **Overview** workspace, click the **Top Data Domains** indicator label.

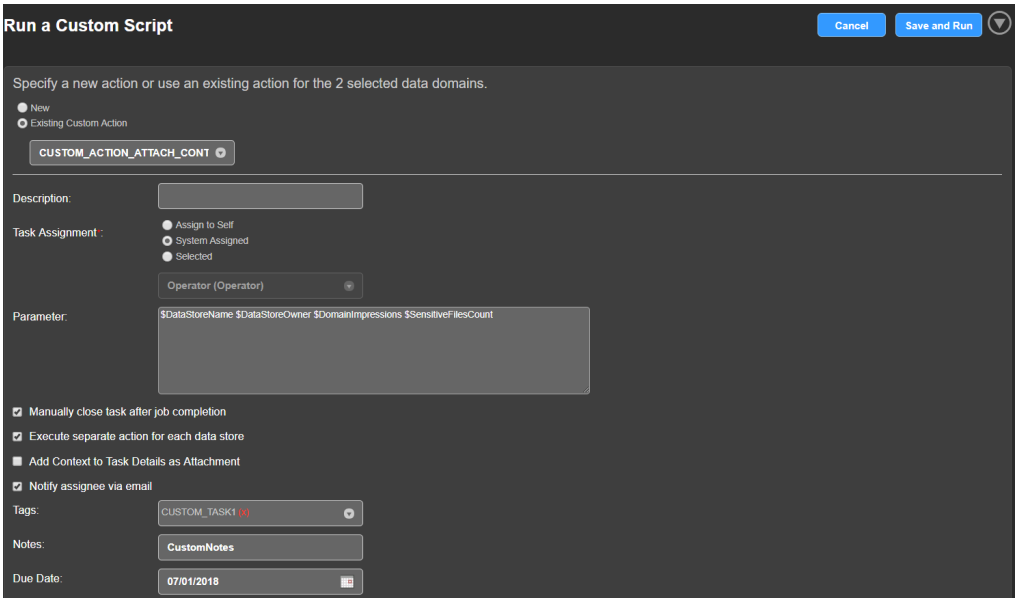
   The **Data Domains** list page appears.
2. Select the check box next to one or more data domains.
3. From the **Actions** menu, select **Take Action** > **Run a Custom Script**.

   Note: The **Run a Custom Script** option might be disabled if you do not have privileges to create a custom task or you do not have access to the data stores included in the selected data domains.

   The **Run a Custom Script** page displays on the **Tasks** workspace.
4. Select whether to create a new custom action or to use an existing custom action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the custom extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Run a Custom Script** page after selecting two data domains that include a total of three data stores:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Optionally, enter or edit a parameter for the action. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
9. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the script runs.
10. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data domain details to the tasks associated with the action.
11. To indicate that Data Privacy Management will notify the user assigned to the custom task once you create it, select the **Notify assignee via email** check box.
12. Optionally, select tags for the custom task and add any notes.
13. Optionally, select a task due date.

14. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to run the custom script. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Running a Custom Script from the Top Data Stores Grid Page

You can run a custom script from the **Top Data Stores** grid page.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.

2. Click **Grid** in the left column of the page.

   The **Data Stores** page displays the list of data stores.

3. Select the check box next to one or more data stores.

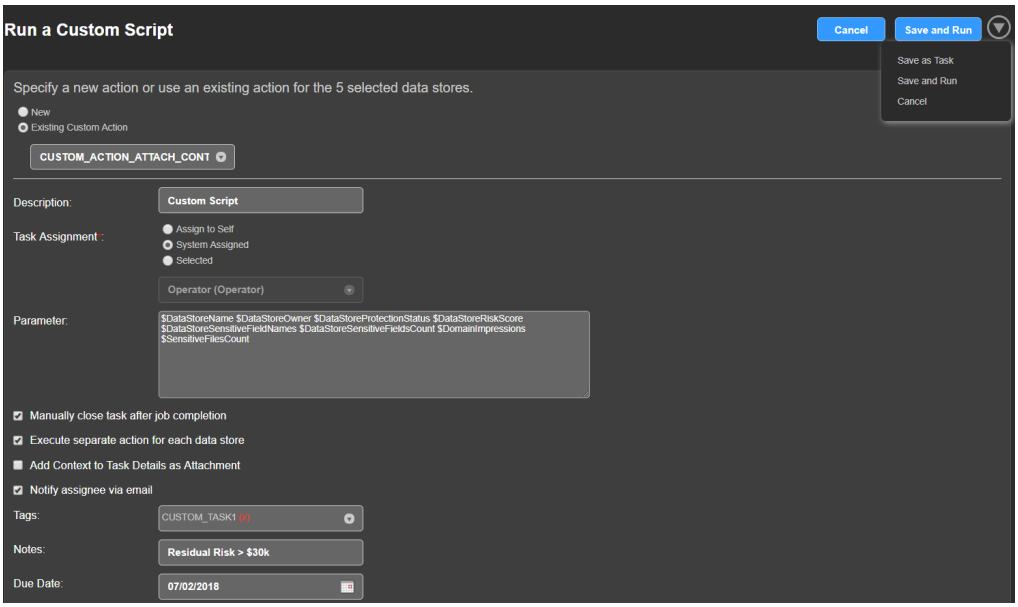
4. From the **Actions** menu, select **Take Action** > **Run a Custom Script**.

   Note: The **Run a Custom Script** option might be disabled if you do not have privileges to create a custom task or you do not have access to the selected data stores.

   The **Run a Custom Script** page displays on the **Tasks** workspace.

5. Select whether to create a new custom action or to use an existing custom action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the custom extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Run a Custom Script** page after selecting five data stores that each have a residual risk value greater than $30k:



6. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
7. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. Optionally, enter or edit a parameter for the action. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
9. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
10. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when the script runs.
11. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
12. To indicate that Data Privacy Management will notify the user assigned to the custom task once you create it, select the **Notify assignee via email** check box.
13. Optionally, select tags for the custom task and add any notes.
14. Optionally, select a task due date.
15. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to run the custom script. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Top Data Stores Indicator" on page 525

Sending an Email

You can create an email task on demand when you identify an issue in Data Privacy Management that requires a notification email to be sent.

Create a new action or select an existing email action to use as a template for creating the ticket. When you create a new action, specify the email extension and all required properties. When you select an existing email action, the extension and other properties such as placeholders appear automatically. Include a description of the action, which can contain placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.

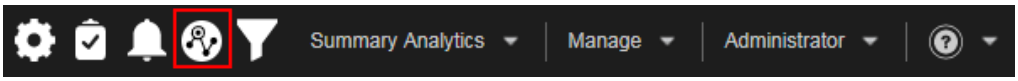
Optionally, edit the sender and recipients of the email. You can also add attachments to the email, specify how Data Privacy Management will run and close the task, add tags and other notes, and edit the due date.

Sending an Email from the Anomaly Detection Workspace

You can send an email from the **Anomaly Detection** workspace.

Important: Data Privacy Management can only perform actions or create tasks for anomalies that are associated with at least one data store. Verify that each anomaly lists at least one data store in the **Data Store** column before sending an email for the anomaly.

1. Click the **Anomalies** icon in the Data Privacy Management header. The following image shows the **Anomalies** icon:

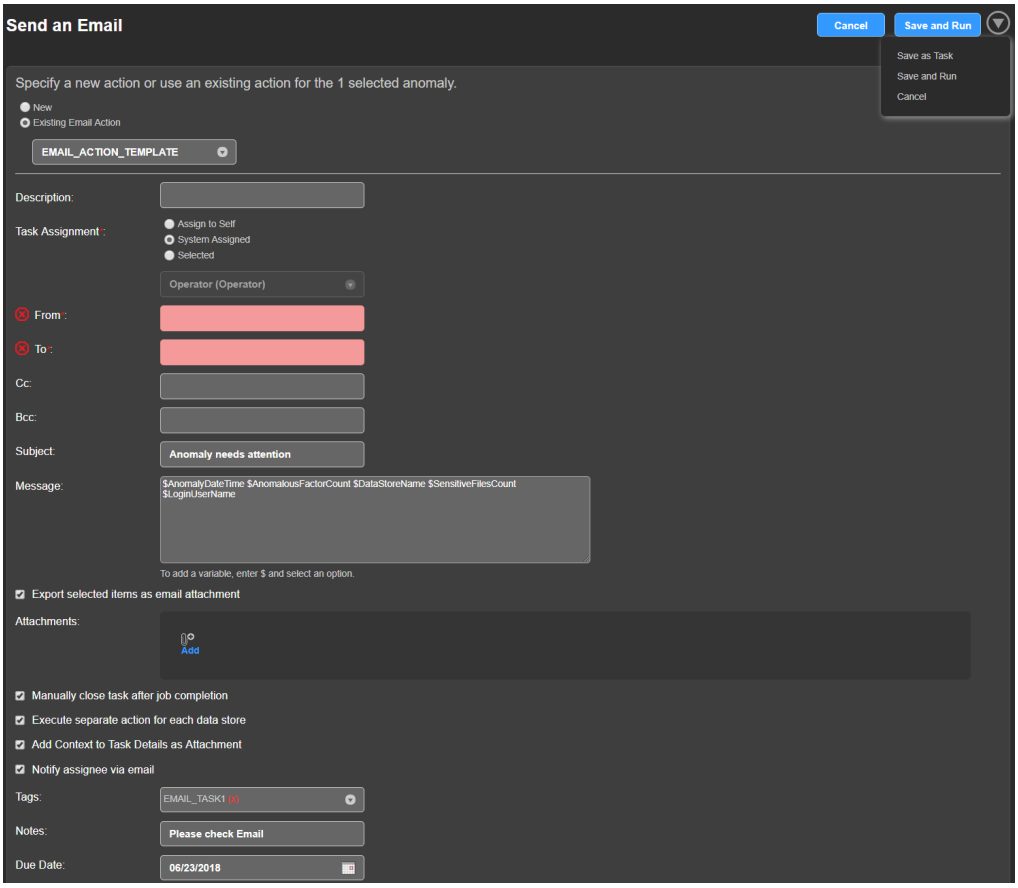
   The **Anomaly Detection** workspace appears.
2. In the **Anomalies** list page, select the check box next to one or more anomalies for which you want to send an email.
3. From the **Actions** menu, select **Take Action** > **Send an Email**.

   Note: The **Send an Email** option might be disabled if you do not have privileges to send an email or you do not have access to the data stores in the selected anomalies.

   The **Send an Email** page displays on the **Tasks** workspace.
4. Select whether to create a new email action or to use an existing email action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the email extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Send an Email** page after selecting an existing email action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. In the **From** field, enter or edit the sender email address.
8. In the **To** field, enter or edit the recipient email address. You can enter multiple addresses separated by a comma.
9. Optionally, enter email addresses for recipients to be copied on the email in the **Cc** and **Bcc** fields.
10. In the **Subject** field, enter or edit a subject for the email.
11. Enter or edit the email message contents. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
12. Optionally, select the **Export selected items as email attachment** check box to attach a CSV file to the email that contains the anomaly details.
13. Optionally, add attachments to the email.
14. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is

successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.

15. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management sends the email.
16. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the details of the anomaly to the tasks associated with the action.
17. To indicate that Data Privacy Management will notify the user assigned to the email task once you create it, select the **Notify assignee via email** check box.
18. Optionally, select tags for the email task and add any notes.
19. Optionally, select a task due date.
20. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to send the email. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository.
    The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Anomaly Detection Workspace" on page 348

Sending an Email from the Proliferation Page

You can send an email from the **Proliferation** page.

1. On the **Overview** workspace, click a data store name from the **Top Data Stores** indicator.

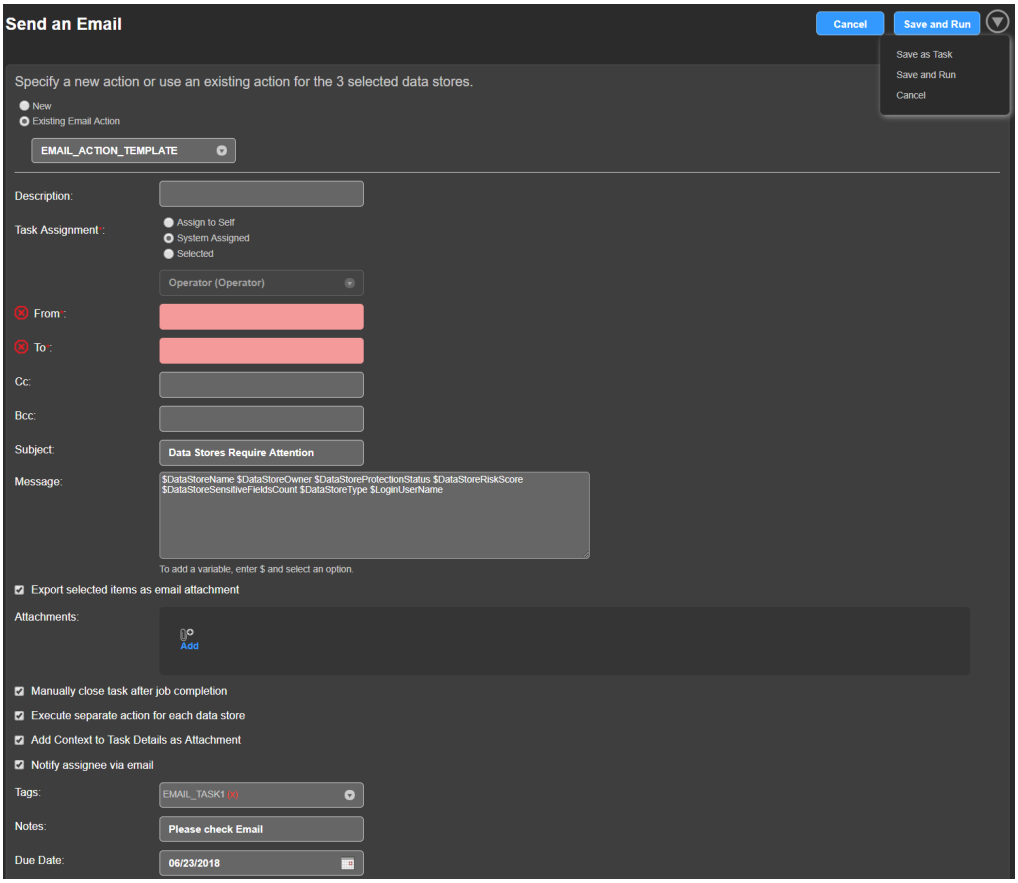
   The **Proliferation** page appears.
2. Select the check box next to one or more data stores on the **Proliferation** page.
3. From the **Actions** menu, select **Take Action** > **Send an Email**.

   Note: The **Send an Email** option might be disabled if you do not have privileges to create an email task or you do not have access to the selected data stores.

   The **Send an Email** page displays on the **Tasks** workspace.
4. Select whether to create a new email action or to use an existing email action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the email extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Send an Email** page after selecting three data stores and an existing email action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. In the **From** field, enter or edit the sender email address.
8. In the **To** field, enter or edit the recipient email address. You can enter multiple addresses separated by a comma.
9. Optionally, enter email addresses for recipients to be copied on the email in the **Cc** and **Bcc** fields.
10. In the **Subject** field, enter or edit a subject for the email.
11. Enter or edit the email message contents. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
12. Optionally, select the **Export selected items as email attachment** check box to attach a CSV file to the email that contains the data store details.
13. Optionally, add attachments to the email.

14. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
15. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management sends the email.
16. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
17. To indicate that Data Privacy Management will notify the user assigned to the email task once you create it, select the **Notify assignee via email** check box.
18. Optionally, select tags for the email task and add any notes.
19. Optionally, select a task due date.
20. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to send the email. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository.
    The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

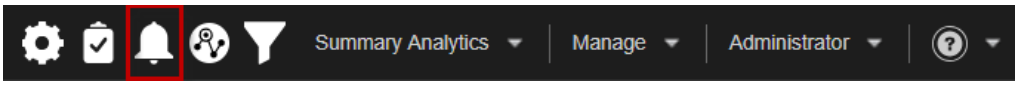
Related Topics:

- "Security Dashboard Overview" on page 513
- "Proliferation Page" on page 542
- "Top Data Stores Indicator" on page 525

Sending an Email from the Security Policy Violations Workspace

You can send an email regarding security policy violations from the **Security Policy Violations** workspace.

1. Click the **Security Policy Violations** icon in the Data Privacy Management header. The following image shows the **Security Policy Violations** icon:



   The **Security Policy Violations** workspace appears.
2. In the **Violations** list panel, select the check box next to one or more security policy violations for which you want to send an email.

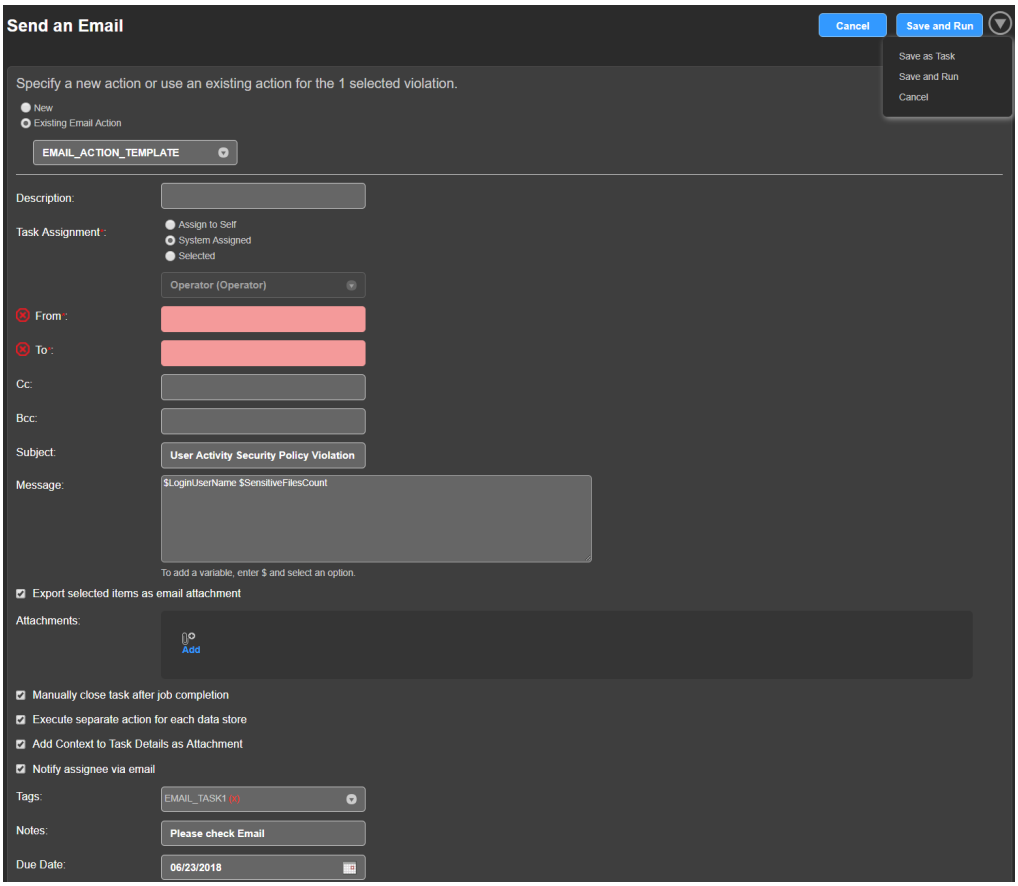
   Important: Data Privacy Management can only perform actions or create tasks for violations that are associated with at least one data store. If you select an anomaly violation, verify that the violation is associated with at least one data store before sending an email.
3. From the **Actions** menu, select **Take Action** > **Send an Email**.

   Note: The **Send an Email** option might be disabled if you do not have privileges to create an email task or you do not have access to the selected data stores.

   The **Send an Email** page displays on the **Tasks** workspace.

4. Select whether to create a new email action or to use an existing email action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the email extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Send an Email** page after selecting a user activity security policy violation:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. In the **From** field, enter or edit the sender email address.
8. In the **To** field, enter or edit the recipient email address. You can enter multiple addresses separated by a comma.
9. Optionally, enter email addresses for recipients to be copied on the email in the **Cc** and **Bcc** fields.
10. In the **Subject** field, enter or edit a subject for the email.
11. Enter or edit the email message contents. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.

12. Optionally, select the **Export selected items as email attachment** check box to attach a CSV file to the email that contains the violation details.
13. Optionally, add attachments to the email.
14. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
15. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management sends the email.
16. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
17. To indicate that Data Privacy Management will notify the user assigned to the email task once you create it, select the **Notify assignee via email** check box.
18. Optionally, select tags for the email task and add any notes.
19. Optionally, select a task due date.
20. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to send the email. A message indicates the number of tasks theData Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository.
    The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Policy Violations Workspace" on page 465

Sending an Email from the Sensitive Fields Page

You can send an email regarding a data store with unprotected sensitive fields from the **Sensitive Fields** page.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.
2. Click **Grid** in the left column of the page.

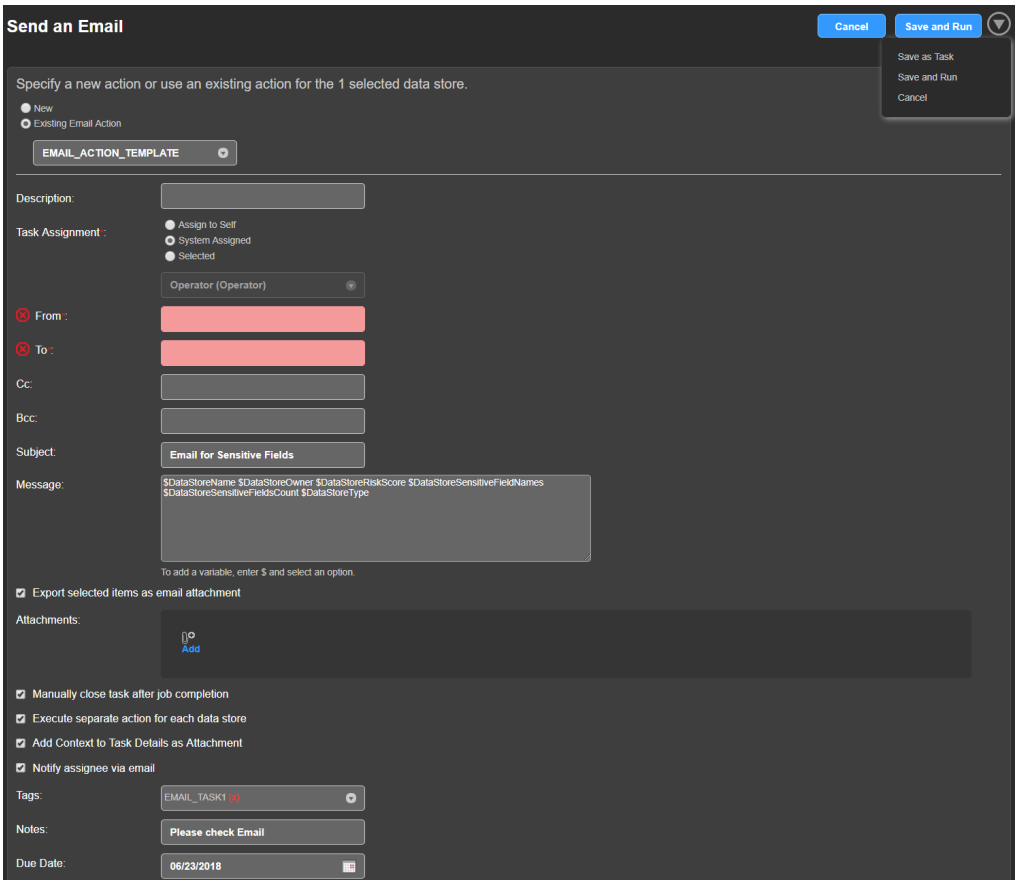
   The **Data Stores** page displays the list of data stores.
3. Click a value in the **Sensitive Fields** column.

   The **Sensitive Fields** page appears.
4. From the **Actions** menu, select **Take Action** > **Send an Email**.

   Note: The **Send an Email** option might be disabled if you do not have privileges to create an email task or you do not have access to the data store that contains the sensitive fields.

   The **Send an Email** page displays on the **Tasks** workspace.
5. Select whether to create a new email action or to use an existing email action that is listed on the **Actions** workspace.

If you select a new action, you first choose the email extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
The following image shows the **Send an Email** page after selecting a data store that has 35 sensitive fields, 33 of which are unprotected, and a risk score of 1:



6. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
7. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. In the **From** field, enter or edit the sender email address.
9. In the **To** field, enter or edit the recipient email address. You can enter multiple addresses separated by a comma.
10. Optionally, enter email addresses for recipients to be copied on the email in the **Cc** and **Bcc** fields.
11. In the **Subject** field, enter or edit a subject for the email.
12. Enter or edit the email message contents. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
13. Optionally, select the **Export selected items as email attachment** check box to attach a CSV file to the email that contains the sensitive field details.

14. Optionally, add attachments to the email.
15. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
16. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management sends the email.
17. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the sensitive field details to the tasks associated with the action.
18. To indicate that Data Privacy Management will notify the user assigned to the email task once you create it, select the **Notify assignee via email** check box.
19. Optionally, select tags for the email task and add any notes.
20. Optionally, select a task due date.
21. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to send the email. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository.
    The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Sensitive Fields Page" on page 549
- "Top Data Stores Indicator" on page 525

Sending an Email from the Top Data Domains List Page

You can send an email from the **Top Data Domains** list page.

1. On the **Overview** workspace, click the **Top Data Domains** indicator label.

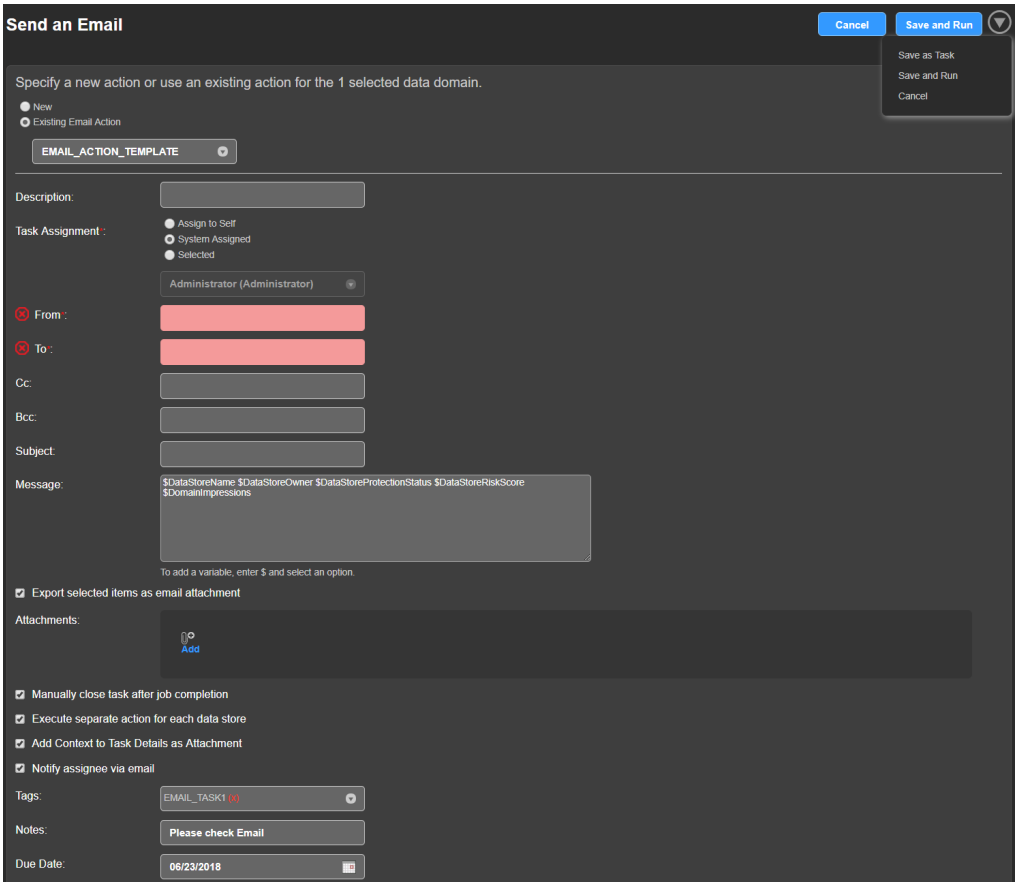
   The **Data Domains** list page appears.
2. Select the check box next to one or more data domains.
3. From the **Actions** menu, select **Take Action** > **Send an Email**.

   Note: The **Send an Email** option might be disabled if you do not have privileges to create an email task or you do not have access to the data stores included in the selected data domains.

   The **Send an Email** page displays on the **Tasks** workspace.
4. Select whether to create a new email action or to use an existing email action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the email extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Send an Email** page after selecting a data domain that includes three data stores and an existing email action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. In the **From** field, enter or edit the sender email address.
8. In the **To** field, enter or edit the recipient email address. You can enter multiple addresses separated by a comma.
9. Optionally, enter email addresses for recipients to be copied on the email in the **Cc** and **Bcc** fields.
10. In the **Subject** field, enter or edit a subject for the email.
11. Enter or edit the email message contents. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
12. Optionally, select the **Export selected items as email attachment** check box to attach a CSV file to the email that contains the data domain details.
13. Optionally, add attachments to the email.

14. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
15. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management sends the email.
16. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data domain details to the tasks associated with the action.
17. To indicate that Data Privacy Management will notify the user assigned to the email task once you create it, select the **Notify assignee via email** check box.
18. Optionally, select tags for the email task and add any notes.
19. Optionally, select a task due date.
20. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to send the email. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository.
    The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Sending an Email from the Top Data Stores Grid Page

You can send an email from the **Top Data Stores** grid page.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.
2. Click **Grid** in the left column of the page.

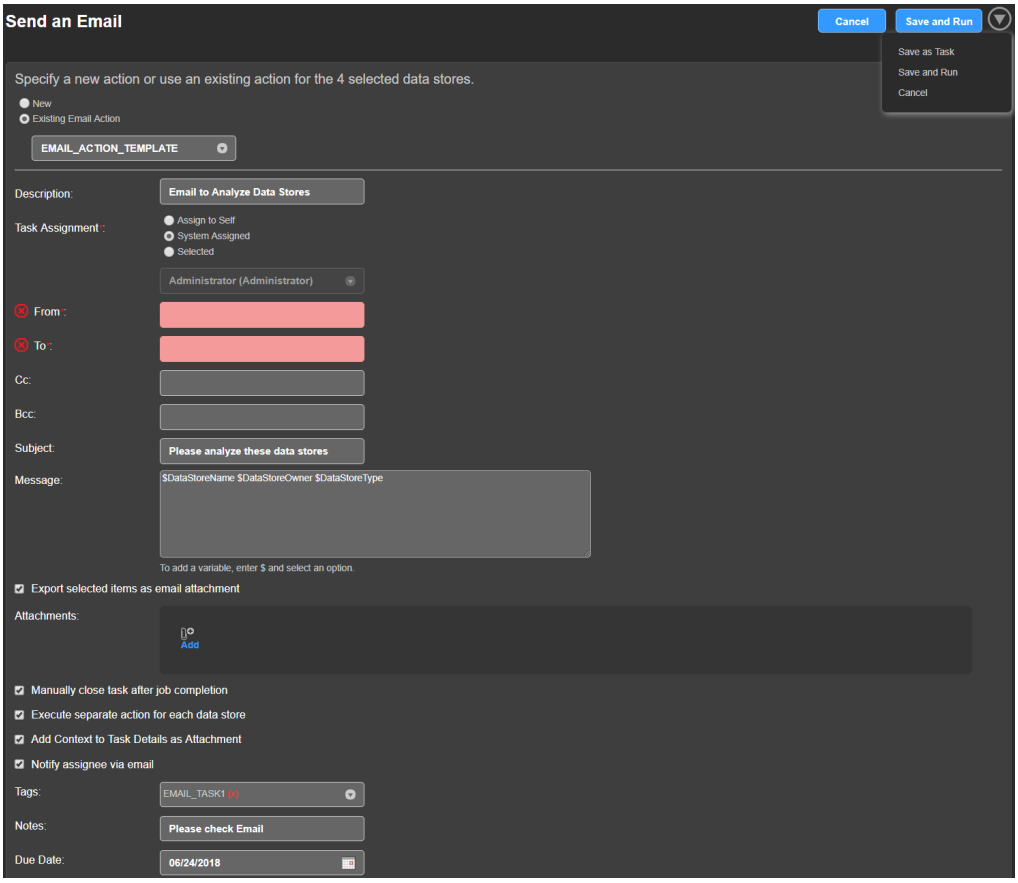
   The **Data Stores** page displays the list of data stores.
3. Select the check box next to one or more data stores.
4. From the **Actions** menu, select **Take Action** > **Send an Email**.

   Note: The **Send an Email** option might be disabled if you do not have privileges to create an email task or you do not have access to the selected data stores.

   The **Send an Email** page displays on the **Tasks** workspace.
5. Select whether to create a new email action or to use an existing email action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the email extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.

The following image shows the **Send an Email** page after selecting four data stores that have not been scanned:



6. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
7. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. In the **From** field, enter or edit the sender email address.
9. In the **To** field, enter or edit the recipient email address. You can enter multiple addresses separated by a comma.
10. Optionally, enter email addresses for recipients to be copied on the email in the **Cc** and **Bcc** fields.
11. In the **Subject** field, enter or edit a subject for the email.
12. Enter or edit the email message contents. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
13. Optionally, select the **Export selected items as email attachment** check box to attach a CSV file to the email that contains the data store details.
14. Optionally, add attachments to the email.

15. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
16. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management sends the email.
17. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
18. To indicate that Data Privacy Management will notify the user assigned to the email task once you create it, select the **Notify assignee via email** check box.
19. Optionally, select tags for the email task and add any notes.
20. Optionally, select a task due date.
21. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to send the email. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository.
    The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Top Data Stores Indicator" on page 525

Writing a System Log Message

You can create a task on demand for Data Privacy Management to write a system log message to a remote server on demand.

Create a new action or select an existing system log action to use as a template for writing the message. When you create a new action, specify the system log extension and all required properties. When you select an existing system log action, the extension and other properties such as placeholders appear automatically. Include a description of the action, which can contain placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.

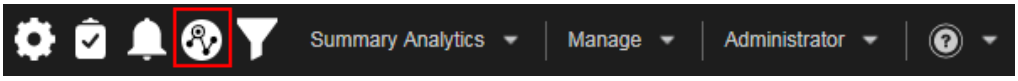
Optionally, edit the log level and specify how Data Privacy Management will run and close the task. You can also add tags and other notes, and edit the due date.

Writing a System Log Message from the Anomaly Detection Workspace

You can write a system log message to a remote system log server from the **Anomaly Detection** workspace.

Important: Data Privacy Management can only perform actions or create tasks for anomalies that are associated with at least one data store. Verify that each anomaly lists at least one data store in the **Data Store** column before writing a system log message for the anomaly.

1. Click the **Anomalies** icon in the Data Privacy Management header. The following image shows the **Anomalies** icon:



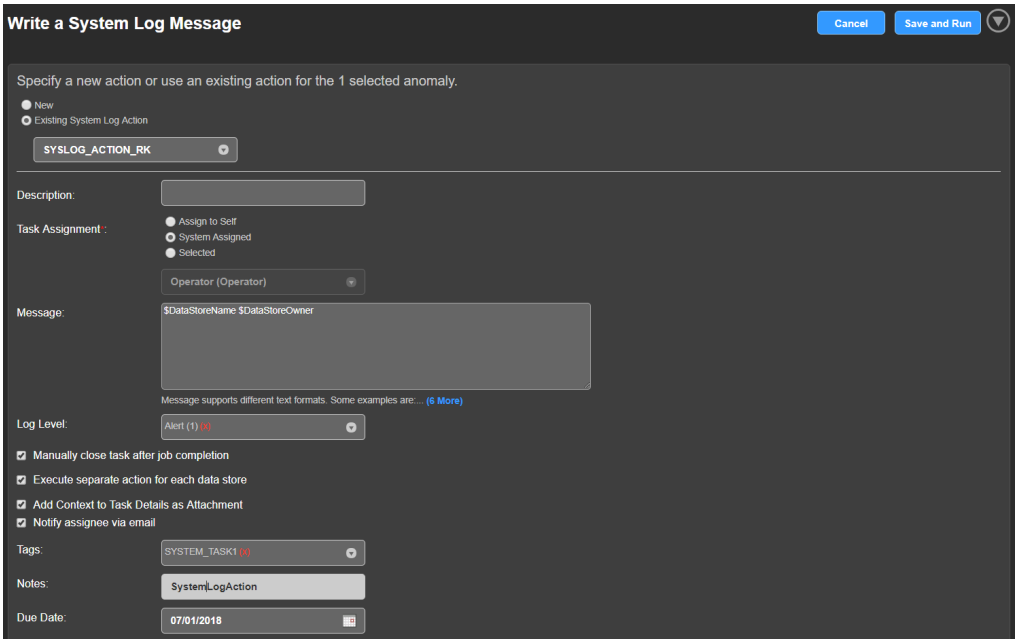
   The **Anomaly Detection** workspace appears.

2. In the **Anomalies** list page, select the check box next to one or more anomalies for which you want to write a system log message.
3. From the **Actions** menu, select **Take Action** > **Write a System Log Message**.

   Note: The **Write a System Log Message** option might be disabled if you do not have privileges to create a system log task or you do not have access to the data stores in the selected anomalies.

   The **Write a System Log Message** page displays on the **Tasks** workspace.
4. Select whether to create a new system log action or to use an existing system log action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the system log extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Write a System Log Message** page after selecting an existing system log action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.

6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Optionally, enter or edit the contents of the system log message that Data Privacy Management will send to the remote system log server configured in the extension. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. Optionally, select one of the following log levels for the message:
   - **Alert (1)**
   - **Critical (2)**
   - **Debug (7)**
   - **Emergency (0)**
   - **Error (3)**
   - **Informational (6)**
   - **Notice (5)**
   - **Warning (4)**
9. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
10. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management writes the log message.
11. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the details of the anomaly to the tasks associated with the action.
12. To indicate that Data Privacy Management will notify the user assigned to the system log task once you create it, select the **Notify assignee via email** check box.
13. Optionally, select tags for the system log task and add any notes.
14. Optionally, select a task due date.
15. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to write the system log message. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Anomaly Detection Workspace" on page 348

Writing a System Log Message from the Proliferation Page

You can write a system log message from the **Proliferation** page.

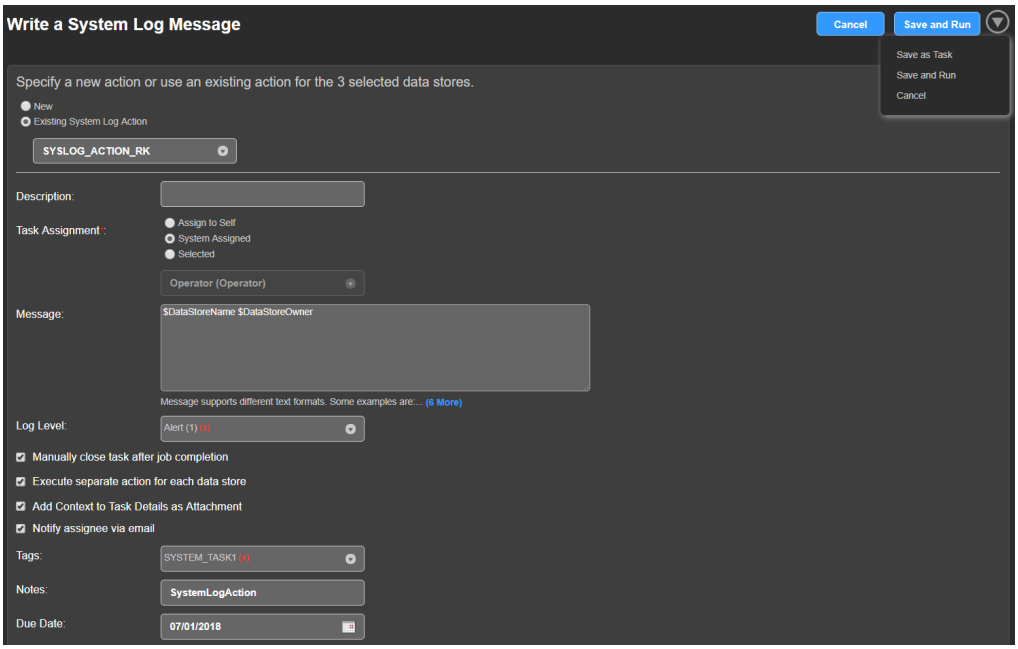
1. On the **Overview** workspace, click a data store name from the **Top Data Stores** indicator.

   The **Proliferation** page appears.
2. Select the check box next to one or more data stores on the **Proliferation** page.
3. From the **Actions** menu, select **Take Action** > **Write a System Log Message**.

   Note: The **Write a System Log Message** option might be disabled if you do not have privileges to create a system log task or you do not have access to the selected data stores.

   The **Write a System Log Message** page displays on the **Tasks** workspace.
4. Select whether to create a new system log action or to use an existing system log action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the system log extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Write a System Log Message** page after selecting three data stores and an existing system log action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.

- **Selected**. Select a user from the list.

7. Optionally, enter or edit the contents of the system log message that Data Privacy Management will send to the remote system log server configured in the extension. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. Optionally, select one of the following log levels for the message:
   - **Alert (1)**
   - **Critical (2)**
   - **Debug (7)**
   - **Emergency (0)**
   - **Error (3)**
   - **Informational (6)**
   - **Notice (5)**
   - **Warning (4)**
9. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
10. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management writes the log message.
11. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
12. To indicate that Data Privacy Management will notify the user assigned to the system log task once you create it, select the **Notify assignee via email** check box.
13. Optionally, select tags for the system log task and add any notes.
14. Optionally, select a task due date.
15. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to write the system log message. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

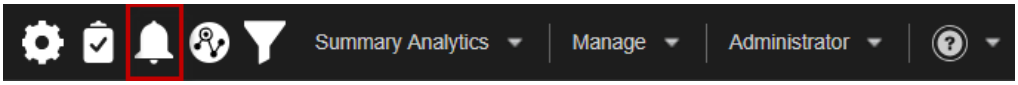
Related Topics:

- "Security Dashboard Overview" on page 513
- "Proliferation Page" on page 542
- "Top Data Stores Indicator" on page 525

Writing a System Log Message from the Security Policy Violations Workspace

You can write a system log message to a remote server from the **Security Policy Violations** workspace.

1. Click the **Security Policy Violations** icon in the Data Privacy Management header. The following image shows the **Security Policy Violations** icon:



   The **Security Policy Violations** workspace appears.

2. In the **Violations** list panel, select the check box next to one or more security policy violations for which you want to write a system log message.

   Important: Data Privacy Management can only perform actions or create tasks for violations that are associated with at least one data store. If you select an anomaly violation, verify that the violation is associated with at least one data store before writing a system log message.

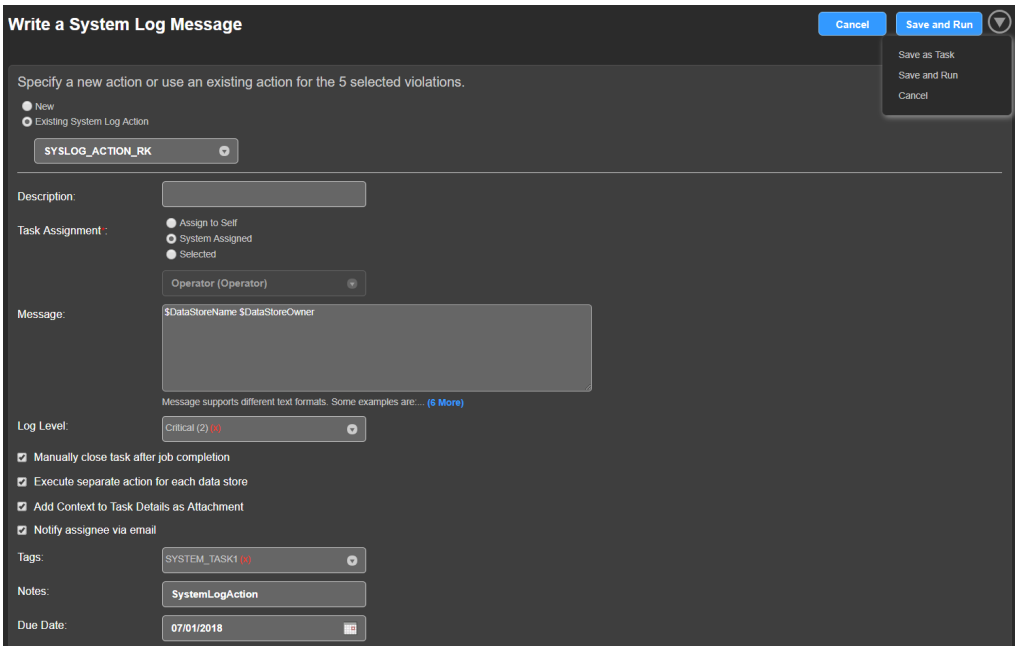
3. From the **Actions** menu, select **Take Action** > **Write a System Log Message**.

   Note: The **Write a System Log Message** option might be disabled if you do not have privileges to create a system log task or you do not have access to the selected data stores.

   The **Write a System Log Message** page displays on the **Tasks** workspace.

4. Select whether to create a new system log action or to use an existing system log action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the system log extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Write a System Log Message** page after selecting five security policy violations and an existing system log action:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.

6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Optionally, enter or edit the contents of the system log message that Data Privacy Management will send to the remote system log server configured in the extension. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
8. Optionally, select one of the following log levels for the message:
   - **Alert (1)**
   - **Critical (2)**
   - **Debug (7)**
   - **Emergency (0)**
   - **Error (3)**
   - **Informational (6)**
   - **Notice (5)**
   - **Warning (4)**
9. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
10. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management writes the log message.
11. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
12. To indicate that Data Privacy Management will notify the user assigned to the system log task once you create it, select the **Notify assignee via email** check box.
13. Optionally, select tags for the system log task and add any notes.
14. Optionally, select a task due date.
15. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to write the system log message. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Policy Violations Workspace" on page 465

Writing a System Log Message from the Sensitive Fields Page

You can write a system log message to a remote server from the **Sensitive Fields** page.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.
2. Click **Grid** in the left column of the page.

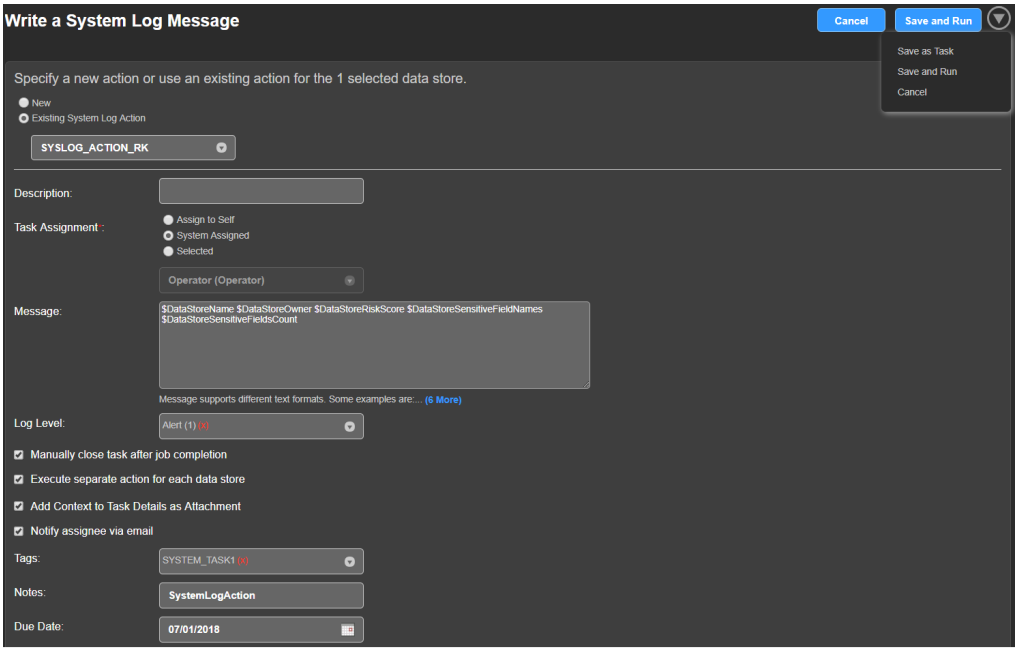
   The **Data Stores** page displays the list of data stores.
3. Click a value in the **Sensitive Fields** column.

   The **Sensitive Fields** page appears.
4. From the **Actions** menu, select **Take Action** > **Write a System Log Message**.

   Note: The **Write a System Log Message** option might be disabled if you do not have privileges to create a system log task or you do not have access to the selected data stores.

   The **Write a System Log Message** page displays on the **Tasks** workspace.
5. Select whether to create a new system log action or to use an existing system log action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the system log extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Write a System Log Message** page after selecting a data store that has 32 sensitive fields, 30 of which are unprotected, and a risk score of 1:



6. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
7. Select or change one of the following task assignments:

- **Assign to Self**. Assign the task to yourself.
- **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
- **Selected**. Select a user from the list.

8. Optionally, enter or edit the contents of the system log message that Data Privacy Management will send to the remote system log server configured in the extension. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
9. Optionally, select one of the following log levels for the message:
   - **Alert (1)**
   - **Critical (2)**
   - **Debug (7)**
   - **Emergency (0)**
   - **Error (3)**
   - **Informational (6)**
   - **Notice (5)**
   - **Warning (4)**
10. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
11. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Managementwill create separate tasks for each data store that is included in the action when Data Privacy Management writes the log message.
12. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
13. To indicate that Data Privacy Management will notify the user assigned to the system log task once you create it, select the **Notify assignee via email** check box.
14. Optionally, select tags for the system log task and add any notes.
15. Optionally, select a task due date.
16. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to write the system log message. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Sensitive Fields Page" on page 549
- "Top Data Stores Indicator" on page 525

Writing a System Log Message from the Top Data Domains List Page

You can write a system log message to a remote server from the **Top Data Domains** list page.

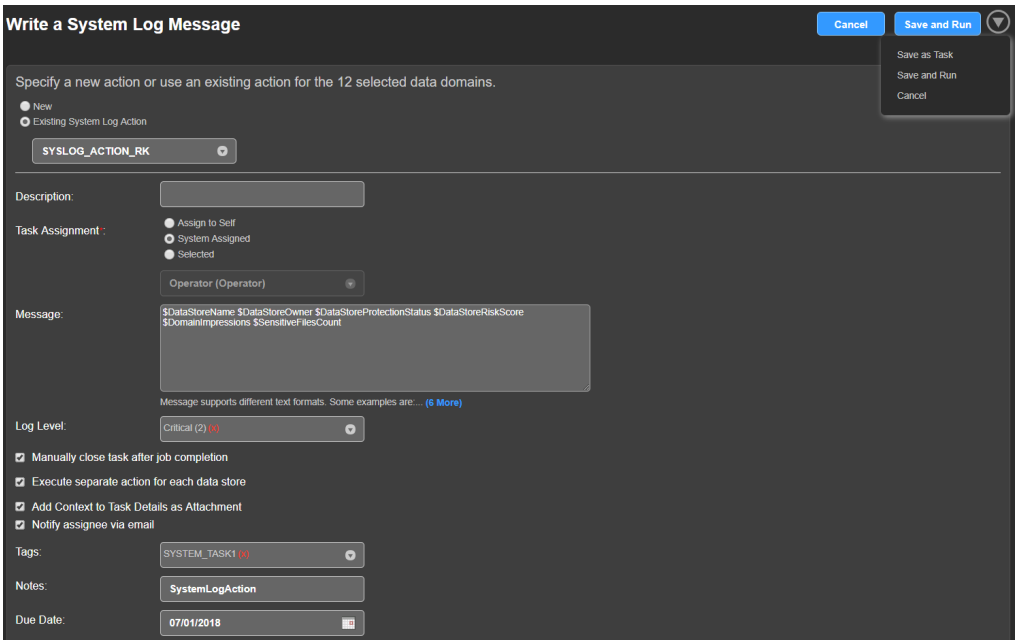
1. On the **Overview** workspace, click the **Top Data Domains** indicator label.

   The **Data Domains** list page appears.
2. Select the check box next to one or more data domains.
3. From the **Actions** menu, select **Take Action** > **Write a System Log Message**.

   Note: The **Write a System Log Message** option might be disabled if you do not have privileges to create a system log task or you do not have access to the data stores in the selected data domains.

   The **Write a System Log Message** page displays on the **Tasks** workspace.
4. Select whether to create a new system log action or to use an existing system log action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the system log extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Write a System Log Message** page after selecting 12 data domains that include 26 data stores:



5. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
6. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
7. Optionally, enter or edit the contents of the system log message that Data Privacy Management will send to the remote system log server configured in the extension. You can add placeholders. For more

information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.

8. Optionally, select one of the following log levels for the message:
   - **Alert (1)**
   - **Critical (2)**
   - **Debug (7)**
   - **Emergency (0)**
   - **Error (3)**
   - **Informational (6)**
   - **Notice (5)**
   - **Warning (4)**
9. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
10. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management writes the log message.
11. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data domain details to the tasks associated with the action.
12. To indicate that Data Privacy Management will notify the user assigned to the system log task once you create it, select the **Notify assignee via email** check box.
13. Optionally, select tags for the system log task and add any notes.
14. Optionally, select a task due date.
15. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to write the system log message. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Writing a System Log Message from the Top Data Stores Grid Page

You can write a system log message to a remote server from the **Top Data Stores** grid page.

1. On the Overview workspace, click the **Top Data Stores** indicator label.

   The **Data Stores** page appears.
2. Click **Grid** in the left column of the page.

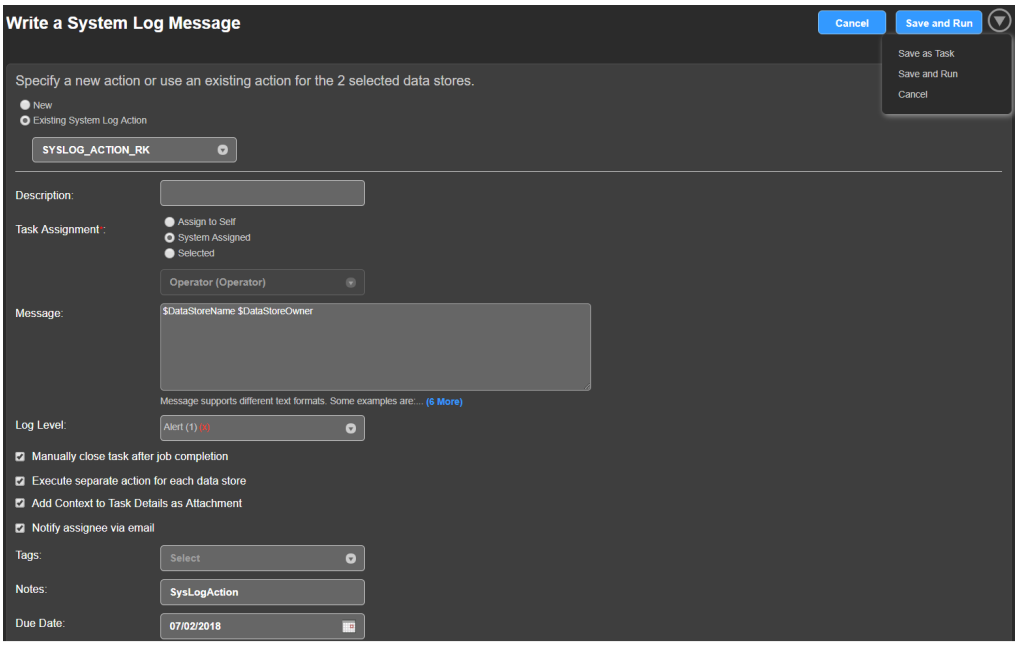
   The **Data Stores** page displays the list of data stores.
3. Select the check box next to one or more data stores.
4. From the **Actions** menu, select **Take Action** > **Write a System Log Message**.

Note: The **Write a System Log Message** option might be disabled if you do not have privileges to create a system log task or you do not have access to the selected data stores.

The **Write a System Log Message** page displays on the **Tasks** workspace.

5. Select whether to create a new system log action or to use an existing system log action that is listed on the **Actions** workspace.

   If you select a new action, you first choose the system log extension for the action and then enter properties for the action. If you select an existing action, all properties for the action are populated. You can edit the existing action properties except for the associated extension.
   The following image shows the **Write a System Log Message** page after selecting two data stores that have a protection status greater than 10%:



6. Optionally, add a description. This description will display under the task number on the **Tasks** workspace to identify the task.
7. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
8. Optionally, enter or edit the contents of the system log message that Data Privacy Management will send to the remote system log server configured in the extension. You can add placeholders. For more information, see "Placeholders in Custom, Email, Service Management, and System Log Actions" on page 383.
9. Optionally, select one of the following log levels for the message:
   - **Alert (1)**
   - **Critical (2)**
   - **Debug (7)**
   - **Emergency (0)**

- **Error (3)**
- **Informational (6)**
- **Notice (5)**
- **Warning (4)**

10. Optionally, select the **Manually close task after job completion** check box to indicate that the user assigned to the task will mark the task as closed on the **Tasks** workspace after the task job is successful. If you do not select this check box, Data Privacy Management will automatically update the task status to Closed after the job completes successfully.
11. Optionally, select the **Execute separate action for each data store** check box to indicate that Data Privacy Management will create separate tasks for each data store that is included in the action when Data Privacy Management writes the log message.
12. Optionally, select the **Add Context to Task Details as Attachment** check box to attach the data store details to the tasks associated with the action.
13. To indicate that Data Privacy Management will notify the user assigned to the system log task once you create it, select the **Notify assignee via email** check box.
14. Optionally, select tags for the system log task and add any notes.
15. Optionally, select a task due date.
16. To save the task on the **Tasks** workspace and run it immediately, click **Save and Run**. To save the task on the **Tasks** workspace to manually run it later, click **Actions** > **Save as Task**.

    Data Privacy Management creates a task to write the system log message. A message indicates the number of tasks the Data Privacy Management Service saved successfully and the number of tasks that encountered an error. The service saves successful tasks in the Data Privacy Management repository. The tasks appear on the **Tasks** list page. Saved tasks have a status of New. Tasks you run display the success or failure of the job in the **Status** column.

Related Topics:

- "Security Dashboard Overview" on page 513
- "Top Data Stores Indicator" on page 525

CHAPTER 20

Security Policies

This chapter includes the following topics:

- Security Policies Overview, 449
- Security Policy Types, 450
- Security Policies Workspace, 452
- Security Policy Properties, 453
- Security Policy Management, 456

Security Policies Overview

A security policy is a rule-based policy designed to detect violations or decrypt data in use. You can create a security policy to detect high-risk data stores, user activity violations, specific system-identified anomalies, or to dynamically decrypt data.

For example, you can create a security policy to detect one of the following situations:

- Sensitive data that moves across countries.
- A user accesses data from a location other than the office location.
- A critical anomaly.
- Encrypted data in specific columns that you want to see in clear.

When you create an anomaly, data store, or user activity security policy, you set the severity level of the violation and specify a rule that defines the violation. Optionally, you can specify an action for Data Privacy Management to perform in the event of a violation. Data Privacy Management can send emails and run an executable script that performs a custom action.

Based on the type of security policy you create, Data Privacy Management evaluates anomalies, data stores, encrypted data, and user activity to determine if the rule in a security policy is met. When the rule criteria for an anomaly, data store, or user activity policy is met, Data Privacy Management creates a security policy violation. You can view violations on the **Security Policy Violations** workspace. When the rule criteria for a decryption policy is met, Data Privacy Management decrypts the encrypted data.

You view and manage security policies from the **Security Policies** workspace. To access the workspace, click **Manage** > **Security Policies** from the header.

Security Policy Types

When you create a new security policy on the **Security Policies** workspace, specify the security policy type based on the information you want Data Privacy Management to evaluate.

You can create the following types of security policies:

- Anomaly policy. Data Privacy Management evaluates anomalies to determine if an anomaly matches the conditions in the anomaly policy.
- Data store policy. Data Privacy Management runs the Evaluate Security Policy job to evaluate data stores that you can access. The job determines if a data store matches the conditions in the data store policy.
- Decryption policy. Data Privacy Management evaluates all data that meets the criteria specified in the conditions for the policy.
- User activity policy. Data Privacy Management evaluates events that stream to the user activity store to determine if an event matches the conditions in the user activity policy. It evaluates only the new events since the previous evaluation.

If an anomaly, data store, or user activity event matches the conditions in the respective security policy, it creates a security policy violation. If the conditions specified in a decryption policy are met, it decrypts the data that meets the conditions.

Anomaly Policy Example

Data Privacy Management evaluates the stream of user activity events and creates an anomaly when unusual user behavior occurs. You want email alerts sent to the data store owner when an anomaly involving data store OracleABC occurs. You want to limit the alerts to anomalies with a high severity status and when the event that triggered the anomaly occurred after business hours.

Perform the following steps:

1. Create an anomaly policy with the following conditions:
   - Data store equals OracleABC.
   - Severity equals High.
   - Time of day is between 11pm and 4am.
2. Specify the email addresses of the data store owner.

Data Store Policy Example

You have production and test data stores in Canada. You want to monitor the risk score of the production data stores. Specifically, you want an alert email when the risk score increases by 20% or more.

Perform the following steps:

1. Edit each production data store and assign a tag such as ProductionDB.
   You tag the production data stores so you can prevent the non-production data stores from triggering a violation.
2. Create a security policy with the following conditions:
   - Data store tag equals ProductionDB.
   - Data store country equals Canada.
   - Risk score percentage is equal to or greater than 20%.

3. Specify the email addresses that you want Data Privacy Management to send the alert to when a security policy violation occurs.
4. Schedule a date and time for Data Privacy Management to evaluate the data stores.

Decryption Policy Example

Encryption extensions, rules, and keys protect sensitive string data in Cloudera Hive data stores so that no users can see the data in clear. You want certain authorized users to view the decrypted sensitive data from one or more sensitive data domains.

To decrypt data based on conditions such as a user group, you create a new decryption security policy. On the **Define Rule** page, add the conditions for decrypting data. For example, you might add some of the following conditions in the **All are true** tab:

- One or more data domains or data domain groups. For example, a data domain group named "Employee Data" includes sensitive data for employees in data domains such as First Name, Last Name, Birthday, Gender, Social Security Numbers, Address, City, Zip Code, and Salary.
- A user group with authorization to view decrypted data. For example, a user group named "Human Resources Managers" that need to view the "Employee Data" in clear.

Add actions for how Data Privacy Management will decrypt the data. Select **Protect Data** and specify an encryption extension. You can send an email to notify users when a policy condition to decrypt data is met. You can write a system log message to the host defined in the system log extension from protection remote agent servers.

Decryption security policies do not generate jobs in Data Privacy Management when the actions to decrypt data run. Instead, Data Privacy Management notifies all protection remote agents of the decryption actions, unless the security policy includes a condition for data stores associated with protection remote agents. In this case, Data Privacy Management notifies only the protection remote agents associated with the data stores specified in the condition.

Related Topics:

- “Security Policy Properties” on page 453
- “Publishing Protection Remote Agents” on page 84

User Activity Policy Example

Your Data Privacy Management installation includes streaming of user activity events to the user activity store. You want to detect any attempt to access PCI data by users outside the Accounting department. In the event of user activity on PCI data, you want to receive email alerts. You also want to run a custom script that immediately encrypts the data that the user is attempting to access.

Perform the following steps:

1. Create a user activity policy with the following conditions:
   - Classification policy equals PCI.
   - User department is not Accounting.
2. Specify the recipient email addresses for the alert.
3. Specify the file path of the custom executable script that Data Privacy Management will run when the policy conditions are met.

Security Policies Workspace

From the **Security Policies** workspace, you can create and manage security policies.

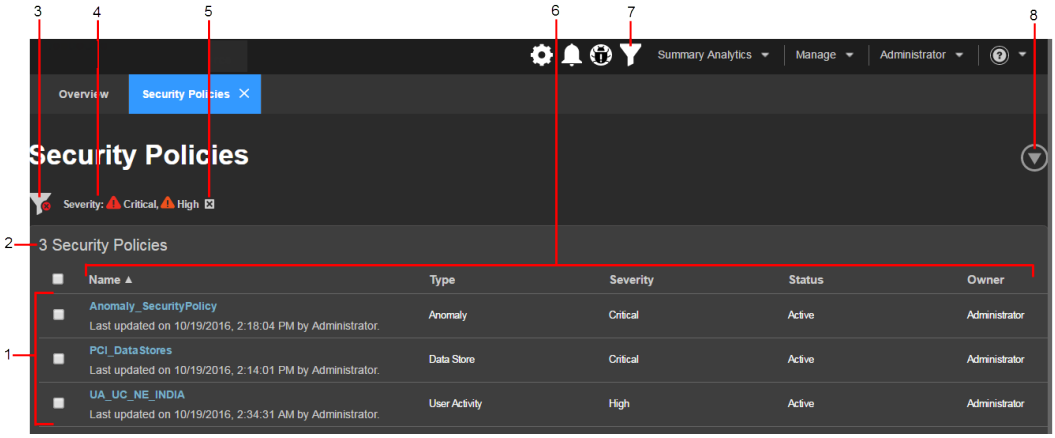
The default workspace view displays a list of security policies. From the list, you can select a security policy to view and edit details. You can use filters to quickly find the security policies that you want to manage.

To access the **Security Policies** workspace, click **Manage** > **Security Policies**.

Security Policies List Page

The **Security Policies** list page shows the security policies that match the active filter. By default, you view the **Security Policies** list page when you access the **Security Policies** workspace.

The following image shows an example of the **Security Policies** list page:



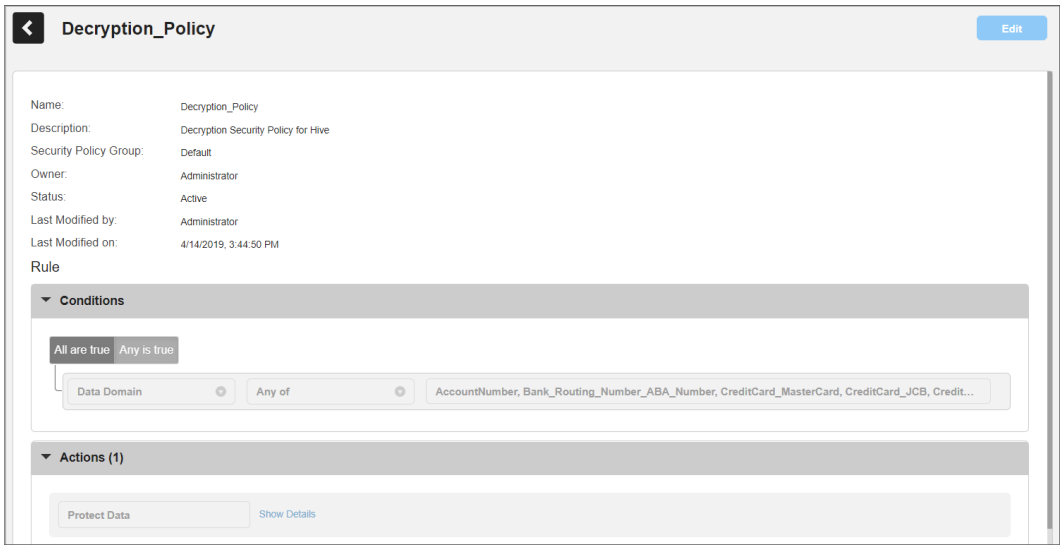
1. Security policies
2. Security policy count
3. **Clear Filter** icon
4. Filter conditions
5. **Clear Filter Condition** icon
6. Security policy properties
7. **Filter** icon
8. **Actions** menu

Security Policy Details Page

To view or edit security policy properties, access the **Security Policy Details** page from the **Security Policies** workspace.

- Click a security policy name. To edit the properties, click **Edit** on the **Remote Agent Details** page.
- Select the check box next to a security policy and then select **Open** or **Edit** from the **Actions** menu.

The following image shows an example of a decryption **Security Policy Details** page:



Security Policy Properties

Security policy properties identify details such as the security policy type, security policy group, conditions, and actions.

Security policies include the following properties:

Security Policy Type

When you create a new security policy, select one of the following options from the **Actions** > **New** menu:

- **Anomaly Policy**
- **Data Store Policy**
- **Decryption Policy**
- **User Activity Policy**

Name

A unique name that easily identifies the security policy.

The name is not case sensitive and cannot exceed 255 characters. Security policy names cannot contain a space or the following special characters: ~ ! $ % ^ & * ( ) +

Description

Optional. A security policy description that differentiates between similarly named security policies. The description cannot exceed 255 characters.

Security Policy Group

The name of the security policy group. The default group is the Default security policy group.

For more information, see Chapter 21, “Security Policy Groups” on page 459.

Severity

For anomaly, data store, and user activity policies, indicates the severity level of the security policy in the event of a violation. Options are:

- **Informational**
- **Low**
- **Medium**
- **High**
- **Critical**

Owner

Optional. Individual, group, or organization name that is responsible for the accuracy and integrity of the security policy. The name is case-sensitive and cannot exceed 100 characters.

Status

Optional. Options are:

- **Active**
- **Inactive**. Default.

Last Modified by

Displays the author of the last update to the security policy.

Last Modified on

Displays the timestamp of the last update to the security policy.

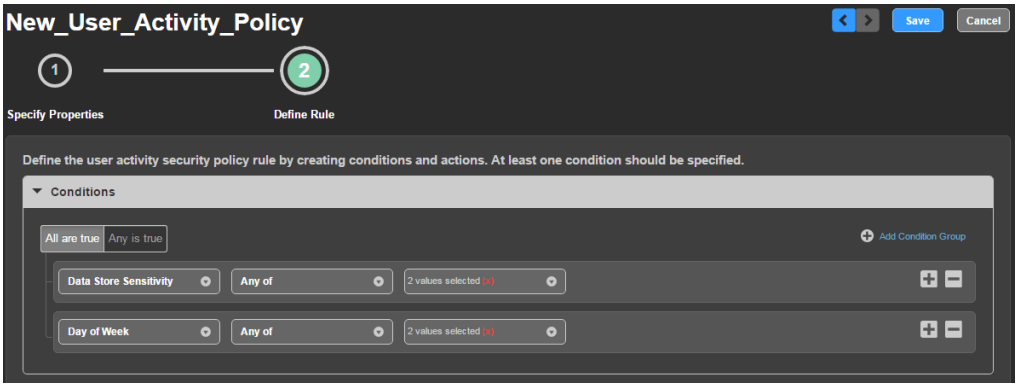
Rule

Specifies the conditions, condition groups, and actions defined for a security policy.

For example, you want to be alerted when data stores with a sensitivity status of Critical or Restricted are accessed over the weekend. You can specify the following conditions:

```
All are true
Data Store Sensitivity - Any of - Critical, Restricted.
Day of week - Any of - Saturday, Sunday.
```

The following image shows the example rule:



Actions

Optional. For decryption policies, specifies the decryption method and optionally, email or system log actions that Data Privacy Management must perform if the conditions in the policy are met. For all other policy types, specifies the action Data Privacy Management must perform in the event of a security policy violation.

You can choose one or more of the following actions, based on the policy type:

- **Create a Service Management Ticket**. For anomaly, data store, or user activity security policies only. Create a new action or use an existing service management action. Specify the urgency, impact, and service category. Optionally add file attachments to a ServiceNow ticket, specify the number of days from now that the action is due, and add tags, notes, and a description, which optionally includes placeholders. Service management actions also include details about how Data Privacy Management will run and close associated tasks, and whether to send an email to notify users when associated tasks are assigned to them. In the event of a security policy violation, the Data Privacy Management Service runs the service management task and creates the service management ticket for one or more data stores.

  For more information, see "Service Management Action Properties" on page 380.

- **Protect Data**. For data store, decryption, or user activity security policies only. Optionally, select tags, specify the number of days until the action is due, add attachments, and enter notes.
  For data store or user activity security policies, in the event of a security policy violation, Data Privacy Management creates a protection task for each data store included in the violation. The new protection tasks are displayed on the **Tasks** workspace, where you configure and run the tasks.

  When the conditions of a decryption security policy are met, Data Privacy Management dynamically decrypts data in all protection remote agent data stores by default. To decrypt data in specific data stores associated with protection remote agents, add a Data Store Name condition to the policy.

  Note: You cannot protect data in an anomaly security policy because the violation for this type of policy might be associated with diverse data stores that cannot be protected by the same protection extension.

  For more information, see "Protecting Data" on page 403.

- **Run a Custom Script**. For anomaly, data store, or user activity security policies only. Create a new action or use an existing custom action. Specify parameters, tags, notes, the number of days from now that the action is due, details about how Data Privacy Management will run and close associated tasks, and whether to send an email to notify users when associated tasks are assigned to them. When you save the security policy, Data Privacy Management validates the path and determines whether the file is executable. In the event of a security policy violation, Data Privacy Management runs the script file.

  For more information, see "Custom Action Properties" on page 377.

- **Send an Email**. Create a new action or use an existing email action. Assign the task to yourself or to a different user. Provide an email address in the **From** field. Provide one or more email addresses in the **To**, **Cc**, and **Bcc** fields. Separate email addresses with a comma. Optionally, provide a subject and message, attachments, tags, notes, details about how Data Privacy Management will run and close associated tasks, and the number of days from now that the action is due. If a security policy violation occurs, it sends an email to the task assignee and recipient email addresses.

  Note: Decryption security policies must use the default email extension.

  For more information, see "Email Action Properties" on page 378.

- **Write a System Log Message**. Create a new action or use an existing system log action. Specify the server connection details, tags, notes, the number of days from now that the action is due, details about how Data Privacy Management will run and close associated tasks, and whether to send an email to notify users when associated tasks are assigned to them. System log actions also include the log message impact level and the contents of the system log message in a text format, which optionally includes placeholders.

  Note: After you add a system log action to a decryption security policy, you must publish to the protection remote agents associated with the data stores in the policy.

  For more information, see "System Log Action Properties" on page 382.

If you do not specify an action in anomaly, data store, or user activity policies, check the **Security Policy Violations** workspace routinely for any violations. Default is no action.

Note: For anomaly security policies, Data Privacy Management can only perform actions or create tasks in the event of a violation if the security policy is associated with at least one data store.

Data Store Policy Evaluation

For data store security policies only. Specifies when Data Privacy Management runs the Evaluate Security Policies job, and indicates whether or not it shows only violations from data stores scanned since the last time the job ran.

Related Topics:

- "Decryption Policy Example" on page 451
- "Publishing Protection Remote Agents" on page 84

Security Policy Management

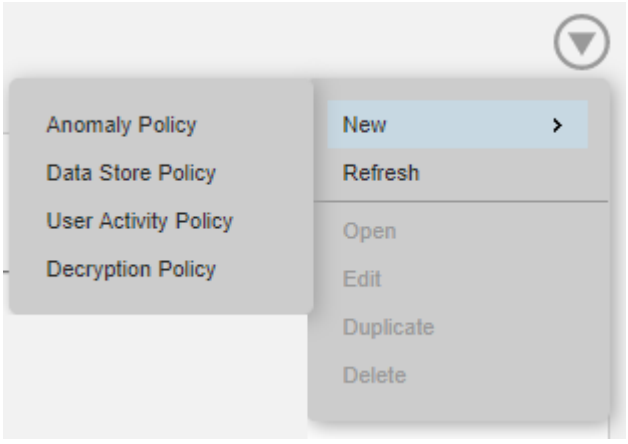
You can copy, create, view, edit, and delete security policies from the **Security Policies** workspace. The Data Privacy Management repository stores security policies. When you create or edit a security policy, the Data Privacy Management Service accesses the information from the Data Privacy Management repository.

Creating a Security Policy

When you create a security policy, you select the type of policy from the **Actions** menu and then you enter basic information, conditions, and security policy actions.

1. Click **Manage** > **Security Policies**.

   The **Security Policies** workspace appears.
2. Click **Actions** > **New** and select a security policy type from the menu.



   The **New_<Policy_Type>_Policy** page appears.
3. On the **Specify Properties** pane, enter a name for the security policy.

   The name cannot contain a space or the following special characters: ~!$%^&*()+.
4. Optionally, enter a description that is no longer than 255 characters.

5. Select a security policy group.
6. For anomaly, data store, and user activity policies, select one of the following severity levels:
   - **Informational**
   - **Low**
   - **Medium**
   - **High**
   - **Critical**
7. Optionally, enter the name of the security policy owner.
8. Select one of the following options:
   - **Active**. Select this option if you want Data Privacy Management to evaluate the conditions in the policy.

     Note: You cannot delete an active security policy.
   - **Inactive**. Select this option if you do not want Data Privacy Management to evaluate the conditions in the policy. For example, you might plan to finish defining the conditions or actions in the policy at a later time, or you might want to retain the policy in the Data Privacy Management repository for future use.

     Note: You can delete an inactive security policy if the policy is not associated with any prior violations. If violations exist, first delete the violations and then delete the inactive security policy.

   Default is Inactive.
9. Click the **Next** icon.

   The **Define Rule** pane appears.
10. Build one or more conditions.

    To group related conditions, click **Add Condition Group**. Specify two or more conditions within each group. You can specify the `And` or `Or` operator between condition groups.
11. Optionally, specify an action that Data Privacy Management will perform in the event that the conditions in the security policy are met.
12. If you are creating an anomalous, decryption, or user activity security policy, skip to step 16. If you are creating a data store security policy, go to the next step.
13. Click the **Next** icon.

    The **Schedule** pane appears.
14. Specify when you want to run the Evaluate Security Policies job to locate scanned data stores that meet the security policy conditions. Select one of the following options:
    - **Run Now**. The Data Privacy Management Service starts the job immediately.
    - **Schedule Evaluation**. Specify the **Start Date** and **Start Time** to run the Evaluate Security Policies job. Optionally, select the **Recurrence** check box to schedule how often Data Privacy Management will run the Evaluate Security Policies job at recurring intervals.

    Default is **Run Now**.
15. Optionally, to see only violations from data stores scanned since the last time the Evaluate Security Policies job ran, select the **Show Only New Results** checkbox.
16. Click **Save**.

Editing a Security Policy

You can edit the properties of a security policy. After you edit a security policy, the violations associated with the prior version of the security policy remain unchanged on the **Security Policy Violations** workspace.

1. Click **Manage** > **Security Policies**.

   A list of security policies appears. You can use the filters to narrow the list of security policies.
2. Select the check box next to the security policy that you want to edit.
3. From the **Actions** menu, click **Edit**.
4. Edit the security policy properties that you want to change.

   The security policy name cannot contain spaces or the following special characters: `~!$%^&*()+`
5. Click **Save**.

   A message indicates if the Data Privacy Management Service saved the changes to the security policy or if an error occurred. If there are no errors, the service saves the security policy changes in the Data Privacy Management repository.

Copying a Security Policy

To create a new security policy with similar properties to an existing security policy, you can copy the existing policy. After you copy the security policy, edit the security policy properties.

1. Click **Manage** > **Security Policies**.

   The **Security Policies** workspace appears.
2. Select the checkbox next to the security policy that you want to copy.
3. From the **Actions** menu, select **Duplicate**.

   The **Security Policy Details** page appears with the following name: `Copy_of_<security_policy_name>`
4. Edit the security policy name and other properties you want to change.

   The security policy name cannot contain spaces or the following special characters: `~!$%^&*()+`
5. Click **Save**.

Deleting a Security Policy

You cannot delete an active security policy. You can delete an inactive security policy if there are no violations of the policy. If violations exist, delete the violations first and then delete the security policy.

1. Click **Manage** > **Security Policies**.

   A list of security policies appears.
2. Select the check box next to the security policy that you want to delete.
3. From the **Actions** menu, select **Delete**.

   You receive a message to confirm that you want to delete the security policy.
4. Click **Yes, I'm Sure**.

CHAPTER 21

Security Policy Groups

This chapter includes the following topics:

- Security Policy Groups Overview, 459
- Security Policy Groups Workspace, 459
- Security Policy Groups Properties, 461
- Security Policy Groups Management, 461

Security Policy Groups Overview

A security policy group is a collection of related security policies. A security policy may belong to more than one security policy group.

Organize security policies in logical groups. For example, you can create security policy groups based on the following security policy attributes:

- Content. For example, all PCI security policies.
- Classification level. For example all confidential security policies.
- Business unit. For example, all Finance security policies.
- Location. For example, all North American security policies.

You can also create a security policy group by copying an existing security policy group.

Data Privacy Management includes a default security policy group that has all the security policies. When you create a new security policy, the Data Privacy Management Service adds the security policy to the default security policy group. To remove a security policy from the default group, edit the security policy.

When you review security policy violations, you can filter the violations by one or more security policy groups.

Security Policy Groups Workspace

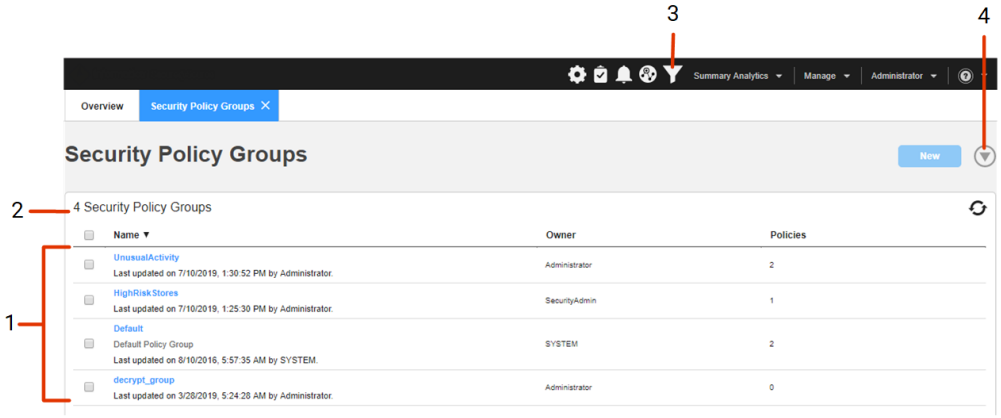
Use the security policy groups workspace to manage security policy groups. The workspace includes a panel that displays a list of security policy groups and a panel that displays the details for a security policy group. You can use filters to quickly find the security policy groups that you want to manage.

To access the Security Policy Groups workspace, click **Manage** > **Security Policy Groups**.

Security Policy Group List Panel

The security policy group list panel shows all of the security policy groups that match the active filter. By default, you view the security policy group list panel when you access the security policy groups workspace.

The following image shows an example of the security policy groups workspace with the security policy group list panel:

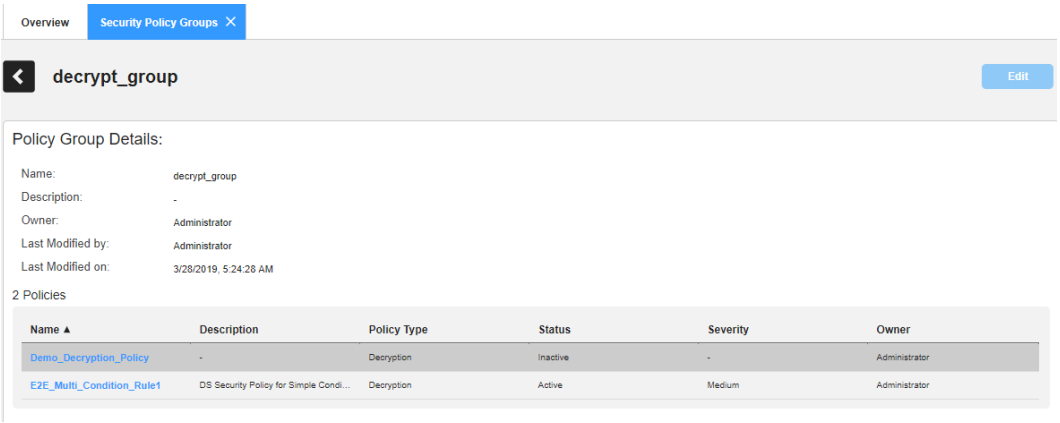


1. Security policy group list
2. Security policy group count
3. Filter
4. Actions menu

Security Policy Group Details Panel

The security policy group details panel shows the properties of a security policy group. Access the security policy group details panel to view or edit the security policy group properties. You access the security policy group details panel when you click a security policy group name from the security policy group list panel, or when you view a security policy group from the Actions menu.

The following image shows an example of a security policy group details panel:



Security Policy Groups Properties

Security policy group properties identify the group, the owner of the group, and the number of policies in the group.

Security policy groups include the following properties:

Name

Required. Short description of the security policy group.

The name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 255 characters. The name cannot contain a space or special characters.

Use a naming convention that easily identifies a security policy group.

Description

Optional. Long description of the security policy group. The description cannot exceed 254 characters.

Use this property to specify unique details to differentiate between similarly named security policy groups.

Owner

Optional. Individual, group, or organization name that is responsible for the accuracy and integrity of the security policy group. The name is case-sensitive and cannot exceed 100 characters.

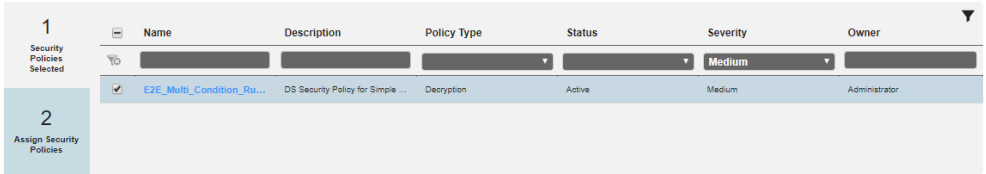
Security Policies

Optional. Shows the following security policies lists:

- Security Policies Selected. The button shows the number of security policies in the security policy group. When you click the button, the list of security policies in the security policy group appear. This list appears by default when you edit a security policy group.
- Assign Security Policies. The button shows the number of security policies available in the Data Privacy Management repository. When you click the button, the list of all security policies appear. This list appears by default when you create a security policy group.
  You can add one or more security policies to the security policy group. To add, click the check box next to the name of the security policy.

  To quickly find specific security policies from the list, click the security policy filter icon and specify the filter conditions.

  The following image shows a sample Securities Policies section with a filter condition enabled:



Security Policy Groups Management

Use the security policy group workspace to create, view, edit, duplicate, and delete security policy groups. The Data Privacy Management repository stores security policy groups. When you create or edit a security policy group, the Data Privacy Management Service accesses the information from the Data Privacy Management repository.

Creating a Security Policy Group

When you create a security policy group, you enter a name and specify the policies to include in the group.

1. Click **Manage** > **Security Policy Groups**.

   The **Security Policy Groups** workspace appears.
2. Click **New**.

   The **New Security Policy Group** page appears.
3. Enter the name for the security policy group.

   The name cannot contain spaces or the following special characters: `~!$%^&*()-+=[]{}\|;:'",./<>?`
4. Optionally, enter a description.
5. Optionally, enter the name of the owner.
6. Optionally, add one or more policies to the security policy group.
7. Click **Save**.

   A message indicates if the Data Privacy Management Service saved the security policy group or if an error occurred. If there are no errors, the service saves the security policy group in the Data Privacy Management repository.

Viewing a Security Policy Group

You can view the security policy group properties.

1. Click **Manage** > **Security Policy Groups**.

   A list of all security policy groups appears.
2. Click the security policy group name to view the details.

   The security policy group details panel appears.
3. To view the details of a security policy in the security policy group, click the security policy name.

   The **Security Policy** details page appears.
4. Optionally, click **Edit** to change the security policy group properties.

Editing a Security Policy Group

You can edit the properties of a security policy group that you created or imported.

1. Click **Manage** > **Security Policy Group**.

   A list of security policy groups appears. You can use the filters to narrow the list of security policy groups.
2. Select the check box next to the security group policy that you want to edit.
3. From the Actions menu, click **Edit**.
4. Edit the security policy group properties that you want to change.

   The security policy group name cannot contain spaces or the following special characters: `~!$%^&*()-+=[]{}\|;:'",./<>?`
5. Click **Save**.

Copying a Security Policy Group

You can copy a security policy group to create a security policy group with similar properties. After you copy the security policy group, edit the security policy group properties.

1. Click **Manage** > **Security Policy Groups**.
2. Select the checkbox next to the security policy group that you want to copy.
3. From the Actions menu, select **Duplicate**.

   The security policy group details page appears with the following name: `Copy of <security policy group name>`
4. Edit the security policy group properties.

   The security policy group name cannot contain spaces or the following special characters: `~!$%^&*()-+=[]{}\|;:'",./<>?`
5. Click **Save**.

Deleting a Security Policy Group

You can delete security policy groups from the Data Privacy Management repository. You cannot delete the system created Default security policy group.

1. Click **Manage** > **Security Policy Groups**.

   A list of security policy groups appears.
2. Select the check box next to the security policy group that you want to delete.
3. From the Actions menu, select **Delete**.

   You receive a message to confirm if you want to delete the security policy group.
4. Click **Yes**.

   You receive a message that indicates the success or failure of the deletion. If no errors occurred, Data Privacy Management deletes the security policy group from the Data Privacy Management repository.

Exporting Security Policy Groups, Security Policies, and Actions

You can use the export and import actions to copy security policy groups, security policies, and actions from one instance to another. When you export from the Security Policy Groups workspace, the Data Privacy Management Service downloads a JSON file to your machine. Use the JSON file to import the list of security policy groups, security policies, and actions to another Data Privacy Management instance.

1. Click **Manage** > **Security Policy Groups**.

   The Security Policy Groups workspace appears.
2. From the Actions menu, click **Export**.

   A dialog box appears with the exported file.

   The export file is named `SecurityPolicyGroups.json`. The Data Privacy Management Service saves the file to the default download directory on your machine. If you export the file again, the service downloads a new file. The file name includes an incremental number such as `SecurityPolicyGroups(1).json`.

   The export file contains all the security policy groups, security policies, and actions. The export file is not affected by any filter conditions you set on the Security Policy Groups page.

Importing Security Policy Groups, Security Policies, and Actions

You can use the export and import actions to copy security policy groups, security policies, and actions from one instance to another.

First, export the security policy groups, security policies, and actions from the instance you want to copy from. The Data Privacy Management Service downloads a JSON file to your machine. The export file is named `SecurityPolicyGroups.json`.

Important: Do not edit the `SecurityPolicyGroups.json` file.

Import the `SecurityPolicyGroups.json` file to the instance you want to copy the security policy groups, security policies, and actions to.

A success or error message appears after Data Privacy Management processes the JSON file. To view details about the import, read the Data Privacy Management log file.

Steps to Import Security Policy Groups, Security Policies, and Actions

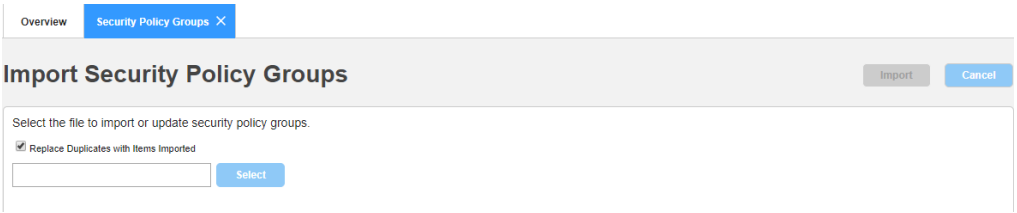
To populate a Data Privacy Management instance with security policy groups, security policies, and actions, import the information from the export JSON file `SecurityPolicyGroups.json`.

1. Click **Manage** > **Security Policy Groups**.

   A list of security policy groups appears.

2. From the **Actions** menu, select **Import**.

   The **Import Security Policy Groups** page appears.
   The following image shows the **Import Security Policy Groups** page:



3. Enable or disable the **Replace Duplicates with Items Imported** option.

   The option determines how the Data Privacy Management Service handles updates for security policy groups, security policies, and actions that exist in the Data Privacy Management repository.
   If you enable the option, the Data Privacy Management Service updates the properties of existing security policy groups, security policies, or actions with the values in the .json file.

   If you disable the option, the service does not update existing security policy groups, security policies, or actions.

4. Select the file from which you want to import security policy groups, security policies, or actions.

   The service validates the import file, such as the file extension.
   If the file is valid, the user interface displays a success message.

   If the file is not valid, the user interface displays an error message.

   To view details about the import, access the Data Privacy Management log.

5. Click **Import**.

6. After the Data Privacy Management Service processes the import file, you can go to the **Security Policy Groups**, **Security Policies**, and **Actions** workspace to view the security policy groups, security policies, and actions that you created or updated.

CHAPTER 22

Security Policy Violations

This chapter includes the following topics:

- Security Policy Violations Overview, 465
- Security Policy Violations Workspace, 465
- Security Policy Violation Details Page, 471
- Security Policy Violations Workspace Views, 477
- Security Policy Violations Management, 479

Security Policy Violations Overview

A security policy violation occurs when one or more properties in an anomaly, data store, or user activity event match the rule specified in a security policy.

For example, you want to detect user activity on fields that contain US social security numbers. You set up a user activity policy that specifies that user activity must impact sensitive data that match the data domain for US social security. Data Privacy Management evaluates the stream of user activity events. When a user activity event impacts fields that contain US social security numbers, Data Privacy Management creates a violation.

Data Privacy Management lists violations on the **Security Policy Violations** workspace. In addition, if you specified an action in the security policy, such as email notification or custom script execution, then Data Privacy Management performs the action.

You can manage violations. For example, you can mark a violation as read or flag a violation to review at a later time. You can add and remove tags to a violation. You can delete violations. You can also export violation details for further analysis.

Click the **Security Policy Violations** icon on the Overview page to access the **Security Policy Violations** workspace.

Security Policy Violations Workspace

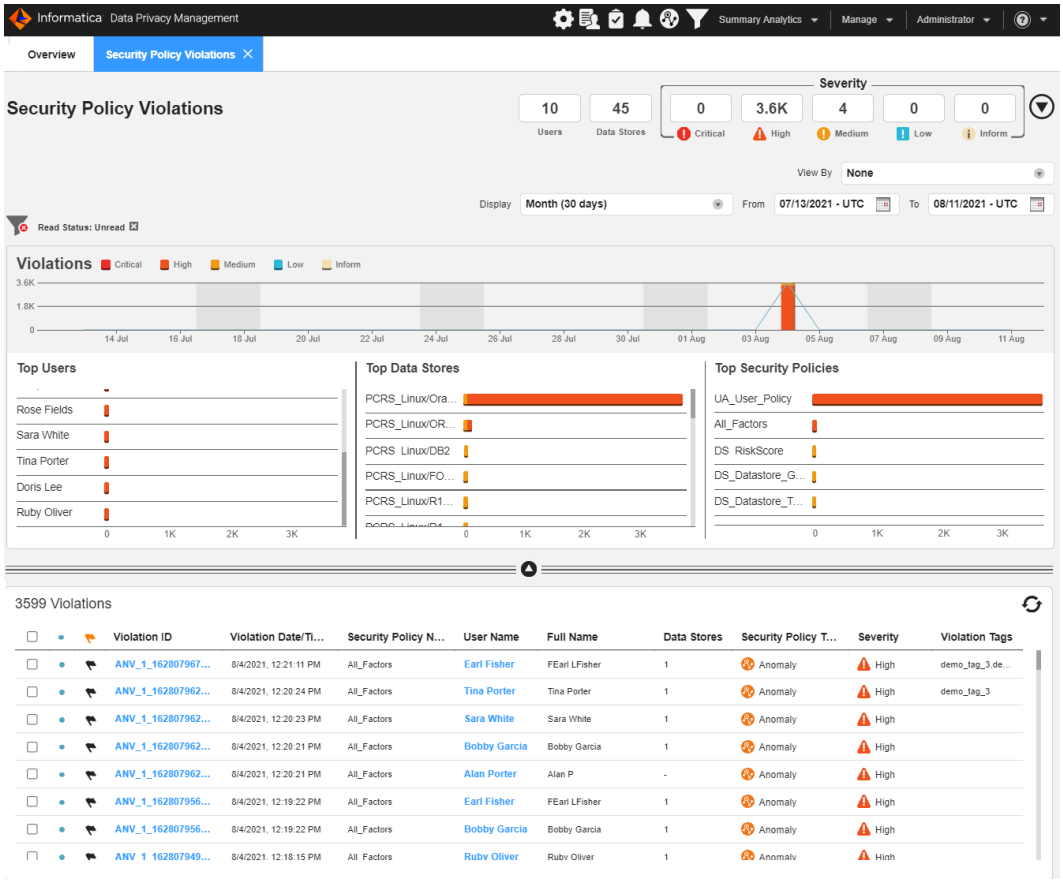
Use the **Security Policy Violations** workspace to view and manage security policy violations. The workspace includes a summary of violation counts, a violation trend graph, violation indicators, and a list of violations.

The information in the workspace changes based on the selected view, time period, and filter conditions. You can show or hide the violation indicators and violations graph on the workspace.

The violations you see on the **Security Policy Violations** workspace are violations that involve at least one data store that you have access to. The workspace includes predefined views that display violation information in a variety of ways. You can view the data stores, users, and anomalies that triggered the most number of violations.

To access the **Security Policy Violations** workspace, click the **Security Policy Violations** icon in the header.

The following image shows a sample **Security Policy Violations** workspace:



The **Security Policy Violations** workspace includes the following sections:

Section	Description
Violations	Displays a list of violations based on the selected view, time period, and filter conditions. For more information, see "Violations List" on page 468.
Violation indicators	Displays the top 25 users, data stores, and security policies that are associated with the most violations based on the selected time period and filter conditions. Click a row in an indicator to filter the workspace by the selected item. Click an indicator title to open a view that shows all information for the indicator category. For more information, see "Security Policy Violation Indicators" on page 468.
Violations graph	Displays a trend of violations based on the selected time period and filter conditions. Click an entry in the graph to filter the workspace by the selected item. All time periods are in the UTC time zone.

Section	Description
Filter	Filters the information on the workspace to match the filter conditions. When you click the **Filter** icon, the filter pane appears with a list of properties. Set the properties to filter information based on violation ID, flag status, read status, severity level, users, data stores and data store tags, violation tags, security policies, security policy groups, security policy types, data domains, data store department, data store country, or data store location. The filter conditions apply to all sections on the workspace. When you apply a filter, the workspace displays the active filter conditions. You can clear all filters or clear specific filter conditions.
Violations summary	Displays key results to help you quickly assess the number and severity of violations across your enterprise. You can view the number of users associated to violations, the aggregated number of data stores associated to the violations, and the violation count based on severity level. For more information see, "Security Policy Violations Summary" on page 467.
Actions menu	From the **Actions** menu, you can mark a violation as read or unread. You can flag, unflag, or delete a violation. You can add or update tags. You can export information related to violations. For more information, see "Security Policy Violations Management" on page 479
Display options	The Views field provides different options to view and display the aggregated violation information. You can view violations by users, data stores, or by the security policies. For more information, see "Security Policy Violations Workspace Views" on page 477. Use the Display and time Period fields to Filter information related to violations detected within the selected time period. All time periods are in the UTC time zone. **Note:** For time periods in days, the workspace displays information for full days. A full day is from 0:00 through 23:59 UTC. The time period you select might not include information from the current day. For example, you select 30 days. The workspace shows violations that were detected within the last 30 days. The count begins from the last completed day in the UTC time zone, which can be yesterday.

Security Policy Violations Summary

Use the security policy violations summary to quickly assess the number and severity of violations that involve data stores that you have access to. You can view the number of users and data stores associated with violations and the number of violations in each severity level.

The following image shows an example of the violations summary:



The violations summary includes the following information:

Users

Aggregated number of users associated with the violations. A user can be associated with multiple violations.

Data Stores

Aggregated number of data stores associated with the violations. A violation can include multiple data stores. A data store can be included in multiple violations.

Note: The data store filter does not affect the data store count. For example, if you filter the workspace by one data store, the data store count might show more than one data store. The violation that includes

the filtered data store can also include other data stores. The count includes all data stores for the violation regardless of the data store filter.

Severity

Total number of violations in each severity level. Data Privacy Management uses the severity level of the security policy as the severity level for the corresponding violation. A violation can only have one severity level.

Security Policy Violation Indicators

The indicators display the top 25 users, data stores, and security policies associated with security policy violations.

Indicators include the following information:

Title

Click the corresponding title to open a list of all data stores, users, or security policies that are associated with at least one violation. For each indicator, the list shows the number of associated users, security policies, data stores, and violations.

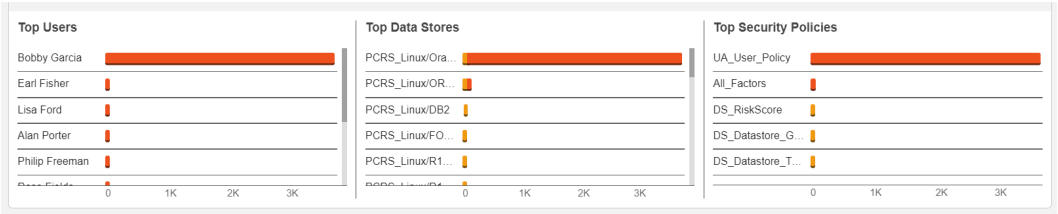
Violation count

Displays the number of violations.

Page icons

Click the back and forward arrows to scroll through the entries.

The following image shows an example of the **Security Policy Violations** workspace indicators:



The **Top Users** indicator displays the name of the user account that performed the user activity event that is associated with the violation. The user account name appears based on the source of the user activity information, such as user activity logs.

Click a user name to filter the workspace data for information related to the selected user.

Note: If you disable **User Activity** in the Data Privacy Management Service properties, you cannot view information in the **Top Users** indicator.

Click a data store row on the **Top Data Stores** indicator to filter the workspace data for information related to the selected data store.

Click a security policy row on the **Top Security Policies** indicator to filter the workspace data for information related to the selected security policy.

Violations List

The violations list panel includes a list of all violations and the corresponding violations details. The list of violations is dynamic and can change based on the time period and filter conditions that you select in the

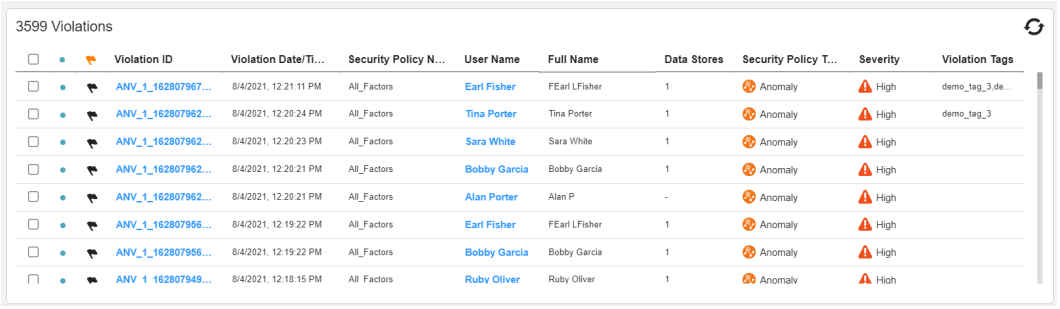
workspace. The list displays violations based on your data store access. You can see a violation if the violation is associated with at least one data store to which you have access.

To view the violations list, select one of the views in the Security Policy Violations workspace. The violations list disappears. Select View By > None to display the violations list again.

By default, the list of violations is sorted by unread status, then by the highest severity level, and then by the violation timestamp. You can click a column header to sort the list of violations by that column.

Tip: Click the filter icon to filter the list of violations. For example, you can filter the list to show only critical severity violations.

The following image shows an example of the violations list:



The violations list includes the following information:

Property	Description
Selection box	You can perform actions on one or more violations. Select one, multiple, or all violations. Then, click the **Actions** menu and select the action. For example, you can mark all violations as read or unread.
Violations count	Number of violations that involve at least one data store to which you have access. This number depends on the selected time period and filter conditions.
Read/Unread	Read or unread status of a violation. You can mark a violation as read or unread. Changes in the status affect the sort order of the violations. The list of violations is first sorted by unread violations. Click the column header to sort by the status. You can filter the workspace by the status to show only the unread or read violations. By default, a violation is marked as unread until you mark the violations as read. For more information, see "Marking a Violation as Read or Unread" on page 480.
Flag/Unflag	Flagged or unflagged status of the violations. Click the flag to change the status. A clear flag indicates a violation that is not flagged. A filled flag indicates a violation that is flagged. You might want to flag the violations that you need to review further or to flag for follow-up. Click the column header to sort by flag status. You can use the flag status as a filter condition for the workspace. By default, a violation is not flagged until you mark the violations as flagged. The flag status does not affect the default violations sort order. For more information, see "Flagging or Unflagging A Security Policy Violation" on page 480.

Property	Description
Violation ID	Violation identification number. When Data Privacy Management detects a violation, it assigns a unique ID to the violation in the following format: `<Abbreviation for violation type>_<Unique number for node ID(s)>_<Violation date and time in Epoch format>` Violation types have the following abbreviations: - Anomaly violation: `ANV` - Data store violation: `DSV` - User activity violation: `UAV` An example of a user activity violation ID is: `UAV_10078_1480379025542`
Violation Date/Time	Date and time that Data Privacy Management detected the violation. The date and time are in the UTC time zone.
Security Policy Name	Name of the security policy that triggered the violation. Click on the violation ID to see details such as the security policy condition that triggered the violation and the anomaly, data store, or user activity that matched the security policy condition.
User Name	Name of the user account that performed the user activity event associated with the violation. Data Privacy Management shows the user account name from the source of the user activity information, such as user activity logs. You can click a user name to view more details about the user's activity on the **User Profile** page.
Full Name	Full name that is associated with the user account name in the Data Privacy Management repository. If the full name is null in the Data Privacy Management repository, the value is a concatenation of the first name and last name. If the first name and last name is null, the value is the user account name. If the user account is not in the repository, the value is the user account name.
Data Stores	Total number of data stores that match the rule in the security policy that triggered the violation. Click on the violation ID to see the data stores associated with the violation. **Note:** The count includes all of the data stores that are associated with the violations, regardless of the data stores to which you have access. When you view the violations details, you only see the data stores name to which you have access.
Security Policy Type	Specifies whether the violation is based on an anomaly, data store, or user activity security policy.
Severity	Severity level of the violations. Data Privacy Management uses the severity level of the security policy as the severity level for the corresponding violation. A violation can have a critical, high, medium, low, or informational severity level.
Refresh	Click to refresh the list of violations based on the selected duration and filter conditions.

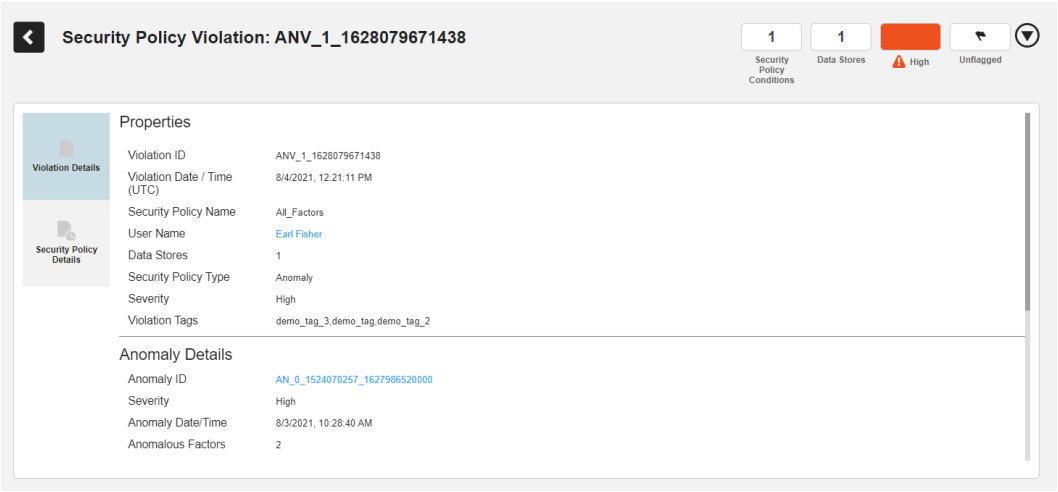
Related Topics:

- "User Profile Page" on page 567

Security Policy Violation Details Page

The Security Policy Violation details page shows the violation details and the security policy details. You access the page when you click a violation ID from the list of violations on the **Security Policy Violation** workspace.

The following image shows a sample Security Policy Violation details page:



The Security Policy Violations details page includes the following sections:

Property	Description
Detail tabs	The detail tabs allow you to switch between the following views: - Violation Details - Security Policy Details For more information, see "Violation Details" on page 472 and "Security Policy Details" on page 477.
Violation ID	Violation identification number that Data Privacy Management assigns when a violation is detected. The highest number indicates the most recent violation.
Violation summary	Displays the following key information about the violation: - **Policy conditions**. The number of conditions configured in the anomaly, data store, or user activity policy. - **Data stores**. Number of data stores associated to the violation. The count only includes the data stores to which you have access. - **Severity level**. The severity level of the violation. Data Privacy Management assigns the severity level of the security policy to the corresponding violation. - **Flag status**. The flagged or unflagged status of the violation. You can change the flag status from the **Actions** menu. You might want to flag the violation to review it at a later time.
Actions menu	From the **Actions** menu, you can mark the violation as read or unread. You can flag or unflag the violation. You can also export details about the violation.

Violation Details

The Violation Details view includes a section for the general properties and a section for the specific properties of an anomaly, data store, or user activity violation.

Violation General Properties

The general properties of the violation include the following properties:

Property	Description
Violation ID	Violation identification number. When Data Privacy Management detects a violation, it assigns a unique ID to the violation in the following format: `<Abbreviation for violation type>_<Unique number for node ID(s)>_<Violation date and time in Epoch format>` Violation types have the following abbreviations: - Anomaly violation: `ANV` - Data store violation: `DSV` - User activity violation: `UAV` An example of a user activity violation ID is: `UAV_10078_1480379025542`
Violation Date and Time	Date and time that Data Privacy Management detected the violation. The date and time are in the UTC time zone.
Security Policy Name	Name of the security policy that triggered the violation. Click on the **Security Policy Details** tab to see the conditions that triggered the violation.
User Name	Name of the user account that performed the user activity event associated with the violation. Data Privacy Management shows the user account name from the source of the user activity information, such as user activity logs. To learn more about a user, click the user name. The **User Profile** page appears. For data store security policies, the user name is not applicable and appears as `Not Applicable`.
Data Stores	Total number of data stores that are associated with the violation. **Note:** The count includes all of the data stores that are associated with the violation, regardless of the data stores to which you have access. When you view the list of data stores, you only see the data stores names to which you have access.
Security Policy Type	Specifies whether the violation is based on an anomaly, data store, or user activity security policy.
Severity	Severity level of the violations. Data Privacy Management uses the severity level of the security policy as the severity level for the corresponding violation. A violation can have a critical, high, medium, low, or informational severity level.

Related Topics:

- "User Profile Page" on page 567

Anomaly Violation Properties

The anomaly violation properties include the following properties:

Property	Description
Anomaly ID	Anomaly identification number. Data Privacy Management uses the following syntax to assign an ID to the anomaly: `AN_<Node ID>_<User hash code>_<Anomaly timestamp in Epoch format>`
Severity	Severity level of the anomaly. Data Privacy Management automatically determines the severity level for each anomaly based on an internal algorithm. An anomaly can have a high, medium, or low severity level.
Anomaly Date and Time	Date and time of the anomalous behavior that is associated with the anomaly. This is not the date and time when Data Privacy Management detected the anomaly. The date and time are in the UTC time zone.
Anomalous Factors	Number of system-defined anomalous factors that are associated to the anomaly. For more information, see "Anomalous Factors" on page 361.
Anomaly Score	A calculated value based on the weight of each anomalous factor specified on the **Settings** page. Indicates the degree of risk of the anomaly.
Full Name	Full name that is associated with the user account name in the Data Privacy Management repository. If the full name is null in the Data Privacy Management repository, the value is a concatenation of the first name and last name. If the first name and last name is null, the value is the user account name. If the user account is not in the repository, the value is the user account name.
Title	Title that is associated with the user account name in the Data Privacy Management repository. If the user account is not in the repository, the title appears as –
User Department	User department that is associated with the user account name in the Data Privacy Management repository.
User Group	Peer group that Data Privacy Management uses to compare user behavior. When Data Privacy Management analyzes user activity events for anomalous behavior, it compares the behavior of the user and the user's peer group to detect an anomaly. It uses the department associated with the user account name in the repository for the peer group. Adds the user from the same department in the peer group. If the user account is not in the repository, the group is blank.
User Location	Location that is associated with the user account in the Data Privacy Management repository. This is the location from where the user works. For example, the corporate office that the user is assigned. **Note:** This location is not necessarily the same location from where the user performs the user activity events.

Property	Description
Data Stores	List of all data stores that are associated with the anomaly. A data store is associated with an anomaly when at least one user activity event in the anomaly included the data store. If you do not have authorization for a data store in the list, Data Privacy Management masks the data store name and displays ?. By default, the list is sorted by data store name in ascending order.
Data Domains	List of all data domains that are associated with an anomaly. A data domain is associated with an anomaly when the user's activity event in the anomaly accessed sensitive data that matches the data domain. By default, the list is sorted by data domain name in ascending order.
Anomalous Factor Details	List of anomalous factors associated with the anomaly that triggered the security policy violation. For each anomalous factor, the list shows the factor name, qualitative degree of difference between the observed and expected behaviors, whether the anomalous factor was suppressed at the time the anomaly was triggered, and the observed and expected values. An anomaly can either include an unusual degree or a highly unusual degree.

Data Store Violation Properties

The data store violation properties include the following properties:

Property	Description
Data Store	List of data stores associated with violations within the selected duration and filter conditions. The list includes only the data stores to that which you have access. Click a data store to see proliferation details.
Risk Score	The number indicates the risk level of a data store. Data Privacy Management calculates the risk score using different risk factors. The higher the number, the greater the risk to sensitive data. Click the value to see the risk factors and their respective weights that Data Privacy Management used to calculate the risk score.
Protection Status	The percentage of sensitive fields in the data store that are protected and unprotected. Green represents the percentage of protected fields. Red represents the percentage of unprotected fields.
Targets	The number of target data stores for the source data store. Click the value to see the proliferation path from the source data store to the target data stores.
Sensitive Fields	Number of sensitive fields in the data store. Click the value to see the list of sensitive fields.
Sensitive Records	Number of sensitive records in the data store.
Risk Cost	The cost that the business will incur in the event sensitive data is exposed.
Owner	Individual, group, or organization that is responsible for the accuracy and integrity of the data store.

User Activity Violation Properties

The user activity violation properties are categorized in the following sections:

- User details
- Data store details
- Violation details

User Details

The user details section includes the following properties:

Property	Description
User Name	Name of the user account that performed the user activity event. Data Privacy Management shows the user account name from the source of the user activity information, such as user activity logs.
Full Name	Full name that is associated with the user account name in the Data Privacy Management repository. If the full name is null in the Data Privacy Management repository, the value is a concatenation of the first name and last name. If the first name and last name is null, the value is the user account name. If the user account is not in the repository, the value is the user account name.
Title	Title that is associated with the user account name in the Data Privacy Management repository. If the user account is not in the repository, the title appears as –
Department	User department that is associated with the user account name in the repository.
Location	The location associated with the user account in the repository. This is the location from where the user works. For example, the corporate office that the user is assigned to. **Note:** The location need not necessarily be same as where the user performs the user activity event.
User IP	The IP address of the device the user-activity event originated from.

Data Store Details

The data store details section includes the following properties:

Property	Description
Data Store	The name of the data store that the user accessed.
Owner	Individual, group, or organization that is responsible for the accuracy and integrity of the data store associated with the violation.
Department	Department that is responsible for the maintenance or ownership of the data store.
Location	Geographical area of the data store.
Data Store Group	Logical grouping of related data stores to which the data store associated with the violation belongs.
Data Store Type	Type of database repository for the data store associated with the violation.

Property	Description
Sensitivity Level	A qualitative degree of data sensitivity assigned to the sensitive data associated with the violation.
Risk Score	The number indicates the risk level of a data store. Data Privacy Management calculates the risk score using different risk factors. The higher the number, the greater is the risk to sensitive data.
Risk Cost	The cost that the business will incur in the event sensitive data in the data store is exposed.
Data store host or IP address	The host or IP address for the data store associated with the violation.

Violation Details

The violation details section includes the following properties:

Property	Description
User Activity Date and Time	Date and time of the user activity event. The date and time are in the UTC time zone.
Schema	The names of the schemas associated with the violation.
Object	The names of the objects, such as tables in a relational database, associated with the violation.
Column	The names of the columns associated with the violation.
Operation	Specifies the nature of the request from the user. Values can be Select, Insert, Update, or Delete.
Request	Details of the query that the user specified. For example: `SELECT IP_ADDRESS, AGE, EMPLOYEE_GENDER, CITY, ZIP_CODE FROM CORP_HR.CONSUMER_HR, CORP_HR.CUSTOMERS_HR, CORP_HR.EMPL_ACQ`
Data Domain Count	The number of data domains that match the sensitive fields which were impacted by the user request.
Data Domains	The list of data domains that match the sensitive fields that were impacted by the user request. By default, the list is sorted by the data domain name in ascending order.
Results	The number of records or rows of sensitive data that were impacted by the user request.
Error Status	Specifies whether the user received an error after submitting the query. Data Privacy Management determines the error status from the source of the user activity information, such as user activity logs. If the user did not receive an error after submitting the query, the value is None. If the user received an error, the value is the value in the corresponding field in the user activity logs.
Error Details	If the user received an error message after submitting the query, displays the error message. Otherwise, displays None. Data Privacy Management determines error details from the source of the user activity information, such as user activity logs.
Event Source	The name of the application that streams user activity information to Data Privacy Management.

Security Policy Details

The security policy details include the security policy name, description, the conditions that the anomaly, data store, or user activity event matched, and the action.

The security policy details include the following properties:

Property	Description
Name	Name of the security policy that the violation is based on.
Description	Description of the security policy that the violation is based on.
Rules	Conditions that the anomaly, data store, or user activity event associated with the violation, matched. The rules property also shows what action Data Privacy Management took after detecting the violation.

Security Policy Violations Workspace Views

You can view violations information from different views or perspectives. Each view displays data on violations within the selected time period and filter conditions.

You can view information from the following views:

Data stores view

Displays a list of all data stores that have one or more violations. The view only includes the data stores to which you have access.

Security policies view

Displays a list of all security policies that have one or more violations.

Users view

Displays a list of all users that have one or more violations.

Related Topics:

- "User Profile Page" on page 567

Security Policy Violation Workspace View Properties

To choose a view, select an option from the **View By** list or click the corresponding indicator title. To go back to the default workspace view, select **View By** > **None**.

You can click a column header on the page to sort the list by the selected column. By default, the list is sorted by the number of violations in descending order.

The following table lists the properties that appear on the Data Stores view:

Property	Description
Data Stores	List of data stores that are associated with violations.
Users	Aggregated number of users that have violations for the data store.
Security Policies	Number of security policies that a violation was based on.
Violations	Total number of violations in all severity levels detected for the data store.
Severity	Total number of violations in the selected severity level.

The following table lists the properties that appear on the Security Policies view:

Property	Description
Data Stores	Total number of data stores associated with the security policy.
Users	Aggregated number of users that are associated with the security policy.
Security Policies	Security policies that are associated to violations.
Violations	Total number of violations detected for the security policy.
Severity	Severity level of the violation.
Security Policy Type	Specifies whether the violation is based on an anomaly, data store, or user activity security policy.

The following table lists the properties that appear on the Users view:

Property	Description
Data Stores	Total number of data stores associated with violations for the user. **Note:** The count includes all of the data stores that are associated with the violation, regardless of the data stores to which you have access. When you click the count to open the data stores view, you only see the data stores to which you have access.
Security Policies	Total number of security policies that were the basis of violations triggered by the user.
Violations	Total number of violations in all severity levels detected for the user.
Severity	Total number of violations in the selected severity level detected for the user.
User Name	Name of the user account that performed the user activity event that is associated with the violation.
Full Name	Full name that is associated with the user account name in the Data Privacy Management repository.

Property	Description
Title	The title associated with the user account name in the Data Privacy Management repository.
User Department	User department that is associated with the user account name in the Data Privacy Management repository.

The following points apply to properties across views:

- Data and counts that appear correspond to the selected period and filter conditions.
- Data Stores. On the Security Policies and Users views, you can click the count to view a list of data stores for a selected policy or user.
- Users. On the Data Stores and Security Policies views, you can click the user count to view users related to violations for a data store or security policy.
- Violations. You can click the violations count to view violation information for a data store, security policy, or user.
- Severity. On the Data Stores and Users views, you can click the violations count to view severity information for a data store or user. Data Privacy Management uses the severity level of the security policy as the severity level for the corresponding violation. A violation can have a critical, high, medium, low, or informational severity level.

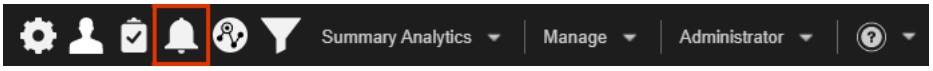
Security Policy Violations Management

Use the **Security Policy Violations** workspace to view all of the violations that Data Privacy Management detected.

You can perform the following tasks in the **Security Policy Violations** workspace:

- Flag or unflag a security policy violation for follow-up.
- Mark a violation as read or unread.
- Filter and view violations.
- Apply and manage security policy violation tags.
- Export information about violations to a CSV file.
- Delete a violation.
- Create a protection task to protect the data stores associated with a violation.

The following image shows the Security Policy Violations icon in the header:



Flagging or Unflagging A Security Policy Violation

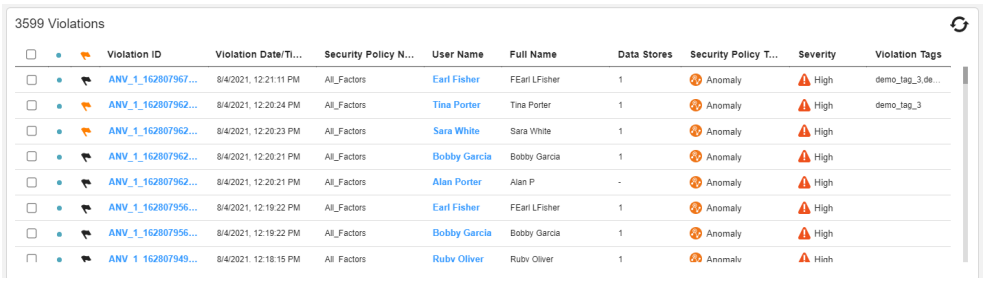
By default, Data Privacy Management creates a violation with an unflagged status. You can flag or unflag violations. For example, you flag violations that you need to review further.

After you flag or unflag a violation, you can sort on the flag status. You can use the status to filter the workspace. For example, you can filter the workspace to only show information for flagged violations. Changes to the flag status do not affect the violation sort order in the violations list panel.

Note: When you flag or unflag a violation, Data Privacy Management updates the violation. The change impacts all users that can see the violation, not just the logged-in user.

1. Click the **Security Policy Violations** icon in the header to open the **Security Policy Violations** workspace.
2. Navigate to the violations list panel and perform one of the following steps:
   - To flag or unflag one violation, click the icon in the flag status column for the associated violation.

     The following image shows an example of the flag status column:



   - To flag or unflag multiple violations, select the check box next to the violations. From the Actions menu, select **Flag** or **Unflag**.

Marking a Violation as Read or Unread

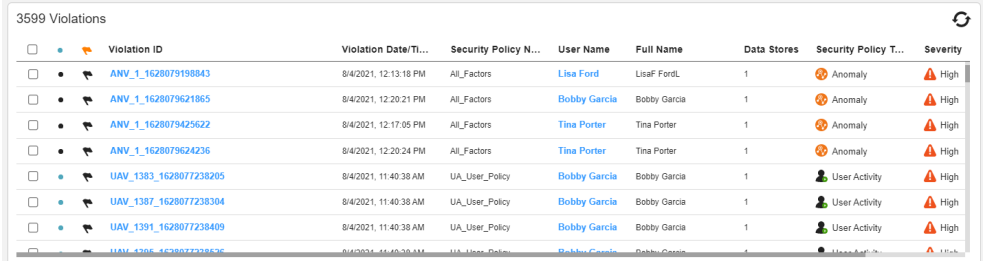
By default, Data Privacy Management creates a violation with an unread status. You can mark a violation as read or unread. When you change the read/unread status of a violation, the sort order of the violations list panel changes. By default, the violations list is first sorted by violations that have an unread status.

After you mark a violation as read or unread, you can use the status to filter the workspace. For example, you can filter the workspace to only show information for unread violations.

Note: When you mark a violation as read or unread, Data Privacy Management updates the violation. The change impacts all users that can see the violation, not just the logged-in user.

1. Click the **Security Policy Violations** icon in the header to open the **Security Policy Violations** workspace.
2. Navigate to the violations list panel and perform one of the following steps:
   - To mark one violation as read or unread, click the icon in the read/unread status column for the associated violation.

     The following image shows an example of the read/unread status column:



- To mark multiple violations as read or unread, select the check box next to the violations. From the Actions menu, select **Mark as Read** or **Mark as Unread**.

Data Privacy Management marks the violation as read or unread and refreshes the sort order based on the status.

Adding or Removing Security Policy Violation Tags

You can apply tags to security policy violations to add identifying information or to help track and manage violations. Add or remove violation tags from the **Security Policy Violations** workspace.

To add or remove tags, select the violations. You can use the workspace filter to filter violations based on tags. If you select multiple violations and then open the **Manage Tags** page, tags that are common to all selected violations appear selected in the **Tags** field. Changes that you make apply to all selected violations.

1. From the list of violations on the **Security Policy Violations** workspace, select the required violations.

   To add and manage tags of a single violation, you can perform the task from the **Security Policy Violation** details page.
2. Click **Actions** > **Manage Tags**.

   The **Manage Tags** page appears.
3. Click the **Tags** field to view a list of tags with a text field to create tags.

   Tags that are common to selected violations appear selected in the list.
4. Perform one of the following steps:
   - To create and apply a tag, enter the tag in the text field and click **Add**.
     The tag is created and applied to the violations.
   - To apply a tag from the list of tags, select the tags.
   - To remove a tag assigned to a violation, clear the tag selection.
     You can remove a tag assignment but cannot delete a tag.

Exporting Violations

You can export information about violations and use the exported file for reporting analysis. You can export information from the Security Policy Violations workspace and from each view in the workspace.

When you export information, Data Privacy Management saves a compressed file to the default download directory on your machine. The compressed file includes CSV files that contain information related to the page you exported from. The information in the CSV files matches the selected filter conditions and time periods at the time of the export. You cannot export information for violations that you manually select in the violation list.

1. Click the **Security Policy Violations** icon in the header to open the **Security Policy Violations** workspace.
2. Optionally, set the filter conditions, time period, and view.
3. From the Actions menu, select **Export**.

   A dialog box appears with the exported file. The export file is named Data.zip. Data Privacy Management saves the file to the default download directory on your machine. If you export the file again, Data Privacy Management downloads a new file. The file name includes an incremental number such as `Data (1).zip`.

Deleting a Violation

You can delete violations in the Security Policy Violations workspace.

Warning: The deletion impacts all Data Privacy Management users, not just the logged-in user. After you delete a violation, it is not possible to restore the violation.

1. Click the **Security Policy Violations** icon in the header to open the **Security Policy Violations** workspace.
2. Select the check box next to the violation that you want to delete.

   You can select one, multiple, or all violations.
3. From the Actions menu, select **Delete**.

   A message appears to confirm that you want to delete the violation.

CHAPTER 23

Tasks

This chapter includes the following topics:

- Tasks Overview, 483
- Tasks Workspace, 484
- Task Status, 495
- Task Management, 497
- Protection Task Management, 502

Tasks Overview

A task performs a specific function depending on the task type and how you configure the task. When a task runs, Data Privacy Management runs an associated job and updates the task status as the job progresses and completes successfully or fails.

DSAR tasks create an automated Data Subject Access Request report as a CSV file.

Protection tasks use one of the following protection extensions to protect sensitive and privacy data:

- Encryption
- Persistent Data Masking - Big Data
- Persistent Data Masking - Remote Domain

System tasks use an extension to perform a corresponding action based on how you configure the action properties. The following task types are system tasks:

- Custom tasks run a script.
- Email tasks send an email.
- Service management tasks create a ServiceNow ticket.
- System log tasks write a system log message to a remote server.

Data subject tasks are custom, email, and service management tasks that fulfill a data subject request.

Data Privacy Management automatically creates some tasks, such as when actions in a security policy include tasks and a violation of the security policy occurs. You can also create protection, system, and data subject tasks manually when you perform manual actions.

You view and manage tasks on the **Tasks** workspace.

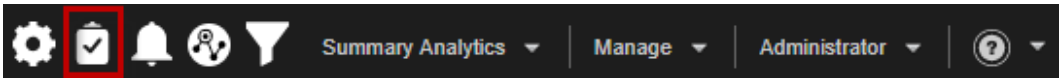
Each task type follows a typical workflow as it progresses through status changes and as the associated job for each task runs.

Tasks Workspace

From the **Tasks** workspace, you can view and manage custom, data subject, DSAR, email, protection, service management, and system log tasks.

The **Tasks** workspace displays a list of tasks and a panel of tabs that displays the number of tasks for the current user, the number of completed tasks, and the total number of tasks. You can also view details and properties pages for each task in the list.

To access the **Tasks** workspace, click the **Tasks** icon in the header:



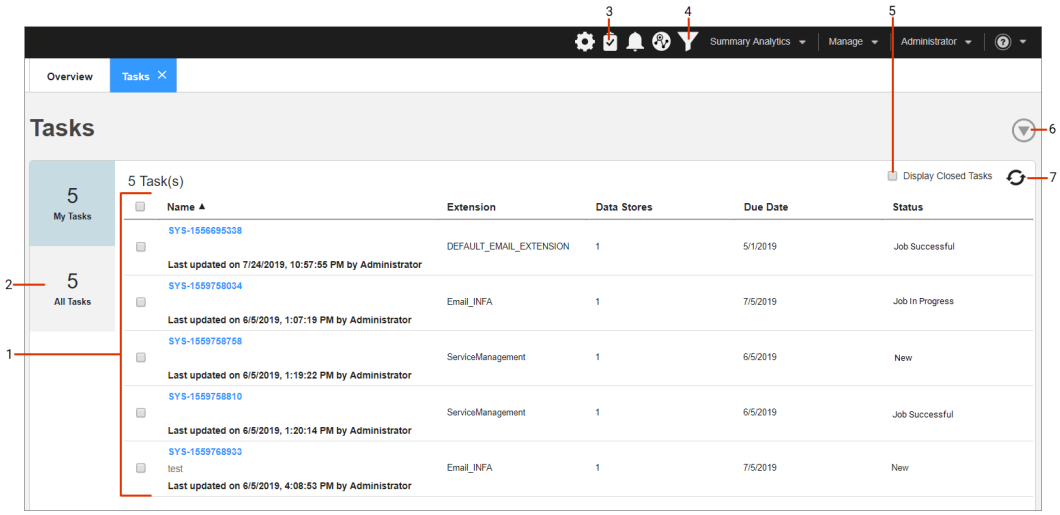
Tasks List Page

The **Tasks** list page displays a list of tasks associated with data stores that you can access. By default, the list of tasks is sorted by tasks that are assigned to you, then by the task name.

To open the **Tasks** workspace, click the **Tasks** icon in the header.

To sort the list of tasks by other than the task name, click a column header or the **Filter** icon to sort the list by specific conditions. To see the list of all tasks that are associated with data stores that you can access, click **All Tasks**.

The following image shows an example of the **Tasks** list page:



1. Task list
2. All Tasks view
3. Tasks icon
4. Filter icon
5. Display Closed Tasks box
6. Actions menu
7. Refresh icon

Task Filter Conditions

To filter the list of tasks on the **Tasks** workspace, click the **Filter** icon on the header. Select or enter filter conditions to narrow the list of tasks.

The following table describes the filter conditions:

Filter Condition	Description
Task Name	Enter a full or partial task name. The workspace displays a list of tasks that match the name or name fragment.
Task Status	Select a task status to view the list of tasks in that status.
Data Store	Enter a full or partial data store name. The workspace displays the list of tasks associated with the data stores that match the name or name fragment.
Data Store Categories	Select one or more of the following data store categories: Application, Big Data, Cloud, Database Management, Data Integration, File Management. To view tasks associated with all data store categories, select All.
Data Store Types	Select one or more of the data store types that Data Privacy Management lists on the **Data Stores** workspace. To view tasks associated with all data store types, select All.
Extension	Select an extension to view the list of tasks associated with the extension.
Data Domains	Select one or more data domains to view the list of tasks associated with the selected domains. To view tasks associated with all data domains, select All.
Current Task Owner	Select a user name to view the list of tasks the user currently owns.
Users Associated with Task Any Time	Select one or more user names to view the list of tasks assigned to the users at any point during the task workflow. To view tasks associated with all users at any time, select All.
Task Tags	Select one or more tags to view the list of tasks associated with the tags.
Due Date	Click in the field to view the **Select Date Range** dialog box. From the calenders in the **From** and **To** panes, select a due date range and then click **Apply**. The workspace displays a list of tasks with due dates in the range you selected.

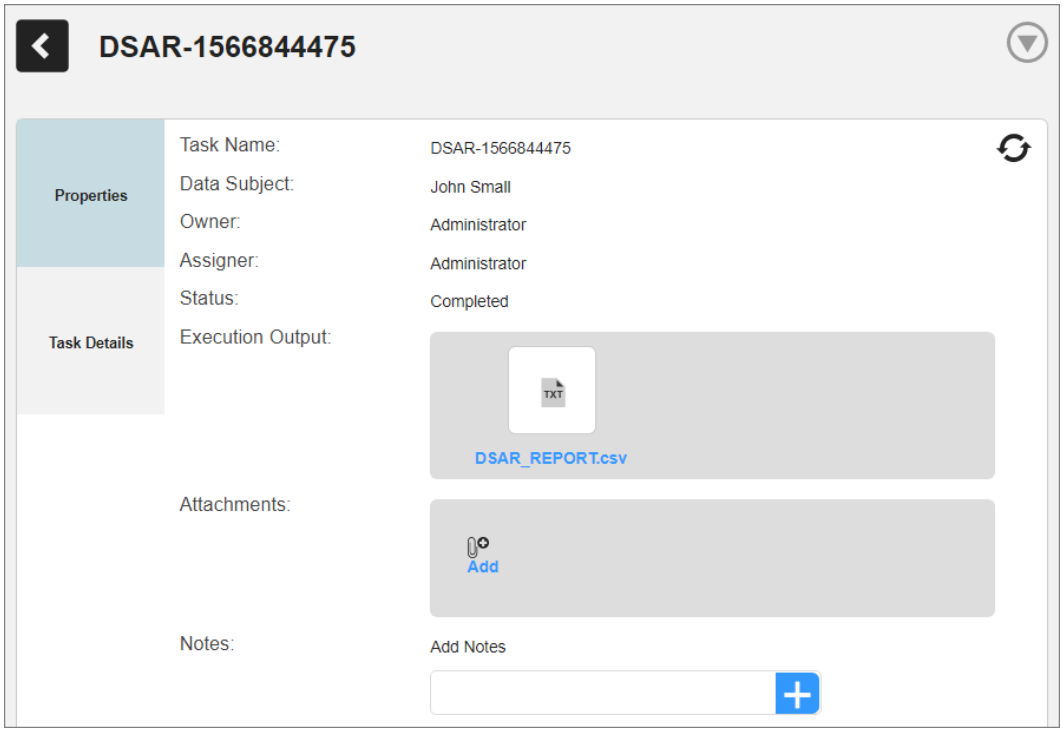
DSAR Task Properties

The DSAR task **Properties** page shows a set of properties about a DSAR task. Data Privacy Management automatically runs the task to generate a Data Subject Access Request report from a data **Subject Details** page when you select **Prepare DSAR** from the **Actions** menu.

The task and the report appear on the **Tasks** workspace.

To access the **Properties** page for a DSAR task, click a task name on the **Tasks** list page.

The <*task name*> **Properties** page appears. The following image shows an example **Properties** page for the task DSAR-1566844475:



The DSAR task **Properties** page includes the following properties:

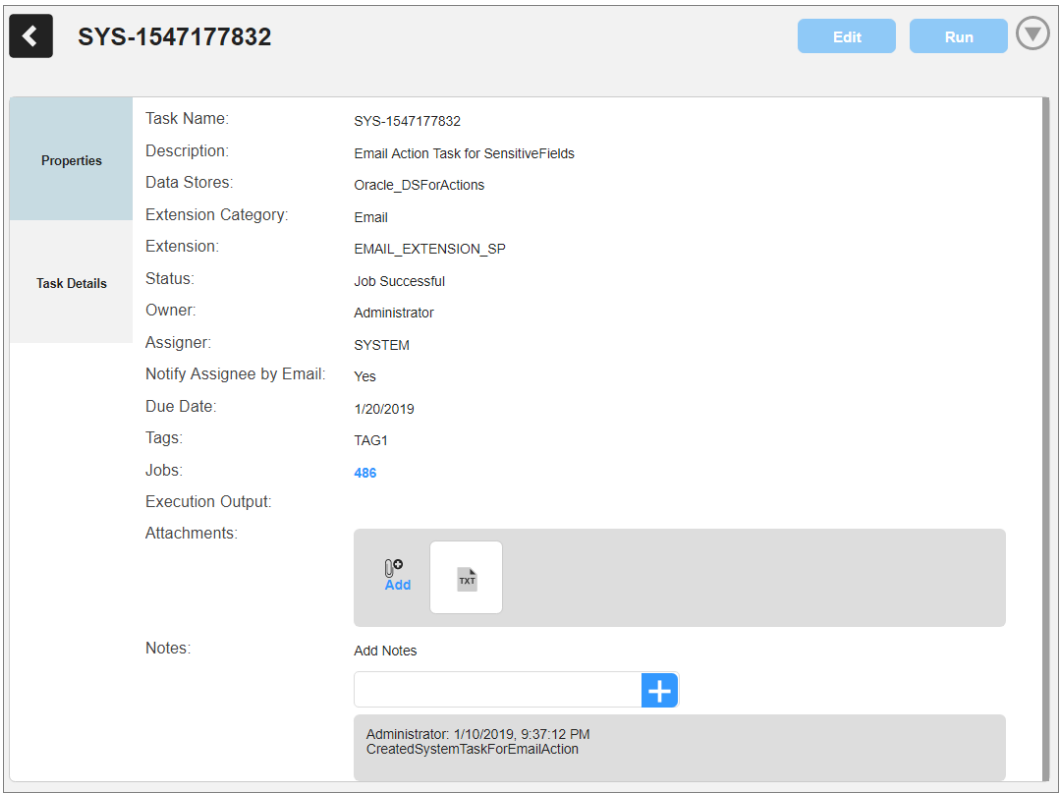
Property	Description
Task Name	The name that Data Privacy Management automatically generates when the task is created, in the following format: *<DSAR>-<10-digit number>*.
Data Subject	The name of the data subject selected on the **Subject Registry** workspace for creating a DSAR report.
Owner	Lists the user name of the current task owner.
Assigner	Shows whether the Data Privacy Management system or a user assigned the task to the current owner.
Status	Lists the status of the task.
Execution Output	Contains a DSAR report as a CSV file attachment. If the task encountered errors, the property also contains a DSAR error report as a CSV file attachment. Click each file name to download one CSV file at a time.
Attachments	Optional. Includes any attachments that users added to the task.
Notes	Includes any notes that the task creator added to the task. To add other notes, type in the text field and then click the **+** icon. Data Privacy Management lists the most recent notes at the top of the display box. The first line contains the name of the user who added the note and the date and timestamp the user entered the note. The second line contains the note text.

Protection, System, and Data Subject Task Properties

Data Privacy Management lists a set of common properties for protection, system, and data subject tasks on the task **Properties** page. Protection tasks mask or encrypt sensitive privacy data with a protection extension. System tasks include running a custom script, sending an email, creating a service management ticket, and writing a system log message. Data subject tasks are custom, email, or service management tasks created on the **Subject Details** page.

To access the **Properties** page for a protection, system, or data subject task, click a task name on the **Tasks** list page.

The *<task name>* **Properties** page appears. The following image shows an example **Properties** page for the email task SYS-1547177832:



The protection, system, and data subject task **Properties** pages include the following properties:

Property	Description
Task Name	The name that is automatically generated when the task is created, in the following format: *<prefix>-<10-digit number>*. - For tasks configured to run with a custom, email, service management, or system log extension, the task name prefix is `SYS`. - For custom, email, and service management tasks configured for a data subject, the task name prefix is `DSR`. - For tasks configured to run with an encryption or other protection extension, the task name prefix is `PRT`.
Description	Optional. A description of the task that uses a maximum of 255 characters. Task descriptions help you find tasks on the **Tasks** list page. You can edit this property when you edit the task.

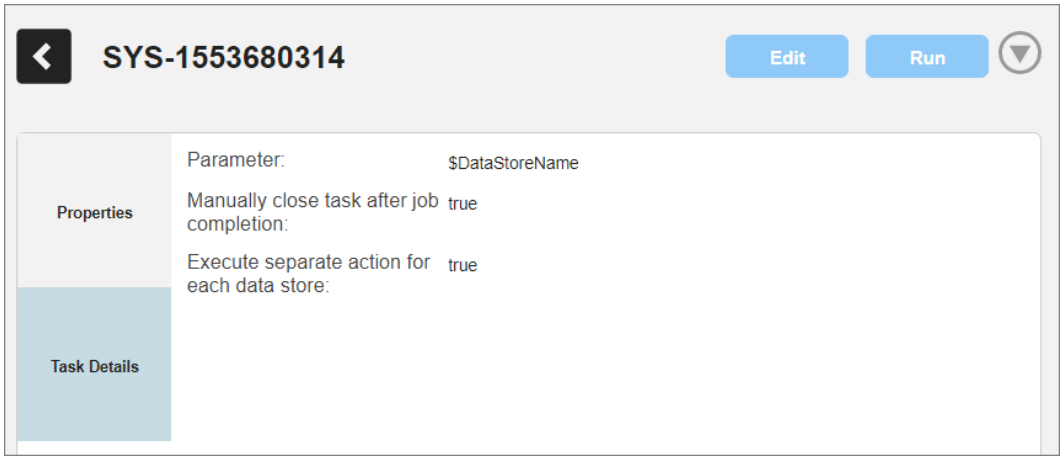
Property	Description
Data Stores	For protection and system tasks only, lists the data stores included in the task. Each protection task is associated with only one data store. System tasks can include more than one data store.
Data Subject	For data subject tasks only, lists the name of the data subject that displays on the **Subject Registry** workspace.
Violations	If a protection or system task runs because a violation of a security policy that contains an action to run the task occurred, this property appears on the task **Properties** page. The property lists the security policy violation IDs associated with the task. You can view the violation details on the **Security Policy Violations** workspace.
Extension Category	Lists one of the following categories for the extension that runs the task: Custom, Email, Protection, Service Management, or System Log.
Extension	Lists the name of the extension plugin configured on the **Extensions** workspace.
Status	Lists the status of the task.
Owner	Lists the user name of the current task owner.
Assigner	Shows whether the Data Privacy Management system or a user assigned the task to the current owner.
Notify Assignee Via Email	Specifies whether or not Data Privacy Management sends an email to the user assigned to the task to notify the user that the task is created. Default value is Yes. You can edit this property when you edit the task.
Due Date	The date the task is due. The default value is the current date. You can edit this property when you edit the task.
Tags	Optional. Lists any tags associated with the task. You can edit this property when you edit the task.
Jobs	For tasks that Data Privacy Management has not run, the property is empty. For tasks that have run, lists the job ID for each Orchestration or Protection job that ran the task. You can click the job ID to view the **Job Details** page on the **Jobs** workspace.
Execution Output	For custom, email, protection, and system log tasks, the property is always empty. For service management tasks that ran successfully, provides a link to the ServiceNow ticket. To view the ticket that the task created, click the URL.
Attachments	Optional. Includes any attachments users added to the task. Also includes a .txt file if the task creator selected the **Add Context to Task Details as Attachment** option.
Notes	For protection tasks that have run and for all system tasks, contains information about the task creation. The first line contains the name of the user who created the task and the date and timestamp the user created the task. On a second line, protection tasks display the text in the Description property. System tasks that are not created for a data subject display the following syntax: `CreatedSystemTaskFor<extension category>Action` For protection, system, and data subject tasks, includes any notes that the task creator added. You can add other notes. Type in the text field and then click the **+** icon. Data Privacy Management lists the most recent notes at the top of the display box. The first line contains the name of the user who added the note and the date and timestamp the user entered the note. The second line contains the note text.

Custom Task Details Page

You can view additional details about a custom task on the custom **Task Details** page. A custom task runs a script based on the properties defined in a custom action and a custom extension.

To access the custom **Task Details** page, click a custom task name on the **Tasks** list page. On the page that appears with the task name at the top, click the **Task Details** tab.

The following image shows the custom **Task Details** page for the task SYS-1553680314:



The custom **Task Details** page includes the following properties:

Property	Description
Parameter	Shows placeholders defined in the custom action associated with the task.
Action Type	For data subject custom tasks only. Displays the type of subject request that the task fulfills.
Report	For data subject custom tasks only. Indicates whether the task provides attributes only or attributes and values.
Manually close task after job completion	If true, indicates that a user must manually close the task after the associated Orchestration job completes successfully. If false, Data Privacy Management closes the task after the task reaches the Job Successful status.
Execute separate action for each data store	For system custom tasks only. If true, Data Privacy Management creates a separate task for each data store included in the action when the custom script runs. If false, Data Privacy Management creates a single task for all affected data stores when the custom script runs.
Edit	Click to edit the task details or properties if the task has a status of New or Job Successful.
Run	Click to run the task if the task has a status of New and the task is assigned to you.

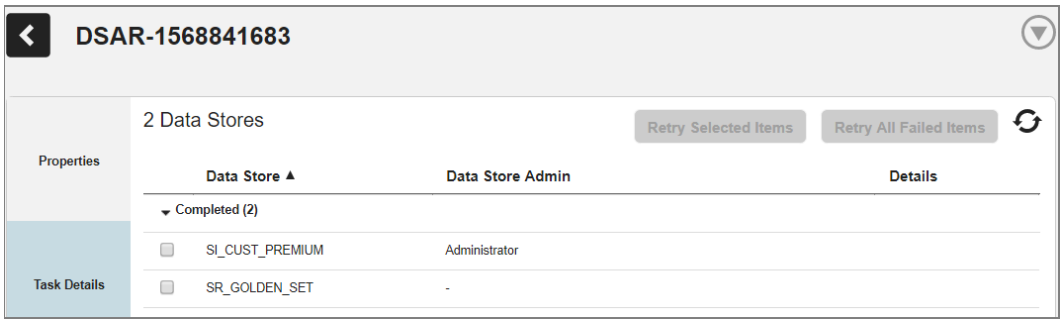
DSAR Task Details Page

The DSAR **Task Details** page shows additional details about a DSAR task. Data Privacy Management automatically runs the task to generate a Data Subject Access Request report from a data **Subject Details**

page when you select **Prepare DSAR** from the **Actions** menu. The task and the report appear on the **Tasks** workspace.

To access the DSAR **Task Details** page, click a DSAR task name on the **Tasks** list page. On the page that appears with the task name at the top, click the **Task Details** tab.

The following image shows the DSAR **Task Details** page for the completed task DSAR-1568841683:



The DSAR **Task Details** page includes the following properties:

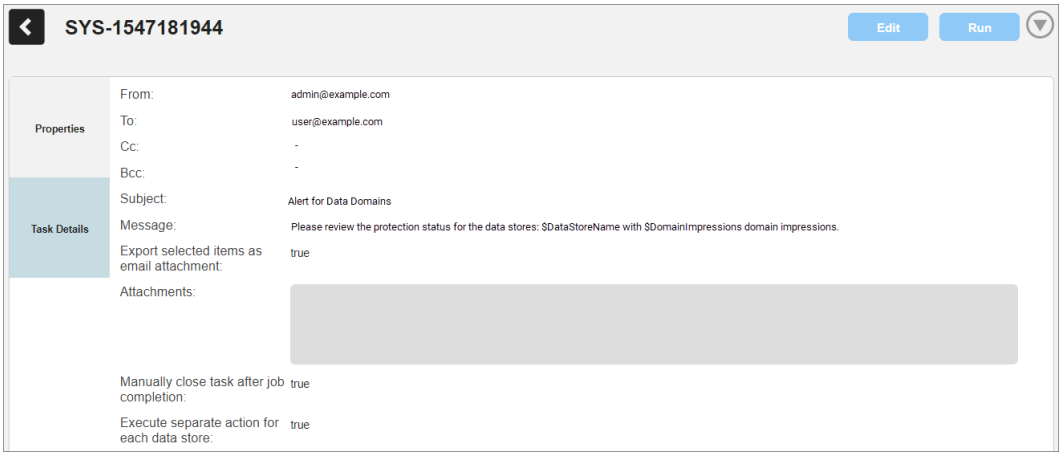
Property	Description
Data Store	Lists the data stores that contain privacy data about the data subject in the DSAR report. Contains a list of the data stores that completed successfully when the task ran and the data stores that failed when the task ran.
Data Store Admin	The user name of the administrator for each data store.
Details	Shows details about each data store, if available.
Retry Selected Items	For DSAR tasks that are not in progress, click after selecting one or more completed or failed data stores to run the DSAR task again for the selected data stores and the data subject.
Retry All Failed Items	For DSAR tasks that are not in progress, click to run the DSAR task again for the failed data stores and the data subject.

Email Task Details Page

You can view additional details for an email task on the email **Task Details** page. An email task sends an email to specified email addresses based on the properties defined in an email action and an email extension.

To access the email **Task Details** page, click an email task name on the **Tasks** list page. On the page that appears with the task name at the top, click the **Task Details** tab.

The following image shows the email task details for the task SYS-1547181944:



The email **Task Details** page includes the following properties:

Property	Description
From	The email address of the sender.
To	The email addresses of the primary recipients.
Cc	The email addresses of the copied recipients.
Bcc	The email addresses of blind copied recipients that are not displayed when the task runs.
Subject	The subject of the email message that Data Privacy Management sends to the specified recipients when the task runs.
Message	The contents of the email message that Data Privacy Management sends to the specified recipients when the task runs. The message can contain placeholders.
Export selected items as email attachment	If true, indicates that Data Privacy Management will attach a CSV file with the task details to the email the task sends to the specified recipients.
Attachments	Contains any attachments that the task creator added to the task.
Action Type	For data subject email tasks only. Displays the type of subject request that the task fulfills.
Report	For data subject email tasks only. Indicates whether the task provides attributes only or attributes and values.
Attach DSAR Report if Available	For data subject email tasks only. If true, indicates that Data Privacy Management attaches a DSAR report, if available, to the email message.
Manually close task after job completion	If true, indicates that a user must manually close the task after the associated Orchestration job completes successfully. If false, Data Privacy Management closes the task after the task reaches the Job Successful status.
Execute separate action for each data store	For system email tasks only. If true, Data Privacy Management creates a separate task for each data store included in the action when the custom script runs. If false, Data Privacy Management creates a single task for all affected data stores when the custom script runs.

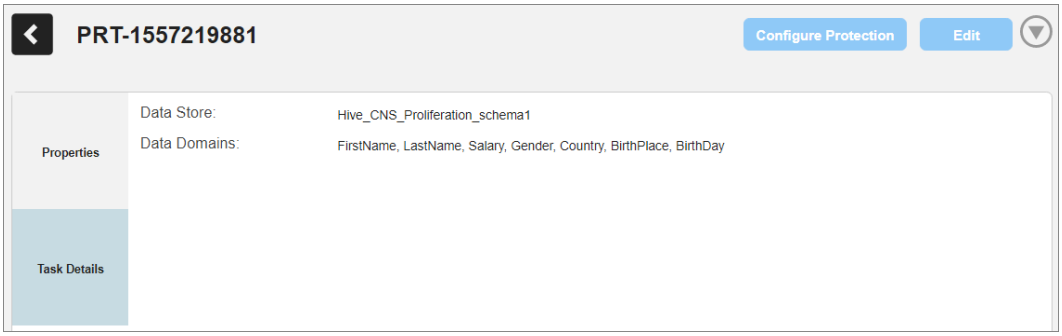
Property	Description
Edit	Click to edit the task details or properties if the task has a status of New or Job Successful.
Run	Click to run the task if the task has a status of New and the task is assigned to you.

Protection Task Details Page

The protection **Task Details** page shows the details for a protection task. Protection tasks use an encryption or Persistent Data Masking extension to protect sensitive data in the data store specified in the task. Data Privacy Management creates a protection task for each data store.

To access the protection **Task Details** page, click a protection task name on the **Tasks** list page. On the page that appears with the task name at the top, click the **Task Details** tab.

The following image shows the protection task details for the task PRT-1547181944, which uses a Persistent Data Masking - Big Data Management protection extension and has a status of New:



The protection **Task Details** page includes the following properties:

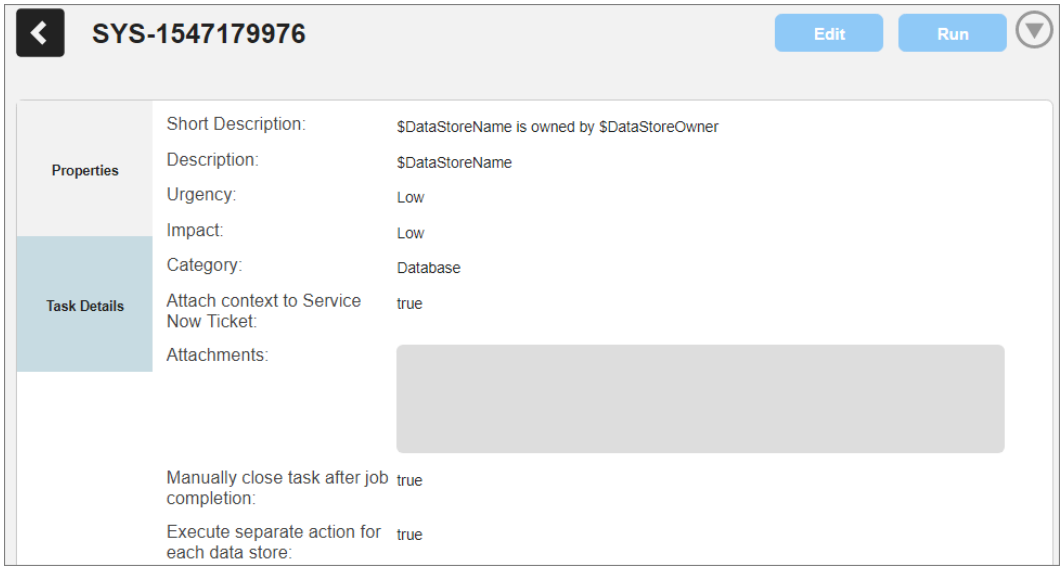
Protection Task Property	Description
Data Store	The data store that contains sensitive data that Data Privacy Management protects when the task runs.
Data Domains	The data domains included in the data store that contain sensitive fields that you will protect when the task runs.
Configure Protection	Click to configure the protection rules for sensitive fields if the task has a status of New.
Edit	Click to edit task details if the task has a status of Job In Progress, Job Failed, or Closed.
Schedule Protection Job	Click to schedule when the task will run if the task has a status of Configured.
View Configuration	Click to view the configuration for a protection task in any status except for New.

Service Management Task Details Page

The service management **Task Details** page shows the details for a service management task. Service management tasks create ServiceNow tickets based on the properties defined in service management actions and extensions.

To access the service management **Task Details** page, click a service management task name on the **Tasks** list page. On the page that appears with the task name at the top, click the **Task Details** tab.

The following image shows the service management task details for the task SYS-1547179976:



The service management **Task Details** page includes the following properties:

Property	Description
Short Description	A description of the task that can contain placeholders.
Description	A description of the task that appears under the task name on the **Tasks** list page. The description can contain placeholders.
Urgency	One of the following urgency levels: High, Low, Medium.
Impact	One of the following impact levels: High, Low, Medium.
Category	One of the following categories: Database, Hardware, Inquiry, Network, Software.
Attach context to Service Now Ticket	For system service management tasks only. If true, indicates that Data Privacy Management will attach a CSV file with the task details to the ServiceNow ticket that the task creates.
Attachments	Contains any attachments that the task creator added to the task.
Wait for Service Now Ticket Status Change	For data subject service management tasks only. If true, indicates that the ServiceNow ticket status must change to In Progress before the service management task in Data Privacy Management can close.
Action Type	For data subject service management tasks only. Displays the type of subject request that the task fulfills.

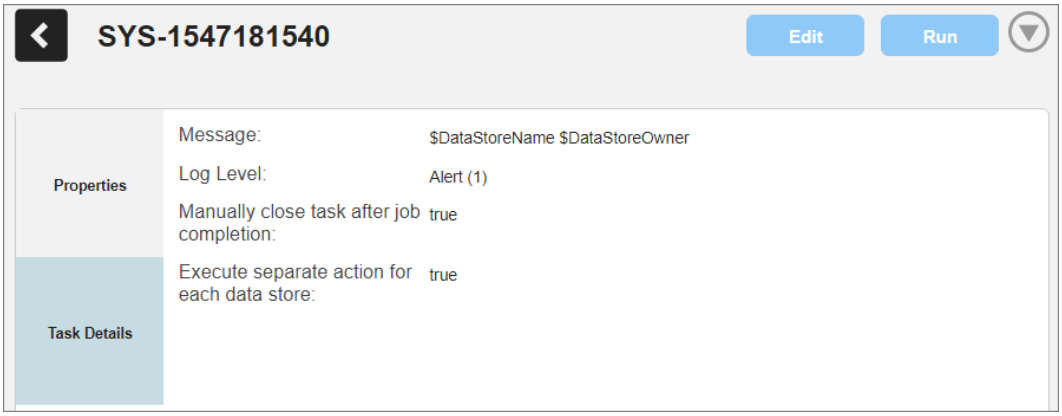
Property	Description
Report	For data subject service management tasks only. Indicates whether the task provides attributes only or attributes and values.
Attach DSAR Report if Available	For data subject service management tasks only. If true, indicates that the task attaches a DSAR report, if available, to the ServiceNow ticket.
Manually close task after job completion	If true, indicates that a user must manually close the task after the associated Orchestration job completes successfully. If false, Data Privacy Management closes the task after the task reaches the Job Successful status.
Execute separate action for each data store	For system service management tasks only. If true, Data Privacy Management creates a separate task for each data store included in the action when the custom script runs. If false, Data Privacy Management creates a single task for all affected data stores when the custom script runs.
Edit	Click to edit the task details or properties if the task has a status of New or Job Successful.
Run	Click to run the task if the task has a status of New and the task is assigned to you.

System Log Task Details Page

The system log **Task Details** page shows the task details for a system log task. System log tasks write system log messages to a remote server based on the properties defined in system log actions and extensions.

To access the system log **Task Details** page, click a system log task name on the **Tasks** list page. On the page that appears with the task name at the top, click the **Task Details** tab.

The following image shows the system log task details for the task SYS-1547181540:



The system log **Task Details** page includes the following properties:

System Log Task Property	Description
Message	The contents of the message that system log task writes to the remote server. The message can contain placeholders.
Log Level	Lists one of the following log severity levels: - Emergency (0) - Alert (1) - Critical (2) - Error (3) - Warning (4) - Notice (5) - Informational (6) - Debug (7)
Manually close task after job completion	If true, indicates that a user must manually close the task after the associated Orchestration job completes successfully. If false, Data Privacy Management closes the task after the task reaches the Job Successful status.
Execute separate action for each data store	If true, Data Privacy Management creates a separate task for each data store included in the action when the custom script runs. If false, Data Privacy Management creates a single task for all affected data stores when the custom script runs.
Edit	Click to edit the task details or properties if the task has a status of New or Job Successful.
Run	Click to run the task if the task has a status of New and the task is assigned to you.

Task Status

Data Privacy Management tasks change status as they progress through the task workflow. The **Tasks** list page shows the status for each task in the **Status** column.

When protection, system, and data subject tasks run, Data Privacy Management runs an associated job that you can view and track on the **Jobs** workspace.

- For protection tasks, Data Privacy Management runs Protection jobs.
- For system and data subject tasks, Data Privacy Management runs Orchestration jobs.

For DSAR tasks, Data Privacy Management runs an associated job that does not appear on the **Jobs** workspace.

Data Privacy Management assigns the following task statuses:

Task Status	Task Type	Description	Next Steps
New	Protection, System, Data Subject	The status of tasks when they first appear on the **Tasks** workspace.	- For protection tasks, assign a protection extension to the task and configure the task. - For system and data subject tasks, run the task. You can also edit the task details and properties before configuring or running the task.
Configured	Protection	The status after you finish configuring a protection task.	Schedule a protection job for the task. You can also select one of the following options from the **Actions** menu: - Reassign Task - Upload Attachment - Mark Task as Completed - Mark Task as Closed - Reject Task - Reconfigure Task - Edit Task - Delete Task - Export
In Progress	DSAR	The status of DSAR tasks when they first appear on the **Tasks** workspace and are running automatically.	Wait for the task to finish with a status of Completed. You can then view the task **Properties** page and download the DSAR report as a CSV file, or download a DSAR Report from the **Subject Details** page.
Job Scheduled	Protection	The Protection job is scheduled to run at a specified time in the future.	Wait until the job runs. You can also select one of the following options from the **Actions** menu: - Reassign Task - Upload Attachment - Reconfigure Task - Edit Task - Export
Job In Progress	Protection, System, Data Subject	The Protection or Orchestration job for the task is running.	Wait until the job finishes. If required, you can pause, stop, or resume a running Protection or Orchestration job on the **Jobs** workspace.
Job Failed	Protection, System, Data Subject	The Protection or Orchestration job is terminated and you can no longer run or resume the job on the **Jobs** workspace.	To delete the task, delete the terminated job on the**Jobs** workspace. Return to the **Tasks** workspace, select the task, and then delete the task.
Job Successful	Protection, System, Data Subject	The job associated with running the task completed successfully.	- For system tasks that enable the **Manually close task after job completion** option, close the task. - For protection tasks, complete the task.

Task Status	Task Type	Description	Next Steps
Completed	DSAR and Protection	For DSAR tasks, Data Privacy Management finished generating the requested DSAR report for a subject. For protection tasks, the Protection job was successful and the task is marked as complete.	Close the task. For protection tasks, you can also select one of the following options from the **Actions** menu: - Reassign Task - Reconfigure Task - Edit Task - Delete Task
Closed	DSAR, Protection, System, Data Subject	For system and data subject tasks that do not enable the **Manually close task after job completion** option, Data Privacy Management automatically closes the task after the job completes successfully. For DSAR and protection tasks, indicates that the task was marked as complete and then closed.	From the **Actions** menu, you can select one of the following options: - Delete Task - Rollback (encryption protection tasks only) When a protection task is closed, Data Privacy Management copies the protection status for the columns in the task to the columns on the data store.
Rejected	Protection	A user rejected the configuration of the protection task.	You can select one of the following options from the **Actions** menu: - Configure Task - Reassign Task - Mark Task as Configured - Mark Task as Closed - Mark Task as Completed - Edit Task - Delete Task - Export

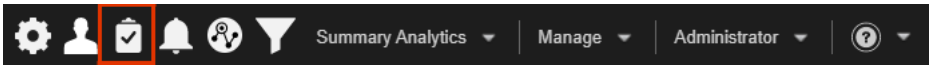
Related Topics:

- "Job Management" on page 307
- "Job Types" on page 287

Task Management

You can manage custom, data subject, DSAR, email, protection, service management, and system log tasks on the **Tasks** workspace.

The following image shows the Tasks icon in the header:



You create new protection and system tasks when you take manual action from the following Data Privacy Management views:

- **Anomaly Detection** workspace.
- **Proliferation** page

- **Security Policy Violations** workspace.
- **Sensitive Fields** page
- **Top Data Domains** list page
- **Top Data Stores** grid page

You create new DSAR tasks when you select **Actions** > **Prepare DSAR** on the **Subject Details** page.

You create data subject tasks that create a service management ticket, run a custom script, or send an email when you select an option in the **Actions** > **Take Action** menu on the **Subject Details** page.

Data Privacy Management creates new tasks when actions are included in an anomaly, data store, or user activity security policy and a violation of that policy occurs.

The topics in this section provide instructions for managing data subject, DSAR, protection, and system tasks.

Adding Attachments to a Task

You can upload attachments to a task if the task is assigned to you. For example, upload evidence reports to a protection task that support the claim that protection is complete for the protected data store columns.

1. On the **Tasks** workspace. select the check box next to a task name.
2. Click **Actions** > **Upload Attachment**.
3. Navigate to the directory on your machine that contains the file. Select the file and click **Open**.

   The **Task Details and Properties** page appears and the file is added in the **Attachments** field. The attachment remains in the task properties.
4. Optionally, click **Add** to attach more files.

Removing Attachments from a Task

You can remove an attachment from a task that is still open and that is assigned to you.

For example, you can remove an existing proof of protection file from a protection task and replace it by adding a new proof of protection file to the task.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Click the name of the task for which you want to remove an attachment.

   The **Task Details and Properties** page appears.
3. In the **Attachments** field, click the **X** at the top right of the attachment file icon to delete the file.
4. Optionally, click **+** to attach a new file.
5. Optionally, you can edit, mark the task as closed, or reassign any task type. You can run custom, data subject, email, service management, or system log tasks. If the task is a protection task, you can view the task configuration, schedule a protection job for the task, mark the task as completed, or reject the task.

Deleting a Task

You can delete a protection task from the **Tasks** workspace if the task status is Configured, Completed, Job Successful, Rejected, or Closed. You can delete a custom, email, service management, or system log task if the task status is Job Successful or Closed. You cannot delete DSAR and DSR tasks.

Before performing these steps, terminate any running jobs for the task.

1. On the **Tasks** workspace, select the check box next to one or more tasks that you want to delete.
2. Click **Actions** > **Delete Task**.

   The **Confirm Delete** prompt asks if you are sure you want to delete the selected tasks.
3. Click **Yes, I'm Sure**.

Editing a Custom Task

You can edit custom task properties from the **Tasks** workspace.

The task must be assigned to you and you must have privileges to edit tasks. You cannot edit a task that has a status of Job In Progress, Job Failed, or Closed.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. To edit a custom task, click the task name to display the **Task Details and Properties** page. Click **Edit**.
3. On the **Properties** view, you can edit the description, the **Notify Assignee Via Email** check box, the due date, tags, and add any notes.
4. On the **Task Details** view, you can edit the parameter and the check boxes that specify how Data Privacy Management will run and close the task.

   For data subject custom tasks, you can also edit the Action Type and Report properties.
5. Click **Save**.

Editing an Email Task

You can edit email task properties from the **Tasks** workspace.

The task must be assigned to you and you must have privileges to edit tasks. You cannot edit a task that has a status of Job In Progress, Job Failed, or Closed.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. To edit an email task, click the task name to display the **Task Details and Properties** page. Click **Edit**.
3. On the **Properties** view, you can edit the description, the **Notify Assignee Via Email** check box, the due date, tags, and notes.
4. On the **Task Details** view, you can edit the sender and recipient email addresses, email subject and message, and the check boxes that specify how Data Privacy Management will run and close the task. You can also add attachments.

   For data subject email tasks, you can also edit the request type, Report property, and the option to attach a DSAR report.
5. Click **Save**.

Editing a Service Management Task

You can edit service management task properties from the **Tasks** workspace.

The task must be assigned to you and you must have privileges to edit tasks. You cannot edit a task that has a status of Job In Progress, Job Failed, or Closed.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. To edit a service management task, click the task name to display the **Task Details and Properties** page. Click **Edit**.
3. On the **Properties** view, you can edit the description, the **Notify Assignee Via Email** check box, the due date, tags, and notes.
4. On the **Task Details** view, you can edit the short description, description, and the urgency, impact and category selections. You can also add attachments and edit the check boxes that specify how Data Privacy Management will run and close the task.

   For data subject service management tasks, you can edit the request type, Report property, the option to wait for a ServiceNow ticket status change, and the option to attach a DSAR report.
5. Click **Save**.

Editing a System Log Task

You can edit system log task properties from the **Tasks** workspace.

The task must be assigned to you and you must have privileges to edit tasks. You cannot edit a task that has a status of Job In Progress, Job Failed, or Closed.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. To edit a service management task, click the task name to display the **Task Details and Properties** page. Click **Edit**.
3. On the **Properties** view, you can edit the description, the **Notify Assignee Via Email** check box, the due date, tags, and notes.
4. On the **Task Details** view, you can edit the message, log level selection, and the check boxes that specify how Data Privacy Management will run and close the task. Optionally, you can use placeholders in the message.
5. Click **Save**.

Exporting a Task

You can export task information on the **Tasks** workspace to a CSV file. You can use the information from the exported file for reporting analysis.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Optionally, sort the tasks by extension, number of data stores, due date, or status.
3. From the **Actions** menu, select **Export**.

   A prompt at the bottom of the **Tasks** workspace asks if you want to open or save the CSV file. The default file name is `Tasks.csv`.

4. Save the file in one of the following ways:
   - Click **Open** to view the CSV file and optionally edit and save the file.
   - Click **Save** to save the CSV file with the default file name to the default download directory on your machine.
   - Click **Save As** to specify a file name and directory in which to save the CSV file.
   - Click **Save and Open** to view the CSV file and save the file with the default file name to the default download directory on your machine at the same time.

   If you export the file again, the Data Privacy Management Service downloads a new file. The default file name includes an incremental number such as `Tasks (2).csv`.

Marking a Task as Closed

You can mark a task as closed after the associated job is successful. You must mark protection tasks as completed before marking them as closed.

Note: For custom, email, service management, and system log actions, if the **Manually close task after job completion** check box is not selected in the action properties, Data Privacy Management will automatically change the status of the associated task to Closed after the job completes successfully. In this case, you do not need to perform these steps to close a task.

On the **Tasks** workspace, mark a task as closed in one of the following ways:

1. Select the check box next to one or more system or data subject tasks that have a status of Job Successful, or one or more protection tasks that have a status of Completed. From the **Actions** menu, select **Mark Task as Closed**.
2. Click a task name to display the **Task Details and Properties** page. From the **Actions** menu, select **Mark Task as Closed**.

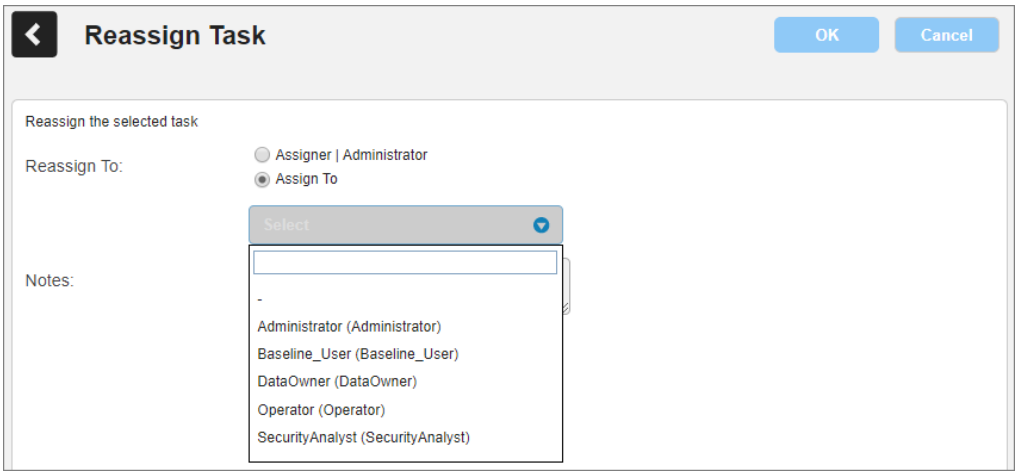
   If no errors occurred, Data Privacy Management updates the task status to Closed. For protection tasks, Data Privacy Management also copies the protection status for the columns in the task to the columns on the data store.
   By default, the **Tasks** workspace does not list closed tasks. To view closed tasks, select the **Display Closed Tasks** check box.

Reassigning a Task

At any stage of the workflow before a protection, system, or data subject task is closed, you can reassign a task to another Data Privacy Management user.

1. On the **Tasks** workspace, select the check box next to one or more tasks you want to reassign.
2. From the **Actions** menu, select **Reassign Task**.

The **Reassign Task** page appears:



3. To assign the task back to the original assigner or the Administrator, select **Assigner | Administrator.** To assign the task to another user, select **Assign To** and then select a user from the list.
4. Optionally, enter notes.
5. Click **OK**.

Running a System or Data Subject Task

When you select **Take Action** from the **Actions** menu in Data Privacy Management to create a task on demand, you can save the action as a task that appears on the **Tasks** workspace with a New status.

System tasks appear with a prefix of `SYS`. Data subject tasks appear with a prefix of `DSR`.

The user assigned to the new task runs the task manually.

1. On the **Tasks** workspace, select the check box next to one or more new custom, data subject, email, service management, or system log tasks that you want to run.
2. Click **Actions** > **Run Task**.

   If the task ran successfully, the status changes to Job Successful. You can run the task again or mark the task as closed.

Protection Task Management

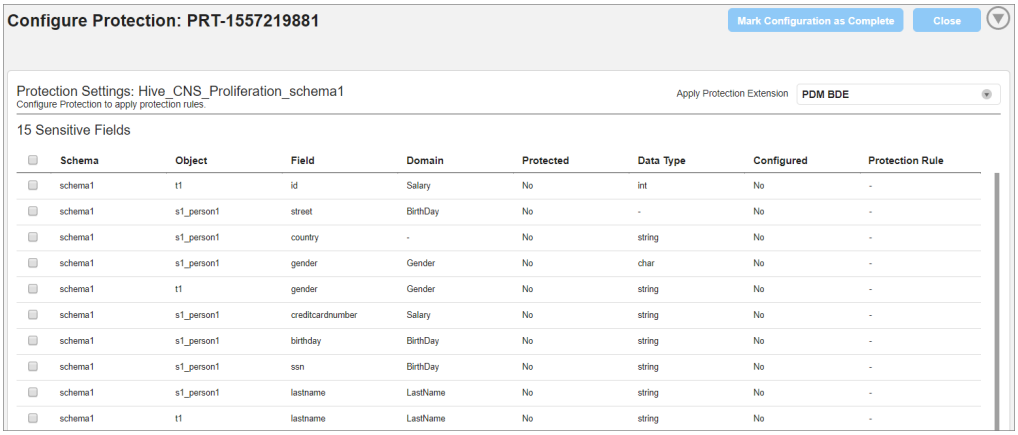
The topics in this section provide instructions for managing protection tasks on the **Tasks** workspace.

Configuring Protection Tasks

You can configure the protection details for tasks assigned to you that have a status of New or Rejected. Configure the protection rules for each sensitive field in the data store that is associated with the task. The protection extension specified in the task determines the protection rules that you configure.

1. On the **Tasks** list page, select the check box next to the protection task you want to configure.
2. Select **Actions** > **Configure Task**.

The **Configure Protection: PRT-<task_number>** page displays the sensitive fields for the data store associated with the task.
The following image shows this view for the task PRT-1557219881 that will apply protection rules from a PDM BDE protection extension to the sensitive fields in a Hive data store:



3. Select the check box next to one or more sensitive fields to configure for protection. You can select fields that have the same data type.

   Important: To configure protection for sensitive fields, the field data type must be defined. You cannot apply protection to sensitive fields that do not have a value in the **Data Type** column. To configure encryption rules for sensitive fields in Cloudera Hive data stores, the **Data Type** column must have a value of string.

   The **Protection Properties** dialog box displays configuration options based on the protection extension defined in the **Apply Protection Extension** field.

4. Optionally, to apply the default protection rules to all sensitive fields in the data store based on the protection extension and the rules specified in data domains, select the top left check box next to the **Schema** column and then select **Apply default rules** from the **Actions** menu.

   A confirmation prompt asks you to confirm that you want to apply default protection rules for all data domains in the data store. To confirm, click **Yes, I'm Sure**. Data Privacy Management displays a message indicating the number of columns for which the default data domain protection rules are applied.

5. To mark one or more selected sensitive fields as protected, select **Mark as protected** from the **Actions** menu.

6. To mark one or more selected sensitive fields as not protected, select **Mark as unprotected** from the **Actions** menu.

7. To erase the configuration settings you applied before completing the configuration, select **Clear confirguration** from the **Actions** menu.

8. To manually configure other sensitive fields in the data store, or to configure protection rules that differ from the default rules defined in data domains, perform the steps that correspond to the protection extension for the task:

   - "Configuring Encryption Rules and Encryption Keys" on page 504
   - "Editing Persistent Data Masking Rules" on page 505

When finished, change the protection task status to Configured in one of the following ways:

- On the **Task Details** page, select **Mark Configuration as Complete**.
- On the **Tasks** workspace, select the task and then select **Actions** > **Mark Task as Configured**.

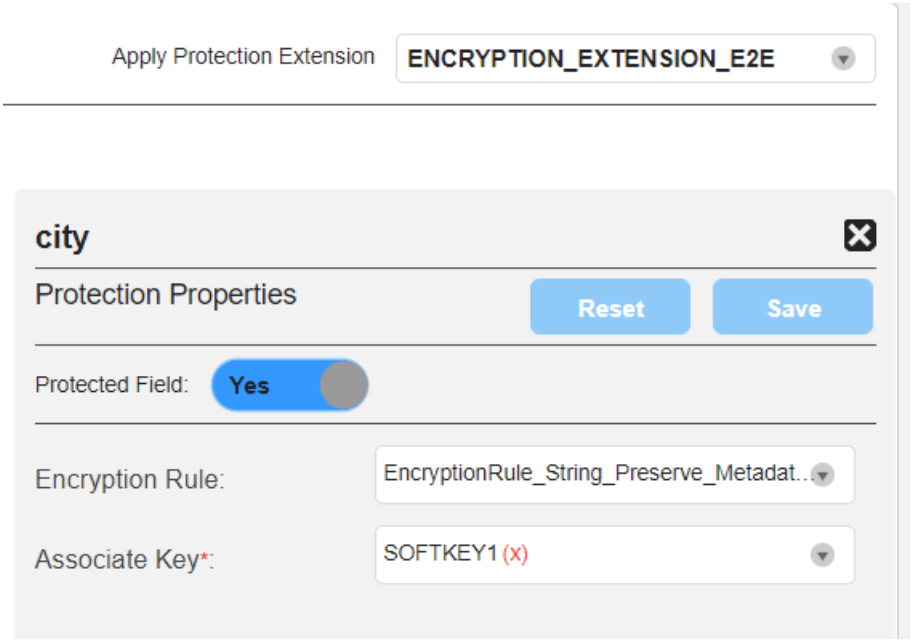
Related Topics:

- "Scheduling a Protection Job for a Task" on page 508
- "Protection Job" on page 302

Configuring Encryption Rules and Encryption Keys

When you configure a protection task that uses an encryption protection extension, you can assign encryption rules and encryption keys to sensitive string fields in data domains that are associated with encryption rules and encryption keys.

1. To designate the sensitive field as protected, in the **Protection Properties** box, toggle the **Protected Field** indicator to display **Yes**.



2. Select an encryption rule from the list.
3. Select an encryption key to associate with the encryption rule.
4. Click **Save**.

   A message indicates the success or failure of the field configuration. If no errors occurred, Data Privacy Management updates the value of the **Protected** column to **Yes**.
5. To encrypt more sensitive fields in the protection task, repeat these steps for each sensitive column that has a string data type and an encryption rule associated with the data domain.
6. To close the **Configure Protection: PRT-<task_number>** page, click **Close**. If you are finished configuring the protection task, click **Mark Configuration as Complete**.

   The protection task status changes to Configured.

Related Topics:

- "Decrypting a Closed Encryption Task" on page 506
- "Adding or Updating a Protection Extension on the Data Domain Details Page" on page 74

Editing Persistent Data Masking Rules

When you configure a protection task that uses a Persistent Data Masking - Big Data or a Persistent Data Masking - Remote Domain protection extension, you can edit the masking rules for sensitive fields in the data store if you do not use the default protection rules for the data domains.

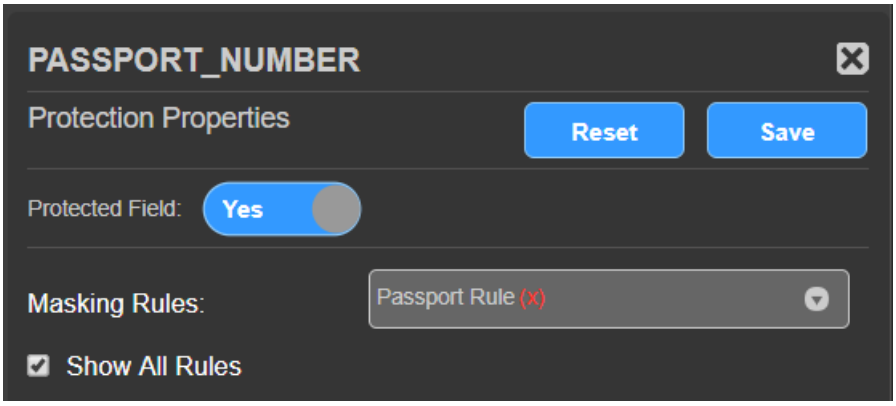
1. In the **Protection Properties** dialog box, click **No** to designate the sensitive field as protected.

   The value for the **Protected Field** changes to **Yes**.

2. Select the masking rule to apply to the protected field.

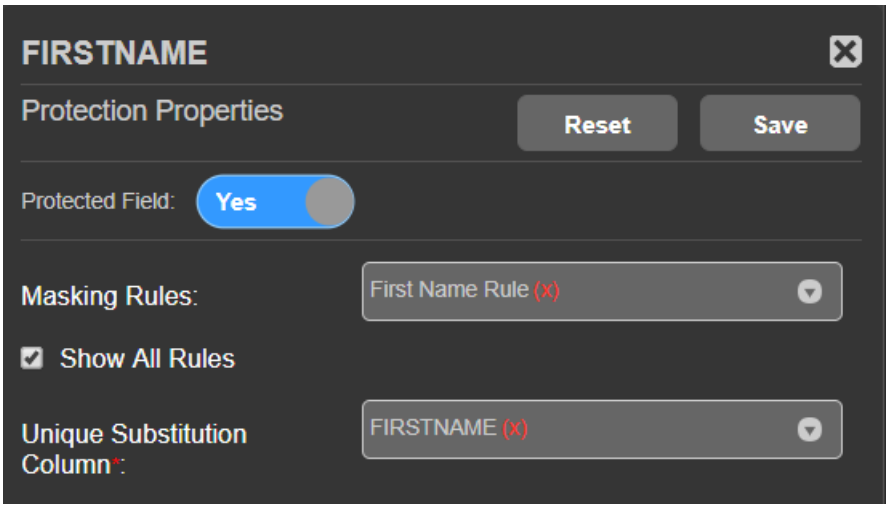
   Note: If the data domain associated with the field is configured with protection rules, the **Masking Rules** list displays only the configured rules, and the **Show All Rules** check box is cleared by default. If you select a masking rule from the list and then select the **Show All Rules** check box, Data Privacy Management clears the assigned rule and you must select the rule again from the **Masking Rules** list. If the data domain associated with the field is not configured with protection rules, the **Show All Rules** check box is selected by default.

   The following image shows the **Protection Properties** dialog box for the sensitive field **PASSPORT_NUMBER**, which uses the **Passport Rule** masking rule:



3. Depending on the masking rule you selected in step 2, the **Unique Substitution Column** field might appear. This field is required. Select a substitution column from the list.

   The following image shows the **Protection Properties** dialog box for the sensitive field **FIRSTNAME**, which uses the **First Name Rule** masking rule and the **FIRSTNAME** unique substitution column:



4. Click **Save**.

You receive a message that indicates the success or failure of the field configuration. If no errors occurred, Data Privacy Management updates the value of the **Protected** and **Configured** columns to **Yes** and lists the masking rule in the **Protection Rule** column on the **Configure Protection: PRT-<task_number>** page.

5. Select additional sensitive fields in the protection task one at a time and perform the previous steps again to configure protection rules for each field.
6. To close the **Configure Protection: <task_name>** page, click **Close**. If you are finished configuring the protection task, click **Mark Configuration as Complete** to change the task status to Configured.

Decrypting a Closed Encryption Task

You can decrypt data at rest in a closed encryption protection task with the **Rollback** option.

1. On the **Tasks** list page, select the check box next to a protection task that uses an encryption extension and has a status of Closed.
2. Click **Actions** > **Rollback**.

Editing a Protection Task

You can edit protection task properties from the **Tasks** workspace.

The task must be assigned to you and you must have privileges to edit tasks. You cannot edit a task that has a status of Job In Progress, Job Failed, or Closed.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. To edit a protection task, click the task name to display the **Task Details and Properties** page. Click **Edit**.
3. On the **Properties** view, you can edit the description, the **Notify Assignee Via Email** check box, the due date, tags, and notes.
4. On the **Task Details** view, you can edit the selections for the data domains to include in the protection task.
5. Click **Save**.

Marking a Protection Task as Completed

You can mark a protection task as completed after it is configured and before the task is closed.

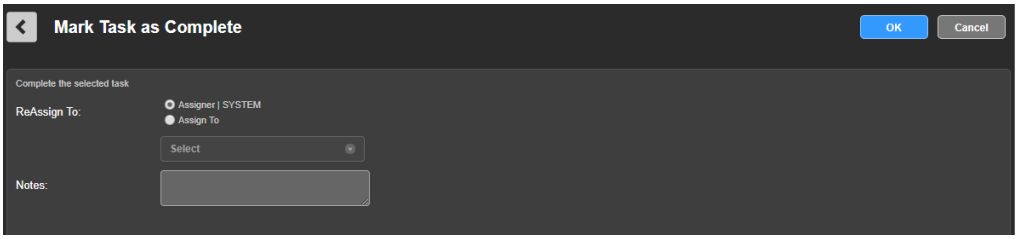
1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Optionally, click the **Refresh** icon to display the current task statuses.
3. Mark a protection task as completed in one of the following ways:
   - Select the check box next to one or more protection tasks that have a status of Configured, Job Successful, or Rejected. From the **Actions** menu, select **Mark Task as Completed**.
   - Click a protection task name to display the task details page. From the **Actions** menu, select **Mark Task as Completed**.

   The **Confirm Complete Action** prompt asks if you are sure you want to mark the selected task(s) as complete.

4. Click **Yes, I'm Sure**.

   The **Mark Task as Complete** page appears:



5. Reassign the task to the Data Privacy Management user who will close the task. The system can assign the task or you can select a user from the list.
6. Optionally, enter any notes.
7. Click **OK**.

   A message indicates if the Data Privacy Management Service marked the task as complete or if an error occurred. If no errors occurred, Data Privacy Management updates the task status to Completed.

You can add attachments and mark the task as closed.

Marking a Protection Task as Configured

After you configure protection for sensitive fields in a protection task, you complete the configuration by marking the task as configured from the **Tasks** workspace or the **Task Details** page.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Mark a task as configured in one of the following ways:
   - Select the check box next to one or more protection tasks. From the **Actions** menu, select **Mark Task as Configured**.
   - Click a task name to display the protection **Task Details** page. Click **Mark Configuration as Complete**.

   Data Privacy Management updates the task status to Configured.

You can now schedule a protection job for the configured task.

Reconfiguring a Protection Task

You can reconfigure a protection task from the **Tasks** workspace if the task has a status of Configured, Job Successful, Job Failed, or Completed.

Before performing these steps, terminate any running protection jobs for the task. For more information, see *Terminating a Job* in the *Jobs* chapter.

If a protection job fails, you might need to change some task configuration settings. Perform the steps in "Configuring Protection Tasks" on page 502 to update the protection properties and then perform the following steps to reconfigure the task:

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Select the check box to the left of the protection task that you want to reconfigure and click **Actions** > **Reconfigure Task**.

   You receive a message that indicates the success or failure of the reconfigure action.

After reconfiguring the task, perform the steps in "Scheduling a Protection Job for a Task" on page 508 to reschedule the protection job.

Rejecting a Protection Task

You can reject a protection task from the **Tasks** workspace or the **Task Details and Properties** page if the task status is Configured or Rejected.

Before performing these steps, terminate any running protection jobs for the task. For more information, see *Terminating a Job* in the *Jobs* chapter.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Reject a protection task in one of the following ways:
   - Select the check box next to one or more protection tasks that you want to reject. Click **Actions** > **Reject Task**.
   - Click a protection task name to display the **Task Details and Properties** page. Click **Actions** > **Reject Task**.

   The **Reject Task** page appears.
3. Optionally, reassign the task to another Data Privacy Management user.
4. In the **Notes** field, enter a reason for rejecting the task.
5. Click **OK**.

   You receive a message that indicates the success or failure of the rejection. If no errors occurred, Data Privacy Management updates the status of the protection task to Rejected.

Scheduling a Protection Job for a Task

After a protection task is configured, you can schedule a protection job to run the task. Data Privacy Management updates the task status when the job is created, when the job is in progress, and when the job is successful or the job fails.

Related Topics:

- "Configuring Protection Tasks" on page 502
- "Protection Job" on page 302

Scheduling a Persistent Data Masking - Big Data Protection Job

You can schedule a protection job for a configured task that is associated with a Persistent Data Masking - Big Data protection extension.

1. On the **Tasks** workspace, schedule a Persistent Data Masking - Big Data protection job in one of the following ways:
   - Select the check box next to a configured protection task that is associated with the Persistent Data Masking - Big Data (PDM BDE) protection extension. From the **Actions** menu, select **Schedule Protection Job**.
   - Click a task name to display the protection **Task Details and Properties** page. Click **Schedule Protection Job**.

The **Schedule Protection Job: PRT-<task_number>** page appears. the settings defined for the protection extension in the . You can edit these settings for the protection job.

2. On the **Runtime Configuration** pane, optionally select the arrow next to the **Advanced Settings** label to edit the following advanced settings:
   - To copy the protection configuration for the sensitive fields in the tasks to the Model Repository Service when the task runs, select **Persist Mappings in MRS**.
   - Optionally, select a substitution dictionary connection for the Persistent Data Masking instance.
3. Click **2**.

   The **Schedule** pane appears.
4. Select **Run Now** to run the protection job immediately. Select **Schedule For** to specify a start date and start time to run the protection job.
5. Click **Save**.

   The **Tasks** workspace appears. If there are no errors, Data Privacy Management updates the protection task status to Job Scheduled or Job In Progress.

After the protection job finishes and the task has a status of Job Successful, the new task configurations and masking rules that were part of the successful job are shown as existing rules when you view or reconfigure the task.

Scheduling a Persistent Data Masking - Remote Domain Protection Job

You can schedule a protection job for a configured task that is associated with a Persistent Data Masking - Remote Domain protection extension.

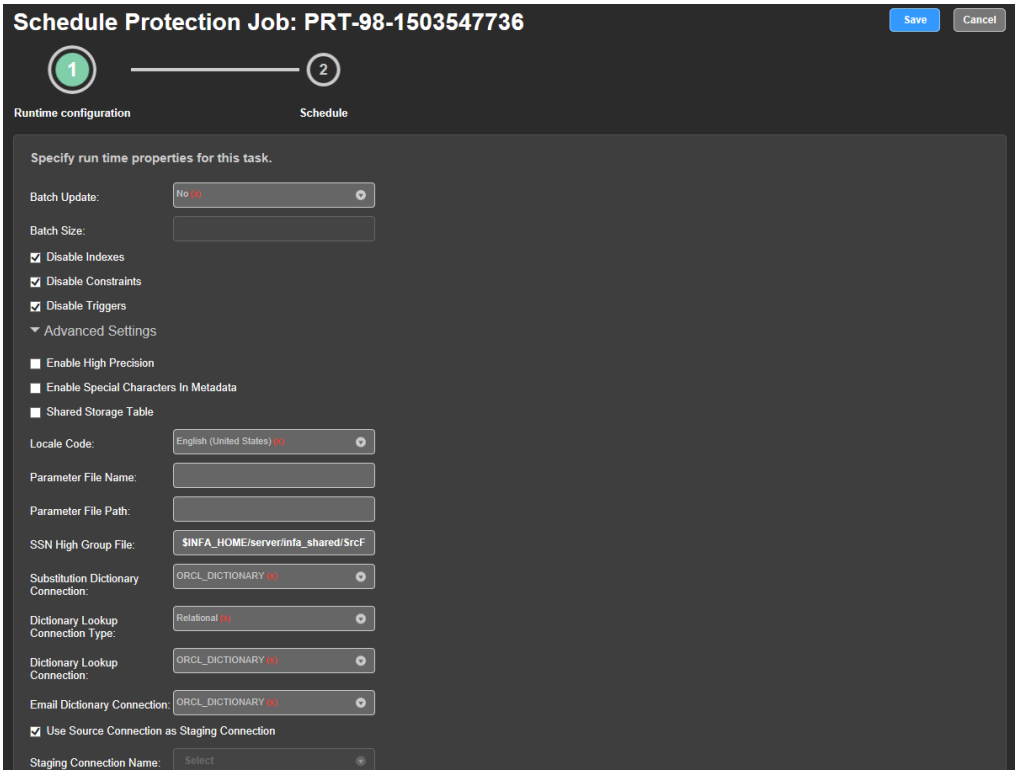
1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Schedule a Persistent Data Masking - Remote Domain protection job in one of the following ways:
   - Select the check box next to a protection task that is associated with the Persistent Data Masking - Remote Domain protection extension. From the **Actions** menu, select **Schedule Protection Job**.
   - Click a task name to display the protection **Task Details and Properties** page. Click **Schedule Protection Job**.

   The **Schedule Protection Job: PRT-<task_number>** page displays the settings defined for the protection extension in the **Runtime Configuration** pane. You can edit these settings for the protection job.
3. In the **Batch Update** field, select **Yes** or **No**.

   If you select Yes, specify a batch size in the **Batch Size** field.
4. Optionally, edit the selections in the check boxes that indicate whether to disable indexes, constraints, and triggers.
5. Optionally, select the arrow next to the **Advanced Settings** label to display the advanced settings.

The following image shows the **Runtime Configuration** pane for a protection task configured with a Persistent Data Masking - Remote Domain protection extension:



6. Optionally, edit any of the advanced settings.
7. Click **2**.

   The **Schedule** pane appears.
8. Select **Run Now** to run the protection job immediately. Select **Schedule For** to specify a start date and start time to run the protection job.
9. Click **Save**.

   A message indicates if the Data Privacy Management Service scheduled the protection job or if an error occurred. The **Tasks** workspace appears. If there are no errors, Data Privacy Management updates the protection task status to Job Scheduled or Job In Progress.

After the protection job finishes and the task has a status of Job Successful, the new task configurations and masking rules that were part of the successful job are shown as existing rules when you view or reconfigure the task.

Scheduling an Encryption Protection Job

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Schedule an encryption protection job in one of the following ways:
   - Select the check box next to a protection task that is associated with an encryption protection extension. From the **Actions** menu, select **Schedule Protection Job**.
   - Click a task name to display the **Task Details and Properties** page. Click **Schedule Protection Job**.

   The **Schedule Protection Job: PRT-<task_number>** page displays.

3. Select **Run Now** to run the protection job immediately. Select **Schedule For** to specify a start date and start time to run the protection job.
4. Click **Save**.

   A message indicates if the Data Privacy Management Service scheduled the protection job or if an error occurred. The **Tasks** workspace appears. If there are no errors, Data Privacy Management updates the protection task status to Job Scheduled or Job In Progress.

After the protection job finishes and the task has a status of Job Successful, the encryption rules and encryption keys that were part of the successful job are shown as existing associations to the protected sensitive fields when you view or reconfigure the task.

Viewing the Configuration for a Protection Task

You can view the configuration details for a protection task that has a completed configuration.

1. Click the **Tasks** icon in the header to open the **Tasks** workspace.

   The **Tasks** workspace displays a list of tasks.
2. Click the protection task name that has a completed configuration you want to view.

   The **Task Details and Properties** page appears.
3. Click **View Configuration**.

   The **View Configuration: PRT-<task_number>** page appears. You can view the sensitive fields in the data store and whether or not each column is protected and configured.

Part IV: Analysis

This part contains the following chapters:

- Security Dashboard, 513
- Summary Workspaces, 581

CHAPTER 24

Security Dashboard

This chapter includes the following topics:

- Security Dashboard Overview, 513
- Enterprise-Level Summary of Sensitive Data, 514
- Security Dashboard Indicators, 519
- Pages Accessed from the Security Dashboard, 532
- Security Dashboard Management, 572

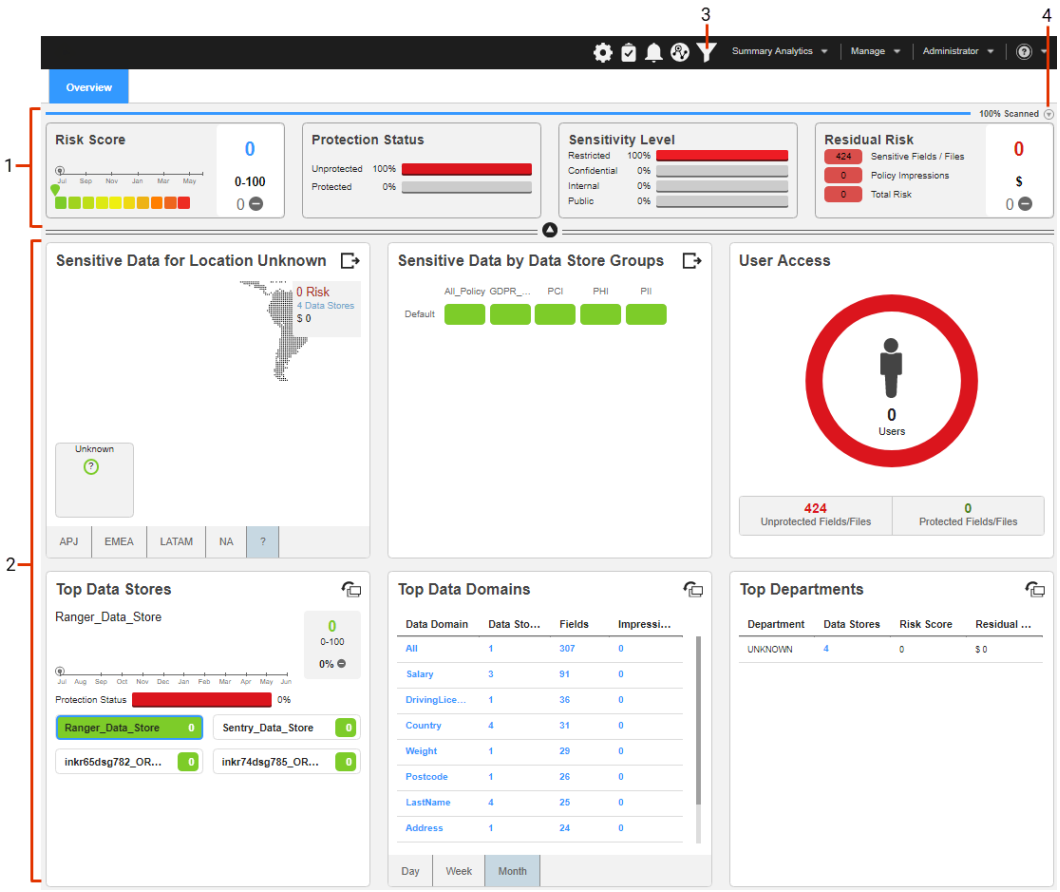
Security Dashboard Overview

You can view dashboard analytics, global visibility of sensitive information in data stores and data domains, user behavior, sensitive data locations and proliferation, and protection metrics on the **Overview** workspace of the **Security Dashboard**.

Each indicator on the **Overview** workspace displays high-level information about a key risk factor. Most dashboard indicators contain links for accessing and managing the metric details from other pages.

For example, click the **Top Data Stores** indicator title to open the **Data Stores** page that lists data stores from highest to lowest risk score. To open the **Proliferation** page for a data store listed in the indicator, click the data store name. To open the **Sensitive Fields** page for a data store listed in the indicator, click the value in the **Fields/Files** column.

The following image shows a sample **Overview** workspace:



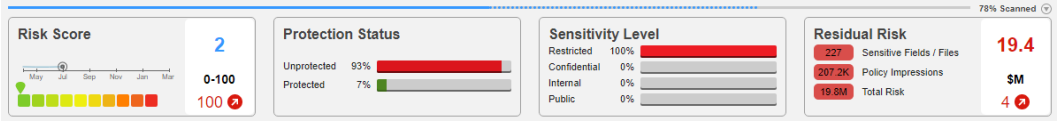
1. Enterprise-level summary metrics
2. Workspace indicators
3. Filter
4. Actions menu

Enterprise-Level Summary of Sensitive Data

To quickly view the status of sensitive data that Data Privacy Management has analyzed, review the enterprise-level summary on the Security Dashboard.

You can view the percentage of scanned data stores, the average risk score for sensitive data, the number of data stores in each protection and sensitivity level, and the estimated cost of unprotected sensitive data. Click an indicator to view the details of the metric.

The following image shows the enterprise-level indicators:



Discovery Bar Indicator

The **Discovery Bar** indicator shows the percentage of fully scanned, partially scanned, and unscanned data stores.

- The solid blue section represents the percentage of fully scanned data stores.
- The dotted blue section represents the percentage of partially scanned data stores.
- The gray section represents the percentage of data stores that Data Privacy Management has not scanned.
- The percentage number to the right of the bar represents the total percentage of data stores that are fully or partially scanned.

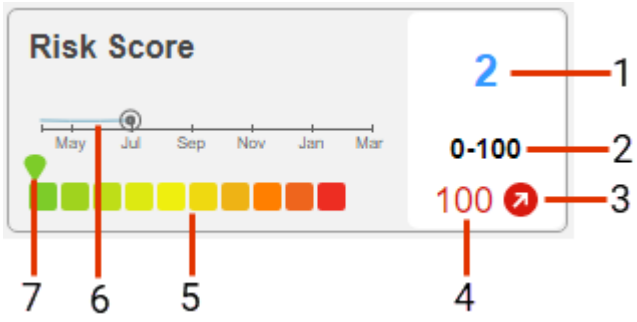
To view the exact percentages of fully or partially scanned data stores, hover over the solid and dotted sections of the blue bar. To view the list of fully scanned, partially scanned, or unscanned data stores, click the respective section of the **Discovery Bar** indicator. The **Data Stores** workspace opens in a new tab, and lists the data stores for the scanned status you selected.

To view more details about a data store from the **Data Stores** list page, click the data store name. The **Data Store Details** page appears.

Risk Score Indicator

The **Risk Score** indicator shows the average risk score of the data stores that Data Privacy Management scanned in the current month. The indicator also shows the trend across previous months.

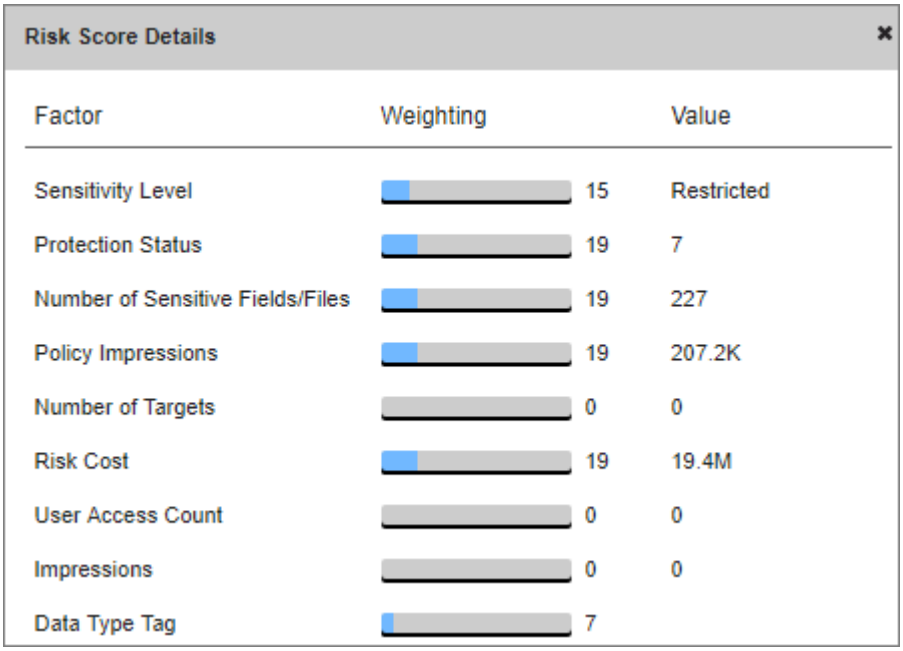
The following image shows the **Risk Score** indicator on the **Overview** workspace:



The following table describes the information displayed on the **Risk Score** indicator:

Number in Image	Metric	Description
1	Average risk score for the current month	Indicates the average risk score of data stores that Data Privacy Management scanned in the current month. Data Privacy Management calculates risk scores based on the following metrics for sensitive data in each scanned data store: - Sensitivity level - Protection status - Number of sensitive fields - Number of sensitive records - Number of targets - Residual risk cost - Number of active users - Number of active and inactive users with activity in Data Privacy Management
2	Risk score range	The minimum and maximum values for the risk score. Defaults are 0 and 100.
3	Trend icon	Indicates the direction of change in the average risk score from the previous month to the current month: - An up arrow indicates an increase. - A down arrow indicates a decrease. - A straight line indicates no change.
4	Change from previous month	Indicates the difference in the average risk score value between the previous month and the current month.
5	Color categories	Indicates the level of the current risk score with 10 squares that are each a shade of green, yellow, orange, and red. Each color square represents 10 units of the risk score range moving left to right. For example, low risk scores between 1-30 are represented by a shade of green, medium risk scores between 40-60 are represented by a shade of yellow, and higher risk scores are represented by a shade of orange or red.
6	Current month color category	Indicates the color category of the risk score for the current month.
7	Line graph	Plots the average risk score for each month over a maximum 12-month period to show the trend change. If there are no scans in the current month, Data Privacy Management reuses the average risk score from the previous month.

To view the weight and details of each factor in the risk score calculation, click the **Risk Score** indicator. The **Risk Score Details** window appears.



You can change the weight of each risk score factor from the **Settings** workspace. Data Privacy Management recalculates the risk score of the scanned data stores and refreshes the values on the **Risk Score** indicator.

Related Topics:

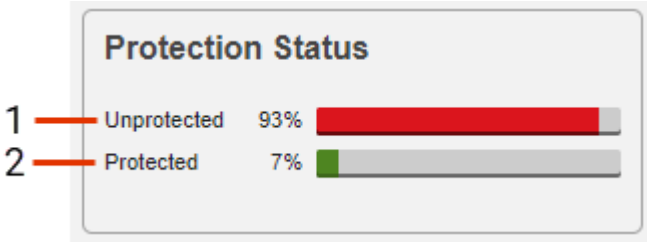
- "Adding Custom Risk Score Factors" on page 43
- "Changing Risk Score Factor Weights" on page 42

Protection Status Indicator

The **Protection Status** indicator shows the percentage and number of sensitive fields and files that are protected and unprotected.

Data Privacy Management determines the protection status of sensitive fields and files when you scan data stores and when you import protection status for a data store on the **Data Stores** workspace.

The following image shows the **Protection Status** indicator on the **Overview** workspace.



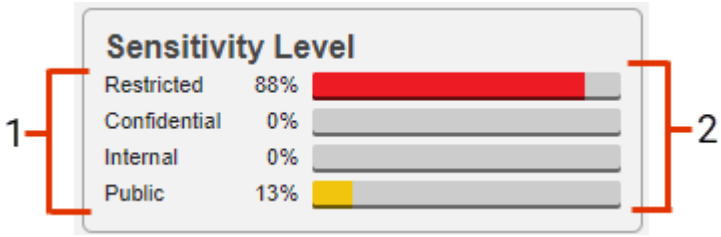
1. The percentage of discovered sensitive fields and files that are not protected. To view the exact percentage and number of unprotected sensitive fields and files, hover over the red bar.
2. The percentage of discovered sensitive fields and files that are protected. To view the exact percentage and number of protected sensitive fields and files, hover over the green bar.

Sensitivity Level Indicator

The **Sensitivity Level** indicator shows the percentage and number of data stores that Data Privacy Management associates with the sensitivity levels defined on the **Settings** workspace.

You can define a maximum of five sensitivity levels on the **Settings** workspace. When you create a classification policy, you specify a sensitivity level for the policy. If a data store meets the classification policy criteria during a scan, Data Privacy Management associates the sensitivity level of the classification policy with the data store.

The following image shows the **Sensitivity Level** indicator on the **Overview** workspace:



1. The names of the data sensitivity level classifications that are defined on the **Settings** workspace.
2. The bar is a visual representation of the percentage of data stores for each sensitivity level, with higher sensitivity levels in red or orange and lower sensitivity levels in yellow or green. To view the percentage and number of data stores that matched a classification policy for each sensitivity level, hover over the bar.

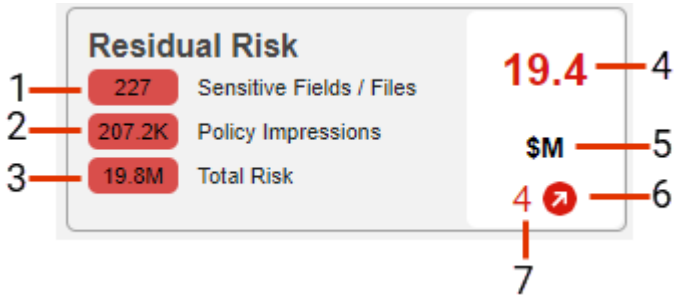
Related Topics:

- "Creating a Classification Policy" on page 229
- "Specifying Data Store Sensitivity Levels" on page 42

Residual Risk Indicator

The **Residual Risk** indicator shows the amount of money it would take to resolve the problems of a data breach involving unprotected sensitive data. The indicator also shows the number of fields, files, and impressions that contain sensitive data.

The following image shows the **Residual Risk** indicator on the **Overview** workspace:



The following table describes the information displayed on the **Residual Risk** indicator:

Number in Image	Metric	Description
1	Number of sensitive fields/files	The total number of sensitive fields and files for all data stores at the current time.
2	Number of classification policy impressions	The total number of sensitive data impressions for all classification policies at the current time. Data Privacy Management calculates the number of policy impressions with a formula for each user query that returns sensitive data records. Policy impressions that a user accesses equal the number of sensitive fields a query returns multiplied by the number of rows returned for each sensitive field in the query. For example, if a user searches a data store for the three sensitive fields `SSN`, `FIRST_NAME`, and `LAST_NAME`, and the search returns 100 rows, Data Privacy Management calculates 300 policy impressions.
3	Total risk	Sum of the cost of unprotected sensitive data in all the data stores in the enterprise. The residual risk cost for each data store is the product of the following factors: - The cost for one impression of sensitive data that you specified in the classification policy. - The number of unprotected sensitive impressions in the data store.
4	Total risk for previous month	Sum of the cost of unprotected sensitive data in all the data stores in the enterprise for the previous month.
5	Currency and unit of cost	Shows the type of currency anc unit of the residual risk cost. For example, the **$** sign indicates the United States dollar and **M** represents units in millions.
6	Trend icon	Indicates the direction of the residual risk cost from the previous month to the current month. The up arrow indicates an increase and the down arrow indicates a decrease in the risk cost. A horizontal line indicates no change from the previous month to the current month.
7	Change from previous month	Indicates the difference in the residual risk between the previous month and the current month.

Related Topics:

- "Classification Policy Properties" on page 223
- "Changing the Default Risk Cost" on page 45

Security Dashboard Indicators

Security Dashboard indicators appear below the enterprise-level summary information on the **Overview** workspace. You can use the indicators to quickly access detailed information about sensitive and private data. To view more information about an indicator, click the indicator name. Data Privacy Management displays the related detail page.

The **Overview** workspace includes the following default indicators:

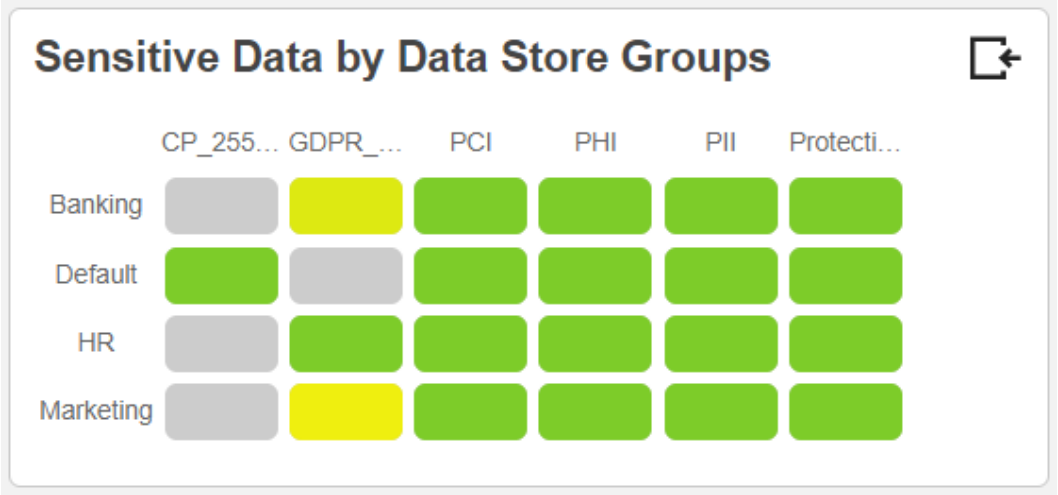
- Sensitive data by data store groups
- Sensitive data for location
- Sensitive data proliferation
- Top data domains
- Top data stores
- Top departments
- Top users
- User access

For information about customizing the indicator display, see "Customizing the Security Dashboard" on page 572.

Sensitive Data by Data Store Groups Indicator

The **Sensitive Data by Data Store Groups** indicator shows a matrix of data store groups and classification policies. The color of each tile represents the average risk score of the data stores that match a classification policy within a specific data store group.

The following image shows the expanded view of the **Sensitive Data by Data Store Groups** indicator:



The **Sensitive Data by Data Store Groups** indicator lists the names of the classification policies and data store groups with sensitive fields/files that match each classification policy. To view the full name of an incomplete data store group or classification policy, hover over the name.

Each tile color represents the level of the average risk score for the data stores that match a classification policy in a data store group. The lower risk scores are shades of green and yellow, and higher risk scores are shades of orange and red.

To expand or contract the indicator view, click the toggle icon in the top right corner of the indicator.

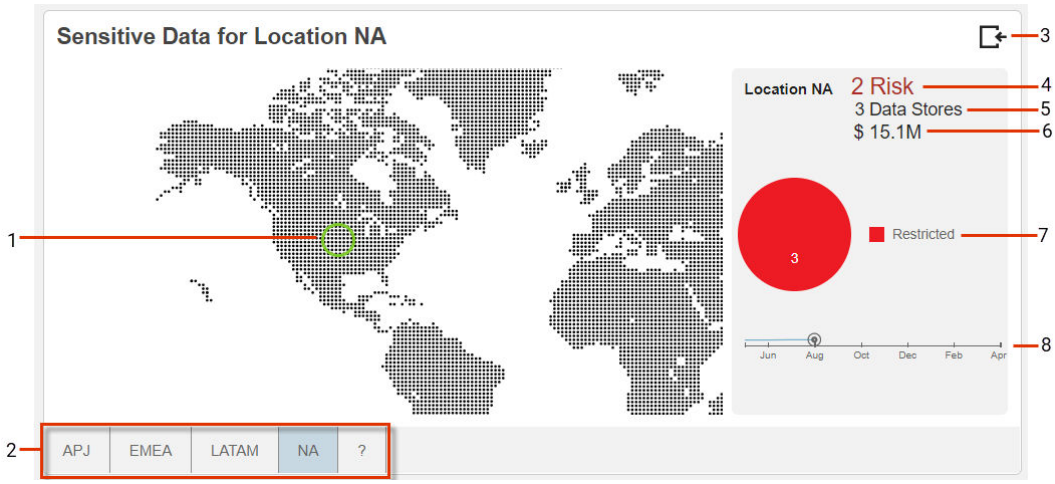
To view more details on the **Sensitive Data by Data Store Group** page, click the indicator label.

Sensitive Data for Location Indicator

When you create a data store on the **Data Stores** workspace, you can associate the data store to a geographic region. The **Sensitive Data for Location** indicator shows data stores that contain sensitive data by region.

The indicator shows the number, protection status, risk score, residual risk cost, and sensitivity level of data stores in each region. By default, the indicator displays the region with the most number of sensitive data stores.

The following image shows the **Sensitive Data for Location** indicator on the **Overview** workspace:



The following table describes the information displayed on the **Sensitive Data for Location** indicator:

Number in Image	Description	Actions
1	Region of the world. The color of the circle represents the risk score level.	View.
2	Region tabs	Click a tab to show the sensitive data information for that location. When you click the **?** tab, the metrics for data stores that are not associated with a region appear.
3	Toggle icon	Click to expand or hide the information in the right pane of the indicator.
4	The aggregate risk score of all the data stores in the region.	View.
5	The number of data stores with at least one sensitive field in the selected region.	Click the label to view the **Data Stores** list page filtered by the data stores included in the selected region.
6	The total residual risk cost of the data stores in the selected region.	View.

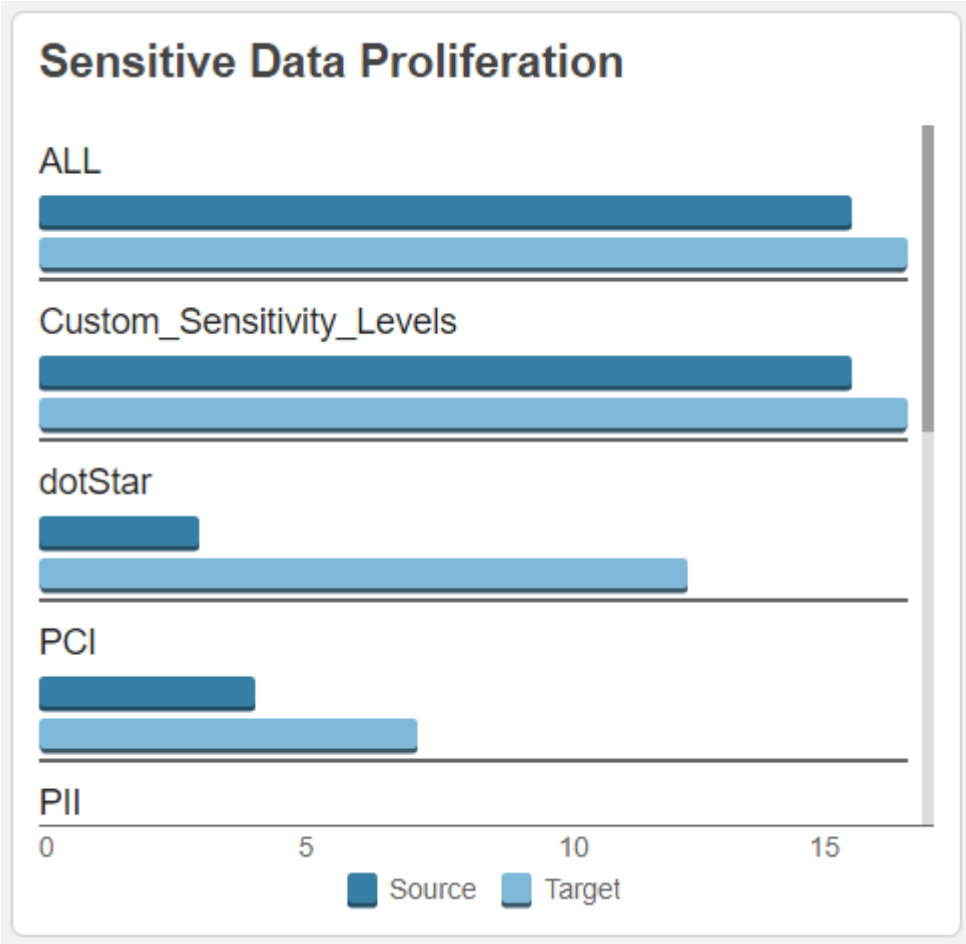
Number in Image	Description	Actions
7	Name of the sensitivity level of the data stores in the selected region, and a pie chart showing the number of data stores for each sensitivity level.	View.
8	Risk score graph. Plots the aggregate risk score for each month over a 12-month period. If there are no scans in the current month, Data Privacy Management reuses the aggregate risk score from the previous month.	-

To view the **Sensitive Data Locations** page, click the indicator label.

Sensitive Data Proliferation Indicator

The **Sensitive Data Proliferation** indicator shows data stores that matched a classification policy and that contain sensitive data that proliferated to one or more target data stores.

The following image shows the **Sensitive Data Proliferation** indicator:



The indicator lists classification policies in descending order according to the number of target data stores. For each classification policy, the indicator shows the number of target data stores that contain sensitive

fields copied from source data stores. The indicator also shows the proliferation status by data store group for each classification policy.

You can perform the following actions from the **Sensitive Data Proliferation** indicator:

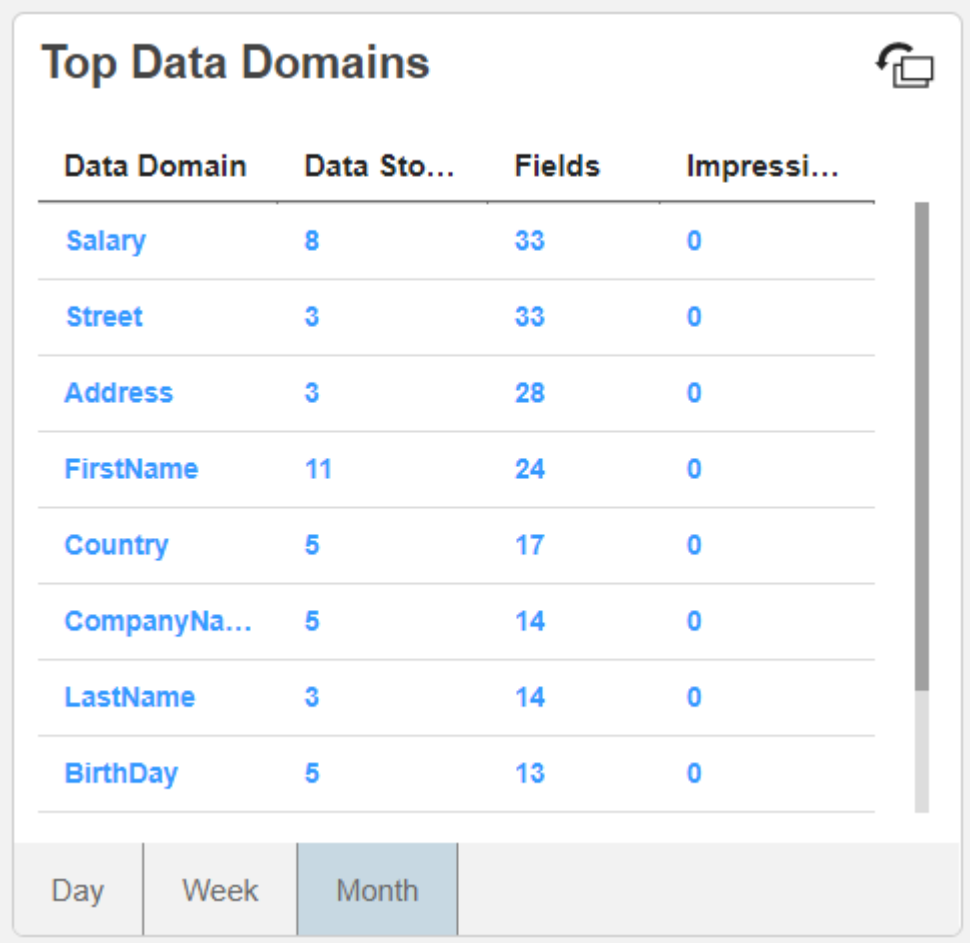
- To view more details about the classification policies and the proliferation of sensitive data, click the indicator label.
- To view the exact number of source or target data stores represented by a bar graph, hover over the source or target bar.
- To view more classification policies with proliferation of sensitive data, click and drag the scroll bar on the right of the indicator.

Top Data Domains Indicator

The **Top Data Domains** indicator provides a quick way to view data domains with the highest number of sensitive fields. You can toggle the display between a list view and a trend view.

Top Data Domains Indicator List View

The following image shows the list view of the **Top Data Domains** indicator:



The list view of the **Top Data Domains** indicator lists data domains from highest to lowest number of sensitive fields.

By default, the indicator lists the number of classification policy impressions for the last 30 days in the UTC time zone, and the **Month** tab at the bottom of the indicator is highlighted blue.

To view the number of impressions for the past seven days in the UTC time zone, click the **Week** tab.

To view the number of impressions for the most recent 24-hour period in the UTC time zone, click the **Day** tab.

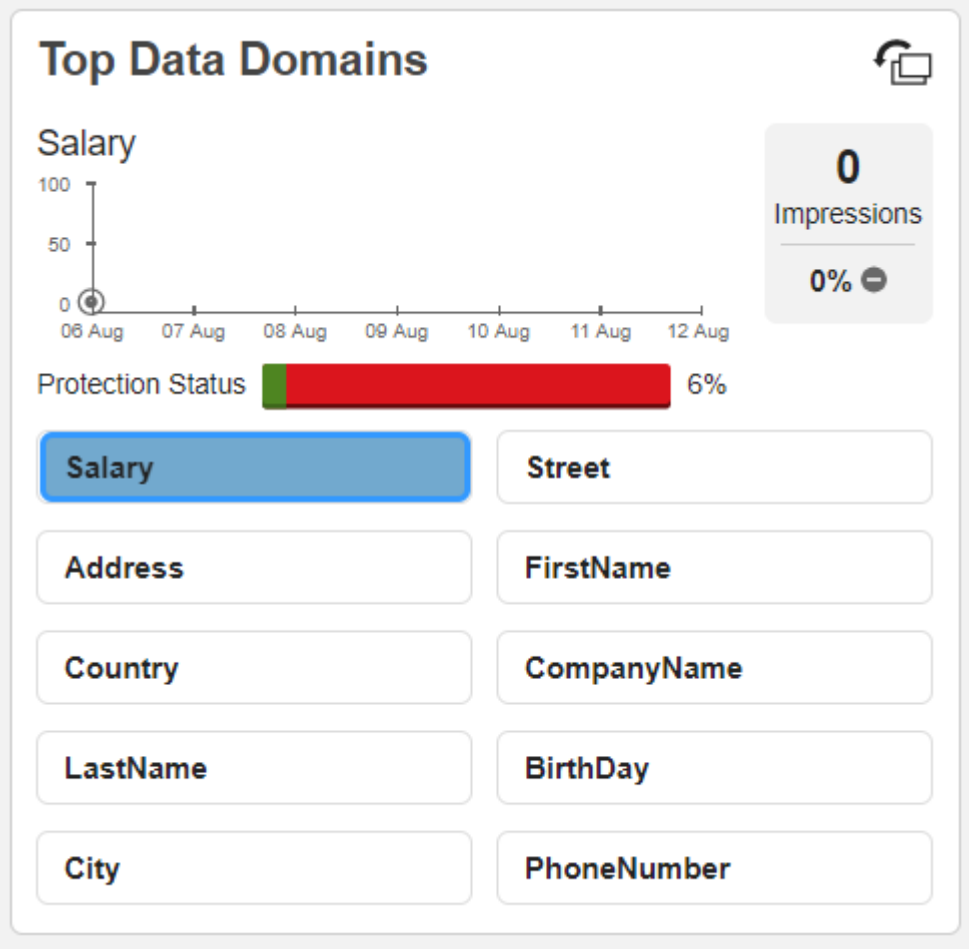
The values in the **Data Stores** and **Fields** columns do not change when you change the time period.

You can perform the following actions from the list view:

- To view the **Data Domains** list page, click the **Top Data Domains** indicator label.
- To view details about a data domain, click a data domain name or the value in the **Fields** column.
- To view the list of data stores included in a data domain, click the **Data Stores** value.
- To view information about users who accessed sensitive records that match classification policies in a data domain, click the **Impressions** value.
- To view more top data domains, click and drag the scroll bar to the right of the list.
- To display the **Top Data Domains** indicator trend view, click the **Toggle** icon in the top right corner.

Top Data Domains Indicator Trend View

The following image shows the trend view of the **Top Data Domains** indicator:



By default, the trend view shows details for the first data domain in the list view. To view details for another data domain, click a data domain name.

The trend view of the **Top Data Domains** indicator shows the following details:

- The number of classification policy impressions that users accessed for the selected data domain and time period.
- Below the number of impressions, the percentage and direction of change in impressions between the previous two time periods. An up arrow indicates an increase, a down arrow indicates a decrease, and a horizontal line indicates no change in impressions accessed between the two time periods.
- A graphical representation of the number of impressions and the direction of change in impressions over a specified time period.
- Below the impression trend graph, the protection status of sensitive fields in the data domain as a percentage from 0-100. The bar graph represents protected fields in green and unprotected fields in red.

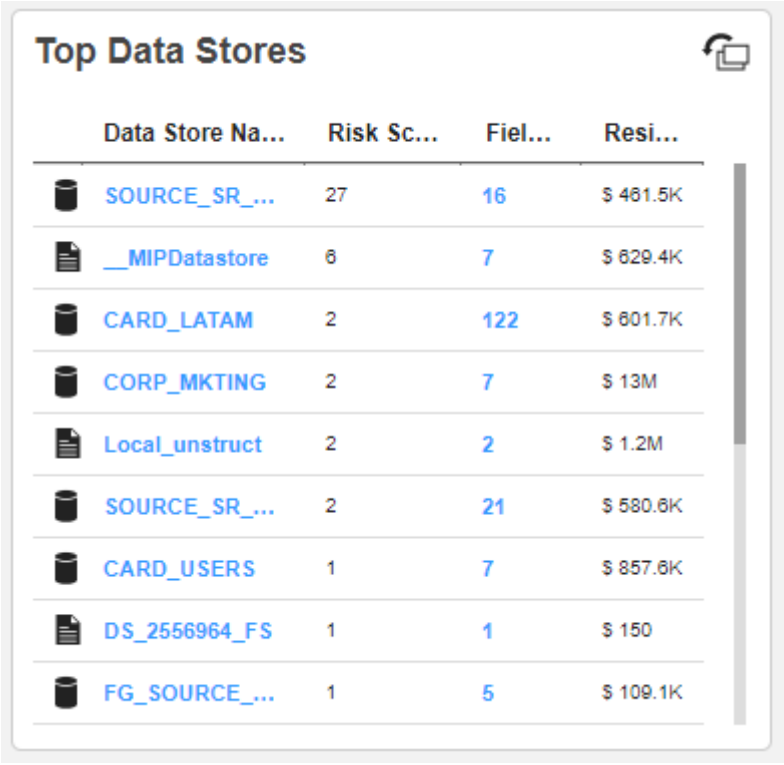
To return to the **Top Data Domains** indicator list view, click the **Toggle** icon in the top right corner.

Top Data Stores Indicator

The **Top Data Stores** indicator provides a quick way to view data stores with the highest number of sensitive fields. You can toggle the display between a list view and a protection status view.

Top Data Stores Indicator List View

The following image shows the list view of the **Top Data Stores** indicator:



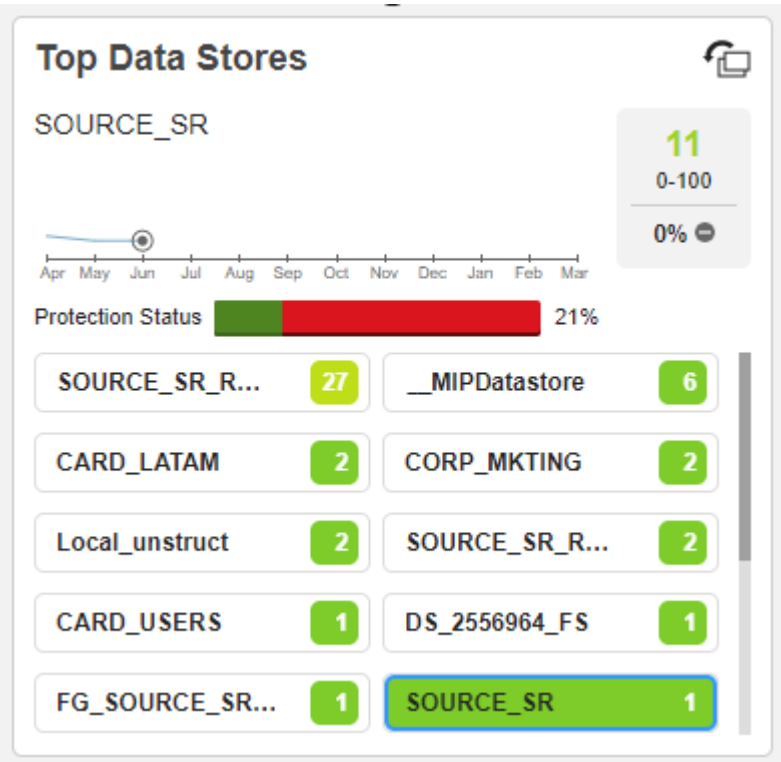
The list view of the **Top Data Stores** indicator lists data stores from highest to lowest risk score. You can perform the following actions from this view:

- To view the **Data Stores Grid** page, click the **Top Data Stores** indicator label.

- To view proliferation details for a data store, click a data store name.
- To view information about the sensitive fields in a data store, click the **Fields** value.
- To view more data stores, click and drag the scroll bar to the right of the list.
- To display the protection status view, click the **Toggle** icon in the top right corner.

Top Data Stores Indicator Protection Status View

The following image shows the protection status view of the **Top Data Stores** indicator:



By default, the protection status view shows details for the first data store in the list view, and each data store name shows the number of sensitive fields. To view protection status details for another data store, click a data store name. To view more data stores in the list, click and drag the scroll bar.

The protection status view of the **Top Data Stores** indicator shows the following details:

- The risk score for each data store as a value from 0-100. If there were no scans of the data store in the current month, Data Privacy Management reuses the risk score from the previous month.
- Below the risk score, the risk score delta from the previous month as a percentage.
- The risk score graph that plots the monthly risk score for the selected data store for a maximum of 12 months.
- The protection status of sensitive fields in the data store as a percentage from 0-100. The protection status percentage appears in a bar graph above the data store list. The bar graph represents protected fields in green and unprotected fields in red.

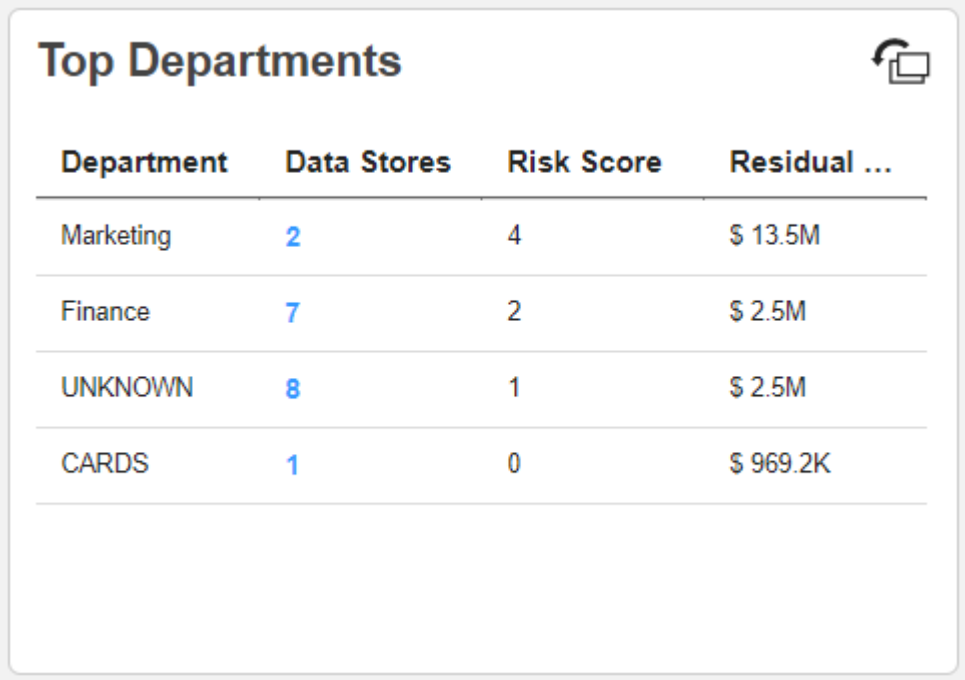
To return to the **Top Data Stores** indicator list view, click the **Toggle** icon in the top right corner.

Top Departments Indicator

The **Top Departments** indicator provides a quick way to assess the departments with the highest risk scores and risk cost. You can toggle the display between a list view and a trend view. Both views display the list of departments associated with at least one data store that has at least one sensitive field.

Top Departments Indicator List View

The following image shows the list view of the **Top Departments** indicator:



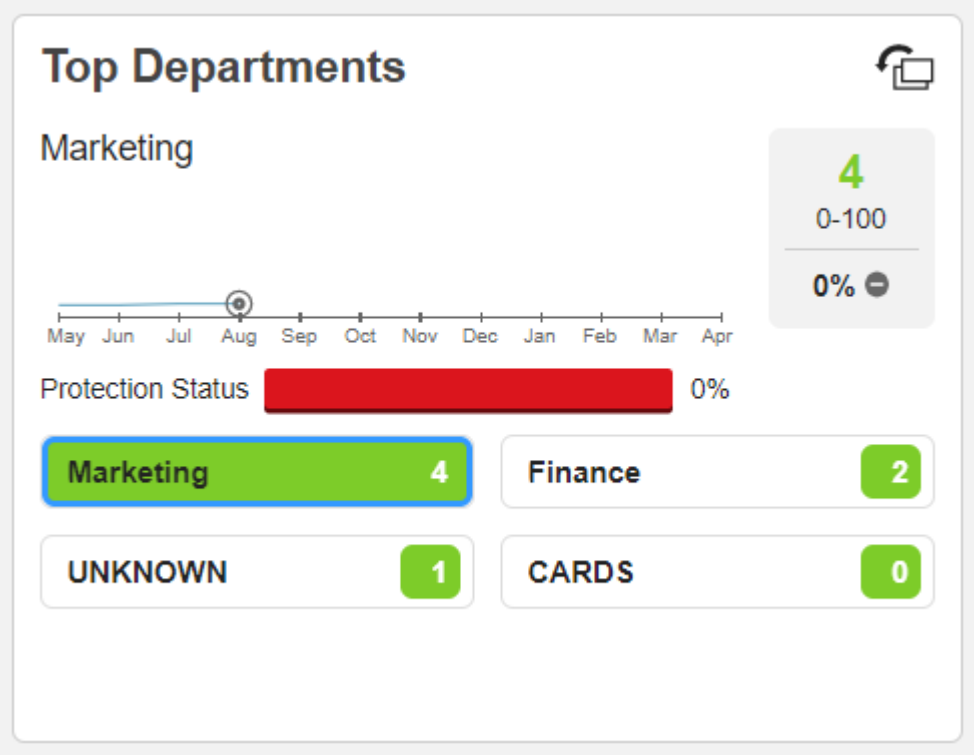
The list view of the **Top Departments** indicator lists departments from highest to lowest risk score. For each department, the indicator lists the name, the number of associated data stores, the average risk score, and the total risk cost.

You can perform the following actions from the list view:

- To view more details about the top departments, click the indicator label.
- To view details about the data stores associated with a department, click the **Data Stores** value.
- To display the trend view, click the **Toggle** icon in the top right corner.

Top Departments Indicator Trend View

The following image shows the trend view of the **Top Departments** indicator:



By default, the trend view shows details for the first department in the list view, and each department name shows the associated risk score. To view details for another department, click a department name.

The trend view of the **Top Departments** indicator shows the following details:

- The risk score as a number between 0 and 100.
- Below the risk score number, the percentage and direction of change in the risk score from the previous month to the current month. An up arrow indicates an increase in the risk score, a down arrow indicates a decrease in risk score, and a horizontal line indicates no change in the risk score.
- A graphical representation of the monthly risk score for the selected department for a maximum of 12 months. If there were no scans in the current month, Data Privacy Management reuses the risk score from the previous month.
- Below the risk score graph, the protection status of sensitive fields for all the data stores in the department, as a percentage from 0-100. The bar graph represents protected fields in green and unprotected fields in red.

To return to the **Top Departments** indicator list view, click the **Toggle** icon in the top right corner.

Top Users Indicator

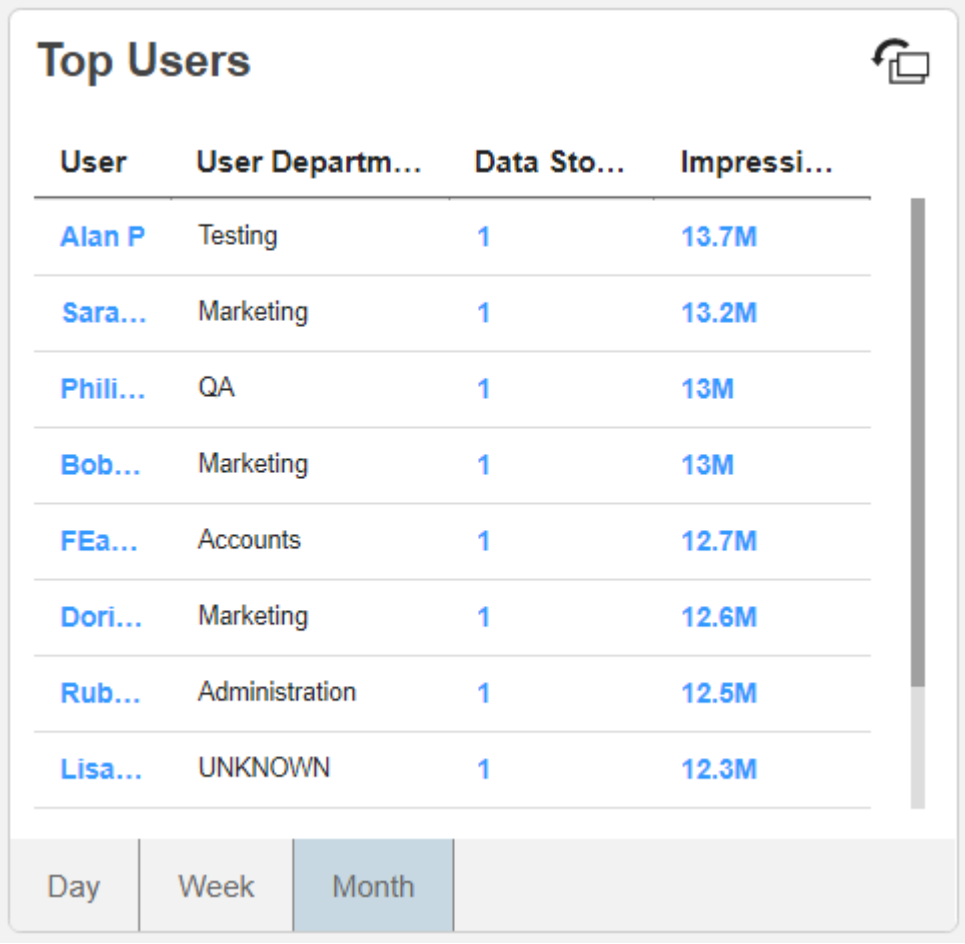
The **Top Users** indicator provides a quick way to detect users with the most activity or unusual patterns of activity on sensitive data.

Top Users Indicator List View

The **Top Users** indicator list view displays information about a maximum of 10 users who accessed the most sensitive data in the specified time period. By default, the indicator lists users in descending order of classification policy impressions in the past 30 days in the UTC time standard.

The indicator also shows the department to which each user belongs, the number of data stores with at least one sensitive field that each user accessed, and the number of records across each sensitive field that each user accessed in the time period selected.

The following image shows the list view of the **Top Users** indicator:



You can perform the following actions on the **Top Users** indicator list view:

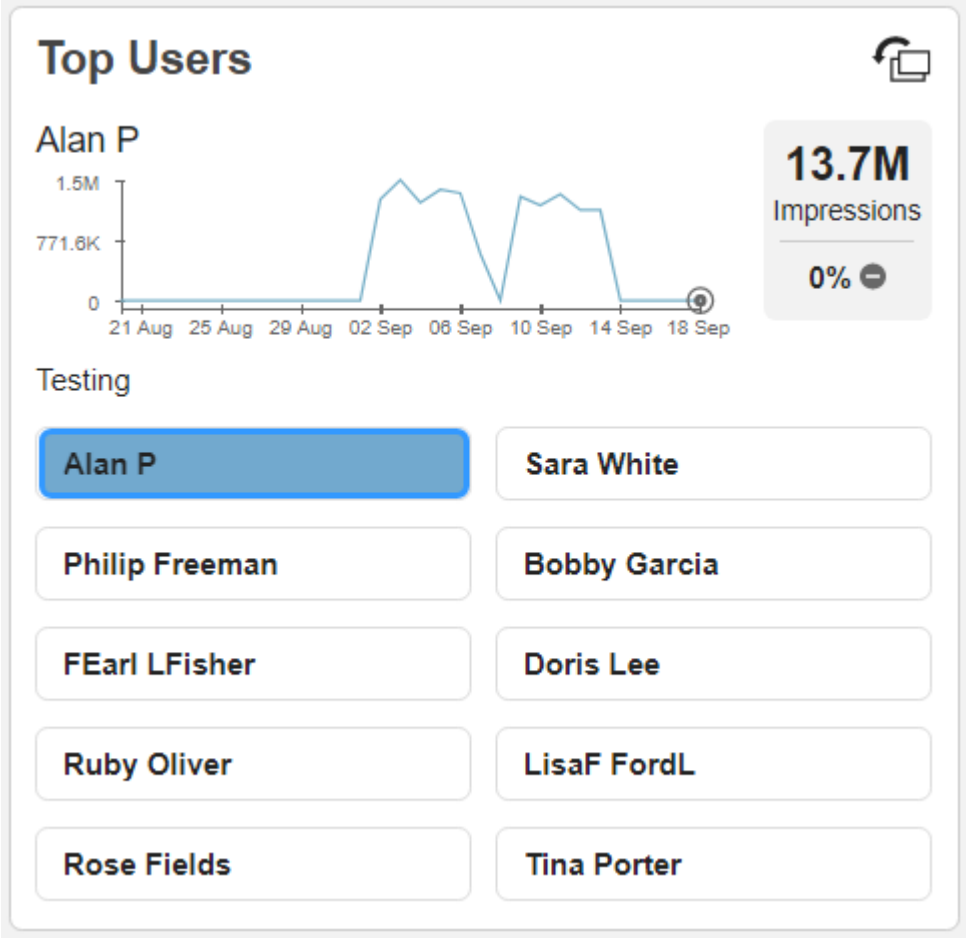
- To view details for each user, click the indicator label.
- To view a graph of impressions over time for a specific user, click a user name.
- To see the list of data stores that a user accessed, click the **Data Stores** value.
- To see impression event details, click the **Impressions** value.
- To scroll through the list of users, click the arrows.

- To view the number of impressions for the past seven days, click the **Week** tab. To view the number of impressions for the most recent 24-hour period, click the **Day** tab. The time period you select in the list view determines the time period for the information you see in the trend view.
- To display the **Top Users** indicator trend view, click the **Toggle** icon in the top right corner.

Top Users Indicator Trend View

The **Top Users** indicator trend view displays information about the number of impressions accessed by each user over the time period you specified in the list view. The trend view includes an impression trend graph and an impressions metrics box that lists the number of impressions that each user accessed in the time period, the difference in the percentage of impressions between the last two time periods, and the direction of user activity between the last two time periods.

The following image shows the trend view of the **Top Users** indicator:



You can perform the following actions on the **Top Users** indicator trend view:

- To view details for each user, click the indicator label.
- To view a graph of impressions over time for a specific user, click a user name.

To return to the **Top Users** indicator list view, click the **Toggle** icon in the top right corner.

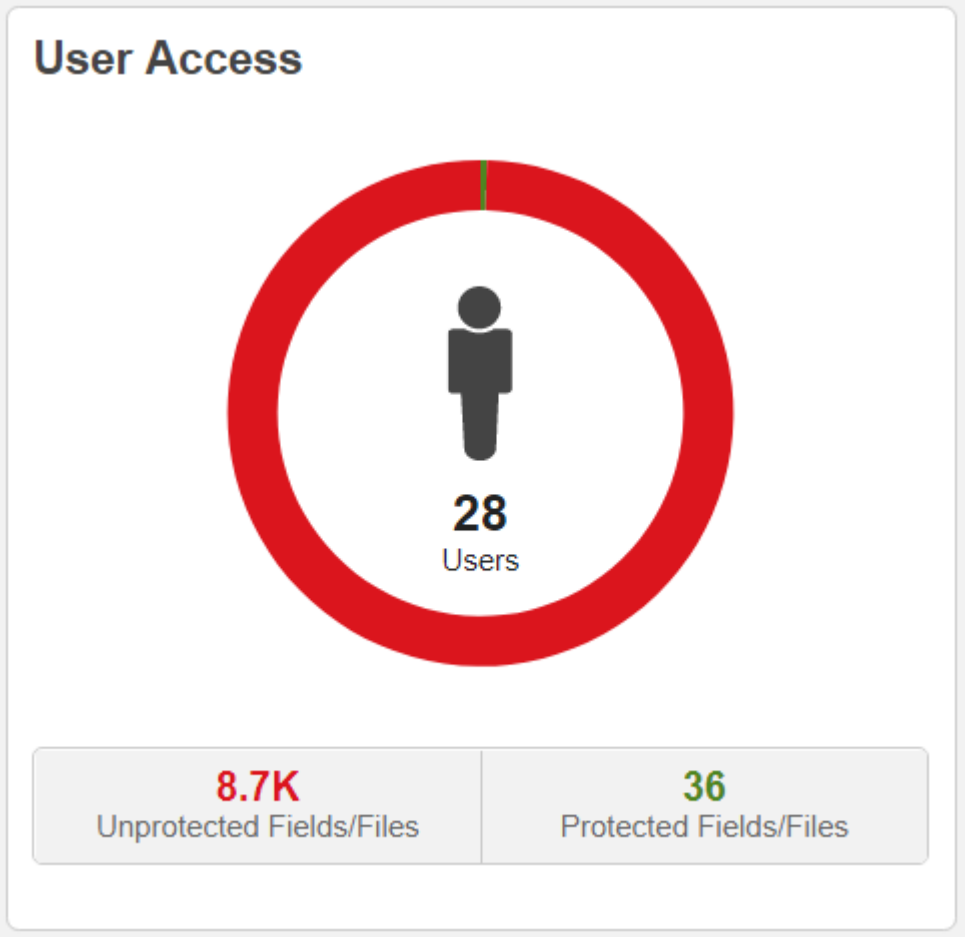
User Access Indicator

The **User Access** indicator provides a quick way to determine the total number of users who can access sensitive fields and files and the number of protected and unprotected sensitive fields and files that users can access.

Information appears on the **User Access** indicator after you run data store scans or import data store details to identify sensitive data and after you import user information.

By default, the user access metrics show the total number of users with access to sensitive fields or files, as represented by the circle. The red arc represents the unprotected sensitive fields or files, and the green arc represents the protected sensitive fields or files. The length of each arc is proportional to the number of sensitive fields or files in that category. The metrics at the bottom of the indicator show the number of protected and unprotected sensitive fields and files.

The following image shows the **User Access** indicator:



You can perform the following actions on the **User Access** indicator:

- To view the **User Access** page, click the indicator label.
- To view the number and percentage of users with access to unprotected or protected sensitive fields or files, click the corresponding portion of the arc.
- To see how many users can access protected or unprotected sensitive fields, click the number above the **Unprotected Fields/Files** and **Protected Fields/Files** labels.

Pages Accessed from the Security Dashboard

When you click an indicator label on the Security Dashboard view of the **Overview** workspace, a new page appears. The page provides more details about the summary information shown on the Security Dashboard indicator.

Some Security Dashboard indicators provide links to two or more pages in Data Privacy Management. The links display as blue values in indicator columns. You can access some pages from more than one indicator. Data Privacy Management filters the page results according to the type of information the indicator displays, such as a data store or data domain name.

You can access the following pages from indicators on the Security Dashboard:

- Data Domain Details
- Data Domains List
- Data Stores Grid
- Data Stores List
- Departments
- Proliferation
- Sensitive Data by Data Store Group
- Sensitive Data Proliferation by Classification Policy
- Sensitive Data Proliferation by Data Store Group
- Sensitive Data Locations
- Sensitive Fields
- User Access
- User Activity
- User Profile
- Users

Data Domain Details Page

The **Data Domain Details** page displays details and key metrics about the sensitive fields in a data domain. The summary metrics in the top right of the page show the number of data stores, sensitive fields/files, and protected fields/files in the data domain.

You can access the **Data Domain Details** page in one of the following ways:

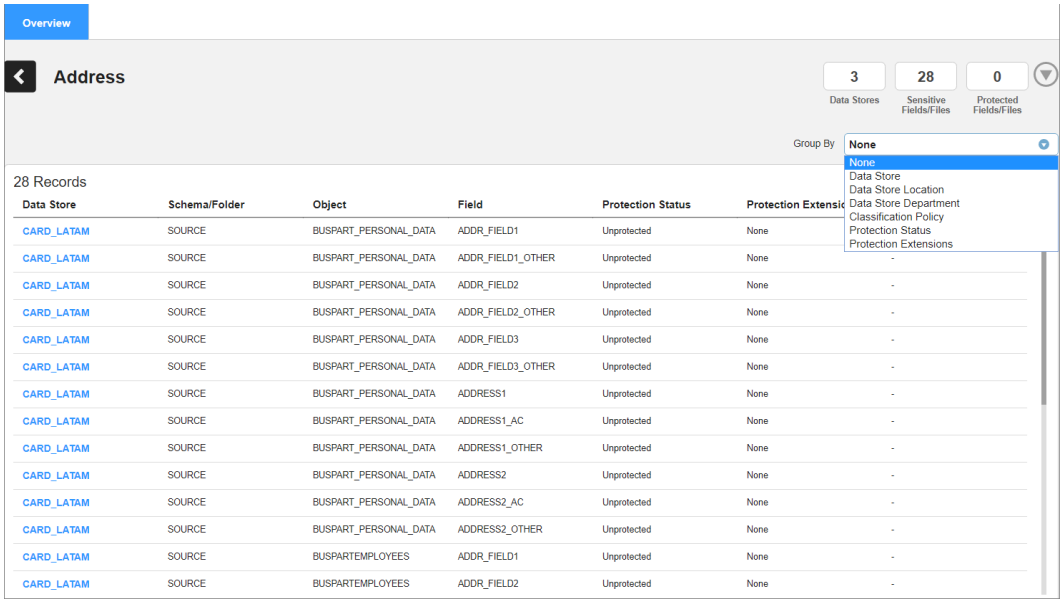
- On the **Top Data Domains** indicator, click a data domain name.
- On the **Data Domains** list page, click a data domain name, a value in the **Sensitive Fields** column, or a value in the **Sensitive Files** column.

The page shows information about the sensitive fields in a data domain in the following columns:

Column Name	Description
Data Store	The data store that contains each sensitive field. By default, the page lists the sensitive fields by data store in alphabetical order.
Schema/Folder	The schema or folder that contains each sensitive field.

Column Name	Description
Object	The object or table name that contains each sensitive field.
Field	The sensitive field name.
Protection Status	Indicates whether the field is protected or unprotected.
Protection Extension	For protected fields, lists the name of the protection extension.
Tags	Lists any tag names associated with the data store.

The following image shows the **Data Domain Details** page for the Address data domain:



Actions

To view the **Proliferation** page for a data store, click the data store name.

To organize the list of sensitive fields by a property other than data store name, select one of the following options from the **Group By** list:

- Data Store Location
- Data Store Department
- Classification Policy
- Protection Status
- Protection Extensions

To filter the list of sensitive fields, click the **Filter** icon in the Data Privacy Management header.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`DataDomainDetails.csv`	For each data domain, lists the data store name, schema name, object name, field name, protection status, protection techniques, number of sensitive fields, and tags.
`DataDomainDetailsSummary.csv`	Lists the information in the summary metrics area of the page: the total number of data stores, sensitive fields, and protected fields.

Data Domains List Page

When you click the **Top Data Domains** indicator label on the **Overview** workspace, the **Data Domains** list page appears. The page displays additional details about the data domains that you can access in Data Privacy Management.

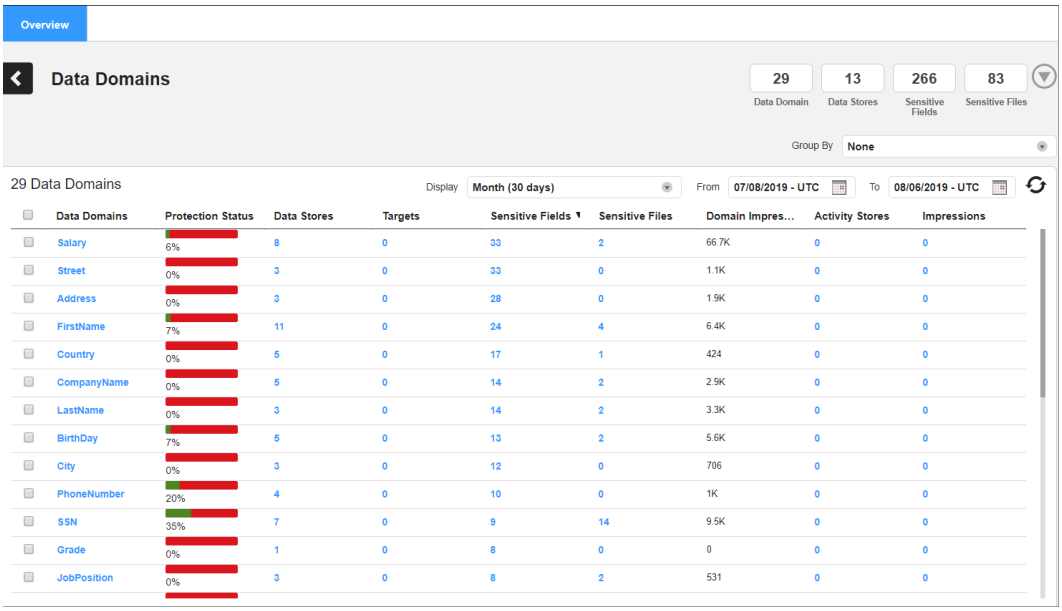
The summary metrics in the top right of the page list the total number of data domains and the total number of data stores, sensitive fields, and sensitive files in the data domains.

The page shows information about each data domain in the following columns:

Column Name	Description
Data Domains	The data domain name.
Protection Status	The protection status of sensitive fields in each data domain, represented as a numeric percentage of protected fields and a bar graph showing the protected percentage in green and the unprotected percentage in red.
Data Stores	The number of source data stores that contain fields that match the data domain.
Targets	The number of target data stores that contain fields that match the data domain and that proliferated from another data store.
Sensitive Fields	The number of sensitive fields in each data domain.
Sensitive Files	The number of sensitive files across all the fields that match each data domain.
Domain Impressions	The total number of classification policy impressions for each data domain.
Activity Stores	The number of data stores that contain fields that match the data domain and impressions that users accessed.
Impressions	The number of impressions accessed by the users.

By default, the page lists the data domains by the number of sensitive fields in descending order and displays information from the past 30 days.

The following image shows the **Data Domains** list page:



Actions

To view details about a data domain, click a data domain name or click the value in the **Sensitive Fields** or **Sensitive Files** column.

To perform a manual action, select one or more data domains and then select an option from the **Actions** > **Take Action** menu.

To view the list of source or target data stores in a data domain, click the value in the **Data Stores** or **Targets** column. The **Data Stores** list page appears, filtered by the data domain name.

To view the list of activity stores in a data domain, click the value in the **Activity Stores** column. The **Data Stores** list page appears, filtered by the user activity data domain name.

To view details about user activity events, click the value in the **Impressions** column. The **User Activity** page appears.

To organize the list of data domains by a property other than the number of sensitive fields, click a column name to sort by that property or select one of the following options from the **Group By** list:

- Data Store Department
- Data Store Group
- Data Store Location
- Classification Policy

To change the default time period of the last Month (30 days), select one of the following options from the **Display** menu:

- Today
- Day (24 hours)
- Week (7 days)
- 60 days
- 90 days
- 1 year

- Custom dates. When you select this option, use the calendars in the **From** and **To** fields to select the custom date range.

To filter the page with a condition common to all pages accessed from the **Overview** workspace, click the **Filter** icon in the Data Privacy Management header.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`DataDomainDrillDown.csv`	For each data domain, lists the data domain name and ID. The file also lists the number of protected and unprotected fields, data stores, targets, sensitive fields, sensitive files, domain impressions, activity stores, and impressions.
`DataDomainDrillDownSummary.csv`	Lists the information in the summary metrics area of the page: the total number of data domains in the list, data stores, sensitive fields, and sensitive files.

Data Stores Grid Page

The **Data Stores** grid page lists data stores and key scan metrics based on the way you accessed the page from the **Overview** workspace. The summary metrics in the top right of the page show the number of data stores and data store locations in the list.

You can access the **Data Stores** grid page in the following ways:

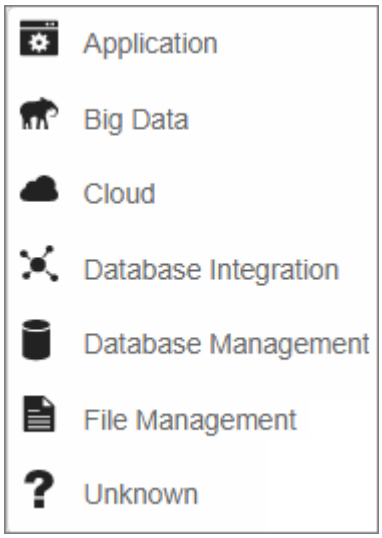
- On the **Overview** workspace, click the **Top Data Stores** indicator label. The page displays the list of top data stores.
- On the **Top Users** indicator, click a value in the **Data Stores** column. The page displays the list of data stores the user can access.
- On the **Sensitive Data Proliferation by Classification Policy** page, click a value in the **Sources** or **Targets** column. The page displays the list of source or target data stores that contain sensitive data that matches the classification policy.

The page shows information about each data store in the following columns:

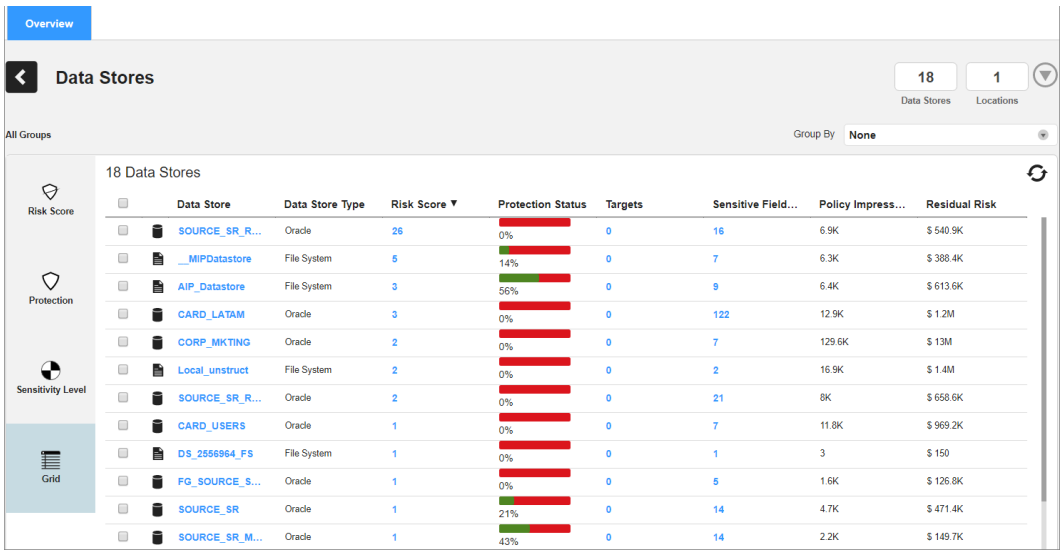
Column Name	Description
Data Store	The data store names. The icon to the left of the name indicates whether the data store is an application, big data, cloud, database integration, database management, file management, or unknown data source.
Data Store Type	The name of the type of data store application, big data, cloud, database integration, database management, file management, or unknown data source. For example, the data store applications in the example image have a data store type of Oracle.
Risk Score	The average risk score of the sensitive fields and files in the data store. By default, the page lists the data stores by the risk score in descending order.
Max Risk Score	The maximum risk score of the sensitive fields and files in the data store. Appears if you configure the dashboard settings.

Column Name	Description
Protection Status	The protection status of sensitive fields in each data store, represented as a numeric percentage of protected fields and a bar graph showing the protected percentage in green and the unprotected percentage in red.
Targets	The number of target data stores that proliferated from the data store.
Sensitive Fields/ Files	The number of sensitive fields/files in the data store.
Policy Impressions	The total number of classification policy impressions for the data store.
Residual Risk	The residual risk cost of the unprotected sensitive fields/files in the data store.

The following image shows the data store icon list and the data source type each icon represents:



The following image shows the **Data Stores** grid page:



Actions

To perform a manual action, select one or more data stores and then select an option from the **Actions** > **Take Action** menu.

To create a new risk simulation plan, select one or more data stores and then select **Actions** > **Simulate Risk**.

To view proliferation details about a data store, click a data store name or the value in the **Targets** column.

To view the **Risk Score Details** window, click the value in the **Risk Score** column.

To view the list of sensitive fields or sensitive files in a data store, click the value in the **Sensitive Fields/Files** column. The **Sensitive Fields** page appears.

To organize the list of data stores by a property other than the risk score, click a column name to sort by that property or select one of the following options from the **Group By** list:

- Department
- Data Owner
- Data Store Group
- Sensitivity Level
- Location

To filter the list, click the **Filter** icon in the Data Privacy Management header.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`DataStores.csv`	For each data store, lists the data store name, protection status, number of unprotected fields, number of sensitive fields and records, number of targets, the risk score, residual risk cost, total risk cost, risk cost unit of measure, severity level, location, department, data owner, and the data store group. If you configure the dashboard Risk Score settings to display the maximum risk score, the column is included in the exported file.
`DataStoresSummary.csv`	Lists the information in the summary metrics area of the page: the total number of data stores and locations.

Data Stores List Page

The **Data Stores** list page displays details and key scan metrics based on the way you accessed the page from the **Overview** workspace. The summary metrics in the top right of the page show the number of data stores and data store locations in the list.

You can access the **Data Stores** list page in one of the following ways:

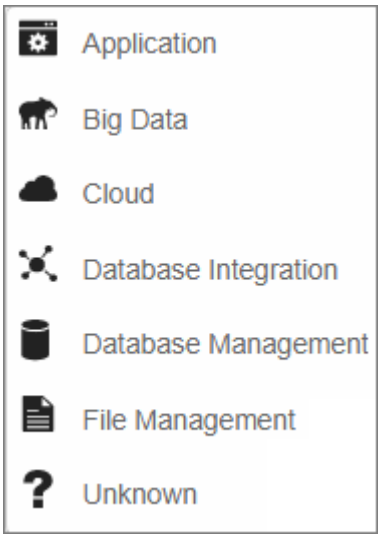
- On the **Top Data Domains** indicator, click a value in the **Data Stores** column. The page displays the list of data stores included in the data domain.
- On the **Top Departments** indicator or the **Departments** page, click a value in the **Data Stores** column. The page displays the list of data stores included in the department.
- On the **Sensitive Data for Location** indicator, click the number of data stores. The page displays the list of data stores included in the region selected on the location indicator.
- On the **Data Domains** list page, click a value in the **Data Stores** or **Targets** column. The page displays the list of data stores included in the data domain.

- On the **Sensitive Data by Data Store Group** page, click a value in the **Data Stores** or **Targets** column in the Scan Results pane. The page displays the list of data stores in a data store group that matches a classification policy or the target data stores to which sensitive fields proliferated.
- On the **Sensitive Data Proliferation by Classification Policy** page, click the value in the **Data Stores** or **Targets** column. The page displays the list of data stores in a data store group that matches a classification policy or the target data stores to which sensitive fields proliferated.
- On the **Sensitive Data Proliferation by Data Store Group** page, click the value in the **Sources** or **Targets** column. The page displays the list of data stores in a data store group that matches a classification policy or the target data stores to which sensitive fields proliferated.
- On the **Users** page, click the value in the **Data Stores** column. The page displays a list of data stores that a user accessed.
- On the **User Profile** page, click the Data Stores count. The page displays a list of data stores that a user accessed.

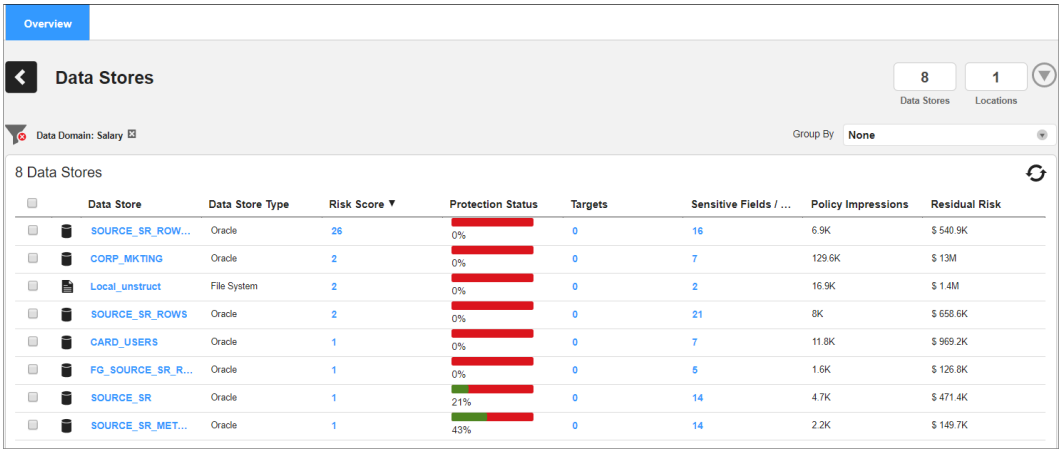
The **Data Stores** list page shows information about each data store in the following columns:

Column Name	Description
Data Store	The data store names. The icon to the left of the name indicates whether the data store is an application, big data, cloud, database integration, database management, file management, or unknown data source.
Data Store Type	The name of the application, big data, cloud, database integration, database management, file management, or unknown data store type. For example, the data store applications in the example image have a data store type of Oracle.
Risk Score	The aggregate risk score of the sensitive fields and files in the data store. By default, the page lists the data stores by the risk score in descending order. If you configure the dashboard Risk Score settings to display the maximum risk score, the column is included in the exported file.
Max Risk Score	The maximum risk score of the sensitive fields and files in the data store. Appears if you configure the dashboard settings.
Protection Status	The protection status of sensitive fields in each data store, represented as a numeric percentage of protected fields and a bar graph showing the protected percentage in green and the unprotected percentage in red.
Targets	The number of target data stores that proliferated from the data store.
Sensitive Fields/ Files	The number of sensitive fields/files in the data store.
Policy Impressions	The total number of classification policy impressions for the data store.
Residual Risk	The residual risk cost of the unprotected sensitive fields/files in the data store.

The following image shows the data store icon list and the data store type each icon represents:



The following image shows the **Data Stores** list page filtered by the Salary data domain:



Actions

To list the data stores by a property other than the risk score, you can click a column name to sort by that property, or you can select one of the following options from the **Group By** menu:

- Department
- Data Owner
- Data Store Group
- Sensitivity Level
- Location

To perform a manual action, select one or more data stores and then select an option from the **Actions** > **Take Action** menu.

To view proliferation details about a data store, click a data store name or the value in the **Targets** column.

To view the **Risk Score Details** window, click the value in the **Risk Score** column.

To view the list of sensitive fields or sensitive files in a data store, click the value in the **Sensitive Fields/Files** column. The **Sensitive Fields** page appears.

To filter the list, click the **Filter** icon in the Data Privacy Management header.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`TargetDataStores.csv`	For each data store shown on the page, lists the protection status, number of protected sensitive fields/files, number of sensitive fields/files, number of targets, risk score, residual risk cost, and total risk cost. If you configure the dashboard Risk Score settings to display the maximum risk score, the column is included in the exported file.
`TargetDataStoresSummary.csv`	Shows the total number of data stores and locations as shown in the summary metrics.

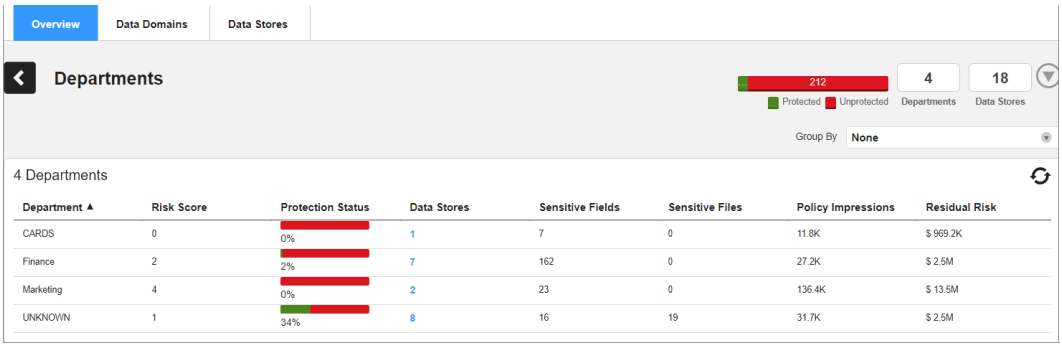
Departments Page

To access the **Departments** page, click the **Top Departments** indicator label on the **Overview** workspace. The summary metrics in the top right of the page show the total number of protected and unprotected fields in the department list, the number of departments, and the total number of data stores in the departments.

The page shows information about each department in the following columns:

Column Name	Description
Department	The name of a department that includes one or more data stores. By default, the page lists the departments by name in alphabetical order.
Risk Score	The average risk score of the sensitive fields and files in the data stores included in the department.
Max Risk Score	The maximum risk score of the sensitive fields and files in the data stores included in the department. Appears if you configure the dashboard settings.
Protection Status	The protection status of sensitive fields in the data stores in each department, shown as a numeric percentage of protected fields and a bar graph that represents protected fields in green and unprotected fields in red.
Data Stores	The number of data stores included in the department.
Sensitive Fields	The number of sensitive fields in the data stores included in the department.
Sensitive Files	The number of sensitive files in the data stores included in the department.
Policy Impressions	The total number of classification policy impressions for the data stores included in the department.
Residual Risk	The residual risk cost of the unprotected sensitive fields/files in the data stores included in the department.

The following image shows the **Departments** page:



Actions

To view the list of data stores in a department, click the value in the **Data Stores** column.

To view the list of departments by a property other than the name, you can click a column name to sort by that property, or you can select **Sensitivity Level** from the **Group By** menu. When you sort the list by sensitivity level, Data Privacy Management groups the departments with data stores from highest to lowest sensitivity level based on the criteria defined on the **Settings** workspace.

To filter the list, click the **Filter** icon in the Data Privacy Management header.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`Departments.csv`	For each department, lists the department name, number of unprotected and protected fields, number of data stores, number of sensitive fields and records, number of targets, the residual risk cost, and the total risk cost. If you configure the dashboard Risk Score settings to display the maximum risk score, the column is included in the exported file.
`DepartmentsSummary.csv`	Lists the total number of departments and locations. The data appears in the summary metrics area of the page.

Proliferation Page

The **Proliferation** page shows the movement of sensitive data from source to target data stores.

You can access the **Proliferation** page in the following ways:

- On the **Top Data Stores** indicator, click the name of a data store in the **Data Stores** column. The page shows the proliferation details for the selected data store.
- On the **Proliferation by Classification Policy** page, click a classification policy name. The page shows the proliferation of sensitive data for all data stores included in the classification policy.
- On the **Data Domain Details** page, click the name of a data store. The page shows the proliferation details for the selected data store.
- On the **Data Stores** grid page and the **Data Stores** list page, click a data store name or a value in the **Targets** column. The page shows the proliferation details for the selected data store.
- On the **User Access** page, select a data store in the **Data Stores** pane and then click **View**. The page shows the proliferation details for the selected data store.

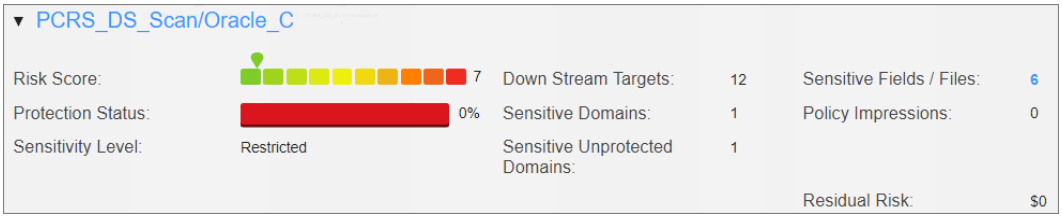
- On the **User Activity** page, click the name of a data store. The page displays the proliferation status of the data store included in the user activity event.

The following image shows the **Proliferation** page for data stores included in the ALL classification policy:



1. Full Screen (left) and Search (right) icons
2. Data store icon legend
3. Navigation icons. Click the top icon to zoom in. Click the middle icon to zoom out. Click the arrows in the lower icon to navigate the page when viewing in full screen mode or when viewing a small area of the page after zooming in.
4. Scan Results pane. Displays key scan metrics for a selected data store. To view the metrics, click the arrow to the left of the data store name.
5. Proliferation pane. Shows the proliferation path of sensitive data from the source data store through upstream sources and downstream target data stores.
6. Page display options. By default, the page shows the data store icon legend, lists the risk score under each data store name, and the colors the data store hexagons to indicate protection status.
7. Actions menu.

The following image shows the **Scan Results** pane for the PCRS_DS_Scan/Oracle_C data store in the example **Proliferation** page:



Actions

To perform a manual action for the selected data store, select an option from the **Actions** > **Take Action** menu.

To remove the data store icon legend, clear the **Data Store Types** box.

To change the default text below data store names, select one of the following options from the **Text** menu:

- Residual Risk
- Sensitivity Level
- Region
- Location

To change the default color code for data store hexagons, select one of the following options from the **Color** menu: Sensitivity Level or Risk Score.

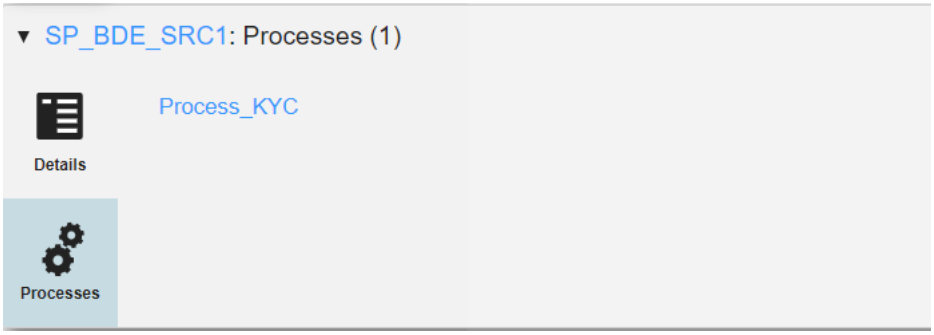
To view the **Scan Results** pane for different data stores, click the data store hexagon in the **Proliferation** pane.

To view details about the sensitive fields/files in the data store shown in the **Scan Results** pane, click the number of sensitive fields/files in blue. .

To view more details about the data store shown in the **Scan Results** pane, click the data store name in blue at the top of the pane.

To view information on Axon Data Governance processes linked to the data store, click the arrow to expand the **Scan Results** pane. The number of linked processes is indicated next to the data store name. Select the **Processes** tab. The names of the processes are links that open the process page in Axon Data Governance.

The following image shows the Processes tab on the **Scan Results** pane with one process listed:



To filter the data stores shown on the page, click the **Filter** icon in the Data Privacy Management header.

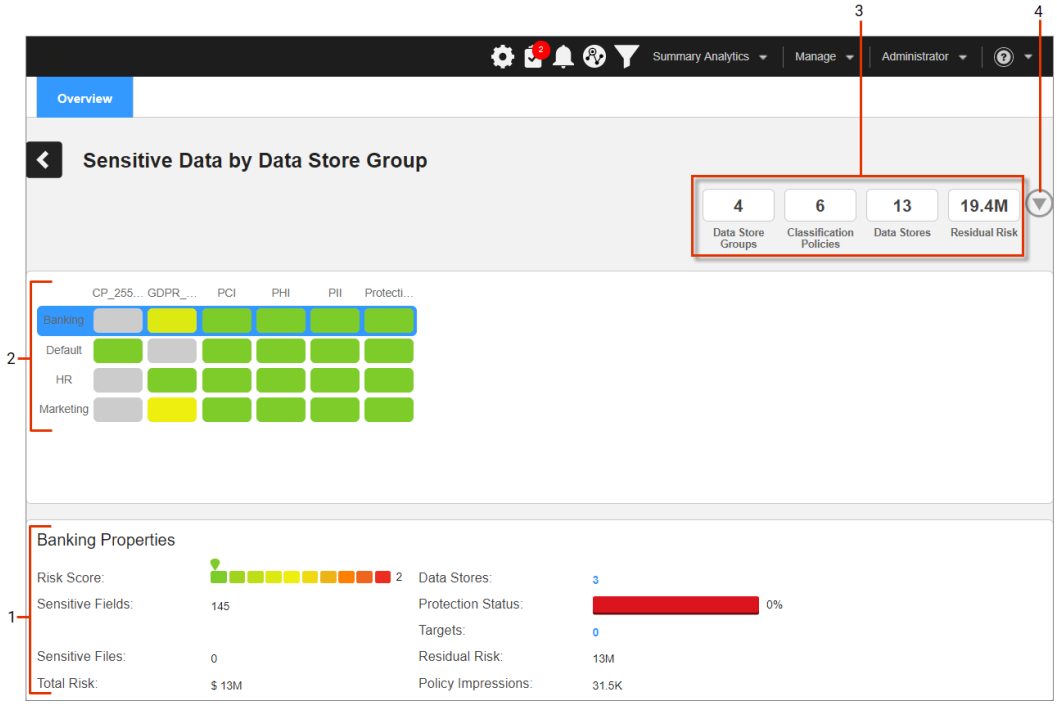
To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`DataStoreProliferation.csv`	For the selected data stores on the page, lists the data store name, ID, parent data store ID, risk score, residual risk cost, total risk cost, risk cost unit of measure, region, location, severity level, number of protected fields, and number of sensitive fields.
`findConnection.csv`	Lists data stores and associated risk score, protection status, the number of sensitive domains, the number of sensitive unprotected domains, and sensitivity level.

Sensitive Data by Data Store Group Page

When you click the **Sensitive Data by Data Store Groups** indicator label on the **Overview** workspace, the **Sensitive Data by Data Store Group** page appears. The page shows a matrix of data store groups sorted by classification policies.

The following image shows the **Sensitive Data by Data Store Group** page:



1. Scan Results pane. Includes key aggregate scan results for the data stores in a selected row or column.
2. Tile pane. Displays data stores that match a classification policy in a tile grid, with rows by data store group and columns by classification policy. To view the Scan Results pane, click a classification policy name or a data store group name.
3. Data store group summary metrics. Shows the total number of data store groups and matching classification policies, the number of data stores in the data store groups, and the total risk cost of the data stores.
4. Actions menu

Actions

On the Scan Results pane, click the value for the **Data Stores** or **Targets** count. The **Data Stores** list page shows details about the data stores in the data store group that matches the classification policy or the target data stores to which sensitive fields proliferated.

To filter the page with a condition common to all pages accessed from the **Overview** workspace, click the **Filter** icon in the Data Privacy Management header.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

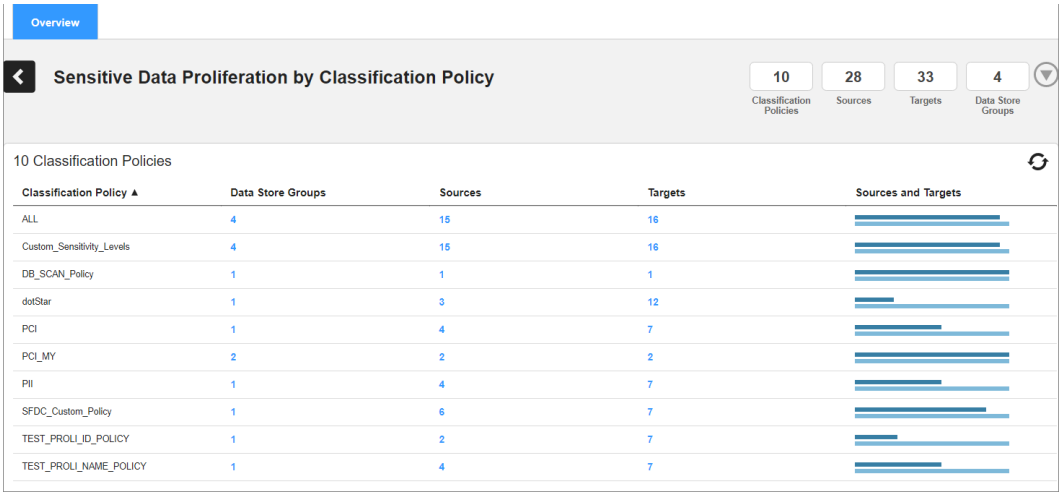
File Name	Description
`SensitiveDataByDataStoreGroup.csv`	Lists the classification policies and associated data store group and risk score.
`SensitiveDataByDataStoreGroupSummary.csv`	Lists the information in the summary metrics area of the page: the total number of data store groups, classification policies, data stores, and residual risk cost.

Sensitive Data Proliferation by Classification Policy Page

When you click the **Sensitive Data Proliferation** indicator label on the **Overview** workspace, the **Sensitive Data Proliferation by Classification Policy** page appears. The summary metrics in the top right of the page show the total number of classification policies, sources, targets, and data store groups associated with sensitive data proliferation.

The **Sensitive Data Proliferation by Classification Policy** page lists the classification policies in alphabetical order by default. For each policy, the page lists the number of data store groups that match the policy, the number of source data stores, the number of target data stores, and a horizontal bar graph that shows the ratio of sources to targets. The source data store bar in dark blue is located above the target data store bar in light blue.

The following image shows the **Sensitive Data Proliferation by Classification Policy** page:



Actions

To sort the list of classification policies by the number of data store groups, sources, or targets, click the column label.

To view details about the data store groups that match a classification policy, click the value in the **Data Store Groups** column. The **Sensitive Data Proliferation by Data Store Group** page appears.

To view details about the data store sources included in the groups or the target data stores to which sensitive fields proliferated, click the value in the **Data Stores** or **Targets** column. The **Data Stores** list page appears.

To filter the page with a condition common to all pages accessed from the **Overview** workspace, click the **Filter** icon in the Data Privacy Management header.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

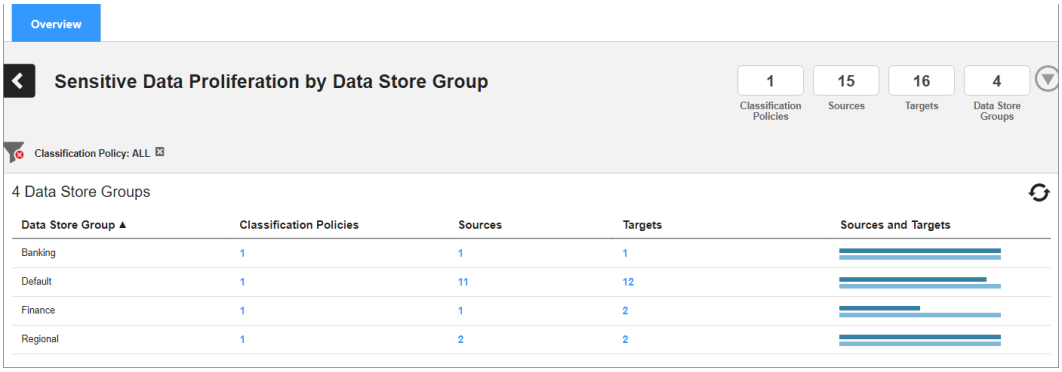
File Name	Description
`SensitiveDataProliferationByPolicy.csv`	Lists each classification policy name and the number of data stores, sources, and targets that match the policy.
`SensitiveDataProliferationSummary.csv`	Lists the information in the summary metrics area of the page: the total number of classification policies, sources, targets, and data store groups.

Sensitive Data Proliferation by Data Store Group Page

When you click a value in the **Data Store Groups** column on the **Sensitive Data Proliferation by Classification Policy** page, the **Sensitive Data Proliferation by Data Store Group** page appears. The summary metrics in the top right of the page show the total number of classification policies, sources, targets, and data store groups associated with sensitive data proliferation.

The **Sensitive Data Proliferation by Data Store Group** page lists the data store groups in alphabetical order by default. For each data store group, the page lists the number of classification policies, the number of source data stores, the number of target data stores, and a horizontal bar graph that shows the ratio of sources to targets. The source data store bar in dark blue is located above the target data store bar in light blue.

The following image shows the **Sensitive Data Proliferation by Data Store Group** page for the ALL classification policy:



Actions

To sort the list of data store groups by the number of sources or targets, click the column label.

To view details about the classification policy that the data store group matches, click the value in the **Classification Policies** column. The **Sensitive Data Proliferation by Classification Policy** page appears.

To view details about the data store sources included in the groups or the target data stores to which sensitive fields proliferated, click the value in the **Sources** or **Targets** column. The **Data Stores** list page appears.

To filter the page with a condition common to all pages accessed from the **Overview** workspace, click the **Filter** icon in the Data Privacy Management header.

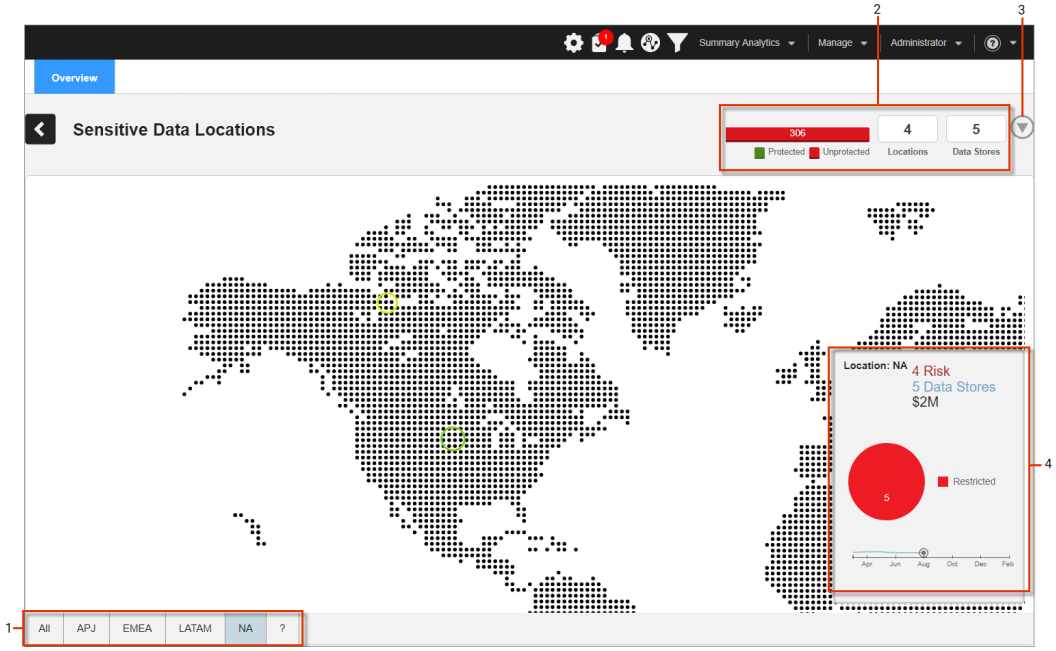
To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`SensitiveDataProliferationByAppGroup.csv`	Lists each data store group name and the number of classification policies, sources, and targets included in the group.
`SensitiveDataProliferationSummary.csv`	Lists the information in the summary metrics area of the page: the total number of classification policies, sources, targets, and data store groups.

Sensitive Data Locations Page

When you click the **Sensitive Data for Location** indicator label on the **Overview** workspace, the **Sensitive Data Locations** page appears. The page shows the number of sensitive fields in each region that are protected and unprotected and the number of locations associated with the data stores in the each region.

The following image shows the **Sensitive Data Locations** page for the North America region:



1. Region tabs.
2. Sensitive data summary.
3. Actions menu.
4. Information shown on the workspace indicator.

Actions

To filter the page with a condition common to all pages accessed from the **Overview** workspace, click the **Filter** icon in the Data Privacy Management header.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`SensitiveDataByLocation.csv`	Shows the number of data stores for the region you selected on the page. Also shows details such as the residual risk cost and risk score for the group of data stores.
`SensitiveDataByLocationRiskTrendSummary.csv`	Shows the monthly risk scores for the group of data stores in the region you selected.
`SensitiveDataByLocationSensitivityLevelSummary.csv`	Lists the sensitivity levels and the number of data stores for each sensitivity level.
`SensitiveDataLocationsSummary.csv`	Shows the number of locations, data stores, protected sensitive fields, and sensitive fields in each region.

Sensitive Fields Page

For structured data stores, the **Sensitive Fields** page lists the fields in a data store that Data Privacy Management identifies as sensitive or protected.

The summary metrics in the top right of the page show the number of sensitive and protected fields associated with the data store, the risk score, the number of scans for the data store, and the number of master categories associated with the fields in the list.

You can access the **Sensitive Fields** page in the following ways:

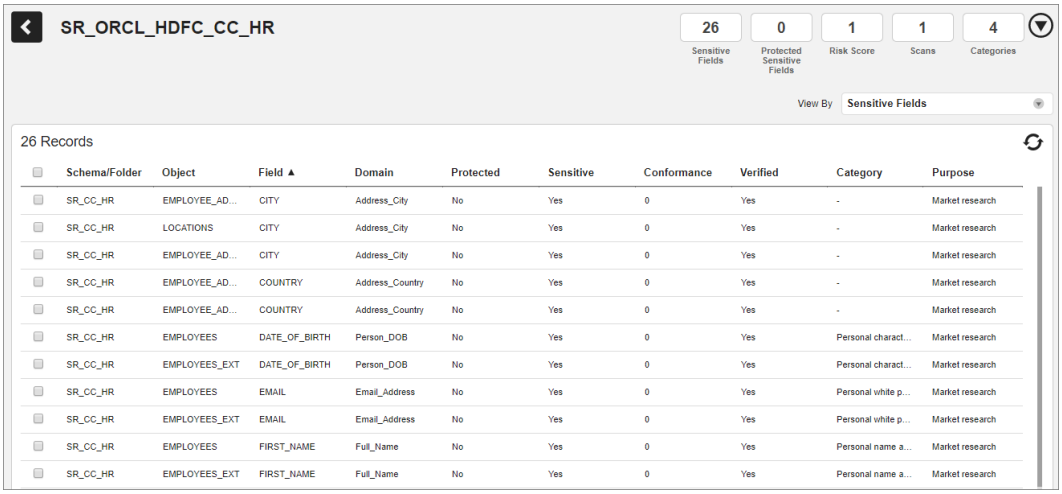
- On the **Top Data Stores** indicator, click a value in the **Fields/Files** column. The page displays the list of sensitive fields identified for the data store.
- On the **Data Stores** grid page and the **Data Stores** list page, click a value in the **Sensitive Fields/Files** column. The page displays the list of sensitive fields identified for the data store.
- On the **Proliferation** page, click the number of sensitive fields/files in the data store **Scan Results** pane.

The **Sensitive Fields** page shows information about each sensitive field in the following columns:

Column	Description
Schema/Folder	The schema or folder name that contains the sensitive field. For Snowflake databases, the name appears in the following format: `DatabaseName.SchemaName` This allows you to distinguish between schemas with the same name across different databases.
Object	The object or table name that contains the sensitive field.
Field	The name of the sensitive field. By default, the page lists the sensitive fields by name in alphabetical order.
Domain	The data domain that includes the sensitive field.
Protected	Indicates whether the sensitive field is protected. Possible values are Yes and No.

Column	Description
Sensitive	Indicates whether Data Privacy Management has identified the field as sensitive. Possible values are Yes and No.
Conformance	The conformance score that represents the percentage of values in the field that meet the data match condition for the data domain. If the Run Profiling option in the scan job specifies a data profile, Data Privacy Management evaluates each value in the sample set to determine the conformance score.
Verified	Indicates whether or not the data store owner verified the sensitivity and protection status of the field. Possible values are Yes and No.
Category	The master category associated with the data domain assigned to the sensitive field. If you override the master category at the field level, the list displays the category that you assign to the field.
Purpose	The master purpose associated with the data store. If you override the master purpose at the field level, the list displays the purpose that you assign to the field.

The following image shows the **Sensitive Fields** page for the data store SR_ORCL_HDFC_CC_HR:



How Data Privacy Management Identifies Fields as Sensitive

A field can appear on the **Sensitive Fields** page for the following reasons:

1. You imported data store details from a CSV file that identified the field as sensitive.
2. You specified a global sensitivity status for the field as always sensitive for a specific data domain. If a classification policy includes the data domain, the field appears on the **Sensitive Fields** page even if the Profiling job step in a scan did not identify the field as sensitive.
3. You specified a global sensitivity status for the field as Never Sensitive for one or more data domains, but the Profiling job step identified the field as sensitive.
4. The conformance score is in the Auto Accept range, and the metadata and data profile results do not conflict.
5. The conformance score is in the Validate range, but you enabled the **Mark columns within the validate range as sensitive** property for the data domain, and the metadata and data profile results do not conflict.
6. The conformance score is in the Validate range, but the scan job identified the field as protected.

7. The metadata profile and the data profile results conflict. Data Privacy Management applies the conflict resolution rule specified in the data domain to determine sensitivity, and the rule determines that the field is sensitive.

The list of sensitive fields in a data store updates when a scan job finishes, when an Import job that imports sensitivity status finishes, or when you change the protection or sensitivity status of a field in the **Properties** pane.

Actions

By default, the page lists sensitive fields imported to Data Privacy Management and sensitive fields identified in a scan job. To view only the sensitive fields imported on the **Data Stores** workspace, select **Imported Fields** in the **View By** menu located below the summary metrics.

To change the sensitivity or protection status for a sensitive field, select the check box in the field row and then click **Edit** in the **Properties** pane.

To manage the primary keys and unique keys for the tables that contain the identified sensitive columns, select **Actions** > **Manage Keys**.

To perform a manual action, select one or more fields and then select an option from the **Actions** > **Take Action** menu.

To filter the list, click the **Filter** icon in the Data Privacy Management header.

To export information about the sensitive fields, select **Export** > **Sensitive Fields** from the **Actions** menu. To export data store details, select **Export** > **Data Store Details** from the **Actions** menu. You can update the protection status information in the exported DataStoreDetails.csv file and then import the edited information on the **Data Stores** workspace.

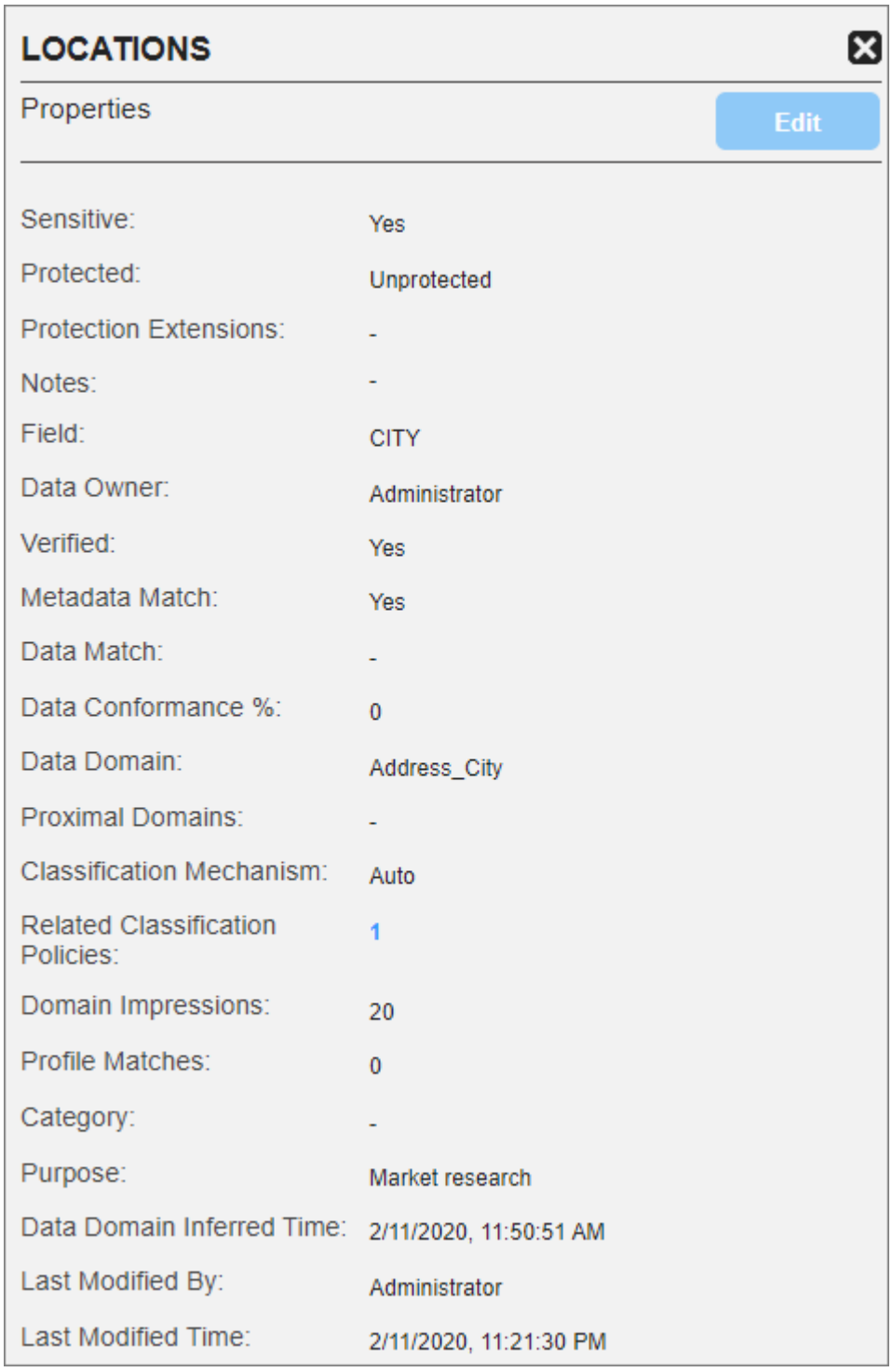
To override the master category and purpose association for a sensitive field, select **Export** > **Categories and Purpose** from the **Actions** menu. You can then edit the exported file and import the updated file on the **Sensitive Fields** page.

Sensitive Field Properties Pane

To view the **Properties** pane for a sensitive field, select the check box next to the field. The **Properties** pane includes field properties that are not listed in the **Sensitive Fields** page columns.

To update the sensitivity and protection status for the field, click **Edit**.

The following image shows an example **Properties** pane for the sensitive field CITY in the LOCATIONS object:



The following table describes the information on the **Properties** pane that is not listed on the **Sensitive Fields** page:

Property	Description
Protection Extensions	If the field is protected, the name of the protection extension.
Notes	Optional. You can enter notes when you edit the protection or sensitivity status of a sensitive field.
Data Owner	The name of the data store owner.
Metadata Match	Indicates if the Metadata Profiling job step of the data store scan identified the field as sensitive for the data domain. Values are Yes, No, or a hyphen (-). The hyphen indicates that the Scan job did not include a metadata profile.
Data Match	Indicates if the Data Profiling job step of the data store scan identified the field as sensitive for the data domain. Values are Yes, No, or a hyphen (-). The hyphen indicates that the Scan job did not include a data profile.
Proximal Domains	A list of any data domains that are related to the data domain that includes the sensitive field, if a Proximity Match condition is specified for the data domain.
Classification Mechanism	The method that determined the current sensitivity status of the selected field. Possible values are: - Auto. The sensitivity status is based on the results of the Profiling job step. - Auto (Always Sensitive). The field is globally sensitive for the data domain. - Auto (Never Sensitive). The field is never sensitive for the data domain. However, the Profiling job step identified that the field is sensitive for the data domain. - Import. The user imported the sensitivity status from a CSV file. - Manual. The user updated the sensitivity status by editing the **Sensitive** property in the **Properties** pane.
Related Classification Policies	The number of classification policies that include the field. To view or manage the policies, click the property value. The **Classification Policies** workspace opens.
Domain Impressions	The number of data domain impressions for the sensitive field and related classification policies.
Profile Matches	The number of data profiles that match the data domain for the sensitive field in the classification policy match conditions.
Data Domain Inferred Time	The current time in the data domain location.
Last Modified By	The user who most recently edited the data domain that contains the sensitive field.
Last Modified Time	The time the data domain that contains the sensitive field was most recently edited in the time zone of the user who edited the data domain.

Updating the Sensitivity and Protection Status for Sensitive Fields

To change the sensitivity and protection status for a field, edit the field properties on the **Sensitive Fields** page.

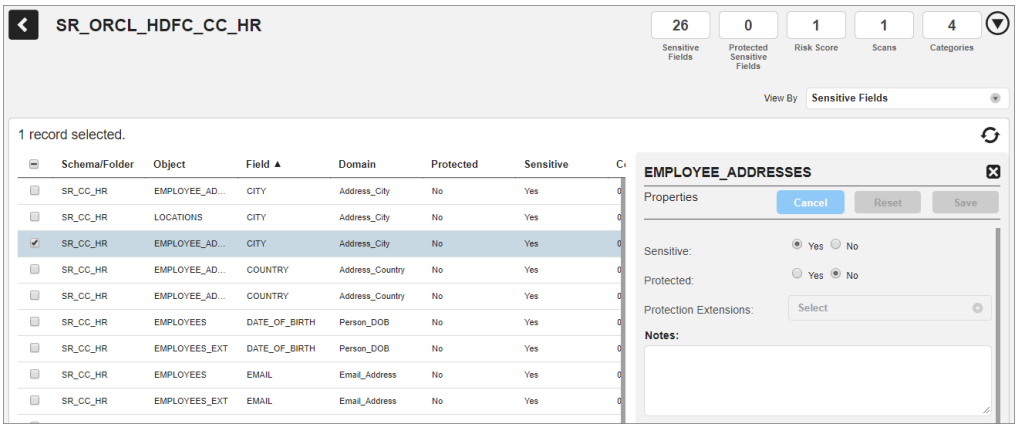
To edit fields that Data Privacy Management identified as protected or sensitive in a data store scan.

1. On the **Sensitive Fields** page, select the check box next to the field you want to update.

   The **Properties** pane for the selected field appears on the right of the page.

2. Click **Edit**.

   The following image shows an example of this view for the sensitive field CITY in the EMPLOYEE_ADDRESSES object:



3. To change the sensitivity status, click **Yes** or **No** for the **Sensitive** property.

   If you update the sensitivity information in Data Privacy Management, the change is updated in Enterprise Data Catalog.

4. To change the protection status, click **Yes** or **No** for the **Protected** property. If you change the protection status to **Yes**, you must select a configured protection extension to specify how the field is protected.

5. Optionally, enter any notes.

6. Click **Save**.

7. To update the values for sensitive and protected fields on the **Overview** workspace, you must first run the Evaluate Classification Policies job. Click **Manage** > **Classification Policies**.

   The **Classification Policies** workspace opens.

8. From the **Actions** menu, select **Evaluate Classification Policies**.

   The Evaluate Classification Policies job updates the results on the **Overview** workspace.

Exporting Sensitive Field Information

You can export details about the sensitive fields and the summary metrics on the **Sensitive Fields** page.

You can export details for one or more selected fields in the list. If you do not select any fields, the exported files provide details for all fields in the list.

1. On the **Sensitive Fields** page, select the check boxes next to the fields that you want to export if you do not want to export information about the entire list.

2. From the **Actions** menu, select **Export** > **Sensitive Fields**.

   Note: If you selected sensitive fields, the **Export** menu option appears below the solid line.

   The **Save As** dialog box appears with a .zip file named `Data.zip`.

3. Optionally, rename the .zip file. Save the file to a directory on your local machine.
4. Extract the file.

   The following CSV files appear: `DataStoreDetails.csv` and `DataStoreDetailsSummary.csv`
5. Open the CSV files to view or update the details.

   The following table describes the information in the CSV files:

File Name	Description
`DataStoreDetails.csv`	For each sensitive field, lists the repository name, schema/folder name, object name, field name/file type, whether the field is sensitive, the data domain name, the conformance score, protection status, protection technique, whether the field is verified, the impression count, the number of matched rows, the data domain inferred time, notes, the user who last updated the field, and the time of the last update. **Note:** The following points apply to unstructured sources: - The **Rows Matched** column is not applicable to scans run through the remote agent. - The Impression Count includes the total number of instances that the data domain was identified in the data and not the unique instances.
`DataStoreDetailsSummary.csv`	Lists the information in the summary metrics area of the page: the total number of sensitive fields, protected sensitive fields, aggregate risk score, and the number of data store scans.

Related Topics:

- "Sensitive Files Page" on page 558

Exporting Data Store Details

On the **Sensitive Fields** page, you can export details about the data stores that include the fields in the list.

1. From the **Actions** menu, select **Export** > **Data Store Details**.

   The **Save As** dialog box appears with a .zip file named `Data.zip`.
2. Optionally, rename the .zip file. Save the file to a directory on your local machine.
3. Extract the file.

   Seven CSV files appear in the directory.
4. Open the CSV files to view or update the details.

The following table describes the CSV files:

File Name	Description
DataStoreDetails.csv	For each sensitive field, lists the repository name, schema/folder name, object name, field name.file type, whether the field is sensitive, the data domain name, the conformance score, protection status, protection technique, whether the field is verified, the impression count, the number of matched rows, the data domain inferred time, notes, the user who last updated the field, and the time of the last update. **Note:** The following points apply to unstructured sources: - The **Rows Matched** column is not applicable to scans run through the remote agent. - The Impression Count includes the total number of instances that the data domain was identified in the data and not the unique instances.
DataStoreDetailsSummary.csv	Lists the information in the summary metrics area of the page: the total number of sensitive fields, protected sensitive fields, aggregate risk score, and the number of data store scans.
PolicyViolations.csv	For security policy violations that included sensitive fields, lists the security policy ID, flag and read status, the date and time the violation occurred, the security policy name, the name of the user associated with the violation, the data stores included in the violation, the security policy type, and the severity level.
Tasks.csv	For tasks that include sensitive fields, lists the task name, data store owner, extension, extension category, data stores, the task due date, and the task status.
UbaAnomaly.csv	For anomalous user behavior incidents that include sensitive fields, lists user details such as the user name, location, title, department, country, and user behavior group name. Lists anomaly details such as the ID, severity level and description, anomaly score, date and time the anomaly occurred, whether the anomaly is unread or flagged, and the total data stores and anomaly factors.
UserActivityDrillDown.csv	For user activity security policy violations that include sensitive fields, lists the event ID, the duration in milliseconds, the name and IP address of the user involved in the violation, the operation that caused the violation, the affected data stores and data store IP addresses, and the total number of data domains included in the violation.
Users.csv	For users associated with sensitive fields, provides details about the user names, title, email address, manager name, location, department, organization, and user status.

Importing Sensitive Fields from a CSV File

You can import a CSV file to add, update, or delete sensitive fields in Data Privacy Management. The content of the CSV file must be in the order and format that the Import job requires.

You can create an import CSV file, or you can update and import one of the following files that you download from Data Privacy Management:

- Scan report
- DataStoreDetails.csv file

Scan Report

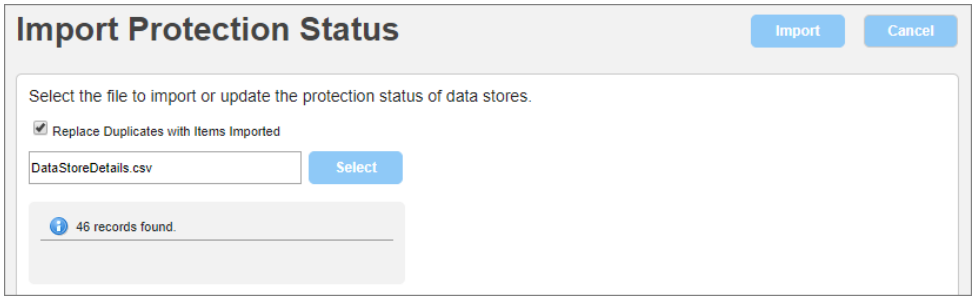
The scan report includes fields that the Scan job could not automatically reject or identify as sensitive.

1. Select **Manage** > **Jobs** to open the **Jobs** workspace.
2. Click the job ID of the data store Scan job.
3. Click the icon in the **Download Report** column of the Profiling step.
4. Download the .zip file and extract the two CSV files that have the following naming conventions: <*Job Step ID*>.csv and . <*Job Step ID*>_Scan.csv
5. Open the <*Job Step ID*>.csv file and update the values in the IsSensitive, ProtectionStatus, and Verified columns.
6. Optionally, add new fields that you identified as sensitive.
7. Save the file.
8. Select **Manage** > **Data Stores** to open the **Data Stores** workspace.
9. On the **Actions** menu, click **Import** > **Protection Status**.
10. On the **Import Protection Status** page, select the updated file you saved.
11. Optionally, select **Replace Duplicates with Items Imported**.
12. Click **Import**.

DataStoreDetails.csv File

The DataStoreDetails.csv file includes fields that the Scan job identified as sensitive or protected.

1. On the **Sensitive Fields** page, select **Export** > **Data Store Details** from the **Actions** menu.
2. Extract the DataStoreDetails.csv file from the .zip file that downloads.
3. Open the DataStoreDetails.csv file.
4. Update the values in the IsSensitive, ProtectionStatus, and Verified columns.
5. Save the file.
6. Select **Manage** > **Data Stores** to open the **Data Stores** workspace.
7. On the **Actions** menu, click **Import** > **Protection Status**.
8. On the **Import Protection Status** page, select the updated file you saved.



9. Optionally, select **Replace Duplicates with Items Imported**.
10. Click **Import**.

Related Topics:

- "Downloading Scan Job Reports" on page 309
- "Importing Protection Statuses" on page 109

Overriding the Master Category and Purpose Association of a Sensitive Field

You can override the master category or master purpose that you assign from the Subject Registry page.

You cannot restore the master category or purpose association to a field after you override it. If you want to update the association to the master category or purpose, you must repeat the steps to update the association of individual fields.

1. On the **Sensitive Fields** page, select **Export** > **Categories and Purpose** from the **Actions** menu.

   Save the `DataPrivacyFieldLinks_Summary.csv` file that downloads.
2. Open the `DataPrivacyFieldLinks_Summary.csv` file.
3. Update the required values in the **DataCategory** and **Purpose** columns.
4. Save the file.
5. Select **Import** > **Categories and Purpose** from the **Actions** menu.
6. On the **Import Categories and Purpose** page, select the updated file.

   If you choose to import a file that you create, the file must include all columns and headers that appear in the downloaded file.
7. Optionally, select **Replace Duplicates with Items Imported**.
8. Click **Import**.

   The import ignores rows that are unchanged and does not consider the values as overrides.

Sensitive Files Page

For unstructured data stores, the **Sensitive Files** page lists the files in a data store that Data Privacy Management identifies as sensitive or protected.

The summary metrics in the top right of the page show the number of sensitive files and data domains associated with the data store, the number of domain impressions for the data store, and the number classification policies and policy impressions associated with the files in the list.

You can access the **Sensitive Files** page in the following ways:

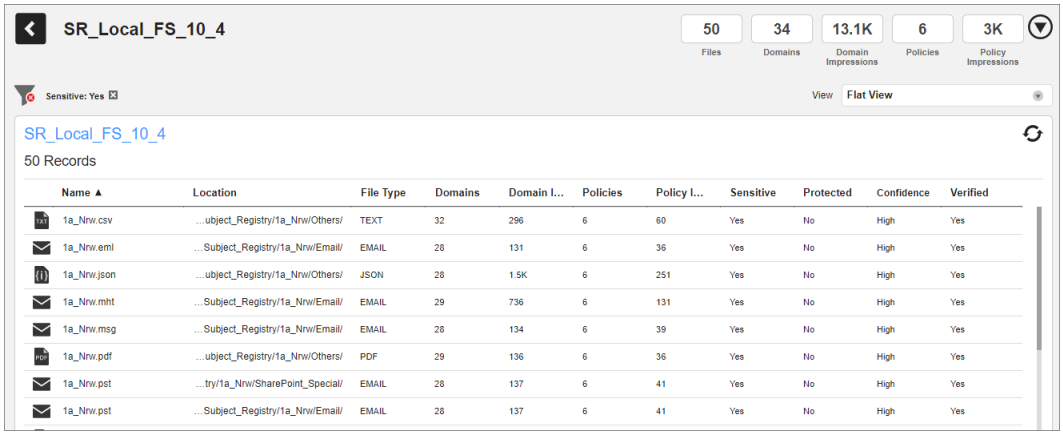
- On the **Top Data Stores** indicator, click a value in the **Fields/Files** column. The page displays the list of sensitive fields identified for the data store.
- On the **Data Stores** grid page and the **Data Stores** list page, click a value in the **Sensitive Fields/Files** column. The page displays the list of sensitive fields identified for the data store.
- On the **Proliferation** page, click the number of sensitive fields/files in the data store **Scan Results** pane.

The **Sensitive Files** page shows information about each sensitive file in the following columns in the default **Flat View**:

Column	Description
Name	The name of the sensitive file.
Location	The file path of the directory that contains the file.
File Type	The type of unstructured file, such as email, text, JSON, PDF, XML, or Compressed.
Domains	The number of data domains that match data in the sensitive file.

Column	Description
Domain Impressions	The number of data domain impressions for the file.
Policies	The number of classification policies that the scan job matched with the data in the sensitive file.
Policy Impressions	The number of classification policy impressions for the file.
Sensitive	Indicates whether Data Privacy Management has identified the file as sensitive. Possible values are Yes and No.
Protected	Indicates whether the sensitive file is protected. Possible values are Yes and No.
Confidence	The confidence level that the data domain is an exact match for the sensitive data in the file. Possible values are High, Medium, and Low.
Verified	Indicates whether or not the Confidence Level is verified. Possible values are Yes and No.

The following image shows the **Sensitive Files** page for the data store DR_Local_FS_10_4:



How Data Privacy Management Calculates Confidence Levels and Verified Status

The value in the **Confidence** column is calculated in the following ways for files that have classification policy matches with data domains:

1. At the data store scan level, the Confidence Level is the maximum value or highest level for each classification policy match.
2. At the classification policy level, the Confidence Level is the minimum value or lowest level for each domain that matched the criteria of the classification policy.
3. At the data domain level, the Confidence Level is the maximum value or highest level of all Confidence Levels for each data domain match for the file.

The value in the **Verified** column of Yes or No is determined by the Confidence level for the file as follows:

- If the Confidence Level is High and the file does not match a classification policy condition, the value is Yes.
- If the Confidence Level is Medium or Love, the value is No.

The list of sensitive files in an unstructured data store updates when a scan job finishes, when an Import job that imports sensitivity status finishes, or when you change the protection or sensitivity status of a file in the **Properties** pane.

Actions

By default, the page lists sensitive files in a single list in **Flat View**. To view the sensitive files in a hierarchy beginning with the first level folder in the directory, select **Tree View** in the **View** menu located below the summary metrics.

To change the sensitivity of data domains in the file or the protection status for a sensitive file, select the check box in the file row and then click **Edit** in the **Properties** pane.

To perform a manual action, a field and then select an option from the **Actions** > **Take Action** menu.

To filter the list, click the **Filter** icon in the Data Privacy Management header.

To export information about the sensitive fields, select **Export** from the **Actions** menu and download the `Data.zip` file to a local directory. Extract the file to view the following `.csv` files:

FileSystemDataStoreSummary.csv

Lists the total number of files scanned, the number of data domains, number of classification policies used, the policy impression count, and the domain impression count.

FileSystemDataStoreDetails.csv

Lists the domain names and the corresponding impression count for each field.

FileSystemClassificationPolicyDetails.csv

Lists the impression count for each classification policy used.

Related Topics:

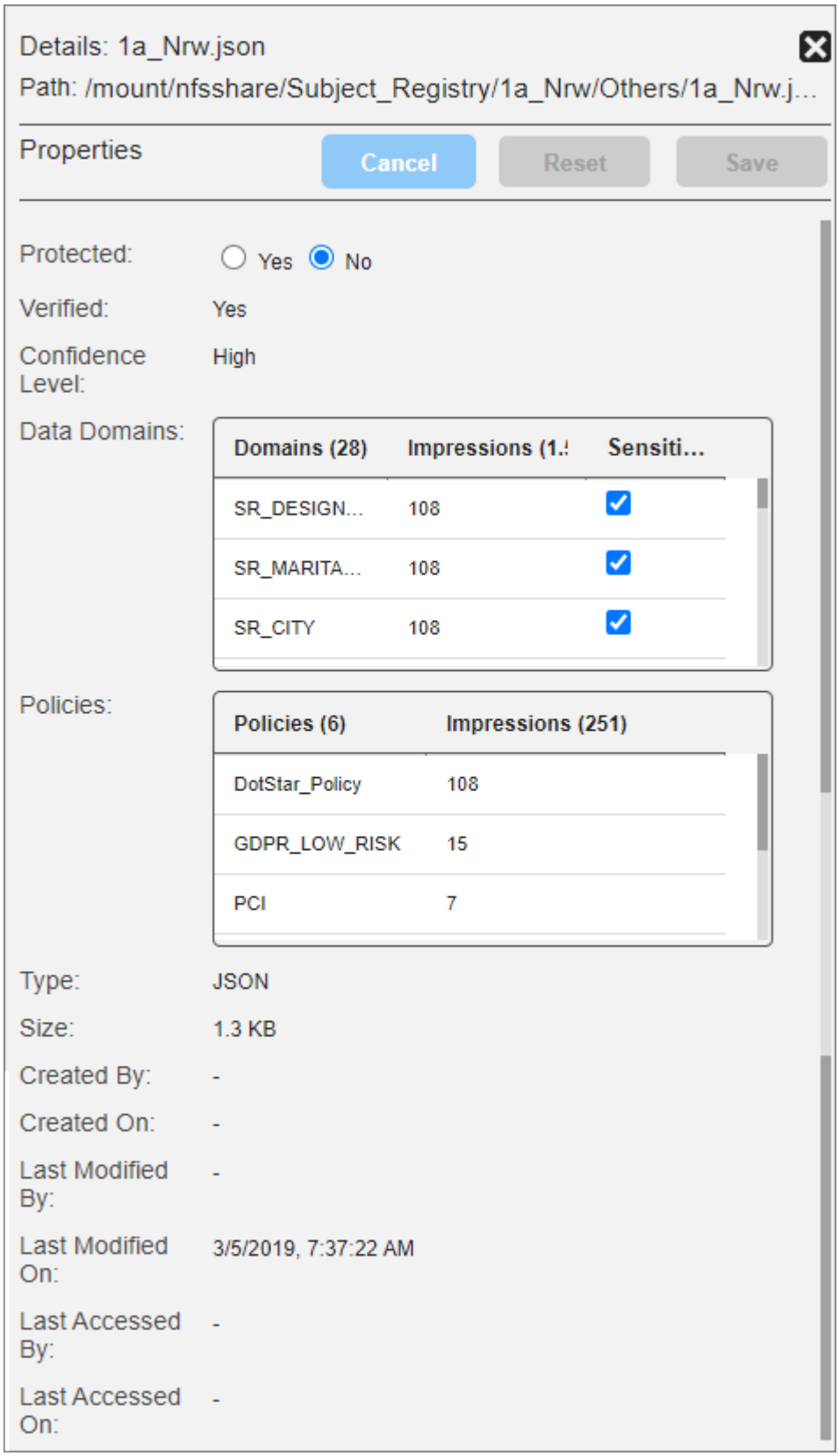
- "Exporting Sensitive Field Information" on page 554

Sensitive File Properties Pane

To view the **Properties** pane for a sensitive file, select the file row. The **Properties** pane includes file properties that are not listed in the **Sensitive Files** page columns.

To update the sensitivity of data domains or the protection status for the file, click **Edit**.

The following image shows an example **Properties** pane for the sensitive file 1a_Nrw.json:



The following table describes the information on the **Properties** pane that is not listed on the **Sensitive Files** page:

Property	Description
Data Domains	Lists the total number of data domains and domain impressions that match the file, and includes the list of each data domain, impressions and sensitivity status for each data domain. You can edit the sensitivity status for each data domain.
Policies	Lists the number of classification policies and their names that match the sensitive file and the number of total policy impressions for the file and the number of impressions for each policy.
Size	The size of the sensitive file.
Created By	The user who created the file.
Created On	The date and time the sensitive file was created in the time zone of the internet browser used to access Data Privacy Management.
Last Modified By	The user who most recently edited the sensitive file.
Last Modified On	The date and time the sensitive file was most recently edited in the time zone of the internet browser used to access Data Privacy Management.
Last Accessed By	The user who most recently accessed the sensitive file.
Last Accessed On	The date and time the sensitive file was most recently accessed in the time zone of the internet browser used to access Data Privacy Management.

Email Server Data Stores on the Sensitive Files Page

Some of the options that appear on the **Sensitive Files** page for Email Server data stores differ from other unstructured files.

The **Sensitive Files** page for Email Server data stores shows information about each sensitive file in the following columns:

Column	Description
Name	The name of the resource.
Created On	Date of creation of the email. **Note:** The date of creation of an email that is attached to another email differs from the actual date of creation of the email.
Resources	The number of resources nested within a resource.
Resource Type	The type or level of the file in the email server. The following resource types apply to email server file types: - **Email ID**. The user email ID scanned. For example, `abc@xyz.com` - **Email Folder**. The email folder or sub folder. For example, `Inbox`, or `Drafts`. - **Email**. The emails in the folder or subfolder. - **Content Body**. The content of an email. - **<File type>**. The type of file attached to an email. For example, `PDF`, `Avro`, or `Email` for email attachments.

Column	Description
Domains	The number of data domains that match data in the sensitive file.
Domain Impressions	The number of data domain impressions for the file.
Policies	The number of classification policies that the scan job matched with the data in the sensitive file.
Policy Impressions	The number of classification policy impressions for the file.
Sensitive	Indicates whether Data Privacy Management has identified the file as sensitive. Possible values are Yes and No.
Protected	Indicates whether the sensitive file is protected. Possible values are Yes and No.

You can click a specific row to open the **Properties** pane. The **Properties** pane includes file properties that are not listed in the **Sensitive Files** page columns. For example, you can check the Confidence Level and Verified status information.

Based on the resource type, you can view some or all of the following details on the **Properties** pane:

Property	Description
Confidence Level	The confidence level that the data domain is an exact match for the sensitive data in the file. Possible values are High, Medium, and Low.
Verified	Indicates whether or not the Confidence Level is verified. Possible values are Yes and No.
Data Domains	Lists the total number of data domains and domain impressions that match the file, and includes the list of each data domain, impressions and sensitivity status for each data domain. You can edit the sensitivity status for each data domain.
Policies	Lists the number of classification policies and their names that match the sensitive file and the number of total policy impressions for the file and the number of impressions for each policy.
Size	The size of the sensitive file. **Note:** The size of emails and attachments differ from the size that appears in the email folder and the disk.
Content Type	The type of content body. For example, HTML or text.
Type	The file type of an attachment.
Last Modified On	The date and time the sensitive file was most recently edited in the time zone of the internet browser used to access Data Privacy Management. **Note:** The **Last Modified Date** values on the **Sensitive Files** page differ based on how you attach the file to a mail. If you attach a file directly, the date appears the same as the email receipt date. If you attach a compressed file, it displays the date that the file was last modified on the disk.

Actions

To change the sensitivity of data domains in the file or the protection status for a sensitive file, click the name in the file row and then click **Edit** in the **Properties** pane.

To perform a manual action, select a field and then select an option from the **Actions** > **Take Action** menu.

To filter the list, click the **Filter** icon in the Data Privacy Management header.

To export information about the sensitive fields, select **Export** from the **Actions** menu.

User Access Page

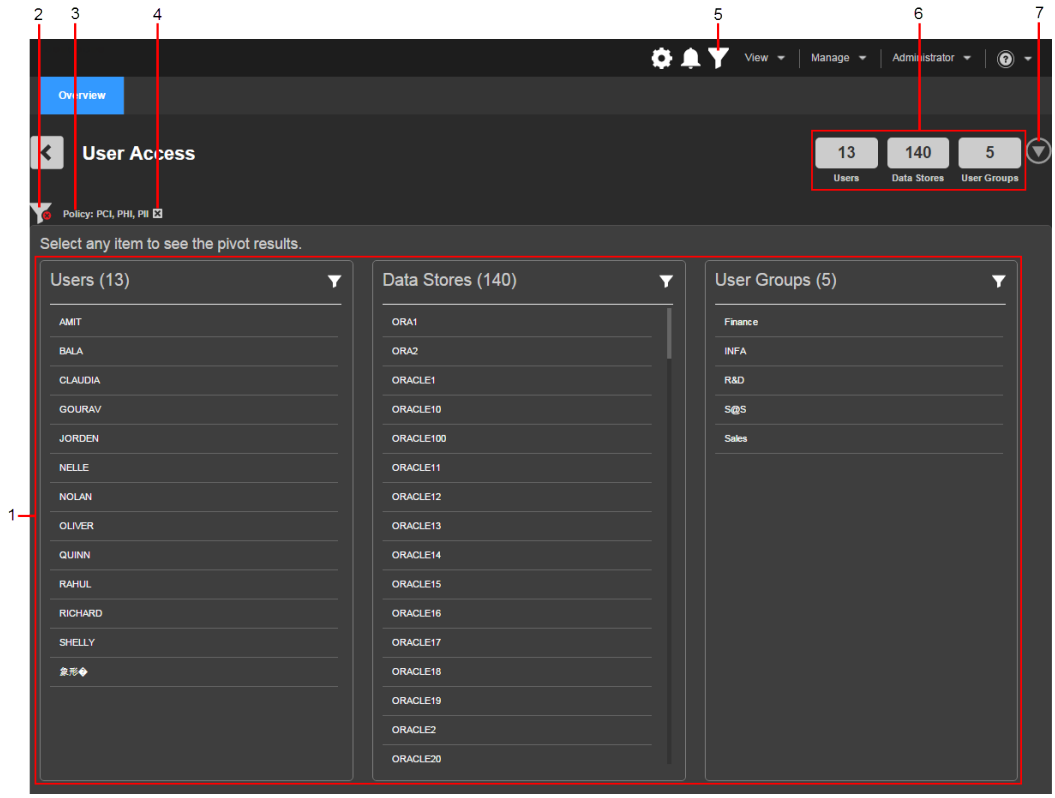
The **User Access** page displays a list of users, data stores, and user groups based on the way you accessed the page from the **Overview** workspace. The summary metrics in the top right of the page show the number of users in the list, the number of data stores the users can access, and the number of user groups that include the users.

You can access the **User Access** page in the following ways:

- On the **Overview** workspace, click the **User Access** indicator label.
- On the **Users** page, click the **User Groups** or **Data Store Access** counts in the expanded view for a user.

When you click an item in one pane on the **User Access** page, the related items in the other two panes dynamically appear. For example, when you click a user name, the page shows the data stores the user can access and the user groups that include the user.

The following image shows and example of the **User Access** page filtered by a classification policy:



1. Users, Data Stores, and User Groups Panels. By default, the lists are displayed in alphabetical order.
2. Clear Filter icon
3. Filter conditions
4. Clear Filter Condition icon
5. Filter
6. User access summary
7. Actions menu

Actions You Can Perform on the User Access Page

Select a user, data store, or user group to see relevant information in the other panels. For example, you click a user name in the Users panel. The Data Stores panel displays the data stores that the selected user can access and the User Groups panel displays the user groups that include the selected user.

To quickly find a user, data store, or user group, click the panel-specific **Filter** icon. Enter a full or partial name in the text box and press `ENTER`.

Hover over a user, data store, or user group name to view more details.

To view additional details about a user, click the user name and then click **View**. The **User Profile** page appears. For more information, see "User Profile Page" on page 567.

To view the **Proliferation** page for a data store, select a data store in the **Data Stores** panel and then click **View**. For more information, see "Proliferation Page" on page 542.

To filter the page with a condition common to all pages accessed from the **Overview** workspace, click the **Filter** icon in the Data Privacy Management header. For more information, see "Filtering Information on the Security Dashboard" on page 573.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`PivotViewUserSummary.csv`	Contains the number of users, data stores, and user groups in Data Privacy Management.
`UserAccessPivotView.csv`	Contains the names of the users, data stores, and user groups in Data Privacy Management.

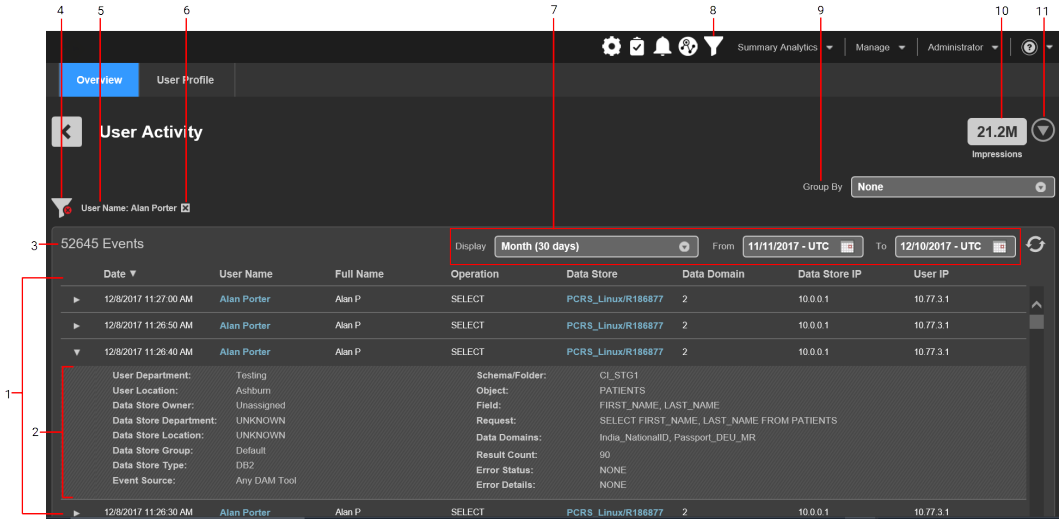
User Activity Page

The **User Activity** page displays a log of events associated with a specific user. You can view details about each event such as the query that the user made to retrieve sensitive data, the data domains that the user accessed, and the IP address of the user. The summary metric at the top right of the page shows the total number of impressions that the user accessed based on the selected filter conditions and time period.

You can access the **User Activity** page in the following ways:

- On the **Data Domains** list page, click the value in the **Impressions** column.
- On the **User Profile** page, click the number of impressions, the **Top Data Stores Accessed** indicator label, or the **Top Data Domains** indicator label.
- On the **Users** page, click the value in the **Impressions** column.

The following image shows the **User Activity** page:



1. Events. Displays a log of events triggered by the user you selected on the Users page. You can cancel the filter condition to see events for all users. By default, the page lists events by most recent to least recent date.
2. Expanded event view
3. Event count
4. Clear Filter icon
5. Filter conditions
6. Clear Filter Condition icon
7. Time period. By default, the time period is the same time period on the Top Users indicator or the Users page, measured by the UTC time standard.
8. Filter
9. Group By menu
10. Summary metric
11. Actions menu

Actions You Can Perform on the User Activity Page

To view the expanded view for an event, click the arrow next to the event date.

To sort the list by a column other than the event date, click the column name.

To learn more about a user, click the user name. The **User Profile** page appears. For more information, see "User Profile Page" on page 567.

To view the proliferation status of the data store involved in the event, click the Data Store name. The **Proliferation** page appears. For more information, see "Proliferation Page" on page 542.

To change the time period, select one of the following options from the **Display** menu:

- Today
- Day (24 hours)
- Week (7 days)
- Month (30 days)
- 60 days
- 90 days
- 1 year
- Custom Dates. Select dates in the **From** and **To** fields.

To organize the list of events into groups, select one of the following options from the **Group By** list:

- Data Domain
- Data Store Group
- Data Store Location
- User Department
- Classification Policy
- Protection Status
- User Name
- User Group
- User Location

To filter the list, click the **Filter** icon in the Data Privacy Management header. For more information, see "User Activity Page Filter Conditions" on page 578.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`UserActivityDrillDownExport.csv`	Shows the details of each event that the user triggered. For example, the CSV file shows the time of the event, the type of operation, and the affected data store and column.
`UserActivityDrillDownImpressionSummary.csv`	Shows the total number of impressions that the user accessed.
`UserActivityDrillDownSummary.csv`	Shows the total number of events that the user triggered.

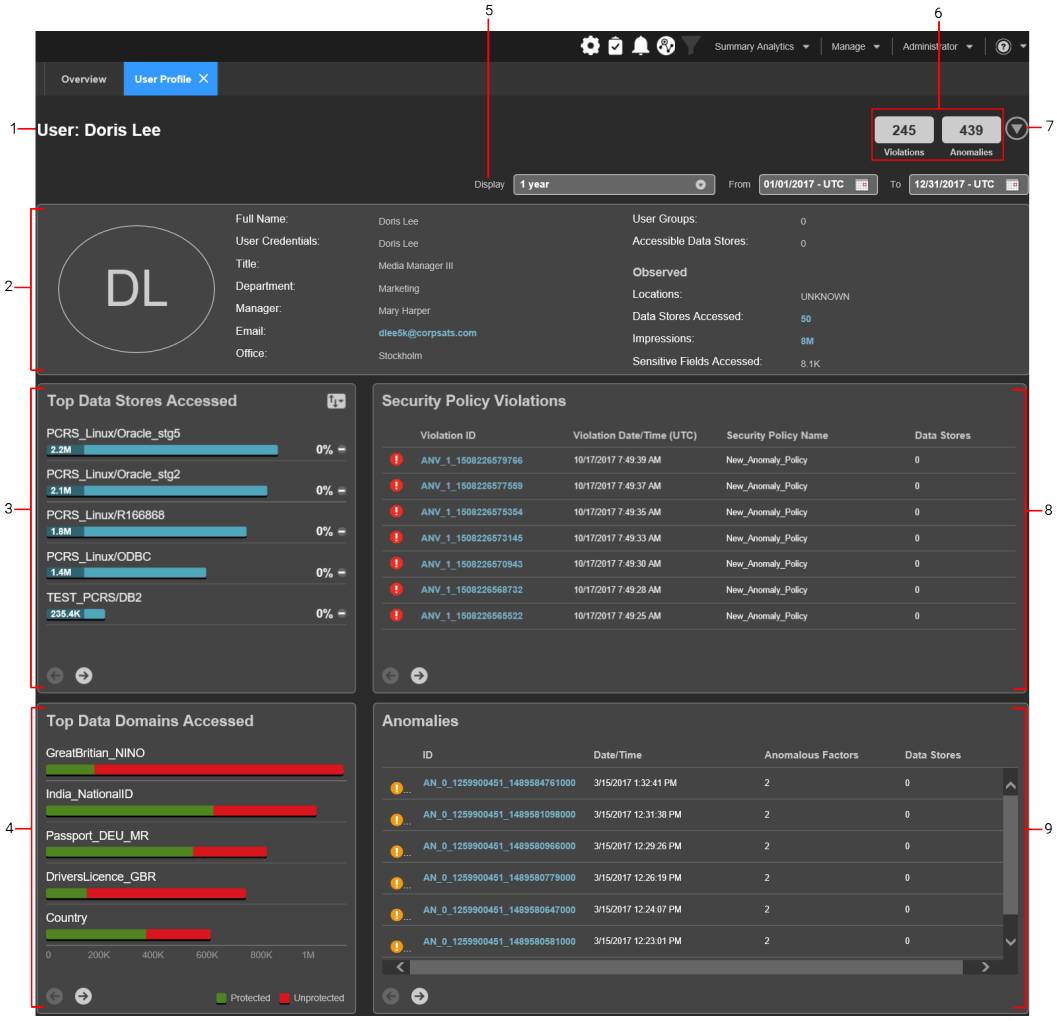
User Profile Page

The **User Profile** page displays detailed information about a Data Privacy Management user, such as the user's title, department, manager, email address, location, the number of data stores and sensitive fields the user has accessed, and the number of impressions the user has made on sensitive data. The summary metrics at the top right of the page show the total number of security policy violations and anomalies associated with the user.

You can access the **User Profile** page in the following ways:

- On the **User Access** page, click a user name and then click **View**.
- On the **User Activity** page, click a user name.
- On the **Users** page, click a user name.
- On the **Suppression Rules** list page, click the name of the user who created a suppression rule.

The following image shows the **User Profile** page:



1. User name
2. User detail summary
3. Top Data Stores Accessed
4. Top Data Domains Accessed
5. Time period. By default, the time period is Month (30 days).
6. Summary metrics
7. Actions menu
8. Security Policy Violations .Lists the security policy violations associated with the user in reverse chronological order.
9. Anomalies. Lists the anomalies associated with the user in reverse chronological order.

Actions You Can Perform on the User Profile Page

To send an email to the user, click the email address in the user detail summary.

To view a complete list of the data stores the user has accessed, click the **Data Stores Accessed** count in the user detail summary. The **Data Stores** list page appears. For more information, see "Data Stores List Page" on page 538.

To view a log of events the user triggered, click the **Impressions** count in the user detail summary, the **Top Data Stores Accessed** indicator label, or the **Top Data Domains Accessed** indicator label. The **User Activity** page appears. For more information, see "User Activity Page" on page 565.

To change the time period, select one of the following options from the **Display** menu:

- Today
- Day (24 hours)
- Week (7 days)
- Month (30 days)
- 60 days
- 90 days
- 1 year
- Custom Dates. Select dates in the **From** and **To** fields.

To view more details about a security policy violation, click the **Violation ID** in the **Security Policy Violations** indicator.

To view more details about an anomaly, click the **ID** in the **Anomalies** indicator.

To filter the page with a condition common to all pages accessed from the **Overview** workspace, click the **Filter** icon in the Data Privacy Management header. For more information, see "Filtering Information on the Security Dashboard" on page 573.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`PolicyViolations.csv`	Shows the details of each security policy that the user violated. For example, the CSV file shows the security policy violation ID, date and time of the violation, the type and severity level of the violation, and the affected security policy and data store.
`PolicyViolationSummary.csv`	Contains a summary of the security policies that the user violated, including the number of affected data stores and the total number of violations for each severity level.
`UbaAnomaly.csv`	Shows the details of each anomaly associated with the user. For example, the CSV file shows the anomaly ID, date and time of the anomaly, the severity level of the anomaly, the anomaly score, the number of affected anomalous factors and data stores, and the user's title, department, and home location.
`UbaSummary.csv`	Contains a summary of anomalies associated with the user, including the number of affected data stores and the total number of anomalies for each severity level.
`UserActivitySummaryByDataDomain.csv`	Contains a list of the data domains the user accessed, ordered from high to low number of impressions. Also includes the number of protected and unprotected sensitive fields.
`UserActivitySummaryByDataStoreAndTrend.csv`	Contains a list of the data stores ordered from high to low number of impressions in the current period. Also includes the total number of impressions and the trend change as a percentage for each data store.

File Name	Description
UserActivitySummaryByLocation.csv	Contains the data store and user location. Also contains the number of events that the user triggered.
UserActivitySummaryProperties.csv	Contains the aggregate risk score for the user, total number of protected fields the user accessed, the user's total risk cost, the number of data stores and targets to which the user has access, the number of sensitive fields and sensitive records the user has accessed, the number of events the user has triggered, and the risk cost per unit.
UsersExport.csv	Shows details for the user such as the user location and the number of data stores and sensitive fields that the user accessed.

Related Topics:

- "Suppression Rules List" on page 370

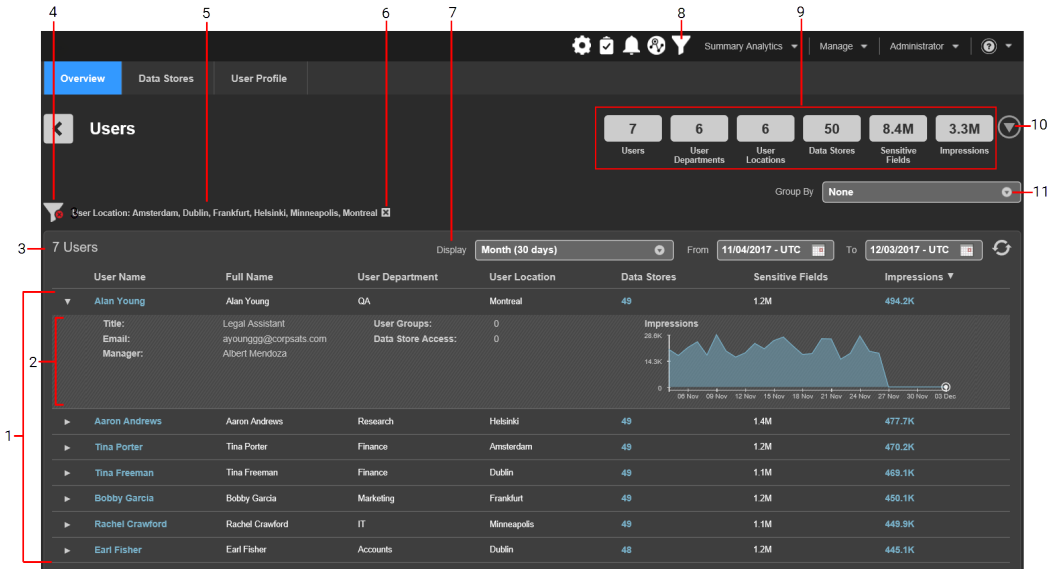
Users Page

To access the **Users** page, click the **Top Users** indicator label on the **Overview** workspace. The page lists users who have accessed sensitive data in data stores that you can access.

For each user, the page lists the user name, full name, department, location, the number of data stores the user can access, the number of sensitive fields in the data stores, and the number of impressions the user accessed for the time period specified on the page.

The summary metrics in the top right of the page show the total number of users in the current list, the number of departments and locations for the users, the total number of data stores the users can access, the number of sensitive fields in the data stores, and the total number of impressions that all users on the page accessed.

The following image shows the **Users** page:



1. Users. By default, the page lists users by the number of impressions in descending order.
2. Expanded view of user. Shows the user's job title, email address, manager name, the number of user groups that include the user, the number of data stores the user can access, and a graph that shows the number of impressions the user accessed over time. You can hover over a section of the graph to see the number of impressions the user accessed on that day or month.
3. User count
4. Clear Filter icon
5. Filter conditions
6. Clear Filter Condition icon
7. Time period. By default, the page displays user activity over the last 30 days. Data Privacy Management uses the UTC time standard for all time periods.
8. Filter
9. Summary metrics
10. Actions menu
11. Group By. The default grouping is None.

Actions You Can Perform on the Users Page

To view the expanded view for a user, click the arrow next to the user name.

To sort the list by a column other than the number of impressions, click the column name.

To view the data stores that a user accessed, click the value in the **Data Stores** column. The **Data Stores** list page appears. For more information, see "Data Stores List Page" on page 538.

To view the log of a user's activity, click the value in the **Impressions** column. The **User Activity** page appears. For more information, see "User Activity Page" on page 565.

To learn more about a user, click the user name. The **User Profile** page appears. For more information, see "User Profile Page" on page 567.

To view the names of the user groups and data stores the user can access, click the **User Groups** or **Data Store Access** counts in the expanded view. The **User Access** page appears. For more information, see "User Access Page" on page 564.

To change the time period, select one of the following options from the **Display** menu:

- Today
- Day (24 hours)

- Week (7 days)
- Month (30 days)
- 60 days
- 90 days
- 1 year
- Custom Dates. Select dates in the **From** and **To** fields.

To organize the list of users into groups, select one of the following options from the **Group By** list:

- User Department
- User Location
- Data Domain
- Data Store Group
- Data Store Location
- Classification Policy

To filter the list, click the **Filter** icon in the Data Privacy Management header. For more information, see "Users Page Filter Conditions" on page 578.

To export the page information, select **Export** from the **Actions** menu. Download a ZIP file named `Data.zip` to your local machine. Extract the ZIP file to view the following CSV files:

File Name	Description
`UsersExport.csv`	Lists the users. Shows details for each user such as the user location and the number of data stores and sensitive fields that the user accessed.
`UsersSummary.csv`	Shows totals such as the number of users, user locations, and sensitive fields that users accessed.

Security Dashboard Management

On the **Overview** workspace, you can customize the indicators that display on the Security Dashboard, filter the information that displays on the workspace, and export a .zip file that contains one CSV file with detailed information for each indicator and summary on the workspace.

Toggle between dashboards from the **Actions** menu. To switch to the Privacy Dashboard, click **Switch to Privacy Dashboard**.

Customizing the Security Dashboard

You can tailor the layout of the **Overview** workspace so that the indicators that are critical to your business needs are immediately visible when you open the Security Dashboard.

You can select which indicators display on the Security Dashboard and customize the order of the indicators. The changes are visible only to the user who is logged in to Data Privacy Management. When you use the

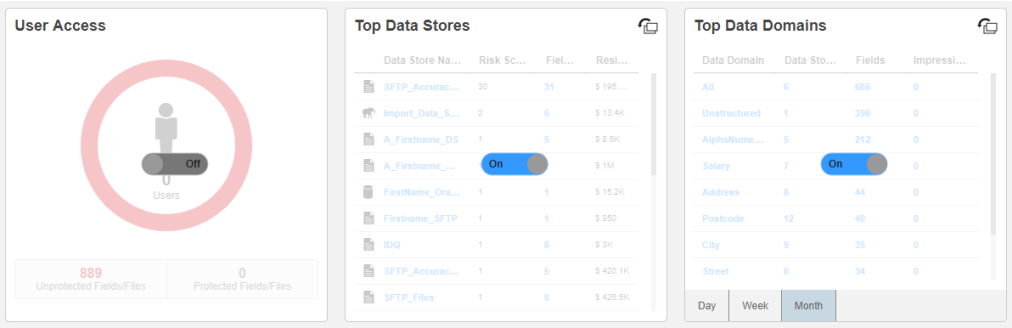
same account to log in, you will continue to see the new layout even if you log in from a different machine or browser.

1. From the **Actions** menu, select **Customize Dashboard**.

   The **Customize Dashboard** page appears.

2. Each indicator on the page contains a toggle that is set to **On** by default. To remove an indicator from the Security Dashboard, click the toggle to change the value to **Off**.

   The following image shows an example of this view:



3. When finished, click **Save**.

   The **Overview** workspace appears. The Security Dashboard view displays only the indicators you selected.

4. To change the order in which the indicators display, hover the cursor over the indicator that you want to move to a different location.

   The **Move** icon appears.

5. Drag and drop the indicator to the new location on the page. Adjacent indicators move to adjust to the new location.

   In some cases, you can expand or contract an indicator.

Filtering Information on the Security Dashboard

To set filter conditions that apply to the Security Dashboard and the drill-down pages, click the **Filter** icon in the Data Privacy Management header and select global filter conditions from the **Filter** pane.

On the **Filter** panes for the drill-down pages, a dashed line separates the global conditions from the page-specific filter conditions. The page-specific filter conditions appear above the dashed line.

Note: The **Filter** pane on the **Sensitive Fields** page displays only two of the global filter conditions: Classification Policy and Data Domain.

The following table describes the filter conditions:

Filter Condition	Description
Risk Score	Two sliders represent risk score ranges from 0 through 100. To view data stores with risk scores that fall within a range, use one slider to specify the minimum value and the other slider to specify the maximum value of the range. To view data stores with a specific risk score, drag both sliders to the same value on the scale.
Sensitivity Level	Select one or more sensitivity levels.

Filter Condition	Description
Classification Policy	Select one or more classification policies. The selections also filter the number of sensitive fields that display in the **Top Data Domains** and **Top Data Stores** indicators.
Data Domain	Select one or more data domains.
Category	Select one or more categories.
Data Store Type	Select one or more data store types.
Data Store Group	Select one or more data store groups.
Process	Select one or more processes. Filters based on data stores and risk associated with Axon Data Governance processes linked to a data store.
Data Store Tag	Select one or more data store tags.
Region	Filters based on one or more geographic regions. When you select a country or data store location without specifying a region, a square appears in the region check box to indicate that the region contains one or more countries or data store locations that you did not select.
Country	When you select a region first, Data Privacy Management automatically selects the countries in the region. You can clear any of the selections. You can also include other countries outside the region.
Data Store Location	When you select a region first, Data Privacy Management automatically selects the data store locations in with the region.

Data Domain Details Page Filter Conditions

To filter the list of sensitive fields on the **Data Domain Details** page, click the **Filter** icon on the Data Privacy Management header.

The following table describes the filter conditions:

Filter Condition	Description
Schema/Folder	Enter a schema or folder name that contains sensitive fields.
Object	Enter an object or table name that contains sensitive fields.
Field	Enter a sensitive field name.
Protection Status	Select Protected or Unprotected.
Protection Extensions	Select a protection extension.

Data Stores Pages Filter Conditions

To filter the list of data stores on the **Data Stores** grid page or the **Data Stores** list page, click the **Filter** icon on the Data Privacy Management header.

The following table describes the filter conditions:

Filter Condition	Description
Data Store	Enter a partial or complete data store name. The page displays the data stores that match the name or name fragment.
Department	The department associated with one or more data stores. Select a department from the list.

Departments Page Filter Conditions

To filter the list of departments on the **Departments** page, click the **Filter** icon on the Data Privacy Management header.

The following table describes the filter conditions:

Filter Condition	Description
Department	Enter a partial or complete department name. The page displays departments that match the name or name fragment.
Data Stores	The number of data stores in a department. To specify minimum and maximum values, drag the left and right sliders. To view departments with a specific number of data stores, drag both sliders to the same value.
Sensitive Fields	The number of sensitive fields in a department. To specify minimum and maximum values, drag the left and right sliders. To view departments with a specific number of sensitive fields, drag both sliders to the same value.
Sensitive Files	The number of sensitive records in a department. To specify minimum and maximum values, drag the left and right sliders. To view departments with a specific number of sensitive records, drag both sliders to the same value.
Policy Impressions	The number of classification policy impressions. To specify minimum and maximum values, drag the left and right sliders. To view departments with a specific number of policy impressions, drag both sliders to the same value.
Residual Risk	The total residual risk cost of the data stores in one or more departments. To specify minimum and maximum values, drag the left and right sliders. To view departments with a specific residual risk cost, drag both sliders to the same value.

Proliferation Page Filter Conditions

To filter the results on the **Proliferation** page, click the **Filter** icon on the Data Privacy Management header.

The following table describes the filter conditions:

Filter Condition	Description
Schema	Select a schema that contains sensitive fields.
Table	Select a table that contains sensitive fields.
Column/Field	Select a column or field name.

Sensitive Fields Page Filter Conditions

To filter the list of fields on the **Sensitive Fields** page, click the **Filter** icon on the Data Privacy Management header and select filter conditions from the **Filter** pane.

The conditions above the dashed line on the **Filter** pane are specific to the **Sensitive Fields** page. The conditions below the dashed line apply to all pages in the **Overview** workspace.

The following table describes the filter conditions:

Filter Condition	Description
Schema/Folder	Enter the full or partial name of a schema or folder. The page displays a list of sensitive fields in schemas or folders that match the name or name fragment.
Object	Enter the full or partial name of a table. The page displays a list of sensitive fields in tables that match the name or name fragment.
Field	Enter the full or partial name of a field. The page displays a list of sensitive fields that match the name or name fragment.
Protected	Select Yes to view only protected sensitive fields. Select No to view only unprotected sensitive fields.
Sensitive	Select Yes to view fields that Data Privacy Management identified as sensitive by the Profiling job step in a data store scan or that were identified as sensitive by importing CSV files. Select No to view fields that you manually identified as not sensitive.
Conformance	Move the sliders on each end of the bar to specify the minimum and maximum values for a conformance score range. The page displays a list of sensitive fields that match the conformance score range.
Verified	Select Yes to display a list of sensitive fields that are verified for one of the following reasons: - The fields have a conformance score in the auto-accept range. - The fields have a conformance score in the validate range, but you changed the verified status value to Yes and imported the changes from a CSV file. Select No to display a list of sensitive fields that are not verified for one of the following reasons: - The fields have a conformance score in the validate range, and the **Mark columns within the validate range as sensitive** property is enabled for the data domain. By default, the verification status for these fields is No. - The fields have a conformance score in the validate range, and you retained the default verified status value of No in an imported CSV file.

Filter Condition	Description
Origin	For Cloudera Navigator, Informatica Data Engineering Integration, Informatica Cloud, and Informatica PowerCenter data stores, select a data store from which the sensitive fields proliferated.
Target	For Cloudera Navigator, Informatica Data Engineering Integration, Informatica Cloud, and Informatica PowerCenter data stores, select a data store to which the sensitive fields proliferated.

Sensitive Files Page Filter Conditions

To filter the list of files on the **Sensitive Files** page for unstructured sources, click the **Filter** icon on the Data Privacy Management header and select filter conditions from the **Filter** pane.

The following table describes the filter conditions:

Filter Condition	Description
Name	Enter the full or partial name of a file. The page displays a list of sensitive fields that match the name or name fragment.
Created On	Select a data range to view files created during a specific period of time.
Last Modified On	Select a data range to view files last modified during a specific period of time.
Last Accessed On	Select a data range to view files last accessed during a specific period of time.
Has Attachment	Applies to the Email Server category. Select Yes to view only files with attachments. Select No to view only files without attachments.
Attachment File Type/File Type	Select one or more file types to view only the selected types of files. For Email Server sources, select one or more file types to view only files with the selected types of attachments. Choose one or more types from the following file types: - Avro - Compressed Files - Email - Image - JSON - Microsoft Excel - Microsoft PowerPoint - Microsoft Word - Parquet - PDF - Web Page Files - XML
Protected	Select Yes to view only protected sensitive files. Select No to view only unprotected sensitive files.
Sensitive	Select Yes to view files that Data Privacy Management identified as sensitive by the Profiling job step in a data store scan or that were identified as sensitive by importing CSV files. Select No to view files that you manually identified as not sensitive.
Classification Policy	Select one or more classification policies.

Filter Condition	Description
Data Domain	Select one or more data domains.
Domain Impressions	Move the sliders on each end of the bar to specify the minimum and maximum values for a domain impressions range. The page displays a list of sensitive files that match the domain impressions range.
Verified	Select Yes to display a list of sensitive files that are verified for one of the following reasons: - The files have a conformance score in the auto-accept range. - The files have a conformance score in the validate range, but you changed the verified status value to Yes and imported the changes from a CSV file. Select No to display a list of sensitive files that are not verified for one of the following reasons: - The files have a conformance score in the validate range, and the **Mark columns within the validate range as sensitive** property is enabled for the data domain. By default, the verification status for these files is No. - The files have a conformance score in the validate range, and you retained the default verified status value of No in an imported CSV file.
Confidence Level	Choose a level to view files with a specific confidence level.

User Activity Page Filter Conditions

To filter the results on the **User Activity** page, click the **Filter** icon on the Data Privacy Management header.

The following table describes the filter conditions:

Filter Condition	Description
User Name	Enter one user name. To open the **User Profile** page, click the user name.
Data Store	Enter one data store name to view the list of users who accessed the data store.
Department	Select a department to view the list of users in the selected department.
User Location	Select a user location to view only users in that location.
Protection Status	Select Protected, Unprotected, or Not Analyzed.
User Activity Data Domain	Select one or more data domains to view the list of users that accessed sensitive data in specific data domains.

Users Page Filter Conditions

To filter the results on the **Users** page, click the **Filter** icon on the Data Privacy Management header.

The following table describes the filter conditions:

Filter Condition	Description
User Name	Enter one user name. To open the **User Profile** page, click the user name.
Data Store	Enter one data store name to view the list of users who accessed the data store.

Filter Condition	Description
Department	Select a department to view the list of users in the selected department.
User Location	Select a user location to view only users in that location.
Protection Status	Select Protected, Unprotected, or Not Analyzed.
User Activity Data Domain	Select one or more data domains to view the list of users that accessed sensitive data in specific data domains.

Exporting Information on the Security Dashboard

You can export information on the Security Dashboard view of the **Overview** workspace. If you filter the workspace, the export files contain information that matches the active filter conditions.

To export information, select **Export** from the **Actions** menu.

Save the `Data.zip` compressed file to a directory on your machine. If you export the file again, the file name includes an incremental number such as `Data(1).zip`.

Extract the `Data.zip` file to view the CSV files that contain information displayed on each indicator on the Security Dashboard.

Security Dashboard Export Files

To export information on the Security Dashboard view of the **Overview** workspace and the pages accessed from the workspace, select **Export** from the **Actions** menu and download the `Data.zip` file to a local directory.

The `Data.zip` file contains CSV files. Each CSV file contains the information in a workspace indicator. If you filter the workspace before you export, the information in the CSV files matches the active filter conditions.

The following table lists the CSV files in the `Data.zip` file that you export from the Security Dashboard:

File Name	Description
`DataRiskCost.csv`	Contains the number of sensitive records and fields. Also contains the risk cost for the current and previous months.
`DataStoreStatusBar.csv`	Contains the number of data stores for each section of the Discovery bar. Also contains the total number of data stores with the database repository type.
`ProtectedData.csv`	Contains the number of protected sensitive fields and the number of unprotected sensitive fields. Also shows the numbers of protected and unprotected sensitive fields for the previous month.
`RiskScoreByMonth.csv`	Contains the risk score for the current and previous months.
`SensitiveDataByLocationDetails.csv`	Contains the number of data stores and key scan results for each region.
`SensitiveDataProliferation.csv`	Contains the number of source and target data stores for each classification policy.

File Name	Description
SensitiveDataVsGroups.csv	Contains the aggregate risk score for data stores in each data store group that match each classification policy.
SensitivityLevelOfDataStores.csv	Contains the number of data stores for each sensitivity level in the current and previous months.
TopDataDomains.csv	Contains a list of the data domains ordered from high to low number of sensitive fields. Also includes the number of data stores in each data domain, the number of protected and unprotected sensitive fields, and the number of events triggered for each data domain.
TopDataStores.csv	Contains a list of the data stores ordered from high to low risk scores. Also includes the risk cost, number of sensitive fields, repository ID and data owner name for each data store.
TopDepartments.csv	Contains a list of departments ordered from high to low risk scores. Also includes risk cost and number of protected and unprotected fields for each department.
TopUsers.csv	Contains a list of users. Contains the name and user name of the user. Also contains the department the user belongs to, the number of data stores the user can access, and the number of events the user triggered.
UserAccess.csv	Contains the number of users with access to sensitive fields, protected sensitive fields, and unprotected sensitive fields. Also includes the number of sensitive fields, protected sensitive fields, and unprotected sensitive fields.

CHAPTER 25

Summary Workspaces

This chapter includes the following topics:

- Summary Workspaces Overview, 581
- Data Stores Summary Workspace, 581
- Department Summary Workspace, 584
- User Activity Summary Workspace, 587
- Summary Workspace Management, 590

Summary Workspaces Overview

The summary workspaces display scan results for data stores, departments, and user activity. The scan results on the summary workspaces are based on data stores with the repository type property set to database.

You access the summary workspaces from the **Summary Analytics** menu.

The summary workspaces include the following workspaces:

- Data Stores Summary
- Department Summary
- User Activity Summary

Each workspace shows scan results organized within a specific indicator for clarity. For example, on the **Department Summary** workspace, the By Proliferation indicator shows the number of target data stores for the sensitive fields across the data stores in each department.

The Data Privacy Management Service updates the scan results on the summary workspaces after a scan job completes. You can export the information from a summary workspace to your local machine.

Data Stores Summary Workspace

The **Data Store Summary** workspace contains indicators that show the number of data stores by classification policy, department, sensitivity level, region, data store group, proliferation status, and location.

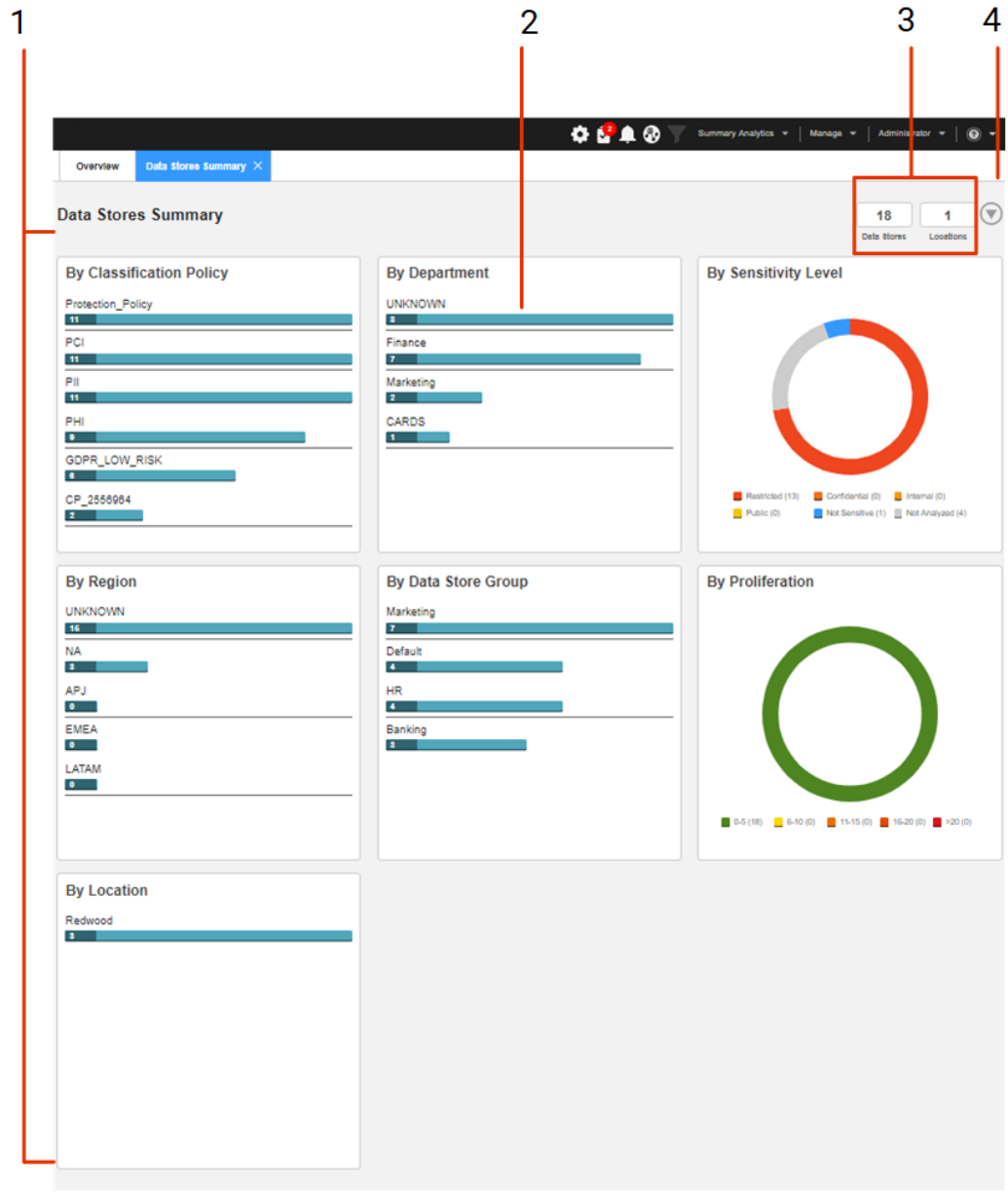
You can click a specific value within an indicator to see the number of data stores and the scan results of the data stores.

For example, from the By Sensitivity Level indicator, you can click the band that represents the Restricted level. The scan results pane appears with the aggregate scan metrics for the data stores that match a restricted classification policy. You can also click on the target count on the scan results pane to see the list of target data stores on the **Data Stores** page. The target data stores contain sensitive data from source data stores. The source data stores match a restricted classification policy.

The indicators on the **Data Store Summary** workspace act as a filter for the page. When you click on a bar or band within one indicator, the Data Privacy Management Servicefilters the information in the other indicators by that category.

You can use the export option to download the results you view on the **Data Store Summary** workspace for further analyses.

The following image shows the **Data Store Summary** workspace:



1. Data store indicators
2. Data store bar
3. Data stores summary
4. Actions menu

The **Data Stores Summary** workspace includes the following sections:

Data store indicators

Each indicator represents a group of data stores that share a common factor such as the same location, proliferation status, sensitivity level, or department. Bars or bands within an indicator represent the different categories for that factor. Numbers on a bar indicate the number of data stores for the category.

For example, the **By Classification Policy** indicator lists the different classification policies. The number of data stores that match each classification policy appears on the bar for that classification policy.

You can filter the information on the **Data Store Summary** workspace by a factor. To filter, you click the appropriate bar or band within an indicator. For example, you want to quickly get information for all data stores that matched the PII classification policy. Click the PII bar in the **By Classification Policy** indicator. The Data Privacy Management Service filters the results in the other indicators to match the active filter. You see the departments, sensitivity levels, geographic areas, data store groups, and proliferation status for the data stores that match the PII classification policy.

Data store bar or band

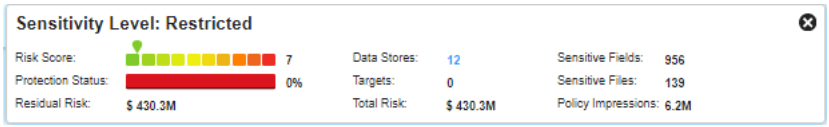
Represents the categories or levels for each factor.

For example, the bands that form the circle in the **By Sensitivity Level** indicator represent the different sensitivity levels that the Data Privacy Management Service can assign to a data store.

You can click a band or bar to filter the entire workspace by that category or level. The Data Privacy Management Service filters the results in the other indicators to match the active filter.

In addition, the Details pane appears with key metrics such as the aggregate risk score, residual risk cost, number of source and target data stores, and overall protection status for the active filter.

The following image shows a sample Details pane:



When you click the **Targets** count, the **Data Stores** page appears. The page shows the list of target data stores to which sensitive fields proliferate from source data stores. The source data stores for these targets belong to the group represented by the bar or band you selected.

Data stores summary

Shows the number of data stores and the number of locations represented on the **Data Stores Summary** workspace.

Actions menu

Use the Actions menu to export the results from the page. When you select the export option, Data Privacy Management downloads a .zip file that contains CSV files. The CSV files contain information that you view in each indicator.

Related Topics:

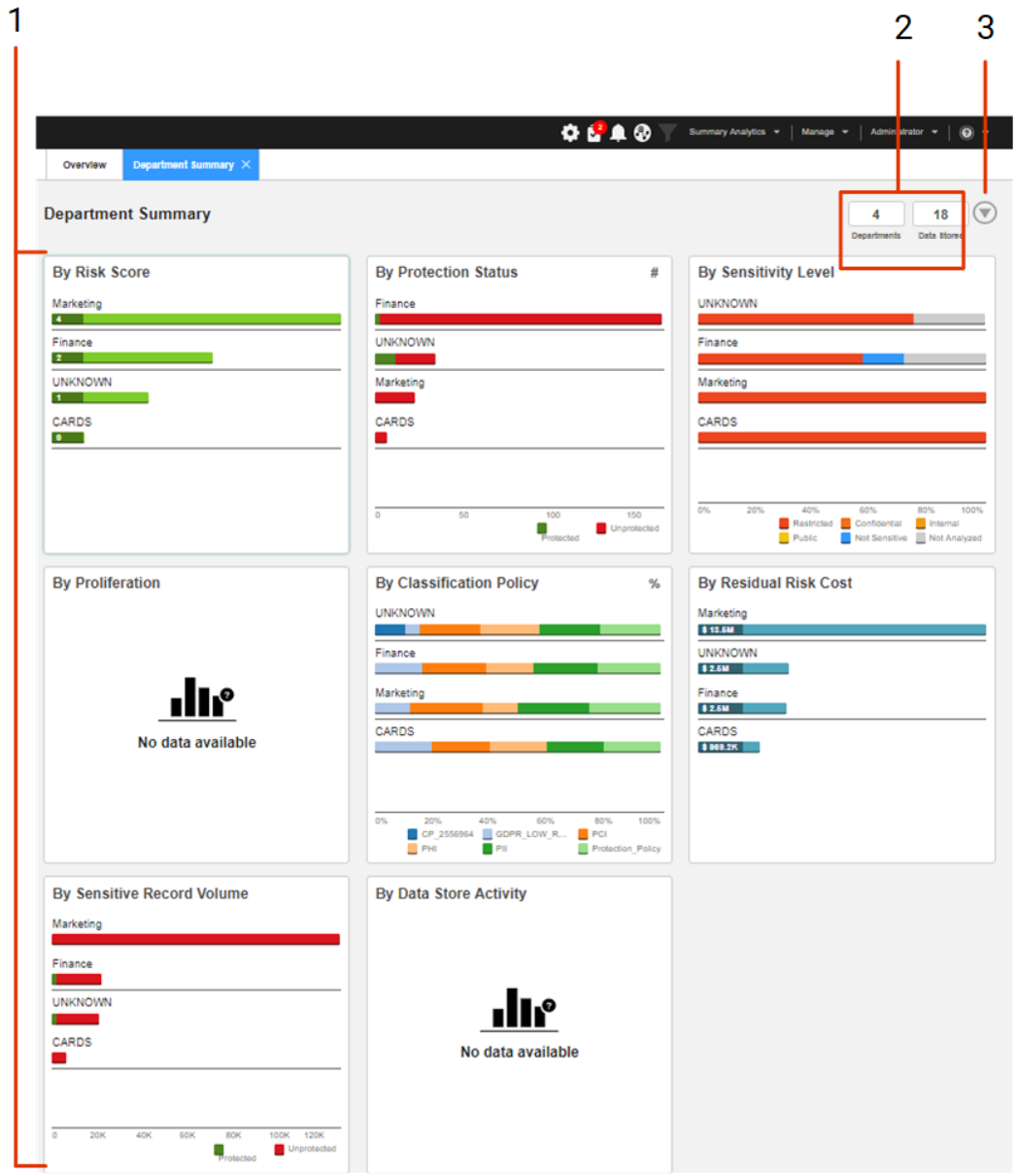
- "Data Stores Grid Page" on page 536

Department Summary Workspace

The **Department Summary** workspace contains indicators that show department scan results by risk score, protection status, sensitivity level, proliferation status, classification policy, risk cost, and volume of sensitive records. You can download the results for further analysis.

To view scan results for a department, click a department within any indicator. The scan results pane appears with the aggregate scan results of the data stores in the department. You can click the target count on the scan results pane to view the list of target data stores associated with the source data stores in the department that you selected.

The following image shows the **Department Summary** workspace:



1. Scan indicators
2. Department summary
3. Actions menu

The **Department Summary** workspace includes the following sections:

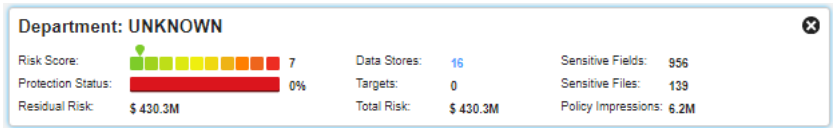
Scan indicators

Shows the scan results by department for a scan factor. The following image shows the **By Residual Risk Cost** indicator with a department selected:



Click a department bar to view details for the data stores in a department. The Details pane appears with key metrics such as the aggregate risk score, residual risk cost, number of source and target data stores, and overall protection status for the active filter.

The following image shows a sample Details pane for a department:



When you click the **Targets** count, the **Data Stores** page appears. The page shows the list of target data stores to which sensitive fields proliferate from source data stores. The source data stores for these targets belong to the department you selected.

Department summary

Shows the number of departments and the number of data stores associated with a department.

Actions menu

Use the Actions menu to export the scan results from the page. When you select the export option, Data Privacy Management downloads a .zip file that contains CSV files. The CSV files contain scan results by department for a scan factor.

Related Topics:

- "Data Stores Grid Page" on page 536

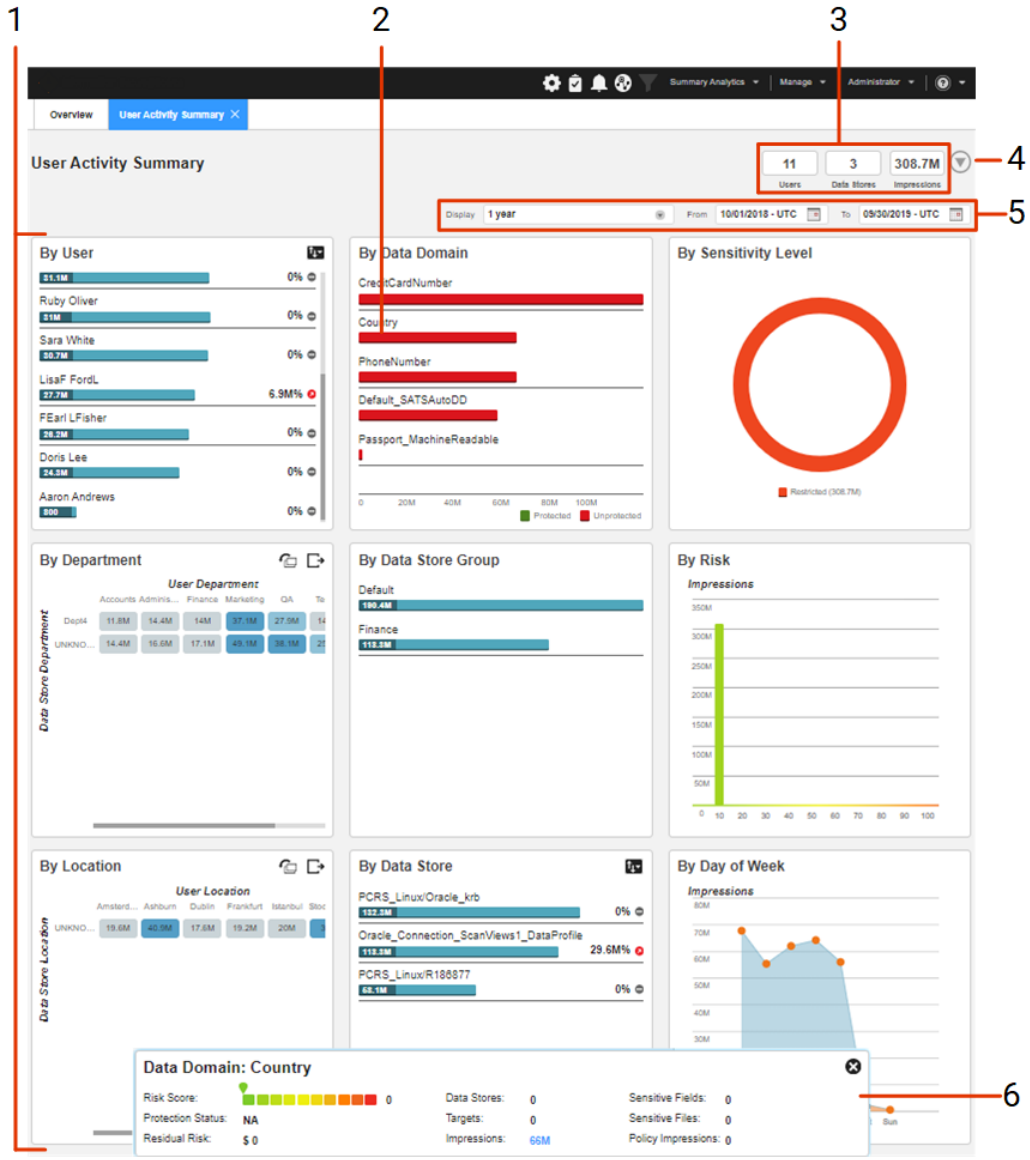
User Activity Summary Workspace

View the **User Activity Summary** workspace to see the number of events by category. For example, you can see the number of events triggered by a user, associated to a data store, or that occur on a specific day of the week.

Event information by a category appears within an indicator. For instance, when you view the By User indicator, you see the number of events created by each user starting with the user with the most events.

You can click a factor within an indicator to filter the results in the other indicators by that factor. For example, you want to see the number of events triggered by user Matt G. When you click on Matt G in the By User indicator, the Data Privacy Management Service updates the **User Activity Summary** workspace to show the number of events by Matt G organized by categories such as data store group, sensitivity level, and location.

The following image shows the **User Activity Summary** workspace:



1. Indicators
2. Filter condition
3. Summary
4. Actions menu
5. Time period
6. Details pane

The **User Activity Summary** workspace includes the following sections:

Indicators

Each indicator shows the number of events by a specific category. Data Privacy Management displays the results within an indicator in different ways. For example, in the By User indicator, you see the number of events next to each user. In the By Sensitivity Level indicator, you see the number of events for a sensitivity level represented in a doughnut chart.

Filter condition

To filter the **User Activity Summary** workspace, click on a factor within an indicator. The Data Privacy Management Service updates the results in the other indicators by that factor. The Data Privacy Management Service also shows the details for the factor in the details pane.

Summary

The summary section shows the following values:

- The number of users that triggered at least one event each.
- The number of data stores with at least one sensitive field that was associated with a read, write, or delete query.
- The number of events across all data stores that you have permissions to.

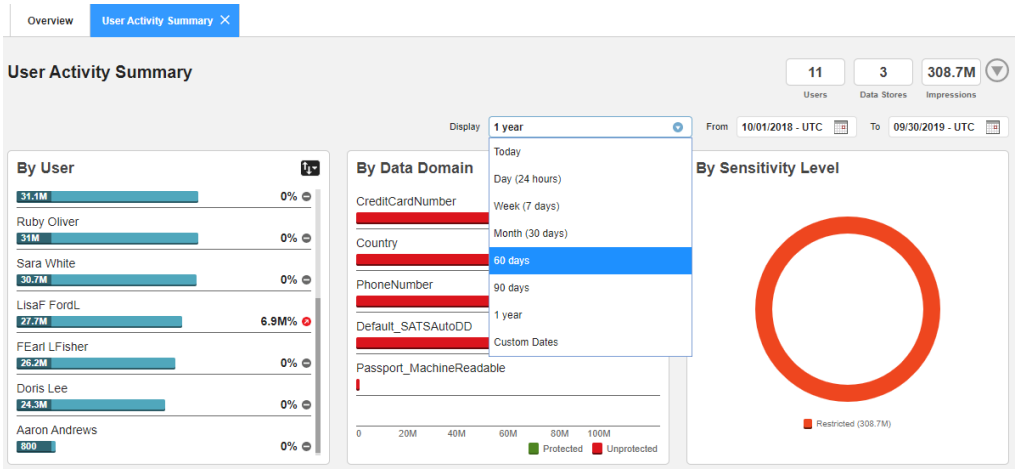
Actions

From the **Actions** menu, you can export the information on the page. When you export, Data Privacy Management downloads a compressed file named `Data.zip` to your local machine. The `Data.zip` file contains CSV files. The CSV files contain the information displayed in the summary, indicator, and details sections of the page.

Time period

By default, the **User Activity Summary** workspace displays user activity over the last 30 days. To change the time period, select an option from the **Display** menu.

The following image shows the time-period section and the options in the **Display** menu:

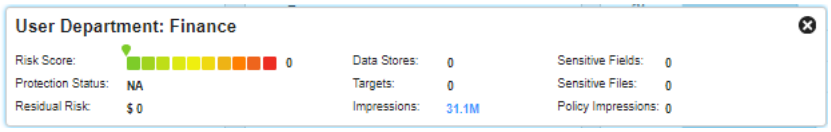


To specify a predefined time period, select an option such as **90 days**, from the **Display** menu. To customize the time period, click **Display** > **Custom Dates** and specify the start and end dates.

Details pane

The Details pane appears when you click a bar or band within an indicator. The pane shows key metrics such as the residual risk score, number of source and target data stores, and protection status. For example, you filter the workspace by the user department Finance. The details pane appears with the aggregate metrics for the data stores that the users in the Finance department accessed.

The following image shows a sample Details pane:



You can click on the following links to view pages that contain more information:

- Data Stores. When you click the link, the Data Stores page appears with a list of the data stores in the group.
- Targets. When you click the link, the Data Stores page appears with a list of source data stores for the targets.
- Impressions: When you click the link, the User Activity page appears with a list of events associated with the data stores in the group.

User Activity Summary Indicators

The indicators on the User Activity Summary workspace display the number of events by category.

The User Activity Summary workspace includes the following indicators:

- By User
- By Data Domain
- By Sensitivity Level
- By Department
- By Data Store Group
- By Risk
- By Location
- By Data Store
- By Day of the Week

By default, the workspace displays all events in every category. You can filter the information on the workspace by any factor within an indicator. For example, you can filter by the PCI data domain to identify any unusual activity on sensitive fields that match the PCI data domain.

Summary Workspace Management

After you scan data stores, view the summary workspaces to filter information by a group, location, or key scan factors. You can export the information for reports and further analysis.

Exporting Information from the Summary Workspaces

You can export information from the **Data Stores Summary**, **Department Summary**, and **User Activity Summary** workspaces.

To export information, select the **Export** option from the **Actions** menu.

When you export information, the Data Privacy Management Service saves a compressed file to the default download directory on your machine. The name of the compressed file is `Data.zip`. If you export the file again or export the file from a different page, the Data Privacy Management Service downloads a new file. The file name includes an incremental number such as `Data(1).zip`.

The `Data.zip` file contains CSV files. The information in the CSV files matches the information on the page you export from. When you export from a summary workspace, the `Data.zip` file contains CSV files. Each CSV file contains the information displayed on an indicator on the workspace.

Part V: Protection

This part contains the following chapters:

- Protection, 592
- Encryption Rules, 597
- Risk Simulation Plans, 604

CHAPTER 26

Protection

This chapter includes the following topics:

- Protection Overview, 592
- Protection Prerequisites, 593
- Protection Process Flow, 594

Protection Overview

In Data Privacy Management, you can protect sensitive data with Persistent Data Masking, and encryption rules and keys. You can also decrypt encrypted data.

In Data Privacy Management, you scan data stores to identify sensitive data. Then you create protection extensions on the **Extensions** workspace. Protection extensions configure connection properties to the protection applications once so that you can easily reuse the properties when you configure protection tasks and security policies.

You can also add protection extensions to data domains and specify the default protection rules that Data Privacy Management applies to the sensitive fields in each data store that is included in the data domain. You specify an active protection extension that Data Privacy Management supports for the protected data store when you configure protection tasks. A protection task applies the default rules or access conditions of the associated protection extension to sensitive fields in a data store. Each protection task is associated with one data store.

Data Privacy Management includes the following protection extensions:

- Encryption
- Persistent Data Masking - Big Data
- Persistent Data Masking - Remote Domain

You create encryption rules on the **Encryption Rules** workspace. To add encryption rules to data domains and encryption protection tasks, you must associate encryption keys with the rules.

You can generate and manage 16-, 24-, or 32-byte PKCS #11 encryption keys with the Soft Hardware Security Module (SoftHSM) key management tool that is included with Data Privacy Management. For more information about generating encryption keys in the SoftHSM key management tool, see the "Encryption and Decryption Management" chapter in the *Informatica Data Privacy Management Installation and Configuration Guide*.

Related Topics:

- "Protecting Data" on page 403
- "Security Policy Properties" on page 453
- "Task Status" on page 495
- "Protection Job" on page 302
- "Adding or Updating a Protection Extension on the Data Domain Details Page" on page 74

Protection Prerequisites

Data Privacy Management uses several components to protect sensitive data and to display protection information.

Verify the following prerequisites for protection:

Verify that your role and permissions enable you to work with protection components.

The following table lists the default protection permissions for Data Privacy Management custom roles:

Custom Role	Protection Permissions
Data Owner	View, add, and manage encryption rules and tasks. View extensions and protection remote agents.
Operator	View and run tasks. View encryption rules and protection remote agents.
Policy Author	View encryption rules.
Security Analyst	View and manage encryption rules and tasks. View extensions and protection remote agents. Manage risk simulation plans.
Security Manager	View and manage tasks. View extensions, encryption rules, protection remote agents, and subject registry. Manage risk simulation plans.
Technical Administrator	View and delete tasks. View and manage extensions, encryption rules, and protection remote agents.

For more information about Data Privacy Management roles and protection permissions, see the *Informatica Data Privacy Management Administrator Guide*.

Create at least one active protection extension.

If you have Data Privacy Management privileges to manage extensions, you can create protection extensions that define rules for protecting sensitive data. You can create protection extensions for the following data protection methods:

- Encryption
- Persistent Data Masking (PDM) - Big Data
- Persistent Data Masking (PDM) - Remote Domain

Protection extensions save connection details to the protection method services for easy re-use in data domains, security policies, and protection tasks.

Discover sensitive data in data stores.

Discover sensitive data in one of the following ways:

- On the **Scans** workspace, run Scan jobs for data stores to detect sensitive columns based on data domain conditions, classification policy configurations, and default conformance criteria settings. After Scan jobs identify sensitive fields, you can create, configure, and schedule protection tasks.
- On the **Data Stores** workspace, import Catalog metadata from a CSV file for the sensitive fields in a data store. Then, import protection status for the sensitive fields from a CSV file. The sensitive fields in the protection status file must contain Catalog metadata.

Related Topics:

- "Supported Data Store Types for Protection Extension Plugins" on page 66
- "How Data Privacy Management Determines Sensitivity Status" on page 252
- "Importing Catalog Metadata" on page 115
- "Importing Protection Statuses" on page 109

Protection Process Flow

Sensitive data protection inData Privacy Management typically follows a sequential process of steps that involves several users with different roles and privileges.

You can protect data on demand from several views in Data Privacy Management. You can also add a protection action to a data store or user activity security policy. In the event of a security policy violation, Data Privacy Management creates a protection task for each data store in the policy that has an active, supported protection extension defined on the **Extensions** workspace.

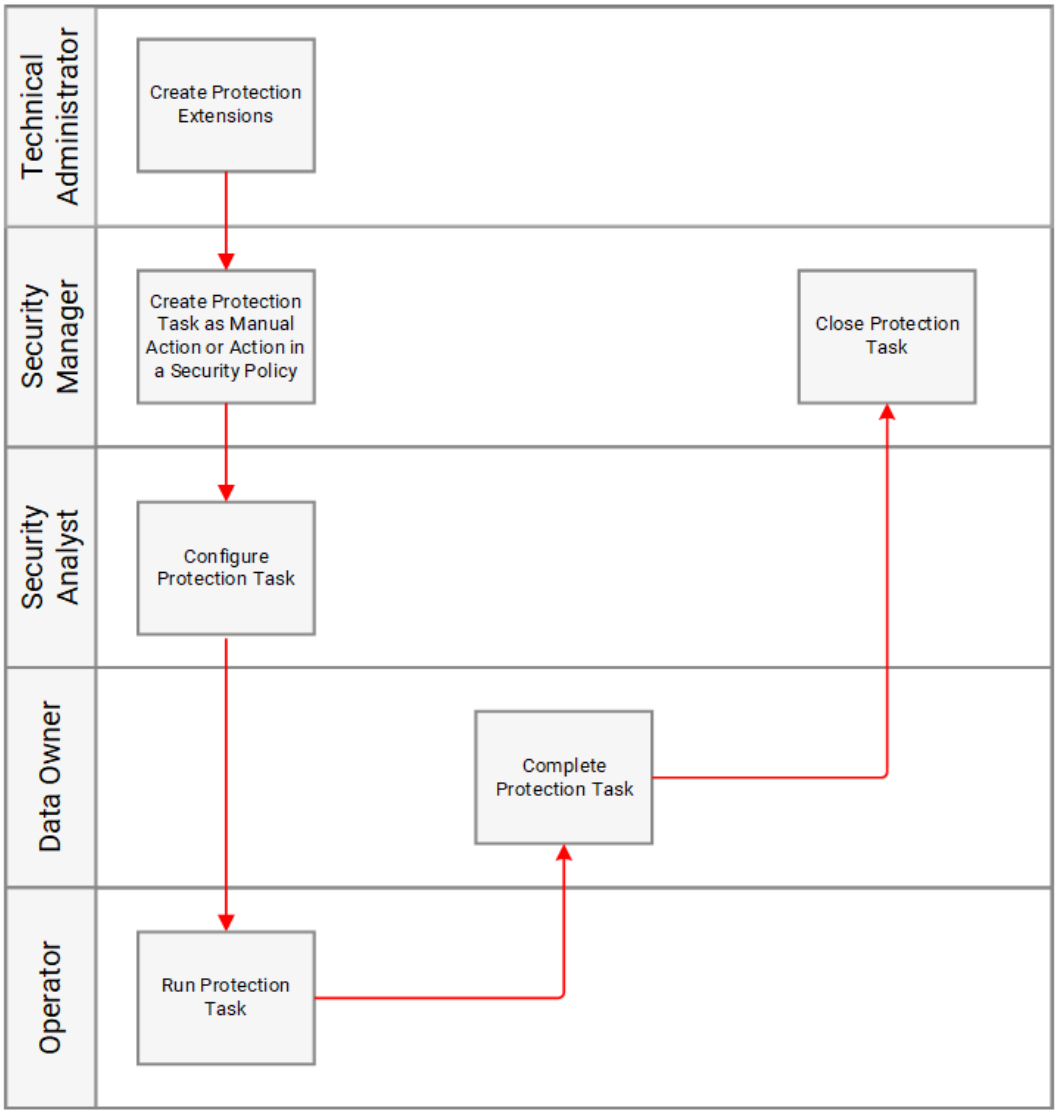
Data protection occurs in the following general sequence in Data Privacy Management:

1. Identify sensitive fields in a data store in one of the following ways:
   - Scan the data store on the **Scans** workspace.
   - On the **Data Stores** workspace, import Catalog metadata for the data store, and then import protection status from a CSV file that also contains the Catalog metadata.
2. The Technical Administrator creates protection extensions on the **Extensions** workspace.
3. The Security Manager adds protection actions in data store or user activity security policies. Security Managers can also create protection tasks on demand by selecting **Protect Data** from the **Actions** > **Take Action** menu option on the following views in Data Privacy Management:
   - **Anomaly Detection** workspace
   - **Proliferation** page
   - **Security Policy Violations** workspace
   - **Sensitive Fields** page
   - **Top Data Domains** list page
   - **Top Data Stores** grid page

   The protection tasks appear on the **Tasks** list page with a status of New. The extension is listed as Unconfigured until the Security Analyst edits the task and selects a protection extension for the data store.

4. The **Tasks** list page now displays the protection extension plugin name. The Security Analyst configures the protection rules for the task, and then marks the configuration as complete when they are finished. The protection task then has a status of Configured.
5. The Operator schedules a protection job to run the task.
6. When a protection job completes successfully, the Security Manager completes and then closes the protection task.

The following image shows an example process flow for several Data Privacy Management roles that have privileges to protect sensitive data:



For more information about data protection roles and privileges, see the *Informatica Data Privacy Management Administrator Guide*.

Related Topics:

- "Tasks Overview" on page 483
- "Task Status" on page 495
- "Extension Management" on page 73

- “Scans Overview” on page 246

CHAPTER 27

Encryption Rules

This chapter includes the following topics:

- Encryption Rules Overview, 597
- Encryption Rules Workspace, 598
- Encryption Rule Properties, 599
- Encryption Rule Management, 601

Encryption Rules Overview

Encryption rules define how Data Privacy Management masks sensitive columns with string data types in Cloudera Hive data stores. You can create and manage encryption rules on the **Encryption Rules** workspace.

For each encryption rule, you can create an encryption technique to change metadata, preserve metadata, and preserve the data format.

When you associate an encryption protection extension with a data domain on the **Data Domains** workspace, you specify default encryption rules and encryption keys to mask sensitive string data in data stores that contain the data domain columns.

When you configure encryption protection tasks on the **Tasks** workspace, you specify encryption extensions, encryption rules, and encryption keys to mask sensitive fields when the encryption tasks run.

You can generate and manage 16-, 24-, or 32-byte PKCS #11 encryption keys with the Soft Hardware Security Module (SoftHSM) key management tool that is included with Data Privacy Management. For more information about generating encryption keys in the SoftHSM key management tool, see the "Configuring Encryption and Decryption" section in the *Informatica Data Privacy Management Administrator Guide*.

Related Topics:

- "Decrypting a Closed Encryption Task" on page 506
- "Adding or Updating a Protection Extension on the Data Domain Details Page" on page 74
- "Configuring Encryption Rules and Encryption Keys" on page 504

Encryption Rules Workspace

On the **Encryption Rules** workspace, you can create and manage encryption rules. The workspace includes a page that displays a list of encryption rules and a detail page for each encryption rule.

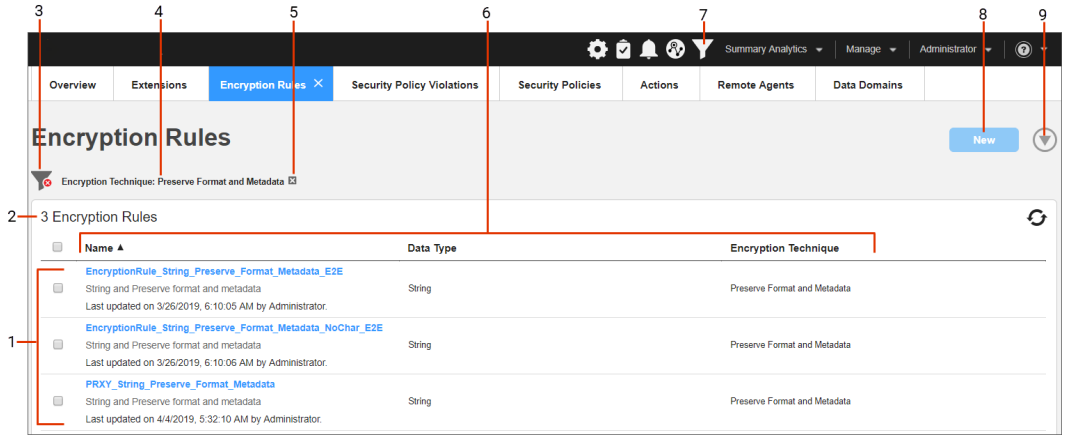
You can use filters to quickly find the encryption rules that you want to manage.

To access the **Encryption Rules** workspace, click **Manage** > **Encryption Rules**.

Encryption Rules List Page

The **Encryption Rules** list page shows the encryption rules that match the active filter. By default, you view the **Encryption Rules** list page when you access the **Encryption Rules** workspace.

The following image shows an example of the **Encryption Rules** list page:



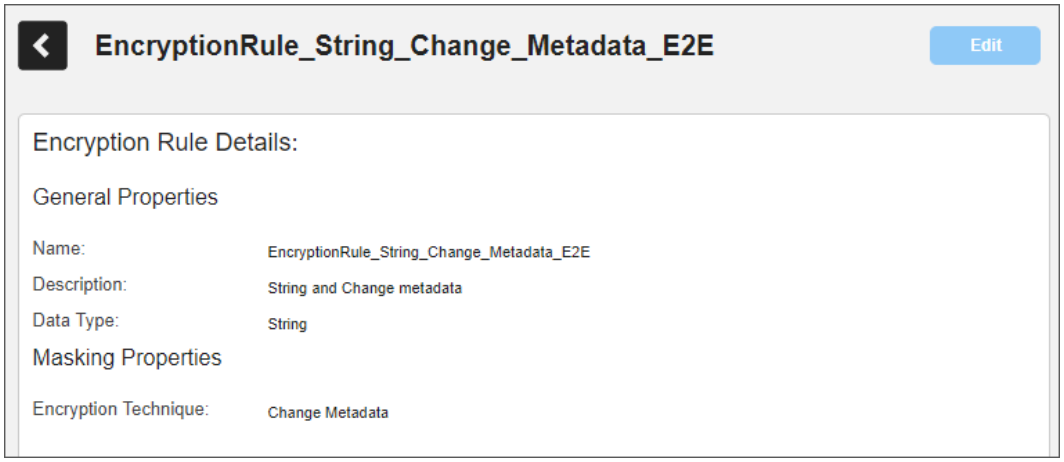
1. Encryption rules
2. Encryption rule count
3. Clear Filter icon
4. Filter conditions
5. Clear Filter Condition icon
6. Encryption rule properties
7. Filter icon
8. New encryption rule button
9. Actions menu

Encryption Rule Details Page

To view or edit encryption rule properties, access the **Encryption Rule Details** page from the **Encryption Rules** workspace.

- Click an encryption rule name. To edit the properties, click **Edit** on the **Encryption Rule Details** page.
- Select the check box next to an encryption rule and then select **Open** or **Edit** from the **Actions** menu.

The following image shows an example of an **Encryption Rule Details** page:



Encryption Rule Properties

Configure encryption rule properties for the encryption technique that masks sensitive data in Cloudera Hive data stores.

You can specify encryption rules in data domains and encryption protection task properties.

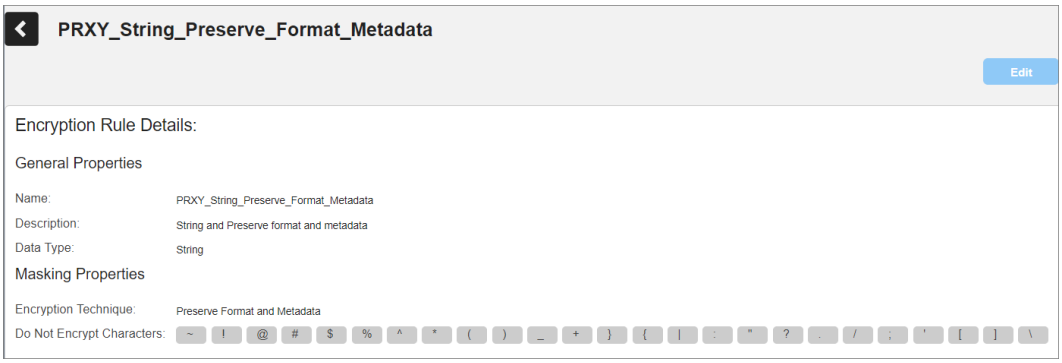
The following image shows the encryption rule properties on the **New_Encryption_Rule** page:



Property	Description
Name	A name for the encryption rule that helps you easily identify the rule. The name cannot exceed 255 characters, contain a space, or contain the following special characters: ~ ! $ % ^ & * ( ) +
Description	Optional. Long description of the encryption rule that cannot exceed 255 characters.
Data Type	Select the string data type.

Property	Description
Encryption Technique	Select one of the following options: - Change metadata - Preserve format and metadata - Preserve metadata
Do Not Encrypt Characters	Optional. Displays when you select the **Preserve format and metadata** encryption technique. Enter a list of characters that encryption protection tasks configured with the encryption rule do not encrypt.

The following image shows a sample configuration of properties for an encryption rule that uses the Preserve Format and Metadata encryption technique:



Encryption Techniques

Encryption techniques specify how encryption rules mask sensitive data when associated encryption protection tasks run.

When you create or edit an encryption rule, you specify an encryption technique. The following table describes the encryption technique options:

Encryption Technique	Description
Change metadata	Changes the length and format of the source data. For example, an encryption rule that changes metadata for the City data domain might change a column value of `London` to `Xuep@8f5`, `fmch529`, or `6ky#ke33h*we`.
Preserve format and metadata	Preserves the format and length of the source data. Replaces each alphanumeric and special character value with another value that retains the same case, numeric format, or special character format. Provides the option to specify characters that the encryption rule will not encrypt. You cannot exclude the following characters from encryption: `<>&` For example, an encryption rule that preserves format and metadata for the Email Address data domain, but does not exclude any characters from encryption, might change a column value of `Abc123@xyz.com` to `Mpz849#dje!kuw`. In the same Email Address example, if you specify that Data Privacy Management cannot encrypt the "`@`" and "`.`" characters, the email address `Abc123@xyz.com` might become `Mpz849@dje.kuw` after encryption.
Preserve metadata	Preserves only the length of the source data. For example, an encryption rule that preserves metadata for the First Name data domain might change a column value of `Alexander` to `jl6#HB91v`.

Encryption Rule Management

You can create and manage encryption rules on the **Encryption Rules** workspace.

Encryption rules specify how Data Privacy Management masks sensitive string data in Cloudera Hive data stores when associated encryption protection tasks run. You can create, edit, copy delete, export, import, and view encryption rules.

Important: You cannot delete an encryption rule that is associated with at least one data domain, an active task, or a task that ran successfully.

Creating an Encryption Rule

Create encryption rules to specify default encryption techniques for masking sensitive data in data domains that include Cloudera Hive data stores. You can edit the encryption rule associated with sensitive fields when you configure encryption protection tasks on the **Tasks** workspace.

1. Click **Manage** > **Encryption Rules**.

   The **Encryption Rules** workspace appears.
2. Click **New**.

   The **New_Encryption_Rule** page appears.

3. Enter a name for the encryption rule.

   The encryption rule name cannot contain spaces or the following special characters: `~!$%^&*()+`

4. Optionally, enter a description.
5. Select the String data type.
6. Select one of the following encryption techniques:
   - Change Metadata
   - Preserve Format and Metadata
   - Preserve Metadata

   If you selected the Preserve Format and Metadata encryption technique, the **Do Not Encrypt Characters** field appears. Optionally, specify characters that Data Privacy Management does not encrypt when encryption tasks associated with the encryption rule run.
7. Click **Save.**

Exporting Encryption Rules

You can export a list of encryption rules to a CSV file to add encryption rules or update encryption rule properties in bulk and then import the file. The file includes a list of encryption rules and the corresponding properties. You can export all encryption rules, a filtered list of encryption rules, or individually selected encryption rules.

The export file contains the same format that Data Privacy Management requires to import encryption rules. When you import the revised CSV file, the Import job adds and updates the encryption rules in the Model repository.

1. Click **Manage** > **Encryption Rules**.

   The **Encryption Rules** workspace appears.
2. From the **Actions** menu, select **Export**.

   A dialog box displays the exported CSV file with the following default file name: `EncryptionRule.csv`. If you export the file again, the Data Privacy Management Service downloads a new file. The file name includes an incremental number such as `EncryptionRule (1).csv`.
3. Save the CSV file to the default download directory on your machine, or browse to a different directory.
4. Open the CSV file.

   The following image shows the spreadsheet view of a sample export file:

	A	B	C	D	E	F	G
1	RuleName	RuleDescription	EncryptionType	DataType	EncryptionTechnique	ExcludedCharList	Action
2	EncryptionRule_String_Change_Metadata_E2E	String and Change metadata	FPE	String	Change Metadata		
3	EncryptionRule_String_Preserve_Format_Metadata_E2E	String and Preserve format and metadata	FPE	String	Preserve Format and Metadata	abc	
4	EncryptionRule_String_Preserve_Format_Metadata_NoChar_E2E	String and Preserve format and metadata	FPE	String	Preserve Format and Metadata		
5	EncryptionRule_String_Preserve_Metadata_E2E	String and Preserve metadata	FPE	String	Preserve Metadata		
6	PRXY_String_Preserve_Format_Metadata	String and Preserve format and metadata	FPE	String	Preserve Format and Metadata	~!@#$%^*()_+{|:"?./;'[]\	

   The following image shows the text view of a sample export file:

```
RuleName,RuleDescription,EncryptionType,DataType,EncryptionTechnique,ExcludedCharList,Action
EncryptionRule_String_Change_Metadata_E2E,String and Change metadata,FPE,String,Change Metadata,,
EncryptionRule_String_Preserve_Format_Metadata_E2E,String and Preserve format and metadata,FPE,String,Preserve Format and Metadata,abc,
EncryptionRule_String_Preserve_Format_Metadata_NoChar_E2E,String and Preserve format and metadata,FPE,String,Preserve Format and Metadata,,
EncryptionRule_String_Preserve_Metadata_E2E,String and Preserve metadata,FPE,String,Preserve Metadata,,
PRXY_String_Preserve_Format_Metadata,String and Preserve format and metadata,FPE,String,Preserve Format and Metadata,"~!@#$%^*()_+{|:""?./;'[]\",
```

5. Optionally, update the encryption rule properties in the CSV file and save the file.

   You can then import the file on the **Encryption Rules** workspace.

Importing Encryption Rules

After you export encryption rule details to a CSV file and add new rules or edit rule properties, you can import the new encryption rule details to update the rules in bulk.

1. Click **Manage** > **Encryption Rules**.

   The **Encryption Rules** workspace displays a list of encryption rules.

2. From the **Actions** menu, select **Import**.

   The **Import Encryption Rules** page appears:



   By default, **Replace Duplicates with Items Imported** is enabled. This option prevents creating duplicate encryption rules by updating existing rules with new details from the imported file.

3. Click in the empty field.

   The **Open** dialog box appears.

4. Browse to the directory that contains the CSV file with the encryption rule details you want to import, select the file, and then click **Open**.

   The file name appears in the field on the **Import Encryption Rules** page. A message indicates the number of encryption rules that Data Privacy Management found in the CSV file.

5. Click **Select**.

6. Click **Import**.

Copying an Encryption Rule

To create a new encryption rule with similar properties to an existing encryption rule, you can copy the existing rule. After you copy the encryption rule, edit the encryption rule properties.

1. Click **Manage** > **Encryption Rules**.

   The **Encryption Rules** workspace appears.

2. Select the checkbox next to the encryption rule that you want to copy.

3. From the **Actions** menu, select **Duplicate**.

   The **Encryption Rule Details** page appears with the following name: `Copy_of_<encryption_rule_name>`

4. Edit the encryption rule name, description, and optionally, the encryption technique.

   The encryption rule name cannot contain spaces or the following special characters: `~!$%^&*()+`

5. Click **Save**.

CHAPTER 28

Risk Simulation Plans

This chapter includes the following topics:

- Risk Simulation Plans Overview, 604
- Risk Simulation Plans Workspace, 604
- Risk Simulation Plan Properties, 606
- Risk Simulation Plan Management, 607

Risk Simulation Plans Overview

On the **Risk Simulation Plans** workspace, you can create and manage plans to reduce the risk score and residual risk cost of unprotected sensitive data domains in one or more data stores.

For each risk simulation plan, you can adjust the protection status of sensitive data domains in scanned data stores to simulate the impact on the potential cost to your organization in the event of a security breach.

The risk simulation indicators at the top of each plan detail page show both the current and estimated risk score, protection status, and residual risk cost in the currency selected for your organization on the **Settings** workspace. The current and estimated values for each data store in the plan appear in the columns on the page.

After you save a risk simulation plan, you can edit the plan to protect more data domains, or select fewer data domains to protect. You cannot remove protection from all data domains in a data store after you apply and save the plan. To simulate and compare risk, create a new plan that includes the data store and do not apply protection to any data domains.

Risk Simulation Plans Workspace

On the **Risk Simulation Plans** workspace, you can create and manage risk simulation plans. The workspace includes a page that displays a list of risk simulation plans and a detail page for each plan.

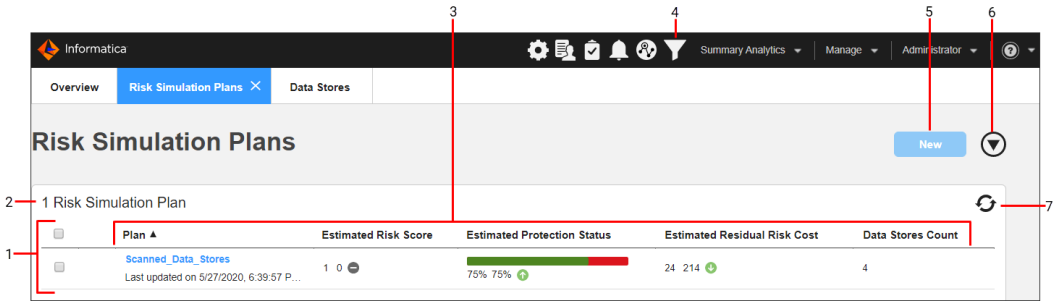
You can use filters to quickly find the risk simulation plan that you want to manage.

To access the **Risk Simulation Plans** workspace, click **Manage** > **Risk Simulation Plans**.

Risk Simulation Plans List Page

The **Risk Simulation Plans** list page shows the risk simulation plans for scanned data stores in your organization. By default, you view the **Risk Simulation Plans** list page when you access the **Risk Simulation Plans** workspace.

The following image shows an example of the **Risk Simulation Plans** list page:



1. Risk simulation plans
2. Risk simulation plan count
3. Risk simulation plan properties
4. Filter icon
5. New risk simulation plan button
6. Actions menu
7. Refresh icon

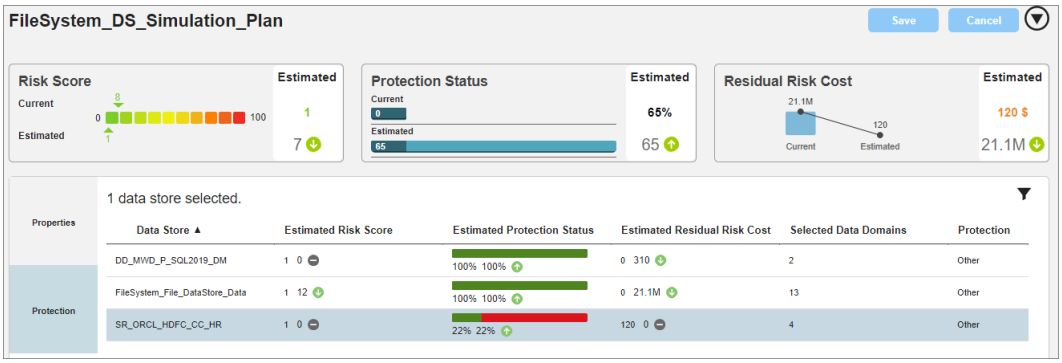
Risk Simulation Plan Details Page

To view or edit a risk simulation plan, access the **Risk Simulation Plan Details** page from the **Risk Simulation Plans** workspace.

Access the **Risk Simulation Plan Details** page in one of the following ways:

- Click a risk simulation plan name. To edit the properties, click **Edit** on the **Risk Simulation Plan Details** page.
- Select the check box next to a risk simulation plan and then select **Open** or **Edit** from the **Actions** menu.

The following image shows an example of a **Risk Simulation Plan Details** page:



The following table describes the information listed on the **Protection** view of the **Risk Simulation Plan Details** page:

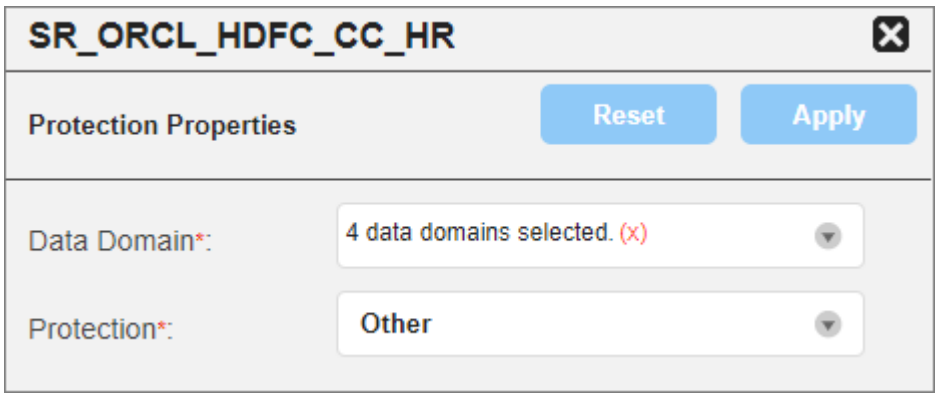
Property	Description
Risk Score Indicator	The Current risk score shows the aggregate risk score of data stores in the plan, with a value between 0 and 100. Data Privacy Management calculates risk scores based on the following metrics for sensitive data in each scanned data store: - Sensitivity level - Protection status - Number of sensitive fields - Number of sensitive records - Number of targets - Residual risk cost - Number of active users - Number of active and inactive users with activity in Data Privacy Management You can change the weight of each risk score factor from the **Settings** workspace. Data Privacy Management recalculates the risk score of the scanned data stores and refreshes the values on the **Risk Score** indicator. The Estimated risk score shows the predicted risk score if you implement the selected data protection changes in the plan.
Protection Status Indicator	The Current protection status shows the percentage of sensitive fields in the data store that are currently protected. The Estimated protection status shows the percentage of sensitive fields in the data store that are protected in the simulation. For data store types that protection extensions do not support, the value is 0.
Residual Risk Cost Indicator	The Current residual risk cost shows the amount of money it would take to resolve the problems of a data breach for sensitive data that is currently unprotected in the data stores included in the simulation plan. The Estimated residual risk cost shows the predicted risk cost if you implement the selected data protection changes in the simulation plan.
Selected Data Domains	The column shows the number of data domains selected in the simulation plan.
Protection	The column shows the method used to protect sensitive data in each data store included in the plan.

Risk Simulation Plan Properties

Configure risk simulation plan protection properties for each data store included in the plan to reduce the risk score and residual risk cost of unprotected sensitive data.

On the **Risk Simulation Plan Details** page, you can select sensitive data domains to protect for each scanned data store included in the plan.

The following image shows the risk simulation plan protection properties on the **Risk Simulation Plan Details** page:



The following table describes the risk simulation plan protection properties:

Property	Description
Data Domain	Required. Select one or more sensitive data domains for the selected data store to simulate the impact that protecting data will have on the risk score, protection status, and residual risk cost indicators.
Protection	Required. Select a protection method to simulate risk impact.

Risk Simulation Plan Management

You can create and manage risk simulation plans on the **Risk Simulation Plans** workspace. You can create, edit, export, copy, filter, delete, and refresh simulation plans.

Creating a Risk Simulation Plan

Create risk simulation plans to reduce the risk score and residual risk cost of unprotected sensitive data domains in one or more data stores.

1. To create a new risk simulation plan, access the **New Risk Simulation Plan** page in one of the following ways:
   - Click **Manage** > **Risk Simulation Plans**. The **Risk Simulation Plans** workspace appears. Click **New**.
   - On the Security Dashboard, click the **Top Data Stores** indicator label. The **Data Stores Grid** page appears. Select one or more data stores to include in the plan. From the **Actions** menu, select **Simulate Risk**.
2. Click the **Properties** tab.
3. Enter a name for the plan.

   The risk simulation plan name cannot contain spaces or the following special characters: `~!$%^&*()-+=[]{}|;:'",./<>?`
4. Optionally, enter a description for the plan.

   The description cannot exceed 255 characters.

5. Optionally, select tags from the list.
6. Click the **Protection** tab.
7. From the **Actions** menu, select **Add Data Stores**.

   A list of available data stores appears.
8. Select the check boxes to the left of the data stores you want to include in the risk simulation plan.
9. Click **OK**.
10. To see how the estimated risk for a data store changes when you protect sensitive data, click the **-** icon in one of the plan columns or click the blue arrow icon (**<**) on the right side of the page.

    The **Properties** pane appears.
11. Select the sensitive data domains that you want to protect, and select a protection method.
12. Click **Apply**.

    The risk simulation indicators update with aggregate totals for the data stores in the plan.
13. Optionally, adjust protection properties to view other risk scenarios for the data stores in the plan.
14. Click **Save.**

    The plan appears on the **Risk Simulation Plans** list page.

To apply protection to sensitive data domains and achieve the reduction in risk score and cost, create and schedule a protection task for each data store in the plan. The enterprise-level summary indicators on the **Security Dashboard** update with the changes after the tasks run.

Exporting a Risk Simulation Plan

You can export risk simulation plan details to a CSV file to view, sort, and analyze the information on the **Risk Simulation Plan Detalis** page for the plans you export. You can export all risk simulation plans, a filtered list of risk simulation plans, or individually selected risk simulation plans.

1. Click **Manage** > **Risk Simulation Plans**.

   The **Risk Simulation Plans** workspace appears.
2. To export details about every simulation plan, select **Actions** > **Export**.

   A dialog box displays the exported CSV file with the following default file name: `Simulations.csv`.
3. To export details about one simulation plan, click the plan name to open the **Risk Simulation Plan Detalis** page. Then, select **Actions** > **Export** and download the CSV file.
4. Save the CSV file to the default download directory on your machine, or browse to a different directory.

Copying a Risk Simulation Plan

To create a new risk simulation plan with similar properties to an existing risk simulation plan, you can copy the existing plan. After you copy the plan, edit the plan properties.

1. Click **Manage** > **Risk Simulation Plans**.

   The **Risk Simulation Plans** workspace appears.
2. Select the checkbox next to the risk simulation plan that you want to copy.
3. From the **Actions** menu, select **Duplicate**.

   The **Risk Simulation Plan Details** page appears with the following name: `Copy_of_<risk_simulation_plan_name>`

4. Edit the plan name, description, and optionally, add or remove data stores included in the plan.

   The plan name cannot contain spaces or the following special characters: `~!$%^&*()-+=[]{}|;:'",./<>?`

5. Optionally, edit the sensitive data domains to protect for each data store included in the plan to compare the risk results with the risk score, protection status, and residual risk cost of the plan you copied.
6. Click **Save**.

Editing a Risk Simulation Plan

You can edit risk simulation plans to preview the estimated risk score and cost for protecting sensitive data domains. You can also add and remove data stores in the simulation plan.

Before you can simulate risk, scan the data stores you want to include in the simulation plan. The Scan job discovers and classifies sensitive and personal data domains in data stores.

1. On the **Risk Simulation Plans** workspace, select the checkbox next to the remote agent that you want to edit.
2. From the **Actions** menu, select **Edit**.

   The **<Risk_Simulation_Plan_Name>** page appears.
3. To edit the plan name, description, or tags, click the **Propeties** tab.
4. On the **Protection** tab, you can add or remove sensitive data domains to protect for each data store in the plan.

   The risk indicators update with the changes.

   Note: You cannot remove protection from all data domains that you selected and saved when you created the plan. You can add data domains to simulate protection and risk scoring if you did not protect all sensitive data domains when you created the plan. You can remove protection simulation for all but one data domain in a data store. To simulate risk for a data store without protecting sensitive data, create a new risk simulation plan and compare the results on the list page.
5. To add data stores to the plan, select **Actions** > **Add Data Stores** and select one or more scanned data stores from the list.
6. To remove a data store from the plan, select the data store row. From the **Actions** menu, select **Remove Data Store**.

   A confirmation prompt appears. Click **Yes, I'm Sure**.
7. When finished, click **Save**.

Part VI: Privacy

This part contains the following chapters:

- Subject Registry, 611
- Privacy Dashboard, 629

CHAPTER 29

Subject Registry

This chapter includes the following topics:

- Subject Registry Overview, 611
- Legal Hold, 612
- Residency, 613
- Category, 613
- Purpose, 613
- Subject Registry Workspace, 614
- Subject Registry Workspace Management, 616
- Subject Details Page, 618
- DSAR Report Formats, 620
- Subject Details Page Management, 623

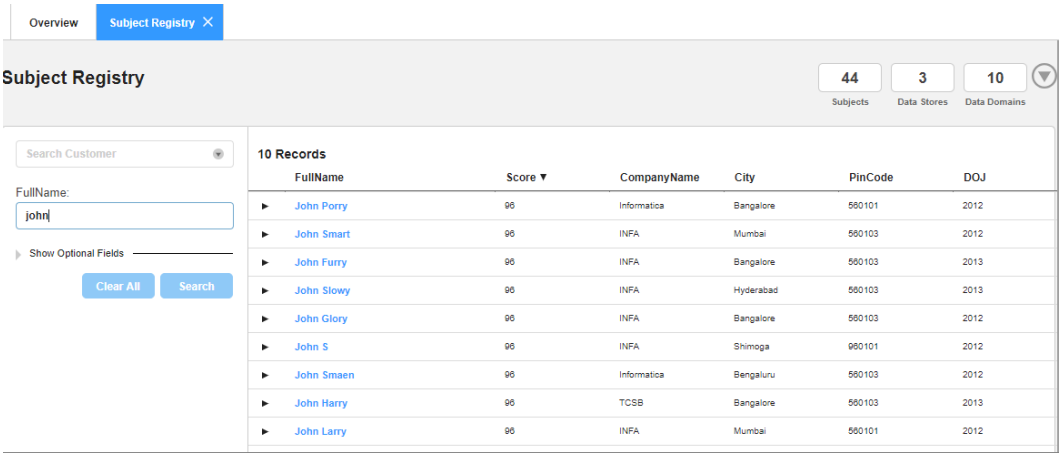
Subject Registry Overview

As the custodian of customer and employee sensitive data, ensuring data privacy is essential to your company's success. To effectively carry out a data privacy plan, you must have the ability to quickly locate an individual's data that might be dispersed across your enterprise data.

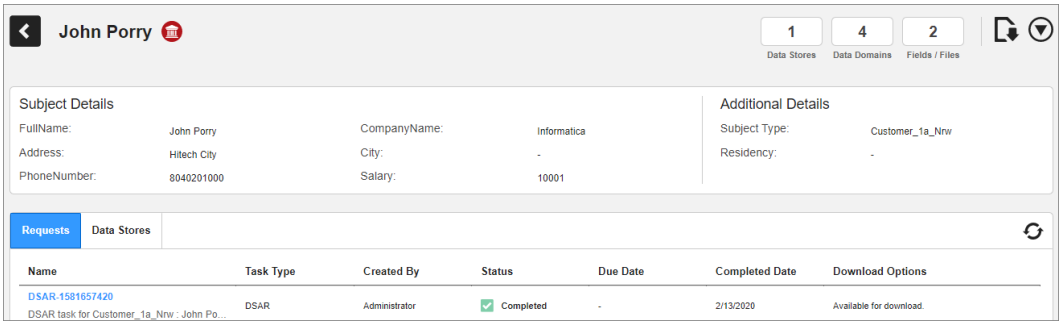
Subject Registry maps individuals with their data and provides a search tool to quickly locate an individual whose data you collect and store. You can search for the individual by name or any other identifying detail. After you identify the correct individual, Subject Registry provides details of the data stores, schemas, tables, fields, and files that contain the individual's data.

Use the results to promptly respond to and stay in compliance with data privacy regulations such as GDPR, CCPA, the right-to-be-forgotten, data portability, and breach notification.

The following image shows the **Subject Registry** workspace with a sample search result:



The following image shows the details for an individual's sensitive data:



Before you use Subject Registry, you must configure the database, define the entity, and define the SQL query. In addition, for unstructured data stores, you must create a connection to a remote agent on the **Remote Agents** workspace. Then, run a Subject Registry scan to build the index and identity map.

For more information on the prerequisite tasks for Subject Registry, see the *Informatica Data Privacy Management Administrator Guide*.

Legal Hold

You can apply a legal hold flag to a data subject in the Subject Registry. A legal hold flag indicates an organization's legal notification to retain specific data.

Based on a legal hold notification, you might choose to reject some subject requests such as the right-to-be-forgotten requests.

You can apply and remove legal hold on a data subject from the **Subject Registry** page. If applied, a legal hold notification appears next to the subject name on search results pages and subject details pages.

You can export legal hold information when you export details from the search results page or the Subject Details page. Legal hold information is included in DSAR reports. If the subject is under legal hold, a prompt appears to confirm if you want to generate a DSAR report.

Residency

Use Residency as criteria to define how you apply specific privacy policies.

For example, some privacy policies might apply to data subjects who are residents of countries in the European Union but not to other data subjects. Identify residency information for data subjects and apply specific privacy policies that are relevant to a data subject.

You can use more than one column as criteria for residency. For example, if you configure city and state columns, both values appear in the Residency field separated by a comma.

To view Residency information on the Subject Registry search results page and on Subject Details pages, configure the Residency attribute in the Subject Registry configuration files. On the Subject Details page, residency information appears in the **Additional Details** section. Residency information also appears in DSAR reports.

The order that you use in the configuration file determines the order of appearance of values.

For information on how to configure Residency in the Subject Registry configuration files, see the *Informatica Data Privacy Management Administrator Guide*.

Category

A category is a kind or a grouping that you can use to classify data.

Data privacy regulations require that you associate subjects and subject information that you collect with a category. For example, you can assign a data subject a subject category such as Employee or Vendor. Personal data that you collect might fall under different categories such as Address, Family, and Beneficiaries.

The subject category appears in the Subject Type field on Subject Details pages that you access from the Subject Registry. Personal data categories appear in data store information on a Subject Details page. Click the **Fields/Files** link to view the information. DSAR reports include the information to indicate the subject type and the category of each personal data field.

You associate a category with a data domain. When a scan associates a sensitive field with a data domain, the field is associated with the category. You import a list of data domain-to-master category assignments from the Subject Registry page. If required, you can override the master category associated with specific fields from the **Sensitive Fields** page.

Purpose

Purpose indicates the objective or reason for storing specific data. You can assign a purpose to a data store.

Data privacy regulations require that you collect data from a subject for a purpose and use the data only for that purpose. For example, for payroll requirements you collect financial information such as bank account details. You can assign the Payroll purpose to the data store and use the data for payroll requirements. The purpose appears in data store information on a Subject Registry **Subject Details** page. Click the **Fields/Files** link to view the information. DSAR reports include the purpose to indicate the reason for collecting the specific data.

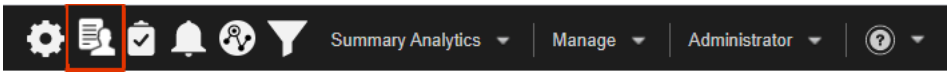
You import a list of data store-to-purpose assignments on the **Subject Registry** workspace. If required, you can override the purpose for specific fields on the **Sensitive Fields** page.

Subject Registry Workspace

On the **Subject Registry** workspace, you can locate sensitive information related to an individual and generate reports to respond to data subject access requests (DSAR).

To access the **Subject Registry** workspace, click the **Subject Registry** icon on the Data Privacy Management header.

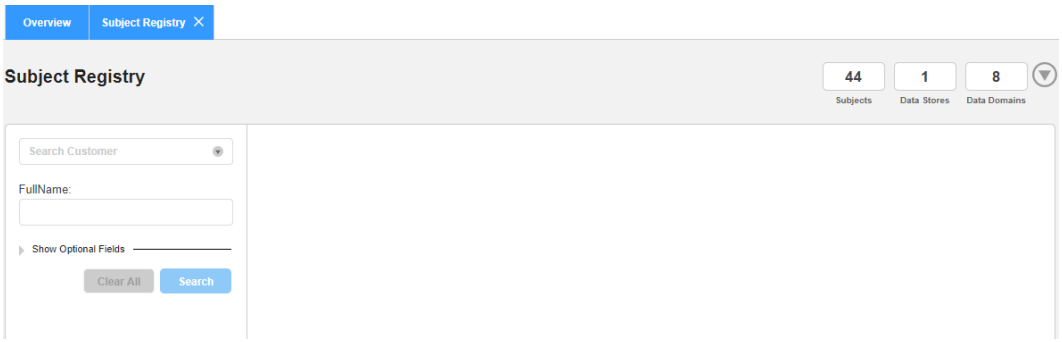
The following image shows the **Subject Registry** icon on the header:



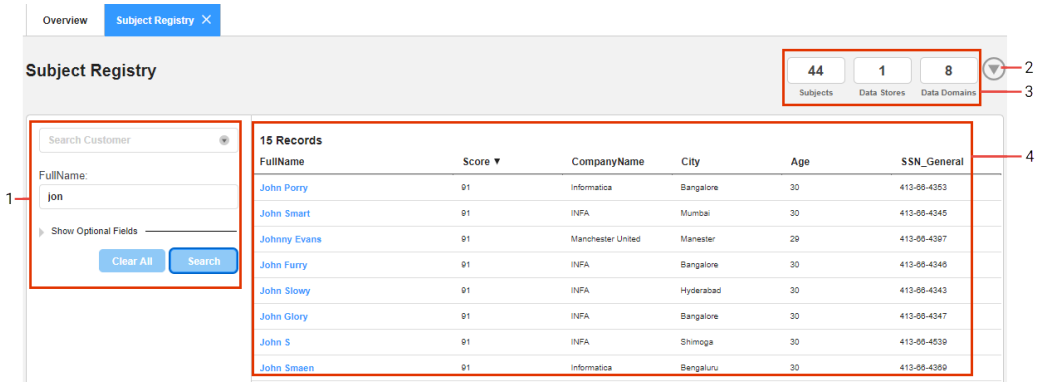
The **Subject Registry** workspace includes the following sections:

- Search
- Search results
- Summary metrics

The following image shows the Subject Registry workspace before you enter the search criteria:



The following image shows the Subject Registry workspace with a sample search and search results:

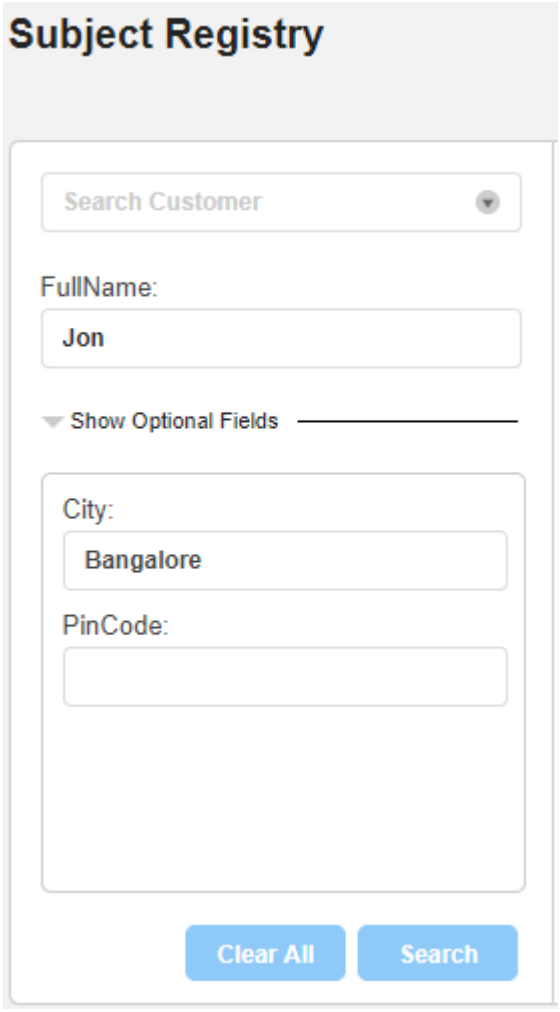


1. Search
2. Actions menu
3. Summary metrics
4. Search results

Subject Registry Search

The **Search** section contains the entities and fields you specified in the JSON file.

The following image shows an example of a Subject Registry search:



The following table describes the fields in the Subject Registry**Search** section:

Search Field	Description
Search <*Entity*>	Select the entity that contains the data subject you want to find. When you configure a JSON file, you specify an entity to include in the search. For example, if you create two JSON files, one for a **Customers** entity and another for an **Employees** entity, both entities appear as options in the list. Entity names also display as the **Subject Type** on the **Subject Details** page.
Main Search Field	Displays the main search criteria that you specify in the JSON file, and any match criteria that you specify as `Exact` in the JSON file. For example, you specified the **FullName** field to search by an individual's whole or partial first or last name.
Show Optional Fields	Displays the optional match criteria that you specified in the JSON file. For example, you specified the **City** and **PinCode** fields to filter individuals by the city and zip or pin code in their home address.

Subject Registry Summary

The Subject Registry Summary provides an overview of the number of individuals in the search pool and the number of related data stores and data domains that contain sensitive data for the individuals in the search pool.

The following image shows an example of a Subject Registry Summary:



Subject Registry Workspace Management

You can import and export information from the **Actions** menu on the **Subject Registry** workspace.

You can import lists of category and purpose associations. You can export the summary metrics, search results, and lists of master purposes and categories.

To export information on the **Subject Registry** workspace, select **Export** from the **Actions** menu.

To export search results, select **Export** > **Search Results** from the **Actions** menu. Download the `Data.zip` file to a local directory. Extract the `Data.zip` file to view the CSV files that contain information about the summary metrics and search results.

The following table describes the CSV files that you extract when you export search results from the **Subject Registry** workspace:

File Name	Description
`SubjectRegistryPreferredRecord.csv`	For the data subjects that appear in the search results, lists the information that you configured to display in columns on the **Subject Registry** workspace. For example, the file might contain columns that list each individual's name, company, city, age, Social Security number, and match score.
`SubjectRegistryPreferredRecordSummary.csv`	Contains three columns that correspond to the summary metrics on the **Subject Registry** workspace: `SubjectCount`, `DataStoreCount`, and `DataDomainCount`. In a single row, displays the numeric value for each column as shown in the summary metrics for Subjects, Data Stores, and Data Domains.

To export data category information, select **Export** > **Categories** from the **Actions** menu. To export data store purpose information, select **Export** > **Purposes** from the **Actions** menu.

The following table describes the CSV files that you extract when you export category and purpose information from the **Subject Registry** workspace:

File Name	Description
DataCategoriesMasterList.csv	Contains a list of data categories and the data domain associated with each data category.
DataPurposeMasterList.csv	Contains a list of data stores and the purpose associated with each data store.

Importing a Master List of Categories

To assign categories to data domains, you can import a list of categories with data domain associations from the **Subject Registry** workspace.

1. Create a .csv file with the following column headers and data:
   - DataCategory. Enter the names of master data categories.
   - DataDomain. For each data category, enter the name of the data domain to which you want to associate the category. The data domain must exist. If the data domain does not exist, the import of the row fails, and the import job continues.
   - Action. Enter D to delete an association with the data domain. If you do not enter anything, the system performs an upsert of associations. An upsert updates or inserts the value based on whether the field contains a value or is blank.
2. From the **Actions** menu, click **Import** > **Categories**.
3. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.
4. Click **Import**.

   The Import job begins.

To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

Importing a Master List of Purposes

To assign purposes to data stores, you can import a list of purposes with data store associations from the **Subject Registry** workspace.

1. Create a CSV file with the following column headers and data:
   - DataStore. Enter the names of data stores.
   - Purpose. For each data store, enter the purpose to which you want to associate the data store. The data store must exist. If the data store does not exist, the import of the row fails, and the import job continues.
2. From the **Actions** menu, click **Import** > **Purpose**.

3. Select the file from which you want to import data.

   The Data Privacy Management Service validates the import file, such as the file extension and column headers.
   If the file is valid, the user interface displays a count of how many records are in the file.

   If the file is not valid, the user interface displays an error message. For example, an error can occur if the file does not include valid column headings.

4. Click **Import**.

   The Import job begins.

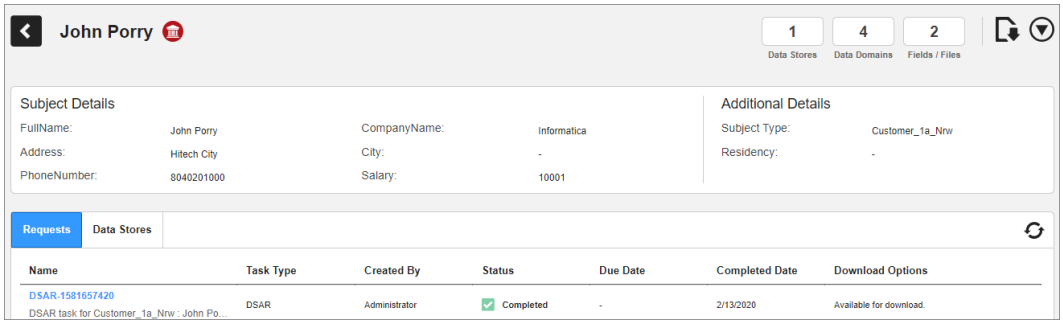
To check the status of the Import job, go to the **Jobs** workspace and view the job log for the Import File job step. The job log shows how many records the job processed, rejected, skipped, deleted, inserted, or updated. If the job rejected one or more rows in the CSV file, you can download a list of the rejected rows.

Subject Details Page

The **Subject Details** page shows additional details about a data subject, including the data stores and data domains that contain sensitive information about the subject. The page also lists the tasks created in Data Privacy Management to respond to subject data requests.

To access the **Subject Details** page, click the name of a data subject that appears in the search results on the **Subject Registry** workspace. If the subject is under legal hold, a legal hold indicator appears to the right of the subject name.

The following image shows the **Subject Details** page for the data subject John Porry, who is under legal hold:



The **Subject Details** page includes the following sections:

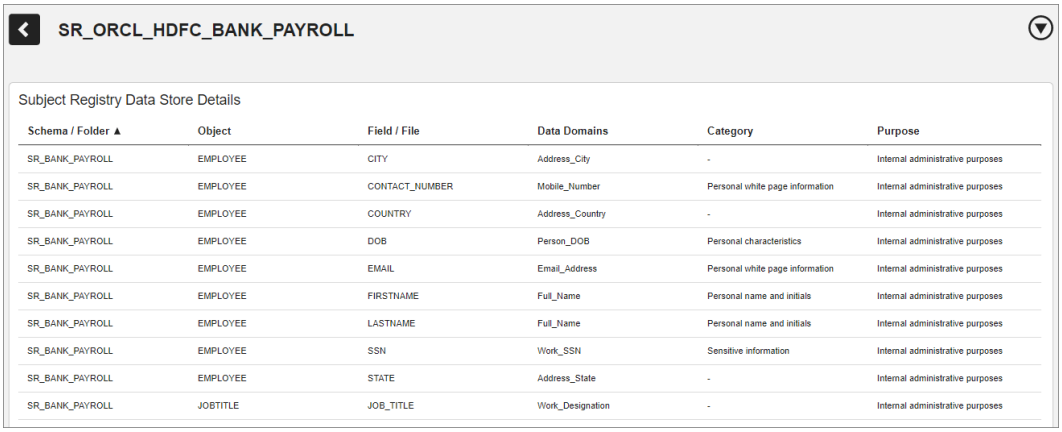
Section	Description
Summary	The Subject Details summary in the top right of the page lists the number of data stores, data domains, and fields or files that contain sensitive data about the individual. Click the **Download DSAR Report** icon to download reports that Data Privacy Management created for the subject. Select one of the following report formats: - Detailed CSV - Summary CSV - Summary PDF - Summary PDF Without Personal Data
Subject Details	Contains information about the data subject that you configured to display in columns on the **Subject Registry** workspace. The example image lists the FullName, CompanyName, Address, City, PhoneNumber, and Salary information for the data subject.
Additional Details	Contains additional information about the subject. Includes the subject type. The subject type is the name of the entity file that you use in the Subject Registry search. Residency information appears if the Subject Registry configuration includes the Residency attribute.
Requests	Displays a list of tasks created about the subject, such as DSAR reports or service management tickets. For each task, displays the task name, description, type, name of the user who created the task, status, due date, completed date, and download options. DSAR report task names begin with the prefix `DSAR`. Data subject task names begin with the prefix `DSR`. To view task details and properties on the **Tasks** workspace, click the task name. For each DSAR report task in the list, the Download Options column displays one of the following messages: - Available for download. - Available for download until *mm/dd/yyyy*. - Not available for download.
Data Stores	Lists the data stores that contain sensitive information about the subject. For each data store, also lists the number of data domains and fields or files with sensitive data about the subject, the country, the data owner name, and the third-party share status. If the information is shared with third parties, lists the names of the third parties. You can use the **Group by** menu to group the list by data domain or data owner name. By default, data stores are listed by name in alphabetical order. To order the list by data domain names, select **Data Domain**. To order the list by data owner names, select **Data Owner**. To view the names of the data domains, click the arrow to the left of the data store name. To view the **Proliferation** page for the data store, click the data store name. To view the **Subject Registry Data Store Details** page, click the value in the Fields/Files column.

Subject Registry Data Store Details Page

You can view details about the sensitive fields and files in the data stores associated with a data subject on the **Subject Registry Data Store Details** page .

You access the **Subject Registry Data Store Details** page from the **Subject Details** page. On the **Data Stores** pane, click the value in the **Fields/Files** column for a data store.

The following image shows an example **Subject Registry Data Store Details** page for the SR_ORCL_HDFC_BANK_PAYROLL data store:



Schema / Folder ▲	Object	Field / File	Data Domains	Category	Purpose
SR_BANK_PAYROLL	EMPLOYEE	CITY	Address_City	-	Internal administrative purposes
SR_BANK_PAYROLL	EMPLOYEE	CONTACT_NUMBER	Mobile_Number	Personal white page information	Internal administrative purposes
SR_BANK_PAYROLL	EMPLOYEE	COUNTRY	Address_Country	-	Internal administrative purposes
SR_BANK_PAYROLL	EMPLOYEE	DOB	Person_DOB	Personal characteristics	Internal administrative purposes
SR_BANK_PAYROLL	EMPLOYEE	EMAIL	Email_Address	Personal white page information	Internal administrative purposes
SR_BANK_PAYROLL	EMPLOYEE	FIRSTNAME	Full_Name	Personal name and initials	Internal administrative purposes
SR_BANK_PAYROLL	EMPLOYEE	LASTNAME	Full_Name	Personal name and initials	Internal administrative purposes
SR_BANK_PAYROLL	EMPLOYEE	SSN	Work_SSN	Sensitive information	Internal administrative purposes
SR_BANK_PAYROLL	EMPLOYEE	STATE	Address_State	-	Internal administrative purposes
SR_BANK_PAYROLL	JOBTITLE	JOB_TITLE	Work_Designation	-	Internal administrative purposes

The following table describes the details about sensitive fields listed on the page:

Column Name	Description
Schema / Folder	Schema or folder name in the data store that includes the sensitive information about the data subject.
Object	Name of the table or object that contains the sensitive field or file.
Field / File	Field or file name that contains sensitive information about the data subject.
Data Domains	Name of the data domain that includes the sensitive field or file.
Category	Name of the personal data category associated with the sensitive field or file.
Purpose	The purpose for collecting the sensitive information about the data subject.

To download a CSV file that contains the information displayed on the page, select **Export** from the **Actions** menu. The default file name is `SubjectRegistryFieldFiles.csv`.

DSAR Report Formats

After you create a DSAR report on the **Subject Details** page and the report is ready to download, Data Privacy Management enables the **Download DSAR Report** icon in the Summary panel.

To verify that the report is ready to download, click the **Refresh** icon in the **Requests** pane on the **Subject Details** page. When the DSAR task to create the report is complete, the task appears in the list and the **Download DSAR Report** icon is enabled in the Summary panel.

Click the **Download DSAR Report** icon and select one of the following report formats:

- Detailed CSV
- Summary CSV
- Summary PDF

- Summary PDF Without Personal Data

  Note: The Summary PDF Without Personal Data report lists the types of information that your organization holds for an individual, but the report does not include the subject data that the Summary CSV and Summary PDF reports contain.

A new browser tab opens with the **Save As** dialog box that contains the report file in the format you selected. The report names use the following format:
<Subject_Name>_<Subject_Type>_<Format_Name>.<FileExtension>

For example, if you download a DSAR report for an Employee named John Porry and you select the Summary CSV report format, the report has the following file name: `John_Porry_Employees_Summary_CSV.csv`

Save the file to a directory on your machine.

The following table describes the data subject information included in the reports:

Subject Detail	Description
Subject Type	The name of the entity file that includes the data subject.
Subject Name	The data subject's name in the format configured for the IsSubjectName attribute in the Subject Registry configuration files. If the IsSubjectName attribute is not configured or the configuration files do not contain the FullName, FirstName, or LastName data domains, DSAR reports do not include the data subject's name.
Residency	The location where the data subject resides, in the format configured for the Residency attribute in the Subject Registry configuration files. If the Residency attribute is set for more than one data domain, the report lists the locations separated by a comma (,). If multiple golden data stores are configured in the Subject Registry entity file and the Residency attribute is empty in the primary golden data store, DSAR reports list the Residency attribute value in the non-primary golden data stores. However, if the data subject does not exist in the non-primary golden data stores, DSAR reports do not list a Residency attribute value.
Legal Hold	Indicates whether the data subject is under a legal hold. Values are `Yes` or `No`.
Data Category	Name of the data category associated with the sensitive field in the data domain in the same row. Category names are case sensitive. Multiple values are separated by a comma (,) in the DSAR reports. For unstructured data stores configured for Subject Registry, Data Privacy Management does not associate data domains with categories. PDF report formats for unstructured data stores list the value of the Data Category property as `Uncategorized`.
Data Domain	Name of the data domain that contains personal data about the individual.
Value	For each data domain, lists the sensitive data associated with the individual. If the value is empty for a data domain, the DSAR reports on the **Subject Details** page do not include the **Value** property. However, the DSAR report file that you access from the DSAR task **Properties** page includes the property with an empty value.
Purpose	The reason that you collected the sensitive data about the individual. If a data domain and associated category match exists in multiple data stores, the DSAR reports list the purposes separated by a comma (,). Unstructured data stores configured for Subject Registry cannot specify a purpose.

Subject Detail	Description
Shared	Indicates whether the data is shared with a third party. Values are `Yes` or `No`.
Shared With	If the data is shared, lists all third parties that have access to the personal data. If a data domain and associated category match exists in multiple data stores, the DSAR reports list the third parties separated by a comma (,).

The following table describes the additional data subject information in the Detailed CSV report format:

Subject Detail	Description
Data Source/File System	Name of the data store that contains subject data.
Schema/Folder	Schema or folder name in the data store that contains subject data.
Table/File	Table or file name in the schema or folder that contains subject data.
Column/File Type	Name of the column or file that contains subject data.
Data Owner	Name of the data store owner.
Time of Retrieval	The date and timestamp that you downloaded the DSAR report in the following format: *Day Mth dd hh:mm:ss* `IST` *yyyy*. For example: `Thu Feb 13 07:44:07 IST 2020`

You can customize the look and style of the PDF report format. Customize the template to change the image that appears in the report or to move or delete subject information fields that appear by default. You can delete or move fields around but cannot add fields that do not appear in the report by default.

Customize the DSAR Report PDF Format

Edit the default report template to customize the style and format of the PDF reports. Changes that you make apply to both PDF formats.

Use JasperStudio 6.10.0 to edit the template.

1. Browse to the following location to access the `DSAR_REPORT.jrxml` file: `<INFA_HOME>/SecureAtSourceService/config/samples`
2. Open the file and make the required changes.

   You can change the logo image, position, the background colors used, and delete or move subject details. You cannot add new subject detail fields.
3. Save the file with a different name.
4. Log in to Informatica Administrator and update the Data Privacy Management Service custom properties to include properties based on what you configure. For example, you might update the images but use the default template.

Add the following properties:

Property	Value
DSAR_HEADER_LOGO_PATH	Enter the complete path and file name of the custom image to use in the template header.
DSAR_POWERED_BY_LOGO_PATH	Add the complete path and file name of the custom image to use in the template footer.
DSAR_TEMPLATE_PATH	Add the complete path and file name of the template file that you create to customize the report format.

For information about how to add and modify custom properties, see the Data Privacy Management Service chapter in the Informatica Data Privacy Management Administrator Guide.

5. Restart the Data Privacy Management Service.

   Download a DSAR PDF report to view the updated template style.

Subject Details Page Management

From the **Actions** menu on the **Subject Details** page, you can export details about a data subject, prepare a DSAR report, and apply or remove a legal hold. You can also run on-demand tasks to create a ServiceNow ticket, run a custom script, or send an email.

To export details about a data subject, select **Export** from the **Actions** menu. A new browser tab opens. A **Save As** dialog box that contains a .zip file named `Data.zip` appears. Download the .zip file to a directory on your machine and extract the CSV files that contain information about the data subject.

The following table describes the CSV files that you export from the **Subject Details** page:

File Name	Description
SubjectRegistryDetailsSummary.csv	Contains three columns that correspond to the following summary metrics on the **Subject Details** page: - DataStoreCount - DataDomainCount - FieldCount In a single row, the file displays the numeric value for each column as shown in the corresponding summary metrics on the page.
SubjectRegistryEntityAttributes.csv	Lists the information about the data subject that you configured to display in columns on the **Subject Registry** workspace. For example, the file might contain columns that list the individual's name, company, city, age, Social Security number, and risk score. The legal hold status also appears as a column in the file.
SubjectRegistryEntityDetails.csv	Lists the following information about the data subject: - Names of the data stores that contain information about the individual - The data owner for each data store - The number and names of the data domains that contain subject information - The number of sensitive fields or files with information about the data subject - Country where the data subject is located - Data store purpose - Third-party share information

Applying and Removing Legal Holds

You can apply or remove legal holds for data subjects on the **Subject Details** page.

1. Search for a data subject on the **Subject Registry** workspace.

   A list of names that match the search criteria appears.

2. Click the name of the data subject for whom you want to apply or remove a legal hold.

   The **Subject Details** page appears.

3. To apply a legal hold to the subject, from the **Actions** menu select **Apply Legal Hold**.

   The option is visible if a legal hold is not applied. When you apply a legal hold, an icon appears next to the name on the **Subject Details** page and in the list of subjects on the **Subject Registry** workspace.

4. To remove a legal hold, from the **Actions** menu select **Clear Legal Hold**.

   The option is visible if the subject is under legal hold. When you remove a legal hold, the legal hold icon next to the data subject's name disappears.

Preparing a DSAR Report

You can prepare an automated DSAR report for a data subject on the **Subject Details** page.

Important: Domain discovery scans for subject registry must include schema and objects that are marked as sensitive in an SQL file. If the SQL file does not include the schema and objects that are marked as sensitive, the information in the DSAR reports might be incorrect.

1. Click the **Subject Registry** icon on the header.

   The **Subject Registry** workspace appears.
2. In the **Search** section, select an entity from the list, enter search criteria, and then click **Search**.

   A list of individuals matching the search criteria in the specified entity appears in the Records list.
3. Click the name of the subject you wanted to find.

   The **Subject Details** page appears.
4. From the **Actions** menu, select **Prepare DSAR**.

   Note: If the data subject is under legal hold, a message appears asking you to confirm that you want to create the report. Click **Yes**.

   A DSAR task appears on the **Tasks** workspace with a status of In Progress until the task completes.

Downloading a DSAR Report

After a DSAR task finishes, you can download the DSAR report for the data subject.

When the DSAR task finishes and the report is ready to view, Data Privacy Management sends an email to the address configured to receive DSAR task notifications. The email contains a link to the DSAR task **Properties** page on the **Tasks** workspace.

You can download the DSAR report in one of the following ways:

- Click the link in the notification email to download the report from the DSAR task **Properties** page on the **Tasks** workspace. The report has the following default file name: `DSAR_REPORT.csv`
- On the **Subject Details** page, click the **DSAR Report Download** icon to select a CSV or PDF file format and download the report. The icon always downloads the most recent DSAR report created for the data subject.

If the DSAR task failed for any data stores, Data Privacy Management sends a failure notification email to the configured email address. The email contains details only about the first data store that failed in the DSAR task.

Perform the following steps to view and fix the errors:

1. On the **Tasks** workspace, go to the DSAR task **Properties** page.

   The page contains a CSV file that describes the reason that the task failed for the data stores. The error report has the following default file name: `DSAR_ERROR.csv`
2. Download the error report to a directory on your machine.
3. Open the error report to view the list of errors that the DSAR task encountered.
4. Resolve the errors for the data stores that failed in the DSAR task.
5. On the DSAR **Task Details** page, retry the task for failed or selected data stores.
6. If the task finishes without errors, you can download the new DSAR report.

Creating a Service Management Ticket

On the **Subject Details** page, you can raise a service management ticket in ServiceNow to fulfill a data subject inquiry.

To create a ServiceNow ticket for a data subject, at least one Active extension that uses the DSR ServiceNow Plugin must be configured on the **Extensions** workspace.

1. Click the **Subject Registry** icon on the header.

   The **Subject Registry** workspace appears.
2. In the **Search** section, select a group, enter search criteria, and then click **Search**.

   A list of individuals matching the search criteria in the specified group appears in the Records list.
3. Click the name of the subject you wanted to find.

   The **Subject Details** page appears.
4. Click the **Data Stores** tab.
5. Select the check boxes next to one or more data stores that contain personal data about the individual.
6. From the **Actions** menu, select **Take Action** > **Create a Service Management Ticket**.

   Note: If the data subject is under legal hold, a message appears asking you to confirm that you want to create the ticket. Click **Yes**.

   The **Create a Service Management Ticket** page appears.
7. Select an existing service management action or create a new action that uses a configured and Active DSR ServiceNow Plugin.

   If you select a new action, you first choose the service management extension for the action and then configure the properties. If you select an existing action, the action properties are configured.
8. Optionally, add a description that will display under the system-generated task number on the **Tasks** workspace.
9. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
10. Configure or edit the remaining properties.
11. To run the task immediately, click **Save and Run**. To save the task on the **Tasks** workspace to run later, click **Actions** > **Save as Task**.

Data Privacy Management creates a ServiceNow task that appears on the **Tasks** workspace with the prefix `DSR-`. Saved tasks have a status of New. Tasks you run immediately display the success or failure of the job in the **Status** column after they run.

Related Topics:

- "Service Management Action Properties" on page 380

Running a Custom Script

On the **Subject Details** page, you can create a task that runs a custom script to fulfill a data subject request.

To run a custom script for a data subject, at least one Active extension that uses the DSR Custom Plugin must be configured on the **Extensions** workspace.

1. Click the **Subject Registry** icon on the header.

   The **Subject Registry** workspace appears.
2. In the **Search** section, select a group, enter search criteria, and then click **Search**.

   A list of individuals matching the search criteria in the specified group appears in the Records list.
3. Click the name of the subject you wanted to find.

   The **Subject Details** page appears.
4. Click the **Data Stores** tab.
5. Select the check boxes next to one or more data stores that contain personal data about the individual.
6. From the **Actions** menu, select **Take Action** > **Run a Custom Script**.

   Note: If the data subject is under legal hold, a message appears asking you to confirm that you want to run the script. Click **Yes**.

   The **Run a Custom Script** page appears.
7. Select an existing custom action or create a new action that uses a configured and Active DSR Custom Plugin.

   If you select a new action, you first choose the custom extension for the action and then configure the properties. If you select an existing action, the action properties are configured.
8. Optionally, add a description that will display under the system-generated task number on the **Tasks** workspace.
9. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
10. Configure or edit the remaining properties.
11. To run the task immediately, click **Save and Run**. To save the task on the **Tasks** workspace to run later, click **Actions** > **Save as Task**.

Data Privacy Management creates a custom task that appears on the **Tasks** workspace with the prefix `DSR-`. Saved tasks have a status of New. Tasks you run immediately display the success or failure of the job in the **Status** column after they run.

Related Topics:

- "Custom Action Properties" on page 377

Sending an Email

On the **Subject Details** page, you can create a task that sends an email to fulfill a data subject request.

To send an email to fulfill a data subject request, at least one Active extension that uses the DSR Email Plugin must be configured on the **Extensions** workspace.

1. Click the **Subject Registry** icon on the header.

   The **Subject Registry** workspace appears.
2. In the **Search** section, select a group, enter search criteria, and then click **Search**.

   A list of individuals matching the search criteria in the specified group appears in the Records list.
3. Click the name of the subject you wanted to find.

   The **Subject Details** page appears.
4. Click the **Data Stores** tab.
5. Select the check boxes next to one or more data stores that contain personal data about the individual.
6. From the **Actions** menu, select **Take Action** > **Send an Email**.

   Note: If the data subject is under legal hold, a message appears asking you to confirm that you want to send the email. Click **Yes**.

   The **Send an Email** page appears.
7. Select an existing custom action or create a new action that uses a configured and Active DSR Email Plugin.

   If you select a new action, you first choose the email extension for the action and then configure the properties. If you select an existing action, the action properties are configured.
8. Optionally, add a description that will display under the system-generated task number on the **Tasks** workspace.
9. Select or change one of the following task assignments:
   - **Assign to Self**. Assign the task to yourself.
   - **System Assigned**. Data Privacy Management assigns the task to the data store owner. If a data store owner is not defined, Data Privacy Management assigns the task to you.
   - **Selected**. Select a user from the list.
10. Configure or edit the remaining properties.
11. To run the task immediately, click **Save and Run**. To save the task on the **Tasks** workspace to run later, click **Actions** > **Save as Task**.

Data Privacy Management creates an email task that appears on the **Tasks** workspace with the prefix `DSR-`. Saved tasks have a status of New. Tasks you run immediately display the success or failure of the job in the **Status** column after they run.

Related Topics:

- "Email Action Properties" on page 378

CHAPTER 30

Privacy Dashboard

This chapter includes the following topics:

- Privacy Dashboard Overview, 629
- Enterprise-Level Summary of Subject Data, 630
- Privacy Dashboard Indicators, 631
- Privacy Dashboard Management, 637

Privacy Dashboard Overview

You can view and access information about subject data in the Subject Registry. The summary data includes indicators for subjects, subject requests, subject types, and data stores in the Subject Registry.

The **Privacy Dashboard** is the default dashboard that opens when you login to Data Privacy Management. The dashboard opens on the **Overview** workspace. The **Overview** workspace displays an enterprise-level summary of subject data and workspace indicators for key elements of subject data in the Subject Registry.

Workspace indicators display high-level information on key elements of subject data. Some dashboard indicators contain links for accessing and managing the metric details from other pages. For example, you can click the name of a subject request in the **Subject Requests by Due Date** indicator to view the task details page.

You can use filters to filter and view relevant information.

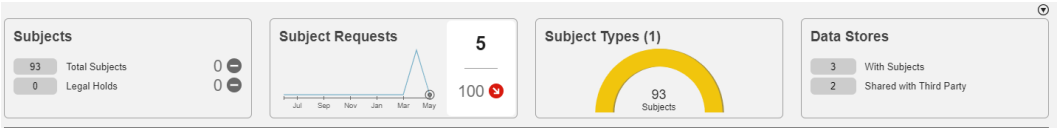
Each indicator drills down to open a page with more detailed information. You can sort, filter and view information, and perform tasks from these pages.

From the **Actions** menu, you can export workspace details as a .zip file that contains a CSV file for each indicator. You can also switch to the Security Dashboard, refresh the view, and customize the dashboard display.

Enterprise-Level Summary of Subject Data

Review the enterprise-level summary to quickly view information on subject data in the Subject Registry.

The following image shows the summary indicators on the **Overview** workspace of the Privacy Dashboard:



You can view the total number of subjects and subject types in the Subject Registry. You can view information on subject requests raised in a 12-month period from the current month. You can also view the distribution of subjects across data stores and high-level information on how much data is shared with a third party.

Rest on an indicator to view more details on each metric.

Some indicators update when you run a scan on golden data stores. To view updated information, from the **Actions** menu click **Refresh**.

Subjects Indicator

The Subjects indicator gives you a quick insight into the total number of subjects in the Subject Registry, and into the trend in subject count and application of legal holds on subjects. It displays the total number of subjects across all golden data stores in the Subject Registry. It also indicates the number of subjects that have legal holds applied.

The data updates when you run a subject registry scan that includes the **Discover Subjects** option.

You can view the following data on the **Subjects** indicator:

- The figure to the right of the total number of subjects indicates the change in the number of subjects from the previous scan. The subjects trend appears as zero if there were no subjects identified in the previous scan.
- The figure to the right of the legal hold information indicates the change in the number of subjects with legal holds applied from the previous scan. The legal holds trend appears as zero if there were no legal holds identified in the previous scan.

Both values are in percent. If you apply a filter, trend values do not appear on the indicator.

To view data in the **Subjects** indicator, you must have run a subject registry scan with the **Discover Subjects** option.

Note: Although you can update legal hold information on the **Subject Details** page, the information in the summary indicator updates only when you run a scan on golden data stores.

Subject Requests Indicator

The **Subject Requests** indicator displays the number of subject requests raised each month over a 12-month period.

The total count appears on the right with the trend in comparison with the previous month. The trend value is in percent.

Rest the mouse over a month in the graph to view the number of requests raised in the month. The count includes DSAR and DSR request types in any status including deleted requests.

Subject Types Indicator

The **Subject Types** indicator displays information on subject types or entities in the Subject Registry.

Each shade on the graph represents a subject type. Rest the mouse over an area on the graph to view the name of the subject type and the number of subjects it contains. The total number of subjects also appears in the indicator.

The subject type information updates when you add or delete an entity file in the Subject Registry. The total number of subjects updates when you run a scan on golden data stores.

Data Stores Indicator

You can use the data in the **Data Stores** indicator to understand how subject data is distributed across data stores and what data is shared outside the organization. The **Data Stores** indicator displays the number of data stores in the Subject Registry that contain subjects. The number includes golden and transaction data stores.

To view data in the Data Stores indicator, you must have run a subject registry scan with the Discover Subjects option. The indicator updates each time you run a subject registry scan with the **Discover Subjects** or **Link Subjects** option.

The indicator also displays how many of these data stores are marked as shared with a third party. You can indicate if a data store contains data that is shared when you create the data store.

Privacy Dashboard Indicators

Use the Overview workspace indicators on the Privacy Dashboard to quickly view and understand the distribution of subject data and the status of subject requests for a Subject Registry.

Overview workspace indicators appear below the summary indicators. You can use the indicators to drill down and access detailed information about subject data. You can click entries in some indicators to open and manage the data in related pages.

The Overview workspace includes the following default indicators:

- Top Data Stores by Subjects
- Subject Requests by Type
- Subject Data by Location
- Subject Requests by Due Date

Top Data Stores by Subjects

The **Top Data Stores by Subjects** indicator lists up to 25 data stores on which you have run a Subject Registry scan.

The indicator lists the data stores with the total number of subjects, data domains, and sensitive fields in each data store.

The list is sorted in descending order of the subject count. The number of data domains and sensitive fields appears as zero if you have not run a domain discovery scan. You can use filters to view specific information or refresh the page to view updated information after you run a scan.

The indicator includes subjects from golden and transaction data stores based on the scans run. Because the **Subjects** and **Subject Types** summary indicators include only golden data stores, you might notice a difference in the subject count.

You can click entries in the indicator to view the following pages:

- Data Store Name. Opens the **Proliferation** page.
- Fields. Opens the **Sensitive Fields** page for the data store.

To view more information about the data stores with subjects, click the indicator label to open the **Data Stores by Subjects** page.

Related Topics:

- "Proliferation Page" on page 542
- "Sensitive Fields Page" on page 549

Data Stores by Subjects Page

The **Data Stores by Subjects** page contains a list of all data stores in the Subject Registry with additional information including the type, region, third-party share information, and targets.

Summary information on the total number of data stores and the number of data stores shared with a third party appears on the upper right corner of the page.

You can perform the following actions on the **Data Stores by Subjects** page:

Access related pages

You can click entries in specific columns to view the following pages:

- Data Store Name or Target. Opens the **Proliferation** page.
- Sensitive Fields. Opens the **Sensitive Fields** page.

Sort the data

You can click the column header to sort the data on any column except the Data Store Type column.

Apply filters

The **Filter** icon opens the **Filter** pane that includes page-specific filters in addition to the global filters.

The **Data Stores by Subjects** page includes the following page-specific filters:

Filter Condition	Description
Data Store Type	Select one or more data store types to view the data stores based on type.
Third-Party Shared	Choose a value to filter data based on whether the data stores are shared with a third party or not.

Export information

Click **Actions** > **Export** to export data store information. The **Export** option downloads a `Data.zip` file that includes the following `.csv` files:

- `DataStoresBySubjectsSummary.csv`. Includes summary information of the total number of data stores and the number of shared data stores.
- `DataStoresBySubjectsDetails.csv`. Includes the information that appears on the page for each data store.

If you apply a filter, select data stores on the page, or sort the data, only the filtered or selected data is exported and in the order of sorting.

Perform manual actions from the Take Action menu

Click **Actions** > **Take Action** to view manual action options. You can choose from the following options:

- Create a Service Management Ticket
- Protect Data
- Run a Custom Script
- Send an Email
- Write a System Log Message

Subject Requests by Type

The **Subject Requests by Type** indicator shows the types of subject requests raised regardless of the current status of the request.

You can choose to view requests by week, month, or year. The X axis tracks the time period, and the Y axis tracks the number of requests. Different shades in the bar graph indicate different subject request types. Move the pointer over each bar to view different request types with the count in each time frame.

For example, the **Week** tab tracks the last seven days on the X axis and the number of requests on the Y axis.

The **Month** tab tracks the month by the current week and the last four weeks. A week is calculated from Sunday through Saturday.

The **Year** tab tracks data over a 12-month period that includes the current month.

You can use filters to view specific information on the indicator.

Click the indicator label to open a page of detailed subject request information sorted by type.

Subject Requests by Type Details Page

The **Subject Request by Type** details page contains a list of all subject requests sorted by the request type.

An indicator of the total number of requests appears on the left upper corner. By default, the list also includes closed and completed requests. You can choose to exclude closed and completed requests in the list.

The page includes the request name, subject name, subject type, request type, and information on when and who last updated the request. The status and due date information also appears.

The page includes an **All Requests** tab that lists requests created and assigned to any user. The **My Requests** tab lists requests created by you and requests assigned to you.

You can perform the following actions on the page:

Filter based on date of request creation

Choose from options in the **Display** list to filter on specified time periods, or enter a From and To date. The dates correspond to the date of creation of the request.

Sort the data

You can click the column header to sort the data on any column.

Apply filters

The **Filter** icon opens the **Filter** pane that includes page-specific filters in addition to the global filters.

The **Subject Requests by Type Details** page includes the following page-specific filters:

Filter Condition	Description
Request Name	Enter the request name. You can enter a partial or complete name. The page updates with data that matches the name or name fragment.
Subject Name	Choose a value to filter data based on whether the data stores are shared with a third party or not.
Request Type	Select one or more request types to view specific request types.
Status	Choose one or more status values to filter data on. If you include closed or completed tasks in the filter, the **Display Closed and Completed Requests** option on the page is ignored.
Due Date	Select a date range to view tasks with a due date within the specified date range.

Export information

Click **Actions** > **Export** to export the information in a `.csv` file. The **Export** option downloads a `SubjectRequests.csv` file. If you apply a filter, select data stores on the page, or sort the data, only the filtered or selected data is exported and in the order of sorting.

Subject Data by Location

You can use the **Subject Data by Location** indicator to quickly understand how subject data in data stores is distributed across different geographical locations.

When you create a data store on the **Data Stores** workspace, you can associate the data store with a geographic region. The **Subject Data by Location** indicator shows data stores that contain subject data by region. The indicator includes tabs for each region. The **?** tab includes data stores that you do not assign to a region.

Use the region tabs to view information on each region. Click the title to expand the indicator. You can click a region on the map to expand it and drill down further to view the location and data store and subject information.

The indicator shows the number of data stores in each region and the number of subjects in the data store. By default, the indicator displays the region with the most number of data stores with subjects.

You can use global filters to filter the data.

Data stores that are not part of the Subject Registry or that do not contain any subjects do not appear in the indicator.

Click the indicator label to expand the view.

You can perform the following actions on the expanded **Subject Data Locations** page:

- To refresh the view, click **Actions** > **Refresh**.
- To export the data that appears on the page, click **Actions** > **Export**. The Export option downloads a `Data.zip` file that includes the following `.csv` files:
  - `SubjectDataLocationSummary.csv`. Contains summary count information for the region selected. Includes the location count, number of data stores, number of subjects, and the number of protected fields and sensitive fields for the region.

- `SubjectDataLocationDetails.csv`. Contains detailed information on the data selected. For example, if you select a region on the indicator, the exported file includes information on each country within the region that has data stores with subjects. The information includes the number of data stores, number of subjects, and the number of protected fields and sensitive fields.

The `Data.zip` file does not include the `SubjectDataLocationDetails.csv` file if there is no detailed information available. For example, if you choose a region that does not include any locations with subject data.

If you apply a filter or select locations on the page, only the filtered or selected data is exported.

- Click the data store count on the lower right corner of the map to open the **Subject Data by Location** details page.

Subject Data by Locations Page

The **Subject Data by Locations** page contains a list of all data stores with subjects, and additional information including the type, region, third-party share information, and targets.

An indicator of the total number of data stores appears on the left upper corner.

Summary information on the total number of data stores and the number of data stores shared with a third party appears on the upper right corner of the page.

You can perform the following actions on the **Subject Data by Locations** page:

Access related pages

You can click entries in specific columns to view the following pages:

- Data Store Name or Target. Opens the **Proliferation** page.
- Sensitive Fields. Opens the **Sensitive Fields** page.

Sort the data

You can click the column header to sort the data on any column except the Data Store Type column.

Apply filters

The **Filter** icon opens the **Filter** pane that includes page-specific filters in addition to the global filters.

The **Subject Data by Locations** page includes the following page-specific filters:

Filter Condition	Description
Data Store Type	Select one or more data store types to view the data stores based on type.
Third-Party Shared	Choose a value to filter data based on whether the data stores are shared with a third party or not.

Export information

Click **Actions** > **Export** to export data store information. The Export option downloads a `Data.zip` file that includes the following `.csv` files:

- `DataStoresBySubjectsSummary.csv`. Includes summary information of the total number of data stores and the number of shared data stores.
- `DataStoresBySubjectsDetails.csv`. Includes the information that appears on the page for each data store.

If you apply a filter, select data stores on the page, or sort the data, only the filtered or selected data is exported and in the order of sorting.

Perform manual actions from the Take Action menu

Click **Actions** > **Take Action** to view manual action options. You can choose from the following options:

- Create a Service Management Ticket
- Protect Data
- Run a Custom Script
- Send an Email
- Write a System Log Message

Subject Requests by Due Date

You can view a list of up to 10 subject requests sorted by the earliest due date by using the **Subject Requests by Due Date** indicator.

The indicator includes the name of the task, type of task, status, due date, and the last updated date. Tasks that are in closed or completed status do not appear in the indicator.

Tasks include DSAR and DSR tasks. The **Type** field indicates the specific action that you choose in the task. For a DSAR task, the **Due Date** column indicates the date of creation of the task. For DSR tasks, the column indicates the due date.

You can perform the following actions on the **Subject Requests by Due Date** indicator:

Open the Task Details page

You can click a task name to open the specific task details page. The task page that opens depends on the type of task. You can review the details on the **Task Details** page and take further action as required.

Apply filters

You can apply global filters to view specific information.

View a detailed list of subject requests

Click the indicator label to open detailed information on subject requests.

Related Topics:

- "DSAR Task Details Page" on page 489
- "Custom Task Details Page" on page 489
- "Service Management Task Details Page" on page 493
- "Email Task Details Page" on page 490

Subject Requests by Due Date Details Page

The Subject Request by Due Date details page contains a list of all subject requests sorted by due date.

An indicator of the total number of requests appears on the left upper corner. By default, the list does not include closed and completed requests. Only requests that are open, in progress, failed, or completed with errors appear in the list. You can choose to include closed and completed requests in the list.

In addition to information that appears on the indicator on the Privacy Dashboard, the page includes the subject name, subject type, and information on when and who last updated the request.

The page includes an **All Requests** tab that lists requests created and assigned to any user. The **My Requests** tab lists requests created by you and requests assigned to you.

You can perform the following actions on the page:

Sort the data

You can click the column header to sort the data on any column.

Apply filters

The **Filter** icon opens the **Filter** pane that includes page-specific filters in addition to the global filters.

The **Subject Requests by Due Date Details** page includes the following page-specific filters:

Filter Condition	Description
Request Name	Enter the request name. You can enter a partial or complete name. The page updates with data that matches the name or name fragment.
Subject Name	Choose a value to filter data based on whether the data stores are shared with a third party or not.
Request Type	Select one or more request types to view requests based on the request type.
Status	Choose one or more status values to filter data on. If you include closed or completed tasks in the filter, the **Display Closed and Completed Requests** option on the page is ignored.
Due Date	Select a date range to view tasks with a due date within the specified date range.

Export information

Click **Actions** > **Export** to export the information in a `.csv` file. The Export option downloads a `SubjectRequests.csv` file. If you apply a filter, select data stores on the page, or sort the data, only the filtered or selected data is exported and in the order of sorting.

Privacy Dashboard Management

You can use the options in the **Actions** menu to manage the Privacy Dashboard.

You can perform the following tasks from the **Actions** menu:

- To customize the workspace indicators that display on the dashboard, click the **Customize Dashboard** option.
  - Click **On** or **Off** to display or hide an indicator and then click **Save** to save the changes.
  - To change the order of display of the indicators, rest the cursor over the indicator that you want to move. You can also click and hold on the indicator. The **Move** icon appears. Drag the indicator to the required location on the page. Adjacent indicators move to adjust to the changed location.
- To export information for each indicator and summary indicator on the workspace as a CSV file, click **Export**. Extract the ZIP file to view the CSV files.
- To refresh the dashboard data, **Refresh**.
- To switch to the Security dashboard, click **Switch to Security Dashboard**.

Filtering Information on the Privacy Dashboard

To set filter conditions that apply to the Privacy Dashboard and the drill-down pages, click the **Filter** icon in the Data Privacy Management header and select global filter conditions from the **Filter** pane.

On the **Filter** panes for the drill-down pages, a dashed line separates the global conditions from the page-specific filter conditions. The page-specific filter conditions appear above the dashed line.

When you apply filters, the subject data in all indicators updates based on the criteria.

The following table describes the global filter conditions:

Filter Condition	Description
Data Store Name	Enter the data store name. You can enter a partial or complete name. The dashboard data updates with data that matches the name or name fragment.
Classification Policy	Select one or more classification policies to view subject data related to classification policies.
Subject Type	Select one or more subject types.
Region	Filters based on one or more geographic regions. When you select a country or data store location without specifying a region, a dash appears in the region check box to indicate that the region contains one or more countries or data store locations that you did not select.
Country	When you select a region first, Data Privacy Management automatically selects the countries in the region. You can clear any of the selections. You can also include other countries outside the region.
Data Store Location	When you select a region first, Data Privacy Management automatically selects the data store locations in with the region.

Export Files From the Privacy Dashboard

To export information on the Privacy Dashboard **Overview** workspace, select **Export** from the **Actions** menu and download the `DPM-PrivacyDashBoardExport<timestamp>.zip` file to a local directory.

The .zip file contains CSV files. Each CSV file contains the information in a workspace indicator. If you customize the workspace before you export, the information in the CSV files matches the active indicators.

The following table lists the CSV files in the `DPM-PrivacyDashBoardExport<timestamp>.zip` file that you export from the **Overview** workspace:

File Name	Description
`DataStores.csv`	Contains the number of data stores that include subject data and the number of these data stores that are shared with a third party. Includes both golden and transaction data stores.
`SubjectDataByLocation.csv`	Contains the number of data stores for each region with the number of subjects in each data store.
`SubjectRequests.csv`	Contains the number of subject requests raised each month for the last year. Also includes the change from the previous month to the current month. The change is indicated in percent.

File Name	Description
SubjectRequestsByDueDate.csv	Contains a list subject requests. Also includes the type of request, status, due date and the date it was last updated.
SubjectRequestsByType_<Week/Month/Year>.csv	Contains a list of all subject request types raised during the week, month, or year. Also includes the number of each request type raised.
Subjects.csv	Contains the total number of subjects in the Subject Registry and the number of subjects under legal hold. Also includes information on the trends in change of subject count and legal holds.
SubjectTypes.csv	Contains a list of subject types in the Subject Registry and the number of subjects in each subject type.
TopDataStoresBySubjects.csv	Contains a list of up to 25 data stores ordered from high to low subject count. Also includes the number of data domains and files/fields.

APPENDIX A

Troubleshooting

This appendix includes the following topics:

- Data Stores, 640
- Jobs, 641
- Protection, 645
- Scans, 646
- Subject Registry, 648

Data Stores

If you encounter a problem when creating, editing, importing, or synching a data store, review the following list of previously reported issues and their solutions before contacting Informatica Global Customer Support.

When I import a data store, the import job issues a warning and the data store is not created or updated.

This issue occurs either if the Enterprise Unified Metadata server is down or the data store already exists in Enterprise Unified Metadata.

Verify that the Enterprise Unified Metadata server is up. Verify that the data store details in the import file are correct. Import the file again.

When I unmerge data stores, I get an error.

This issue occurs when you try to unmerge immediately after you merge the data stores.

Review the list of data stores in Enterprise Unified Metadata to see if the secondary data store is on the list. If the secondary data store is still on the list, then wait for some time and unmerge again.

When I create an Amazon Redshift data store, the test connection fails.

This issue occurs when the Amazon Redshift jar file is not available in the `$INFA_HOME/java/jre/lib/ext` folder.

Copy the Amazon Redshift jar file to the `$INFA_HOME/java/jre/lib/ext` folder. Then, restart the domain.

When I test the connection to an SAP data store, I get the error `Cannot load metadata adapter`.

Verify that you installed SAP JCo correctly. For information on how to install SAP JCo, see the *Informatica Data Privacy Management Administrator Guide*.

When I test the connection to an SAP data store, I get the error `Connect to SAP gateway failed`.

To resolve this error, verify that you entered correct values for the SAP connection properties, especially the **System Number** property.

Jobs

If you encounter a problem with jobs in Data Privacy Management, consult the job logs before contacting Informatica Global Customer Support. A job log contains informational, warning, and error messages that a job encountered.

Each job step has a separate log. The logs include information about the tasks performed in each job step and messages that determine the cause of errors. You can view and download the logs for each job step from the **Job Details** page on the **Jobs** workspace.

The following list describes previously reported job errors and solutions:

The profiling job step in a Database Scan job fails due to an error in the underlying transport layer.

The profiling job step of a database scan job fails when the Informatica domain and services use the Secure Sockets Layer (SSL) protocol for secure communication.

The following excerpt shows some of the errors in the Data Privacy Management log files:

```
2015-12-09 15:57:35,444 INFO  [DpmProfilerImpl]
[JOB_TYPE=SCAN:JOB_ID=2:JOB_STEP_ID=8:REPO_ID=1]--[Administrator] Data Store
Name=Oracle1_12c: Starting the data profile for the data store.
2015-12-09 15:57:35,771 INFO  [EdrConnectionManager]
[JOB_TYPE=SCAN:JOB_ID=2:JOB_STEP_ID=8:REPO_ID=1]--[Administrator] Data Store
Name=DS_1_DPM: Creating an EDR connection named DS_1_DPM.
2015-12-09 15:57:35,834 INFO  [JobProcessor] [Administrator] Get job step details
successful.
2015-12-09 15:57:36,127 INFO  [JobProcessor] [Administrator] Getting job step logs for
the job step id {0} successful.
2015-12-09 15:57:37,420 INFO  [JobProcessor] [Administrator] Get job step details
successful.
2015-12-09 15:57:37,880 INFO  [JobProcessor] [Administrator] Getting job step logs for
the job step id {0} successful.
2015-12-09 15:57:43,735 INFO  [JobProcessor] [Administrator] Get job step details
successful.
2015-12-09 15:57:43,803 INFO  [JobProcessor] [Administrator] Getting job step logs for
the job step id {0} successful.
2015-12-09 15:57:45,460 ERROR [OSGIClientFactoryImpl] [DTF_0001] An error occurred in
the underlying transport layer: [[sendMSG]: Channel is in [7 : CLOSED] state.]
com.informatica.pcsf.datatransport.DataTransportException: [DTF_0001] An error occurred
in the underlying transport layer: [[sendMSG]: Channel is in [7 : CLOSED] state.]
at
com.informatica.pcsf.datatransport.impl.DataTransportChannelImpl.sendRequest(DataTranspor
tChannelImpl.java:75)
at
com.informatica.pcsf.servicesframework.client.impl.ClientFactoryImpl.sendHeartbeatToRecip
ient(ClientFactoryImpl.java:979)
at com.informatica.pcsf.servicesframework.client.impl.ClientFactoryImpl.access
$3(ClientFactoryImpl.java:957)
at com.informatica.pcsf.servicesframework.client.impl.ClientFactoryImpl
$HeartbeatSender.run(ClientFactoryImpl.java:939)
at java.lang.Thread.run(Thread.java:745)
Caused by: org.beepcore.beep.core.BEEPException: [sendMSG]: Channel is in [7 : CLOSED]
state.
at org.beepcore.beep.core.ChannelImpl.sendMsgInternal(ChannelImpl.java:399)
```

```
at org.beepcore.beep.core.ChannelImpl.sendMSG(ChannelImpl.java:447)
at
com.informatica.pcsf.datatransport.impl.DataTransportChannelImpl.sendRequest(DataTranspor
tChannelImpl.java:73)
```

The following excerpt shows some of the errors in the corresponding Data Integration Service log files:

```
2015-12-09 11:39:20.607 <ProfilingScheduler> INFO: [IDP_1102] Unable to query database
for ResultPurgeRequest status: [IDP_0010] Unable to fetch profile results
2015-12-09 11:43:02.787 <Thread-31> SEVERE: Throwable on DTMProcess:
com.informatica.dtm.transport.DTFUncheckedException: [JDTM_10005] Internal error. The
Integration Service could not create the DTM process. Contact Informatica Global
Customer Support.
at com.informatica.jdtm.oop.DTMProcess.spawnProcess(DTMProcess.java:868)
at com.informatica.jdtm.oop.DTMProcess.init(DTMProcess.java:597)
at com.informatica.jdtm.oop.DTMProcess.access$7(DTMProcess.java:584)
at com.informatica.jdtm.oop.DTMProcess$DTMProcessInitializer.run(DTMProcess.java:1127)
at java.lang.Thread.run(Thread.java:745)
Caused by: java.io.IOException: Cannot run program "pmdtmsvc2": error=13, Permission
denied
at java.lang.ProcessBuilder.start(ProcessBuilder.java:1047)
at com.informatica.jdtm.oop.DTMProcess.spawnProcess(DTMProcess.java:865)
... 4 more
Caused by: java.io.IOException: error=13, Permission denied
at java.lang.UNIXProcess.forkAndExec(Native Method)
at java.lang.UNIXProcess.<init>(UNIXProcess.java:187)
at java.lang.ProcessImpl.start(ProcessImpl.java:130)
at java.lang.ProcessBuilder.start(ProcessBuilder.java:1028)
... 5 more
2015-12-09 11:43:02.789 <Thread-31> SEVERE: Internal error. The DTM process encountered
an exception because of the following error:
[com.informatica.dtm.transport.DTFUncheckedException: [JDTM_10005] Internal error. The
Integration Service could not create the DTM process. Contact Informatica Global
Customer Support.]. Contact Informatica Global Customer Support.
2015-12-09 11:43:02.821 <MappingServiceFutureNotifyThreadPool_Thread_0> WARNING:
[JDTM_10014] Internal error. The DTM process encountered an exception because of the
following error: [com.informatica.dtm.transport.DTFUncheckedException: [JDTM_10005]
Internal error. The Integration Service could not create the DTM process. Contact
Informatica Global Customer Support.]. Contact Informatica Global Customer Support.
2015-12-09 11:43:02.828 <MappingServiceFutureNotifyThreadPool_Thread_0> WARNING:
[CORE_0003] An internal exception occurred with message:
java.util.concurrent.ExecutionException:
com.informatica.jdtm.oop.OOPPPUncheckedException: [JDTM_10014] Internal error. The DTM
process encountered an exception because of the following error:
[com.informatica.dtm.transport.DTFUncheckedException: [JDTM_10005] Internal error. The
Integration Service could not create the DTM process. Contact Informatica Global
Customer Support.]. Contact Informatica Global Customer Support.
java.util.concurrent.ExecutionException:
com.informatica.jdtm.oop.OOPPPUncheckedException: [JDTM_10014] Internal error. The DTM
process encountered an exception because of the following error:
[com.informatica.dtm.transport.DTFUncheckedException: [JDTM_10005] Internal error. The
Integration Service could not create the DTM process. Contact Informatica Global
Customer Support.]. Contact Informatica Global Customer Support.
at java.util.concurrent.FutureTask.report(FutureTask.java:122)
at java.util.concurrent.FutureTask.get(FutureTask.java:188)
at com.informatica.platform.ldtm.executor.ExecutionEngine
$LdtmFuture.get(ExecutionEngine.java:909)
at
com.informatica.ds.server.impl.TransformationMachineDISFuture.get(TransformationMachineDI
SFuture.java:103)
at
com.informatica.ds.ms.service.MappingExecutionCompletedState.notifyStatus(MappingExecutio
nCompletedState.java:94)
at
com.informatica.ds.ms.service.MappingExecutionCompletedState.onEvent(MappingExecutionComp
letedState.java:63)
at
com.informatica.ds.ms.service.StateMachine.changeStateAndProcessEvent(StateMachine.java:6
5)
at com.informatica.ds.ms.service.State.changeStateAndProcessEvent(State.java:57)
at
```

```
com.informatica.ds.ms.service.MappingExecutionWaitForCompletionState.onEvent(MappingExecu
tionWaitForCompletionState.java:50)
at com.informatica.ds.ms.service.StateMachine.dispatch(StateMachine.java:32)
at com.informatica.ds.ms.service.MappingServiceImpl
$FutureNotificationTask.run(MappingServiceImpl.java:1416)
at java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:471)
at java.util.concurrent.FutureTask.run(FutureTask.java:262)
at java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1145)
at java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:615)
at java.lang.Thread.run(Thread.java:745)
Caused by: com.informatica.jdtm.oop.OOPPPUncheckedException: [JDTM_10014] Internal
error. The DTM process encountered an exception because of the following error:
[com.informatica.dtm.transport.DTFUncheckedException: [JDTM_10005] Internal error. The
Integration Service could not create the DTM process. Contact Informatica Global
Customer Support.]. Contact Informatica Global Customer Support.
at com.informatica.jdtm.oop.DTMProcess.throwSevereException(DTMProcess.java:1016)
at com.informatica.jdtm.oop.DTMProcess.joinOnCreationTask(DTMProcess.java:512)
at
com.informatica.jdtm.oop.MappingDispatcher.findOrCreateProcess(MappingDispatcher.java:144
)
at com.informatica.jdtm.oop.MappingDispatcher.createDTM(MappingDispatcher.java:201)
at com.informatica.powercenter.sdk.dtm.OOPPPDTM.<init>(OOPPPDTM.java:45)
at
com.informatica.platform.ldtm.executor.edtm.ExecutorRuntime.init(ExecutorRuntime.java:168
)
at
com.informatica.platform.ldtm.executor.edtm.EdtmExecutor.initializeRuntime(EdtmExecutor.j
ava:517)
at
com.informatica.platform.ldtm.executor.edtm.EdtmExecutor.createAndExeJdtm(EdtmExecutor.ja
va:408)
at com.informatica.platform.ldtm.executor.edtm.EdtmExecutor.run(EdtmExecutor.java:267)
at com.informatica.platform.ldtm.executor.ExecutionEngine
$SubmittedRunnable.run(ExecutionEngine.java:830)
... 5 more
2015-12-09 11:43:02.830 <MappingServiceFutureNotifyThreadPool_Thread_0> WARNING:
[MPSVCCMN_10093] The Mapping Service Module failed to run the job with ID
[5znE9J47EeWi5PHEi22VVw].
2015-12-09 11:57:19.252 <pool-6-thread-5> WARNING: [Worchcc_10001] The workflow database
connection is not configured.
2015-12-09 11:57:19.452 <pool-6-thread-4> WARNING: [Worchcc_10001] The workflow database
connection is not configured.
2015-12-09 11:57:47.569 <pool-6-thread-2> WARNING: [Worchcc_10001] The workflow database
connection is not configured.
2015-12-09 11:57:47.613 <pool-6-thread-5> WARNING: [Worchcc_10001] The workflow database
connection is not configured.
2015-12-09 11:58:38.700 <pool-6-thread-4> WARNING: [Worchcc_10001] The workflow database
connection is not configured.
2015-12-09 11:58:38.713 <pool-6-thread-2> WARNING: [Worchcc_10001] The workflow database
connection is not configured.
2015-12-09 11:58:56.718 <Update and Monitor Service Process Controller Thread> INFO:
[DS_10067] Deinitializing the Integration Service web application.
2015-12-09 11:58:56.723 <Update and Monitor Service Process Controller Thread> INFO:
[MPSVCCMN_10000] [MappingService] Mapping Service cancel operation completed..
2015-12-09 11:58:56.757 <Update and Monitor Service Process Controller Thread> INFO:
[WorchSS_10006] The stop action on the application listener was aborted.
2015-12-09 11:58:57.197 <Update and Monitor Service Process Controller Thread> SEVERE:
Internal error. The DTM process encountered an exception because of the following error:
[com.informatica.jdtm.oop.OOPPPUncheckedException: [JDTM_10014] Internal error. The DTM
process encountered an exception because of the following error:
[com.informatica.dtm.transport.DTFUncheckedException: [JDTM_10005] Internal error. The
Integration Service could not create the DTM process. Contact Informatica Global
Customer Support.]. Contact Informatica Global Customer Support.]. Contact Informatica
Global Customer Support.
2015-12-09 11:58:57.199 <Update and Monitor Service Process Controller Thread> INFO:
Shutting down statistics manager. Forced= 'true'.
2015-12-09 11:58:57.200 <Update and Monitor Service Process Controller Thread> INFO:
Statistics manager shut down.
2015-12-09 11:59:06.194 <MC_Thread_Factory-Thread_1> WARNING: [MonSDK_10045] Thread
[MC_Thread_Factory-Thread_1] is interrupted.
```

```
java.lang.InterruptedException
at java.lang.Object.wait(Native Method)
at com.informatica.isp.monitoringservice.sdk.common.mrspool.StatsMRSServiceConnectionPool
$StatsRepoConnectionPoolMonitoringTask.run(StatsMRSServiceConnectionPool.java:169)
at java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:471)
at java.util.concurrent.FutureTask.run(FutureTask.java:262)
at java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1145)
at java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:615)
at java.lang.Thread.run(Thread.java:745)
2015-12-09 11:59:06.197 <MC_Thread_Factory-Thread_1> INFO: [MonSDK_10049] Received a
request to stop thread [MC_Thread_Factory-Thread_1]. The thread will  stop now.
2015-12-09 11:59:07.236 <Update and Monitor Service Process Controller Thread> INFO:
[DS_10047] Deinitializing the Integration Service.
2015-12-09 11:59:07.654 <Update and Monitor Service Process Controller Thread> INFO:
[DS_10048] Finished deinitializing the Integration Service.
2015-12-09 11:59:07.728 <Update and Monitor Service Process Controller Thread> INFO:
[DS_10069] Deinitializing the web application client library.
2015-12-09 11:59:07.746 <Update and Monitor Service Process Controller Thread> INFO:
[DS_10068] Finished deinitializing the Integration Service web application.
2015-12-09 12:00:02.904 <localhost-startStop-1> INFO: [DS_10119] Integration Service
process version [10.0.0], build [848], date [11/13/15 5:34 PM].
Copyright (c) 1993-2015 Informatica LLC. All Rights Reserved.
```

Grant execute permissions to the following file:

```
$INFA_HOME/services/shared/bin/pmdtmsvc2
```

The profiling job step of a Database Scan job fails with an incorrect syntax message.

The profiling job step of a database scan job fails under the following conditions:

- The data source table or column names include dashes or hyphens.
- The Data Privacy Management Service uses ODBC drivers to connect to the data source.

Update the `odbc.ini` files on the machines that host the Data Integration Service. Configure the following entry:

```
EnableQuotedIdentifiers=1
```

Then, resume the job.

The SAP Scan job step fails because the user cannot open a file in the following directory: `/usr/sap/interfaces/`.

The Scan job fails because the user does not have the complete set of authorizations. The following excerpt shows a sample error message:

```
2017-08-09 22:12:19,102 [MonitorExecutionStatus] ERROR
com.infa.products.edc.scanners.profilescanner.BatchProfileExecutor- ParentName:
SAPSchema , SourceName: T006J , ErrorMessage: (java.util.concurrent.ExecutionException:
com.informatica.sdk.dtm.ExecutionException: [LDTM_0072] The following error occurred
while executing the RFC module: [     No authorization to open file /usr/sap/
interfaces/                  ])
2017-08-09 22:12:19,102 [MonitorExecutionStatus] ERROR
com.infa.products.edc.scanners.profilescanner.BatchProfileExecutor- ParentName:
SAPSchema , SourceName: T006T , ErrorMessage: (java.util.concurrent.ExecutionException:
```

To resolve this issue, verify that the SAP user has the required authorizations. For the required list of user authorizations, see the *Informatica Data Privacy Management Administrator Guide*.

The SAP Scan job step fails because the user cannot open a table.

The Scan job fails because the user does not have the authorization to access tables. The following excerpt shows a sample error message:

```
2017-08-09 17:21:23,698 [MonitorExecutionStatus] ERROR
com.infa.products.edc.scanners.profilescanner.BatchProfileExecutor- ParentName:
```

```
SAPSchema , SourceName: T006A , ErrorMessage: (java.util.concurrent.ExecutionException:
com.informatica.sdk.dtm.ExecutionException: [LDTM_0072] The following error occurred
while executing the RFC module: [User ( DPMUSER ) is not authorized to access the
specified table T006A])
```

To resolve this issue, verify that the SAP user has the required authorizations. For the required list of user authorizations, see the *Informatica Data Privacy Management Administrator Guide*.

Troubleshooting Subject Registry Scan Jobs

Read the following information before you troubleshoot or reset a Subject Registry Scan job:

- The number and types of child jobs created for a master Subject Registry job differ based on the scan configuration.
- If one or more domain discovery scans fail, the status of the poll domain discovery job and the master Subject Registry Scan job are set to failed.
- If the master job fails because a poll domain discovery job fails, identify and fix issues in the domain discovery configuration and manually resume individual domain discovery jobs. After they finish successfully, resume the master Subject Registry job.
- If a golden data store subject scan fails, the status of the subject scan job, the Scan Golden Data Stores job, and the master job are set to failed. Identify and fix the issues in the subject scan and then resume the master job. The master job triggers the required jobs to finish the scan.
- If a transaction data store subject scan fails, the status of the specific subject scan and the Scan Transaction Data Stores job are set to failed and the rest of the scans continue. The status of the master job is then set to failed. Identify and fix the issue in the subject scan and then resume the master job. The master job triggers the failed jobs.

Protection

If you encounter a problem while performing tasks to protect sensitive data inData Privacy Management, review the following list of previously reported issues and their solutions before contacting Informatica Global Customer Support.

A protection job that uses a Persistent Data Masking protection extension fails at the ExportProjectXML job step.

The ExportProjectXML job step log file lists the following error or a similar error: `cannot insert NULL into (ILM_PLAN_PROPERTY&quot ;.& quot;VALUE")`. This problem might occur for one of the following reasons:

- The TDM_CONNECTION is not created in the Test Data Management (TDM) application.
  Before running the first protection job that includes a Persistent Data Masking protection extension login to TDM. The first time you log in, TDM creates the TDM_CONNECTION and sets it as a storage connection.
- A primary key or unique index is not configured for the sources on which data masking is applied.
  Verify that the sources with data masking all contain a primary key or unique index.
- One or all of the following data attributes might be missing: precision, datatype, and scale.
  Verify that the source data contains the required attributes for masking.

A protection job that uses a Persistent Data Masking protection extension and a Microsoft Azure SQL database, Microsoft Azure SQL data warehouse, Microsoft SQL Server, or Sybase source fails when unique substitution rules or representative constrained shuffle rules are specified in the job.

This problem might occur because the storage connection is a type other than SQL Server and the dictionary connection is a Relational SQL Server type. If the SQL environment variable is set to SQL Server, the `SET QUOTED_IDENTIFIER ON` connection is then set to Oracle, which causes the protection job to fail.

In the PowerCenter Workflow Manager, edit the ODBC connections and remove the `SET QUOTED_IDENTIFIER ON` property from the SQL environment variable if the Table section in the job step log file lists the following error:

```
ERROR : Sat Sep 09 02:06:20 2017 [pmilmcmnlibs_1762] : TRANSF_1_1_1 [ERROR] SET
QUOTED_IDENTIFIER ON
ORA-00922: missing or invalid option
Database driver error...
Function Name : executeDirect
SQL Stmt : SET QUOTED_IDENTIFIER ON
Oracle Fatal Error
Database driver error...
Function Name : ExecuteDirect
Oracle Fatal Error
```

A protection job that uses a Persistent Data Masking protection extension fails at the ExecuteWorkflow job step with invalid data.

In the Data Privacy Management **Jobs** workspace, view the Job Details and select the ExecuteWorkflow job step. In the **Tables** pane for the job ID, download the session log for the table that caused the job to fail. The session log contains the following message: `The session reached the error threshold for the Data Masking transformation.`

This error occurs because some data in the table is not in a valid format. PowerCenter validates data before applying the masking rules for columns with Email address, Phone, IP address, SSN, Credit Card, and URL datatypes. To remedy this error, apply one of the following methods:

- Verify that columns with these datatypes contain data that is in the correct format.
- Update the Stop on Error value to a higher number if ignoring errors is an acceptable option.
- Update the Stop on Error value to 0 to ignore all data errors.
- Update the error and null handling parameters in the masking rule so that the protection extension will proceed when it encounters this problem.

A protection job that uses a Persistent Data Masking protection extension fails when the job includes metadata with special characters or letters in camel case.

This problem might occur because the protection job was not configured to enable special characters or letters in camel case.

In the Data Privacy Management **Tasks** workspace, select the task and then select **Actions** > **Schedule Protection Job**. In the **Advanced Settings** section on the **Runtime Configuration** page, ensure that the **Enable Special Characters in Metadata** check box is selected.

Scans

If you encounter a problem during a scan, review the following list of previously reported issues and their solutions before contacting Informatica Global Customer Support.

When scanning an unstructured data store, the scan job fails at the Load job step even though the test connection for the data store is successful.

This issue occurs because the source directory is not mounted on the Data Privacy Management and Hadoop nodes.

Mount the source directory on the Data Privacy Management and Hadoop nodes. Verify that the Path property on the data store UI page shows the correct mount point.

When scanning an unstructured data store, the Load job step keeps running even though the test connection for the data store is successful.

This issue might be because of one of the following reasons:

- The CIFS or NFS mount servers that hosts the source might be down.
- If the server is CIFS, the password for the user account that accesses the unstructured data might have changed.

Check if the CIFS and NFS mount server is down. If the server is CIFS and the password has changed, remount the directory on the Hadoop cluster.

When scanning a Microsoft OneDrive or SharePoint data store with a remote agent, I get a 429 error.

This issue occurs when there is more than one request to scan the data store at the same time.

For Microsoft OneDrive data stores, perform the following steps to increase the retry count and retry interval of the Scan job, and then run the scan again:

1. Create a file named `onedrive-connector.properties` and save the file in the following directory: `siagent/WEB-INF/classes/`
2. Add the following two properties in the file:

```
od.retryCount=<number of times to retry the failed request>
od.retryWaitTime=<number of milliseconds between retries>
```

For Microsoft SharePoint data stores, perform the following steps to increase the retry count and retry interval of the Scan job, and then run the scan again:

1. Create a file named `sharepoint-connector.properties` and save the file in the following directory: `siagent/WEB-INF/classes/`
2. Add the following two properties in the file:

```
sp.retryCount=<number of times to retry the failed request>
sp.retryWaitTime=<number of milliseconds between retries>
```

Also, if files or folders in Microsoft OneDrive or SharePoint data stores contain a pound symbol (#) at the beginning of a file or folder name, the Browse job step might fail or enter an infinite loop.

Remove the pound symbol (#) from all file and folder names included in the Scan job, and run the scan again.

When I run a PCRS scan, the Collect Connections job step completes with a warning. The data store is not created in Enterprise Unified Metadata.

This issue occurs when Data Privacy Management cannot connect to Enterprise Unified Metadata.

Verify that the connection properties to Enterprise Unified Metadata on the data store UI page are correct. Then scan the data store again.

Subject Registry

If you encounter a problem while creating or updating the Subject Registry configuration, review the following list of previously reported issues and their solutions before contacting Informatica Global Customer Support.

When I select the Discover Subject or Link Subject options from the Scan Type list, I get a warning message.

Data Privacy Management validates the Subject Registry configuration in the following ways:

- Data Privacy Management verifies settings that have a predefined set of values and numeric value settings that are bound by a specific range. For example, Data Privacy Management verifies that the MatchType setting is specified with a valid value such as "Fuzzy." If the setting is not valid, a warning message appears.
- Data Privacy Management verifies references to other settings in the entity file. For example, it verifies that the field name that is specified in the MatchFields section of the entity file, such as FieldName, matches a field that is defined in EntityFields section of the entity file. If Data Privacy Management cannot find a match, a warning message appears.
- Data Privacy Management verifies the configuration against other objects defined in Data Privacy Management. For example, it verifies that the FieldName specified in the EntityFields section of the entity file is a valid data domain name. If the data domain that matches the field name is not present, a warning message appears.

When a warning message appears, review the following Subject Registry configuration settings in the entity file:

- Verify that settings that have predefined values are set to a valid value.
- Verify that numeric value settings are specified within the valid range.
- Verify references to other settings within the entity file.
- Verify that data domain names and other object names are valid.

APPENDIX B

Updating Keystore and Truststore Certificates to Maintain Secure Communication

This appendix includes the following topics:

- Overview, 649
- Updating the Keystore and Truststore with Informatica Certificates, 649
- Updating the Keystore and Truststore with OpenSSL-Generated Certificates, 650

Overview

When you need to change the current keystore and truststore certificates for a business reason or because the certificates have expired, import the new certificates to maintain secure communication. You can import either the default certificate from Informatica or a certificate you created using OpenSSL.

Note: To create new keystore and truststore files, refer to the Informatica How-To Article *How to Create Keystore and Truststore Files for Secure Communication in the Informatica Domain*.

Updating the Keystore and Truststore with Informatica Certificates

Perform the following steps to update the keystore and truststore with Informatica certificates.

1. Shut down the Informatica domain.
2. Backup the existing keystore certificates and PEM files in the following locations:
   - `$INFA_HOME/tomcat/conf/Default.keystore`
   - `$INFA_HOME/services/shared/security/`
3. Delete the expired or old certificates from the keystore and truststore.

Use the following commands:

- `keytool -delete -alias infa_dflt -keystore $INFA_HOME/tomcat/conf/Default.keystore -storepass changeit`
- `keytool -delete -alias infa_dflt -keystore $INFA_HOME/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL`

4. Generate or locate the new certificate.

   Use the following command to generate the new certificate:

   ```
   keytool -genkey -alias infa_dflt -keyalg RSA -keypass changeit -storepass changeit -keystore $INFA_HOME/tomcat/conf/Default.keystore -dname CN=<host name>,OU=<organizational unit>,O=<organization>,L=<city>,S=<state>,C=<two-letter country code>
   ```

   Example:

   ```
   keytool -genkey -alias infa_dflt -keyalg RSA -keypass changeit -storepass changeit -keystore $INFA_HOME/tomcat/conf/Default.keystore -dname CN=psvilxilmt01.informatica.com,OU=Data Privacy Management,O=Informatica,L=RedwoodCity,S=California,C=US
   ```

5. Import the new certificate to the `Default.keystore` file.

   Use the following command:

   ```
   keytool -import -noprompt -file <new certificate name> -alias infa_dflt -keystore $INFA_HOME/tomcat/conf/Default.keystore -storepass changeit
   ```

6. Export the certificate from the `Default.keystore` file.

   Use the following command:

   ```
   keytool -export -keystore $INFA_HOME/tomcat/conf/Default.keystore -alias infa_dflt -storepass changeit -file $INFA_HOME/tomcat/conf/infa_dflt.cert
   ```

7. Import the certificate to the truststore.

   Use the following command:

   ```
   keytool -import -noprompt -file $INFA_HOME/tomcat/conf/infa_dflt.cert -alias infa_dflt -keystore $INFA_HOME/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL
   ```

8. For all the Hadoop nodes, delete the existing `infa_truststore.jks` file from the `/opt/Data Privacy Management` folder.
9. For all the Hadoop nodes, copy the new truststore file `$INFA_HOME/services/shared/security/infa_truststore.jks` to the `/opt/Data Privacy Management` folder.
10. Start the Informatica domain.

Updating the Keystore and Truststore with OpenSSL-Generated Certificates

Perform the following steps to update the keystore and truststore with certificates generated by OpenSSL.

1. Shut down the Informatica domain.
2. Backup the existing keystore certificates and PEM files in the following locations:
   - `$INFA_HOME/tomcat/conf/Default.keystore`

- `$INFA_HOME/services/shared/security/`

3. Delete the expired or old certificates from the keystore and truststore.

   Use the following commands:

   - `keytool -delete -alias infa_dflt -keystore $INFA_HOME/tomcat/conf/Default.keystore -storepass changeit`
   - `keytool -delete -alias infa_dflt -keystore $INFA_HOME/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL`

4. Generate the new certificate in PEM format.

   Use the following command:

   ```
   openssl req -newkey rsa:2048 -nodes -keyout key.pem -x509 -days 365 -out certificate.pem -sha256 -x509
   ```

5. Convert the certificate from PEM to PKCS12 format.

   Use the following command:

   ```
   openssl pkcs12 -inkey key.pem -in certificate.pem -export -out certificate.p12 -name <name>
   ```

6. Import the new certificate to the `Default.keystore` file.

   Use the following command:

   ```
   keytool -v -importkeystore -srckeystore certificate.p12 -srcstoretype PKCS12 -destkeystore $INFA_HOME/tomcat/conf/Default.keystore -deststoretype JKS
   ```

7. Export the certificate from the `Default.keystore` file.

   Use the following command:

   ```
   keytool -export -keystore $INFA_HOME/tomcat/conf/Default.keystore -alias <alias_name> -storepass changeit -file $INFA_HOME/tomcat/conf/infa_dflt.cert
   ```

8. Import the certificate to the truststore.

   Use the following command:

   ```
   keytool -import -noprompt -file $INFA_HOME/tomcat/conf/infa_dflt.cert -alias <alias_name> -keystore $INFA_HOME/services/shared/security/infa_truststore.jks -storepass pass2038@infaSSL
   ```

9. For all the Hadoop nodes, delete the existing `infa_truststore.jks` file from the `/opt/Secure@Source` folder.
10. For all the Hadoop nodes, copy the new truststore file `$INFA_HOME/services/shared/security/infa_truststore.jks` to the `/opt/Secure@Source` folder.
11. Start the Informatica domain.

INDEX

A

actions
  common properties 377
  copying 386
  creating 385
  custom properties 377
  deleting 386
  editing 386
  email properties 378
  managing 385
  overview 375
  placeholders 383
  service management properties 380
  system log properties 382
agent scan
  settings 41
Always Sensitive
  scope 238
Always Sensitive/Never Sensitive
  guidelines for rules 239
  list page 237
  overview 235
Always Sensitive/Never Sensitive rule
  management 237
Always Sensitive/Never Sensitive rules
  activating 244
  deleting 245
  editing 244
  inactivating 244
anomalies
  anomalous factors 361
anomalous factors
  anomalies 361
anomaly detection
  prerequisites 347
  workspace 348

C

classification policies
  copying 234
  creating 229
  data domain match conditions 225
  data domains 224
  deleting 234
  details page 222
  editing 233
  exporting 229
  in scan jobs 227
  managing 228
  overview 221
  properties 223
connection properties
  active directory 129
connection properties *(continued)*
  additional 128
  Amazon Redshift 131
  Amazon S3 133
  Apache Cassandra 136
  Cloudera Navigator 138
  data store 127
  database management 139
    email server
      Office 365 Outlook 144
  file system 145
  Google BigQuery 148
  Google Drive 149
  HDFS 150
  Hive 155
  Informatica Cloud 158
  Informatica Data Engineering Integration 160
  Informatica PowerCenter on IBM Db2 162
  Informatica PowerCenter on Microsoft SQL Server 166
  Informatica PowerCenter on Oracle 169
  Microsoft Azure Blob Storage 173
  Microsoft Azure Data Lake 175
  Microsoft Azure SQL Data Warehouse 181
  Microsoft Azure SQL Database 178
  Microsoft OneDrive 183
  Microsoft SharePoint 187
  Salesforce 190
  SAP 191
  SAP S/4HANA 194
  Snowflake 196
  SQL Server Integration Services 199
custom task
  subject registry 627

D

dashboard
  data domain details page 532
  data domains list page 534
  departments page 541
  indicators, discovery bar 515
  indicators, protection status 517
  indicators, residual risk 518
  indicators, risk score 515
  indicators, sensitivity level 518
  proliferation page 542
  sensitive data by data store group page 545
  sensitive data locations page 548
  sensitive data proliferation by classification policy page 546
  sensitive data proliferation by data store group page 547
  user access page 564
  user activity page 565
  user profile page 567
  users page 570

data breach reports
exporting 117
data domains
conformance score 212
copying 219
creating 214
data match condition 209
deleting 220
delinking associations 219
details page 204
editing 219
exporting 215
importing 215
management 213
metadata match condition 207
overview 202
properties 205
synchronizing 220
workspace 203
data privacy
data breach reports 117
data privacy management
overview 23
data store details
exporting 555
data store groups
creating 62
managing 63
managing assignments 63
system-defined 62
data stores
bulk update 96
copying 93
creating 91
deleting 124
details page 88
editing 94
exporting aliases 119
exporting configurations 119
fixing 94
importing 97
importing aliases 106
importing Catalog metadata 115
importing catalog resources
importing catalog resources
importing catalog resources
data stores 116
importing connection assignments 114
importing data owners 108
importing lineage 112
importing protection status 109
list page 87
location assignments 59
management 90
overview 86
proliferation 542
protection status 525
resetting classification results 95
status filters 89
summary 89
testing a remote agent connection 92
decryption
rollback task 506
DSAR
task properties 485
DSAR report
downloading 625
errors 625
formats 620
DSAR report *(continued)*
preparing 625
DSAR task
details 490

E

email task
subject registry 628
encryption
overview 592
tasks, configuring 504
tasks, decrypting 506
techniques 601
encryption rules
copying 603
creating 601
deleting 601
details 598
editing 601
exporting 602
importing 603
managing 601
overview 597
properties 599
workspace 598
encryption tasks
importing protection status 109
extensions
common properties 68
creating 74
custom 69
deleting 75
editing 75
email 69
managing 73
overview 65
plugins 66
protection, adding to data domains 74
protection, encryption 70
protection, PDM - Big Data 70
protection, PDM - Remote Domain 71
protection, supported data stores 66
service management 72
system log 73
types 66
workspace 68

G

global sensitivity status
exporting 245
importing 240–242

I

import
locations 55
importing data store connections 114

J

jobs
big data 291

jobs *(continued)*
cloud 293
data integration 294
database management 296
downloading rejected records 310
evaluate security policy 298
exporting 309
file management 298
filters 285
import catalog results 300
incremental scan 301
orchestration 301
overview 284
pausing 308
protection 302
protection, scheduling 508
remote agent scan job report 310
resuming 308
scan job report 309
stopping 308
subject data report purge 305
sync catalog updates 307
task status 495
terminating 309
troubleshooting 641
types 287
workspace 284

L

legal hold 612
legal holds
applying 624
removing 624
locations
copying 58
data store assignments 59
editing 58
importing 55
properties 52
sensitive data 548
workspace 51

M

management
privacy dashboard 637
manual actions
custom 412
email 423
from Anomaly Detection workspace 389, 403, 412, 424, 437
from Proliferation page 391, 405, 414, 426, 439
from Security Policy Violations workspace 393, 406, 416, 428, 441
from Sensitive Fields page 395, 408, 418, 430, 443
from Top Data Domains list page 398, 409, 420, 432, 445
from Top Data Stores grid page 411, 434, 446
from Top Data Stores Grid page 400, 422
overview 388
protection 403
service management 389
system log 436
master category
assign 617
import categories 617
override 558
master purpose
assign 617
import purpose 617
override 558

N

Never Sensitive
scope 238

O

override
category 558
purpose 558
overview workspace
data domain details page 532
data domains list page 534
data stores grid page 536
data stores list page 538
departments page 541
export files 579
exporting data 579
indicators, discovery bar 515
indicators, protection status 517
indicators, residual risk 518
indicators, risk score 515
indicators, sensitivity level 518
proliferation page 542
sensitive data by data store group page 545
sensitive data locations page 548
sensitive data proliferation by classification policy page 546
sensitive data proliferation by data store group page 547
user access page 564
user activity page 565
user profile page 567
users page 570

P

privacy
DSAR report 490
DSAR tasks 485
privacy dashboard
export files 638
filtering 638
overview 629
summary indicators 630
workspace indicators 631
workspace management 637
Privacy dashboard
data stores by subjects 632
privacy dashboard indicators
subject data by location 634
subject requests by due date 636
subject requests by type 633
top data stores by subjects 631
privacy dashboard summary indicators
data stores 631
subject requests 630
subject types 631
subjects 630
proliferation page 542
protection
extensions, supported data stores 66

protection *(continued)*
overview 592
permissions 593
prerequisites 593
process flow 594
remote agents 78
protection extensions
adding to data domains 74
protection task
scheduling, PDM BDE 508
protection tasks
configuring 502
creating 388, 403
details 492
encryption rule properties 599
encryption techniques 601
extension plugins 66
extensions 66
list page 484
managing 502
overview 483
permissions 593
prerequisites 593
process flow 594
properties 487
reassigning 501
sensitive fields page 549
status 495
troubleshooting 645

R

remote agents
copying 82
creating 81
editing 82
exporting 82
filter 78
guidelines 77
importing 84
list page 78
managing 81
overview 76
properties 80
protection 78, 81
publishing 84
subject registry 79, 81
risk simulation plans
copying 608
creating 607
creating, data stores grid page 536
details page 605
editing 609
exporting 608
list page 605
managing 607
overview 604
properties 606
workspace 604
rules
subset 238
super 238
rworkspaces
remote agents 77

S

scans
Amazon Redshift 267
Amazon S3 267
Azure Data Lake 267
big data 263
Cassandra 279
cloud 267
Cloudera Navigator 263
copying 258
creating 254
data integration 270
database management 274
deleting 258
editing 258
exporting 258
file management 276
file system 276
Google Drive 276
HDFS 263
Hive 263
IBM Db2 274
Informatica Cloud 270
Informatica Data Engineering Integration 270
Informatica PowerCenter 270
JDBC 274
managing 254
Microsoft Azure Blob Storage 267
Microsoft Azure Data Warehouse 267
Microsoft Azure Database 267
Microsoft OneDrive 276
Microsoft SharePoint 276
Microsoft SQL Server 274
Netezza 274
noSQL 279
Oracle 274
overview 246
properties 251
running 255
Salesforce 267
SAP 260
SAP HANA 274
Snowflake 267
SQL Server Integration Services 270
Sybase 274
Teradata 274
scope
Always Sensitive 238
Never Sensitive 238
security dashboard
customizing 572
data stores grid page 536
data stores list page 538
enterprise-level summary 514
export files 579
exporting data 579
filtering 573
indicators, sensitive data by data store groups 520
indicators, sensitive data by location 521
indicators, sensitive data proliferation 522
indicators, top data domains 523
indicators, top data stores 525
indicators, top departments 527
indicators, top users 529
indicators, user access 531
managing 572
sensitive fields page 549, 558

security policies
action placeholders 383
copying 458
creating 456
decryption 451
details 452
editing 458
properties 453
types 450
workspace 452
security policy violations
workspace 466
sensitive columns
managing 235
sensitive data
locations 548
sensitive data by data store group page 545
sensitive fields
editing properties 554
exporting 554
importing 556
proliferation 542
properties pane 551
sensitive fields page
email server 562
filter 576
sensitive files
properties pane 560
sensitive files page
filter 577
service management task
subject registry 626
settings
agent scan 41
anomaly factor weights 44
anomaly severity ranges 46
Axon integration 47
conformance ranges 45
custom risk score factors 43
dashboard 41
data privacy 41
risk cost 45
risk score factors 42
sensitivity levels 42
theme color 40
settings workspace
properties 39
subect details page
actions 623
subject details
exporting 623
subject details page 618
subject registry
categories 616
category 613
custom task 627
data store details page 619
email task 628
exporting data 616
legal hold 612
overview 611
purpose 613, 616
remote agent details 79
residency 613
search 615
ServiceNow task 626
subject details 618
summary 616
subject registry *(continued)*
troubleshooting 648
subset rules 238
super rules 238
suppression rules
workspace 369
system tasks
properties 487

T

tasks
attachments, adding 498
attachments, deleting 498
custom 489
data subject, running 502
deleting 499
DSAR details 490
editing 499, 500, 506
email 490
encryption, configuring 504
exporting 500
filter 485
list page 484
managing 497
marking as closed 501
overview 483
PDM - Big Data, editing 505
PDM - Remote Domain, editing 505
properties 487
protection 492
protection, configuring 502
protection, decrypting 506
protection, marking as completed 506
protection, marking as configured 507
protection, reconfiguring 507
protection, rejecting 508
protection, scheduling jobs 508
protection, viewing configuration 511
reassigning 501
service management 493
statuses 495
system log 494
system, running 502
workspace 484
troubleshooting
subject registry 648

U

user interface 29
users
page 570
user access page 564
user activity page 565
user profile page 567
Users
workspace 315

W

workspaces
actions 376
always sensitive/never sensitive 236
anomaly detection 348

workspaces *(continued)*
classification policies 221
data domains 203
data store groups 60
data stores 87
encryption rules 598
extensions 68
jobs 284
locations 51
overview 513
workspaces *(continued)*
risk simulation plans 604
scans 249
security policies 452
security policy violations 466
settings 36
subject registry 614
suppression rules 369
tasks 484